

Algebra: Chapter 0

Paolo Aluffi

**Graduate Studies
in Mathematics**

Volume 104



American Mathematical Society

Algebra:

Chapter 0

Algebra: Chapter 0

Paolo Aluffi

Graduate Studies
in Mathematics

Volume 104



American Mathematical Society
Providence, Rhode Island

Editorial Board

David Cox (Chair)

Steven G. Krantz

Rafe Mazzeo

Martin Scharlemann

2010 *Mathematics Subject Classification*. Primary 00–01; Secondary 12–01, 13–01, 15–01, 18–01, 20–01.

For additional information and updates on this book, visit
www.ams.org/bookpages/gsm-104

Library of Congress Cataloging-in-Publication Data

Aluffi, Paolo, 1960–

Algebra: chapter 0 / Paolo Aluffi.

p. cm. — (Graduate studies in mathematics ; v. 104)

Includes index.

ISBN 978-0-8218-4781-7 (alk. paper)

1. Algebra—Textbooks. I. Title.

QA154.3.A527 2009

512—dc22

2009004043

Copying and reprinting. Individual readers of this publication, and nonprofit libraries acting for them, are permitted to make fair use of the material, such as to copy select pages for use in teaching or research. Permission is granted to quote brief passages from this publication in reviews, provided the customary acknowledgment of the source is given.

Republication, systematic copying, or multiple reproduction of any material in this publication is permitted only under license from the American Mathematical Society. Permissions to reuse portions of AMS publication content are handled by Copyright Clearance Center's RightsLink® service. For more information, please visit: <http://www.ams.org/rightslink>.

Send requests for translation rights and licensed reprints to reprint-permission@ams.org.

Excluded from these provisions is material for which the author holds copyright. In such cases, requests for permission to reuse or reprint material should be addressed directly to the author(s). Copyright ownership is indicated on the copyright page, or on the lower right-hand corner of the first page of each article within proceedings volumes.

© 2009 by the American Mathematical Society. All rights reserved.

Reprinted with corrections by the American Mathematical Society, 2016.

The American Mathematical Society retains all rights
except those granted to the United States Government.

Printed in the United States of America.

∞ The paper used in this book is acid-free and falls within the guidelines
established to ensure permanence and durability.

Visit the AMS home page at <http://www.ams.org/>

10 9 8 7 6 5 4 3 2 21 20 19 18 17 16

Contents

Preface to the second printing	xv
Introduction	xvii
Chapter I. Preliminaries: Set theory and categories	1
§1. Naive set theory	1
1.1. Sets	1
1.2. Inclusion of sets	3
1.3. Operations between sets	4
1.4. Disjoint unions, products	5
1.5. Equivalence relations, partitions, quotients	6
Exercises	8
§2. Functions between sets	8
2.1. Definition	8
2.2. Examples: Multisets, indexed sets	10
2.3. Composition of functions	10
2.4. Injections, surjections, bijections	11
2.5. Injections, surjections, bijections: Second viewpoint	12
2.6. Monomorphisms and epimorphisms	14
2.7. Basic examples	15
2.8. Canonical decomposition	15
2.9. Clarification	16
Exercises	17
§3. Categories	18
3.1. Definition	18
3.2. Examples	20
Exercises	26

§4. Morphisms	27
4.1. Isomorphisms	27
4.2. Monomorphisms and epimorphisms	29
Exercises	30
§5. Universal properties	31
5.1. Initial and final objects	31
5.2. Universal properties	33
5.3. Quotients	33
5.4. Products	35
5.5. Coproducts	36
Exercises	38
Chapter II. Groups, first encounter	41
§1. Definition of group	41
1.1. Groups and groupoids	41
1.2. Definition	42
1.3. Basic properties	43
1.4. Cancellation	45
1.5. Commutative groups	45
1.6. Order	46
Exercises	48
§2. Examples of groups	49
2.1. Symmetric groups	49
2.2. Dihedral groups	52
2.3. Cyclic groups and modular arithmetic	54
Exercises	56
§3. The category $\mathbf{Grp}$	58
3.1. Group homomorphisms	58
3.2. $\mathbf{Grp}$: Definition	59
3.3. Pause for reflection	60
3.4. Products et al.	61
3.5. Abelian groups	62
Exercises	63
§4. Group homomorphisms	64
4.1. Examples	64
4.2. Homomorphisms and order	66
4.3. Isomorphisms	66
4.4. Homomorphisms of abelian groups	68
Exercises	69
§5. Free groups	70
5.1. Motivation	70
5.2. Universal property	71
5.3. Concrete construction	72
5.4. Free <i>abelian</i> groups	75

Exercises	78
§6. Subgroups	79
6.1. Definition	79
6.2. Examples: Kernel and image	80
6.3. Example: Subgroup generated by a subset	81
6.4. Example: Subgroups of cyclic groups	82
6.5. Monomorphisms	84
Exercises	85
§7. Quotient groups	88
7.1. Normal subgroups	88
7.2. Quotient group	89
7.3. Cosets	90
7.4. Quotient by normal subgroups	92
7.5. Example	94
7.6. $\text{kernel} \iff \text{normal}$	95
Exercises	95
§8. Canonical decomposition and Lagrange's theorem	96
8.1. Canonical decomposition	97
8.2. Presentations	99
8.3. Subgroups of quotients	100
8.4. HK/H vs. $K/(H \cap K)$	101
8.5. The index and Lagrange's theorem	102
8.6. Epimorphisms and cokernels	104
Exercises	105
§9. Group actions	108
9.1. Actions	108
9.2. Actions on sets	109
9.3. Transitive actions and the category $G\text{-Set}$	110
Exercises	113
§10. Group objects in categories	115
10.1. Categorical viewpoint	115
Exercises	117
Chapter III. Rings and modules	119
§1. Definition of ring	119
1.1. Definition	119
1.2. First examples and special classes of rings	121
1.3. Polynomial rings	124
1.4. Monoid rings	126
Exercises	127
§2. The category Ring	129
2.1. Ring homomorphisms	129
2.2. Universal property of polynomial rings	130
2.3. Monomorphisms and epimorphisms	132

2.4. Products	133
2.5. $\text{End}_{\text{Ab}}(G)$	134
Exercises	136
§3. Ideals and quotient rings	138
3.1. Ideals	138
3.2. Quotients	139
3.3. Canonical decomposition and consequences	141
Exercises	143
§4. Ideals and quotients: Remarks and examples. Prime and maximal ideals	144
4.1. Basic operations	144
4.2. Quotients of polynomial rings	146
4.3. Prime and maximal ideals	150
Exercises	153
§5. Modules over a ring	156
5.1. Definition of (left-) R -module	156
5.2. The category $R\text{-Mod}$	158
5.3. Submodules and quotients	160
5.4. Canonical decomposition and isomorphism theorems	162
Exercises	163
§6. Products, coproducts, etc., in $R\text{-Mod}$	164
6.1. Products and coproducts	164
6.2. Kernels and cokernels	166
6.3. Free modules and free algebras	167
6.4. Submodule generated by a subset; Noetherian modules	169
6.5. Finitely generated vs. finite type	171
Exercises	172
§7. Complexes and homology	174
7.1. Complexes and exact sequences	174
7.2. Split exact sequences	177
7.3. Homology and the snake lemma	178
Exercises	183
Chapter IV. Groups, second encounter	187
§1. The conjugation action	187
1.1. Actions of groups on sets, reminder	187
1.2. Center, centralizer, conjugacy classes	189
1.3. The Class Formula	190
1.4. Conjugation of subsets and subgroups	191
Exercises	193
§2. The Sylow theorems	194
2.1. Cauchy's theorem	194
2.2. Sylow I	196
2.3. Sylow II	197

2.4. Sylow III	199
2.5. Applications	200
Exercises	202
§3. Composition series and solvability	205
3.1. The Jordan-Hölder theorem	205
3.2. Composition factors; Schreier's theorem	207
3.3. The commutator subgroup, derived series, and solvability	210
Exercises	213
§4. The symmetric group	214
4.1. Cycle notation	214
4.2. Type and conjugacy classes in S_n	216
4.3. Transpositions, parity, and the alternating group	219
4.4. Conjugacy in A_n ; simplicity of A_n and solvability of S_n	220
Exercises	224
§5. Products of groups	226
5.1. The direct product	226
5.2. Exact sequences of groups; extension problem	228
5.3. Internal/semidirect products	230
Exercises	233
§6. Finite abelian groups	234
6.1. Classification of finite abelian groups	234
6.2. Invariant factors and elementary divisors	237
6.3. Application: Finite subgroups of multiplicative groups of fields	239
Exercises	240
Chapter V. Irreducibility and factorization in integral domains	243
§1. Chain conditions and existence of factorizations	244
1.1. Noetherian rings revisited	244
1.2. Prime and irreducible elements	246
1.3. Factorization into irreducibles; domains with factorizations	248
Exercises	249
§2. UFDs, PIDs, Euclidean domains	251
2.1. Irreducible factors and greatest common divisor	251
2.2. Characterization of UFDs	253
2.3. $\text{PID} \implies \text{UFD}$	254
2.4. Euclidean domain $\implies$ PID	255
Exercises	258
§3. Intermezzo: Zorn's lemma	261
3.1. Set theory, reprise	261
3.2. Application: Existence of maximal ideals	264
Exercises	265
§4. Unique factorization in polynomial rings	267
4.1. Primitivity and content; Gauss's lemma	268

4.2. The field of fractions of an integral domain	270
4.3. R UFD $\implies R[x]$ UFD	273
Exercises	276
§5. Irreducibility of polynomials	280
5.1. Roots and reducibility	281
5.2. Adding roots; algebraically closed fields	283
5.3. Irreducibility in $\mathbb{C}[x]$, $\mathbb{R}[x]$, $\mathbb{Q}[x]$	285
5.4. Eisenstein's criterion	288
Exercises	289
§6. Further remarks and examples	291
6.1. Chinese remainder theorem	291
6.2. Gaussian integers	293
6.3. Fermat's theorem on sums of squares	297
Exercises	300
Chapter VI. Linear algebra	305
§1. Free modules revisited	305
1.1. R -Mod	305
1.2. Linear independence and bases	306
1.3. Vector spaces	308
1.4. Recovering B from $F^R(B)$	309
Exercises	312
§2. Homomorphisms of free modules, I	314
2.1. Matrices	314
2.2. Change of basis	318
2.3. Elementary operations and Gaussian elimination	320
2.4. Gaussian elimination over Euclidean domains	323
Exercises	324
§3. Homomorphisms of free modules, II	327
3.1. Solving systems of linear equations	327
3.2. The determinant	328
3.3. Rank and nullity	333
3.4. Euler characteristic and the Grothendieck group	334
Exercises	338
§4. Presentations and resolutions	340
4.1. Torsion	340
4.2. Finitely presented modules and free resolutions	341
4.3. Reading a presentation	344
Exercises	347
§5. Classification of finitely generated modules over PIDs	349
5.1. Submodules of free modules	350
5.2. PIDs and resolutions	353
5.3. The classification theorem	354
Exercises	357

§6. Linear transformations of a free module	359
6.1. Endomorphisms and similarity	359
6.2. The characteristic and minimal polynomials of an endomorphism	361
6.3. Eigenvalues, eigenvectors, eigenspaces	365
Exercises	368
§7. Canonical forms	371
7.1. Linear transformations of free modules; actions of polynomial rings	371
7.2. $k[t]$ -modules and the rational canonical form	373
7.3. Jordan canonical form	377
7.4. Diagonalizability	380
Exercises	381
Chapter VII. Fields	385
§1. Field extensions, I	385
1.1. Basic definitions	385
1.2. Simple extensions	387
1.3. Finite and algebraic extensions	391
Exercises	397
§2. Algebraic closure, Nullstellensatz, and a little algebraic geometry	400
2.1. Algebraic closure	400
2.2. The Nullstellensatz	404
2.3. A little affine algebraic geometry	406
Exercises	414
§3. Geometric impossibilities	417
3.1. Constructions by straightedge and compass	417
3.2. Constructible numbers and quadratic extensions	422
3.3. Famous impossibilities	425
Exercises	427
§4. Field extensions, II	428
4.1. Splitting fields and normal extensions	429
4.2. Separable polynomials	433
4.3. Separable extensions and embeddings in algebraic closures	436
Exercises	438
§5. Field extensions, III	440
5.1. Finite fields	441
5.2. Cyclotomic polynomials and fields	445
5.3. Separability and simple extensions	449
Exercises	452
§6. A little Galois theory	454
6.1. The Galois correspondence and Galois extensions	454
6.2. The fundamental theorem of Galois theory, I	459
6.3. The fundamental theorem of Galois theory, II	461
6.4. Further remarks and examples	464
Exercises	466

§7. Short march through applications of Galois theory	468
7.1. Fundamental theorem of algebra	468
7.2. Constructibility of regular n -gons	469
7.3. Fundamental theorem on symmetric functions	471
7.4. Solvability of polynomial equations by radicals	474
7.5. Galois groups of polynomials	478
7.6. Abelian groups as Galois groups over $\mathbb{Q}$	479
Exercises	480
Chapter VIII. Linear algebra, reprise	483
§1. Preliminaries, reprise	483
1.1. Functors	483
1.2. Examples of functors	485
1.3. When are two categories 'equivalent'?	487
1.4. Limits and colimits	489
1.5. Comparing functors	492
Exercises	496
§2. Tensor products and the Tor functors	500
2.1. Bilinear maps and the definition of tensor product	501
2.2. Adjunction with Hom and explicit computations	504
2.3. Exactness properties of tensor; flatness	507
2.4. The Tor functors.	509
Exercises	511
§3. Base change	515
3.1. Balanced maps	515
3.2. Bimodules; adjunction again	517
3.3. Restriction and extension of scalars	518
Exercises	520
§4. Multilinear algebra	522
4.1. Multilinear, symmetric, alternating maps	522
4.2. Symmetric and exterior powers	525
4.3. Very small detour: Graded algebra	527
4.4. Tensor algebras	529
Exercises	532
§5. Hom and duals	535
5.1. Adjunction again	536
5.2. Dual modules	537
5.3. Duals of free modules	538
5.4. Duality and exactness	539
5.5. Duals and matrices; biduality	541
5.6. Duality on vector spaces	542
Exercises	543
§6. Projective and injective modules and the Ext functors	545
6.1. Projectives and injectives	546

6.2. Projective modules	547
6.3. Injective modules	548
6.4. The Ext functors	551
6.5. $\text{Ext}_{\mathbb{Z}}^*(G, \mathbb{Z})$	554
Exercises	555
Chapter IX. Homological algebra	559
§1. (Un)necessary categorical preliminaries	560
1.1. Undesirable features of otherwise reasonable categories	560
1.2. Additive categories	561
1.3. Abelian categories	564
1.4. Products, coproducts, and direct sums	567
1.5. Images; canonical decomposition of morphisms	570
Exercises	574
§2. Working in abelian categories	576
2.1. Exactness in abelian categories	576
2.2. The snake lemma, again	578
2.3. Working with ‘elements’ in a small abelian category	581
2.4. What is missing?	587
Exercises	589
§3. Complexes and homology, again	591
3.1. Reminder of basic definitions; general strategy	591
3.2. The category of complexes	594
3.3. The long exact cohomology sequence	597
3.4. Triangles	600
Exercises	602
§4. Cones and homotopies	605
4.1. The mapping cone of a morphism	605
4.2. Quasi-isomorphisms and derived categories	607
4.3. Homotopy	611
Exercises	614
§5. The homotopic category. Complexes of projectives and injectives	616
5.1. Homotopic maps are identified in the derived category	616
5.2. Definition of the homotopic category of complexes	618
5.3. Complexes of projective and injective objects	619
5.4. Homotopy equivalences vs. quasi-isomorphisms in $K(A)$	620
5.5. Proof of Theorem 5.9	624
Exercises	626
§6. Projective and injective resolutions and the derived category	628
6.1. Recovering A	629
6.2. From objects to complexes	631
6.3. Poor man’s derived category	635
Exercises	638

§7. Derived functors	641
7.1. Viewpoint shift	641
7.2. Universal property of the derived functor	643
7.3. Taking cohomology	645
7.4. Long exact sequence of derived functors	647
7.5. Relating $\mathcal{F}$, $L_i\mathcal{F}$, $R^i\mathcal{F}$	653
7.6. Example: A little group cohomology	655
Exercises	658
§8. Double complexes	661
8.1. Resolution by acyclic objects	662
8.2. Complexes of complexes	665
8.3. Exactness of the total complex	670
8.4. Total complexes and resolutions	672
8.5. Acyclic resolutions again and balancing Tor and Ext	675
Exercises	677
§9. Further topics	680
9.1. Derived categories	681
9.2. Triangulated categories	683
9.3. Spectral sequences	686
Exercises	695
Index	699

Preface to the second printing

In the occasion of the second printing of this book I have corrected the errors of which I am aware and implemented several other minor adjustments to the exposition. I take this opportunity to collectively thank all who had the patience to suggest corrections. Special thanks are due to Cihan Bahran and Amit Shah for particularly extensive comments. Ettore Aldrovandi, Selcan Aksoy, Maarten Bergvelt, Joseph Berner, Henk Brozius, Alex Cameron, Owen Colman, Izzet Coskun, Lou van den Dries, Georges Elencwajg, Neil Epstein, David Feuer, Thomas Frey, Mark Girard, William Giuliano, Guillaume Gouge, Venkata Krishna Kishore Gangavarapu, Hamza Hajji, Joseph Hoisington, Tyler Holden, Amanda Hussey, Stephan Kirchner, Jay Kopper, Jeremy Kun, Janis Lazovskis, Xia Liao, Trevor Leslie, Michael Maloney, Justin Mohr, Christopher Perez, Vanessa Radzinski, Aamir Rasheed, Matt Sourisseau, Avery St. Dizier, Brennan Vincent, Jon Weed, Dylan Wilson, Jonathan Wolf, Bei Ye, Mirroslav Yotov, Hui Yu, Chris Ziemko all contributed errata that have been incorporated in the current edition.

October 2015

Paolo Aluffi

Introduction

This text presents an introduction to algebra suitable for upper-level undergraduate or beginning graduate courses. While there is a very extensive offering of textbooks at this level, in my experience teaching this material I have invariably felt the need for a self-contained text that would start ‘from zero’ (in the sense of not assuming that the reader has had substantial previous exposure to the subject) but that would impart from the very beginning a rather modern, categorically minded viewpoint and aim at reaching a good level of depth. Many textbooks in algebra brilliantly satisfy some, but not all, of these requirements. This book is my attempt at providing a working alternative.

There is a widespread perception that categories should be avoided at first blush, that the abstract language of categories should not be introduced until a student has toiled for a few semesters through example-driven illustrations of the nature of a subject like algebra. According to this viewpoint, categories are only tangentially relevant to the main topics covered in a beginning course, so they can simply be mentioned occasionally for the general edification of the reader, who will in time learn about them (by osmosis?). Paraphrasing a reviewer of a draft of the present text, ‘Discussions of categories at this level are the reason why God created appendices.’

It will be clear from a cursory glance at the table of contents that I think otherwise. In this text, categories are introduced on page 18, after a scant reminder of the basic language of naive set theory, for the main purpose of providing a context for universal properties. These are in turn evoked constantly as basic definitions are introduced. The word ‘universal’ appears at least 100 times in the first three chapters.

I believe that awareness of the categorical language, and especially some appreciation of universal properties, is particularly helpful in approaching a subject such as algebra ‘from the beginning’. The reader I have in mind is someone who has reached a certain level of mathematical maturity—for example, who needs no

special assistance in grasping an induction argument—but may have only been exposed to algebra in a very cursory manner. My experience is that many upper-level undergraduates or beginning graduate students at Florida State University and at comparable institutions fit this description. For these students, seeing the many introductory concepts in algebra as instances of a few powerful ideas (encapsulated in suitable universal properties) helps to build a comforting unifying context for these notions. The amount of categorical language needed for this catalyzing function is very limited; for example, functors are not really necessary in this acclimatizing stage.

Thus, in my mind the benefit of this approach is precisely that it helps a true beginner, if it is applied with due care. This is my experience in the classroom, and it is the main characteristic feature of this text. The very little categorical language introduced at the outset informs the first part of the book, introducing in general terms groups, rings, and modules. This is followed by a (rather traditional) treatment of standard topics such as Sylow theorems, unique factorization, elementary linear algebra, and field theory. The last third of the book wades into somewhat deeper waters, dealing with tensor products and Hom (including a first introduction to Tor and Ext) and including a final chapter devoted to homological algebra. Some familiarity with categorical language appears indispensable to me in order to appreciate this latter material, and this is hopefully uncontroversial. Having developed a feel for this language in the earlier parts of the book, students find the transition into these more advanced topics particularly smooth.

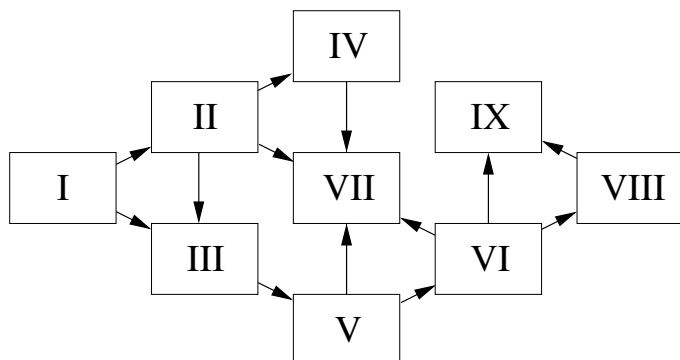
A first version of this book was essentially a careful transcript of my lectures in a run of the (three-semester) algebra sequence at FSU. The chapter on homological algebra was added at the instigation of Ed Dunne, as were a very substantial number of the exercises. The main body of the text has remained very close to the original ‘transcript’ version: I have resisted the temptation of expanding the material when revising it for publication. I believe that an effective introductory textbook (this is Chapter 0, after all...) should be realistic: it must be possible to cover in class what is covered in the book. Otherwise, the book veers into the ‘reference’ category; I never meant to write a reference book in algebra, and it would be futile (of me) to try to ameliorate excellent available references such as Lang’s ‘Algebra’.

The problem sets will give an opportunity to a teacher, or any motivated reader, to get quite a bit beyond what is covered in the main text. To guide in the choice of exercises, I have marked with a $\triangleright$ those problems that are directly referenced from the text, and with a $\neg$ those problems that are referenced from other problems. A minimalist teacher may simply assign all and only the $\triangleright$ problems; these do nothing more than anchor the understanding by practice and may be all that a student can realistically be expected to work out while juggling TA duties and two or three other courses of similar intensity as this one. The main body of the text, together with these exercises, forms a self-contained presentation of essential material. The other exercises, and especially the threads traced by those marked with $\neg$, will offer the opportunity to cover other topics, which some may well consider just as essential: the modular group, quaternions, nilpotent groups, Artinian rings, the Jacobson radical, localization, Lagrange’s theorem on four squares, projective space and

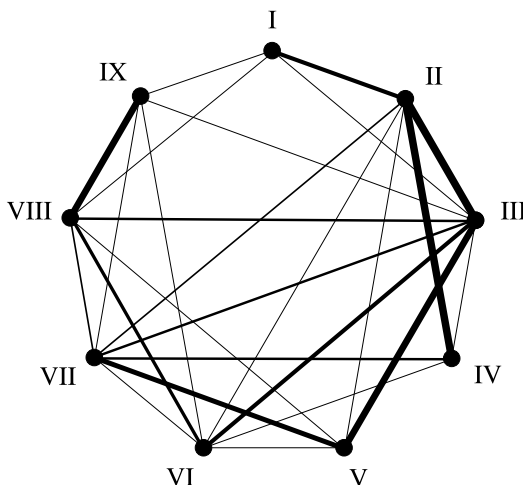
Grassmannians, Nakayama's lemma, associated primes, the spectral theorem for normal operators, etc., are some examples of topics that make their appearance in the exercises. Often a topic is presented over the course of several exercises, placed in appropriate sections of the book. For example, 'Wedderburn's little theorem' is mentioned in Remark III.1.16 (that is: Remark 1.16 in Chapter III); particular cases are presented in Exercises III.2.11 and IV.2.17, and the reader eventually obtains a proof in Exercise VII.5.14, following preliminaries given in Exercises VII.5.12 and VII.5.13. The $\neg$ label and perusal of the index should facilitate the navigation of such topics. To help further in this process, I have decorated every exercise with a list (added in square brackets) of the places in the book that refer to it. For example, an instructor evaluating whether to assign Exercise V.2.25 will be immediately aware that this exercise is quoted in Exercise VII.5.18, proving a particular case of Dirichlet's theorem on primes in arithmetic progressions, and that this will in turn be quoted in §VII.7.6, discussing the realization of abelian groups as Galois groups over $\mathbb{Q}$.

I have put a high priority on the requirement that this should be a self-contained text which essentially crosses all t's and dots all i's, and does not require that the reader have access to other texts while working through it. I have therefore made a conscious effort to *not* quote other references: I have avoided as much as possible the exquisitely tempting escape route 'For a proof, see . . . ' This is the main reason why this book is as thick as it is, even if so many topics are *not* covered in it. Among these, commutative algebra and representation theory are perhaps the most glaring omissions. The first is represented to the extent of the standard basic definitions, which allow me to sprinkle a little algebraic geometry here and there (for example, see §VII.2), and of a few slightly more advanced topics in the exercises, but I stopped short of covering, e.g., primary decompositions. The second is missing altogether. It is my hope to complement this book with a 'Chapter 1' in an undetermined future, where I will make amends for these and other shortcomings.

By its nature, this book should be quite suitable for self-study: readers working on their own will find here a self-contained starting point which should work well as a prelude to future, more intensive, explorations. Such readers may be helped by the following '9-fold way' diagram of logical interdependence of the chapters:



This may however better reflect my original intention than the final product. For a more objective gauge, this alternative diagram captures the web of references from a chapter to earlier chapters, with the thickness of the lines representing (roughly) the number of references:



With the self-studying reader especially in mind, I have put extra effort into providing an extensive index. It is not realistic to make a fanfare for each and every new term introduced in a text of this size by an official ‘definition’; the index should help a lone traveler find the way back to the source of unfamiliar terminology.

Internal references are handled in a hopefully transparent way. For example, Remark III.1.16 refers to Remark 1.16 in Chapter III; if the reference is made from within Chapter III, the same item is called Remark 1.16. The list in brackets following an exercise indicates other exercises or sections in the book referring to that exercise. For example, Exercise 3.1 in Chapter I is followed by [5.1, §VIII.1.1, §IX.1.2, IX.1.10]: this alerts the reader that there are references to this problem in Exercise 5.1 in Chapter I, section 1.1 in Chapter VIII, section 1.2 in Chapter IX, and Exercise 1.10 in Chapter IX (and nowhere else).

Acknowledgments. My debt to Lang’s book, to David Dummit and Richard Foote’s ‘Abstract Algebra,’ or to Artin’s ‘Algebra’ will be evident to anyone who is familiar with these sources. The chapter on homological algebra owes much to David Eisenbud’s appendix on the topic in his ‘Commutative Algebra’, to Gelfand-Manin’s ‘Methods of homological algebra’, and to Weibel’s ‘An introduction to homological algebra’. But in most cases it would simply be impossible for me to retrace the original source of an expository idea, of a proof, of an exercise, or of a specific pedagogical emphasis: these are all likely offsprings of ideas from any one of these and other influential references and often of associations triggered by following the manifold strands of the World Wide Web. This is another reason why, in a spirit of equanimity, I resolved to essentially avoid references altogether. In any case, I believe all the material I have presented here is standard, and I only retain absolute ownership of every error left in the end product.

I am very grateful to my students for the constant feedback that led me to write this book in this particular way and who contributed essentially to its success in my classes. Some of the students provided me with extensive lists of typos and outright mistakes, and I would especially like to thank Kevin Meek, Jay Stryker, and Yong Jae Cha for their particularly helpful comments. I had the opportunity to try out the material on homological algebra in a course given at Caltech in the fall of 2008, while on a sabbatical from FSU, and I would like to thank Caltech and the audience of the course for their hospitality and the friendly atmosphere. Thanks are also due to MSRI for hospitality during the winter of 2009, when the last fine-tuning of the text was performed.

A few people spotted big and small mistakes in preliminary versions of this book, and I will mention Georges Elencwajg, Xia Liao, and Mirroslav Yotov for particularly precious contributions. I also commend Arlene O'Sean and the staff at the AMS for the excellent copyediting and production work.

Special thanks go to Ettore Aldrovandi for expert advice, to Matilde Marcolli for her encouragement and indispensable help, and to Ed Dunne for suggestions that had a great impact in shaping the final version of this book.

Support from the Max-Planck-Institut in Bonn, from the NSA, and from Caltech, at different stages of the preparation of this book, is gratefully acknowledged.

Preliminaries: Set theory and categories

Set theory is a mathematical field in itself, and its proper treatment (say via the famous ‘Zermelo-Fränkel’ axioms) goes well beyond the scope of this book and the competence of this writer. We will only deal with so-called ‘naive’ set theory, which is little more than a system of notation and terminology enabling us to express precisely mathematical definitions, statements, and their proofs.

Familiarity with this language is essential in approaching a subject such as algebra, and indeed the reader is assumed to have been previously exposed to it. In this chapter we first review some of the language of naive set theory, mainly in order to establish the notation we will use in the rest of the book. We will then get a small taste of the language of *categories*, which plays a powerful unifying role in algebra and many other fields. Our main objective is to convey the notion of ‘universal property’, which will be a constant refrain throughout this book.

1. Naive set theory

1.1. Sets. The notion of *set* formalizes the intuitive idea of ‘collection of objects’. A set is determined by the *elements* it contains: two sets A, B are equal (written $A = B$) if and only if they contain precisely the same elements. ‘What is an *element*?’ is a forbidden question in naive set theory: the buck must stop somewhere. We can conveniently pretend that a ‘universe’ of elements is available to us, and we draw from this universe to construct the elements and sets we need, implicitly assuming that all the operations we will explore can be performed within this universe. (This is the tricky point!) In any case, we specify a *set* by giving a precise recipe determining which elements are in it. This definition is usually put between braces and may consist of a simple, complete, list of elements:

$$A := \{1, 2, 3\}$$

is¹ the set consisting of the integers 1, 2, and 3. By convention, the order² in which the elements are listed, or repetitions in the list, are immaterial to the definition. Thus, the same set may be written out in many ways:

$$\{1, 2, 3\} = \{1, 3, 2\} = \{1, 2, 1, 3, 3, 2, 3, 1, 1, 2, 1, 3\}.$$

This way of denoting sets may be quite cumbersome and in any case will only really work for *finite* sets. For infinite sets, a popular way around this problem is to write a list in which some of the elements are understood as being part of a pattern—for example, the set of even integers may be written

$$E = \{\dots, -2, 0, 2, 4, 6, \dots\},$$

but such a definition is inherently ambiguous, so this leaves room for misinterpretation. Further, some sets are simply ‘too big’ to be listed, even in principle: for example (as one hopefully learns in advanced calculus) there are simply too many *real numbers* to be able to ‘list’ them as one may ‘list’ the integers.

It is often better to adopt definitions that express the elements of a set as elements s of some larger (and already known) set S , satisfying some property P . One may then write

$$A = \{s \in S \mid s \text{ satisfies } P\}$$

($\in$ means *element of*...) and this is in general precise and unambiguous³.

We will occasionally encounter a variation on the notion of set, called ‘multiset’. A multiset is a set in which the elements are allowed to appear ‘with multiplicity’: that is, a notion for which $\{2, 2\}$ would be *distinct* from $\{2\}$. The correct way to define a multiset is by means of *functions*, which we will encounter soon (see Example 2.2).

A few famous sets are

- $\emptyset$: the *empty set*, containing no elements;
- $\mathbb{N}$: the set of *natural numbers* (that is, nonnegative integers);
- $\mathbb{Z}$: the set of *integers*;
- $\mathbb{Q}$: the set of *rational numbers*;
- $\mathbb{R}$: the set of *real numbers*;
- $\mathbb{C}$: the set of *complex numbers*.

Also, the term *singleton* is used to refer to any set consisting of precisely one element. Thus $\{1\}$, $\{2\}$, $\{3\}$ are different sets, but they are all *singletons*.

Here are a few useful symbols (called *quantifiers*):

- $\exists$ means *there exists...* (the *existential* quantifier);

¹ $:=$ is a notation often used to mean that the symbol on the left-hand side is defined by whatever is on the right-hand side. Logically, this is just expressing the equality of the two sides and could just as well be written ‘=’; the extra $:$ is a psychologically convenient decoration inherited from computer science.

²Ordered lists are denoted with round parentheses: $(1, 2, 3)$ is not the same as $(1, 3, 2)$.

³But note that there exist pathologies such as *Russell’s paradox*, showing that even this style of definitions can lead to nonsense. All is well so long as S is indeed known to be a *set* to begin with.

- $\forall$ means *for all...* (the *universal* quantifier).

Also, $\exists!$ is used to mean *there exists a unique...*

For example, the set of even integers may be written as

$$E = \{a \in \mathbb{Z} \mid (\exists n \in \mathbb{Z}) a = 2n\} :$$

in words, “all integers a such that there exists an integer n for which $a = 2n$ ”. In this case we could replace $\exists$ by $\exists!$ without changing the set—but that has to do with properties of $\mathbb{Z}$, not with mathematical syntax. Also, it is common to adopt the shorthand

$$E = \{2n \mid n \in \mathbb{Z}\},$$

in which the existential quantifier is understood.

Being able to parse such strings of symbols effortlessly, and being able to write them out fluently, is extremely important. The reader of this book is assumed to have already acquired this skill.

Note that the *order* in which things are written may make a big difference. For example, the statement

$$(\forall a \in \mathbb{Z}) (\exists b \in \mathbb{Z}) \quad b = 2a$$

is *true*: it says that the result of doubling an arbitrary integer yields an integer; but

$$(\exists b \in \mathbb{Z}) (\forall a \in \mathbb{Z}) \quad b = 2a$$

is *false*: it says that there exists a fixed integer b which is ‘simultaneously’ twice as much as *every* integer—there is no such thing.

Note also that writing simply

$$b = 2a$$

by itself does not convey enough information, unless the context makes it completely clear what quantifiers are attached to a and b : indeed, as we have just seen, different quantifiers may make this into a true or a false statement.

1.2. Inclusion of sets. As mentioned above, two sets are equal if and only if they contain the same elements. We say that a set S is a *subset* of a set T if every element of S is an element of T , in symbols,

$$S \subseteq T.$$

By convention, $S \subset T$ means the same thing: that is (unlike $<$ vs. $\leq$), it does *not* exclude the possibility that S and T may be equal. To avoid any confusion, I will consistently use $\subseteq$ in this book. One adopts $S \subsetneq T$ to mean that S is ‘properly’ contained in T : that is, $S \subseteq T$ and $S \neq T$.

We can think of ‘inclusion of sets’ in terms of logic: $S \subseteq T$ means that

$$s \in S \implies s \in T$$

(the quantifier $\forall s$ is understood); that is, ‘if s is an element of S , then s is an element of T ’; that is, all elements of S are elements of T ; that is, $S \subseteq T$ as promised.

Note that for all sets S , $\emptyset \subseteq S$ and $S \subseteq S$.

If $S \subseteq T$ and $T \subseteq S$, then $S = T$.

The symbol $|S|$ denotes the *number of elements* of S , if this number is finite; otherwise, one writes $|S| = \infty$. If S and T are finite, then

$$S \subseteq T \implies |S| \leq |T|.$$

The subsets of a set S form a set, called the *power set*, or the *set of parts* of S . For example, the power set of the empty set $\emptyset$ consists of one element: $\{\emptyset\}$. The power set of S is denoted $\mathcal{P}(S)$; a popular alternative is 2^S , and indeed $|\mathcal{P}(S)| = 2^{|S|}$ if S is finite (cf. Exercise 2.11).

1.3. Operations between sets. Once we have a few sets to play with, we can obtain more by applying certain standard operations. Here are a few:

- $\cup$: the *union*;
- $\cap$: the *intersection*;
- $\setminus$: the *difference*;
- $\amalg$: the *disjoint union*;
- $\times$: the (Cartesian) *product*;
- and the important notion of ‘quotient by an equivalence relation’.

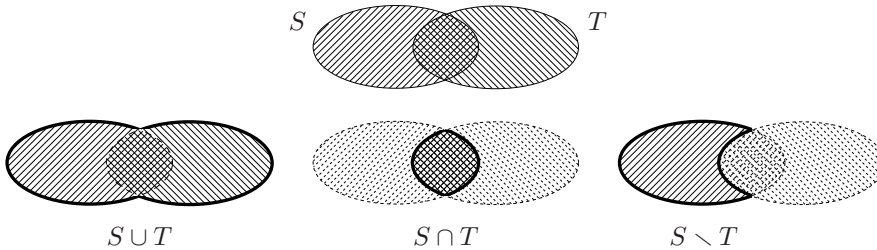
Most of these operations should be familiar to the reader: for example,

$$\{1, 2, 4\} \cup \{3, 4, 5\} = \{1, 2, 3, 4, 5\}$$

while

$$\{1, 2, 4\} \setminus \{3, 4, 5\} = \{1, 2\}.$$

In terms of *Venn diagrams* of infamous ‘new math’ memory:



(the solid black contour indicates the set included in the operation).

Several of these operations may be written out in a transparent way in terms of logic: thus, for example,

$$s \in S \cap T \iff (s \in S \text{ and } s \in T).$$

Two sets S and T are *disjoint* if $S \cap T = \emptyset$, that is, if *no* element is ‘simultaneously’ in both of them.

The *complement* of a subset T in a set S is the difference set $S \setminus T$ consisting of all elements of S which are *not* in T . Thus, for example, the complement of the set of even integers in $\mathbb{Z}$ is the set of odd integers.

The operations $\amalg$, $\times$, and quotients by equivalence relations are slightly more mysterious, and it is very instructive to contemplate them carefully. We will look

at them in a particularly naive way first and come back to them in a short while when we have acquired more language and can view them from a more sophisticated viewpoint.

1.4. Disjoint unions, products. One problem with these operations is that their output may not be defined *as a set*, but rather as a set *up to isomorphisms of sets*, that is, up to bijections. To make sense out of this, we have to talk about *functions*, and we will do that in a moment.

Roughly speaking, the *disjoint union* of two sets S and T is a set $S \amalg T$ obtained by first producing ‘copies’ S' and T' of the sets S and T , with the property that $S' \cap T' = \emptyset$, and then taking the (ordinary) union of S' and T' . The careful reader will feel uneasy, since this ‘recipe’ does not *define* one set: whatever it means to produce a ‘copy’ of a set, surely there are many ways to do so. This ambiguity will be clarified below.

Nevertheless, note that we can say something about $S \amalg T$ even on these very shaky grounds: for example, if S consists of 3 elements and T consists of 4 elements, the reader should expect (correctly) that $S \amalg T$ consists of 7 elements.

Products are marred by the same kind of ambiguity, but fortunately there is a convenient convention that allows us to write down ‘one’ set representing the product of two sets S and T : given S and T , we let $S \times T$ be the set whose elements are the *ordered pairs*⁴ (s, t) of elements of S and T :

$$S \times T := \{(s, t) \text{ such that } s \in S, t \in T\}.$$

Thus, if $S = \{1, 2, 3\}$ and $T = \{3, 4\}$, then

$$S \times T = \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4)\}.$$

For a more sophisticated example, $\mathbb{R} \times \mathbb{R}$ is the set of pairs of real numbers, which (as we learn in calculus) is a good way to represent a *plane*. The set $\mathbb{Z} \times \mathbb{Z}$ could be represented by considering the points in this plane that happen to have integer coordinates. Incidentally, it is common to denote these sets $\mathbb{R}^2$, $\mathbb{Z}^2$; and similarly, the product $A \times A$ of a set by itself is often denoted A^2 .

If S and T are finite sets, clearly $|S \times T| = |S| |T|$.

Also note that we can use products to obtain explicit ‘copies’ of sets as needed for the disjoint union: for example, we could let $S' = \{0\} \times S$, $T' = \{1\} \times T$, guaranteeing that S' and T' are disjoint (why?); and there is an evident way to ‘identify’ S and S' , T and T' . Again, making this precise requires a little more vocabulary.

The operations $\cup$, $\cap$, $\amalg$, $\times$ extend to operations on whole ‘families’ of sets: for example, if $S_1, \dots, S_n$ are sets, we write

$$\bigcap_{i=1}^n S_i = S_1 \cap S_2 \cap \dots \cap S_n$$

⁴One can define the ordered pair (s, t) *as a set* by setting $(s, t) = \{s, \{s, t\}\}$: this carries the information of the elements s, t , as well as conveying the fact that s is special (= the first element of the pair).

for the set whose elements are those elements which are simultaneously elements of all sets $S_1, \dots, S_n$; and similarly for the other operations. But note that while it is clear from the definitions that, for example,

$$S_1 \cup S_2 \cup S_3 = (S_1 \cup S_2) \cup S_3 = S_1 \cup (S_2 \cup S_3),$$

it is not so clear in what sense the sets

$$S_1 \times S_2 \times S_3, \quad (S_1 \times S_2) \times S_3, \quad S_1 \times (S_2 \times S_3)$$

should be ‘identified’ (where we can define the leftmost set as the set of ‘ordered triples’ of elements of S_1, S_2, S_3 , by analogy with the definition for two sets). In fact, again, we can really make sense of such statements only after we acquire the language of functions. However, all such statements do turn out to be true, as the reader probably expects; by virtue of this fortunate circumstance, we can be somewhat cavalier and gloss over such subtleties.

More generally, if $\mathcal{S}$ is a *set of sets*, we may consider sets

$$\bigcup_{S \in \mathcal{S}} S, \quad \bigcap_{S \in \mathcal{S}} S, \quad \coprod_{S \in \mathcal{S}} S, \quad \prod_{S \in \mathcal{S}} S,$$

for the union, intersection, disjoint union, product of all sets in $\mathcal{S}$. There are important subtleties concerning these definitions: for example, if all $S \in \mathcal{S}$ are nonempty, does it follow that $\prod_{S \in \mathcal{S}} S$ is nonempty? The reader probably thinks so, but (if $\mathcal{S}$ is infinite) this is a rather thorny issue, amounting to the *axiom of choice*.

By and large, such subtleties do not affect the material in this course; we will partly come to terms with them in due time⁵, when they become more relevant to the issues at hand (cf. §V.3).

1.5. Equivalence relations, partitions, quotients. Intuitively, a *relation* on elements of a set S is some special affinity among selections of elements of S . For example, the relation $<$ on the set $\mathbb{Z}$ is a way to compare the size of two integers: since $2 < 5$, 2 ‘is related to’ 5 in this sense, while 5 is not related to 2 in the same sense.

For all practical purposes, what a relation ‘means’ is completely captured by which elements are related to which elements in the set. We would really know all there is to know about $<$ on $\mathbb{Z}$ if we had a complete list of all pairs (a, b) of integers such that $a < b$. For example, $(2, 5)$ is such a pair, while $(5, 2)$ is not.

This leads to a completely straightforward definition of the notion of relation: a *relation* on a set S is simply a *subset* R of the *product* $S \times S$. If $(a, b) \in R$, we say that a and b are ‘related by R ’ and write

$$a R b.$$

Often we use fancier symbols for relations, such as $<, \leq, =, \sim, \dots$

⁵The reader will have to employ the axiom of choice in some exercises every now and then, even before we come back to these issues, but this will probably happen below the awareness level, and so it should.

The prototype of a well-behaved relation is ‘=’, which corresponds to the ‘diagonal’

$$\{(a, b) \in S \times S \mid a = b\} = \{(a, a) \mid a \in S\} \subseteq S \times S.$$

Three properties of this very special relation turn out to be particularly important: if $\sim$ denotes for a moment the relation = of equality, then $\sim$ satisfies

- *reflexivity*: $(\forall a \in S) a \sim a$;
- *symmetry*: $(\forall a \in S) (\forall b \in S) a \sim b \implies b \sim a$;
- *transitivity*: $(\forall a \in S) (\forall b \in S) (\forall c \in S), (a \sim b \text{ and } b \sim c) \implies a \sim c$.

That is, every a is equal to itself; if a is equal to b , then b is equal to a ; etc.

Definition 1.1. An *equivalence relation* on a set S is any relation $\sim$ satisfying these three properties. $\lrcorner$

In terms of the corresponding subset R of $S \times S$, ‘reflexivity’ says that the diagonal is contained in R ; ‘symmetry’ says that R is unchanged if flipped about the diagonal (that is, if every (a, b) is interchanged with (b, a)); while unfortunately ‘transitivity’ does not have a similarly nice pictorial translation.

The datum of an equivalence relation on S turns out to be equivalent to a type of information which looks a little different at first, that is, a *partition* of S . A partition of S is a family of *disjoint* nonempty subsets of S , whose union is S : for example,

$$\mathcal{P} = \{\{1, 4, 7\}, \{2, 5, 8\}, \{3, 6\}, \{9\}\}$$

is a partition of the set

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Here is how to get a partition of S from a relation $\sim$ on S : for every element $a \in S$, the *equivalence class* of a (w.r.t. $\sim$) is the subset of S defined by

$$[a]_{\sim} := \{b \in S \mid b \sim a\};$$

then the equivalence classes form a partition $\mathcal{P}_{\sim}$ of S (Exercise 1.2).

Conversely (Exercise 1.3) every partition $\mathcal{P}$ is the partition corresponding in this fashion to an equivalence relation. Therefore, the notions of ‘equivalence relation on S ’ and ‘partition of S ’ are really equivalent.

Now we can view $\mathcal{P}_{\sim}$ as a *set* (whose elements are the equivalence classes with respect to $\sim$). This is the quotient operation mentioned in §1.3.

Definition 1.2. The *quotient* of the set S with respect to the equivalence relation $\sim$ is the set

$$S/\sim := \mathcal{P}_{\sim}$$

of equivalence classes of elements of S with respect to $\sim$. $\lrcorner$

Example 1.3. Take $S = \mathbb{Z}$, and let $\sim$ be the relation defined by

$$a \sim b \iff a - b \text{ is even.}$$

Then $\mathbb{Z}/\sim$ consists of two equivalence classes:

$$\mathbb{Z}/\sim = \{[0]_{\sim}, [1]_{\sim}\}.$$

Indeed, every integer b is either even (and hence $b-0$ is even, so $b \sim 0$, and $b \in [0]_{\sim}$) or odd (and hence $b-1$ is even, so $b \sim 1$, and $b \in [1]_{\sim}$). This is of course the starting point of *modular arithmetic*, which we will cover in due detail later on (§II.2.3). $\perp$

One way to think about this operation is that the equivalence relation ‘becomes equality in the quotient’: that is, two elements of the quotient $S/\sim$ are equal if and only if the corresponding elements in S are related by $\sim$. In other words, taking a quotient is a way to turn any equivalence relation into an equality. This observation will be further formalized in ‘categorical terms’ in a short while (§5.3).

Exercises

Exercises marked with a $\triangleright$ are referred to from the text; exercises marked with a $\neg$ are referred to from other exercises. These referring exercises and sections are listed in brackets following the current exercise; see the introduction for further clarifications, if necessary.

1.1. Locate a discussion of Russell’s paradox, and understand it.

1.2. $\triangleright$ Prove that if $\sim$ is an equivalence relation on a set S , then the corresponding family $\mathcal{P}_{\sim}$ defined in §1.5 is indeed a partition of S : that is, its elements are nonempty, disjoint, and their union is S . [§1.5]

1.3. $\triangleright$ Given a partition $\mathcal{P}$ on a set S , show how to define a relation $\sim$ on S such that $\mathcal{P}$ is the corresponding partition. [§1.5]

1.4. How many different equivalence relations may be defined on the set $\{1, 2, 3\}$?

1.5. Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you attempt to use this relation to define a partition on the set? (Hint: Thinking about the second question will help you answer the first one.)

1.6. $\triangleright$ Define a relation $\sim$ on the set $\mathbb{R}$ of real numbers by setting $a \sim b \iff b-a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a ‘compelling’ description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by declaring $(a_1, a_2) \approx (b_1, b_2) \iff b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$. [§II.8.1, II.8.10]

2. Functions between sets

2.1. Definition. A common thread we will follow for just about every structure introduced in this book will be to try to understand both the type of structures *and* the ways in which different instances of a given structure may interact.

Sets interact with each other through *functions*. It is tempting to think of a function f from a set A to a set B in ‘dynamic’ terms, as a way to ‘go from A to B ’. Similarly to the business with relations, it is straightforward to formalize this notion in ways that do not need to invoke any deep ‘meaning’ of any given f : everything that can be known about a function f is captured by the information of

which element b of B is the image of any given element a of A . This information is nothing but a subset of $A \times B$:

$$\Gamma_f := \{(a, b) \in A \times B \mid b = f(a)\} \subseteq A \times B.$$

This set Γ_f is the *graph* of f ; officially, a function really ‘is’ its graph⁶.

Not all subsets $\Gamma \subseteq A \times B$ correspond to (‘are’) functions: we need to put one requirement on the graphs of *functions*, which can be expressed as follows:

$$(\forall a \in A) (\exists! b \in B) \quad (a, b) \in \Gamma_f,$$

or (‘in functional notation’)

$$(\forall a \in A) (\exists! b \in B) \quad f(a) = b.$$

That is, a function must send each element a of A to exactly one element of B , depending on a . ‘Multivalued functions’ such as $\pm\sqrt{x}$ (which are very important in, e.g., the study of Riemann surfaces) are *not* functions in this sense.

To announce that f is a function from a set A to a set B , one writes $f : A \rightarrow B$ or draws the following picture (‘diagram’):

$$A \xrightarrow{f} B.$$

The action of a function $f : A \rightarrow B$ on an element $a \in A$ is sometimes indicated by a ‘decorated’ arrow, as in

$$a \mapsto f(a).$$

The collection of all functions from a set A to a set B is itself a set⁷, denoted B^A . If we take seriously the notion that a function is really the same thing as its graph, then we can view B^A as a (special) subset of the power set of $A \times B$.

Every set A comes equipped with a very special function, whose graph is the diagonal in $A \times A$: the *identity function* on A

$$\text{id}_A : A \rightarrow A$$

defined by $(\forall a \in A) \text{id}_A(a) = a$. More generally, the inclusion of any subset S of a set A determines a function $S \rightarrow A$, simply sending every element s of S to ‘itself’ in A .

If S is a subset of A , we denote by $f(S)$ the subset of B defined by

$$f(S) := \{b \in B \mid (\exists a \in S) b = f(a)\}.$$

That is, $f(S)$ is the subset of B consisting of all elements that are images of elements of S by the function f . The largest such subset, that is, $f(A)$, is called the *image* of f , denoted ‘ $\text{im } f$ ’.

Also, $f|_S$ denotes the ‘restriction’ of f to the subset S : this is the function $S \rightarrow B$ defined by

$$(\forall s \in S) : \quad f|_S(s) = f(s).$$

⁶To be precise, it is the graph Γ_f *together with* the information of the source A and the target B of f . These are part of the data of the function.

⁷This is another ‘operation among sets’, not listed in §1.3. Can you see why we use B^A for this set? (Cf. Exercise 2.10.)

That is, $f|_S$ is the composition (in the sense explained in §2.3) $f \circ i$, where $i : S \rightarrow A$ is the inclusion. Note that $f(S) = \text{im}(f|_S)$.

2.2. Examples: Multisets, indexed sets. The ‘multisets’ mentioned briefly in §1.1 are a simple example of a notion easily formalized by means of functions. A multiset may be defined by giving a function from a (regular) set A to the set $\mathbb{N}^*$ of positive⁸ integers; if $m : A \rightarrow \mathbb{N}^*$ is such a function, the corresponding multiset consists of the elements $a \in A$, each taken $m(a)$ times. Thus, the *multiset* $\{a, a, a, b, b, b, b, b, c\}$ is really the function $m : \{a, b, c\} \rightarrow \mathbb{N}^*$ for which $m(a) = 3$, $m(b) = 5$, $m(c) = 1$. As with ordinary sets, the order in which the elements are listed is not part of the information carried by a multiset. Simple set-theoretic notions such as inclusion, union, etc., extend in a straightforward way to multisets. For another viewpoint on multisets, see Exercise 3.9.

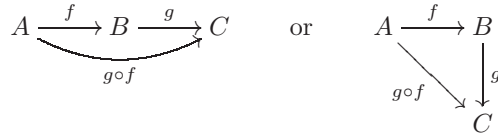
Another example is given by the use of ‘indices’. If we write *let* $a_1, \dots, a_n$ *be integers...*, we really mean *consider a function* $a : \{1, \dots, n\} \rightarrow \mathbb{Z} \dots$, with the understanding that a_i is shorthand for the value $a(i)$ (for $i = 1, \dots, n$). It is tempting to think of an *indexed set* $\{a_i\}_{i \in I}$ simply as a set whose elements happen to be denoted a_i , for i ranging over some ‘set of indices’ I ; but such an indexed set is more properly a *function* $I \rightarrow A$, where A is some set from which we draw the elements a_i . For example, this allows us to consider a_0 and a_1 as distinct elements of $\{a_i\}_{i \in \mathbb{N}}$, even if by coincidence $a_0 = a_1$ as elements of the target set A .

It is easy to miss such subtleties, and some abuse of notation is common and usually harmless. These distinctions play a role in (for example) discussions of *linear independence* of sets of vectors; cf. §VI.1.2.

2.3. Composition of functions. Functions may be *composed*: if $f : A \rightarrow B$ and $g : B \rightarrow C$ are functions, then so is the operation $g \circ f$ defined by

$$(*) \quad (\forall a \in A) \quad (g \circ f)(a) := g(f(a)) :$$

that is, we use f to go from A to B , then apply g to reach C . Graphically we may draw pictures such as



Such graphical representations of collections of (for example) sets connected by functions are called *diagrams*. We will draw many diagrams, and in contexts substantially more general than the one at hand right now.

We say that the diagrams drawn above ‘commute’, or ‘are commutative’, meaning that if we start from A and travel to C in either of the two possible ways prescribed by the diagram, the result of applying the functions one encounters is the same. This is precisely the content of the statement (*).

⁸Some references allow 0 as a possible multiplicity.

Composition is associative: that is, if $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$ are functions, then $h \circ (g \circ f) = (h \circ g) \circ f$. Graphically, the diagram

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D \\ & & \searrow^{h \circ g} & & \nearrow_{g \circ f} & & \\ & & & & & & \end{array}$$

commutes. This important observation should be completely evident from the definition of composition.

The identity function is very special with respect to compositions: if $f : A \rightarrow B$ is any function, then $\text{id}_B \circ f = f$ and $f \circ \text{id}_A = f$. Graphically, the diagrams

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow^{f} & \nearrow_{\text{id}_B} \\ & & B \end{array}, \quad \begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ & \searrow^{f} & \nearrow_f \\ & & B \end{array}$$

commute.

2.4. Injections, surjections, bijections. Special kinds of functions deserve highlighting:

- A function $f : A \rightarrow B$ is *injective* (or an *injection* or *one-to-one*) if

$$(\forall a' \in A) (\forall a'' \in A) \quad a' \neq a'' \implies f(a') \neq f(a') :$$

that is, if f sends different elements to different elements⁹.

- A function $f : A \rightarrow B$ is *surjective* (or a *surjection* or *onto*) if

$$(\forall b \in B) (\exists a \in A) \quad b = f(a) :$$

that is, if f ‘covers the whole of B ’; more precisely, if $\text{im } f = B$.

Injections are often drawn $\hookrightarrow$; surjections are often drawn $\twoheadrightarrow$.

If f is *both* injective and surjective, we say it is *bijective* (or a *bijection* or a *one-to-one correspondence* or an *isomorphism of sets*.) In this case we often write $f : A \xrightarrow{\sim} B$, or

$$A \cong B,$$

and we say that A and B are ‘isomorphic’ sets.

Of course the identity function $\text{id}_A : A \rightarrow A$ is a bijection.

If $A \cong B$, that is, if there is a bijection $f : A \rightarrow B$, then the sets A and B may be ‘identified’ through f , in the sense that we can match precisely the elements a of A with the corresponding elements $f(a)$ of B . For example, if A is a finite set and $A \cong B$, then B is necessarily also a finite set and $|A| = |B|$.

This terminology allows us to make better sense of the considerations on ‘disjoint union’ given in §1.3: the ‘copies’ A' , B' of the given sets A , B should simply

⁹Often one checks this definition in the contrapositive (hence equivalent) formulation, that is,

$$(\forall a' \in A) (\forall a'' \in A) \quad f(a') = f(a'') \implies a' = a''.$$

be *isomorphic* sets to A , B , respectively. The proposal given at the end of §1.4 to produce such disjoint ‘copies’ works, because (for example) the function

$$f : A \rightarrow \{0\} \times A$$

defined by

$$(\forall a \in A) \quad f(a) = (0, a)$$

is manifestly a bijection.

2.5. Injections, surjections, bijections: Second viewpoint. There is an alternative and instructive way to think about these notions.

If $f : A \rightarrow B$ is a bijection, then we can ‘flip its graph’ and define a function

$$g : B \rightarrow A :$$

that is, we can let $a = g(b)$ precisely when $b = f(a)$. (The fact that f is both injective and surjective guarantees that the flip of Γ_f is the graph of a *function* according to the definition given in §2.1. Check this!)

This function g has a very interesting property: graphically,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \xrightarrow{g} A, \\ \text{id}_A \swarrow & & \searrow \text{id}_B \\ & & \end{array} \quad \begin{array}{ccc} B & \xrightarrow{g} & A \xrightarrow{f} B \\ \text{id}_B \swarrow & & \searrow \text{id}_A \\ & & \end{array}$$

commute; that is, $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. The first identity tells us that g is a ‘left-inverse’¹⁰ of f ; the second tells us that g is a ‘right-inverse’ of f . We simply say that it is *the inverse of f* , denoted f^{-1} . Thus, ‘bijections have inverses’.

What about the converse? If a function has an inverse, is it a bijection? This is true, but in fact we can be much more precise.

Proposition 2.1. *Assume $A \neq \emptyset$, and let $f : A \rightarrow B$ be a function. Then*

- (1) *f has a left-inverse if and only if it is injective.*
- (2) *f has a right-inverse if and only if it is surjective.*

Proof. Let’s prove (1).

($\Rightarrow$) If $f : A \rightarrow B$ has a left-inverse, then there exists a $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$. Now assume that $a' \neq a''$ are arbitrary different elements in A ; then

$$g(f(a')) = \text{id}_A(a') = a' \neq a'' = \text{id}_A(a'') = g(f(a''));$$

that is, g sends $f(a')$ and $f(a'')$ to different elements. This forces $f(a')$ and $f(a'')$ to be different, showing that f is injective.

($\Leftarrow$) Now assume $f : A \rightarrow B$ is injective. In order to construct a function $g : B \rightarrow A$, we have to assign a unique value $g(b) \in A$ for each element $b \in B$. For this, choose any fixed element $s \in A$ (which we can do because $A \neq \emptyset$); then set

$$g(b) := \begin{cases} a & \text{if } b = f(a) \text{ for some } a \in A, \\ s & \text{if } b \notin \text{im } f. \end{cases}$$

¹⁰Never mind that g is *drawn* to the right of f in the diagram—we say that g is a *left-inverse* of f because it is *written* to the left of f : $g \circ f = \text{id}_A$.

In words, if b is the image of an element a of A , send it back to a ; otherwise, send it to your fixed element s .

The given assignment defines a function, precisely because f is injective: indeed, this guarantees that every b that is the image of some $a \in A$ by f is the image of a *unique* a (two distinct elements of A cannot be simultaneously sent to b by f , since f is injective). Thus every $b \in B$ is sent to a unique well-defined element of A , as is required of functions.

Finally, the function $g : B \rightarrow A$ is a left-inverse of f . Indeed, if $a \in A$, then $b = f(a)$ is of the first type, so it is sent back to a by g ; that is, $g \circ f(a) = a = \text{id}_A(a)$ for all $a \in A$, as needed.

The proof of (2) is left as an exercise (Exercise 2.2). $\square$

Corollary 2.2. *A function $f : A \rightarrow B$ is a bijection if and only if it has a (two-sided) inverse.*

This is not completely innocent: if f has both a left-inverse and a right-inverse, why should it have *one* inverse that works as both on the left and on the right? Try to prove this by yourself now. We will come back to this issue soon (in §4).

If a function is injective *but not surjective*, then it will not have a right-inverse, and if the source has at least two elements, it will necessarily have more than one left-inverse (this should be clear from the argument given in the proof of Proposition 2.1). Similarly, a surjective function will in general have many right-inverses; they are often called *sections*.

Proposition 2.1 hints that something deep is going on here. The definition of injective and surjective maps given in §2.4 relied crucially on working directly with the *elements* of our sets; Proposition 2.1 shows that in fact these properties are detected by the way *functions* are ‘organized’ among sets. Even if we did not know what ‘elements’ means, still we could make sense of the notions of injectivity and surjectivity (and hence of isomorphisms of sets) by exclusively referring to properties of functions.

This is a more ‘mature’ point of view and one that will be championed when we talk about categories. To some extent, it should cure the reader from the discomfort of talking about ‘elements’, as we did in our informal introduction to sets, without defining what these mysterious entities are supposed to be.

The standard notation for *the* inverse of a bijection f is f^{-1} . This symbol is also used for functions that are not bijections, but in a slightly different context: if $f : A \rightarrow B$ is any function and $T \subseteq B$ is a subset of B , then $f^{-1}(T)$ denotes the subset of A of ‘all elements that map to T ’; that is,

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}.$$

If $T = \{q\}$ consists of a single element of B , $f^{-1}(T)$ (abbreviated $f^{-1}(q)$) is called the *fiber* of f over q . Thus a function $f : A \rightarrow B$ is a bijection if it has nonempty fibers over all elements of B (that is, f is surjective), and these fibers are in fact singletons (that is, f is injective). In this case, this notation f^{-1} matches nicely with the notation of ‘inverse’ mentioned above.

2.6. Monomorphisms and epimorphisms. There is *yet another* way to express injectivity and surjectivity, which appears at first more complicated than what we have seen so far but which is in fact even more basic.

A function $f : A \rightarrow B$ is a *monomorphism* (or *monic*) if the following holds:

$$\begin{aligned} &\text{for all sets } Z \text{ and all functions } \alpha', \alpha'' : Z \rightarrow A \\ &f \circ \alpha' = f \circ \alpha'' \implies \alpha' = \alpha''. \end{aligned}$$

Proposition 2.3. *A function is injective if and only if it is a monomorphism.*

Proof. ($\implies$) By Proposition 2.1, if a function $f : A \rightarrow B$ is injective, then it has a left-inverse $g : B \rightarrow A$. Now assume that α', α'' are arbitrary functions from another set Z to A and that

$$f \circ \alpha' = f \circ \alpha'';$$

compose on the left by g , and use associativity of composition:

$$(g \circ f) \circ \alpha' = g \circ (f \circ \alpha') = g \circ (f \circ \alpha'') = (g \circ f) \circ \alpha'';$$

since g is a left-inverse of f , this says

$$\text{id}_A \circ \alpha' = \text{id}_A \circ \alpha'',$$

and therefore

$$\alpha' = \alpha'',$$

as needed to conclude that f is a monomorphism.

($\Leftarrow$) Now assume that f is a monomorphism. This says something about arbitrary sets Z and arbitrary functions $Z \rightarrow A$; we are going to use a microscopic portion of this information, choosing Z to be any singleton $\{p\}$. Then assigning functions $\alpha', \alpha'' : Z \rightarrow A$ amounts to choosing to which elements $a' = \alpha'(p)$, $a'' = \alpha''(p)$ we should send the single element p of Z . For this particular choice of Z , the property defining monomorphisms, $f \circ \alpha' = f \circ \alpha'' \implies \alpha' = \alpha''$, becomes

$$f \circ \alpha'(p) = f \circ \alpha''(p) \implies \alpha' = \alpha'',$$

that is,

$$f(a') = f(a'') \implies \alpha' = \alpha''.$$

Now two functions from $Z = \{p\}$ to A are equal if and only if they send p to the same element, so this says

$$f(a') = f(a'') \implies a' = a''.$$

This has to be true for all α', α'' , that is, for all choices of distinct a', a'' in A . In other words, f has to be injective, as was to be shown. $\square$

The reader should now expect that there be a definition in the style of the one given for monomorphisms and which will turn out to be equivalent to ‘surjective’. This is the case: such a notion is called *epimorphism*. Finding it, and proving the equivalence with the ordinary definition of ‘surjective’, is left to the reader¹¹ (Exercise 2.5).

¹¹This is a particularly important exercise, and I recommend that the reader write out all the gory details carefully.

2.7. Basic examples. The basic operations on sets provided us with several important examples of injective and surjective functions.

Example 2.4. Let A, B be sets. Then there are *natural projections* π_A, π_B :

$$\begin{array}{ccc} & A \times B & \\ \pi_A \swarrow & & \searrow \pi_B \\ A & & B \end{array}$$

defined by

$$\pi_A((a, b)) := a, \quad \pi_B((a, b)) := b$$

for all $(a, b) \in A \times B$. Both of these maps are (clearly) surjective. $\lrcorner$

Example 2.5. Similarly, there are natural injections from A and B to the disjoint union:

$$\begin{array}{ccc} A & & B \\ & \searrow & \swarrow \\ & A \amalg B & \end{array}$$

obtained by sending $a \in A$ (resp., $b \in B$) to the corresponding element in the isomorphic copy A' of A (resp., B' of B) in $A \amalg B$. $\lrcorner$

Example 2.6. If $\sim$ is an equivalence relation on a set A , there is a (clearly surjective) canonical projection

$$A \twoheadrightarrow A/\sim$$

obtained by sending every $a \in A$ to its equivalence class $[a]_\sim$. $\lrcorner$

2.8. Canonical decomposition. The reason why we focus our attention on injective and surjective maps is that they provide the basic ‘bricks’ out of which *any* function may be constructed.

To see this, we observe that every function $f : A \rightarrow B$ determines an equivalence relation $\sim$ on A as follows: for all $a', a'' \in A$,

$$a' \sim a'' \iff f(a') = f(a'').$$

(The reader should check that this is indeed an *equivalence* relation.)

Theorem 2.7. Let $f : A \rightarrow B$ be any function, and define $\sim$ as above. Then f decomposes as follows:

$$\begin{array}{ccccc} & & f & & \\ & \searrow & \curvearrowright & \swarrow & \\ A & \twoheadrightarrow & (A/\sim) & \xrightarrow[\tilde{f}]{\sim} & \text{im } f \hookrightarrow B \end{array}$$

where the first function is the canonical projection $A \rightarrow A/\sim$ (as in Example 2.6), the third function is the inclusion $\text{im } f \subseteq B$, and the bijection $\tilde{f}$ in the middle is defined by

$$\tilde{f}([a]_\sim) := f(a)$$

for all $a \in A$.

The formula defining $\tilde{f}$ shows immediately that the diagram commutes; so all we have to verify in order to prove this theorem is that

- that formula *does* define a function;
- that function is in fact a bijection.

The first item is an instance of a class of verifications of the utmost importance. The formula given for $\tilde{f}$ has a colossal built-in ambiguity: *the same* element in $A/\sim$ may be the equivalence class of *many* elements of A ; applying the formula for $\tilde{f}$ requires *choosing* one of these elements and applying f to it. We have to prove that the result of this operation is *independent* from this choice: that is, that all possible choices of representatives for that equivalence class lead to the same result.

We encode this type of situation by saying that we have to verify that $\tilde{f}$ is *well-defined*. We will often have to check that the operations we consider are well-defined, in contexts very similar to the one epitomized here.

Proof. Spelling out the first item discussed above, we have to verify that, for all a', a'' in A ,

$$[a']_{\sim} = [a'']_{\sim} \implies f(a') = f(a'').$$

Now $[a']_{\sim} = [a'']_{\sim}$ means that $a' \sim a''$, and the definition of $\sim$ has been engineered precisely so that this would mean $f(a') = f(a'')$ as required here. So $\tilde{f}$ is indeed well-defined.

To verify the second item, that is, that $\tilde{f} : A/\sim \rightarrow \text{im } f$ is a bijection, we check explicitly that $\tilde{f}$ is injective and surjective.

Injective: If $\tilde{f}([a']_{\sim}) = \tilde{f}([a'']_{\sim})$, then $f(a') = f(a'')$ by definition of $\tilde{f}$; hence $a' \sim a''$ by definition of $\sim$, and then $[a']_{\sim} = [a'']_{\sim}$. Therefore

$$\tilde{f}([a']_{\sim}) = \tilde{f}([a'']_{\sim}) \implies [a']_{\sim} = [a'']_{\sim}$$

proving injectivity.

Surjective: Given any $b \in \text{im } f$, there is an element $a \in A$ such that $f(a) = b$. Then

$$\tilde{f}([a]_{\sim}) = f(a) = b$$

by definition of $\tilde{f}$. Since b was arbitrary in $\text{im } f$, this shows that $\tilde{f}$ is surjective, as needed. $\square$

Theorem 2.7 shows that *every* function is the composition of a surjection, followed by an isomorphism, followed by an injection. While its proof is trivial, this is a result of some importance, since it is the prototype of a situation that will occur several times in this book. It will resurface every now and then, with names such as ‘the first isomorphism theorem’.

2.9. Clarification. Finally, we can begin to clarify one comment about *disjoint unions*, products, and quotients, made in §1.4. Our definition of $A \amalg B$ was the (conventional) union of two disjoint sets A', B' isomorphic to A, B , respectively. It is easy to provide a way to effectively produce such isomorphic copies (as we did in §1.4); but it is in fact a little too easy—many other choices are possible, and one does not look any better than any other. It is in fact more sensible *not* to

make a fixed choice once and for all and simply accept the fact that all of them produce acceptable candidates for $A \amalg B$. From this egalitarian standpoint, the result of the operation $A \amalg B$ is *not* ‘well-defined’ as a set in the sense specified above. However, it is easy to see (Exercise 2.9) that $A \amalg B$ is well-defined *up to isomorphism*: that is, that any two choices for the copies A' , B' lead to *isomorphic* candidates for $A \amalg B$. The same considerations apply to products and quotients.

The main feature of sets obtained by taking disjoint unions, products, or quotients is not really ‘what elements they contain’ but rather ‘their relationship with all other sets’. This will be (even) clearer when we revisit these operations and others¹² in the context of categories.

Exercises

2.1. ▷ How many different bijections are there between a set S with n elements and itself? [§II.2.1]

2.2. ▷ Prove statement (2) in Proposition 2.1. You may assume that given a family of disjoint nonempty subsets of a set, there is a way to choose one element in each member of the family¹³. [§2.5, V.3.3]

2.3. Prove that the inverse of a bijection is a bijection and that the composition of two bijections is a bijection.

2.4. ▷ Prove that ‘isomorphism’ is an equivalence relation (on any set of sets). [§4.1]

2.5. ▷ Formulate a notion of *epimorphism*, in the style of the notion of *monomorphism* seen in §2.6, and prove a result analogous to Proposition 2.3, for epimorphisms and surjections. [§2.6, §4.2]

2.6. With notation as in Example 2.4, explain how any function $f : A \rightarrow B$ determines a section of π_A .

2.7. Let $f : A \rightarrow B$ be any function. Prove that the graph Γ_f of f is isomorphic to A .

2.8. Describe as explicitly as you can all terms in the canonical decomposition (cf. §2.8) of the function $\mathbb{R} \rightarrow \mathbb{C}$ defined by $r \mapsto e^{2\pi i r}$. (This exercise matches one assigned previously. Which one?)

2.9. ▷ Show that if $A' \cong A''$ and $B' \cong B''$, and further $A' \cap B' = \emptyset$ and $A'' \cap B'' = \emptyset$, then $A' \cup B' \cong A'' \cup B''$. Conclude that the operation $A \amalg B$ (as described in §1.4) is well-defined *up to isomorphism* (cf. §2.9). [§2.9, 5.7]

2.10. ▷ Show that if A and B are finite sets, then $|B^A| = |B|^{|A|}$. [§2.1, 2.11, §II.4.1]

¹²The reader should also be aware that there are important variations on the operations we have seen so far—particularly important are the *fibred* flavors of products and disjoint unions.

¹³This (reasonable) statement is the *axiom of choice*; cf. §V.3.

2.11. ▷ In view of Exercise 2.10, it is not unreasonable to use 2^A to denote the set of functions from an arbitrary set A to a set with 2 elements (say $\{0, 1\}$). Prove that there is a bijection between 2^A and the *power set* of A (cf. §1.2). [§1.2, III.2.3]

3. Categories

The language of categories is affectionately known as *abstract nonsense*, so named by Norman Steenrod. This term is essentially accurate and not necessarily derogatory: categories refer to *nonsense* in the sense that they are all about the ‘structure’, and not about the ‘meaning’, of what they represent. The emphasis is less on how you run into a specific set you are looking at and more on how that set may sit in relationship with all other sets. Worse (or better) still, the emphasis is less on studying sets, and functions between sets, than on studying ‘things, and things that go from things to things’ without necessarily being explicit about what these things are: they may be sets, or groups, or rings, or vector spaces, or modules, or other objects that are so exotic that the reader has no right whatsoever to know about them (yet).

‘Categories’ will intuitively look like sets at first, and in multiple ways. Categories may make you think of sets, in that they are ‘collections of objects’, and further there will be notions of ‘functions from categories to categories’ (called *functors*¹⁴). At the same time, every category may make you think of the collection of all sets, since there will be analogs of ‘functions’ among the things it contains.

3.1. Definition. The definition of a category looks complicated at first, but the gist of it may be summarized quickly: a category consists of a collection of ‘objects’, and of ‘morphisms’ between these objects, satisfying a list of natural conditions.

The reader will note that I refrained from writing a *set* of objects, opting for the more generic ‘collection’. This is an annoying, but unavoidable, difficulty: for example, we want to have a ‘category of sets’, in which the ‘objects’ are sets and the ‘morphisms’ are functions between sets, and the problem is that there simply is not a *set of all sets*¹⁵. In a sense, the collection of all sets is ‘too big’ to be a set. There are however ways to deal with such ‘collections’, and the technical name for them is *class*. There *is* a ‘class’ of all sets (and there will be classes taking care of groups, rings, etc.).

An alternative would be to define a large enough set (called a *universe*) and then agree that all objects of all categories will be chosen from this gigantic entity.

In any case, all the reader needs to know about this is that there is a way to make it work. We will use the term ‘class’ in the definition, but this will not affect any proof or any other definition in this book. Further, in some of the examples considered below the class in question *is* a set (we say that the category is *small* in this case), so the reader will feel perfectly at home when contemplating these examples.

¹⁴However, we will not consider functors until later chapters: our first formal encounter with functors will be in Chapter VIII.

¹⁵That is one thing we learn from Russell’s paradox.

Definition 3.1. A *category* $\mathbf{C}$ consists of

- a class $\text{Obj}(\mathbf{C})$ of *objects* of the category; and
- for every two objects A, B of $\mathbf{C}$, a set $\text{Hom}_{\mathbf{C}}(A, B)$ of *morphisms*, with the properties listed below. $\lrcorner$

As a prototype to keep in mind, think of the objects as ‘sets’ and of morphisms as ‘functions’. This one example should make the defining properties of morphisms look natural and easy to remember:

- For every object A of $\mathbf{C}$, there exists (at least) one morphism $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$, the ‘identity’ on A .
- One can compose morphisms: two morphisms $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}}(B, C)$ determine a morphism $gf \in \text{Hom}_{\mathbf{C}}(A, C)$. That is, for every triple of objects A, B, C of $\mathbf{C}$ there is a function (of sets)

$$\text{Hom}_{\mathbf{C}}(A, B) \times \text{Hom}_{\mathbf{C}}(B, C) \rightarrow \text{Hom}_{\mathbf{C}}(A, C),$$

and the image of the pair (f, g) is denoted gf .

- This ‘composition law’ is associative: if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$, and $h \in \text{Hom}_{\mathbf{C}}(C, D)$, then

$$(hg)f = h(gf).$$

- The identity morphisms are identities with respect to composition: that is, for all $f \in \text{Hom}_{\mathbf{C}}(A, B)$ we have

$$f1_A = f, \quad 1_B f = f.$$

This is really a mouthful, but again, to remember all this, just think of functions of sets. One further requirement is that the sets

$$\text{Hom}_{\mathbf{C}}(A, B), \quad \text{Hom}_{\mathbf{C}}(C, D)$$

be *disjoint* unless $A = C$, $B = D$; this is something you do not usually think about, but again it holds for ordinary set-functions¹⁶. That is, if two functions are one and the same, then necessarily they have the same source and the same target: source and target are part of the datum of a set-function.

A morphism of an object A of a category $\mathbf{C}$ to itself is called an *endomorphism*; $\text{Hom}_{\mathbf{C}}(A, A)$ is denoted $\text{End}_{\mathbf{C}}(A)$. One of the axioms of a category tells us that this is a ‘pointed’ set, as $1_A \in \text{End}_{\mathbf{C}}(A)$. The reader should note that composition defines an ‘operation’ on $\text{End}_{\mathbf{C}}(A)$: if f, g are elements of $\text{End}_{\mathbf{C}}(A)$, so is their composition gf .

Writing ‘ $f \in \text{Hom}_{\mathbf{C}}(A, B)$ ’ gets tiresome in the long run. If the category is understood, one may safely drop the index $\mathbf{C}$, or even use arrows as we do with set-functions: $f : A \rightarrow B$. This also allows us to draw *diagrams* of morphisms in any category; a diagram is said to ‘commute’ (or to be a ‘commutative’ diagram) if all ways to traverse it lead to the same results of composing morphisms along the way, just as explained for diagrams of functions of sets in §2.3.

¹⁶I will often use the term ‘set-function’ to emphasize that we are dealing with a function in the context of sets.

In fact, I will now feel free to use *diagrams* as possible objects of categories. The official definition of a diagram in this context would be a set of objects of a category $\mathbf{C}$, along with prescribed morphisms between these objects; the diagram commutes if it does in the sense specified above. The specifics of the visual representation of a diagram are of course irrelevant.

3.2. Examples. The reader should note that 90% of the definition of the notion of category goes into explaining the properties of its *morphisms*; it is fair to say that the morphisms *are* the important constituents of a category. Nevertheless, it is psychologically irresistible to think of a category in terms of its *objects*: for example, one talks about the ‘category of sets’. The point is that usually the kind of ‘morphisms’ one may consider are (psychologically at least) determined by the objects: if one is talking about sets, what can one possibly mean for ‘morphism’ other than a function of sets? In other situations (cf. Example 3.5 below or Exercise 3.9) it is a little less clear what the morphisms should be, and looking for the ‘right’ notion may be an interesting project.

Example 3.2. It is hopefully crystal clear by now that sets (as objects), together with set-functions (as morphisms), form a category; if not, the reader must stop here and go no further until this assertion sheds any residual mystery¹⁷.

There is no universally accepted, official notation for this important category. It is customary to write the word ‘Set’ or ‘Sets’, with some fancy decoration for emphasis. For example, in the literature one may encounter $\mathbf{SET}$, **Sets**, $\mathfrak{Set}$, (*Sets*), and many amusing variations on these themes. We will use ‘sans-serif’ fonts to denote categories; thus, $\mathbf{Set}$ will denote the category of sets. Thus

- $\mathbf{Obj}(\mathbf{Set})$ = the class of all sets;
- for A, B in $\mathbf{Obj}(\mathbf{Set})$ (that is, for A, B sets) $\mathbf{Hom}_{\mathbf{Set}}(A, B) = B^A$.

Note that the presence of the operations recalled in §§1.3–1.5 is *not* part of the definition of category: these operations highlight interesting features of $\mathbf{Set}$, which may or may not be shared by other categories. We will soon come back to some of these operations and understand more precisely what they say about $\mathbf{Set}$. $\square$

Example 3.3. Here is a completely different example.

Suppose S is a set and $\sim$ is a relation on S satisfying the reflexive and transitive properties. Then we can encode this data into a category:

- *objects*: the elements of S ;
- *morphisms*: if a, b are objects (that is, if $a, b \in S$), then let $\mathbf{Hom}(a, b)$ be the set consisting of the element $(a, b) \in S \times S$ if $a \sim b$, and let $\mathbf{Hom}(a, b) = \emptyset$ otherwise.

¹⁷I will give the reader such prompts every now and then: at key times, it is more useful to take stock of what one knows than blindly march forward hoping for the best. A difficulty at this time signals the need to reread the previous material carefully. If the mystery persists, that’s what office hours are there for. But typically you should be able to find your way out on your own, based on the information I have given you, and you will most likely learn more this way. You should give it your best try before seeking professional help.

Note that (unlike in **Set**) there are *very few* morphisms: at most one for any pair of objects, and no morphisms at all between ‘unrelated’ objects.

We have to define ‘composition of morphisms’ and verify that the conditions specified in §3.1 are satisfied. First of all, do we have ‘identities’? If a is an object (that is, if $a \in S$), we need to find an element

$$1_a \in \text{Hom}(a, a).$$

This is precisely why we are assuming that $\sim$ is reflexive: this tells us that $\forall a, a \sim a$; that is, $\text{Hom}(a, a)$ consists of the single element (a, a) . So we have no choice: we must let

$$1_a = (a, a) \in \text{Hom}(a, a).$$

As for composition, let a, b, c be objects (that is, elements of S) and

$$f \in \text{Hom}(a, b), \quad g \in \text{Hom}(b, c);$$

we have to define a corresponding morphism $gf \in \text{Hom}(a, c)$. Now,

$$f \in \text{Hom}(a, b)$$

tells us that $\text{Hom}(a, b)$ is nonempty, and according to the definition of morphisms in this category that means that $a \sim b$, and f is in fact the element (a, b) of $S \times S$. Similarly, $g \in \text{Hom}(b, c)$ tells us $b \sim c$ and $g = (b, c)$. Now

$$a \sim b \text{ and } b \sim c \implies a \sim c$$

since we are assuming that $\sim$ is transitive. This tells us that $\text{Hom}(a, c)$ consists of the single element (a, c) . Thus we again have no choice: we must let

$$gf := (a, c) \in \text{Hom}(a, c).$$

Is this operation associative? If $f \in \text{Hom}(a, b)$, $g \in \text{Hom}(b, c)$, and $h \in \text{Hom}(c, d)$, then necessarily

$$f = (a, b), \quad g = (b, c), \quad h = (c, d)$$

and

$$gf = (a, c), \quad hg = (b, d)$$

and hence

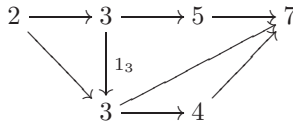
$$h(gf) = (a, d) = (hg)f,$$

proving associativity.

The reader will have no difficulties checking that 1_a is an identity with respect to this composition, as needed (Exercise 3.3).

The most trivial instance of this construction is the category obtained from a set S taken with the equivalence relation ‘=’; that is, the only morphisms are the identity morphisms. These categories are called *discrete*.

As another example, consider the category corresponding to endowing $\mathbb{Z}$ with the relation $\leq$. For example,



is a (randomly chosen) commutative diagram in this category. It would still be a (commutative) diagram in this category if we reversed the vertical arrow $3 \rightarrow 3$ or if we added an arrow from 3 to 4, while we are not allowed to draw an arrow *from 4 to 3*, since $4 \not\leq 3$.

These categories are very special—for example, every diagram drawn in them is necessarily commutative, and this is *very far* from being the case in, e.g., **Set**. Also note that these categories are all *small*. $\lrcorner$

Example 3.4. Here is another¹⁸ example of a small category.

Let S again be a set. Define a category $\hat{S}$ by setting

- $\text{Obj}(\hat{S}) = \mathcal{P}(S)$, the power set S (cf. §1.2 and Exercise 2.11);
- for A, B objects of $\hat{S}$ (that is, $A \subseteq S$ and $B \subseteq S$) let $\text{Hom}_{\hat{S}}(A, B)$ be the pair (A, B) if $A \subseteq B$, and let $\text{Hom}_{\hat{S}}(A, B) = \emptyset$ otherwise.

The identity 1_A consists of the pair (A, A) (which is one, and in fact the only one, morphism from A to A , since $A \subseteq A$). Composition is obtained by stringing inclusions: if there are morphisms

$$A \rightarrow B, \quad B \rightarrow C$$

in $\hat{S}$, then $A \subseteq B$ and $B \subseteq C$; hence $A \subseteq C$ and there is a morphism $A \rightarrow C$. Checking the axioms specified in §3.1 should be routine (make sure this is the case!).

Examples in this style (but employing more sophisticated structures, such as the family of *open* subsets of a topological space) are hugely important in well-established fields such as algebraic geometry. $\lrcorner$

Example 3.5. The next example is very abstract, but thinking about it will make you rather comfortable with everything we have seen so far; and it is a very common construction, variations of which will abound in the course.

Let $\mathbf{C}$ be a category, and let A be an object of $\mathbf{C}$. We are going to define a category $\mathbf{C}_A$ whose objects are certain *morphisms* in $\mathbf{C}$ and whose morphisms are certain *diagrams* of $\mathbf{C}$ (surprise!).

- $\text{Obj}(\mathbf{C}_A) =$ all morphisms from any object of $\mathbf{C}$ to A ; thus, an object of $\mathbf{C}_A$ is a morphism $f \in \text{Hom}_{\mathbf{C}}(Z, A)$ for some object Z of $\mathbf{C}$. Pictorially, an object of $\mathbf{C}_A$ is an arrow $Z \xrightarrow{f} A$ in $\mathbf{C}$; these are often drawn ‘top-down’, as in

$$\begin{array}{c} Z \\ \downarrow f \\ A \end{array}$$

What are morphisms in $\mathbf{C}_A$ going to be? There really is only one sensible way to assign morphisms to a category with objects as above. The brave reader will want to stop reading here and continue only after having come up with the definition independently. There will be many similar examples lurking behind constructions

¹⁸Actually, this is again an instance of the categories considered in Example 3.3. Do you see why? (Exercise 3.5.)

we will encounter in this book, and ideally speaking, they should appear completely *natural* when the time comes. A bit of effort devoted now to understanding this prototype situation will have ample reward in the future. Spoiler follows, so put these notes away *now* and jot down the definition of morphism in $\mathbf{C}_A$.

Welcome back.

- Let f_1, f_2 be objects of $\mathbf{C}_A$, that is, two arrows

$$\begin{array}{ccc} Z_1 & & Z_2 \\ f_1 \downarrow & & \downarrow f_2 \\ A & & A \end{array}$$

in $\mathbf{C}$. Morphisms $f_1 \rightarrow f_2$ are defined to be *commutative diagrams*

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ f_1 \searrow & & \swarrow f_2 \\ & A & \end{array}$$

in the ‘ambient’ category $\mathbf{C}$.

That is, morphisms $f_1 \rightarrow f_2$ correspond precisely to those morphisms $\sigma : Z_1 \rightarrow Z_2$ in $\mathbf{C}$ *such that* $f_1 = f_2 \sigma$.

Once you understand what morphisms have to be, checking that they satisfy the axioms spelled out in §3.1 is straightforward. The identities are inherited from the identities in $\mathbf{C}$: for $f : Z \rightarrow A$ in $\mathbf{C}_A$, the identity 1_f corresponds to the diagram

$$\begin{array}{ccc} Z & \xrightarrow{1_Z} & Z \\ f \searrow & & \swarrow f \\ & A & \end{array}$$

which commutes by virtue of the fact that $\mathbf{C}$ is a category. Composition is also a subproduct of composition in $\mathbf{C}$. Two morphisms $f_1 \rightarrow f_2 \rightarrow f_3$ in $\mathbf{C}_A$ correspond to putting two commutative diagrams side-by-side:

$$\begin{array}{ccccc} Z_1 & \xrightarrow{\sigma} & Z_2 & \xrightarrow{\tau} & Z_3 \\ f_1 \searrow & & f_2 \downarrow & & \swarrow f_3 \\ & & A & & \end{array}$$

and then it follows (again because $\mathbf{C}$ is a category!) that the diagram obtained by removing the central arrow, i.e.,

$$\begin{array}{ccc} Z_1 & \xrightarrow{\tau\sigma} & Z_3 \\ f_1 \searrow & & \swarrow f_3 \\ & A & \end{array}$$

also commutes. Check all this(!), and verify that composition in $\mathbf{C}_A$ is associative (again, this follows immediately from the fact that composition is associative in $\mathbf{C}$).

Categories constructed in this fashion are called *slice categories* in the literature; they are particular cases of *comma categories*. $\lrcorner$

Example 3.6. For the sake of concreteness, let's apply the construction given in Example 3.5 to the category constructed in Example 3.3, say for $S = \mathbb{Z}$ and $\sim$ the relation $\leq$. Call $\mathbf{C}$ this category, and choose an object A of $\mathbf{C}$ —that is, an integer, for example, $A = 3$. Then the objects of $\mathbf{C}_A$ are morphisms in $\mathbf{C}$ with target 3, that is, pairs $(n, 3) \in \mathbb{Z} \times \mathbb{Z}$ with $n \leq 3$. There is a morphism

$$(m, 3) \rightarrow (n, 3)$$

if and only if $m \leq n$. In this case $\mathbf{C}_A$ may be harmlessly identified with the 'subcategory' of integers ≤ 3 , with 'the same' morphisms as in $\mathbf{C}$. $\lrcorner$

Example 3.7. An entirely similar example to the one explored in Example 3.5 may be obtained by considering morphisms in a category $\mathbf{C}$ from a fixed object A to all objects in $\mathbf{C}$, again with morphisms defined by suitable commutative diagrams. This leads to *coslice categories*. The reader should provide details of this construction (Exercise 3.7). $\lrcorner$

Example 3.8. As a 'concrete' instance of a category as in Example 3.7, let $\mathbf{C} = \mathbf{Set}$ and $A =$ a fixed singleton $\{*\}$. Call the resulting category $\mathbf{Set}^*$.

An object in $\mathbf{Set}^*$ is then a morphism $f : \{*\} \rightarrow S$ in $\mathbf{Set}$, where S is any set. The information of an object in $\mathbf{Set}^*$ consists therefore of the choice of a nonempty set S and of an element $s \in S$ —that is, the element $f(*)$: this element determines, and is determined by, f .

Thus, we may denote objects of $\mathbf{Set}^*$ as pairs (S, s) , where S is any set and $s \in S$ is any element of S .

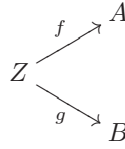
A morphism between two such objects, $(S, s) \rightarrow (T, t)$, corresponds then (check this!) to a set-function $\sigma : S \rightarrow T$ such that $\sigma(s) = t$.

Objects of $\mathbf{Set}^*$ are called 'pointed sets'. Many of the structures we will study in this book will be pointed sets. For example (as we will see) a 'group' is a set G with, among other requirements, a distinguished element e_G (its 'identity'); 'group homomorphisms' will be functions which, among other properties, send identities to identities; thus, they are morphisms of pointed sets in the sense considered above. $\lrcorner$

Example 3.9. It is useful to contemplate a few more 'abstract' examples in the style of Examples 3.5 and 3.7. These will be essential ingredients in the promised revisitation of some of the operations mentioned in §1.3. Their definition will appear disappointingly simple-minded to the reader who has mastered Examples 3.5 and 3.7.

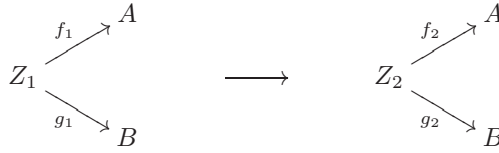
This time we start from a given category $\mathbf{C}$ and *two* objects A, B of $\mathbf{C}$. We can define a new category $\mathbf{C}_{A,B}$ by essentially the same procedure that we used in order to define $\mathbf{C}_A$:

- $\text{Obj}(\mathbf{C}_{A,B}) = \text{diagrams}$

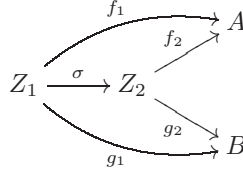


in $\mathbf{C}$; and

- morphisms



are *commutative* diagrams



I will leave to the reader the task of formalizing this rough description. This example is really nothing more than a mixture of $\mathbf{C}_A$ and $\mathbf{C}_B$, where the two structures interact because of the stringent requirement that the same σ must make both sides of the diagram commute:

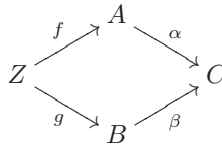
$$f_1 = f_2\sigma \quad \underline{\text{and}} \quad g_1 = g_2\sigma$$

‘simultaneously’.

Flipping most of the arrows gives an analogous variation of Example 3.7, producing a category which we may¹⁹ denote $\mathbf{C}^{A,B}$; details are left to the reader. $\square$

Example 3.10. As a final variation on these examples, we conclude by considering the *fibered* version of $\mathbf{C}_{A,B}$ (and $\mathbf{C}^{A,B}$). Take this as a test to see if you have really understood $\mathbf{C}_{A,B}$ —experts would tell you that this looks fairly sophisticated for students just learning categories, so don’t get disheartened if it does not flow too well at first (but pat yourself on the shoulder if it does!). Start with a given category $\mathbf{C}$, and this time choose two fixed *morphisms* $\alpha : A \rightarrow C$, $\beta : B \rightarrow C$ in $\mathbf{C}$, with the same target C . We can then consider a category $\mathbf{C}_{\alpha,\beta}$ as follows:

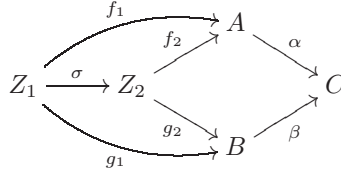
- $\text{Obj}(\mathbf{C}_{\alpha,\beta}) = \text{commutative diagrams}$



in $\mathbf{C}$, and

¹⁹There does not seem to be an established notation for these commonplace categories.

- morphisms correspond to commutative diagrams



A solid understanding of Example 3.9 will make this example look just as tame; at this point the reader should have no difficulties formalizing it (that is, explaining how composition works, what identities are, etc.).

Also left to the reader is the construction of the ‘mirror’ example $C^{\alpha,\beta}$, starting from two morphisms $\alpha : C \rightarrow A$, $\beta : C \rightarrow B$ with common *source*. $\lrcorner$

Exercises

3.1. $\triangleright$ Let C be a category. Consider a structure C^{op} with

- $\text{Obj}(C^{op}) := \text{Obj}(C)$;
- for A, B objects of C^{op} (hence objects of C), $\text{Hom}_{C^{op}}(A, B) := \text{Hom}_C(B, A)$.

Show how to make this into a category (that is, define composition of morphisms in C^{op} and verify the properties listed in §3.1).

Intuitively, the ‘opposite’ category C^{op} is simply obtained by ‘reversing all the arrows’ in C . [5.1, §VIII.1.1, §IX.1.2, IX.1.10]

3.2. If A is a finite set, how large is $\text{End}_{\text{Set}}(A)$?

3.3. $\triangleright$ Formulate precisely what it means to say that 1_a is an identity with respect to composition in Example 3.3, and prove this assertion. [§3.2]

3.4. Can we define a category in the style of Example 3.3 using the relation $<$ on the set $\mathbb{Z}$?

3.5. $\triangleright$ Explain in what sense Example 3.4 is an instance of the categories considered in Example 3.3. [§3.2]

3.6. $\triangleright$ (Assuming some familiarity with linear algebra.) Define a category V by taking $\text{Obj}(V) = \mathbb{N}$ and letting $\text{Hom}_V(n, m) =$ the set of $m \times n$ matrices with real entries, for all $n, m \in \mathbb{N}$. (I will leave the reader the task of making sense of a matrix with 0 rows or columns.) Use product of matrices to define composition. Does this category ‘feel’ familiar? [§VI.2.1, §VIII.1.3]

3.7. $\triangleright$ Define carefully objects and morphisms in Example 3.7, and draw the diagram corresponding to composition. [§3.2]

3.8. $\triangleright$ A *subcategory* C' of a category C consists of a collection of objects of C with sets of morphisms $\text{Hom}_{C'}(A, B) \subseteq \text{Hom}_C(A, B)$ for all objects A, B in $\text{Obj}(C')$, such that identities and compositions in C make C' into a category. A subcategory C' is

full if $\text{Hom}_{C'}(A, B) = \text{Hom}_C(A, B)$ for all A, B in $\text{Obj}(C')$. Construct a category of *infinite sets* and explain how it may be viewed as a full subcategory of **Set**. [4.4, §VI.1.1, §VIII.1.3]

3.9. ▷ An alternative to the notion of *multiset* introduced in §2.2 is obtained by considering sets endowed with equivalence relations; equivalent elements are taken to be multiple instances of elements ‘of the same kind’. Define a notion of morphism between such enhanced sets, obtaining a category **MSet** containing (a ‘copy’ of) **Set** as a full subcategory. (There may be more than one reasonable way to do this! This is intentionally an open-ended exercise.) Which objects in **MSet** determine ordinary multisets as defined in §2.2 and how? Spell out what a morphism of multisets would be from this point of view. (There are several natural notions of morphisms of multisets. Try to define morphisms in **MSet** so that the notion you obtain for ordinary multisets captures your intuitive understanding of these objects.) [§2.2, §3.2, 4.5]

3.10. Since the objects of a category **C** are not (necessarily interpreted as) sets, it is not clear how to make sense of a notion of ‘subobject’ in general. In some situations it *does* make sense to talk about subobjects, and the subobjects of any given object A in **C** are in one-to-one correspondence with the morphisms $A \rightarrow \Omega$ for a fixed, special object Ω of **C**, called a *subobject classifier*. Show that **Set** has a subobject classifier.

3.11. ▷ Draw the relevant diagrams and define composition and identities for the category $C^{A,B}$ mentioned in Example 3.9. Do the same for the category $C^{\alpha,\beta}$ mentioned in Example 3.10. [§5.5, 5.12]

4. Morphisms

Just as in **Set** we highlight certain types of functions (injective, surjective, bijective), it is useful to try to do the same for morphisms in an arbitrary category. The reader should note that defining qualities of morphisms by their actions on ‘elements’ is not an option in the general setting, because objects of an arbitrary category do not (in general) have ‘elements’.

This is why we spent some time analyzing injectivity, etc., from different viewpoints in §§2.4-2.6. It turns out that the other viewpoints on these notions do transfer nicely into the categorical setting.

4.1. Isomorphisms. Let **C** be a category.

Definition 4.1. A morphism $f \in \text{Hom}_C(A, B)$ is an *isomorphism* if it has a (two-sided) inverse under composition: that is, if $\exists g \in \text{Hom}_C(B, A)$ such that

$$gf = 1_A, \quad fg = 1_B. \quad \lrcorner$$

Recall that in §2.5 the inverse of a bijection of sets f was defined ‘elementwise’; in particular, there was no ambiguity in its definition, and we introduced the notation f^{-1} for this function. By contrast, the ‘inverse’ g produced in Definition 4.1

does not appear to have this uniqueness explicitly built into its definition. Luckily, its defining property *does* guarantee its uniqueness, but this requires a verification:

Proposition 4.2. *The inverse of an isomorphism is unique.*

Proof. We have to verify that if both g_1 and $g_2 : B \rightarrow A$ act as inverses of a given isomorphism $f : A \rightarrow B$, then $g_1 = g_2$. The standard trick for this kind of verification is to compose f on the left by one of the morphisms, and on the right by the other one; then apply associativity. The whole argument can be compressed into one line:

$$g_1 = g_1 1_B = g_1(fg_2) = (g_1f)g_2 = 1_A g_2 = g_2$$

as needed. $\square$

Note that the argument really proves that if f is a morphism with a left-inverse g_1 and a right-inverse g_2 , then necessarily f is an isomorphism, $g_1 = g_2$, and this morphism is the (unique) inverse of f . Look back at Corollary 2.2.

Since the inverse of f is uniquely determined by f , there is no ambiguity in denoting it by f^{-1} .

Proposition 4.3. *With notation as above:*

- Each identity 1_A is an isomorphism and is its own inverse.
- If f is an isomorphism, then f^{-1} is an isomorphism and further $(f^{-1})^{-1} = f$.
- If $f \in \text{Hom}_{\mathcal{C}}(A, B)$, $g \in \text{Hom}_{\mathcal{C}}(B, C)$ are isomorphisms, then the composition gf is an isomorphism and $(gf)^{-1} = f^{-1}g^{-1}$.

Proof. These all ‘prove themselves’. For example, it is immediate to verify that $f^{-1}g^{-1}$ is a left-inverse of gf : indeed²⁰,

$$(f^{-1}g^{-1})(gf) = f^{-1}((g^{-1}g)f) = f^{-1}(1_B f) = f^{-1}f = 1_A.$$

The verification that $f^{-1}g^{-1}$ is also a right-inverse of gf is analogous. $\square$

Note that taking the inverse reverses the order of composition: $(gf)^{-1} = f^{-1}g^{-1}$.

Two objects A, B of a category are *isomorphic* if there is an isomorphism $f : A \rightarrow B$. An immediate corollary of Proposition 4.3 is that ‘isomorphism’ is an equivalence relation²¹. If two objects A, B are isomorphic, one writes $A \cong B$.

Example 4.4. Of course, the isomorphisms in the category **Set** are precisely the bijections; this was observed at the beginning of §2.5. $\lrcorner$

Example 4.5. As noted in Proposition 4.3, identities are isomorphisms. They may be the *only* isomorphisms in a category: for example, this is the case in the category **C** obtained from the relation $\leq$ on $\mathbb{Z}$, as in Example 3.3. Indeed, for a, b objects of **C** (that is, $a, b \in \mathbb{Z}$), there is a morphism $f : a \rightarrow b$ and a

²⁰Associativity of composition implies that parentheses may be shuffled at will in longer expressions, as done here (cf. Exercise 4.1).

²¹The reader should have checked this in Exercise 2.4, for **Set**; the same proof will work in any category.

morphism $g : b \rightarrow a$ only if $a \leq b$ and $b \leq a$, that is, if $a = b$. So an isomorphism in $\mathbf{C}$ necessarily acts from an object a to itself; but in $\mathbf{C}$ there is only one such morphism, that is, 1_a . $\lrcorner$

Example 4.6. On the other hand, there are categories in which *every* morphism is an isomorphism; such categories are called *groupoids*. The reader ‘already knows’ many examples of groupoids; cf. Exercise 4.2. $\lrcorner$

An *automorphism* of an object A of a category $\mathbf{C}$ is an isomorphism from A to itself. The set of automorphisms of A is denoted $\text{Aut}_{\mathbf{C}}(A)$; it is a subset of $\text{End}_{\mathbf{C}}(A)$. By Proposition 4.3, composition confers on $\text{Aut}_{\mathbf{C}}(A)$ a remarkable structure:

- the composition of two elements $f, g \in \text{Aut}_{\mathbf{C}}(A)$ is an element $gf \in \text{Aut}_{\mathbf{C}}(A)$;
- composition is associative;
- $\text{Aut}_{\mathbf{C}}(A)$ contains the element 1_A , which is an identity for composition (that is, $f1_A = 1_Af = f$);
- every element $f \in \text{Aut}_{\mathbf{C}}(A)$ has an inverse $f^{-1} \in \text{Aut}_{\mathbf{C}}(A)$.

In other words, $\text{Aut}_{\mathbf{C}}(A)$ is a *group*, for all objects A of all categories $\mathbf{C}$.

We will soon devote all our attention to groups!

4.2. Monomorphisms and epimorphisms. As pointed out above, we do not have the option of defining for morphisms of an arbitrary category a notion such as ‘injective’ in the same way as we do for set-functions in §2.4: that definition requires a notion of ‘element’, and in general no such notion is available for objects of a category. But nothing prevents us from defining *monomorphisms* as we did in §2.6, in an arbitrary category:

Definition 4.7. Let $\mathbf{C}$ be a category. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is a *monomorphism* if the following holds:

$$\begin{aligned} &\text{for all objects } Z \text{ of } \mathbf{C} \text{ and all morphisms } \alpha', \alpha'' \in \text{Hom}_{\mathbf{C}}(Z, A), \\ &f \circ \alpha' = f \circ \alpha'' \implies \alpha' = \alpha''. \end{aligned} \quad \lrcorner$$

Similarly, *epimorphisms* are defined as follows:

Definition 4.8. Let $\mathbf{C}$ be a category. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is an *epimorphism* if the following holds:

$$\begin{aligned} &\text{for all objects } Z \text{ of } \mathbf{C} \text{ and all morphisms } \beta', \beta'' \in \text{Hom}_{\mathbf{C}}(B, Z), \\ &\beta' \circ f = \beta'' \circ f \implies \beta' = \beta''. \end{aligned} \quad \lrcorner$$

Example 4.9. As proven in Proposition 2.3, in the category **Set** the monomorphisms are precisely the injective functions. The reader should have by now checked that, likewise, in **Set** the epimorphisms are precisely the *surjective* functions (cf. Exercise 2.5). Thus, while the definitions given in §2.6 may have looked counterintuitive at first, they work as natural ‘categorical counterparts’ of the ordinary notions of injective/surjective functions. $\lrcorner$

Example 4.10. In the categories of Example 3.3, *every* morphism is both a monomorphism and an epimorphism. Indeed, recall that there is *at most one* morphism between any two objects in these categories; hence the conditions defining monomorphisms and epimorphisms are vacuous. $\lrcorner$

Contemplating Example 4.10 reveals a few unexpected twists in these definitions, which defy our intuition as set-theorists. For instance, in **Set**, a function is an isomorphism if and only if it is both injective and surjective, hence if and only if it is both a monomorphism and an epimorphism. But in the category defined by $\leq$ on $\mathbb{Z}$, *every* morphism is both a monomorphism and an epimorphism, while the only isomorphisms are the identities (Example 4.5). Thus this property is a special feature of **Set**, and we should not expect it to hold automatically in every category; it will not hold in the category **Ring** of *rings* (cf. §III.2.3). It *will* hold in every *abelian category* (of which **Set** is *not* an example!), but that is a story for a very distant future (Lemma IX.1.9).

Similarly, in **Set** a function is an epimorphism, that is, surjective, if and only if it has a right-inverse (Proposition 2.1); this may fail in general, even in respectable categories such as the category **Grp** of groups (cf. Exercise II.8.24).

Exercises

4.1. $\triangleright$ Composition is defined for *two* morphisms. If more than two morphisms are given, e.g.,

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{i} E,$$

then one may compose them in several ways, for example:

$$(ih)(gf), \quad (i(hg))f, \quad i((hg)f), \quad \text{etc.}$$

so that at every step one is only composing two morphisms. Prove that the result of any such nested composition is independent of the placement of the parentheses. (Hint: Use induction on n to show that any such choice for $f_n f_{n-1} \cdots f_1$ equals

$$((\cdots((f_n f_{n-1})f_{n-2})\cdots)f_1).$$

Carefully working out the case $n = 5$ is helpful.) [§4.1, §II.1.3]

4.2. $\triangleright$ In Example 3.3 we have seen how to construct a category from a set endowed with a relation, provided this latter is reflexive and transitive. For what types of relations is the corresponding category a groupoid (cf. Example 4.6)? [§4.1]

4.3. Let A, B be objects of a category **C**, and let $f \in \text{Hom}_{\mathbf{C}}(A, B)$ be a morphism.

- Prove that if f has a right-inverse, then f is an epimorphism.
- Show that the converse does not hold, by giving an explicit example of a category and an epimorphism without a right-inverse.

4.4. Prove that the composition of two monomorphisms is a monomorphism. Deduce that one can define a subcategory $\mathbf{C}_{\text{mono}}$ of a category **C** by taking the same

objects as in $\mathbf{C}$ and defining $\text{Hom}_{\mathbf{C}_{\text{mono}}}(A, B)$ to be the subset of $\text{Hom}_{\mathbf{C}}(A, B)$ consisting of monomorphisms, for all objects A, B . (Cf. Exercise 3.8; of course, in general $\mathbf{C}_{\text{mono}}$ is not full in $\mathbf{C}$.) Do the same for epimorphisms. Can you define a subcategory $\mathbf{C}_{\text{nonmono}}$ of $\mathbf{C}$ by restricting to morphisms that are *not* monomorphisms?

4.5. Give a concrete description of monomorphisms and epimorphisms in the category $\mathbf{MSet}$ you constructed in Exercise 3.9. (Your answer will depend on the notion of morphism you defined in that exercise!)

5. Universal properties

The ‘abstract’ examples in §3 may have left the reader with the impression that one can produce at will a large number of minute variations of the same basic ideas, without really breaking any new ground. This may be fun in itself, but *why* do we really want to explore this territory?

Categories offer a rich unifying language, giving us a bird’s eye view of many constructions in algebra (and other fields). In this course, this will be most apparent in the steady appearance of constructions satisfying suitable *universal properties*. For instance, we will see in a moment that products and disjoint unions (as reviewed in §1.3 and following) are characterized by certain universal properties having to do with the categories $\mathbf{C}_{A,B}$ and $\mathbf{C}^{A,B}$ considered in Example 3.9.

Many of the concepts introduced in this course will have an explicit description (such as the definition of product of sets given in §1.4) and an accompanying description in terms of a universal property (such as the one we will see in §5.4). The ‘explicit’ description may be very useful in concrete computations or arguments, but as a rule it is the universal property that clarifies the true nature of the construction. In some cases (such as for the disjoint union) the explicit description may turn out to depend on a seemingly arbitrary choice, while the universal property will have no element of arbitrariness. In fact, viewing the construction in terms of its corresponding universal property clarifies why one can only expect it to be defined ‘up to isomorphism’.

Also, deeper relationships become apparent when the constructions are viewed in terms of their universal properties. For example, we will see that products of sets and disjoint unions of sets are really ‘mirror’ constructions (in the sense that reversing arrows transforms the universal property for one into that for the other). This is not so clear (to this writer, anyway) from the explicit descriptions in §1.4.

5.1. Initial and final objects.

Definition 5.1. Let $\mathbf{C}$ be a category. We say that an object I of $\mathbf{C}$ is *initial* in $\mathbf{C}$ if for every object A of $\mathbf{C}$ there exists *exactly one* morphism $I \rightarrow A$ in $\mathbf{C}$:

$$\forall A \in \text{Obj}(\mathbf{C}) : \quad \text{Hom}_{\mathbf{C}}(I, A) \text{ is a singleton.}$$

We say that an object F of $\mathbf{C}$ is *final* in $\mathbf{C}$ if for every object A of $\mathbf{C}$ there exists *exactly one* morphism $A \rightarrow F$ in $\mathbf{C}$:

$$\forall A \in \text{Obj}(\mathbf{C}) : \quad \text{Hom}_{\mathbf{C}}(A, F) \text{ is a singleton.} \quad \lrcorner$$

One may use *terminal* to denote either possibility, but in general I would advise the reader to be explicit about which ‘end’ of $\mathbf{C}$ one is considering.

A category need not have initial or final objects, as the following example shows.

Example 5.2. The category obtained by endowing $\mathbb{Z}$ with the relation $\leq$ (see Example 3.3) has no initial or final object. Indeed, an initial object in this category would be an integer i such that $i \leq a$ for all integers a ; there is no such integer. Similarly, a final object would be an integer f larger than every integer, and there is no such thing.

By contrast, the category considered in Example 3.6 *does* have a final object, namely the pair $(3, 3)$; it still has no initial object. $\lrcorner$

Also, initial and final objects, when they exist, may or may not be unique:

Example 5.3. In **Set**, the empty set $\emptyset$ is initial (the ‘empty graph’ defines the unique function from $\emptyset$ to any given object!), and clearly it is the unique set that fits this requirement (Exercise 5.2).

Set also has final objects: for every set A , there is a unique function from A to a singleton $\{p\}$ (that is, the ‘constant’ function). *Every* singleton is final in **Set**; thus, final objects are not unique in this category. $\lrcorner$

However, I claim that if initial/final objects exist, then they are unique *up to a unique isomorphism*. I will invoke this fact frequently, so here is its official statement and its (immediate) proof:

Proposition 5.4. *Let $\mathbf{C}$ be a category.*

- *If I_1, I_2 are both initial objects in $\mathbf{C}$, then $I_1 \cong I_2$.*
- *If F_1, F_2 are both final objects in $\mathbf{C}$, then $F_1 \cong F_2$.*

Further, these isomorphisms are uniquely determined.

Proof. Recall that (by definition of category!) for every object A of $\mathbf{C}$ there is at least one element in $\text{Hom}_{\mathbf{C}}(A, A)$, namely the identity 1_A . If I is initial, then there is a *unique* morphism $I \rightarrow I$, which therefore must be the identity 1_I .

Now assume I_1 and I_2 are both initial in $\mathbf{C}$. Since I_1 is initial, there is a *unique* morphism $f : I_1 \rightarrow I_2$ in $\mathbf{C}$; we have to show that f is an isomorphism. Since I_2 is initial, there is a unique morphism $g : I_2 \rightarrow I_1$ in $\mathbf{C}$. Consider $gf : I_1 \rightarrow I_1$; as observed, necessarily

$$gf = 1_{I_1}$$

since I_1 is initial. By the same token

$$fg = 1_{I_2}$$

since I_2 is initial. This proves that $f : I_1 \rightarrow I_2$ is an isomorphism, as needed.

The proof for final objects is entirely analogous (Exercise 5.3). $\square$

Proposition 5.4 “explains” why, while not unique, the final objects in **Set** are all isomorphic: no singleton is more ‘special’ than any other singleton; this is the typical situation. There may be psychological reasons why *one* initial or final object looks more compelling than others (for example, the singleton $\{\emptyset\} = 2^\emptyset$ may look to some like the most ‘natural’ choice among all singletons), but this plays no role in how these objects sit in their category.

5.2. Universal properties. The most natural context in which to introduce *universal properties* requires a good familiarity with the language of *functors*, which we will only introduce at a later stage (cf. §VIII.1.1). For the purpose of the examples we will run across in (most of) this book, the following ‘working definition’ should suffice.

We say that a construction *satisfies a universal property* (or ‘is the solution to a universal problem’) when it may be viewed as a terminal object of a category. The category depends on the context and is usually explained ‘in words’ (and often without even mentioning the word *category*).

In particularly simple cases this may take the form of a statement such as $\emptyset$ *is universal with respect to the property of mapping to sets*; this is synonymous with the assertion that $\emptyset$ is initial in the category **Set**.

More often, the situation is more complex. Since being initial/final amounts to the existence and uniqueness of certain morphisms, the ‘explanation’ of a universal property may follow the pattern, “object X is universal with respect to the following property: for any Y such that..., there exists a unique morphism $Y \rightarrow X$ such that...”

The not-so-naïve reader will recognize that this explanation hides the definition of an accessory category and the statement that X is terminal (probably final in this case) in this new category. It is useful to learn how to translate such wordy explanations into what they really mean. Also, the reader should keep in mind that it is not uncommon to sweep under the rug part of the essential information about the solution to a universal problem (usually some key morphism): this information is presumably implicit in any given set-up. This will be apparent from the examples that follow.

5.3. Quotients. Let $\sim$ be an equivalence relation defined on a set A . Let’s parse the assertion:

“The quotient $A/\sim$ is universal with respect to the property of mapping A to a set in such a way that equivalent elements have the same image.”

What can this possibly mean, and is it true?

The assertion is talking about functions

$$A \xrightarrow{\varphi} Z$$

with Z any set, satisfying the property

$$a' \sim a'' \implies \varphi(a') = \varphi(a'').$$

These morphisms are objects of a category (very similar to the category defined in Example 3.7); for convenience, let's denote such an object by (φ, Z) . The only reasonable way to define morphisms $(\varphi_1, Z_1) \rightarrow (\varphi_2, Z_2)$ is as commutative diagrams

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ \varphi_1 \swarrow & & \nearrow \varphi_2 \\ & A & \end{array}$$

This is the same definition considered in Example 3.7.

Does this category have initial objects?

Claim 5.5. *Denoting by π the ‘canonical projection’ defined in Example 2.6, the pair $(\pi, A/\sim)$ is an initial object of this category.*

This is what our writer meant by the mysterious assertion copied above. Once this is understood, it is very easy to prove that the assertion is indeed correct.

Proof. Consider any (φ, Z) as above. We have to prove that there exists a unique morphism $(\pi, A/\sim) \rightarrow (\varphi, Z)$, that is, a unique commutative diagram

$$\begin{array}{ccc} A/\sim & \xrightarrow{\bar{\varphi}} & Z \\ \pi \swarrow & & \nearrow \varphi \\ & A & \end{array}$$

that is, a unique function $\bar{\varphi}$ making this diagram commute.

Let $[a]_\sim$ be an arbitrary element of $A/\sim$. If the diagram is indeed going to commute, then necessarily

$$\bar{\varphi}([a]_\sim) = \varphi(a);$$

this tells us that $\bar{\varphi}$ is indeed *unique*, if it exists at all—that is, *if* this prescription does define a function $A/\sim \rightarrow Z$.

Hence, all we have to check is that $\bar{\varphi}$ is well-defined, that is, that if $[a_1]_\sim = [a_2]_\sim$, then $\varphi(a_1) = \varphi(a_2)$; and indeed

$$[a_1]_\sim = [a_2]_\sim \implies a_1 \sim a_2 \implies \varphi(a_1) = \varphi(a_2).$$

This is precisely the condition that morphisms in our category satisfy. $\square$

Note the several levels of sloppiness in the assertion considered above: it does not tell us very explicitly what category to consider; it does not tell us that we should especially pay attention to *initial* objects in this category. Worst of all, the solution to the universal problem is *not really* $A/\sim$, but rather *the morphism* $\pi : A \rightarrow A/\sim$.

The reader should practice the skill of translating loose assertions such as the one given above into precise statements; it is not at all uncommon to run into examples at the same level of ‘abuse of language’ as this one.

The reason why we get away with writing such assertions is that the context really allows the experienced reader to parse them effectively, and they are substantially more concise than their spelled-out version. After all, there is in general no

conceivable choice for a morphism $A \rightarrow A/\sim$ other than the canonical projection; hence, neglecting to mention it is forgivable. Also, the *final* object in the category considered above is supremely uninteresting (what is it? cf. Exercise 5.5), so surely we must have meant the *initial* one.

What do we learn by viewing quotients in terms of their universal property? For example, suppose $\sim$ is the equivalence relation defined starting from a function $f : A \rightarrow B$, as in §2.8. Then the reader will realize easily that $\text{im } f$ *also* satisfies the universal property given above for $A/\sim$; therefore (by Proposition 5.4) $\text{im } f$ and $A/\sim$ must be isomorphic. This is precisely the content of Theorem 2.7; thus, the universal property sheds some light on the ‘canonical decomposition’ studied in §2.8.

5.4. Products. It is also a very good exercise to stare at a familiar construction and try to see the universal property which may be behind it. I will now encourage you, dear reader, to contemplate the notion of the *product of two sets* given in §1.4 and to see if the universal property it satisfies jumps out at you. Spoiler follows, so this is a good time to stop reading these notes and to try on your own.

Here is the universal property. Let A, B be sets, and consider the product $A \times B$, with the two natural projections:

$$\begin{array}{ccc} & & A \\ & \nearrow \pi_A & \\ A \times B & & \\ & \searrow \pi_B & \\ & & B \end{array}$$

(see Example 2.4). Then for every set Z and morphisms

$$\begin{array}{ccc} & & A \\ & \nearrow f_A & \\ Z & & \\ & \searrow f_B & \\ & & B \end{array}$$

there exists a unique morphism $\sigma : Z \rightarrow A \times B$ such that the diagram

$$\begin{array}{ccccc} & & f_A & \rightarrow & A \\ & \nearrow & & \nearrow \pi_A & \\ Z & \xrightarrow{\sigma} & A \times B & & \\ & \searrow & & \searrow \pi_B & \\ & & f_B & \rightarrow & B \end{array}$$

commutes.

In this situation, σ is usually denoted $f_A \times f_B$.

Proof. Define $\forall z \in Z$

$$\sigma(z) = (f_A(z), f_B(z)).$$

This function²² manifestly makes the diagram commute: $\forall z \in Z$

$$\pi_A \sigma(z) = \pi_A(f_A(z), f_B(z)) = f_A(z),$$

²²Note that there is no ‘well-definedness’ issue this time.

showing that $\pi_A \sigma = f_A$ and similarly $\pi_B \sigma = f_B$.

Further, the definition is forced by the commutativity of the diagram; so σ is unique, as claimed. $\square$

In other words, products of sets (or, more precisely, products of sets together with the information of their natural projections to the factors) are *final* objects in the category $\mathbf{C}_{A,B}$ considered in Example 3.9, for $\mathbf{C} = \mathbf{Set}$.

What is the advantage of viewing products this way? The main advantage is that the universal property may be stated in *any* category, while the definition of products given in §1.4 only makes sense in $\mathbf{Set}$ (and possibly in other categories where one has a notion of ‘elements’). We say that a category $\mathbf{C}$ *has (finite) products*, or is a category ‘with (finite) products’, if for all objects A, B in $\mathbf{C}$ the category $\mathbf{C}_{A,B}$ considered in Example 3.9 has final objects. Such a final object consists of the data of an object of $\mathbf{C}$, usually denoted $A \times B$, and of two morphisms $A \times B \rightarrow A$, $A \times B \rightarrow B$.

Note that a ‘product’ from this perspective does not need to ‘look like a product’. Consider our recurring example of the category obtained from $\leq$ on $\mathbb{Z}$, as in Example 3.3. Does this category have products? Objects of this category are simply integers $a, b \in \mathbb{Z}$; call $a \times b$ for a moment the ‘categorical’ product of a and b . The universal property written out above becomes, in this case, *for all $z \in \mathbb{Z}$ such that $z \leq a$ and $z \leq b$, we have $z \leq a \times b$* .

This universal problem *does* have a solution $\forall a, b$: it is conventionally not called $a \times b$, but rather $\min(a, b)$. It is immediate to see that $\min(a, b)$ satisfies the property. Thus this category *has products*, and in fact we see that the product in this category amounts to the familiar operation of taking the minimum of two integers.

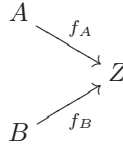
Thus there is an unexpected connection between ‘the Cartesian product of two sets’ and ‘the minimum of two integers’: both are examples of products, taken in different categories; they both satisfy ‘the same’ universal property, in different contexts.

5.5. Coproducts. The prefix *co-* usually indicates that one is ‘reversing all arrows’. Just as *products* are final objects in the categories $\mathbf{C}_{A,B}$ obtained by considering morphisms in $\mathbf{C}$ with common *source*, whose *targets* are A and B , *coproducts* will be *initial* objects in the categories²³ $\mathbf{C}^{A,B}$ of morphisms with common *target*, whose *sources* are A and B . Dear reader, look away and spell this universal property out before we do.

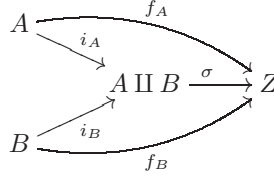
Here it is. Let A, B be objects of a category $\mathbf{C}$. A *coproduct* $A \amalg B$ of A and B will be an object of $\mathbf{C}$, endowed with two morphisms $i_A : A \rightarrow A \amalg B$, $i_B : B \rightarrow A \amalg B$ and satisfying the following universal property: *for all objects Z*

²³These categories were also considered in Example 3.9; cf. Exercise 3.11.

and morphisms



there exists a unique morphism $\sigma : A \amalg B \rightarrow Z$ such that the diagram



commutes.

The symmetry with the universal property of products is hopefully completely apparent. We say that a category $\mathbf{C}$ has *coproducts* if this universal problem has a solution for all pairs of objects A and B .

Is the reader familiar with any coproduct? Yes!

Proposition 5.6. *The disjoint union is a coproduct in $\mathbf{Set}$.*

Proof. Recall (§1.4) that the disjoint union $A \amalg B$ is defined as the union of two disjoint isomorphic copies A' , B' of A , B , respectively; for example, we may let $A' = \{0\} \times A$, $B' = \{1\} \times B$. The functions i_A, i_B are defined by

$$i_A(a) = (0, a), \quad i_B(b) = (1, b),$$

where we view these elements as elements of $(\{0\} \times A) \cup (\{1\} \times B)$.

Now let $f_A : A \rightarrow Z$, $f_B : B \rightarrow Z$ be arbitrary morphisms to a common target. Define

$$\sigma : A \amalg B = (\{0\} \times A) \cup (\{1\} \times B) \rightarrow Z$$

by

$$\sigma(c) = \begin{cases} f_A(a) & \text{if } c = (0, a) \in \{0\} \times A, \\ f_B(b) & \text{if } c = (1, b) \in \{1\} \times B. \end{cases}$$

This definition makes the relevant diagram commute and is in fact forced upon us by this commutativity, proving that σ exists and is unique. $\square$

This observation tells us that the category $\mathbf{Set}$ has coproducts and further sheds considerable light on the mysteries of disjoint unions. For example, there was an element of arbitrariness in our choice of ‘a’ disjoint union, although different choices led to isomorphic notions. Now we see why: terminal objects of a category are not unique in general, although they are unique up to isomorphism (Proposition 5.4); there is not a ‘most beautiful’ disjoint union of two sets just as there is not a ‘most beautiful’ singleton in $\mathbf{Set}$ (cf. §5.1).

Also, an unexpected ‘symmetry’ between products and disjoint unions becomes suddenly apparent from the point of view of universal properties.

The reader is invited to contemplate the notion of coproduct in the other categories we have encountered. For example (and probably not surprisingly at this point) the category obtained from $\leq$ on $\mathbb{Z}$ *does* have coproducts: the coproduct of two objects (i.e., integers) a, b is simply the *maximum* of a and b .

Exercises

5.1. Prove that a final object in a category $\mathbf{C}$ is initial in the opposite category $\mathbf{C}^{op}$ (cf. Exercise 3.1).

5.2. $\triangleright$ Prove that $\emptyset$ is the *unique* initial object in **Set**. [§5.1]

5.3. $\triangleright$ Prove that final objects are unique up to isomorphism. [§5.1]

5.4. What are initial and final objects in the category of ‘pointed sets’ (Example 3.8)? Are they unique?

5.5. $\triangleright$ What are the final objects in the category considered in §5.3? [§5.3]

5.6. $\triangleright$ Consider the category corresponding to endowing (as in Example 3.3) the set $\mathbb{Z}^+$ of positive integers with the *divisibility* relation. Thus there is exactly one morphism $d \rightarrow m$ in this category if and only if d divides m without remainder; there is no morphism between d and m otherwise. Show that this category has products and coproducts. What are their ‘conventional’ names? [§VII.5.1]

5.7. Redo Exercise 2.9, this time using Proposition 5.4.

5.8. Show that in every category $\mathbf{C}$ the products $A \times B$ and $B \times A$ are isomorphic, if they exist. (Hint: Observe that they both satisfy the universal property for the product of A and B ; then use Proposition 5.4.)

5.9. Let $\mathbf{C}$ be a category with products. Find a reasonable candidate for the universal property that the product $A \times B \times C$ of *three* objects of $\mathbf{C}$ ought to satisfy, and prove that both $(A \times B) \times C$ and $A \times (B \times C)$ satisfy this universal property. Deduce that $(A \times B) \times C$ and $A \times (B \times C)$ are necessarily isomorphic.

5.10. Push the envelope a little further still, and define products and coproducts for *families* (i.e., indexed sets) of objects of a category.

Do these exist in **Set**?

It is common to denote the product $\underbrace{A \times \cdots \times A}_{n \text{ times}}$ by A^n .

5.11. Let A , resp. B be a set, endowed with an equivalence relation $\sim_A$, resp. $\sim_B$. Define a relation $\sim$ on $A \times B$ by setting

$$(a_1, b_1) \sim (a_2, b_2) \iff a_1 \sim_A a_2 \text{ and } b_1 \sim_B b_2.$$

(This is immediately seen to be an equivalence relation.)

- Use the universal property for quotients (§5.3) to establish that there are functions $(A \times B)/\sim \rightarrow A/\sim_A$, $(A \times B)/\sim \rightarrow B/\sim_B$.

- Prove that $(A \times B)/\sim$, with these two functions, satisfies the universal property for the product of $A/\sim_A$ and $B/\sim_B$.
- Conclude (without further work) that $(A \times B)/\sim \cong (A/\sim_A) \times (B/\sim_B)$.

5.12. $\neg$ Define the notions of *fibered products* and *fibered coproducts*, as terminal objects of the categories $\mathbf{C}_{\alpha,\beta}$, $\mathbf{C}^{\alpha,\beta}$ considered in Example 3.10 (cf. also Exercise 3.11), by stating carefully the corresponding universal properties.

As it happens, **Set** has both fibered products and coproducts. Define these objects ‘concretely’, in terms of naive set theory. [II.3.9, III.6.10, III.6.11]

Groups, first encounter

In this chapter we introduce groups, we observe they form a category (called $\mathbf{Grp}$), and we study ‘general’ features of this category: what are the monomorphisms and the epimorphisms in this category? what is the appropriate notion of ‘equivalence relation’ and ‘quotients’ for a group? does a ‘decomposition theorem’ hold in $\mathbf{Grp}$? and other analogous questions.

In Chapter III we will acquire a similar degree of familiarity with *rings* and *modules*. A more object-oriented analysis of $\mathbf{Grp}$ (for example, a treatment of the famous *Sylow theorems*, ‘composition series’, or the classification of finite abelian groups) is deferred to Chapter IV.

1. Definition of group

1.1. Groups and groupoids.

Joke 1.1. *Definition:* A group is a groupoid with a single object. ┘

This is actually a perfectly viable definition, since groupoids have been defined already (in Example I.4.6); but most mathematicians would find it ludicrous to introduce groups in this fashion, or they will at the very least politely express doubts on the pedagogical effectiveness of doing so. In order to redeem myself, I will parse this definition right away to show what it really says. If $*$ is the lone object of such a groupoid $\mathbf{G}$,

$$\mathrm{Hom}_{\mathbf{G}}(*, *) = \mathrm{Aut}_{\mathbf{G}}(*)$$

(because $\mathbf{G}$ is a groupoid!), and this set carries all the information about $\mathbf{G}$. Call this set G . Then (by definition of category) there is an associative operation on G , with an identity 1_* , and (by definition of groupoid, which says that every morphism in $\mathbf{G}$ is an isomorphism) every $g \in G$ has an inverse $g^{-1} \in G$.

That is what a group is¹: a set G with a composition law satisfying a few key axioms, i.e., associativity, existence of identity, and existence of inverses.

1.2. Definition. Now for the official definition. Let G be a nonempty set, endowed with a *binary operation*, that is, a ‘multiplication’ map

$$\bullet : G \times G \rightarrow G.$$

Our notation will be

$$\bullet(g, h) =: g \bullet h$$

or simply gh if the name of the operation can be understood. The careful reader may have expected that we should write $h \bullet g$, in the style of what we have done for categories, but this is what common conventions dictate².

Definition 1.2. The set G , endowed with the binary operation $\bullet$ (briefly, $(G, \bullet)$, or simply G if the operation can be understood) is a *group* if

(i) the operation $\bullet$ is *associative*, that is,

$$(\forall g, h, k \in G) : (g \bullet h) \bullet k = g \bullet (h \bullet k);$$

(ii) there exists an *identity element* e_G for $\bullet$, that is,

$$(\exists e_G \in G) (\forall g \in G) : g \bullet e_G = g = e_G \bullet g;$$

(iii) every element in G has an *inverse* with respect to $\bullet$, that is,

$$(\forall g \in G) (\exists h \in G) : g \bullet h = e_G = h \bullet g. \quad \lrcorner$$

Example 1.3. Since we explicitly require G to be nonempty, the most economical way to concoct a group is by letting $G = \{e\}$ be a singleton. There is only one function $G \times G \rightarrow G$ in this case, so there is only one possible binary operation on G , defined by

$$e \bullet e := e.$$

The three axioms trivially hold for this example, so $\{e\}$ is equipped with a unique group structure.

This is usually called the *trivial group*; purists should call any such group a trivial group, since every singleton gives rise to one. \lrcorner

Example 1.4. The reader should check carefully (cf. Exercise 1.2) that $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, and several variations using $\cdot$ (for example, the subset $\{+1, -1\}$ of $\mathbb{Z}$, with ordinary multiplication) all give examples of groups. While very interesting in themselves, these examples do not really capture at any intuitive level ‘what’ a group really is, because they are too special. For example, all these examples are *commutative* (see §1.5). \lrcorner

¹From this perspective, Joke 1.1 is a little imprecise: the group is not the groupoid $\mathbf{G}$, but rather the set of isomorphisms in $\mathbf{G}$, endowed with the operation of composition of morphisms.

²Not without exceptions; see for example permutation groups, discussed in §2.

Example 1.5. My readers are likely familiar with an extremely important *non-commutative* example, namely the group of *invertible*, $n \times n$ matrices with (say) real entries, $n \geq 2$. I will generally shy away from this class of examples in these early chapters, since we will have ample opportunities to think about matrices when we approach linear algebra (starting in Chapter VI). But we may occasionally borrow a matrix or two before then. The reader should now check that 2×2 matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with real entries, and such that $ad - bc \neq 0$, form a group under the ordinary matrix multiplication:

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \cdot \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{pmatrix}.$$

(The condition $ad - bc \neq 0$ guarantees that the matrix is *invertible*. What is its inverse?) Since, for example,

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

this group is indeed *not* commutative.

The group of invertible $n \times n$ matrices with real entries is denoted $\mathrm{GL}_n(\mathbb{R})$. $\lrcorner$

We will encounter more representative examples in §2.

1.3. Basic properties. From the ‘groupoid’ point of view, the identity e_G would be denoted 1_* . It is not uncommon to omit G from the notation (when the group is understood) and to use different symbols rather than e to denote this element: 1 and 0 are popular alternatives, depending on the context. In any case, for any given group G this element is unique. That is, no other element of G can work as an identity:

Proposition 1.6. *If $h \in G$ is an identity of G , then $h = e_G$.*

Incidentally, this makes groups *pointed sets* in the sense of Example I.3.8: every group has a well-defined distinguished element.

Proof. Using first that e_G is an identity and then that h is an identity, one gets³

$$h = e_G h = e_G.$$

(Amusingly, this argument only uses that e_G is a ‘left’ identity and h is a ‘right’ identity.) $\square$

Proposition 1.7. *The inverse is also unique: if h_1, h_2 are both inverses of g in G , then $h_1 = h_2$.*

³As previously announced, we may omit the symbol $\bullet$ for the operation, since for the time being we are only considering *one* operation. This will be done without warning in the future.

Proof. This actually follows from Proposition I.4.2 (by viewing G as the set of isomorphisms of a groupoid with a single object). The reader should construct a stand-alone proof, using the same trick, but carefully hiding any reference to morphisms. $\square$

Proposition 1.7 authorizes us to give a name to *the* inverse of g : this is usually⁴ denoted g^{-1} .

One more notational item is in order. The definition of a group only contemplates the ‘product’ of *two* elements; in multiplying a string of elements, one may in principle have a choice as to the order in which products are executed. For example,

$$(g_1 \bullet g_2) \bullet g_3$$

stands for: apply the operation $\bullet$ to g_1 and g_2 , and then apply $\bullet$ again to the result of this operation and g_3 ; while

$$g_1 \bullet (g_2 \bullet g_3)$$

stands for: apply $\bullet$ to g_2 and g_3 , and then apply it again to g_1 and to the result of this operation.

Associativity tells us precisely that the result of the operation on three elements does not depend on the way in which we perform it. With this in mind, we *are* authorized to write

$$g_1 \bullet g_2 \bullet g_3;$$

this expression is not ambiguous, *by associativity*. What about four or more elements? The reader should have checked in Exercise I.4.1 that all ways to associate any number of elements leads to the same result. So we are also authorized to write things like

$$g_1 \bullet g_2 \bullet g_3 \bullet \cdots \bullet g_{17};$$

this is also unambiguous. The reader should keep in mind, however, that of course the order in which the *elements* are listed *is* important: in general,

$$g_1 \bullet g_2 \bullet g_3 \bullet \cdots \bullet g_{17} \neq g_2 \bullet g_1 \bullet g_3 \bullet \cdots \bullet g_{17}.$$

Of course no such care is necessary if all g_i coincide; the conventional ‘power’ notation can then be used⁵: $g^0 = e_G$, and for a positive integer n

$$g^n = \underbrace{g \cdots g}_{n \text{ times}}, \quad g^{-n} = \underbrace{g^{-1} \cdots g^{-1}}_{n \text{ times}}.$$

It is easy to check that then $\forall g \in G$ and $\forall m, n \in \mathbb{Z}$

$$g^{m+n} = g^m g^n.$$

⁴But note the ‘abelian’ case, discussed in §1.5.

⁵In the abelian case one uses ‘multiples’; cf. §1.5.

1.4. Cancellation. ‘Cancellation’ holds in groups. That is,

Proposition 1.8. *Let G be a group. Then $\forall a, g, h \in G$*

$$ga = ha \implies g = h, \quad ag = ah \implies g = h.$$

Proof. Both statements are proven by multiplying (on the appropriate side) by a^{-1} and applying associativity. For example,

$$\begin{aligned} ga = ha &\implies (ga)a^{-1} = (ha)a^{-1} \implies g(aa^{-1}) = h(aa^{-1}) \implies ge_G = he_G \\ &\implies g = h. \end{aligned}$$

The proof for the other implication follows the same pattern. $\square$

Examples of operations which do *not* satisfy cancellation, and hence do *not* define groups, abound. For instance, the operation of ordinary multiplication does *not* make the set $\mathbb{R}$ of real numbers into a group: indeed, 0 ‘cannot be cancelled’ since $1 \cdot 0 = 2 \cdot 0$ even if $1 \neq 2$. Of course, the problem here is that 0 does not have an inverse in $\mathbb{R}$, with respect to multiplication. As it happens, this is the *only* problem with this example: ordinary multiplication *does* make

$$\mathbb{R}^* := \mathbb{R} \setminus \{0\}$$

into a group, as the reader should check immediately.

1.5. Commutative groups. One axiom *not* appearing in the definition of group is *commutativity*: we say that the operation $\bullet$ is ‘commutative’ if

$$(iv) \quad (\forall g, h \in G) : \quad g \bullet h = h \bullet g.$$

We say that two elements g, h ‘commute’ if $gh = hg$. Thus, in a commutative group any two elements commute.

‘Commutative groups’ are important objects: they arise naturally in several contexts, especially as ‘modules over the ring $\mathbb{Z}$ ’ (about which we will have a lot to say in Chapter III and beyond). When they do so, they are usually called *abelian groups*.

The notation used in treating abelian groups differs somewhat from the standard notation for groups. This is to emphasize the ‘ $\mathbb{Z}$ -module structure’, and it is helpful when an abelian group coexists with other operations—a situation which we will encounter frequently.

Thus, the *operation* in an abelian group A is, as a rule, denoted by $+$ and is called ‘addition’; the *identity* is then called 0_A ; and the *inverse* of an element $a \in A$ is denoted $-a$ (and maybe should be called the ‘opposite’?). The ‘power’ notation is of course replaced by ‘multiple’: $0a = 0$, and for a positive integer n

$$na = \underbrace{a + \cdots + a}_{n \text{ times}}, \quad (-n)a = \underbrace{(-a) + \cdots + (-a)}_{n \text{ times}}.$$

The reader should keep in mind that at this stage ‘ na ’ is a *notation*, not the result of applying a binary operation to two elements n, a of A . Indeed, $n \in \mathbb{Z}$ may very well not be an element of A in any reasonable sense. Moreover, it may very

well be that $na = ma$ even if $n \neq m$, in spite of the fact that ‘cancellation’ works in groups.

The qualifier ‘abelian’ and the notation 0_A , $-a$, etc., are mostly used for commutative groups arising in certain standard situations: for example, the notions of rings, modules, vector spaces are defined by suitably enriching a commutative group, which is then promoted to ‘abelian’ for notational convenience.

There are other situations in which commutative groups arise naturally, without triggering the ‘abelian’ notation. For example, the group $(\mathbb{R}^*, \cdot)$ mentioned at the end of §1.4 *is* commutative, but its operation is indicated by $\cdot$ (if at all), and its identity element is written 1. History, rather than logic, is often the main factor determining notation.

1.6. Order.

Definition 1.9. An element g of a group G has *finite order* if⁶ $g^n = e$ for some positive integer n . In this case, the *order* $|g|$ is the *smallest* positive n such that $g^n = e$. One writes $|g| = \infty$ if g does not have finite order. $\lrcorner$

By definition, if $g^n = e$ for some positive integer n , then $|g| \leq n$. One can be more precise:

Lemma 1.10. *If $g^n = e$ for some positive integer n , then $|g|$ is a divisor of n .*

Proof. As observed, $n \geq |g|$ by definition of order, that is, $n - |g| \geq 0$. There must then exist⁷ a positive integer m such that

$$r = n - |g| \cdot m \geq 0 \quad \text{and} \quad n - |g| \cdot (m + 1) < 0,$$

that is, $r < |g|$. Note that

$$g^r = g^{n - |g| \cdot m} = g^n \cdot (g^{|g|})^{-m} = e \cdot e^{-m} = e.$$

By definition of order, $|g|$ is the smallest *positive* integer such that $g^{|g|} = e$. Since r is smaller than $|g|$ and $g^r = e$, r cannot be positive; hence $r = 0$ necessarily. This says

$$0 = n - |g| \cdot m,$$

that is, $n = |g| \cdot m$, proving that n is indeed an integer multiple of $|g|$ as claimed. $\square$

This lemma has the following immediate and useful consequence, which we encourage the reader to keep firmly in mind:

Corollary 1.11. *Let g be an element of finite order, and let $N \in \mathbb{Z}$. Then*

$$g^N = e \iff N \text{ is a multiple of } |g|.$$

⁶Of course in an abelian group we would write the following prescription as $ng = 0$.

⁷Purists may object that here I am surreptitiously using fairly sophisticated information about $\mathbb{Z}$, namely the ‘division algorithm’, hence essentially the fact that $\mathbb{Z}$ is a Euclidean domain! This is material that will have to wait until Chapter V to be given some justice. I may as well be open about it and admit that yes, I am assuming that my readers have already acquired a thorough familiarity with the operations of addition and multiplication among integers. Shame on me!

Definition 1.12. If G is finite as a set, its *order* $|G|$ is the number of its elements; we write $|G| = \infty$ if G is infinite. $\lrcorner$

Cancellation implies that $|g| \leq |G|$ for all $g \in G$. Indeed, this is vacuously true if $|G| = \infty$; if G is finite, consider the $|G| + 1$ powers

$$g^0 = e, g, g^2, g^3, \dots, g^{|G|}$$

of g . These cannot all be distinct; hence

$$(\exists i, j) \quad 0 \leq i < j \leq |G| \quad \text{such that } g^i = g^j.$$

By cancellation (that is, multiplying on the right by g^{-i})

$$g^{j-i} = e,$$

showing $|g| \leq (j - i) \leq |G|$.

We will soon be able to formulate a much more precise statement concerning the relation between the order of a group and the order of its elements: if $g \in G$ and $|G|$ is finite, then the order of g *divides* the order of G . This will be an immediate consequence of *Lagrange's theorem*; cf. Example 8.15.

Another general remark concerning orders is that their behavior with respect to the operation of the group is not always predictable: it may very well happen that g, h have finite order in a group G , and yet $|gh| = \infty$, or $|gh| =$ your favorite positive integer: work out Exercise 1.12 and Exercise 2.6 if you don't believe it.

On the other hand, the situation is more constrained if g and h commute. In the extreme case in which $g = h$, it is easy to obtain a very precise statement:

Proposition 1.13. *Let $g \in G$ be an element of finite order. Then g^m has finite order $\forall m \geq 0$, and in fact⁸*

$$|g^m| = \frac{\text{lcm}(m, |g|)}{m} = \frac{|g|}{\text{gcd}(m, |g|)}.$$

Proof. The equality of the two numbers $\frac{\text{lcm}(m, |g|)}{m}$ and $\frac{|g|}{\text{gcd}(m, |g|)}$ follows from elementary properties of gcd and lcm: $\text{lcm}(a, b) = ab / \text{gcd}(a, b)$ for all a and b . So we only need to prove that $|g^m| = \frac{\text{lcm}(m, |g|)}{m}$.

The order of g^m is the least positive d for which

$$g^{md} = e,$$

that is (by Corollary 1.11) for which md is a multiple of $|g|$. In other words, $m|g^m|$ is the smallest multiple of m which is also a multiple of $|g|$:

$$m|g^m| = \text{lcm}(m, |g|).$$

The stated formula follows immediately from this. $\square$

In general, for commuting elements,

Proposition 1.14. *If $gh = hg$, then $|gh|$ divides $\text{lcm}(|g|, |h|)$.*

⁸The notation lcm stands for 'least common multiple'. I am also assuming that the reader is familiar with simple properties of gcd and lcm.

Proof. Let $|g| = m$, $|h| = n$. If N is any common multiple of m and n , then $g^N = h^N = e$ by Corollary 1.11. Since g and h commute,

$$(gh)^N = \underbrace{(gh)(gh) \cdots (gh)}_{N \text{ times}} = \underbrace{gg \cdots g}_{N \text{ times}} \cdot \underbrace{hh \cdots h}_{N \text{ times}} = g^N h^N = e.$$

As this holds for every common multiple N of m and n , in particular

$$(gh)^{\text{lcm}(m,n)} = e.$$

The statement then follows from Lemma 1.10. $\square$

One cannot say more about $|gh|$ in general, even if g and h commute (Exercise 1.13). But see Exercise 1.14 for an important special case.

Exercises

1.1. $\triangleright$ Write a careful proof that every group is the group of isomorphisms of a groupoid. In particular, every group is the group of automorphisms of some object in some category. [§2.1]

1.2. $\triangleright$ Consider the ‘sets of numbers’ listed in §1.1, and decide which are made into groups by conventional operations such as $+$ and $\cdot$. Even if the answer is negative (for example, $(\mathbb{R}, \cdot)$ is not a group), see if variations on the definition of these sets lead to groups (for example, $(\mathbb{R}^*, \cdot)$ is a group; cf. §1.4). [§1.2]

1.3. Prove that $(gh)^{-1} = h^{-1}g^{-1}$ for all elements g, h of a group G .

1.4. Suppose that $g^2 = e$ for all elements g of a group G ; prove that G is commutative.

1.5. The ‘multiplication table’ of a group is an array compiling the results of all multiplications $g \bullet h$:

$\bullet$	e	$\cdots$	h	$\cdots$
e	e	$\cdots$	h	$\cdots$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
g	g	$\cdots$	$g \bullet h$	$\cdots$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

(Here e is the identity element. Of course the table depends on the order in which the elements are listed in the top row and leftmost column.) Prove that every row and every column of the multiplication table of a group contains all elements of the group exactly once (like Sudoku diagrams!).

1.6. $\neg$ Prove that there is only *one* possible multiplication table for G if G has exactly 1, 2, or 3 elements. Analyze the possible multiplication tables for groups with exactly 4 elements, and show that there are *two* distinct tables, up to re-ordering the elements of G . Use these tables to prove that all groups with ≤ 4 elements are commutative.

(You are welcome to analyze groups with 5 elements using the same technique, but you will soon know enough about groups to be able to avoid such brute-force approaches.) [2.19]

1.7. Prove Corollary 1.11.

1.8. $\neg$ Let G be a finite abelian group with exactly one element f of order 2. Prove that $\prod_{g \in G} g = f$. [4.16]

1.9. Let G be a finite group, of order n , and let m be the number of elements $g \in G$ of order exactly 2. Prove that $n - m$ is odd. Deduce that if n is even, then G necessarily contains elements of order 2.

1.10. Suppose the order of g is odd. What can you say about the order of g^2 ?

1.11. Prove that for all g, h in a group G , $|gh| = |hg|$. (Hint: Prove that $|aga^{-1}| = |g|$ for all a, g in G .)

1.12. $\triangleright$ In the group of invertible 2×2 matrices, consider

$$g = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

Verify that $|g| = 4$, $|h| = 3$, and $|gh| = \infty$. [§1.6]

1.13. $\triangleright$ Give an example showing that $|gh|$ is not necessarily equal to $\text{lcm}(|g|, |h|)$, even if g and h commute. [§1.6, 1.14]

1.14. $\triangleright$ As a counterpoint to Exercise 1.13, prove that if g and h commute *and* $\gcd(|g|, |h|) = 1$, then $|gh| = |g| |h|$. (Hint: Let $N = |gh|$; then $g^N = (h^{-1})^N$. What can you say about this element?) [§1.6, 1.15, §IV.2.5]

1.15. $\neg$ Let G be a commutative group, and let $g \in G$ be an element of maximal *finite* order, that is, such that if $h \in G$ has finite order, then $|h| \leq |g|$. Prove that in fact if h has finite order in G , then $|h|$ *divides* $|g|$. (Hint: Argue by contradiction. If $|h|$ is finite but does not divide $|g|$, then there is a prime integer p such that $|g| = p^m r$, $|h| = p^n s$, with r and s relatively prime to p and $m < n$. Use Exercise 1.14 to compute the order of $g^{p^m} h^s$.) [§2.1, 4.11, IV.6.15]

2. Examples of groups

2.1. Symmetric groups. In §I.4.1 we have already observed that every object A of every category $\mathbf{C}$ determines a group, called $\text{Aut}_{\mathbf{C}}(A)$, namely the group of *automorphisms* of A . In a somewhat artificial sense it is clear that every group arises in this fashion (cf. Exercise 1.1); this fact is true in more ‘meaningful’ ways, which will become apparent when we discuss *group actions* (§9): cf. especially Theorem 9.5 and Exercise 9.17.

In any case, this observation provides the reader with an infinite class of very important examples:

Definition 2.1. Let A be a set. The *symmetric group*, or *group of permutations* of A , denoted S_A , is the group $\text{Aut}_{\text{Set}}(A)$. The group of permutations of the set $\{1, \dots, n\}$ is denoted by S_n . $\square$

The terminology is easily justified: the automorphisms of a set A are the set-isomorphisms, that is, the bijections, from A to itself; applying such a bijection amounts precisely to permuting (‘scrambling’) the elements of A . This operation may be viewed as a transformation of A which does not change it (as a set), hence a ‘symmetry’.

The groups S_A are famously large: as the reader checked in Exercise I.2.1, $|S_n| = n!$. For example, $|S_{70}| > 10^{100}$, which is substantially larger than the estimated number of elementary particles in the observable universe.

Potentially confusing point: The various conventions clash in the way the operation in S_A should be written. From the ‘automorphism’ point of view, elements of S_A are functions and should be composed as such; thus, if $f, g \in S_A = \text{Aut}_{\text{Set}}(A)$, then the ‘product’ of f and g should be written $g \circ f$ and should act as follows:

$$(\forall p \in A) : \quad g \circ f(p) = g(f(p)).$$

But the prevailing style of notation in group theory would write this element as fg , apparently reversing the order in which the operation is performed.

Everything would fall back into agreement if we adopted the convention of writing functions *after* the elements on which they act rather than before: $(p)f$ rather than $f(p)$. But one cannot change century-old habits, so we have no alternative but to live with both conventions and to state carefully which one we are using at any given time.

Contemplating the groups S_n for small values of n is an exercise of inestimable value. Of course S_1 is a trivial group; S_2 consists of the two possible permutations:

$$\begin{cases} 1 \mapsto 1 \\ 2 \mapsto 2 \end{cases} \quad \text{and} \quad \begin{cases} 1 \mapsto 2 \\ 2 \mapsto 1 \end{cases}$$

which we could call e (identity) and f (flip), with operation

$$ee = ff = e, \quad ef = fe = f.$$

In practice we cannot give a new name to every different element of every permutation group, so we have to develop a more flexible notation. There are in fact several possible choices for this; for the time being, we will indicate an element $\sigma \in S_n$ by listing the effect of applying σ underneath the list $1, \dots, n$, as a matrix⁹. Thus the elements e, f in S_2 may be denoted by

$$e = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \quad f = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

⁹This is only a notational device—these matrices should not be confused with the matrices appearing in linear algebra.

In the same notational style, S_3 consists of

$$\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \right\}.$$

For the multiplication, I will adopt the sensible (but not very standard) convention mentioned above and have permutations act ‘on the right’: thus, for example,

$$\mathbf{1} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \mathbf{2} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \mathbf{1}$$

and similarly

$$\mathbf{2} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \mathbf{3}, \quad \mathbf{3} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \mathbf{2}.$$

That is,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

since the permutations on both sides of the equal sign act in the same way on $\mathbf{1}, \mathbf{2}, \mathbf{3}$. The reader should now check that

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

That is, letting

$$x = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix},$$

then

$$yx \neq xy,$$

showing that the operation in S_3 does not satisfy the commutative axiom. Thus, S_3 is a noncommutative group; the reader will immediately realize that in fact S_n is noncommutative for all $n \geq 3$.

While the commutation relation does not hold, other interesting relations do hold in S_3 . For example,

$$x^2 = e, \quad y^3 = e,$$

showing that S_3 contains elements of order 1 (the identity e), 2 (the element x), and 3 (the element y); cf. Exercise 2.2. (Incidentally, this shows that the result of Exercise 1.15 does require the commutativity hypothesis.) Also,

$$yx = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = xy^2$$

as the reader may check. Using these relations, we see that every product of any assortment of x and y , $x^{i_1}y^{i_2}x^{i_3}y^{i_4}\dots$, may be reduced to a product $x^i y^j$ with $0 \leq i \leq 1$, $0 \leq j \leq 2$, that is, to one of the six elements

$$e, \quad y, \quad y^2, \quad x, \quad xy, \quad xy^2:$$

for example,

$$y^7 x^{13} y^5 = (y^3)^2 y (x^2)^6 x y^3 y^2 = (yx) y^2 = (xy^2) y^2 = xy^3 y = xy.$$

On the other hand, these six elements are all distinct—this may be checked by cancellation and order considerations¹⁰. For example, if we had $xy^2 = y$, then we would get $x = y^{-1}$ by cancellation, and this cannot be since the relations tell us that x has order 2 and y^{-1} has order 3.

The conclusion is that the six products displayed above must be *all* six elements of S_3 :

$$S_3 = \{e, x, y, xy, y^2, xy^2\}.$$

In the process we have verified that S_3 may also be described as the group ‘generated’ by two elements x and y , with the ‘relations’ $x^2 = e$, $y^3 = e$, $yx = xy^2$.

More generally, a subset A of a group G ‘generates’ G if every element of G may be written as a product of elements of A and of inverses of elements of A . We will deal with this notion more formally in §6.3 and with descriptions of groups in terms of generators and relations in §8.2.

2.2. Dihedral groups. A ‘symmetry’ is a transformation which preserves a structure. This is of course just a loose way to talk about automorphisms, when we may be too lazy to define rigorously the relevant category. As automorphisms of objects of a category, symmetries will naturally form groups.

One context in which this notion may be visualized vividly is that of ‘geometric figures’ such as polygons in the plane or polyhedra in space. The relevant category could be defined as follows: let the objects be subsets of an ordinary plane $\mathbb{R}^2$ and let morphisms between two subsets A, B consist of the ‘rigid motions’ of the plane (such as translations, rotations, or reflections about a line) which map A to a subset of B . A rigorous treatment of these notions would be too distracting at this point, so I will appeal to the intuition of the reader, as I do every now and then.

From this perspective, the ‘symmetries’ of a subset of the plane are the rigid motions which map it onto itself; they clearly form a group.

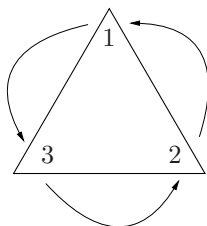
The *dihedral* groups may be defined as these groups of symmetries for the *regular polygons*. Placing the polygon so that it is centered at the origin (thereby excluding translations as possible symmetries), we see that the dihedral group for a regular n -sided polygon consists of the n rotations by $2\pi/n$ radians about the origin and the n distinct reflections about lines through the origin and a vertex or a midpoint of a side. Thus, the dihedral group for a regular n -sided polygon consists of $2n$ elements; I will denote¹¹ this group by the symbol D_{2n} .

Again, contemplating these groups, at least for small values of n , is a wonderful exercise. There is a simple way to relate the dihedral groups to the symmetric groups of §2.1, capturing the fact that a symmetry of a regular polygon P is determined by the fate of the *vertices* of P . For example, label the vertices of an equilateral triangle clockwise by **1**, **2**, **3**; then a counterclockwise rotation by an angle of $2\pi/3$ sends vertex **1** to vertex **3**, **3** to **2**, and **2** to **1**, and no other symmetry of the triangle does the same.

¹⁰It may of course also be checked by explicit computation of the corresponding permutations, but I am trying to illustrate the fact that the relations are ‘all we need to know’.

¹¹Unfortunately there does not seem to be universal agreement on this notation: some references use the symbol D_n for what I call D_{2n} here.

Visually, this looks like



In other words, we can associate with the counterclockwise rotation of an equilateral triangle by $2\pi/3$ the permutation

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \in S_3.$$

Such a labeling defines a *function*

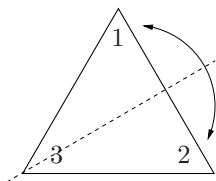
$$D_6 \rightarrow S_3;$$

further, this function is injective (since a symmetry is determined by the permutation of vertices it induces). In fancier language, which we will master in due time, we say that D_6 *acts (faithfully) on the set $\{1, 2, 3\}$* .

It is clear that this can be done in several ways (for example, we could label the vertices in different ways). However, any such assignment will have the property that composition of symmetries in D_{2n} corresponds to composition of permutations in S_n ; the reader should carefully work this out for several examples involving $D_6 \rightarrow S_3$ and $D_8 \rightarrow S_4$.

A concise way to describe the situation is that these functions are (*group*) *homomorphisms* (cf. §4). Since both D_6 and S_3 have 6 elements and the function $D_6 \rightarrow S_3$ given above is injective, it must also be surjective. Thus there are bijective homomorphisms between D_6 and S_3 ; we say that these groups are *isomorphic* (cf. §4.3). We will study these concepts very carefully in the next several sections.

As an alternative (and more abstract) way to draw the same conclusion, denote by y the counterclockwise rotation considered above and by x the reflection about the line through the center and vertex 3 of our equilateral triangle:



Reflecting twice gives the identity, as does rotating three times; thus

$$x^2 = e, \quad y^3 = e.$$

Further, yx (rotating counterclockwise by $2\pi/3$, then flipping about the slanted line) is the same symmetry as xy^2 (flipping first, then rotating *clockwise* by $2\pi/3$). That is,

$$yx = xy^2.$$

In other words, the group D_6 is *also* generated by two elements x, y , subject to the relations $x^2 = e$, $y^3 = e$, $yx = xy^2$ —precisely as we found for S_3 . Since D_6 and S_3 admit matching descriptions in terms of generators and relations (that is, matching *presentations*; cf. §8.2), ‘of course’ they are isomorphic.

2.3. Cyclic groups and modular arithmetic. Let n be a positive integer. Consider the equivalence relation on $\mathbb{Z}$ defined by¹²

$$(\forall a, b \in \mathbb{Z}) : \quad a \equiv b \pmod{n} \iff n \mid (b - a).$$

This is called *congruence modulo n* . We have encountered this relation already, for $n = 2$, in Example I.1.3. It is very easy to check that it is an equivalence relation, for all n ; the set of equivalence classes is often denoted by $\mathbb{Z}_n$, $\mathbb{Z}/(n)$, $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{F}_n$. We will opt for $\mathbb{Z}/n\mathbb{Z}$, which is not preempted by other notions¹³. I will denote by $[a]_n$ the equivalence class of the integer a modulo n , or simply $[a]$ if no ambiguity arises.

The reader should check carefully that $\mathbb{Z}/n\mathbb{Z}$ consists of exactly n elements, namely

$$[0]_n, [1]_n, \dots, [n-1]_n.$$

We can use the group structure on $\mathbb{Z}$ to induce an (abelian) group structure on $\mathbb{Z}/n\mathbb{Z}$. In order to do this, we define an operation $+$ on $\mathbb{Z}/n\mathbb{Z}$, by setting $\forall a, b \in \mathbb{Z}$

$$[a] + [b] := [a + b].$$

Of course we have to check that this prescription is well-defined; luckily, this is very easy: the following small lemma does the job, as it shows that the result of the operation does not depend on the representatives chosen for the classes.

Lemma 2.2. *If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then*

$$(a + b) \equiv (a' + b') \pmod{n}.$$

Proof. By hypothesis $n \mid (a' - a)$ and $n \mid (b' - b)$; therefore $\exists k, \ell \in \mathbb{Z}$ such that

$$(a' - a) = kn, \quad (b' - b) = \ell n.$$

Then

$$(a' + b') - (a + b) = (a' - a) + (b' - b) = kn + \ell n = (k + \ell)n,$$

proving that n divides $(a' + b') - (a + b)$, as needed. $\square$

Therefore, we have a binary operation $+$ on $\mathbb{Z}/n\mathbb{Z}$. It is immediately checked that the resulting structure is a group. Associativity is inherited from $\mathbb{Z}$:

$$([a] + [b]) + [c] = [a + b] + [c] = [(a + b) + c] = [a + (b + c)] = [a] + [b + c] = [a] + ([b] + [c]);$$

and so are the identity $[0]$ and ‘inverse’ $-[a] = [-a]$.

It is also immediately checked that the resulting groups $\mathbb{Z}/n\mathbb{Z}$ are commutative, as the abelian-style notation suggests:

$$[a] + [b] = [a + b] = [b + a] = [b] + [a].$$

¹²The notation $n \mid m$ stands for n is a divisor of m ; that is, $m = nk$ for some integer k .

¹³When $n = p$ is prime, $\mathbb{Z}_p$ is the official notation for ‘ p -adic integers’, which are a completely different concept; see Exercise VIII.1.19.

I trust that this material is not new to the reader, who should in any case check all these assertions carefully.

The abelian groups thus obtained, together with $\mathbb{Z}$, are called *cyclic groups*; a popular alternative notation for the group $(\mathbb{Z}/n\mathbb{Z}, +)$ is C_n . This is adopted especially when one wants to use the ‘multiplicative’ rather than ‘additive’ notation; thus we can say that C_n is generated by one element x , with the relation $x^n = e$.

Cyclic groups are tremendously important, and we will come back to them in later sections. For the time being we record the fact that the element

$$[1]_n \in \mathbb{Z}/n\mathbb{Z}$$

generates the group, in the sense that every other element may be obtained as a multiple of this element. For example, if $m \geq 0$ is an integer, then

$$[m]_n = \underbrace{[1 + \cdots + 1]_n}_{m \text{ times}} = \underbrace{[1]_n + \cdots + [1]_n}_{m \text{ times}} = m \cdot [1]_n.$$

Equivalently, we may phrase this fact by observing that the order of $[1]_n$ in $\mathbb{Z}/n\mathbb{Z}$ is n : this implies that the n multiples $0 \cdot [1]_n, 1 \cdot [1]_n, \dots, (n-1) \cdot [1]_n$ must all be distinct, and hence they must fill up $\mathbb{Z}/n\mathbb{Z}$.

Proposition 2.3. *The order of $[m]_n$ in $\mathbb{Z}/n\mathbb{Z}$ is 1 if $n \mid m$, and more generally*

$$|[m]_n| = \frac{n}{\gcd(m, n)}.$$

Proof. If $n \mid m$, then $[m]_n = [0]_n$. If n does not divide m , observe again that $[m]_n = m[1]_n$ and apply Proposition 1.13. $\square$

Remark 2.4. As a consequence, the order of every element of $\mathbb{Z}/n\mathbb{Z}$ divides $n = |\mathbb{Z}/n\mathbb{Z}|$, the order of the group. We will see later (Example 8.15) that this is a general feature of the order of elements in any finite group. $\lrcorner$

Corollary 2.5. *The class $[m]_n$ generates $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(m, n) = 1$.*

This simple result is quite important. For example, if $n = p$ is a prime integer, it shows that every *nonzero* class in the group $\mathbb{Z}/p\mathbb{Z}$ generates it. In any case, it allows us to construct more examples of interesting groups. The reader should check (or recall; cf. Exercise 2.14) that there also is a well-defined *multiplication* on $\mathbb{Z}/n\mathbb{Z}$, given by

$$[a]_n \cdot [b]_n := [ab]_n.$$

This operation does *not* define a group structure on $\mathbb{Z}/n\mathbb{Z}$: indeed, the class $[0]_n$ does *not* have a multiplicative inverse. On the other hand, for any positive n denote by $(\mathbb{Z}/n\mathbb{Z})^*$ the subset of $\mathbb{Z}/n\mathbb{Z}$ consisting of classes $[m]_n$ such that $\gcd(m, n) = 1$:

$$(\mathbb{Z}/n\mathbb{Z})^* := \{[m]_n \in \mathbb{Z}/n\mathbb{Z} \mid \gcd(m, n) = 1\}.$$

This subset is clearly well-defined: if $m \equiv m' \pmod{n}$, then $\gcd(m, n) = 1 \iff \gcd(m', n) = 1$ (Exercise 2.17), so the defining property of classes in $(\mathbb{Z}/n\mathbb{Z})^*$ is independent of representatives.

Proposition 2.6. *Multiplication makes $(\mathbb{Z}/n\mathbb{Z})^*$ into a group.*

Proof. Simple properties of gcd's show that if $\gcd(m_1, n) = \gcd(m_2, n) = 1$, then $\gcd(m_1 m_2, n) = 1$. (For example, if a prime integer divided both n and $m_1 m_2$, then it would necessarily divide m_1 or m_2 , and one of the two gcd's would not be 1.) Therefore, the product of two elements in $(\mathbb{Z}/n\mathbb{Z})^*$ is an element of $(\mathbb{Z}/n\mathbb{Z})^*$, and $\cdot$ does define a binary operation

$$(\mathbb{Z}/n\mathbb{Z})^* \times (\mathbb{Z}/n\mathbb{Z})^* \rightarrow (\mathbb{Z}/n\mathbb{Z})^*.$$

It is clear that this operation is associative (because multiplication is associative in $\mathbb{Z}$); $[1]_n$ is an element of $(\mathbb{Z}/n\mathbb{Z})^*$ and is an identity with respect to multiplication; so all we have to check is that elements of $(\mathbb{Z}/n\mathbb{Z})^*$ have multiplicative inverses in $(\mathbb{Z}/n\mathbb{Z})^*$.

This follows from Corollary 2.5. If $\gcd(m, n) = 1$, then $[m]_n$ generates the additive group $\mathbb{Z}/n\mathbb{Z}$, and hence some multiple of $[m]_n$ must equal $[1]_n$:

$$(\exists a \in \mathbb{Z}) : \quad a \cdot [m]_n = [1]_n;$$

this implies

$$[a]_n [m]_n = [1]_n.$$

Therefore $[m]_n$ does have a multiplicative inverse in $\mathbb{Z}/n\mathbb{Z}$, namely $[a]_n$. The reader will verify that $\gcd(a, n) = 1$, completing the proof. $\square$

For instance, $[8]_{15}$ has a multiplicative inverse in $(\mathbb{Z}/15\mathbb{Z})^*$. Tracing the argument given in the proof,

$$2 \cdot 8 + (-1) \cdot 15 = 1,$$

and hence $[2]_{15} \cdot [8]_{15} = [1]_{15}$: the multiplicative inverse of $[8]_{15}$ is $[2]_{15}$.

For $n = p$ a positive prime integer, the group $(\mathbb{Z}/p\mathbb{Z})^*, \cdot$ has order $(p - 1)$. We will have more to say about these groups in later sections (cf. Example 4.6).

Exercises

2.1. $\neg$ One can associate an $n \times n$ matrix M_σ with a permutation

$\sigma \in S_n$ by letting the entry at $(i, (i)\sigma)$ be 1 and letting all other entries be 0. For example, the matrix corresponding to the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \in S_3$$

would be

$$M_\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Prove that, with this notation,

$$M_{\sigma\tau} = M_\sigma M_\tau$$

for all $\sigma, \tau \in S_n$, where the product on the right is the ordinary product of matrices. [IV.4.13]

- 2.2.** ▷ Prove that if $d \leq n$, then S_n contains elements of order d . [§2.1]
- 2.3.** For every positive integer n find an element of order n in $S_{\mathbb{N}}$.
- 2.4.** Define a homomorphism $D_8 \rightarrow S_4$ by labeling vertices of a square, as we did for a triangle in §2.2. List the 8 permutations in the image of this homomorphism.
- 2.5.** ▷ Describe generators and relations for all dihedral groups D_{2n} . (Hint: Let x be the reflection about a line through the center of a regular n -gon and a vertex, and let y be the counterclockwise rotation by $2\pi/n$. The group D_{2n} will be generated by x and y , subject to three relations¹⁴. To see that these relations really determine D_{2n} , use them to show that any product $x^{i_1}y^{i_2}x^{i_3}y^{i_4}\dots$ equals x^iy^j for some i, j with $0 \leq i \leq 1, 0 \leq j < n$.) [8.4, §IV.2.5]
- 2.6.** ▷ For every positive integer n construct a group containing elements g, h such that $|g| = 2, |h| = 2$, and $|gh| = n$. (Hint: For $n > 1$, D_{2n} will do.) [§1.6]
- 2.7.** ¬ Find all elements of D_{2n} that commute with every other element. (The parity of n plays a role.) [IV.1.2]
- 2.8.** Find the orders of the groups of symmetries of the five ‘platonic solids’.
- 2.9.** Verify carefully that ‘congruence mod n ’ is an equivalence relation.
- 2.10.** Prove that $\mathbb{Z}/n\mathbb{Z}$ consists of precisely n elements.
- 2.11.** ▷ Prove that the square of every odd integer is congruent to 1 modulo 8. [§VII.5.1]
- 2.12.** Prove that there are no nonzero integers a, b, c such that $a^2 + b^2 = 3c^2$. (Hint: By studying the equation $[a]_4^2 + [b]_4^2 = 3[c]_4^2$ in $\mathbb{Z}/4\mathbb{Z}$, show that a, b, c would all have to be even. Letting $a = 2k, b = 2\ell, c = 2m$, you would have $k^2 + \ell^2 = 3m^2$. What’s wrong with that?)
- 2.13.** ▷ Prove that if $\gcd(m, n) = 1$, then there exist integers a and b such that
- $$am + bn = 1.$$
- (Use Corollary 2.5.) Conversely, prove that if $am + bn = 1$ for some integers a and b , then $\gcd(m, n) = 1$. [2.15, §V.2.1, V.2.4]
- 2.14.** ▷ State and prove an analog of Lemma 2.2, showing that the multiplication on $\mathbb{Z}/n\mathbb{Z}$ is a well-defined operation. [§2.3, §III.1.2]
- 2.15.** ¬ Let $n > 0$ be an odd integer.
- Prove that if $\gcd(m, n) = 1$, then $\gcd(2m + n, 2n) = 1$. (Use Exercise 2.13.)
 - Prove that if $\gcd(r, 2n) = 1$, then $\gcd(\frac{r+n}{2}, n) = 1$. (Ditto.)
 - Conclude that the function $[m]_n \rightarrow [2m + n]_{2n}$ is a bijection between $(\mathbb{Z}/n\mathbb{Z})^*$ and $(\mathbb{Z}/2n\mathbb{Z})^*$.

¹⁴Two relations are evident. To ‘see’ the third one, hold your right hand in front of and away from you, pointing your fingers at the vertices of an imaginary regular pentagon. Flip the pentagon by turning the hand toward you; rotate it counterclockwise w.r.t. the line of sight by 72° ; flip it again by pointing it away from you; and rotate it counterclockwise a second time. This returns the hand to the initial position. What does this tell you?

The number $\phi(n)$ of elements of $(\mathbb{Z}/n\mathbb{Z})^*$ is *Euler's ϕ -function*. The reader has just proved that if n is odd, then $\phi(2n) = \phi(n)$. Much more general formulas will be given later on (cf. Exercise V.6.8). [VII.5.11]

2.16. Find the last digit of $1238237^{18238456}$. (Work in $\mathbb{Z}/10\mathbb{Z}$.)

2.17. ▷ Show that if $m \equiv m' \pmod{n}$, then $\gcd(m, n) = 1$ if and only if $\gcd(m', n) = 1$. [§2.3]

2.18. For $d \leq n$, define an injective function $\mathbb{Z}/d\mathbb{Z} \rightarrow S_n$ preserving the operation, that is, such that the sum of equivalence classes in $\mathbb{Z}/d\mathbb{Z}$ corresponds to the product of the corresponding permutations.

2.19. ▷ Both $(\mathbb{Z}/5\mathbb{Z})^*$ and $(\mathbb{Z}/12\mathbb{Z})^*$ consist of 4 elements. Write their multiplication tables, and prove that no re-ordering of the elements will make them match. (Cf. Exercise 1.6.) [§4.3]

3. The category Grp

Groups will be the objects of the category **Grp**. In this section we define the *morphisms* in the category and deal with simple properties of these morphisms.

3.1. Group homomorphisms. As we know, a group consists of two distinct types of information: a set G and an operation¹⁵

$$m_G : G \times G \rightarrow G$$

satisfying certain properties. For two groups (G, m_G) and (H, m_H) , a *group homomorphism*

$$\varphi : (G, m_G) \rightarrow (H, m_H)$$

is first of all a function (usually given the same name, φ in this case) between the underlying sets; but this function must ‘know about’ the operations m_G on G , m_H on H . What is the most natural requirement of this sort?

Note that the set-function $\varphi : G \rightarrow H$ determines a function

$$(\varphi \times \varphi) : G \times G \rightarrow H \times H :$$

we could invoke the universal property of products to obtain this function (cf. Exercise 3.1), but since we are dealing with sets, there is no need for fancy language here—just define the function by

$$(\forall (a, b) \in G \times G) : \quad (\varphi \times \varphi)(a, b) = (\varphi(a), \varphi(b)).$$

There is a diagram combining all these maps:

$$\begin{array}{ccc} G \times G & \xrightarrow{\varphi \times \varphi} & H \times H \\ m_G \downarrow & & \downarrow m_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

¹⁵In §1, m_G was denoted $\bullet$; here we need to keep track of operations on different groups, so for a moment I will use a symbol recording the group (and evoking ‘multiplication’).

What requirement could be more natural than asking that this diagram *commute*?

Definition 3.1. The set-function $\varphi : G \rightarrow H$ defines a *group homomorphism* if the diagram displayed above commutes. $\lrcorner$

This is a seemingly complicated way of saying something simple: since φ and m_G, m_H are functions of sets, commutativity means the following. For all $a, b \in G$, the two ways to travel through the diagram give

$$\begin{array}{ccc} (a, b) & \xrightarrow{\quad \cdot \quad} & (a \cdot b) \\ \downarrow m_G & & \downarrow m_H \\ a & \xrightarrow{\quad \cdot \quad} & b \end{array} \quad \begin{array}{ccc} (a, b) & \xrightarrow{\quad \cdot \quad} & (\varphi(a), \varphi(b)) \\ \downarrow m_H & & \downarrow m_K \\ \varphi(a) & \xrightarrow{\quad \cdot \quad} & \varphi(b) \end{array}$$

where I now write $\cdot$ for both operations: in G on the left, in H on the right. Commutativity of the diagram means that we must get the same result in both cases; therefore, Definition 3.1 can be rephrased as

the set-function $\varphi : G \rightarrow H$ is a group homomorphism if $\forall a, b \in G$

$$\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b).$$

In other words, φ is a homomorphism if it ‘preserves the structure’. This may sound more familiar to our reader. As usual, the reason to bring in diagrams (as in Definition 3.1) is that this would make it easy to transfer part of the discussion to other categories.

If the context is clear, one may simply write ‘homomorphism’, omitting the qualifier ‘group’.

3.2. Grp: Definition. For G, H groups¹⁶ we define

$$\text{Hom}_{\text{Grp}}(G, H)$$

to be the set of group homomorphisms $G \rightarrow H$.

If G, H, K are groups and $\varphi : G \rightarrow H, \psi : H \rightarrow K$ are two group homomorphisms, it is easy to check that the composition $\psi \circ \varphi : G \rightarrow K$ is a group homomorphism: from the diagram point of view, this amounts to observing that the ‘outer rectangle’ in

$$\begin{array}{ccccc} G \times G & \xrightarrow{\varphi \times \varphi} & H \times H & \xrightarrow{\psi \times \psi} & K \times K \\ \downarrow m_G & & \downarrow m_H & & \downarrow m_K \\ G & \xrightarrow{\varphi} & H & \xrightarrow{\psi} & K \end{array}$$

$\xrightarrow{\psi \circ \varphi}$

¹⁶I am yielding to the usual abuse of language that lets us omit explicit mention of the operation.

must commute if the two ‘inner rectangles’ commute. For arbitrary $a, b \in G$, this means,

$$\begin{aligned} (\psi \circ \varphi)(a \cdot b) &= \psi(\varphi(a \cdot b)) \stackrel{(1)}{=} \psi(\varphi(a) \cdot \varphi(b)) \stackrel{(2)}{=} \psi(\varphi(a)) \cdot \psi(\varphi(b)) \\ &= (\psi \circ \varphi)(a) \cdot (\psi \circ \varphi)(b) \end{aligned}$$

where (1) holds because φ is a homomorphism and (2) holds because ψ is a homomorphism.

Further, it is clear that composition is associative (because it is for set-functions) and that the identity function $\text{id}_G : G \rightarrow G$ is a group homomorphism. Therefore, $\mathbf{Grp}$ is indeed a category.

3.3. Pause for reflection. The careful reader might raise an objection: the group axioms prescribe the existence of an identity element e_G and of an ‘inverse’, that is, a specific function

$$\iota_G : G \rightarrow G, \quad \iota_G(g) := g^{-1}.$$

Shouldn’t the definition of morphism in $\mathbf{Grp}$ keep track of this type of data? The definition we have given only keeps track of the multiplication map m_G .

The reason why we can get away with this is that preserving m automatically preserves e and ι :

Proposition 3.2. *Let $\varphi : G \rightarrow H$ be a group homomorphism. Then*

- $\varphi(e_G) = e_H$;
- $\forall g \in G, \varphi(g^{-1}) = \varphi(g)^{-1}$.

In terms of diagrams, the second assertion amounts to saying that

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & H \\ \downarrow \iota_G & & \downarrow \iota_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

must commute.

Proof. The first item follows from the definition of homomorphism and cancellation: since $e_H = e_H \cdot e_H$,

$$e_H \cdot \varphi(e_G) = \varphi(e_G) = \varphi(e_G \cdot e_G) = \varphi(e_G) \cdot \varphi(e_G),$$

which implies $e_H = \varphi(e_G)$ by ‘cancelling $\varphi(e_G)$ ’.

The proof of the second assertion is similar: $\forall g \in G$,

$$\varphi(g^{-1}) \cdot \varphi(g) = \varphi(g^{-1} \cdot g) = \varphi(e_G) = e_H = \varphi(g)^{-1} \cdot \varphi(g),$$

implying $\varphi(g^{-1}) = \varphi(g)^{-1}$ by cancellation. □

3.4. Products et al. The categories **Grp** and **Set** look rather alike at first: given a group, we can ‘forget’ the information of multiplication and we are left with a set; given a group homomorphism, we can forget that it preserves the multiplication and we are left with a set-function. A concise way to express this fact is that there is a ‘functor’ $\mathbf{Grp} \leadsto \mathbf{Set}$ (called in fact a ‘forgetful’ functor); we will deal with functors more extensively much later on, starting in Chapter VIII.

However, there are important differences between these two categories. For example, recall that **Set** has a unique initial object (that is, $\emptyset$) and this is not the same as the final objects (that is, the singletons). Also recall that a *trivial* group is a group consisting of a single element (Example 1.3).

Proposition 3.3. *Trivial groups are both initial and final in Grp.*

This makes trivial groups ‘zero-objects’ of the category **Grp**.

Proof. It should be clear that trivial groups are final: there is only one function from a set to a singleton, that is, the constant function; this is vacuously a group homomorphism.

To see that trivial groups are initial, let $T = \{e\}$ be a trivial group; for any group G , define $\varphi : T \rightarrow G$ by $T(e) = e_G$. This is clearly a group homomorphism, and it is the only possible one since every group homomorphism must send the identity to the identity (Proposition 3.2). $\square$

Here is a similarity: **Grp** has products; in fact, the product of two groups G, H is supported on the product $G \times H$ of the underlying sets.

To see this, we need to define a multiplication on $G \times H$; the catchword here is *componentwise*: define the operation in $G \times H$ by performing the operation on each component separately. Explicitly, define $\forall g_1, g_2 \in G, \forall h_1, h_2 \in H$

$$(g_1, h_1) \cdot (g_2, h_2) := (g_1 g_2, h_1 h_2).$$

This operation defines a group structure on $G \times H$: the operation is associative, the identity is (e_G, e_H) , and the inverse of (g, h) is (g^{-1}, h^{-1}) . All needed verifications are left to the reader. The group $G \times H$ is called the *direct product* of the groups G and H .

Also note that the natural projections

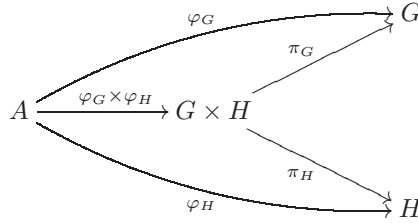
$$\begin{array}{ccc} & G \times H & \\ \pi_G \swarrow & & \searrow \pi_H \\ G & & H \end{array}$$

(defined as set-functions as in §I.2.4) are group homomorphisms: again, this follows immediately from the definitions.

Proposition 3.4. *With operation defined componentwise, $G \times H$ is a product in Grp.*

Proof. Recall (§I.5.4) that this means that $G \times H$ satisfies the following universal property: for any group A and any choice of group homomorphisms $\varphi_G : A \rightarrow G$,

$\varphi_H : A \rightarrow H$, there exists a unique group homomorphism $\varphi_G \times \varphi_H$ making the diagram



commute.

Now, a unique *set-function* $\varphi_G \times \varphi_H$ exists making the diagram commute, because the *set* $G \times H$ is a product of G and H in **Set**. So we only need to check that $\varphi_G \times \varphi_H$ is a group homomorphism, and this is immediate (if cumbersome): $\forall a, b \in A$,

$$\begin{aligned} \varphi_G \times \varphi_H(ab) &= (\varphi_G(ab), \varphi_H(ab)) = (\varphi_G(a)\varphi_G(b), \varphi_H(a)\varphi_H(b)) \\ &= (\varphi_G(a), \varphi_H(a))(\varphi_G(b), \varphi_H(b)) = (\varphi_G \times \varphi_H(a))(\varphi_G \times \varphi_H(b)). \end{aligned}$$

□

What about *coproducts*? They *do* exist in **Grp**, but their construction requires handling presentations more proficiently than we do right now, and general coproducts of groups will not be used in the rest of the book; so the reader will have to deal with them on his or her own. For an important example, see Exercise 3.8; more will show up in Exercises 5.6 and 5.7, since *free groups* are themselves particular cases of coproducts. The reader will finally produce the coproduct of any two groups explicitly in Exercise 8.7. For now, just realize that the *disjoint union*, which works as a coproduct in **Set** (Proposition I.5.6), is not an option in **Grp**: there is no reasonable group structure on the disjoint union. The coproduct of G and H in **Grp** is denoted $G * H$ and is called the *free product* of G and H .

3.5. Abelian groups. The category **Ab** whose objects are *abelian* groups, and whose morphisms are group homomorphisms, will in a sense be more important for us than the category **Grp**. In many ways, as we will see, **Ab** is a ‘nicer’ category¹⁷ than **Grp**. Again the trivial groups are both initial and final (that is, ‘zero’) objects; products exist and coincide with products in **Grp**. But here is a difference: unlike in **Grp**, coproducts in **Ab** coincide with products. That is, if G and H are abelian groups, then the product $G \times H$ (with the two natural homomorphisms $G \rightarrow G \times H$, $H \rightarrow G \times H$) satisfies the universal property for coproducts in **Ab** (cf. Exercise 3.3). When working as a coproduct, the product $G \times H$ of two abelian groups is often called their *direct sum* and is denoted $G \oplus H$.

There is a pretty subtlety here, which may highlight the power of the language: even if G and H are commutative, the product $G \times H$ does *not* (necessarily) satisfy

¹⁷As we will see in due time (Proposition III.5.3), **Ab** is one instance of a general class of categories of ‘modules over a commutative ring R ’ (for $R = \mathbb{Z}$). Unlike **Grp**, these categories are *abelian*, which makes them very well-behaved. We will learn two or three things about abelian categories in Chapter IX.

the universal property for coproducts *in Grp*, even if it does *in Ab*. For an explicit example, see Exercise 3.6.

Exercises

3.1. ▷ Let $\varphi : G \rightarrow H$ be a morphism in a category $\mathbf{C}$ with products. Explain why there is a unique morphism $(\varphi \times \varphi) : G \times G \rightarrow H \times H$ compatible in the evident way with the natural projections.

(This morphism is defined explicitly for $\mathbf{C} = \mathbf{Set}$ in §3.1.) [§3.1, 3.2]

3.2. Let $\varphi : G \rightarrow H$, $\psi : H \rightarrow K$ be morphisms in a category with products, and consider morphisms between the products $G \times G$, $H \times H$, $K \times K$ as in Exercise 3.1. Prove that

$$(\psi\varphi) \times (\psi\varphi) = (\psi \times \psi)(\varphi \times \varphi).$$

(This is part of the commutativity of the diagram displayed in §3.2.)

3.3. ▷ Show that if G , H are *abelian* groups, then $G \times H$ satisfies the universal property for coproducts in $\mathbf{Ab}$ (cf. §I.5.5). [§3.5, 3.6, §III.6.1]

3.4. Let G , H be groups, and assume that $G \cong H \times G$. Can you conclude that H is trivial? (Hint: No. Can you construct a counterexample?)

3.5. Prove that $\mathbb{Q}$ is not the direct product of two nontrivial groups.

3.6. ▷ Consider the product of the cyclic groups C_2 , C_3 (cf. §2.3): $C_2 \times C_3$. By Exercise 3.3, this group is a coproduct of C_2 and C_3 in $\mathbf{Ab}$. Show that it is *not* a coproduct of C_2 and C_3 in $\mathbf{Grp}$, as follows:

- find injective homomorphisms $C_2 \rightarrow S_3$, $C_3 \rightarrow S_3$;
- arguing by contradiction, assume that $C_2 \times C_3$ is a coproduct of C_2 , C_3 , and deduce that there would be a group homomorphism $C_2 \times C_3 \rightarrow S_3$ with certain properties;
- show that there is no such homomorphism.

[§3.5]

3.7. Show that there is a *surjective* homomorphism $\mathbb{Z} * \mathbb{Z} \rightarrow C_2 * C_3$. ($*$ denotes coproduct in $\mathbf{Grp}$; cf. §3.4.)

One can think of $\mathbb{Z} * \mathbb{Z}$ as a group with two generators x , y , subject to no relations whatsoever. (We will study a general version of such groups in §5; see Exercise 5.6.)

3.8. ▷ Define a group G with two generators x, y , subject (only) to the relations $x^2 = e_G$, $y^3 = e_G$. Prove that G is a coproduct of C_2 and C_3 *in Grp*. (The reader will obtain an even more concrete description for $C_2 * C_3$ in Exercise 9.14; it is called the *modular group*.) [§3.4, 9.14]

3.9. Show that *fiber* products and coproducts exist in $\mathbf{Ab}$. (Cf. Exercise I.5.12. For coproducts, you may have to wait until you know about *quotients*.)

4. Group homomorphisms

4.1. Examples. For any two groups G, H , the set $\text{Hom}_{\mathbf{Grp}}(G, H)$ is certainly *nonempty*: at the very least we can define a homomorphism $G \rightarrow H$ by sending every element of G to the identity e_H of H . Thus, $\text{Hom}_{\mathbf{Grp}}(G, H)$ is a ‘pointed set’ (cf. Example I.3.8).

There is a purely categorical way to think about this: since $\mathbf{Grp}$ has zero-objects $\{*\}$ (recall that trivial groups are both initial and final in $\mathbf{Grp}$; cf. Proposition 3.3), there are unique morphisms

$$G \rightarrow \{*\}, \quad \{*\} \rightarrow H$$

in $\mathbf{Grp}$; their composition is the distinguished element of $\text{Hom}_{\mathbf{Grp}}(G, H)$ mentioned above; we will call this element the *trivial* morphism.

More ‘meaningful’ examples can be constructed by considering the groups encountered in §2: for instance, the function $D_6 \rightarrow S_3$ defined in §2.2 is a homomorphism. Such examples will likely all be instances of the notion of *group action*. In general, an ‘action’ of a group G on an object A of a category $\mathbf{C}$ is a homomorphism

$$G \rightarrow \text{Aut}_{\mathbf{C}}(A);$$

that is, a group G ‘acts’ on an object if the elements of G determine isomorphisms of that object to itself, in a way compatible with compositions. If $\mathbf{C} = \mathbf{Set}$, this means that elements of G determine specific permutations of the set A . For example, the symmetries of an equilateral triangle (that is, elements of D_6) determine permutations of the vertices of that triangle (that is, permutations of a set with three elements), and they do so compatibly with composition; this is what gives us homomorphisms $D_6 \rightarrow S_3$. We say that D_6 ‘acts on the set of vertices’ of the triangle.

The reader can (and should) construct many more examples of this kind. We will have much more to say about actions of groups and other algebraic entities in later sections.

Here is an example with a different flavor: the exponential function is a homomorphism from $(\mathbb{R}, +)$ to the group $(\mathbb{R}^{>0}, \cdot)$ of positive real numbers, with ordinary multiplication as operation. Indeed, $e^{a+b} = e^a e^b$. A similar (and very important) class of examples may be obtained as follows: let G be any group and $g \in G$ any element of G ; define an ‘exponential map’ $\epsilon_g : \mathbb{Z} \rightarrow G$ by

$$(\forall a \in \mathbb{Z}) : \quad \epsilon_g(a) := g^a.$$

Then ϵ_g is (clearly) a group homomorphism. The element g generates G if and only if ϵ_g is surjective.

One concrete instance of this homomorphism (in the abelian environment, thus using multiples rather than powers) is the ‘quotient’ function $\pi_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$,

$$a \mapsto a \cdot [1]_n = [a]_n :$$

with the notation introduced above, this is $\epsilon_{[1]_n}$. This function is surjective; hence $[1]_n$ generates $\mathbb{Z}/n\mathbb{Z}$. In fact, as observed in §2.3 (Corollary 2.5), $[m]_n$ generates $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(m, n) = 1$.

If $m \mid n$, there is a homomorphism

$$\pi_m^n : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$$

making the diagram

$$\begin{array}{ccc} \mathbb{Z} & & \\ \pi_n \downarrow & \searrow \pi_m & \\ \mathbb{Z}/n\mathbb{Z} & \xrightarrow{\pi_m^n} & \mathbb{Z}/m\mathbb{Z} \end{array}$$

commute: that is,

$$\pi_m^n([a]_n) = [a]_m;$$

the reader should check carefully that this function is well-defined (Exercise 4.1).

If m_1 and m_2 are both divisors of n , we have homomorphisms $\pi_{m_1}^n, \pi_{m_2}^n$ from $\mathbb{Z}/n\mathbb{Z}$ to both $\mathbb{Z}/m_1\mathbb{Z}$ and $\mathbb{Z}/m_2\mathbb{Z}$ and hence to their direct product. For instance, since $6 = 2 \cdot 3$, there is a homomorphism

$$\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

(or, in ‘multiplicative notation’, $C_6 \rightarrow C_2 \times C_3$). Explicitly,

$$\begin{aligned} [0]_6 &\mapsto ([0]_2, [0]_3), & [1]_6 &\mapsto ([1]_2, [1]_3), & [2]_6 &\mapsto ([0]_2, [2]_3), \\ [3]_6 &\mapsto ([1]_2, [0]_3), & [4]_6 &\mapsto ([0]_2, [1]_3), & [5]_6 &\mapsto ([1]_2, [2]_3). \end{aligned}$$

Note that this homomorphism is a *bijection*; as we will see in a moment (§4.3), this makes it an *isomorphism*; in particular, C_6 is *also* a product of C_2 and C_3 in $\mathbf{Grp}$.

One can concoct a homomorphism $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ also if $n \mid m$: for example, the function $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ defined by

$$[0]_2 \rightarrow [0]_4, \quad [1]_2 \rightarrow [2]_4$$

is clearly a group homomorphism. Unlike π_m^n , this homomorphism is not nicely compatible¹⁸ with the homomorphisms π_n .

On the other hand, is there a nontrivial group homomorphism (for example) $C_4 \rightarrow C_7$? Note that there are $7^4 = 2,401$ *set-functions* from C_4 to C_7 (cf. Exercise I.2.10); the question is whether any of these functions (besides the trivial homomorphism sending everything to e) preserves the operation. We already know that a homomorphism must send the identity to the identity (Proposition 3.2), and that already rules out all but 343 functions (why?); still, it is unrealistic to write all of them out explicitly to see if any is a homomorphism.

The reader should think about this before we spill the beans in the next subsection.

¹⁸Also, note that while π_m^n preserves *multiplication* as well as sum, this new homomorphism does not; that is, it is not a ‘ring homomorphism’. This is immediately visible in the given example: $[1]_2 \cdot [1]_2 = [1]_2$, but $[2]_4 \cdot [2]_4 = [0]_4$.

4.2. Homomorphisms and order. Group homomorphisms are set-functions preserving the group structure; as such, they must preserve many features of the theory. Proposition 3.2 is an instance of this principle: group homomorphisms must preserve identities and inverses. It is also clear that if $\varphi : G \rightarrow H$ is a group homomorphism and g is an element of *finite order* in G , then $\varphi(g)$ must be an element of finite order in H : indeed, if $g^n = e_G$ for some $n > 0$, then

$$\varphi(g)^n = \varphi(g^n) = \varphi(e_G) = e_H.$$

In fact, this observation establishes a more precise statement:

Proposition 4.1. *Let $\varphi : G \rightarrow H$ be a group homomorphism, and let $g \in G$ be an element of finite order. Then $|\varphi(g)|$ divides $|g|$.*

Proof. As observed, $\varphi(g)^{|g|} = e_H$; applying Lemma 1.10 gives the statement. $\square$

Example 4.2. There are no nontrivial homomorphisms $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}$: indeed, the image of every element of $\mathbb{Z}/n\mathbb{Z}$ must have finite order, and the only element with finite order in $(\mathbb{Z}, +)$ is 0.

There are no nontrivial homomorphisms $\varphi : C_4 \rightarrow C_7$. Indeed, the orders of elements in C_4 divide 4 (cf. Proposition 2.3), and the orders of elements in C_7 divides 7. Thus, the order of each $\varphi(g)$ must divide *both* 4 and 7; this forces $|\varphi(g)| = 1$ for all g , that is, $\varphi(g) = e$ for all $g \in C_4$. $\lrcorner$

Of course the order itself is not preserved: for example, $1 \in \mathbb{Z}$ has infinite order, while $[1]_n = \pi_n(1) \in \mathbb{Z}/n\mathbb{Z}$ has order n (with notation as in §4.1). Order *is* preserved through isomorphisms, as we will see in a moment.

4.3. Isomorphisms. An isomorphism of groups $\varphi : G \rightarrow H$ is (of course) an isomorphism in **Grp**, that is, a group homomorphism admitting an inverse

$$\varphi^{-1} : H \rightarrow G$$

which is also a group homomorphism. Taking our cue from **Set**, if a homomorphism of groups is an isomorphism, then it must in particular be a bijection between the underlying sets. Luckily, the converse also holds:

Proposition 4.3. *Let $\varphi : G \rightarrow H$ be a group homomorphism. Then φ is an isomorphism of groups if and only if it is a bijection.*

Proof. One implication is immediate, as pointed out above. For the other implication, assume $\varphi : G \rightarrow H$ is a bijective group homomorphism. As a bijection, φ has an inverse in **Set**:

$$\varphi^{-1} : H \rightarrow G;$$

we simply need to check that this is a group homomorphism. Let h_1, h_2 be elements of H , and let $g_1 = \varphi^{-1}(h_1), g_2 = \varphi^{-1}(h_2)$ be the corresponding elements of G . Then

$$\varphi^{-1}(h_1 \cdot h_2) = \varphi^{-1}(\varphi(g_1) \cdot \varphi(g_2)) = \varphi^{-1}(\varphi(g_1 \cdot g_2)) = g_1 \cdot g_2 = \varphi^{-1}(h_1) \cdot \varphi^{-1}(h_2)$$

as needed. $\square$

Example 4.4. The function $D_6 \rightarrow S_3$ defined in §2.2 is an isomorphism of groups, since it is a bijective group homomorphism. So is the exponential function $(\mathbb{R}, +) \rightarrow (\mathbb{R}^{>0}, \cdot)$ mentioned in §4.1. If the exponential function $\epsilon_g : \mathbb{Z} \rightarrow G$ determined by an element $g \in G$ (as in §4.1) is an isomorphism, we say that G is an ‘infinite cyclic’ group.

The function $\pi_2^6 \times \pi_3^6 : C_6 \rightarrow C_2 \times C_3$ studied (‘additively’) in §4.1 is an isomorphism. ┘

Definition 4.5. Two groups G, H are *isomorphic* if they are isomorphic in $\mathbf{Grp}$ in the sense of §I.4.1, that is (by Proposition 4.3), if there is a bijective group homomorphism $G \rightarrow H$. ┘

We have observed once and for all in §I.4.1 that ‘isomorphic’ is automatically an equivalence relation. We write $G \cong H$ if G and H are isomorphic.

Automorphisms of a group G are isomorphisms $G \rightarrow G$; these form a group $\text{Aut}_{\mathbf{Grp}}(G)$ (cf. §I.4.1), usually denoted $\text{Aut}(G)$.

Example 4.6. We have introduced our template of *cyclic groups* in §2.3. The notion of isomorphism allows us to give a formal definition:

Definition 4.7. A group G is *cyclic* if it is isomorphic to $\mathbb{Z}$ or to $C_n = \mathbb{Z}/n\mathbb{Z}$ for some¹⁹ n . ┘

Thus, $C_2 \times C_3$ is cyclic, of order 6, since $C_2 \times C_3 \cong C_6$. More generally (Exercise 4.9) $C_m \times C_n$ is cyclic if $\gcd(m, n) = 1$.

The reader will check easily (Exercise 4.3) that a group of order n is cyclic if and only if it contains an element of order n .

There is a somewhat surprising source of cyclic groups: *if p is prime, the group $(\mathbb{Z}/p\mathbb{Z})^*, \cdot$ is cyclic*. We will prove a more general statement when we have accumulated more machinery (Theorem IV.6.10), but the adventurous reader can already enjoy a proof by working out Exercise 4.11. This is a relatively deep fact; note that, for example, $(\mathbb{Z}/12\mathbb{Z})^*$ is *not* cyclic (cf. Exercise 2.19 and Exercise 4.10). The fact that $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic for p prime means that there must be integers a such that *every* nonmultiple of p is congruent to a power of a ; the usual proofs of this fact are not *constructive*, that is, they do not explicitly produce an integer with this property. There is a very pretty connection between the order of an element of the cyclic group $(\mathbb{Z}/p\mathbb{Z})^*$ and the so-called ‘cyclotomic polynomials’; but that will have to wait for a little field theory (cf. Exercise VII.5.15).

As we have seen, the groups D_6 and S_3 are isomorphic. Are C_6 and S_3 isomorphic? There are 46,656 functions between the sets C_6 and S_3 , of which 720 are bijections and 120 are bijections preserving the identity. The reader is welcome to list all 120 and attempt to verify by hand if any of them is a homomorphism. But maybe there is a better strategy to answer such questions. . . . ┘

Isomorphic objects of a category are essentially indistinguishable in that category. Thus, isomorphic *groups* share every group-theoretic structure. In particular,

¹⁹This includes the possibility that $n = 1$, that is, trivial groups are cyclic.

Proposition 4.8. *Let $\varphi : G \rightarrow H$ be an isomorphism.*

- $(\forall g \in G) : |\varphi(g)| = |g|$;
- G is commutative if and only if H is commutative.

Proof. The first assertion follows from Proposition 4.1: the order of $\varphi(g)$ divides the order of g , and on the other hand the order of $g = \varphi^{-1}(\varphi(g))$ must divide the order of $\varphi(g)$; thus the two orders must be equal.

The proof of the second assertion is left to the reader. $\square$

Further instances of this principle will be assumed without explicit mention.

Example 4.9. $C_6 \not\cong S_3$, since one is commutative and the other is not. Here is another reason: in C_6 there is 1 element of order one, 1 of order two, 2 of order three, and 2 of order six; in S_3 the situation is different: 1 element of order one, 3 of order two, 2 of order three. Thus, *none* of the 120 bijections $C_6 \rightarrow S_3$ preserving the identity is a group homomorphism.

Note: Two finite *commutative* groups are isomorphic *if and only if* they have the same number of elements of any given order, but we are not yet in a position to prove this; the reader will verify this fact in due time (Exercise IV.6.13). The commutativity hypothesis is necessary: there *do* exist pairs of nonisomorphic finite groups with the same number of elements of any given order (same exercise). $\dashv$

4.4. Homomorphisms of abelian groups. I have already mentioned that **Ab** is in some ways ‘better behaved’ than **Grp**, and I am ready to highlight another instance of this observation. As we have seen, $\text{Hom}_{\text{Grp}}(G, H)$ is a *pointed set* for any two groups G, H . In **Ab**, we can say much more: $\text{Hom}_{\text{Ab}}(G, H)$ is a group (in fact, an abelian group) for any two abelian groups G, H .

The operation in $\text{Hom}_{\text{Ab}}(G, H)$ is ‘inherited’ from the operation in H : if $\varphi, \psi : G \rightarrow H$ are two group homomorphisms, let $\varphi + \psi$ be the function defined by

$$(\forall a \in G) : (\varphi + \psi)(a) := \varphi(a) + \psi(a).$$

Is $\varphi + \psi$ a group homomorphism? Yes, because $\forall a, b \in G$

$$\begin{aligned} (\varphi + \psi)(a + b) &= \varphi(a + b) + \psi(a + b) = (\varphi(a) + \varphi(b)) + (\psi(a) + \psi(b)) \\ &\stackrel{!}{=} (\varphi(a) + \psi(a)) + (\varphi(b) + \psi(b)) = (\varphi + \psi)(a) + (\varphi + \psi)(b). \end{aligned}$$

Note that the equality marked by ! uses crucially the fact that H is commutative.

With this operation, $\text{Hom}_{\text{Ab}}(G, H)$ is clearly a group: the associativity of $+$ is inherited from that of the operation in H ; the trivial homomorphism is the identity element, and the inverse²⁰ of $\varphi : G \rightarrow H$ is defined (not surprisingly) by

$$(\forall a \in G) : (-\varphi)(a) = -\varphi(a).$$

In fact, note that these conclusions may be drawn as soon as H is commutative: $\text{Hom}_{\text{Grp}}(G, H)$ is a group if H is commutative (even if G is not). In fact, if H is

²⁰Unfortunate clash of terminology! I mean the ‘inverse’ as in ‘group inverse’, not as a possible function $H \rightarrow G$.

a commutative group, then $H^A = \text{Hom}_{\text{Set}}(A, H)$ is a commutative group for all sets A ; we will come back to this group in §5.4.

Exercises

4.1. ▷ Check that the function π_m^n defined in §4.1 is well-defined and makes the diagram commute. Verify that it is a group homomorphism. Why is the hypothesis $m \mid n$ necessary? [§4.1]

4.2. Show that the homomorphism $\pi_2^4 \times \pi_2^4 : C_4 \rightarrow C_2 \times C_2$ is *not* an isomorphism. In fact, is there *any* isomorphism $C_4 \rightarrow C_2 \times C_2$?

4.3. ▷ Prove that a group of order n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ if and only if it contains an element of order n . [§4.3]

4.4. Prove that no two of the groups $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are isomorphic to one another. Can you decide whether $(\mathbb{R}, +)$, $(\mathbb{C}, +)$ are isomorphic to one another? (Cf. Exercise VI.1.1.)

4.5. Prove that the groups $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$ are not isomorphic.

4.6. We have seen that $(\mathbb{R}, +)$ and $(\mathbb{R}^{>0}, \cdot)$ are isomorphic (Example 4.4). Are the groups $(\mathbb{Q}, +)$ and $(\mathbb{Q}^{>0}, \cdot)$ isomorphic?

4.7. Let G be a group. Prove that the function $G \rightarrow G$ defined by $g \mapsto g^{-1}$ is a homomorphism if and only if G is abelian. Prove that $g \mapsto g^2$ is a homomorphism if and only if G is abelian.

4.8. ¬ Let G be a group, and let $g \in G$. Prove that the function $\gamma_g : G \rightarrow G$ defined by $(\forall a \in G) : \gamma_g(a) = gag^{-1}$ is an automorphism of G . (The automorphisms γ_g are called ‘inner’ automorphisms of G .) Prove that the function $G \rightarrow \text{Aut}(G)$ defined by $g \mapsto \gamma_g$ is a homomorphism. Prove that this homomorphism is trivial if and only if G is abelian. [6.7, 7.11, IV.1.5]

4.9. ▷ Prove that if m, n are positive integers such that $\gcd(m, n) = 1$, then $C_{mn} \cong C_m \times C_n$. [§4.3, 4.10, §IV.6.1, V.6.8]

4.10. ▷ Let $p \neq q$ be odd prime integers; show that $(\mathbb{Z}/pq\mathbb{Z})^*$ is not cyclic. (Hint: Use Exercise 4.9 to compute the order N of $(\mathbb{Z}/pq\mathbb{Z})^*$, and show that no element can have order N .) [§4.3]

4.11. ▷ In due time we will prove the easy fact that if p is a prime integer, then the equation $x^d = 1$ can have at most d solutions in $\mathbb{Z}/p\mathbb{Z}$. Assume this fact, and prove that the multiplicative group $G = (\mathbb{Z}/p\mathbb{Z})^*$ is cyclic. (Hint: Let $g \in G$ be an element of maximal order; use Exercise 1.15 to show that $h^{|g|} = 1$ for all $h \in G$. Therefore. . . .) [§4.3, 4.15, 4.16, §IV.6.3]

4.12. ¬ • Compute the order of $[9]_{31}$ in the group $(\mathbb{Z}/31\mathbb{Z})^*$.

• Does the equation $x^3 - 9 = 0$ have solutions in $\mathbb{Z}/31\mathbb{Z}$? (Hint: Plugging in all 31 elements of $\mathbb{Z}/31\mathbb{Z}$ is too laborious and will not teach you much. Instead,

use the result of the first part: if c is a solution of the equation, what can you say about $|c|$? [VII.5.15]

4.13. $\neg$ Prove that $\text{Aut}_{\text{Grp}}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \cong S_3$. [IV.5.14]

4.14. $\triangleright$ Prove that the order of the group of automorphisms of a cyclic group C_n is the number of positive integers $r \leq n$ that are *relatively prime* to n . (This is called *Euler's ϕ -function*; cf. Exercise 6.14.) [§IV.1.4, IV.1.22, §IV.2.5]

4.15. $\neg$ Compute the group of automorphisms of $(\mathbb{Z}, +)$. Prove that if p is prime, then $\text{Aut}_{\text{Grp}}(C_p) \cong C_{p-1}$. (Use Exercise 4.11.) [IV.5.12]

4.16. $\neg$ Prove *Wilson's theorem*: an integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

(For one direction, use Exercises 1.8 and 4.11. For the other, assume d is a proper divisor of p , and note that d divides $(p-1)!$; therefore...) [IV.4.11]

4.17. For a few small (but not too small) primes p , find a generator of $(\mathbb{Z}/p\mathbb{Z})^*$.

4.18. Prove the second part of Proposition 4.8.

5. Free groups

5.1. Motivation. Having become more familiar with homomorphisms, we can now contemplate one fancier example of a group. The motivation underlying this new construction may be summarized as follows: given a set A , whose elements have no special ‘group-theoretic’ property, we want to construct a group $F(A)$ containing A ‘in the most efficient way’.

For example, if $A = \emptyset$, then a trivial group will do. If $A = \{a\}$ is a singleton, then a trivial group will *not* do: because although a trivial group $\{a\}$ would itself be a singleton, that one element a in it would have to be the identity, and that is certainly a very special group-theoretic property. Instead, I propose that we construct an infinite cyclic group $\langle a \rangle$ whose elements are ‘formal powers’ a^n , $n \in \mathbb{Z}$, and we identify a with the power a^1 :

$$\langle a \rangle := \{\dots, a^{-2}, a^{-1}, a^0 = e, a^1 = a, a^2, a^3, \dots\};$$

we take all these powers to be distinct and define multiplication in the evident way—so that the exponential map

$$\epsilon_a : \mathbb{Z} \rightarrow \langle a \rangle, \quad \epsilon_a(n) := a^n$$

is an isomorphism. The fact that ‘all powers are distinct’ is the formal way to implement the fact that there is nothing special about a : in the group $F(\{a\}) = \langle a \rangle$, a obeys no condition other than the inevitable $a^0 = e$.

Summarizing: if A is a singleton, then we may take $F(A)$ to be an infinite cyclic group.

The task is to formalize the heuristic motivation given above and construct a group $F(A)$ for *every* set A . As I often do, I will now ask the reader to put away this book and to try to figure out on his or her own what this may mean and how it may be accomplished.

5.2. Universal property. Hoping that the reader has now acquired an individual viewpoint on the issue, here is the standard answer: the heuristic motivation is formalized by means of a suitable universal property. Given a set A , our group $F(A)$ will have to ‘contain’ A ; therefore it is natural to consider the category $\mathcal{F}^A$ whose objects are pairs (j, G) , where G is a group and

$$j : A \rightarrow G$$

is a set-function²¹ from A to G and morphisms

$$(j_1, G_1) \rightarrow (j_2, G_2)$$

are *commutative* diagrams of set-functions

$$\begin{array}{ccc} G_1 & \xrightarrow{\varphi} & G_2 \\ j_1 \uparrow & \nearrow j_2 & \\ A & & \end{array}$$

in which φ is required to be a *group homomorphism*.

The reader will be reminded of the categories we considered in Example I.3.7: the only difference here is that we are mixing objects and morphisms of one category (that is, **Grp**) with objects and morphisms of *another* (related) category (that is, **Set**). The fact that we are considering all possible functions $A \rightarrow G$ is a way to implement the fact that we have no *a priori* group-theoretic information about A : we do not want to put any restriction on what may happen to the elements of A once they are mapped to a group G ; hence we consider all possibilities at once.

A *free group* $F(A)$ on A will be (the group component of) an *initial* object in $\mathcal{F}^A$. This choice implements the fact that A should map to $F(A)$ in the ‘most efficient way’: any other way to map A to a group can be reconstructed from this one, by composing with a group homomorphism. In the language of universal properties, we can state this as follows: $F(A)$ is a free group on the set A if there is a set-function $j : A \rightarrow F(A)$ such that, for all groups G and set-functions $f : A \rightarrow G$, *there exists a unique* group homomorphism $\varphi : F(A) \rightarrow G$ such that the diagram

$$\begin{array}{ccc} F(A) & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ A & & \end{array}$$

commutes. By general nonsense (Proposition I.5.4), this universal property defines $F(A)$ up to isomorphism, *if this group exists*. But does $F(A)$ exist?

Before giving a ‘concrete’ construction of $F(A)$, let’s check that if $A = \{a\}$ is a singleton, then $F(A) \cong \mathbb{Z}$, as proposed in §5.1. The function $j : A \rightarrow \mathbb{Z}$ will send a to $1 \in \mathbb{Z}$. For any group G , giving a set-function $f : A \rightarrow G$ amounts to choosing

²¹We could assume that j is *injective*, identifying A with a subset of G ; the construction would be completely analogous, and the resulting group would be the same. However, considering arbitrary functions leads to a stronger, more useful, universal property.

one element $g = f(a) \in G$. Now *there is a unique* homomorphism $\varphi : \mathbb{Z} \rightarrow G$ making the diagram

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ \{a\} & & \end{array}$$

commute: because this forces $\varphi(1) = \varphi \circ j(a) = f(a) = g$, and then the homomorphism condition forces $\varphi(n) = g^n$. That is, φ is necessarily the exponential map ϵ_g considered in §4.1. Therefore, infinite cyclic groups do satisfy the universal property for free groups over a singleton.

5.3. Concrete construction. As we know, terminal objects of a category need not exist. So I have to convince the reader that free groups $F(A)$ exist, for every set A .

Given any set A , we are going to think of A as an ‘alphabet’ and construct ‘words’ whose letters are elements of A or ‘inverses’ of elements of A . To formalize this, consider a set A' isomorphic to A and disjoint from it; call a^{-1} the element in A' corresponding to $a \in A$. A *word* on the set A is an ordered list

$$(a_1, a_2, \dots, a_n),$$

which we denote by the juxtaposition

$$w = a_1 a_2 \cdots a_n,$$

where each ‘letter’ a_i is either an element $a \in A$ or an element $a^{-1} \in A'$. I will denote the set of words on A by $W(A)$; the number n of letters is the ‘length’ of w ; I include in $W(A)$ the ‘empty word’ $w = ()$, consisting of *no* letters.

For example, if $A = \{a\}$ is a singleton, then an element of $W(A)$ may look like

$$a^{-1} a^{-1} a a a^{-1} a a^{-1}.$$

An element of $W(\{x, y\})$ may look like

$$x x x^{-1} y y^{-1} x x y^{-1} x^{-1} y y^{-1} x y^{-1} x.$$

Now the notation I have chosen hints that elements in $W(A)$ may be redundant: for example,

$$x y y^{-1} x \quad \text{and} \quad x x$$

are distinct words, but they ought to end up being the same element of a group having to do with words. Therefore, we want to have a process of ‘reduction’ which takes a word and cleans it up by performing all cancellations. Note that we have to do this ‘by hand’, since we have not come close yet to defining an operation or making formal sense of considering a^{-1} to be the ‘inverse’ of a .

Describing the reduction process is invariably awkward—it is a completely evident procedure, but writing it down precisely and elegantly is a challenge. I will settle for the following.

- Define an ‘elementary’ reduction $r : W(A) \rightarrow W(A)$: given $w \in W(A)$, search for the first occurrence (from left to right) of a pair aa^{-1} or $a^{-1}a$, and let $r(w)$ be the word obtained by removing such a pair. In the two examples given above,

$$\begin{aligned} r(a^{-1}\underline{a^{-1}aaaa^{-1}aa^{-1}}) &= a^{-1}aaa^{-1}aa^{-1}, \\ r(\underline{xxx^{-1}yy^{-1}xy^{-1}x^{-1}yy^{-1}xy^{-1}x}) &= xyy^{-1}xyy^{-1}x^{-1}yy^{-1}xy^{-1}x. \end{aligned}$$

- Note that $r(w) = w$ precisely when ‘no cancellation is possible’; We say that w is a ‘reduced word’ in this case.

Lemma 5.1. *If $w \in W(A)$ has length n , then²² $r^{\lfloor \frac{n}{2} \rfloor}(w)$ is a reduced word.*

Proof. Indeed, either $r(w) = w$ or the length of $r(w)$ is less than the length of w ; but one cannot decrease the length of w more than $n/2$ times, since each non-identity application of r decreases the length by two. $\square$

- Now define the ‘reduction’ $R : W(A) \rightarrow W(A)$ by setting $R(w) = r^{\lfloor \frac{n}{2} \rfloor}(w)$, where n is the length of w . By the lemma, $R(w)$ is always a reduced word. For example, $R(a^{-1}a^{-1}aaaa^{-1}aa^{-1})$ is the empty word, since

$$\begin{aligned} r^4(a^{-1}a^{-1}aaaa^{-1}aa^{-1}) &= r^3(a^{-1}aaa^{-1}aa^{-1}) = r^2(aa^{-1}aa^{-1}) = r(aa^{-1}) = (); \\ \text{and } R(\underline{xxx^{-1}yy^{-1}xy^{-1}x^{-1}yy^{-1}xy^{-1}x}) &= xxxy^{-1}y^{-1}x, \text{ as the reader may check.} \end{aligned}$$

Let $F(A)$ be the set of reduced words on A , that is, the image of the reduction map R we have just defined.

We are ready to (finally) define free groups ‘concretely’. Define a binary operation on $F(A)$ by *juxtaposition* $\mathcal{E}$ *reduction*: for reduced words w, w' , define $w \cdot w'$ as the reduction of the juxtaposition of w and w' ,

$$w \cdot w' := R(ww').$$

It is essentially evident that $F(A)$ is a group under this operation:

- The operation is associative.
- The empty word $e = ()$ is the identity in $F(A)$, since $ew = we = w$ (no reduction is necessary).
- If w is a reduced word, the inverse of w is obtained by reversing the order of the letters of w and replacing each $a \in A$ by $a^{-1} \in A'$ and each a^{-1} by a .

The most cumbersome of these statements to prove formally is associativity; it follows easily from (for example) Exercise 5.4.

There is a function $j : A \rightarrow F(A)$, defined by sending the element $a \in A$ to the word consisting of the single ‘letter’ a .

Proposition 5.2. *The pair $(j, F(A))$ satisfies the universal property for free groups on A .*

²² $\lfloor q \rfloor$ denotes the largest integer $\leq q$.

Proof. This is also essentially evident, once one has absorbed all the notation. Any function $f : A \rightarrow G$ to a group extends uniquely to a map $\varphi : F(A) \rightarrow G$, determined by the homomorphism condition and by the requirement that the diagram commutes, which fixes its value on one-letter words $a \in A$ (as well as on $a^{-1} \in A'$).

To check more formally that φ exists as a homomorphism, one can proceed as follows. If $f : A \rightarrow G$ is any function, we can extend f to a set-function

$$\tilde{\varphi} : W(A) \rightarrow G$$

by insisting that on one-letter words a or a^{-1} (for $a \in A$),

$$\tilde{\varphi}(a) = f(a), \quad \tilde{\varphi}(a^{-1}) = f(a)^{-1},$$

and that $\tilde{\varphi}$ is compatible with juxtaposition:

$$\tilde{\varphi}(ww') = \tilde{\varphi}(w)\tilde{\varphi}(w')$$

for any two words w, w' . The key point now is that reduction is invisible for $\tilde{\varphi}$:

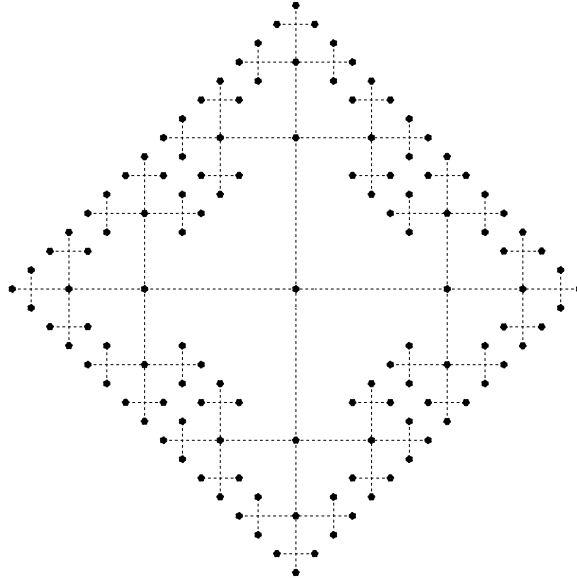
$$\tilde{\varphi}(R(w)) = \tilde{\varphi}(w),$$

since this is clearly the case for *elementary* reductions; therefore, since $\varphi : F(A) \rightarrow G$ agrees with $\tilde{\varphi}$ on reduced words, we have for $w, w' \in F(A)$

$$\varphi(w \cdot w') = \tilde{\varphi}(w \cdot w') = \tilde{\varphi}(R(ww')) = \tilde{\varphi}(ww') = \tilde{\varphi}(w)\tilde{\varphi}(w') = \varphi(w)\varphi(w') :$$

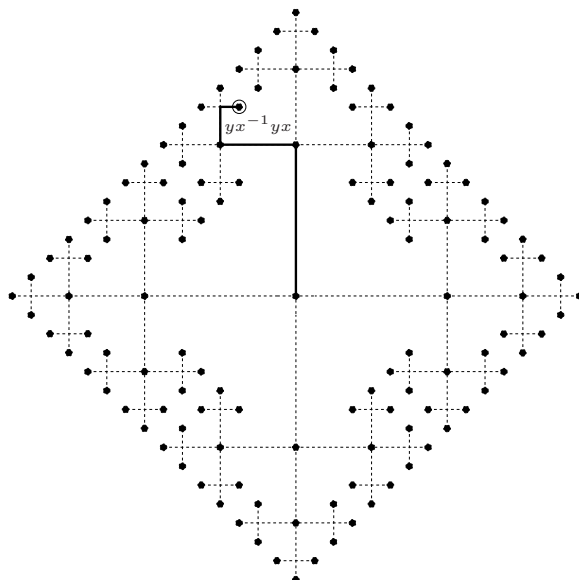
that is, φ is a homomorphism, as needed. $\square$

Example 5.3. It is easy to ‘visualize’ $F(\{a\}) \cong \mathbb{Z}$; but it is already somewhat challenging for the free group on *two* generators, $F(\{x, y\})$. The best I can do is the following: behold the infinite graph²³



²³This is an example of the *Cayley graph* of a group (cf. Exercise 8.6): a graph whose vertices correspond to the elements of the group and whose edges connect vertices according to the action of generators.

obtained by starting at a point (the center of the picture), then branching out in four directions by a length of 1, then branching out similarly by a length of $1/2$, then by $1/4$, then by $1/8$, then... (and I stopped there, to avoid cluttering the picture too much). Then every element of $F(\{x, y\})$ corresponds in a rather natural way to exactly one dot in this diagram. Indeed, we can place the empty word at the center; and we can agree that every x in a word takes us one step to the right, every x^{-1} to the left, every y up, and every y^{-1} down. For example, the word $yx^{-1}yx$ takes us here:



The reader will surely encounter this group elsewhere: it is the *fundamental group* of the ‘figure 8’. ┘

5.4. Free *abelian* groups. We can pose in **Ab** the same question answered above for **Grp**: that is, ask for the *abelian* group $F^{ab}(A)$ which most efficiently contains the set A , provided that we do not have any additional information on the elements of A . Of course we *do* know something about the elements of A this time: they will have to commute with each other in $F^{ab}(A)$. This plays no role if $A = \{a\}$ is a singleton, and therefore $F^{ab}(\{a\}) = F(\{a\}) \cong \mathbb{Z}$; but the requirement is different for larger sets, so we should expect a different answer in general.

The formalization of the heuristic requirement is precisely the same universal property that gave us free groups, but (of course) stated in **Ab**: $F^{ab}(A)$ is a *free abelian group* on the set A if there is a set-function $j : A \rightarrow F^{ab}(A)$ such that, for all *abelian* groups G and set-functions $f : A \rightarrow G$, *there exists a unique* group homomorphism $\varphi : F^{ab}(A) \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} F^{ab}(A) & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ A & & \end{array}$$

Again, Proposition I.5.4 guarantees that $F^{ab}(A)$ is unique up to isomorphism, if it exists; but we have to prove it exists! This is in some way simpler than for $\mathbf{Grp}$, in the sense that $F^{ab}(A)$ is easier to understand, at least for finite sets A .

To fix ideas, I will first describe the answer for a finite set, say $A = \{1, \dots, n\}$. I will denote by $\mathbb{Z}^{\oplus n}$ the direct sum

$$\underbrace{\mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{n\text{-times}}$$

recall (§3.5) that this group ‘is the same as’ the product²⁴ $\mathbb{Z}^n$ (but we view it as a coproduct). There is a function $j : A \rightarrow \mathbb{Z}^{\oplus n}$, defined by

$$j(i) := (0, \dots, 0, \underset{i\text{-th place}}{1}, 0, \dots, 0) \in \mathbb{Z}^{\oplus n}.$$

Claim 5.4. For $A = \{1, \dots, n\}$, $\mathbb{Z}^{\oplus n}$ is a free abelian group on A .

Proof. Note that every element of $\mathbb{Z}^{\oplus n}$ can be written uniquely in the form $\sum_{i=1}^n m_i j(i)$: indeed,

$$\begin{aligned} (m_1, \dots, m_n) &= (m_1, 0, \dots, 0) + (0, m_2, 0, \dots, 0) + \dots + (0, \dots, 0, m_n) \\ &= m_1(1, 0, \dots, 0) + m_2(0, 1, 0, \dots, 0) + \dots + m_n(0, \dots, 0, 1) \\ &= m_1 j(1) + \dots + m_n j(n), \end{aligned}$$

and $(m_1, \dots, m_n) = (0, \dots, 0)$ if and only if all m_i are 0.

Now let $f : A \rightarrow G$ be any function from $A = \{1, \dots, n\}$ to an abelian group G . I define $\varphi : \mathbb{Z}^{\oplus n} \rightarrow G$ by

$$\varphi \left(\sum_{i=1}^n m_i j(i) \right) := \sum_{i=1}^n m_i f(i) :$$

indeed, we have no choice—this definition is forced by the needed commutativity of the diagram

$$\begin{array}{ccc} \mathbb{Z}^{\oplus n} & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ A & & \end{array}$$

and by the homomorphism condition. Thus φ is certainly uniquely determined, and we just have to check that it is a homomorphism. This is where the commutativity of G enters:

$$\varphi \left(\sum_{i=1}^n m'_i j(i) \right) + \varphi \left(\sum_{i=1}^n m''_i j(i) \right) = \sum_{i=1}^n m'_i f(i) + \sum_{i=1}^n m''_i f(i) \stackrel{!}{=} \sum_{i=1}^n (m'_i + m''_i) f(i)$$

because G is commutative,

$$= \varphi \left(\sum_{i=1}^n (m'_i + m''_i) j(i) \right) = \varphi \left(\sum_{i=1}^n m'_i j(i) + \sum_{i=1}^n m''_i j(i) \right)$$

as needed. □

²⁴Indeed, it is common to denote this group by $\mathbb{Z}^n$, omitting the $\oplus$. No confusion is likely, but I will try to distinguish the two to emphasize that they play different categorical roles.

Remark 5.5. A less hands-on, more high-brow argument can be given by contemplating the universal property defining free abelian groups vis-à-vis the universal property for coproducts; cf. Exercise 5.7. $\lrcorner$

Now for the general case: let A be any set. As we have seen, $H^A = \text{Hom}_{\text{Set}}(A, H)$ has a natural abelian group structure if H is an abelian group (§4.4); elements of H^A are arbitrary set-functions $\alpha : A \rightarrow H$. We can define a subset $H^{\oplus A}$ of H^A as follows:

$$H^{\oplus A} := \{\alpha : A \rightarrow H \mid \alpha(a) \neq e_H \text{ for only finitely many elements } a \in A\}.$$

The operation in H^A induces an operation in $H^{\oplus A}$, which makes $H^{\oplus A}$ into a group²⁵.

The reader should note that $H^{\oplus A}$ is the whole of H^A if A is a finite set; and that $\mathbb{Z}^{\oplus A} \cong \mathbb{Z}^{\oplus n}$ if $A = \{1, \dots, n\}$: indeed, $(m_1, \dots, m_n) \in \mathbb{Z}^{\oplus n}$ may be identified with the function $\{1, \dots, n\} \rightarrow \mathbb{Z}$ sending i to m_i .

For $H = \mathbb{Z}$ there is a natural function $j : A \rightarrow \mathbb{Z}^{\oplus A}$, obtained by sending $a \in A$ to the function $j_a : A \rightarrow \mathbb{Z}$ defined by

$$(\forall x \in A) : \quad j_a(x) := \begin{cases} 1 & \text{if } x = a, \\ 0 & \text{if } x \neq a. \end{cases}$$

Note that for $A = \{1, \dots, n\}$ and identifying $\mathbb{Z}^{\oplus A} \cong \mathbb{Z}^{\oplus n}$, this function j is *the same function* denoted j earlier.

Proposition 5.6. *For every set A , $F^{ab}(A) \cong \mathbb{Z}^{\oplus A}$.*

Proof. The key point is again that every element of $\mathbb{Z}^{\oplus A}$ may be written uniquely as a *finite* sum

$$\sum_{a \in A} m_a j(a), \quad m_a \neq 0 \text{ for only finitely many } a;$$

once this is understood, the argument is precisely the same as for Claim 5.4. $\square$

²⁵Thus $H^{\oplus A}$ is a *subgroup* of H^A ; cf. §6.

Exercises

5.1. Does the category $\mathcal{F}^A$ defined in §5.2 have final objects? If so, what are they?

5.2. Since trivial groups T are initial in $\mathbf{Grp}$, one may be led to think that (e, T) should be initial in $\mathcal{F}^A$, for every A : e would be defined by sending every element of A to the (only) element in T ; and for any other group G , *there is a unique* homomorphism $T \rightarrow G$. Explain why (e, T) is *not* initial in $\mathcal{F}^A$ (unless $A = \emptyset$).

5.3. $\triangleright$ Use the universal property of free groups to prove that the map $j : A \rightarrow F(A)$ is injective, for all sets A . (Hint: It suffices to show that for every two elements a, b of A there is a group G and a set-function $f : A \rightarrow G$ such that $f(a) \neq f(b)$. Why? How do you construct f and G ?) [§III.6.3]

5.4. $\triangleright$ In the ‘concrete’ construction of free groups, one can try to reduce words by performing cancellations in any order; the process of ‘elementary reductions’ used in the text (that is, from left to right) is only one possibility. Prove that the result of iterating cancellations on a word is independent of the order in which the cancellations are performed. Deduce the associativity of the product in $F(A)$ from this. [§5.3]

5.5. Verify explicitly that $H^{\oplus A}$ is a group.

5.6. $\triangleright$ Prove that the group $F(\{x, y\})$ (visualized in Example 5.3) is a coproduct $\mathbb{Z} * \mathbb{Z}$ of $\mathbb{Z}$ by itself in the category $\mathbf{Grp}$. (Hint: With due care, the universal property for one turns into the universal property for the other.) [§3.4, 3.7, 5.7]

5.7. $\triangleright$ Extend the result of Exercise 5.6 to free groups $F(\{x_1, \dots, x_n\})$ and to free *abelian* groups $F^{ab}(\{x_1, \dots, x_n\})$. [§3.4, §5.4]

5.8. Still more generally, prove that $F(A \amalg B) = F(A) * F(B)$ and that $F^{ab}(A \amalg B) = F^{ab}(A) \oplus F^{ab}(B)$ for all sets A, B . (That is, the constructions F, F^{ab} ‘preserve coproducts’.)

5.9. Let $G = \mathbb{Z}^{\oplus \mathbb{N}}$. Prove that $G \times G \cong G$.

5.10. $\neg$ Let $F = F^{ab}(A)$.

- Define an equivalence relation $\sim$ on F by setting $f' \sim f$ if and only if $f - f' = 2g$ for some $g \in F$. Prove that $F/\sim$ is a finite set if and only if A is finite, and in that case $|F/\sim| = 2^{|A|}$.
- Assume $F^{ab}(B) \cong F^{ab}(A)$. If A is finite, prove that B is also, and that $A \cong B$ as sets. (This result holds for free groups as well, and without any finiteness hypothesis. See Exercises 7.13 and VI.1.20.)

[7.4, 7.13]

6. Subgroups

6.1. Definition. Let $(G, \cdot)$ be a group, and let $(H, \bullet)$ be another group, whose underlying set H is a subset of G .

Definition 6.1. $(H, \bullet)$ is a *subgroup* of G if the inclusion function $i : H \hookrightarrow G$ is a group homomorphism. $\square$

For example, the trivial group consisting of the single element e_G is a subgroup of G .

If $(H, \bullet)$ is a subgroup of $(G, \cdot)$, then $\forall h_1, h_2 \in H$:

$$(*) \quad i(h_1 \bullet h_2) = i(h_1) \cdot i(h_2).$$

We say that the operation $\bullet$ on H is ‘induced’ from the operation $\cdot$ on G ; in practice one omits explicit mention of i and of the operations, and $(*)$ guarantees that no ambiguity will arise from this.

The subgroup condition may be streamlined. A subset H of a group G determines a subgroup if the operation $\cdot$ in G induces (by $(*)$) a binary operation in H (we say that H is *closed* with respect to the operation in G), satisfying the group axioms. Since the identity and inverses are preserved through homomorphisms (Proposition 3.2), the identity e_H of H will have to coincide with the identity e_G of G and the inverse of an element $h \in H$ has to be the same as the inverse of that element in G . The most economical way to say all this is

Proposition 6.2. A nonempty subset H of a group G is a subgroup if and only if

$$(\forall a, b \in H) : \quad ab^{-1} \in H.$$

Proof. It is clear that if H is a subgroup, then the stated condition holds: indeed, if $b \in H$, then the inverse of b must also be in H and H is closed under the operation of G .

Conversely, assume the stated condition holds; we have to check that H is closed under the operation of G , the induced operation on H is associative, and it admits an identity element and inverses (that is, it contains e_G and is closed under taking inverses in G). Since H is nonempty, we can find an element $h \in H$. Choosing $a = b = h$, we see that

$$e_G = hh^{-1} = ab^{-1} \in H;$$

thus H contains the identity. Given any $h \in H$, choosing $a = e_G$ and $b = h$ shows that

$$h^{-1} = e_G h^{-1} = ab^{-1} \in H;$$

thus H contains the inverse of any of its elements. Given any $h_1, h_2 \in H$, choose $a = h_1$, $b = h_2^{-1}$; the stated condition says that

$$h_1 h_2 = h_1 ((h_2)^{-1})^{-1} = ab^{-1} \in H,$$

proving that H is closed under the operation.

Finally, the fact that the operation is associative in G implies immediately that the induced operation is associative in H , concluding the proof that H , with the induced operation, is a group. $\square$

This criterion makes it particularly straightforward to check simple facts concerning subgroups. For example,

Lemma 6.3. *If $\{H_\alpha\}_{\alpha \in A}$ is any family of subgroups of a group G , then*

$$H = \bigcap_{\alpha \in A} H_\alpha$$

is a subgroup of G .

Proof. This follows right away from Proposition 6.2: H is nonempty, because $e \in H_\alpha$ for all α , so $e \in H$; and

$$a, b \in H \implies (\forall \alpha \in A) : a, b \in H_\alpha \implies (\forall \alpha \in A) : ab^{-1} \in H_\alpha \implies ab^{-1} \in H,$$

proving that H is a subgroup of G . $\square$

Similarly,

Lemma 6.4. *Let $\varphi : G \rightarrow G'$ be a group homomorphism, and let H' be a subgroup of G' . Then $\varphi^{-1}(H')$ is a subgroup of G .*

Proof. Recall (end of §I.2.5) that $\varphi^{-1}(H')$ consists of all $g \in G$ such that $\varphi(g) \in H'$. Since $\varphi(e_G) = e_{G'} \in H'$, this set is nonempty. If $a, b \in \varphi^{-1}(H')$, then $\varphi(a)$ and $\varphi(b)$ are in H' , and hence

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in H' :$$

thus, $ab^{-1} \in \varphi^{-1}(H')$. This implies that $\varphi^{-1}(H')$ is a subgroup of G , by Proposition 6.2. $\square$

6.2. Examples: Kernel and image. Every group homomorphism $\varphi : G \rightarrow G'$ determines two interesting subgroups:

- the *kernel* of φ , $\ker \varphi \subseteq G$; and
- the *image* of φ , $\operatorname{im} \varphi \subseteq G'$.

Definition 6.5. The *kernel* of $\varphi : G \rightarrow G'$ is the subset of G consisting of elements mapping to the identity in G' :

$$\ker \varphi := \{g \in G \mid \varphi(g) = e_{G'}\} = \varphi^{-1}(e_{G'}).$$

⌋

Since $\{e_{G'}\}$ is a subgroup of G' , Lemma 6.4 shows that $\ker \varphi$ is indeed a subgroup of G . For an (even) more explicit argument, note that $\ker \varphi$ is nonempty, since $e_G \in \ker \varphi$; and if a, b are in $\ker \varphi$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_{G'}e_{G'}^{-1} = e_{G'},$$

proving that $ab^{-1} \in \ker \varphi$. This shows that $\ker \varphi$ is a subgroup of G , by Proposition 6.2.

The verification that $\operatorname{im} \varphi$ is a subgroup is left to the reader. In fact, the reader should check that the image of *any* subgroup of G is a subgroup of G' .

We will soon (§7.1) see that kernels are ‘special’ subgroups. As with most constructions of importance in algebra, they satisfy a universal property, which may be expressed as follows.

Proposition 6.6. *Let $\varphi : G \rightarrow G'$ be a homomorphism. Then the inclusion $i : \ker \varphi \hookrightarrow G$ is final in the category²⁶ of group homomorphisms $\alpha : K \rightarrow G$ such that $\varphi \circ \alpha$ is the trivial map.*

In other words, every group homomorphism $\alpha : K \rightarrow G$ such that $\varphi \circ \alpha$ is the trivial homomorphism (denoted ‘0’ in the diagram) factors uniquely through $\ker \varphi$:

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \curvearrowright & & \curvearrowright & \\
 K & \xrightarrow{\alpha} & G & \xrightarrow{\varphi} & G' \\
 & \searrow \exists! \bar{\alpha} & \uparrow & & \\
 & & \ker \varphi & &
 \end{array}$$

Proof. If $\alpha : K \rightarrow G$ is such that $\varphi \circ \alpha$ is the trivial map, then $\forall k \in K$

$$\varphi \circ \alpha(k) = \varphi(\alpha(k)) = e_{G'},$$

that is, $\alpha(k) \in \ker \varphi$. We can (and must) then let $\bar{\alpha} : K \rightarrow \ker \varphi$ simply be α itself, with restricted target. $\square$

Proposition 6.6 indicates how one might define a notion analogous to ‘kernel’ in very general settings. This viewpoint will be championed much later in this book, especially in Chapter IX.

Remark 6.7. The argument shows that in fact kernels of group homomorphisms satisfy a somewhat stronger universal property: any *set-function* $\alpha : K \rightarrow G$ such that the image of $\varphi \circ \alpha$ is the identity in G' must factor (as a set-function) through $\ker \varphi$. $\lrcorner$

6.3. Example: Subgroup generated by a subset. If $A \subseteq G$ is *any* subset, we have a unique group homomorphism

$$\varphi_A : F(A) \rightarrow G$$

extending this inclusion, by the universal property of free groups. The image of this homomorphism is a subgroup of G , the *subgroup generated by A* in G , often denoted²⁷ $\langle A \rangle$.

Of course, if G is abelian, then φ_A factors through $F^{ab}(A)$, so we may replace $F(A)$ by $F^{ab}(A)$ in this case.

The ‘concrete’ description of free groups (§5.3) leads to the following description of $\langle A \rangle$: it consists of all products in G of the form

$$a_1 a_2 a_3 \cdots a_n$$

where each a_i is either an element of A , the inverse of an element of A , or the identity. This is clearly the most ‘economical’ way to manufacture a subgroup of G , given the elements of A .

²⁶The reader should specify what the morphisms are in this category.

²⁷If $A = \{g_1, \dots, g_r\}$ is a finite set, one writes $\langle g_1, \dots, g_r \rangle$.

The reader who has not (yet) developed a taste for free groups may prefer the following alternative description: $\langle A \rangle$ is the intersection of all subgroups of G containing A ,

$$\langle A \rangle = \bigcap_{H \text{ subgroup of } G, H \supseteq A} H.$$

Indeed, the intersection on the right-hand side is a subgroup of G by Lemma 6.3, it contains A , and it is clearly the smallest subgroup satisfying this condition.

If $A = \{g\}$ consists of a single element, then $F(A) = \mathbb{Z}$ and $\varphi_A : \mathbb{Z} \rightarrow G$ is nothing but the ‘exponential map’ ϵ_g (cf. §4.1); $\langle A \rangle = \langle g \rangle$ is then the image of this map:

$$\langle g \rangle = \text{im}(\epsilon_g) = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}.$$

The subgroup $\langle g \rangle$ is the ‘cyclic subgroup generated by g ’: indeed, $\langle g \rangle$ is *cyclic* in the sense of Definition 4.7; the reader can easily check this fact already (Exercise 6.4); it will also be recovered as an immediate consequence of the construction of quotients (cf. §7.5).

Definition 6.8. A group G is *finitely generated* if there exists a *finite* subset $A \subseteq G$ such that $G = \langle A \rangle$. ┘

For examples, cyclic groups are finitely generated (in fact, they are generated by a singleton). By definition, a group is finitely generated if and only if there is a surjective homomorphism

$$F(\{1, \dots, n\}) \twoheadrightarrow G$$

for some n . One of the most memorable results proven in this book will give a *classification of finitely generated abelian groups*: we will be able to prove that every such group is a *direct sum of cyclic groups* (Theorem IV.6.6, Exercise VI.2.19, and the generalization given in Theorem VI.5.6). The situation for general groups is considerably more complex. The classification of *finite* (simple) groups is one of the major achievements of twentieth-century mathematics, and it is spread over at least 10,000 pages of research articles. To appreciate the difference in complexity, note that there are *42 abelian* groups of order 1024 up to isomorphism (as the reader will be able to establish in due time: Exercise IV.6.6); allegedly, there are *49,487,365,402* if we count noncommutative ones as well²⁸.

6.4. Example: Subgroups of cyclic groups. We are ready to determine *all* subgroups of *all* cyclic groups, that is, all subgroups of $\mathbb{Z}$ and of $\mathbb{Z}/n\mathbb{Z}$, for all $n > 0$ (because every cyclic group is isomorphic to one of these; cf. Definition 4.7). The result is easy to remember: *subgroups of cyclic groups are themselves cyclic groups*.

It is convenient to start from $\mathbb{Z}$. For $d \in \mathbb{Z}$ we let

$$d\mathbb{Z} := \langle d \rangle = \{m \in \mathbb{Z} \mid \exists q \in \mathbb{Z}, m = dq\};$$

that is, $d\mathbb{Z}$ denotes the set of *integer multiples* of d . Of course this is nothing but the ‘cyclic subgroup of $\mathbb{Z}$ generated by d ’.

Proposition 6.9. *Let $G \subseteq \mathbb{Z}$ be a subgroup. Then $G = d\mathbb{Z}$ for some $d \geq 0$.*

²⁸This comparison is a little unfair, however, since it so happens that more than 99% of all groups of order < 2000 have order 1024.

The proof will actually show that if $G \subseteq \mathbb{Z}$ is nontrivial, then d is the *smallest positive* element of G , and the reader is invited to remember this useful fact.

Remark 6.10. By Proposition 6.9, every nontrivial subgroup of $\mathbb{Z}$ is in fact isomorphic to $\mathbb{Z}$. Putting this a little strangely, it says that every subgroup of the *free* group on one generator is free. It is in fact true that *every subgroup of a* (finitely generated) *free group is free*; we will not prove this fact, although the diligent reader will get a taste of the argument in Exercise 9.16. In any case, beware that free groups on *two* generators already contain subgroups isomorphic to free groups on *arbitrarily many* generators. Indeed, the commutator subgroup (cf. Exercise 7.12) $[F, F]$ for $F = F(\{x, y\})$ is isomorphic to a free group on *infinitely many* generators (unfortunately, we will not prove this beautiful statement either). $\lrcorner$

Proof of Proposition 6.9. If $G = \{0\}$, then $G = 0\mathbb{Z}$. If not, note that G must contain *positive* integers: indeed, if $a \in G$ and $a < 0$, then $-a \in G$ and $-a > 0$. We can then let d be the *smallest positive integer*²⁹ in G , and I claim $G = d\mathbb{Z}$.

The inclusion $d\mathbb{Z} \subseteq G$ is clear. To verify the inclusion $G \subseteq d\mathbb{Z}$, let $m \in G$, and apply ‘division with remainder’ to write

$$m = dq + r,$$

with $0 \leq r < d$. Since $m \in G$ and $d\mathbb{Z} \subseteq G$ and since G is a subgroup, we see that

$$r = m - dq \in G.$$

But d is the smallest *positive* integer in G , and $r \in G$ is smaller than d ; so r cannot be positive. This shows $r = 0$, that is, $m = qd \in d\mathbb{Z}$; $G \subseteq d\mathbb{Z}$ follows, and we are done. $\square$

The ‘quotient’ homomorphism $\pi_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (cf. §4.1) allows us to establish the analogous result for *finite* cyclic groups:

Proposition 6.11. *Let $n > 0$ be an integer and let $G \subseteq \mathbb{Z}/n\mathbb{Z}$ be a subgroup. Then G is the cyclic subgroup of $\mathbb{Z}/n\mathbb{Z}$ generated by $[d]_n$, for some divisor d of n .*

Proof. Let $\pi_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ be the quotient map, and consider $G' := \pi_n^{-1}(G)$. By Lemma 6.4, G' is a subgroup of $\mathbb{Z}$; by Proposition 6.9, G' is a *cyclic* subgroup of $\mathbb{Z}$, generated by a nonnegative integer d . It follows that

$$G = \pi_n(G') = \pi_n(\langle d \rangle) = \langle [d]_n \rangle;$$

thus G is indeed a cyclic subgroup of $\mathbb{Z}/n\mathbb{Z}$, generated by a class $[d]_n$. Further, since $n \in G'$ (because $\pi_n(n) = [n]_n = [0]_n \in G$) and $G' = d\mathbb{Z}$, we see that d divides n , as claimed. $\square$

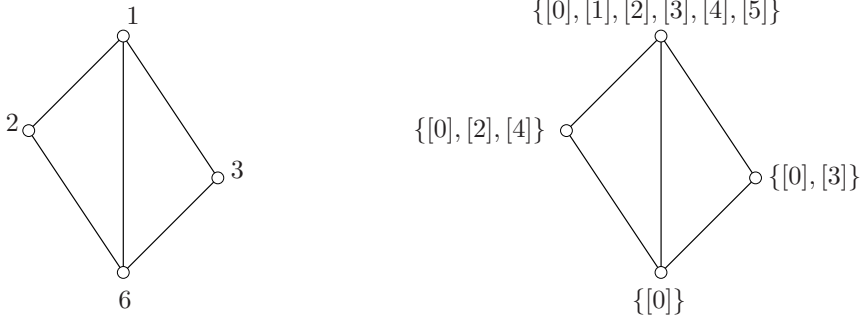
As a consequence of Proposition 6.11, there is a bijection between the set of subgroups of $\mathbb{Z}/n\mathbb{Z}$ and the set of positive divisors of n . For example, $\mathbb{Z}/12\mathbb{Z}$ has

²⁹I am secretly appealing to the ‘well-ordering principle’. That every set of positive integers should have a smallest element is one of those facts about $\mathbb{Z}$ —like the availability of division-with-remainder—that I am assuming the reader is already familiar with.

exactly 6 subgroups, because 12 has 6 positive divisors: 1, 2, 3, 4, 6, and 12. Here is the corresponding list of subgroups:

$$\begin{aligned}\langle [1]_{12} \rangle &= \{[0]_{12}, [1]_{12}, [2]_{12}, [3]_{12}, [4]_{12}, [5]_{12}, [6]_{12}, [7]_{12}, [8]_{12}, [9]_{12}, [10]_{12}, [11]_{12}\}, \\ \langle [2]_{12} \rangle &= \{[0]_{12}, [2]_{12}, [4]_{12}, [6]_{12}, [8]_{12}, [10]_{12}\}, \\ \langle [3]_{12} \rangle &= \{[0]_{12}, [3]_{12}, [6]_{12}, [9]_{12}\}, \\ \langle [4]_{12} \rangle &= \{[0]_{12}, [4]_{12}, [8]_{12}\}, \\ \langle [6]_{12} \rangle &= \{[0]_{12}, [6]_{12}\}, \\ \langle [12]_{12} \rangle &= \{[0]_{12}\}.\end{aligned}$$

Also note that if d_1, d_2 are both divisors of n , and $d_1 \mid d_2$, then $\langle [d_1]_n \rangle \supseteq \langle [d_2]_n \rangle$. That is, the correspondence between subgroups of $\mathbb{Z}/n\mathbb{Z}$ and divisors of n preserves the natural *lattice* structure carried by these sets. We can draw these lattices for $\mathbb{Z}/6\mathbb{Z}$ as follows:



where lines connect multiples in one picture and subsets in the other. The reader will draw the lattice of subgroups of S_3 , noting that it looks completely different from the one for $\mathbb{Z}/6\mathbb{Z}$.

Contemplating subgroups of cyclic groups has pretty (and useful) ‘number-theoretic’ consequences; cf. Exercise 6.14.

6.5. Monomorphisms. I end this section with some categorical considerations.

If H is a subgroup of G , the inclusion $H \hookrightarrow G$ is an example of a *monomorphism* in $\mathbf{Grp}$ in the ‘categorical’ sense of §I.4.2. In fact, it is easy to characterize all monomorphisms $\varphi \in \mathbf{Hom}_{\mathbf{Grp}}(G, G')$ (where G, G' are any groups):

Proposition 6.12. *The following are equivalent:*

- (a) φ is a monomorphism;
- (b) $\ker \varphi = \{e_G\}$;
- (c) $\varphi : G \rightarrow G'$ is injective (as a set-function).

Proof. (a) $\implies$ (b): Assume (a) holds, and consider the two parallel compositions

$$\ker \varphi \xrightarrow[e]{i} G \xrightarrow{\varphi} G',$$

where i is the inclusion and e is the trivial map. Both $\varphi \circ i$ and $\varphi \circ e$ are the trivial map; since φ is a monomorphism, this implies $i = e$. But $i = e$ implies that $\ker \varphi$ is trivial, that is, (b) holds.

(b) $\implies$ (c): Assume $\ker \varphi = \{e_G\}$. Then

$$\begin{aligned} \varphi(g_1) = \varphi(g_2) &\implies \varphi(g_1)\varphi(g_2)^{-1} = e_{G'} \implies \varphi(g_1g_2^{-1}) = e_{G'} \\ &\implies g_1g_2^{-1} \in \ker \varphi \implies g_1g_2^{-1} = e_G \implies g_1 = g_2. \end{aligned}$$

This shows that φ is injective, as needed.

(c) $\implies$ (a): If φ is injective, then it satisfies the defining property for monomorphisms in **Set**: that is, for any set Z and any two set-functions $\alpha', \alpha'' : Z \rightarrow G$,

$$\varphi \circ \alpha' = \varphi \circ \alpha'' \iff \alpha' = \alpha''.$$

This must hold in particular if Z has a group structure and α', α'' are group homomorphisms, so φ is a monomorphism in **Grp**. $\square$

The equivalence (a) $\iff$ (c) may lead the reader to think that from the point of view of monomorphisms, **Grp** and **Set** are pretty much alike. This is not quite so: while it is true that homomorphisms with a left-inverse are necessarily monomorphisms, as in **Set** (cf. Exercise 6.15), the converse is not true in **Grp** (cf. Exercise 6.16).

Exercises

6.1. $\neg$ (If you know about matrices.) The group of invertible $n \times n$ matrices with entries in $\mathbb{R}$ is denoted $\mathrm{GL}_n(\mathbb{R})$ (Example 1.5). Similarly, $\mathrm{GL}_n(\mathbb{C})$ denotes the group of $n \times n$ invertible matrices with *complex* entries. Consider the following sets of matrices:

- $\mathrm{SL}_n(\mathbb{R}) = \{M \in \mathrm{GL}_n(\mathbb{R}) \mid \det(M) = 1\}$;
- $\mathrm{SL}_n(\mathbb{C}) = \{M \in \mathrm{GL}_n(\mathbb{C}) \mid \det(M) = 1\}$;
- $\mathrm{O}_n(\mathbb{R}) = \{M \in \mathrm{GL}_n(\mathbb{R}) \mid MM^t = M^tM = I_n\}$;
- $\mathrm{SO}_n(\mathbb{R}) = \{M \in \mathrm{O}_n(\mathbb{R}) \mid \det(M) = 1\}$;
- $\mathrm{U}(n) = \{M \in \mathrm{GL}_n(\mathbb{C}) \mid MM^\dagger = M^\dagger M = I_n\}$;
- $\mathrm{SU}(n) = \{M \in \mathrm{U}(n) \mid \det(M) = 1\}$.

Here I_n stands for the $n \times n$ *identity matrix*, M^t is the *transpose* of M , $M^\dagger$ is the *conjugate transpose* of M , and $\det(M)$ denotes the *determinant*³⁰ of M . Find all possible inclusions among these sets, and prove that in every case the smaller set is a subgroup of the larger one.

These sets of matrices have compelling geometric interpretations: for example, $\mathrm{SO}_3(\mathbb{R})$ is the group of ‘rotations’ in $\mathbb{R}^3$. [8.8, 9.1, III.1.4, VI.6.16]

³⁰If you are not familiar with some of these notions, that’s ok: leave this exercise and similar ones alone if that is the case. We will come back to linear algebra and matrices in Chapter VI and following.

6.2. $\neg$ Prove that the set of 2×2 matrices

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$$

with a, b, d in $\mathbb{C}$ and $ad \neq 0$ is a subgroup of $\text{GL}_2(\mathbb{C})$. More generally, prove that the set of $n \times n$ complex matrices $(a_{ij})_{1 \leq i, j \leq n}$ with $a_{ij} = 0$ for $i > j$ and $a_{11} \cdots a_{nn} \neq 0$ is a subgroup of $\text{GL}_n(\mathbb{C})$. (These matrices are called ‘upper triangular’, for evident reasons.) [IV.1.20]

6.3. $\neg$ Prove that every matrix in $\text{SU}(2)$ may be written in the form

$$\begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix}$$

where $a, b, c, d \in \mathbb{R}$ and $a^2 + b^2 + c^2 + d^2 = 1$. (Thus, $\text{SU}(2)$ may be realized as a three-dimensional sphere embedded in $\mathbb{R}^4$; in particular, it is *simply connected*.) [8.9, III.2.5]

6.4. $\triangleright$ Let G be a group, and let $g \in G$. Verify that the image of the exponential map $\epsilon_g : \mathbb{Z} \rightarrow G$ is a cyclic group (in the sense of Definition 4.7). [§6.3, §7.5]

6.5. Let G be a *commutative* group, and let $n > 0$ be an integer. Prove that $\{g^n \mid g \in G\}$ is a subgroup of G . Prove that this is not necessarily the case if G is not commutative.

6.6. Prove that the union of a family of subgroups of a group G is not necessarily a subgroup of G . In fact:

- Let H, H' be subgroups of a group G . Prove that $H \cup H'$ is a subgroup of G only if $H \subseteq H'$ or $H' \subseteq H$.
- On the other hand, let $H_0 \subseteq H_1 \subseteq H_2 \subseteq \cdots$ be subgroups of a group G . Prove that $\bigcup_{i \geq 0} H_i$ is a subgroup of G .

6.7. $\neg$ Show that *inner* automorphisms (cf. Exercise 4.8) form a subgroup of $\text{Aut}(G)$; this subgroup is denoted $\text{Inn}(G)$. Prove that $\text{Inn}(G)$ is cyclic if and only if $\text{Inn}(G)$ is trivial if and only if G is abelian. (Hint: Assume that $\text{Inn}(G)$ is cyclic; with notation as in Exercise 4.8, this means that there exists an element $a \in G$ such that $\forall g \in G \exists n \in \mathbb{Z} \gamma_g = \gamma_a^n$. In particular, $gag^{-1} = a^n aa^{-n} = a$. Thus a commutes with every g in G . Therefore....) Deduce that if $\text{Aut}(G)$ is cyclic, then G is abelian. [7.10, IV.1.5]

6.8. Prove that an *abelian* group G is finitely generated if and only if there is a surjective homomorphism

$$\underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{n \text{ times}} \rightarrow G$$

for some n .

6.9. Prove that every finitely generated subgroup of $\mathbb{Q}$ is cyclic. Prove that $\mathbb{Q}$ is not finitely generated.

6.10. $\neg$ The set of 2×2 matrices with integer entries and determinant 1 is denoted $\mathrm{SL}_2(\mathbb{Z})$:

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ such that } a, b, c, d \in \mathbb{Z}, ad - bc = 1 \right\}.$$

Prove that $\mathrm{SL}_2(\mathbb{Z})$ is generated by the matrices

$$s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad t = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(Hint: This is a little tricky. Let H be the subgroup generated by s and t . Given a matrix $m = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $\mathrm{SL}_2(\mathbb{Z})$, it suffices to show that you can obtain the identity

by multiplying m by suitably chosen elements of H . Prove that $\begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ -q & 1 \end{pmatrix}$ are in H , and note that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b - qa \\ c & d - qc \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -q & 1 \end{pmatrix} = \begin{pmatrix} a - qb & b \\ c - qd & d \end{pmatrix}.$$

Note that if c and d are both nonzero, one of these two operations may be used to decrease the absolute value of one of them. Argue that suitable applications of these operations reduce to the case in which $c = 0$ or $d = 0$. Prove directly that $m \in H$ in that case.) [7.5]

6.11. Since direct sums are coproducts in $\mathbf{Ab}$, the classification theorem for abelian groups mentioned in the text says that every finitely generated *abelian group* is a coproduct of cyclic groups in $\mathbf{Ab}$. The reader may be tempted to conjecture that every finitely generated *group* is a coproduct in $\mathbf{Grp}$. Show that this is not the case, by proving that S_3 is not a coproduct of cyclic groups.

6.12. Let m, n be positive integers, and consider the subgroup $\langle m, n \rangle$ of $\mathbb{Z}$ they generate. By Proposition 6.9,

$$\langle m, n \rangle = d\mathbb{Z}$$

for some positive integer d . What is d , in relation to m, n ?

6.13. $\neg$ Draw and compare the lattices of subgroups of $C_2 \times C_2$ and C_4 . Draw the lattice of subgroups of S_3 , and compare it with the one for C_6 . [7.1]

6.14. $\triangleright$ If m is a positive integer, denote by $\phi(m)$ the number of positive integers $r \leq m$ that are *relatively prime* to m (that is, for which the gcd of r and m is 1); this is called *Euler's ϕ - (or 'totient') function*. For example, $\phi(12) = 4$. In other words, $\phi(m)$ is the order of the group $(\mathbb{Z}/m\mathbb{Z})^*$; cf. Proposition 2.6.

Put together the following observations:

- $\phi(m)$ = the number of generators of C_m ,
- every element of C_n generates a subgroup of C_n ,
- the discussion following Proposition 6.11 (in particular, every subgroup of C_n is isomorphic to C_m , for some $m \mid n$),

to obtain a proof of the formula

$$\sum_{m>0, m|n} \phi(m) = n.$$

(For example, $\phi(1) + \phi(2) + \phi(3) + \phi(4) + \phi(6) + \phi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12$.)
[4.14, §6.4, 8.15, V.6.8, §VII.5.2]

6.15. ▷ Prove that if a group homomorphism $\varphi : G \rightarrow G'$ has a left-inverse, that is, a group homomorphism $\psi : G' \rightarrow G$ such that $\psi \circ \varphi = \text{id}_G$, then φ is a monomorphism. [§6.5, 6.16]

6.16. ▷ Counterpoint to Exercise 6.15: the homomorphism $\varphi : \mathbb{Z}/3\mathbb{Z} \rightarrow S_3$ given by

$$\varphi([0]) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad \varphi([1]) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \quad \varphi([2]) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

is a monomorphism; show that it has *no* left-inverse in **Grp**. (Knowing about *normal* subgroups will make this problem particularly easy.) [§6.5]

7. Quotient groups

7.1. Normal subgroups. Before tackling ‘quotient groups’, I should clarify in what sense kernels are *special* subgroups, as claimed in §6.2.

Definition 7.1. A subgroup N of a group G is *normal* if $\forall g \in G, \forall n \in N$,

$$gng^{-1} \in N. \quad \lrcorner$$

Note that *every* subgroup of a commutative group is normal (because then $\forall g \in G, gng^{-1} = n \in N$). However, in general *not all* subgroups are normal: examples may be found already in S_3 (cf. Exercise 7.1). There exist noncommutative groups in which every subgroup is normal (one example is the ‘quaternionic group’ Q_8 ; cf. Exercise III.1.12 (iv)), but they are very rare.

Lemma 7.2. *If $\varphi : G \rightarrow G'$ is any group homomorphism, then $\ker \varphi$ is a normal subgroup of G .*

Proof. We already know that $\ker \varphi$ is a subgroup of G ; to verify it is normal note that $\forall g \in G, \forall n \in \ker \varphi$

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g^{-1}) = \varphi(g)e_{G'}\varphi(g)^{-1} = e_{G'},$$

proving that $gng^{-1} \in \ker \varphi$. □

Loosely speaking, therefore, *kernel* $\implies$ *normal*. In fact more is true, as we will see in a little while; for now I don’t want to spoil the surprise for the reader. (Can the reader guess?)

There is a convenient shorthand to express conditions such as normality: if $g \in G$ and $A \subseteq G$ is any subset, we denote by gA , Ag , respectively, the following subsets of G :

$$gA := \{h \in G \mid (\exists a \in A) : h = ga\},$$

$$Ag := \{h \in G \mid (\exists a \in A) : h = ag\}.$$

Then the normality condition can be expressed by

$$(\forall g \in G) : \quad gNg^{-1} \subseteq N,$$

or in a number of other ways:

$$gNg^{-1} = N \quad \text{or} \quad gN \subseteq Ng \quad \text{or} \quad gN = Ng$$

for all $g \in G$. The reader should check that these are indeed equivalent conditions (Exercise 7.3) and keep in mind that ‘ $gN = Ng$ ’ does *not* mean that g commutes with every element of N ; it means that if $n \in N$, then there are elements $n', n'' \in N$, in general different from n , such that $gn = n'g$ (so that $gN \subseteq Ng$) and $ng = gn''$ (so that $Ng \subseteq gN$).

7.2. Quotient group. Recall that we have the notion of a quotient of a *set* by an equivalence relation (§I.1.5) and that this notion satisfies a universal property (clumsily stated in §I.5.3). It is natural to investigate this notion in **Grp**.

We consider then an equivalence relation $\sim$ on (the set underlying) a group G ; we seek a group $G/\sim$ and a group homomorphism $\pi : G \rightarrow G/\sim$ satisfying the appropriate universal property, that is, initial with respect to group homomorphisms $\varphi : G \rightarrow G'$ such that $a \sim b \implies \varphi(a) = \varphi(b)$.

It is natural to try to construct the *group* $G/\sim$ by defining an operation $\bullet$ on the *set* $G/\sim$. The situation is tightly constrained by the requirement that the quotient map $\pi : G \rightarrow G/\sim$ (as in §I.2.6) be a group homomorphism: for if $[a] = \pi(a)$, $[b] = \pi(b)$ are elements of $G/\sim$ (that is, equivalence classes with respect to $\sim$), then the homomorphism condition *forces*

$$[a] \bullet [b] = \pi(a) \bullet \pi(b) = \pi(ab) = [ab].$$

But is this operation well-defined? This amounts to conditions on the equivalence relation, which we proceed to unearth.

For the operation to be well-defined ‘in the first factor’, it is necessary that if $[a] = [a']$, then $[ab] = [a'b]$ regardless of what b is; that is,

$$(\forall g \in G) : \quad a \sim a' \implies ag \sim a'g.$$

Similarly, for the operation to be well-defined in the second factor we need

$$(\forall g \in G) : \quad a \sim a' \implies ga \sim ga'.$$

Luckily, this is all that there is to it:

Proposition 7.3. *With notation as above, the operation*

$$[a] \bullet [b] := [ab]$$

defines a group structure on $G/\sim$ if and only if $\forall a, a', g \in G$

$$a \sim a' \implies ga \sim ga' \text{ and } ag \sim a'g.$$

In this case the quotient function $\pi : G \rightarrow G/\sim$ is a homomorphism and is universal with respect to homomorphisms $\varphi : G \rightarrow G'$ such that $a \sim a' \implies \varphi(a) = \varphi(a')$.

Proof. We have already noted that the condition is necessary. To prove it is sufficient, with the stated consequences, assume $\forall a, a', g \in G$

$$a \sim a' \implies ga \sim ga' \text{ and } ag \sim a'g.$$

Then the operation

$$[a] \bullet [b] := [ab]$$

is well-defined, and we have to verify that it defines a group structure on $G/\sim$. The associativity of $\bullet$ is inherited from the associativity of G : $\forall a, b, c \in G$

$$([a] \bullet [b]) \bullet [c] = [ab] \bullet [c] = [(ab)c] = [a(bc)] = [a] \bullet [bc] = [a] \bullet ([b] \bullet [c]).$$

The class $[e_G]$ is an identity with respect to this operation: $\forall g \in G$

$$[g] \bullet [e_G] = [ge_G] = [g], \quad [e_G] \bullet [g] = [e_Gg] = [g].$$

The class $[g^{-1}]$ is the inverse of $[g]$:

$$[g^{-1}] \bullet [g] = [g^{-1}g] = [e_G], \quad [g] \bullet [g^{-1}] = [gg^{-1}] = [e_G].$$

This shows $G/\sim$ is indeed a group, and we have already observed that $\pi : G \rightarrow G/\sim$ is a homomorphism: this is what led us to the definition of $\bullet$.

To prove that $G/\sim$ satisfies the universal property, assume

$$\varphi : G \rightarrow G'$$

is a group homomorphism such that $a \sim a' \implies \varphi(a) = \varphi(a')$. Since (cf. §I.5.3) the set $G/\sim$ satisfies the corresponding universal property in **Set**, we know that there exists a unique set-function

$$\tilde{\varphi} : G/\sim \rightarrow G',$$

defined³¹ by $\tilde{\varphi}([a]) := \varphi(a)$. So we only need to check that this function $\tilde{\varphi}$ is in fact a group homomorphism, and this is immediate:

$$\tilde{\varphi}([a] \bullet [b]) = \tilde{\varphi}([ab]) = \varphi(ab) = \varphi(a)\varphi(b) = \tilde{\varphi}([a])\tilde{\varphi}([b])$$

for all $[a], [b] \in G/\sim$, as needed. $\square$

I will say that $\sim$ is *compatible with the group structure* of G if the condition given in Proposition 7.3 holds. Since the operation $\bullet$ on the quotient $G/\sim$ is uniquely determined by the operation on G , I yield to the usual abuse of language and omit it. If $\sim$ is compatible, I will call $G/\sim$ the *quotient group* of G by $\sim$.

7.3. Cosets. The conditions obtained in Proposition 7.3,

$$(\dagger) \quad (\forall g \in G) : \quad a \sim b \implies ga \sim gb,$$

$$(\dagger\dagger) \quad (\forall g \in G) : \quad a \sim b \implies ag \sim bg,$$

lead to a complete description of *all* compatible relations on a group G . In fact, each of these two conditions leads to a description of the relations satisfying it, and we will analyze them separately; the reader should keep in mind that we have a group structure on $G/\sim$ only if *both* are satisfied.

Let's begin with $(\dagger)$. Here is the description:

³¹The point of §I.5.3 is precisely that this function is well-defined.

Proposition 7.4. *Let $\sim$ be an equivalence relation on a group G , satisfying $(\dagger)$. Then*

- *the equivalence class of e_G is a subgroup H of G ; and*
- *$a \sim b \iff a^{-1}b \in H \iff aH = bH$.*

Proof. Let $H \subseteq G$ be the equivalence class of the identity; $H \neq \emptyset$ as $e_G \in H$. For $a, b \in H$, we have $e_G \sim b$ and hence $b^{-1} \sim e_G$ (applying $(\dagger)$, multiplying on the left by b^{-1}); hence $ab^{-1} \sim a$ (by $(\dagger)$ again, multiplying on the left by a); and hence

$$ab^{-1} \sim a \sim e_G$$

by the transitivity of $\sim$ and since $a \in H$. This shows that $ab^{-1} \in H$ for all $a, b \in H$, proving that H is a subgroup (by Proposition 6.2).

Next, assume $a, b \in G$ and $a \sim b$. Multiplying on the left by a^{-1} , $(\dagger)$ implies $e_G \sim a^{-1}b$, that is, $a^{-1}b \in H$. Since H is closed under the operation, this implies $a^{-1}bH \subseteq H$, hence $bH \subseteq aH$; as $\sim$ is symmetric, the same reasoning gives $aH \subseteq bH$; and hence $aH = bH$. Thus, we have proved

$$a \sim b \implies a^{-1}b \in H \implies aH = bH.$$

Finally, assume $aH = bH$. Then $a = ae_G \in bH$, and hence $a^{-1}b \in H$. By definition of H , this means $e_G \sim a^{-1}b$. Multiplying on the left by a shows (by $(\dagger)$ again) that $a \sim b$, completing the proof. $\square$

Proposition 7.4 shows that the equivalence classes of an equivalence relation satisfying $(\dagger)$ are in fact all of the form

$$aH$$

for a fixed subgroup H , as a ranges in G . These important subsets determined by a subgroup H deserve a name.

Definition 7.5. The *left-cosets* of a subgroup H in a group G are the sets aH , for $a \in G$. The *right-cosets* of H are the sets Ha , $a \in G$. $\lrcorner$

Now, a ‘converse’ to Proposition 7.4 holds:

Proposition 7.6. *If H is any subgroup of a group G , the relation $\sim_L$ defined by*

$$(\forall a, b \in G) : \quad a \sim_L b \iff a^{-1}b \in H$$

is an equivalence relation satisfying $(\dagger)$.

Proof. This is straightforward and is mostly left to the reader (Exercise 7.8). To see that the relation satisfies $(\dagger)$, note that

$$a \sim_L b \implies a^{-1}b \in H \implies a^{-1}(g^{-1}g)b \in H \implies (ga)^{-1}(gb) \in H \implies ga \sim_L gb$$

for all $g \in G$. $\square$

Taken together, Propositions 7.4 and 7.6 show

Proposition 7.7. *There is a one-to-one correspondence between subgroups of G and equivalence relations on G satisfying $(\dagger)$; for the relation $\sim_L$ corresponding to a subgroup H , $G/\sim_L$ may be described as the set of left-cosets aH of H .*

The reader should have no difficulty producing the mirror statements (and proofs) giving a similarly exhaustive description of all equivalence relations satisfying $(\dagger\dagger)$. The end result will be

Proposition 7.8. *There is a one-to-one correspondence between subgroups of G and equivalence relations on G satisfying $(\dagger\dagger)$; for the relation $\sim_R$ corresponding to a subgroup H , $G/\sim_R$ may be described as the set of right-cosets Ha of H .*

The relation corresponding to H in this *second* way is defined by

$$a \sim_R b \iff ab^{-1} \in H \iff Ha = Hb.$$

What may be surprising at first is that the relations $\sim_L$ and $\sim_R$ corresponding to the *same* subgroup H may very well *not* be the same relation. That is, left-cosets and right-cosets of a subgroup need not coincide. Of course $eH = He = H$, and more generally

$$(\forall h \in H) : \quad hH = Hh = H.$$

Further

$$(\forall a \in G) : \quad a \in aH \cap Ha;$$

hence, if $aH = Hb$ for any b , then in fact necessarily $aH = Ha$. This is of course automatically true if G is commutative, but it is simply not the case in general.

Example 7.9. Let $G = S_3$, and let H be the subgroup consisting of the identity and the $1 \leftrightarrow 2$ switch:

$$H = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \right\}.$$

Then

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} H = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\},$$

while

$$H \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \right\}.$$

This state of affairs simply reflects the fact that the two conditions $(\dagger)$ and $(\dagger\dagger)$ are different: there is no reason to expect that if one holds, the other one should also hold (unless G is commutative, of course). Once more, keep in mind that *both* have to hold for the quotient $G/\sim$ to be a group, compatibly with the operation in G ; cf. Proposition 7.3.

7.4. Quotient by normal subgroups. To stress the main point again, an arbitrary subgroup H of G leads to two partitions of G , which we have denoted

$$G/\sim_L = \{aH \mid a \in G\}, \quad G/\sim_R = \{Ha \mid a \in G\}.$$

The relation $\sim_L$ satisfies property $(\dagger)$ listed at the beginning of §7.3; $\sim_R$ satisfies $(\dagger\dagger)$. A priori, these relations, and hence the corresponding partitions, are different.

The condition that $\sim_L$ and $\sim_R$ *coincide* (as is necessarily the case, for example, if G is commutative) translates into a condition on H : for such ‘special’ subgroups, $(\dagger)$ and $(\dagger\dagger)$ get back together. The good news is that this condition is easy to identify and is not new to the reader.

Proposition 7.10. *The relations $\sim_L, \sim_R$ corresponding to a subgroup H coincide if and only if H is normal.*

Proof. Two relations coincide if the corresponding partitions agree. Therefore

$$\sim_L = \sim_R \iff \text{left- and right-cosets of } H \text{ coincide} \iff (\forall g \in G) : gH = Hg.$$

But this is one of the equivalent conditions defining the notion of normal subgroup (cf §7.1), proving the statement. $\square$

The innocent-looking Proposition 7.10 is of fundamental importance. If H is normal, then the *one* equivalence relation $\sim = \sim_L = \sim_R$ corresponding to H satisfies both $(\dagger)$ and $(\dagger\dagger)$ (by Proposition 7.4 and its mirror statement), and hence (by Proposition 7.3) the quotient

$$G/\sim = \{aH \mid a \in G\} = \{Ha \mid a \in G\}$$

has a natural group structure.

Definition 7.11. Let H be a normal subgroup of a group G . The *quotient group of G modulo H* , denoted³² G/H , is the group $G/\sim$ obtained from the relation $\sim$ defined above. In terms of (left-) cosets, the product in G/H is defined by

$$(aH)(bH) := (ab)H.$$

The identity element $e_{G/H}$ of the quotient group G/H is the coset of the identity, $e_G H = H$. $\lrcorner$

By Proposition 7.3, the quotient function

$$\pi : G \rightarrow G/H$$

sending $g \in G$ to $gH = Hg$ is a group homomorphism and is universal with respect to group homomorphisms $\varphi : G \rightarrow G'$ such that $aH = bH \implies \varphi(a) = \varphi(b)$. This universal property is extremely useful, so I will grace it with theorem status:

Theorem 7.12. *Let H be a normal subgroup of a group G . Then for every group homomorphism $\varphi : G \rightarrow G'$ such that $H \subseteq \ker \varphi$ there exists a unique group homomorphism $\tilde{\varphi} : G/H \rightarrow G'$ so that the diagram*

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & G' \\ & \searrow \pi & \nearrow \exists! \tilde{\varphi} \\ & G/H & \end{array}$$

commutes.

Proof. We only need to match the stated universal property with the one we proved in Proposition 7.3, and indeed,

$$H \subseteq \ker \varphi \iff (\forall h \in H) : \varphi(h) = e_{G'}$$

is equivalent to

$$(\forall a, b \in G) : ab^{-1} \in H \implies \varphi(ab^{-1}) = e_{G'}$$

³²In a large display I sometime use the full ‘fraction’ notation $\frac{G}{H}$.

that is, to

$$(\forall a, b \in G) : ab^{-1} \in H \implies \varphi(a) = \varphi(b)$$

and finally, keeping in mind how the relation $\sim$ corresponding to H is defined,

$$(\forall a, b \in G) : a \sim b \implies \varphi(a) = \varphi(b),$$

the condition giving the universal property in Proposition 7.3. $\square$

7.5. Example. The reader is already very familiar with an important class of examples: the cyclic groups $\mathbb{Z}/n\mathbb{Z}$. Indeed, in §2.3 we defined $\mathbb{Z}/n\mathbb{Z}$ as the set of equivalence classes in $\mathbb{Z}$ with respect to the congruence equivalence relation

$$(\forall a, b \in \mathbb{Z}) : a \equiv b \pmod{n} \iff n \mid (b - a).$$

Now we recognize that $n \mid (b - a)$ is equivalent to

$$b - a \in n\mathbb{Z},$$

which is the relation $\sim_L$ corresponding (in ‘abelian’ notation) to the subgroup $n\mathbb{Z}$ of $\mathbb{Z}$. This subgroup is of course normal, since $\mathbb{Z}$ is abelian. The ‘congruence classes mod n ’ are nothing but the cosets of the subgroup $n\mathbb{Z}$ in $\mathbb{Z}$; using abelian notation for cosets, we could write

$$[a]_n = a + (n\mathbb{Z}).$$

Of course the operation defined on $\mathbb{Z}/n\mathbb{Z}$ in §2.3 matches precisely the one defined above for quotient groups. This justifies the notation $\mathbb{Z}/n\mathbb{Z}$ introduced in §2.3.

The reader can already appreciate in this simple context the usefulness of Theorem 7.12. Let $g \in G$ be an element of order n and consider the exponential map

$$\epsilon_g : \mathbb{Z} \rightarrow G, \quad N \mapsto g^N.$$

By Corollary 1.11,

$$\ker \epsilon_g = \{N \in \mathbb{Z} \mid N \text{ is a multiple of } |g|\} = n\mathbb{Z}.$$

Theorem 7.12 then implies right away that ϵ_g factors through the quotient:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\epsilon_g} & G \\ & \searrow \pi_n & \nearrow \exists! \tilde{\epsilon}_g \\ & \mathbb{Z}/n\mathbb{Z} & \end{array}$$

That is, there is an induced map

$$\mathbb{Z}/n\mathbb{Z} \rightarrow \langle g \rangle.$$

In fact, the ‘canonical decomposition’ of §I.2.8 implies that this is an isomorphism (verifying that $\langle g \rangle$ is cyclic in the sense of Definition 4.7, as the reader should have checked ‘by hand’ already in Exercise 6.4). We will formalize this observation in general in the next section.

Also note that $|g| = n = |\langle g \rangle|$ in this case.

7.6. kernel $\iff$ normal. If H is a *normal* subgroup, we have now constructed in gory detail a group G/H and a surjective homomorphism

$$\pi : G \rightarrow G/H.$$

What is the kernel of π ? The identity of G/H is the coset $e_G H$, that is, H itself. Therefore

$$\ker \pi = \{g \in G \mid gH = H\} = H.$$

This observation completes the circle of ideas begun in §7.1: there we had noticed that every kernel (of a group homomorphism) is a *normal* subgroup; and now we have verified that every normal subgroup is in fact a *kernel* (of some group homomorphism). I encapsulate this in the slogan

$$\text{kernel} \iff \text{normal} :$$

in group theory³³, ‘kernel’ and ‘normal subgroup’ are equivalent concepts.

For example, every subgroup in an *abelian* group is the kernel of some homomorphism: yet another indication that life is simpler in **Ab** than in **Grp**.

Exercises

7.1. $\triangleright$ List all subgroups of S_3 (cf. Exercise 6.13) and determine which subgroups are normal and which are not normal. [§7.1]

7.2. Is the *image* of a group homomorphism necessarily a *normal* subgroup of the target?

7.3. $\triangleright$ Verify that the equivalent conditions for normality given in §7.1 are indeed equivalent. [§7.1]

7.4. Prove that the relation defined in Exercise 5.10 on a free abelian group $F = F^{ab}(A)$ is compatible with the group structure. Determine the quotient $F/\sim$ as a better known group.

7.5. $\neg$ Define an equivalence relation $\sim$ on $\text{SL}_2(\mathbb{Z})$ by letting $A \sim A' \iff A' = \pm A$. Prove that $\sim$ is compatible with the group structure. The quotient $\text{SL}_2(\mathbb{Z})/\sim$ is denoted $\text{PSL}_2(\mathbb{Z})$ and is called the *modular group*; it would be a serious contender in a contest for ‘the most important group in mathematics’, due to its role in algebraic geometry and number theory. Prove that $\text{PSL}_2(\mathbb{Z})$ is generated by the (cosets of the) matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

(You will not need to work very hard, if you use the result of Exercise 6.10.) Note that the first has order 2 in $\text{PSL}_2(\mathbb{Z})$, the second has order 3, and their product has infinite order. [9.14]

³³We will run into analogous observations in *ring* theory, where we will verify that kernels and *ideals* coincide, and for *modules*, as kernels and submodules again coincide.

7.6. Let G be a group, and let n be a positive integer. Consider the relation

$$a \sim b \iff (\exists g \in G) ab^{-1} = g^n.$$

- Show that in general $\sim$ is *not* an equivalence relation.
- Prove that $\sim$ is an equivalence relation if G is commutative, and determine the corresponding subgroup of G .

7.7. Let G be a group, n a positive integer, and let $H \subseteq G$ be the subgroup generated by all elements of order n in G . Prove that H is normal.

7.8. $\triangleright$ Prove Proposition 7.6. [§7.3]

7.9. State and prove the ‘mirror’ statements of Propositions 7.4 and 7.6, leading to the description of relations satisfying $(\dagger\dagger)$.

7.10. $\neg$ Let G be a group, and $H \subseteq G$ a subgroup. With notation as in Exercise 6.7, show that H is normal in G if and only if $\forall \gamma \in \text{Inn}(G), \gamma(H) \subseteq H$.

Conclude that if H is normal in G , then there is an interesting homomorphism $\text{Inn}(G) \rightarrow \text{Aut}(H)$. [8.25]

7.11. $\triangleright$ Let G be a group, and let $[G, G]$ be the subgroup of G generated by all elements of the form $aba^{-1}b^{-1}$. (This is the *commutator* subgroup of G ; we will return to it in §IV.3.3.) Prove that $[G, G]$ is normal in G . (Hint: With notation as in Exercise 4.8, $g \cdot aba^{-1}b^{-1} \cdot g^{-1} = \gamma_g(aba^{-1}b^{-1})$.) Prove that $G/[G, G]$ is commutative. [7.12, §IV.3.3]

7.12. $\triangleright$ Let $F = F(A)$ be a free group, and let $f : A \rightarrow G$ be a set-function from the set A to a *commutative* group G . Prove that f induces a unique homomorphism $F/[F, F] \rightarrow G$, where $[F, F]$ is the commutator subgroup of F defined in Exercise 7.11. (Use Theorem 7.12.) Conclude that $F/[F, F] \cong F^{ab}(A)$. (Use Proposition I.5.4.) [§6.4, 7.13, VI.1.20]

7.13. $\neg$ Let A, B be sets and $F(A), F(B)$ the corresponding free groups. Assume $F(A) \cong F(B)$. If A is finite, prove that B is also and $A \cong B$. (Use Exercise 7.12 to upgrade Exercise 5.10.) [5.10, VI.1.20]

7.14. Let G be a group. Prove that $\text{Inn}(G)$ is a *normal* subgroup of $\text{Aut}(G)$.

8. Canonical decomposition and Lagrange’s theorem

I will collect in this section a number of observations on the structure of quotient groups. All these results are straightforward, given the background work done so far. Some of them are often given fancy names such as *first isomorphism theorem* in the literature; I am not too fond of such terminology: the universal property proven in Theorem 7.12 is really the only thing I need to take along, and it serves me wonderfully well. The ‘isomorphism theorems’ are all immediate applications of this universal property.

8.1. Canonical decomposition. The first observation comes from the *canonical decomposition* for set-functions, obtained in §I.2.8: every set-functions may be viewed as the composition of a surjective map, followed by a bijective map, followed by an injective map. We now know enough to state the corresponding (very useful) results in Grp:

Theorem 8.1. *Every group homomorphism $\varphi : G \rightarrow G'$ may be decomposed as follows:*

$$G \xrightarrow{\quad \varphi \quad} G' \\ G \twoheadrightarrow G/\ker \varphi \xrightarrow[\tilde{\varphi}]{\sim} \operatorname{im} \varphi \hookrightarrow G'$$

where the isomorphism $\tilde{\varphi}$ in the middle is the homomorphism induced by φ (as in Theorem 7.12).

It is important that the reader agree that we have already proved anything that deserves to be proven here. We know that the projection on the left and the inclusion on the right are homomorphisms and $\tilde{\varphi}$ comes from Theorem 7.12. The decomposition is the same one obtained at the level of set-functions in §I.2.8; in particular, the function in the middle is a bijection. Since bijective homomorphisms are isomorphisms (Proposition 4.3), it is an isomorphism.

Theorem 8.1 should induce the following Pavlovian reaction: exposed to any group homomorphism $\varphi : G \rightarrow G'$, the reader should instantaneously view $G/\ker \varphi$ as (canonically identified with) a subgroup of G' . What is usually called the ‘*first isomorphism theorem*’ is the particular case corresponding to surjective homomorphisms:

Corollary 8.2. *Suppose $\varphi : G \rightarrow G'$ is a surjective group homomorphism. Then*

$$G' \cong \frac{G}{\ker \varphi}.$$

Proof. $\operatorname{im} \varphi = G'$ in Theorem 8.1. □

This result is very useful—it comes in extremely handy when proving that two groups are isomorphic, both in theoretical contexts (as we will see in the rest of this section) and in concrete instances.

Example 8.3. If $H_1 \subseteq G_1$ and $H_2 \subseteq G_2$, then the product $H_1 \times H_2$ may be viewed as a subset of $G_1 \times G_2$. It is clear that if G_1, G_2 are groups and H_1, H_2 are subgroups, then $H_1 \times H_2$ is a subgroup of $G_1 \times G_2$. The following claim is a prototype application of Corollary 8.2:

Claim 8.4. *If $H_1 \subseteq G_1$ and $H_2 \subseteq G_2$ are normal subgroups, then $H_1 \times H_2$ is a normal subgroup of the group $G_1 \times G_2$ and*

$$\frac{G_1 \times G_2}{H_1 \times H_2} \cong \frac{G_1}{H_1} \times \frac{G_2}{H_2}.$$

Indeed, composing the projections

$$\pi_1 : G_1 \times G_2 \rightarrow G_1, \quad \pi_2 : G_1 \times G_2 \rightarrow G_2$$

with the morphisms to the quotients gives surjective homomorphisms

$$\pi_1 : G_1 \times G_2 \rightarrow \frac{G_1}{H_1}, \quad \pi_2 : G_1 \times G_2 \rightarrow \frac{G_2}{H_2}$$

and hence a homomorphism

$$\pi : G_1 \times G_2 \rightarrow \frac{G_1}{H_1} \times \frac{G_2}{H_2}$$

by the universal property of products. Explicitly,

$$\pi(g_1, g_2) = (g_1 H_1, g_2 H_2) :$$

in particular, π is surjective and

$$\begin{aligned} \ker \pi &= \{(g_1, g_2) \in G_1 \times G_2 \mid (g_1 H_1, g_2 H_2) = (H_1, H_2)\} \\ &= \{(g_1, g_2) \in G_1 \times G_2 \mid g_1 \in H_1, g_2 \in H_2\} \\ &= H_1 \times H_2. \end{aligned}$$

The claim then follows immediately from Corollary 8.2.

The result (of course) extends to more factors in the product. Any such check should become second nature and is usually left to the reader. $\lrcorner$

Example 8.5. As a particular case of Claim 8.4, take $H_1 = \{e_{G_1}\} \subseteq G_1$ and $H_2 = G_2 \subseteq G_2$:

$$\frac{G_1 \times G_2}{G_2} \cong \frac{G_1}{\{e_{G_1}\}} \times \frac{G_2}{G_2} \cong G_1,$$

where on the left we identify G_2 with the subgroup $\{e_{G_1}\} \times G_2$. For instance³⁴ (cf. §4.1)

$$\frac{C_6}{C_3} \cong \frac{C_2 \times C_3}{C_3} \cong C_2. \quad \lrcorner$$

Example 8.6. The cyclic group C_3 may be viewed as a subgroup of the dihedral group D_6 : the rotations of a triangle give a copy of C_3 inside D_6 . Then C_3 is normal in D_6 , and

$$\frac{D_6}{C_3} \cong C_2.$$

This can of course be checked ‘by hand’. But note that there is an evident surjective homomorphism $D_6 \rightarrow C_2$, whose kernel is C_3 : map an element σ of D_6 to the identity in C_2 if it does *not* flip the triangle (that is, precisely when $\sigma \in C_3$), and map it to the other element if it does. Corollary 8.2 implies the stated facts immediately. $\lrcorner$

Example 8.7. One can give a *circle* (denoted S^1) a group structure by identifying its points with rotations of a plane about a point and adding them accordingly. The function

$$\rho : \mathbb{R}^1 \rightarrow S^1$$

³⁴Abuses of language such as the formula which follows—in which one is not explicitly specifying how to realize C_3 as a subgroup of C_6 , because there is really only one way to do it—aren’t unfortunately commonplace.

mapping a number r to the result of a rotation by $2\pi r$ radians is then a surjective group homomorphism; this is the identity precisely when we rotate by an *integer* multiple of 2π . Hence

$$\ker \rho = \mathbb{Z} \subseteq \mathbb{R}.$$

By Corollary 8.2, therefore,

$$\frac{\mathbb{R}}{\mathbb{Z}} \cong S^1.$$

(Cf. Exercise I.1.6.) Geometrically, this amounts to ‘wrapping’ $\mathbb{R}$ infinitely many times around the circle, realizing $\mathbb{R}$ as the ‘universal cover’ of S^1 ; here, $\mathbb{Z}$ plays the role of ‘fundamental group’ of S^1 . $\lrcorner$

8.2. Presentations. Every group is a quotient of a free group, and every abelian group is a quotient of a free abelian group. Indeed, every group G can be surjected upon by a free group, and in many ways (at the very least, $F(G)$ will do!). Abelian groups may be likewise surjected upon by free abelian groups. Then Corollary 8.2 produces the needed isomorphism of G with a quotient of a free group.

A *presentation* of a group G is an explicit isomorphism

$$G \cong \frac{F(A)}{R}$$

where A is a set and R is a subgroup of ‘relations’. In other words, a presentation is an explicit surjection

$$\rho : F(A) \twoheadrightarrow G$$

of which R is the kernel. This is especially useful if A is small, and R may be described very explicitly; usually this is done by listing ‘enough’ *relations*, that is, a set $\mathcal{R}$ of words $r_\alpha \in R = \ker \rho$ generating it in the sense that R is the smallest normal subgroup³⁵ of $F(A)$ containing $\mathcal{R}$.

Thus, a presentation of a group G is usually encoded as a pair $(A|\mathcal{R})$, where A is a set and $\mathcal{R} \subseteq F(A)$ is a set of words, such that $G \cong F(A)/R$ with R as above.

A group is *finitely presented* if it admits a presentation $(A|\mathcal{R})$ in which both A and $\mathcal{R}$ are finite. Finitely presented groups are not (necessarily) ‘small’: for example, the free group on finitely many generators is (trivially) finitely presented.

We have already run into several examples of presentations. For instance, the free group $F(A)$ is presented by $(A|\emptyset)$. More interestingly, the description of S_3 given in §2.1 ‘presents’ S_3 as a quotient of the free group $F(\{x, y\})$ (cf. Example 5.3) by the smallest normal subgroup containing x^2 , y^3 , and $yx = xy^2$: $(x, y|x^2, y^3, xyxy)$ in shorthand. From this point of view, it is clear that groups admitting the same presentation (example: S_3 and D_6) are isomorphic.

The situation is less idyllic than it may seem at first, though: even if a presentation of a group G is known, it may be very hard to establish whether two explicit

³⁵Note that this is a different requirement than the one adopted in §6.3, in which normality plays no role.

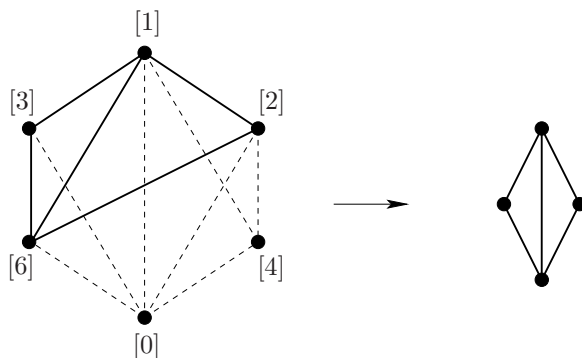
combinations of the generators coincide in G . This is known as the *word problem*, and it has been shown to be undecidable in general³⁶.

In any case, now that we know about presentations of groups, finding coproducts in $\mathbf{Grp}$ should be straightforward: see Exercise 8.7.

There is a ‘mirror’ statement analogous to the fact that all groups are quotients of free groups: every group may be realized as a *subgroup* of a symmetric group. This elementary observation goes under the name of *Cayley’s theorem*; its natural place is within the discussion of *group actions* (cf. Theorem 9.5).

8.3. Subgroups of quotients. The lattice of subgroups (cf. §6.4) of a quotient G/H can be described very explicitly in terms of the lattice of subgroups of the group G : simply keep the part of the lattice of subgroups of G corresponding to subgroups which *contain* H .

Example 8.8. Here is the effect of this operation on the lattice of subgroups of $C_{12} \cong \mathbb{Z}/12\mathbb{Z}$ (labeled by generators; cf. §6.4), after quotienting by $H = \langle [6] \rangle \cong C_2$:



The result matches the lattice of subgroups of $C_6 \cong C_{12}/C_2$. ┘

Here is why this works. First note that if $H \subseteq K$ are subgroups of a group G and H is normal in G , then H is normal in K .

Proposition 8.9. *Let H be a normal subgroup of a group G . Then for every subgroup K of G containing H , K/H may be identified with a subgroup of G/H . The function*

$$u : \{\text{subgroups } K \text{ of } G \text{ containing } H\} \rightarrow \{\text{subgroups of } G/H\}$$

defined by $u(K) = K/H$ is a bijection preserving inclusions.

Proof. The group K/H consists of the cosets $aH \in G/H$ with $a \in K$, and in this sense it is a subset (and clearly a subgroup) of G/H . It is also clear that if $H \subseteq K \subseteq L$, then $u(K) = K/H \subseteq L/H = u(L)$; that is, u preserves inclusions.

³⁶That is, there is no general algorithm that, given a presentation of a group G and two words in the generators, will establish (in a finite time) whether those two words represent the same element of G .

Thus we simply have to verify that u is a bijection, and for this it suffices to produce an inverse function

$$v : \{\text{subgroups of } G/H\} \rightarrow \{\text{subgroups } K \text{ of } G \text{ containing } H\}.$$

Then let K' be a subgroup of G/H ; define $v(K')$ to be the subset of G :

$$K := \pi^{-1}(K') = \{a \in G \mid aH \in K'\},$$

where $\pi : G \rightarrow G/H$ is the canonical projection. Then K is a subgroup of G (by Lemma 6.4) and contains H (because $H = \pi^{-1}(e)$ and $e \in K'$). The reader will check that u and v are inverses of each other. $\square$

In fact, the correspondence is even nicer, in the sense that it preserves *normality*. The following statement is often called the *third isomorphism theorem*:

Proposition 8.10. *Let H be a normal subgroup of a group G , and let N be a subgroup of G containing H . Then N/H is normal in G/H if and only if N is normal in G , and in this case*

$$\frac{G/H}{N/H} \cong \frac{G}{N}.$$

Proof. If N is normal, then consider the projection

$$G \rightarrow \frac{G}{N} :$$

the subgroup H is contained in N , which is the kernel of this homomorphism, so we get (by the universal property of quotients, Theorem 7.12) an induced homomorphism

$$\frac{G}{H} \rightarrow \frac{G}{N}.$$

The subgroup N/H of G/H is the kernel of this homomorphism; therefore it is normal.

Conversely, if N/H is normal in G/H , consider the composition

$$G \twoheadrightarrow \frac{G}{H} \twoheadrightarrow \frac{G/H}{N/H}.$$

The kernel of this homomorphism is N ; therefore N is normal. Further, this homomorphism is surjective; hence the stated isomorphism $(G/H)/(N/H) \cong G/N$ follows immediately from Corollary 8.2. $\square$

8.4. HK/H vs. $K/(H \cap K)$. Section 8.3 deals with two ‘nested’ subgroups $H \subseteq K$ of a group G . What if H, K are *not* nested?

The notation introduced in §7.1 extends to subsets of G : if $A \subseteq G$, $B \subseteq G$, then AB denotes the subset

$$AB := \{ab \mid a \in A, b \in B\}.$$

It would be nice if HK were guaranteed to be a subgroup of G as soon as H and K are subgroups, but this is simply not the case in general, if G is not commutative. It *is*, however, the case if one of the subgroups is normal. The following is often called the *second isomorphism theorem*.

Proposition 8.11. *Let H, K be subgroups of a group G , and assume that H is normal in G . Then*

- HK is a subgroup of G , and H is normal in HK ;
- $H \cap K$ is normal in K , and

$$\frac{HK}{H} \cong \frac{K}{H \cap K}.$$

Proof. To verify that HK is a subgroup of G when H is normal, note that HK is the union of all cosets Hk , with $k \in K$; that is,

$$HK = \pi^{-1}(\pi(K)),$$

where $\pi : G \rightarrow G/H$ is the canonical projection. Since $\pi(K)$ is a subgroup of G/H , HK is a subgroup by Lemma 6.4. It is clear that H is normal in HK .

For the second part, consider the homomorphism

$$\varphi : K \rightarrow HK/H$$

sending $k \in K$ to the coset Hk (that is, the inclusion $K \hookrightarrow HK$ followed by the canonical projection to the quotient). This is *surjective*: indeed, every element of HK/H may be written as a coset

$$Hhk, \quad h \in H, k \in K;$$

but $Hhk = Hk$, so $Hhk = \varphi(k)$ is in the image of φ . By the omnipresent Corollary 8.2,

$$\frac{HK}{H} \cong \frac{K}{\ker \varphi}.$$

What is $\ker \varphi$?

$$\ker \varphi = \{k \in K \mid \varphi(k) = e\} = \{k \in K \mid Hk = H\} = \{k \in K \mid k \in H\} = H \cap K,$$

with the stated result. $\square$

8.5. The index and Lagrange's theorem. The notation G/H is used to denote the set of left-cosets³⁷ of H , even when H is not normal in G . Thus G/H is a set in general, and it is a group when H is in fact normal in G .

Definition 8.12. The *index* of H in G , denoted $[G : H]$, is the number of elements $|G/H|$ of G/H , when this is finite, and ∞ otherwise. $\lrcorner$

Thus, $[G : H]$ (if finite) denotes the number of left-cosets of H in G , regardless of whether H is normal in G .

Lemma 8.13. *Let H be a subgroup of a group G . Then $\forall g \in G$ the functions*

$$H \rightarrow gH, \quad h \mapsto gh,$$

$$H \rightarrow Hg, \quad h \mapsto hg$$

are bijections.

³⁷This may seem an arbitrary choice (why not *right*-cosets?). It is. Writing from left to right gives us a bias towards *left*-actions, and G acts nicely on the *left* on the set of *left*-cosets; this will make better sense when we get to Example 9.4. In any case, there is a bijection between the set of left-cosets and the set of right-coset: Exercise 9.10.

Proof. Both functions are surjective by definition of coset. Cancellation implies that they are injective. $\square$

Corollary 8.14 (Lagrange's theorem). *If G is a finite group and $H \subseteq G$ is a subgroup, then $|G| = [G : H] \cdot |H|$. In particular, $|H|$ is a divisor of $|G|$.*

Proof. Indeed, G is the disjoint union of $|G/H|$ distinct cosets gH , and $|gH| = |H|$ by Lemma 8.13. $\square$

Lagrange's theorem is more useful than it may appear at first.

Example 8.15. The order $|g|$ of any element g of a finite group G is a divisor of $|G|$: indeed, $|g|$ equals the order of the subgroup $\langle g \rangle$ generated by g .

Note: Therefore, $g^{|G|} = e_G$ for all finite groups G , all $g \in G$. $\lrcorner$

Example 8.16. If $|G|$ is a prime integer p , then necessarily $G \cong \mathbb{Z}/p\mathbb{Z}$.

Indeed, let $g \in G$ be any element other than the identity; then $\langle g \rangle$ is a subgroup of G , of order > 1 . By Lagrange's theorem, $|\langle g \rangle| = p = |G|$; that is, $G \cong \langle g \rangle$ is cyclic of order p , as claimed. $\lrcorner$

Example 8.17 (Fermat's little theorem). Let p be a prime integer, and let a be any integer. Then $a^p \equiv a \pmod{p}$.

Indeed, this is immediate if a is a multiple of p ; if a is not a multiple of p , then the class $[a]_p$ modulo p is nonzero, so it is an element of the group $(\mathbb{Z}/p\mathbb{Z})^*$, which has order $p - 1$. Thus

$$[a]_p^{p-1} = [1]_p$$

(Example 8.15); hence $[a]_p^p = [a]_p$ as claimed. $\lrcorner$

Warning: However, do not ask too much of Lagrange's theorem. For example, it does *not* say that if d is a divisor of $|G|$, then there exists a subgroup of G of order d (the smallest counterexample is A_4 , a group of order 12, which does not contain subgroups of order 6; the reader will verify this in Exercise IV.4.17); it does not even say that if p is a prime divisor of $|G|$, then there is an element of order p in G . This latter statement happens to be true, but for 'deeper' reasons. The abelian case is easy (cf. Exercise 8.17). The general case is called *Cauchy's theorem*, and we will deal with it later on (cf. Theorem IV.2.1).

The index is a well-behaved invariant. It is clearly *multiplicative*, in the sense that if $H \subseteq K$ are subgroups of G , then

$$[G : H] = [G : K] \cdot [K : H],$$

provided that these numbers are finite. Also, if H and K are subgroups of G and H is normal (so that HK is a subgroup as well; cf. Proposition 8.11), then

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$$

(again, provided this has a chance of making sense, that is, if the orders are finite): this follows immediately from the isomorphism in Proposition 8.11 and index considerations. In fact, the formula holds even without assuming that one of the subgroups is normal in G . Do you see why? (Exercise 8.21.)

Further, if H and G are finite, then Lemma 8.13 implies immediately that the index of H in G , defined as the number of *left*-cosets of H in G , also equals the number of *right*-cosets of H . It is in fact easy to show that there always is a bijection between the set G/H of left-cosets and the set of right-cosets (cf. Exercise 9.10), regardless of finiteness hypotheses. The set of right-cosets of H in G is often (reasonably) denoted $H \backslash G$.

8.6. Epimorphisms and cokernels. The reader may expect that a mirror statement to Proposition 6.12 should hold for group *epimorphisms*. This is almost true: a homomorphism $\varphi : G \rightarrow H$ is an *epimorphism* (in the category **Grp**) if and only if it is *surjective*. However, while one implication is easy, the proofs I know for epimorphism $\implies$ surjective in **Grp** are somewhat cumbersome.

The situation is leaner (as usual) in **Ab**: there is in **Ab** a good notion of *cokernel*; this is part of what makes **Ab** an ‘abelian category’.

As is often the case, the reader may now want to pause a moment and try to guess the right definition. Keeping in mind the universal property for kernels (Proposition 6.6), can the reader come up with the universal property defining ‘cokernels’? Can the reader prove that these exist in **Ab** and detect epimorphisms? Don’t look ahead!

Here is how the story goes. The universal property is (of course) obtained by reversing the arrows in the property for kernels: given a homomorphism $\varphi : G \rightarrow G'$ of *abelian* groups, we want an abelian group *coker* φ equipped with a homomorphism

$$\pi : G' \rightarrow \text{coker } \varphi$$

which is initial with respect to all morphisms α such that $\alpha \circ \varphi = 0$. That is, every homomorphism $\alpha : G' \rightarrow L$ such that $\alpha \circ \varphi$ is the trivial map must factor (uniquely) through *coker* φ :

$$\begin{array}{ccccc} & & 0 & & \\ & \curvearrowright & & \searrow & \\ G & \xrightarrow{\varphi} & G' & \xrightarrow{\alpha} & L \\ & & \downarrow \pi & \nearrow \exists! \bar{\alpha} & \\ & & \text{coker } \varphi & & \end{array}$$

Cokernels exist in **Ab**: because the image of φ is a subgroup of G' , hence a *normal* subgroup of G' since G' is abelian; the condition that $\alpha \circ \varphi$ is trivial says that $\text{im } \varphi \subseteq \ker \alpha$, and hence

$$\frac{G'}{\text{im } \varphi} \cong \text{coker } \varphi$$

satisfies the universal property, by Theorem 7.12.

The ‘problem’ in **Grp** is that $\text{im } \varphi$ is not guaranteed to be normal in G' ; thus the situation is more complex.

Also note that, in the abelian case, $G'/\text{im } \varphi$ automatically satisfies a stronger universal property: as stated, but with respect to any *set-function* $G' \rightarrow L$ which is constant on cosets of $\text{im } \varphi$.

We can now state a true mirror of Proposition 6.12, in **Ab**:

Proposition 8.18. *Let $\varphi : G \rightarrow G'$ be a homomorphism of abelian groups. The following are equivalent:*

- (a) φ is an epimorphism;
- (b) $\text{coker } \varphi$ is trivial;
- (c) $\varphi : G \rightarrow G'$ is surjective (as a set-function).

Proof. (a) $\implies$ (b): Assume (a) holds, and consider the two parallel compositions

$$G \xrightarrow{\varphi} G' \xrightarrow[e]{\pi} \text{coker } \varphi,$$

where π is the canonical projection and e is the trivial map. Both $\pi \circ \varphi$ and $e \circ \varphi$ are the trivial map; since φ is an epimorphism, this implies $\pi = e$. But $\pi = e$ implies that $\text{coker } \varphi$ is trivial, that is, (b) holds.

(b) $\implies$ (c): If $\text{coker } \varphi = G'/\text{im } \varphi$ is trivial, then $\text{im } \varphi = G'$; hence φ is surjective.

(c) $\implies$ (a): If φ is surjective, then it satisfies the universal property for epimorphisms in **Set**: for any set Z and any two set-functions α' and $\alpha'' : G' \rightarrow Z$,

$$\alpha' \circ \varphi = \alpha'' \circ \varphi \iff \alpha' = \alpha''.$$

This must hold in particular if Z is endowed with a group structure and α', α'' are group homomorphisms, so φ is an epimorphism in **Grp**. $\square$

A cokernel may be defined in **Grp**: the universal property for the cokernel of $\varphi : G \rightarrow G'$ is satisfied by G'/N , where N is the smallest³⁸ normal subgroup of G' containing $\text{im } \varphi$ (Exercise 8.22). But Proposition 8.18 fails, because the implication (b) $\implies$ (c) does not hold: in **Grp** it is no longer true that only surjective homomorphisms have trivial cokernel (cf. Exercise 8.23).

Exercises

8.1. If a group H may be realized as a subgroup of two groups G_1 and G_2 and if

$$\frac{G_1}{H} \cong \frac{G_2}{H},$$

does it follow that $G_1 \cong G_2$? Give a proof or a counterexample.

8.2. $\neg$ Extend Example 8.6 as follows. Suppose G is a group and $H \subseteq G$ is a subgroup of index 2, that is, such that there are precisely two (say, left-) cosets of H in G . Prove that H is normal in G . [9.11, IV.1.16]

8.3. Prove that every finite group is finitely presented.

³⁸The intersection of any family of normal subgroups is normal, as the reader may readily check, so this subgroup exists.

8.4. Prove that $(a, b | a^2, b^2, (ab)^n)$ is a presentation of the dihedral group D_{2n} . (Hint: With respect to the generators defined in Exercise 2.5, set $a = x$ and $b = xy$; prove you can get the relations given here from the ones you obtained in Exercise 2.5, and conversely.)

8.5. Let a, b be distinct elements of order 2 in a group G , and assume that ab has finite order $n \geq 3$. Prove that the subgroup generated by a and b in G is isomorphic to the dihedral group D_{2n} . (Use the previous exercise.)

8.6. $\neg$ Let G be a group, and let A be a set of generators for G ; assume A is finite. The corresponding *Cayley graph*³⁹ is a directed graph whose set of vertices is in one-to-one correspondence with G , and two vertices g_1, g_2 are connected by an edge if $g_2 = g_1 a$ for an $a \in A$; this edge may be labeled a and oriented from g_1 to g_2 . For example, the graph drawn in Example 5.3 for the free group $F(\{x, y\})$ on two generators x, y is the corresponding Cayley graph (with the convention that horizontal edges are labeled x and point to the right and vertical edges are labeled y and point up).

Prove that if a Cayley graph of a group is a tree, then the group is free. Conversely, prove that free groups admit Cayley graphs that are trees. [§5.3, 9.15]

8.7. $\triangleright$ Let $(A | \mathcal{R})$, resp., $(A' | \mathcal{R}')$, be a presentation for a group G , resp., G' (cf. §8.2); we may assume that A, A' are disjoint. Prove that the group $G * G'$ presented by

$$(A \cup A' | \mathcal{R} \cup \mathcal{R}')$$

satisfies the universal property for the *coproduct* of G and G' in $\mathbf{Grp}$. (Use the universal properties of both free groups and quotients to construct natural homomorphisms $G \rightarrow G * G', G' \rightarrow G * G'$.) [§3.4, §8.2, 9.14]

8.8. $\neg$ (If you know about matrices (cf. Exercise 6.1).) Prove that $\mathrm{SL}_n(\mathbb{R})$ is a *normal subgroup* of $\mathrm{GL}_n(\mathbb{R})$, and ‘compute’ $\mathrm{GL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{R})$ as a well-known group. [VI.3.3]

8.9. $\neg$ (Ditto.) Prove that $\mathrm{SO}_3(\mathbb{R}) \cong \mathrm{SU}(2)/\{\pm I_2\}$, where I_2 is the identity matrix. (Hint: It so happens that every matrix in $\mathrm{SO}_3(\mathbb{R})$ can be written in the form

$$\begin{pmatrix} a^2 + b^2 - c^2 - d^2 & 2(bc - ad) & 2(ac + bd) \\ 2(ad + bc) & a^2 - b^2 + c^2 - d^2 & 2(cd - ab) \\ 2(bd - ac) & 2(ab + cd) & a^2 - b^2 - c^2 + d^2 \end{pmatrix}$$

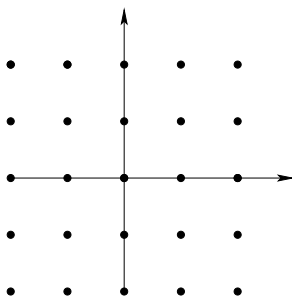
where $a, b, c, d \in \mathbb{R}$ and $a^2 + b^2 + c^2 + d^2 = 1$. Proving this fact is not hard, but at this stage you will probably find it computationally demanding. Feel free to assume this, and use Exercise 6.3 to construct a surjective homomorphism $\mathrm{SU}(2) \rightarrow \mathrm{SO}_3(\mathbb{R})$; compute the kernel of this homomorphism.)

If you know a little topology, you can now conclude that the fundamental group⁴⁰ of $\mathrm{SO}_3(\mathbb{R})$ is C_2 . [9.1, VI.1.3]

³⁹Warning: This is one of several alternative conventions.

⁴⁰If you really want to believe this fact, remember that $\mathrm{SO}_3(\mathbb{R})$ parametrizes rotations in $\mathbb{R}^3$. Hold a tray with a glass of water on top of your extended right hand. You should be able to rotate the tray clockwise by a full 360° without spilling the water, and your muscles will tell you that the corresponding loop in $\mathrm{SO}_3(\mathbb{R})$ is *not* trivial. But then you will be able to rotate the tray *again*

8.10. View $\mathbb{Z} \times \mathbb{Z}$ as a subgroup of $\mathbb{R} \times \mathbb{R}$:



Describe the quotient

$$\frac{\mathbb{R} \times \mathbb{R}}{\mathbb{Z} \times \mathbb{Z}}$$

in terms analogous to those used in Example 8.7. (Can you ‘draw a picture’ of this group? Cf. Exercise I.1.6.)

8.11. (Notation as in Proposition 8.10.) Prove ‘by hand’ (that is, without invoking universal properties) that N is normal in G if and only if N/H is normal in G/H .

8.12. (Notation as in Proposition 8.11.) Prove ‘by hand’ (that is, by using Proposition 6.2) that HK is a subgroup of G if H is normal.

8.13. $\neg$ Let G be a finite group, and assume $|G|$ is odd. Prove that every element of G is a square. [8.14]

8.14. Generalize the result of Exercise 8.13: if G is a group of order n and k is an integer relatively prime to n , then the function $G \rightarrow G$, $g \mapsto g^k$ is surjective.

8.15. Let a , n be positive integers, with $a > 1$. Prove that n divides $\phi(a^n - 1)$, where ϕ is Euler’s ϕ -function; see Exercise 6.14. (Hint: Example 8.15.)

8.16. Generalize Fermat’s little theorem to congruences modulo arbitrary (that is, possibly nonprime) integers. Note that it is *not* true that $a^n \equiv a \pmod n$ for all a and n : for example, 2^4 is not congruent to 2 modulo 4. *What* is true? (This generalization is known as *Euler’s theorem*.)

8.17. $\triangleright$ Assume G is a finite abelian group, and let p be a prime divisor of $|G|$. Prove that there exists an element in G of order p . (Hint: Let g be any element of G , and consider the subgroup $\langle g \rangle$; use the fact that this subgroup is cyclic to show that there is an element $h \in \langle g \rangle$ in G of *prime* order q . If $q = p$, you are done; otherwise, use the quotient $G/\langle h \rangle$ and induction.) [§8.5, 8.18, 8.20, §IV.2.1]

8.18. Let G be an abelian group of order $2n$, where n is odd. Prove that G has *exactly one* element of order 2. (It has at least one, for example by Exercise 8.17. Use Lagrange’s theorem to establish that it cannot have more than one.) Does the same conclusion hold if G is not necessarily commutative?

a full 360° clockwise without spilling any water, taking it back to the original position. Thus, the square of the loop *is* (homotopically) trivial, as it should be if the fundamental group is cyclic of order 2.

8.19. Let G be a finite group, and let d be a proper divisor of $|G|$. Is it necessarily true that there exists an element of G of order d ? Give a proof or a counterexample.

8.20. $\triangleright$ Assume G is a finite abelian group, and let d be a divisor of $|G|$. Prove that there exists a *subgroup* $H \subseteq G$ of order d . (Hint: induction; use Exercise 8.17.) [§IV.2.2]

8.21. $\triangleright$ Let H, K be subgroups of a group G . Construct a bijection between the set of cosets hK with $h \in H$ and the set of left-cosets of $H \cap K$ in H . If H and K are finite, prove that

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

[§8.5, §IV.4.4]

8.22. $\triangleright$ Let $\varphi : G \rightarrow G'$ be a group homomorphism, and let N be the smallest normal subgroup containing $\text{im } \varphi$. Prove that G'/N satisfies the universal property of $\text{coker } \varphi$ in $\mathbf{Grp}$. [§8.6]

8.23. $\triangleright$ Consider the subgroup

$$H = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \right\}$$

of S_3 . Show that the cokernel of the inclusion $H \hookrightarrow S_3$ is trivial, although $H \hookrightarrow S_3$ is not surjective. [§8.6]

8.24. $\triangleright$ Show that epimorphisms in $\mathbf{Grp}$ do not necessarily have right-inverses. [§I.4.2]

8.25. Let H be a commutative normal subgroup of G . Construct an interesting homomorphism from G/H to $\text{Aut}(H)$. (Cf. Exercise 7.10.)

9. Group actions

9.1. Actions. As mentioned in §4.1, an *action* of a group G on an object A of a category $\mathbf{C}$ is simply a homomorphism

$$\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A).$$

The way to interpret this is that every element $g \in G$ determines a ‘transformation of A into itself’, i.e., an isomorphism of A in $\mathbf{C}$, and this happens compatibly with the operation of G and composition in $\mathbf{C}$.

In a rather strong sense, we really only care about groups because they act on things: knowing that G acts on A tells us something about A ; group actions are one key tool in the study of geometric and algebraic entities.

In fact, group actions are one key tool in the study of *groups* themselves: one of the best ways to ‘understand’ a group is to let it act on an object A , hoping that the corresponding homomorphism σ is an isomorphism, or at least an injective monomorphism. For example, we were lucky with D_6 in §2.2: we let D_6 act on a set with three elements (the vertices of an equilateral triangle) and observed that the resulting σ is an isomorphism. Thus $D_6 \cong S_3$. We would be almost as lucky

by letting D_8 act on the vertices of a square: then σ would at least realize D_8 as an explicit *subgroup* of S_4 , which simplifies its analysis.

Definition 9.1. An action of a group G on an object A of a category $\mathbf{C}$ is *faithful* (or *effective*) if the corresponding $\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A)$ is injective. $\lrcorner$

The case $\mathbf{C} = \mathbf{Set}$ is already very rich, and we focus on it in this chapter.

9.2. Actions on sets. Spelling out our definition of action in case A is a *set*, so that $\text{Aut}_{\mathbf{C}}(A)$ is the symmetric group S_A , we get the following:

Definition 9.2. An *action* of a group G on a set A is a set-function

$$\rho : G \times A \rightarrow A$$

such that $\rho(e_G, a) = a$ for all $a \in A$ and

$$(\forall g, h \in G), (\forall a \in A) : \quad \rho(gh, a) = \rho(g, \rho(h, a)). \quad \lrcorner$$

Indeed, given a function ρ satisfying these conditions, we can define $\sigma : G \rightarrow \text{Hom}_{\mathbf{Set}}(A, A)$ by $\sigma(g)(a) = \rho(g, a)$. (This defines $\sigma(g)$ as a set-function $A \rightarrow A$, as needed.) This function preserves the operation, because

$$\begin{aligned} \sigma(gh)(a) &= \rho(gh, a) = \rho(g, \rho(h, a)) = \sigma(g)(\rho(h, a)) = \sigma(g)(\sigma(h)(a)) \\ &= \sigma(g) \circ \sigma(h)(a). \end{aligned}$$

In particular, this verifies that $\sigma(g^{-1})$ acts as the inverse of $\sigma(g)$: because $(\forall a \in A)$

$$\sigma(g^{-1}) \circ \sigma(g)(a) = \sigma(g^{-1}g)(a) = \sigma(e_G)(a) = \rho(e_G, a) = a.$$

Thus the image of σ consists of invertible set-functions; σ is acting as a function

$$\sigma : G \rightarrow S_A,$$

and we have verified that this is a homomorphism, as needed.

Conversely, given a homomorphism $\sigma : G \rightarrow S_A$, define $\rho : G \times A \rightarrow A$ by $\rho(g, a) = \sigma(g)(a)$; the same argument (read backwards) shows that ρ satisfies the needed properties.

It is unpleasant to carry ρ along. In practice, one just writes ga for $\rho(g, a)$; the requirements in Definition 9.2 amount then to $e_G a = a$ for all $a \in A$ and

$$(\forall g, h \in G), (\forall a \in A) : \quad (gh)a = g(ha),$$

‘as if’ ρ defined an *associative* operation.

If G acts on A , then $e_G a = a$ for all $a \in A$; the action of a group G on a set A is faithful if and only if the identity e_G is the *only* element g of G such that $ga = a$ for all $a \in A$, that is, ‘fixing’ every element of A . An action is *free* if the identity e_G is the only element fixing *any* element of A .

Example 9.3. Every *group* G acts in a natural way on the underlying *set* G . The function $\rho : G \times G \rightarrow G$ is simply the operation in the group:

$$(\forall g, a \in G) : \quad \rho(g, a) = ga.$$

In this case the defining property *really* is associativity. This is referred to as the action *by left-multiplication*⁴¹ of G on itself. There is (at least) another very natural way to act with G on itself, by *conjugation*: define $\rho : G \times G \rightarrow G$ by

$$\rho(g, h) = ghg^{-1}.$$

This is indeed an action: $\forall g, h, k \in G$,

$$\rho(g, \rho(h, k)) = g\rho(h, k)g^{-1} = g(hkh^{-1})g^{-1} = (gh)k(gh)^{-1} = \rho(gh, k). \quad \lrcorner$$

Example 9.4. More generally, G acts by left-multiplication on the set G/H of *left-cosets* (cf. §8.5) of any subgroup H : act by $g \in G$ on $aH \in G/H$ by sending it to $(ga)H$. ┐

These examples of actions are extremely useful in studying groups, as we will see in Chapter IV. For instance, an immediate consequence is the following counterpart to §8.2:

Theorem 9.5 (Cayley's theorem). *Every group acts faithfully on some set. That is, every group may be realized as a subgroup of a permutation group.*

Proof. Indeed, simply observe that the left-multiplication action of G on itself is manifestly faithful. □

The notion defined in Definition 9.2 is, for the sake of precision, called a *left-action*. A *right-action* would associate to each pair (g, a) with $g \in G$ and $a \in A$ an element $ag \in A$; our make-believe associativity would now say

$$a(gh) = (ag)h$$

for all $a \in A$ and $g, h \in G$. This is a different requirement than the one given in Definition 9.2; multiplication *on the right* in a group G gives a prototypical example of a right-action of G (on itself).

Every right-action may be turned into a left-action with due care (cf. Exercise 9.3). Therefore it is not restrictive to just consider left-actions; from now on, an 'action' will be understood to be a *left-action*, unless stated otherwise.

9.3. Transitive actions and the category $G\text{-Set}$.

Definition 9.6. An action of a group G on a (nonempty) set A is *transitive* if $\forall a, b \in A \exists g \in G$ such that $b = ga$. ┐

For example, the left-multiplication action of a group on itself is transitive. Transitive actions are the basic ingredients making up *every* action; this is seen by means of the following important concepts.

Definition 9.7. The *orbit* of $a \in A$ under an action of a group G is the set

$$O_G(a) := \{ga \mid g \in G\}. \quad \lrcorner$$

⁴¹This is *left-multiplication* in the sense that the 'acting' element g of G is placed to the left of the element a 'acted upon'.

Definition 9.8. Let G act on a set A , and let $a \in A$. The *stabilizer* subgroup of a consists of the elements of G which fix a :

$$\text{Stab}_G(a) := \{g \in G \mid ga = a\}. \quad \lrcorner$$

Orbits of an action of a group G on a set A form a partition of A ; and we have an induced, *transitive* action of G on each orbit. Therefore we can, in a sense, ‘understand’ all actions if we understand transitive actions. This will be accomplished in a moment, by studying actions related to stabilizers.

For any group G , sets endowed with a (left) G -action form in a natural way a category $G\text{-Set}$: objects are pairs (ρ, A) , where $\rho : G \times A \rightarrow A$ is an action (as in Definition 9.2) and morphisms between two objects are set-functions which are compatible with the actions. That is, a morphism

$$(\rho, A) \rightarrow (\rho', A')$$

in $G\text{-Set}$ amounts to a set-function $\varphi : A \rightarrow A'$ such that the diagram

$$\begin{array}{ccc} G \times A & \xrightarrow{\text{id}_G \times \varphi} & G \times A' \\ \rho \downarrow & & \downarrow \rho' \\ A & \xrightarrow{\varphi} & A' \end{array}$$

commutes. In the usual shorthand notation omitting the ρ 's, this means that $\forall g \in G, \forall a \in A$,

$$g\varphi(a) = \varphi(ga);$$

that is, the action ‘commutes’ with φ . Such functions are called *(G-)equivariant*.

We therefore have a notion of *isomorphism* of G -sets (defined as in §I.4.1); the reader should expect (and should verify) that these are nothing but the *equivariant bijections*.

Among G -sets we single out the sets G/H of left-cosets of subgroups H of G ; as noted in Example 9.4, G acts on G/H by left-multiplication.

Proposition 9.9. *Every transitive left-action of G on a nonempty set A is isomorphic to the left-multiplication of G on G/H , for $H =$ the stabilizer of any $a \in A$.*

Proof. Let G act transitively on a set A , let $a \in A$ be any element, and let $H = \text{Stab}_G(a)$. I claim that there is an equivariant bijection

$$\varphi : G/H \rightarrow A$$

defined by

$$\varphi(gH) := ga$$

for all $g \in G$.

Indeed, first of all φ is well-defined: if $g_1H = g_2H$, then $g_1^{-1}g_2 \in H$, hence $(g_1^{-1}g_2)a = a$, and it follows that $g_1a = g_2a$ as needed. To verify that φ is bijective, define a function $\psi : A \rightarrow G/H$ by sending an element ga of A to gH ; ψ is well-defined because if $g_1a = g_2a$, then $g_1^{-1}(g_2a) = a$, so $g_1^{-1}g_2 \in H$ and $g_1H = g_2H$. It is clear that φ and ψ are inverses of each other; hence φ is a bijection.

Equivariance is immediate: $\varphi(g'(gH)) = g'ga = g'\varphi(gH)$. □

Corollary 9.10. *If O is an orbit of the action of a finite group G on a set A , then O is a finite set and*

$$|O| \text{ divides } |G|.$$

Proof. By Proposition 9.9 there is a bijection between O and $G/\text{Stab}_G(a)$ for any element $a \in O$; thus

$$|O| \cdot |\text{Stab}_G(a)| = |G|$$

by Corollary 8.14. □

Corollary 9.10 upgrades Lagrange's theorem to orbits of any action; it is extremely useful, as it provides a very strong constraint on group actions.

Example 9.11. There are *no* transitive actions of S_3 on a set with 5 elements.

Indeed, 5 does not divide 6. ⌋

Ultimately, almost everything we will prove in Chapter IV on the structure of finite groups will be a consequence of 'counting arguments' stemming from applications of Corollary 9.10 to actions of a group by conjugation or left-multiplication.

There may seem to be an element of arbitrariness in the statement of Proposition 9.9: what if we change the element a of which we are taking the stabilizer? The stabilizer may change, but it does so in a controlled way:

Proposition 9.12. *Suppose a group G acts on a set A , and let $a \in A$, $g \in G$, $b = ga$. Then*

$$\text{Stab}_G(b) = g \text{Stab}_G(a) g^{-1}.$$

Proof. Indeed, assume $h \in \text{Stab}_G(a)$; then

$$(ghg^{-1})(b) = gh(g^{-1}g)a = gha = ga = b :$$

thus $ghg^{-1} \in \text{Stab}_G(b)$. This proves the $\supseteq$ inclusion; $\subseteq$ follows by the same argument, noting that $a = g^{-1}b$. □

For example, if $\text{Stab}_G(a)$ happens to be normal, then it is really independent of a (in any given orbit). In any case, there is an isomorphism of G -sets between G/H and $G/(gHg^{-1})$, as follows from these considerations (and as the reader will independently check in Exercise 9.13).

Exercises

9.1. (Once more, if you are already familiar with a little linear algebra...) The matrix groups listed in Exercise 6.1 all come with evident actions on a vector space: if M is an $n \times n$ matrix with (say) real entries, multiplication to the right by a column n -vector $\mathbf{v}$ returns a column n -vector $M\mathbf{v}$, and this defines a left-action on $\mathbb{R}^n$ viewed as the space of column n -vectors.

- Prove that, through this action, matrices $M \in O_n(\mathbb{R})$ preserve lengths and angles in $\mathbb{R}^n$.
- Find an interesting action of $SU(2)$ on $\mathbb{R}^3$. (Hint: Exercise 8.9.)

9.2. The effect of the matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

on the plane is to respectively flip the plane about the y -axis and to rotate it 90° clockwise about the origin. With this in mind, construct an action of D_8 on $\mathbb{R}^2$.

9.3. If $G = (G, \cdot)$ is a group, we can define an ‘opposite’ group $G^\circ = (G, \bullet)$ supported on the same set G , by prescribing

$$(\forall g, h \in G) : \quad g \bullet h := h \cdot g.$$

- Verify that G° is indeed a group.
- Show that the ‘identity’: $G^\circ \rightarrow G, g \mapsto g$ is an isomorphism if and only if G is commutative.
- Show that $G^\circ \cong G$ (even if G is not commutative!).
- Show that giving a *right*-action of G on a set A is the same as giving a homomorphism $G^\circ \rightarrow S_A$, that is, a *left*-action of G° on A .
- Show that the notions of left- and right-actions coincide ‘on the nose’ for *commutative* groups. (That is, if $(g, a) \mapsto ag$ defines a right-action of a commutative group G on a set A , then setting $ga = ag$ defines a left-action).
- For any group G , explain how to turn a right-action of G into a left-action of G . (Note that the simple ‘flip’ $ga = ag$ does *not* work in general if G is not commutative.)

9.4. As mentioned in the text, *right*-multiplication defines a right-action of a group on itself. Find *another* natural right-action of a group on itself.

9.5. Prove that the action by left-multiplication of a group on itself is free.

9.6. Let O be an orbit of an action of a group G on a set. Prove that the induced action of G on O is transitive.

9.7. Prove that stabilizers are indeed subgroups.

9.8. For G a group, verify that $G\text{-Set}$ is indeed a category, and verify that the isomorphisms in $G\text{-Set}$ are precisely the equivariant bijections.

9.9. Prove that $G\text{-Set}$ has products and coproducts and that every object of $G\text{-Set}$ is a coproduct of objects of the type $G/H = \{\text{left-cosets of } H\}$, where H is a subgroup of G and G acts on G/H by left-multiplication.

9.10. Let H be any subgroup of a group G . Prove that there is a bijection between the set G/H of *left*-cosets of H and the set $H\backslash G$ of *right*-cosets of H in G . (Hint: G acts on the right on the set of right-cosets; use Exercise 9.3 and Proposition 9.9.)

9.11. $\neg$ Let G be a finite group, and let H be a subgroup of index p , where p is the *smallest prime dividing* $|G|$. Prove that H is normal in G , as follows:

- Interpret the action of G on G/H by left-multiplication as a homomorphism $\sigma : G \rightarrow S_p$.
- Then $G/\ker \sigma$ is (isomorphic to) a subgroup of S_p . What does this say about the index of $\ker \sigma$ in G ?
- Show that $\ker \sigma \subseteq H$.
- Conclude that $H = \ker \sigma$, by index considerations.

Thus H is a kernel, proving that it is normal. (This exercise generalizes the result of Exercise 8.2.) [9.12]

9.12. $\neg$ Generalize the result of Exercise 9.11, as follows. Let G be a group, and let $H \subseteq G$ be a subgroup of index n . Prove that H contains a subgroup K that is normal in G and such that $[G : K]$ divides the gcd of $|G|$ and $n!$. (In particular, $[G : K] \leq n!$.) [IV.2.23]

9.13. $\triangleright$ Prove ‘by hand’ that for all subgroups H of a group G and $\forall g \in G$, G/H and $G/(gHg^{-1})$ (endowed with the action of G by left-multiplication) are isomorphic in $G\text{-Set}$. [§9.3]

9.14. $\neg$ Prove that the modular group $\text{PSL}_2(\mathbb{Z})$ is isomorphic to the coproduct $C_2 * C_3$. (Recall that the modular group $\text{PSL}_2(\mathbb{Z})$ is generated by $x = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $y = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$, satisfying the relations $x^2 = y^3 = e$ in $\text{PSL}_2(\mathbb{Z})$ (Exercise 7.5). The task is to prove that x and y satisfy *no* other relation: this will show that $\text{PSL}_2(\mathbb{Z})$ is presented by $(x, y | x^2, y^3)$, and we have agreed that this is a presentation for $C_2 * C_3$ (Exercise 3.8 or 8.7). Reduce this to verifying that no products

$$(y^{\pm 1}x)(y^{\pm 1}x) \cdots (y^{\pm 1}x) \quad \text{or} \quad (y^{\pm 1}x)(y^{\pm 1}x) \cdots (y^{\pm 1}x)y^{\pm 1}$$

with one or more factors can equal the identity. This latter verification is traditionally carried out by cleverly exploiting an action⁴². Let the modular group act on the set of *irrational* real numbers by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} (r) = \frac{ar + b}{cr + d}.$$

Check that this does define an action of $\text{PSL}_2(\mathbb{Z})$, and note that

$$y(r) = 1 - \frac{1}{r}, \quad y^{-1}(r) = \frac{1}{1-r}, \quad yx(r) = 1 + r, \quad y^{-1}x(r) = \frac{r}{1+r}.$$

⁴²The modular group acts on $\mathbb{C} \cup \{\infty\}$ by *Möbius transformations*. The observation that it suffices to act on $\mathbb{R} \setminus \mathbb{Q}$ for the purpose of this verification is due to Roger Alperin.

Now complete the verification with a case-by-case analysis. For example, a product $(y^{\pm 1}x)(y^{\pm 1}x) \cdots (y^{\pm 1}x)y$ cannot equal the identity in $\mathrm{PSL}_2(\mathbb{Z})$ because if it did, it would act as the identity on $\mathbb{R} \setminus \mathbb{Q}$, while if $r < 0$, then $y(r) > 0$, and both yx and $y^{-1}x$ send positive irrationals to positive irrationals.) [3.8]

9.15. $\neg$ Prove that every (finitely generated) group G acts freely on any corresponding Cayley graph. (Cf. Exercise 8.6. Actions on a directed graph are defined as actions on the set of vertices preserving incidence: if the vertices v_1, v_2 are connected by an edge, then so must be gv_1, gv_2 for every $g \in G$.) In particular, conclude that every free group acts freely on a tree. [9.16]

9.16. $\triangleright$ The converse of the last statement in Exercise 9.15 is also true: only free groups can act freely on a tree. Assuming this, prove that every subgroup of a free group (on a finite set) is free. [§6.4]

9.17. $\triangleright$ Consider G as a G -set, by acting with left-multiplication. Prove that $\mathrm{Aut}_{G\text{-Set}}(G) \cong G$. [§2.1]

9.18. Show how to construct a *groupoid* carrying the information of the action of a group G on a set A . (Hint: A will be the set of objects of the groupoid. What will be the morphisms?)

10. Group objects in categories

10.1. Categorical viewpoint. The definition of *group* (Definition 1.2) is firmly grounded on the category **Set**: *A group is a set G endowed with a binary operation...* However, we have noticed along the way (for example, in §3) that what is really behind it is a pair of functions:

$$m : G \times G \rightarrow G, \quad \iota : G \rightarrow G$$

satisfying certain properties (which translate into associativity, existence of inverses, etc.). Much of what we have seen could be expressed exclusively in terms of these functions, systematically replacing considerations on ‘elements’ by suitable commutative diagrams and enforcing universal properties as a means to define key notions such as the quotient of a group by a subgroup. For example, homomorphisms may be defined purely in terms of the commutativity of a diagram: cf. Definition 3.1.

This point of view may be transferred easily to categories other than **Set**, and the corresponding notions are very important in modern mathematics.

Definition 10.1. Let $\mathbf{C}$ be a category with (finite) products and with a final object 1 .

A *group object* in $\mathbf{C}$ consists of an object G of $\mathbf{C}$ and of morphisms

$$m : G \times G \rightarrow G, \quad e : 1 \rightarrow G, \quad \iota : G \rightarrow G$$

in $\mathbf{C}$ such that the diagrams

$$\begin{array}{ccc}
 (G \times G) \times G & \xrightarrow{m \times \text{id}_G} & G \times G \xrightarrow{m} G \\
 \downarrow \cong & & \parallel \\
 G \times (G \times G) & \xrightarrow{\text{id}_G \times m} & G \times G \xrightarrow{m} G
 \end{array}$$

$$\begin{array}{ccc}
 1 \times G & \xrightarrow{e \times \text{id}_G} & G \times G \\
 \searrow \cong & & \downarrow m \\
 & & G
 \end{array}
 \qquad
 \begin{array}{ccc}
 G \times 1 & \xrightarrow{\text{id}_G \times e} & G \times G \\
 \searrow \cong & & \downarrow m \\
 & & G
 \end{array}$$

$$\begin{array}{ccc}
 G & \xrightarrow{\Delta} & G \times G \xrightarrow{\text{id}_G \times \iota} G \times G \\
 \downarrow & & \downarrow m \\
 1 & \xrightarrow{e} & G
 \end{array}
 \qquad
 \begin{array}{ccc}
 G & \xrightarrow{\Delta} & G \times G \xrightarrow{\iota \times \text{id}_G} G \times G \\
 \downarrow & & \downarrow m \\
 1 & \xrightarrow{e} & G
 \end{array}$$

commute. ┘

Comments. The morphism $\Delta = \text{id}_G \times \text{id}_G$ is the ‘diagonal’ morphism $G \rightarrow G \times G$ induced by the universal property for products and the identity map(s) $G \rightarrow G$. Likewise, the other unnamed morphisms in these diagrams are all uniquely determined by suitable universal properties. For example, there is a unique morphism $\epsilon : G \rightarrow 1$ because 1 is final. The composition with the projection

$$G \xrightarrow{\epsilon \times \text{id}_G} 1 \times G \longrightarrow G$$

is the identity; so is

$$1 \times G \longrightarrow G \xrightarrow{\epsilon \times \text{id}_G} 1 \times G$$

(why?); therefore the projection $1 \times G \rightarrow G$ is indeed an isomorphism, as indicated.

The reader will hopefully realize immediately (Exercise 10.2) that our original definition of groups given in §1 is precisely equivalent to the definition of group object *in Set*: the commutativity of the given diagrams codifies associativity and the existence of two-sided identity and inverses.

Most interesting categories the reader will encounter (not necessarily in this book), such as the category of topological spaces, differentiable manifolds, algebraic varieties, schemes, etc., will carry ‘their own’ notion of group object. For example, a *topological group* is a group object in the category of topological spaces; a *Lie group* is a group object in the category of differentiable manifolds, etc.

Exercises

10.1. Define all the unnamed maps appearing in the diagrams in the definition of group object, and prove they are indeed isomorphisms when so indicated. (For the projection $1 \times G \rightarrow G$, what is left to prove is that the composition

$$1 \times G \rightarrow G \rightarrow 1 \times G$$

is the identity, as mentioned in the text.)

10.2. ▷ Show that *groups*, as defined in §1.2, are ‘group objects in the category of sets’. [§10.1]

10.3. Let $(G, \cdot)$ be a group, and suppose $\circ : G \times G \rightarrow G$ is a group homomorphism (w.r.t. $\cdot$) such that $(G, \circ)$ is *also* a group. Prove that $\circ$ and $\cdot$ coincide. (Hint: First prove that the identity with respect to the two operations must be the same.)

10.4. Prove that every *abelian* group has exactly one structure of group object *in the category* **Ab**.

10.5. By the previous exercise, a group object in **Ab** is nothing other than an abelian group. What is a group object in **Grp**?

Rings and modules

1. Definition of ring

In this chapter we will do for *rings* and *modules* what we have done in Chapter II for groups: describe them in general terms, with particular attention to distinguished subobjects and quotients. More detailed information on these structures will be deferred to later chapters: in Chapter V we will look more carefully at several interesting classes of rings, and modules (over commutative rings) will take center stage in our rapid overview of linear algebra in Chapter VI and following. In this chapter I will also include a brief jaunt into homological algebra, a topic that will entertain us greatly in Chapter IX.

1.1. Definition. Rings (and modules) are defined by ‘decorating’ abelian groups with additional data. As motivation for the introduction of such structures, note that all number-based examples of groups that we have encountered, such as $\mathbb{Z}$ or $\mathbb{R}$, are endowed with an operation of *multiplication* as well as the ‘addition’ making them into (abelian) groups. The ‘ring axioms’ will reflect closely the properties and compatibilities of these two operations in such examples.

These examples are, however, very special. A more sophisticated motivation for the introduction of rings arises by analyzing further the structure of homomorphisms of abelian groups. Recall (§II.4.4) that if G, H are abelian groups, then $\text{Hom}_{\text{Ab}}(G, H)$ is also an abelian group. In particular, if G is an abelian group, then so is the set of *endomorphisms* $\text{End}_{\text{Ab}}(G) = \text{Hom}_{\text{Ab}}(G, G)$. More is true: morphisms from an object of a category to itself may be composed with each other (by definition of category!). Thus, *two* operations coexist in $\text{End}_{\text{Ab}}(G)$: addition (inherited from G , making $\text{End}_{\text{Ab}}(G)$ an abelian group), and composition. These two operations are compatible with each other in a sense captured by the *ring axioms*:

Definition 1.1. A *ring* $(R, +, \cdot)$ is an abelian group $(R, +)$ endowed with a *second* binary operation $\cdot$, satisfying on its own the requirements of being associative and having a two-sided identity, i.e.,

- $(\forall r, s, t \in R) : (r \cdot s) \cdot t = r \cdot (s \cdot t),$
- $(\exists 1_R \in R) (\forall r \in R) : r \cdot 1_R = r = 1_R \cdot r$

(which make $(R, \cdot)$ a *monoid*), and further interacting with $+$ via the following *distributive* properties:

- $(\forall r, s, t \in R) : (r + s) \cdot t = r \cdot t + s \cdot t \text{ and } t \cdot (r + s) = t \cdot r + t \cdot s. \quad \lrcorner$

The notation $\cdot$ is often omitted in formulas, and we usually refer to rings by the name of the underlying set.

Warning: What I am calling a ‘ring’, others may call a ‘ring with identity’ or a ‘ring with 1’: it is not uncommon to exclude the axiom of existence of a ‘multiplicative identity’ from the list of axioms defining a ring. Rings without identity are sometimes called *rngs*, but I am not sure this should be encouraged¹. The reader should check conventions carefully when approaching the literature. Examples of structures *without* a multiplicative identity abound: for example, the set $2\mathbb{Z}$ of *even* integers, with the usual addition and multiplication, satisfies all the ring axioms given above with the exception of the existence of 1 (and is therefore a *rng*). But in these notes all rings will have 1.

Of course the multiplicative identity is necessarily unique: the argument given for Proposition II.1.6 works *verbatim*.

The identity element of the abelian group underlying a ring is denoted 0_R (or simply 0, in context) and is called the ‘additive’ identity. This is a special element with respect to multiplication:

Lemma 1.2. *In a ring R ,*

$$0 \cdot r = 0 = r \cdot 0$$

for all $r \in R$.

Proof. Indeed, $0 = 0 + 0$; hence, applying distributivity,

$$r \cdot 0 = r \cdot (0 + 0) = r \cdot 0 + r \cdot 0,$$

from which $r \cdot 0 = 0$ by cancellation (in the *group* $(R, +)$). The equality $0 \cdot r = 0$ is proven similarly. $\square$

It is equally easy to check that multiplication behaves as expected on ‘subtraction’. In fact, if -1 denotes the additive inverse of 1, then the additive inverse $-r$ of any $r \in R$ is the result of the multiplication $(-1) \cdot r$: indeed, using distributivity,

$$r + (-1) \cdot r = 1 \cdot r + (-1) \cdot r = (1 - 1) \cdot r = 0 \cdot r = 0$$

(by Lemma 1.2) from which $(-1)r = -r$ follows by (additive) cancellation.

¹The term ‘rng’ was introduced with this meaning by Jacobson; but essentially at the same time Mac Lane introduced **Rng** as the name for the category of rings *with* identity. Hoping to steer clear of this clash of terminology I have opted to call this category ‘Ring’.

1.2. First examples and special classes of rings.

Example 1.3. We can define a ring structure on a trivial group $\{*\}$ by letting $* \cdot * = *$ (as well as $* + * = *$); this is often called the *zero-ring*. Note that $0 = 1$ in this ring (cf. Exercise 1.1). $\lrcorner$

Example 1.4. More interesting examples are the number-based groups such as $\mathbb{Z}$ or $\mathbb{R}$, with the usual operations. These are very well known to our reader, who will realize immediately that they satisfy the requirements given in Definition 1.1; but they are very special. Why? $\lrcorner$

To begin with, note that multiplication is commutative in these examples; this is not among the requirements we have posed on rings in the official definition given above.

Definition 1.5. A ring R is *commutative* if

$$\bullet (\forall r, s \in R) : \quad r \cdot s = s \cdot r. \quad \lrcorner$$

Commutative rings (with identity) form an extremely important class of rings; *commutative algebra* is the subfield of algebra studying them. We will focus on commutative rings in later chapters; in this chapter we will develop some of the basic theory for the more general case of arbitrary rings (with 1).

Example 1.6. An example of a noncommutative ring that is (likely) familiar to the readers is the ring of 2×2 matrices with, say, real entries: matrices can be added ‘componentwise’, and they can be multiplied as recalled in Example II.1.5; the two operations satisfy the requirements in Definition 1.1.

Square matrices of any size, and with entries in any ring, form a ring (Exercise 1.4). $\lrcorner$

Example 1.7. The reader is already familiar with a large class of (commutative) rings: the groups $\mathbb{Z}/n\mathbb{Z}$, endowed with the multiplication defined in §II.2.3 (that is: $[a]_n \cdot [b]_n := [ab]_n$; this is well-defined (cf. Exercise II.2.14)) satisfy the ring axioms listed above. $\lrcorner$

The rings $\mathbb{Z}/n\mathbb{Z}$ prompt me to highlight an important point. Another reason why rings such as $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, ... are special is that *multiplicative* cancellation by nonzero elements holds in these rings. Of course *additive* cancellation is automatic, since rings are in particular (abelian) *groups*; and multiplicative cancellation clearly fails in general since one cannot ‘cancel 0’ (by Lemma 1.2). But even the fact that

$$(\forall a \in R, a \neq 0) : \quad a \cdot b = a \cdot c \implies b = c,$$

which holds, for example, in $\mathbb{Z}$, does *not* follow from the ring axioms.

Indeed, this cancellation property does not hold in *all* rings; it may well fail in the rings $\mathbb{Z}/n\mathbb{Z}$. For example,

$$[2]_6 \cdot [4]_6 = [8]_6 = [2]_6 = [2]_6 \cdot [1]_6$$

even though $[4]_6 \neq [1]_6$.

The problem here is that in $\mathbb{Z}/6\mathbb{Z}$ there are elements $a \neq 0$ such that $a \cdot b = 0$ for some $b \neq 0$ (take $a = [2]_6$, $b = [3]_6$).

Definition 1.8. An element a in a ring R is a *left-zero-divisor* if there exist elements $b \neq 0$ in R for which $ab = 0$. $\lrcorner$

The reader will have no difficulty figuring out what a *right-zero-divisor* should be. The element 0 is a zero-divisor in all nonzero rings R ; the zero ring is the only ring without zero-divisors(!).

Proposition 1.9. In a ring R , $a \in R$ is not a left- (resp., right-) zero-divisor if and only if left (resp., right) multiplication by a is an injective function $R \rightarrow R$.

In other words, a is not a left- (resp., right-) zero-divisor if and only if multiplicative left- (resp., right-) cancellation by the element a holds in R .

Proof. Let's verify the 'left' statement (the 'right' statement is of course entirely analogous). Assume a is not a left-zero-divisor and $ab = ac$ for $b, c \in R$. Then, by distributivity,

$$a(b - c) = ab - ac = 0,$$

and this implies $b - c = 0$ since a is not a left-zero-divisor; that is, $b = c$. This proves that left-multiplication is injective in this case.

Conversely, if a is a left-zero-divisor, then $\exists b \neq 0$ such that $ab = 0 = a \cdot 0$; this shows that left-multiplication is not injective in this case, concluding the proof. $\square$

Rings such as $\mathbb{Z}$, $\mathbb{Q}$, etc., are commutative rings *without* (nonzero) zero-divisors. Such rings are very special, but very important, and they deserve their own terminology:

Definition 1.10. An *integral domain* is a nonzero commutative ring R (with 1) such that

$$(\forall a, b \in R) : \quad ab = 0 \implies a = 0 \text{ or } b = 0. \quad \lrcorner$$

Chapter V will be entirely devoted to integral domains.

An element which is *not* a zero-divisor is called a *non-zero-divisor*. Thus, integral domains are those nonzero commutative rings in which every nonzero element is a non-zero-divisor. By Proposition 1.9, multiplicative cancellation by nonzero elements holds in integral domains. The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are all integral domains. As we have seen, some $\mathbb{Z}/n\mathbb{Z}$ are *not* integral domains.

Here is one of those places where the reader can do him/herself a great favor by pausing a moment and figuring something out: answer the question, *which $\mathbb{Z}/n\mathbb{Z}$ are integral domains?* This is entirely within reach, given what the reader knows already. Don't read ahead before figuring this out—this question will be answered within a few short paragraphs, spoiling all the fun.

There are even subtler reasons why $\mathbb{Z}$ is a very special ring: we will see in due time that it is a 'UFD' (*unique factorization domain*); in fact, it is a 'PID' (*principal ideal domain*); in fact, it is more special still!, as it is a 'Euclidean domain'. All of this will be discussed in Chapter V, particularly §V.2.

However, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are more special than all of that and then some, since they are *fields*.

Definition 1.11. An element u of a ring R is a *left-unit* if $\exists v \in R$ such that $uv = 1$; it is a *right-unit* if $\exists v \in R$ such that $vu = 1$. *Units* are two-sided units. $\lrcorner$

Proposition 1.12. *In a ring R :*

- u is a left- (resp., right-) unit if and only if left- (resp., right-) multiplication by u is a surjective functions $R \rightarrow R$;
- if u is a left- (resp., right-) unit, then right- (resp., left-) multiplication by u is injective; that is, u is not a right- (resp., left-) zero-divisor;
- the inverse of a two-sided unit is unique;
- two-sided units form a group under multiplication.

Proof. These assertions are all straightforward. For example, denote by $\rho_u : R \rightarrow R$ right-multiplication by u , so that $\rho_u(r) = ru$. If u is a right-unit, let $v \in R$ be such that $vu = 1$; then $\forall r \in R$

$$\rho_u \circ \rho_v(r) = \rho_u(rv) = (rv)u = r(vu) = r1_R = r.$$

That is, ρ_v is a right-inverse to ρ_u , and therefore ρ_u is surjective (Proposition I.2.1).

Conversely, if ρ_u is surjective, then there exists a v such that $1_R = \rho_u(v) = vu$, so that u is a right-unit.

This checks the first statement, for *right-units*.

For the second statement, denote by $\lambda_u : R \rightarrow R$ left-multiplication by u : $\lambda_u(r) = ur$. Assume u is a *right-unit*, and let v be such that $vu = 1_R$; then $\forall r \in R$

$$\lambda_v \circ \lambda_u(r) = \lambda_v(ur) = v(ur) = (vu)r = 1_R r = r.$$

That is, λ_v is a left-inverse to λ_u , so λ_u is injective (Proposition I.2.1 again).

The rest of the proof is left to the reader (Exercise 1.9). $\square$

Since the inverse of a two-sided unit u is unique, we can give it a name; of course we denote it by u^{-1} . The reader should keep in mind that inverses of left- or right-units are *not* unique in general, so the ‘inverse notation’ is not appropriate for them.

Definition 1.13. A *division ring* is a ring in which every nonzero element is a two-sided unit. $\lrcorner$

We will mostly be concerned with the commutative case, which has its own name:

Definition 1.14. A *field* is a nonzero commutative ring R (with 1) in which every nonzero element is a unit. $\lrcorner$

The whole of Chapter VII will be devoted to studying fields.

By Proposition 1.12 (second part), every field is an integral domain, but not conversely: indeed, $\mathbb{Z}$ is an integral domain, *but* it is not a field. Remember:

$$\text{field} \Rightarrow \text{integral domain},$$

$$\text{integral domain} \not\Rightarrow \text{field}.$$

There is a situation, however, in which the two notions coincide:

Proposition 1.15. *Assume R is a finite commutative ring; then R is an integral domain if and only if it is a field.*

Proof. One implication holds for all rings, as pointed out above; thus we only have to verify that if R is a finite integral domain, then it is a field. This amounts to verifying that if a is a non-zero-divisor in a *finite* (commutative) ring R , then it is a unit in R .

Now, if a is a non-zero-divisor, then multiplication by a in R is injective (Proposition 1.9); hence it is surjective, as the ring is finite, by the pigeon-hole principle; hence a is a unit, by Proposition 1.12. $\square$

Remark 1.16. A little surprisingly, the hypothesis of commutativity in Proposition 1.15 is actually superfluous: a theorem known as *Wedderburn's little theorem* shows that *finite division rings are necessarily commutative*. The reader will prove this fact in a distant future (Exercise VII.5.14). $\lrcorner$

Example 1.17. The group of units in the ring $\mathbb{Z}/n\mathbb{Z}$ is precisely the group $(\mathbb{Z}/n\mathbb{Z})^*$ introduced in §II.2.3: indeed, a class $[m]_n$ is a unit if and only if (right-) multiplication by $[m]_n$ is surjective (by Proposition 1.12), if and only if the map $a \mapsto a[m]_n$ is surjective, if and only if $[m]_n$ generates $\mathbb{Z}/n\mathbb{Z}$, if and only if $\gcd(m, n) = 1$ (Corollary II.2.5), if and only if $[m]_n \in (\mathbb{Z}/n\mathbb{Z})^*$.

In particular, those n for which all nonzero elements of $\mathbb{Z}/n\mathbb{Z}$ are units (that is, for which $\mathbb{Z}/n\mathbb{Z}$ is a *field*) are precisely those $n \in \mathbb{Z}$ for which $\gcd(m, n) = 1$ for all m that are *not* multiples of n ; this is the case if and only if n is prime. Putting this together with Proposition 1.15, we get the pretty classification (for integers $p \neq 0$)

$$\mathbb{Z}/p\mathbb{Z} \text{ integral domain} \iff \mathbb{Z}/p\mathbb{Z} \text{ field} \iff p \text{ prime},$$

which the reader is well advised to remember firmly. $\lrcorner$

Example 1.18. The rings $\mathbb{Z}/p\mathbb{Z}$, with p prime, are *not* the only finite fields. In fact, for every prime p and every integer $r > 0$ there is a (unique, in a suitable sense) multiplication on the product group

$$\underbrace{\mathbb{Z}/p\mathbb{Z} \times \cdots \times \mathbb{Z}/p\mathbb{Z}}_{r \text{ times}}$$

making it into a field. A discussion of these fields will have to wait until we have accumulated much more material (cf. §VII.5.1), but the reader could already try to construct small examples ‘by hand’ (cf. Exercise 1.11). $\lrcorner$

1.3. Polynomial rings. We will study polynomial rings in some depth, especially over fields; they are another class of examples that is to some extent already familiar to our reader. I will capitalize on this familiarity and avoid a truly formal (and truly tedious) definition.

Definition 1.19. Let R be a ring. A *polynomial* $f(x)$ in the *indeterminate* x and with *coefficients* in R is a *finite* linear combination of nonnegative ‘powers’ of x with coefficients in R :

$$f(x) = \sum_{i \geq 0} a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots,$$

where all a_i are elements of R (the *coefficients*) and we require $a_i = 0$ for $i \gg 0$. Two polynomials are taken to be equal if all the coefficients are equal:

$$\sum_{i \geq 0} a_i x^i = \sum_{i \geq 0} b_i x^i \iff (\forall i \geq 0) : a_i = b_i. \quad \lrcorner$$

The set of polynomials in x over R is denoted by $R[x]$. Since all but finitely many a_i are assumed to be 0, one usually employs the notation

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n$$

for $\sum_{i \geq 0} a_i x^i$, if $a_i = 0$ for $i > n$.

At this point the reader should just view all of this as a *notation*: a polynomial really stands for an element of an infinite direct sum of the group $(R, +)$. The ‘polynomial’ notation is more suggestive as it hints at what *operations* we are going to impose on $R[x]$: if

$$f(x) = \sum_{i \geq 0} a_i x^i \quad \text{and} \quad g(x) = \sum_{i \geq 0} b_i x^i,$$

then we define

$$f(x) + g(x) := \sum_{i \geq 0} (a_i + b_i) x^i$$

and

$$f(x) \cdot g(x) := \sum_{k \geq 0} \sum_{i+j=k} a_i b_j x^{i+j}.$$

To clarify this latter definition, see how it works for small k : $f(x) \cdot g(x)$ equals

$$a_0 b_0 + (a_0 b_1 + a_1 b_0)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 + (a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0)x^3 + \cdots,$$

that is, business as usual.

It is essentially straightforward (Exercise 1.13) to check that $R[x]$, with these operations, is a ring; the identity 1 of R is the identity of $R[x]$, when viewed as a polynomial (that is, $1_{R[x]} = 1_R + 0x + 0x^2 + \cdots$).

The *degree* of a nonzero polynomial $f(x) = \sum_{i \geq 0} a_i x^i$, denoted $\deg f(x)$, is the largest integer d for which $a_d \neq 0$. This notion is very useful, but really behaves well (Exercise 1.14) only if R is an integral domain: for example, note that over $R = \mathbb{Z}/6\mathbb{Z}$

$$\deg([1] + [2]x) = 1, \quad \deg([1] + [3]x) = 1, \quad \text{but}$$

$$\deg([1] + [2]x) \cdot ([1] + [3]x) = \deg([1] + [5]x) = 1 \neq 1 + 1.$$

Polynomials of degree 0 (together with 0) are called *constants*; they form a ‘copy’ of R in $R[x]$, since the operations $+$, $\cdot$ on constant polynomials are nothing but the original operations in R , up to this identification. It is sometimes convenient to assign to the polynomial 0 the degree $-\infty$.

Polynomial rings in more indeterminates may be obtained by iterating this construction:

$$R[x, y, z] := R[x][y][z];$$

elements of this ring may be written as ‘ordinary’ polynomials in three indeterminates and are manipulated as usual. It can be checked easily that the construction does not really depend on the order in which the indeterminates are listed, in the

sense that different orderings lead to *isomorphic* rings (in the sense soon to be defined officially). Different indeterminates commute with each other; constructions analogous to polynomial rings, but with noncommuting indeterminates, are also very important, but we will not develop them in this book (we will glance at one such notion in Example VIII.4.17).

We will occasionally consider a polynomial ring in *infinitely many* indeterminates: for example, we will denote by $R[x_1, x_2, \dots]$ the case of countably many indeterminates. Keep in mind, however, that polynomials are *finite* linear combinations of *finite* products of the indeterminates; in particular, every given element of $R[x_1, x_2, \dots]$ only involves *finitely many* indeterminates. An honest definition of this ring involves *direct limits*, which await us in §VIII.1.4.

Rings of *power series* may be defined and are very useful; the ring of series $\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + a_2 x^2 + \dots$ in x with coefficients in R , and evident operations, is denoted $R[[x]]$. Regrettably, we will only occasionally encounter these rings in this book.

The ring $R[x]$ is (clearly) commutative if R is commutative; it is an integral domain if R is an integral domain (Exercise 1.15); but it has no chances of being a field even if R is a field, since x has no inverse in $R[x]$. The question of which properties of R are ‘inherited’ by $R[x]$ is subtle and important, and we will give it a great deal of attention in later sections.

1.4. Monoid rings. The polynomial ring is an instance of a rather general construction, which is occasionally very useful. A *semigroup* is a set endowed with an associative operation; a *monoid* is a semigroup with an identity element. Thus a group is a monoid in which every element has an inverse; positive integers with ordinary addition form a semigroup, while the set $\mathbb{N}$ of *natural numbers* (that is, nonnegative integers²) is a monoid under addition.

Given a monoid $(M, \cdot)$ and a ring R , we can obtain a new ring $R[M]$ as follows. Elements of $R[M]$ are formal linear combinations

$$\sum_{m \in M} a_m \cdot m$$

where the ‘coefficients’ a_m are elements of R and $a_m \neq 0$ for at most finitely many summands (hence, as in §1.3, as an abelian group $R[M]$ is nothing but the direct sum $R^{\oplus M}$). Operations in $R[M]$ are defined by

$$\begin{aligned} \left(\sum_{m \in M} a_m \cdot m \right) + \left(\sum_{m \in M} b_m \cdot m \right) &= \sum_{m \in M} (a_m + b_m) \cdot m, \\ \left(\sum_{m \in M} a_m \cdot m \right) \cdot \left(\sum_{m \in M} b_m \cdot m \right) &= \sum_{m \in M} \sum_{m_1 m_2 = m} (a_{m_1} b_{m_2}) \cdot m. \end{aligned}$$

The identity in $R[M]$ is $1_R \cdot 1_M$, viewed as a formal sum in which all other summands have 0 as coefficient.

²Some disagree, and insist that $\mathbb{N}$ should not include 0.

The reader will hopefully see the similarity with the construction of the polynomial ring $R[x]$ in §1.3; in fact (Exercise 1.17) the polynomial ring $R[x]$ may be interpreted as $R[\mathbb{N}]$.

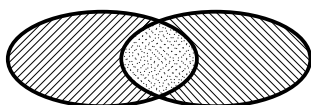
Group rings are the result of this construction when M is in fact a group. The group ring $R[\mathbb{Z}]$ is a ring of ‘Laurent polynomials’ $R[x, x^{-1}]$, allowing for negative as well as positive exponents.

Exercises

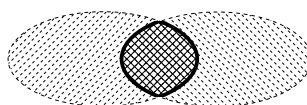
1.1. ▷ Prove that if $0 = 1$ in a ring R , then R is a zero-ring. [§1.2]

1.2. ¬ Let S be a set, and define operations on the power set $\mathcal{P}(S)$ of S by setting $\forall A, B \in \mathcal{P}(S)$

$$A + B := (A \cup B) \setminus (A \cap B), \quad A \cdot B = A \cap B :$$



$A + B$



$A \cdot B$

(where the solid black contour indicates the set included in the operation). Prove that $(\mathcal{P}(S), +, \cdot)$ is a commutative ring. [2.3, 3.15]

1.3. ¬ Let R be a ring, and let S be any set. Explain how to endow the set R^S of set-functions $S \rightarrow R$ of two operations $+$, $\cdot$ so as to make R^S into a ring, such that R^S is just a copy of R if S is a singleton. [2.3]

1.4. ▷ The set of $n \times n$ matrices with entries in a ring R is denoted $\mathcal{M}_n(R)$. Prove that componentwise addition and matrix multiplication make $\mathcal{M}_n(R)$ into a ring, for any ring R . The notation $\mathfrak{gl}_n(R)$ is also commonly used, especially for $R = \mathbb{R}$ or $\mathbb{C}$ (although this indicates one is considering them as *Lie algebras*) in parallel with the analogous notation for the corresponding groups of units; cf. Exercise II.6.1. In fact, the parallel continues with the definition of the following sets of matrices:

- $\mathfrak{sl}_n(\mathbb{R}) = \{M \in \mathfrak{gl}_n(\mathbb{R}) \mid \text{tr}(M) = 0\};$
- $\mathfrak{sl}_n(\mathbb{C}) = \{M \in \mathfrak{gl}_n(\mathbb{C}) \mid \text{tr}(M) = 0\};$
- $\mathfrak{so}_n(\mathbb{R}) = \{M \in \mathfrak{sl}_n(\mathbb{R}) \mid M + M^t = 0\};$
- $\mathfrak{su}(n) = \{M \in \mathfrak{sl}_n(\mathbb{C}) \mid M + M^\dagger = 0\}.$

Here $\text{tr}(M)$ is the *trace* of M , that is, the sum of its diagonal entries. The other notation matches the notation used in Exercise II.6.1. Can we make rings of these sets by endowing them with ordinary addition and multiplication of matrices? (These sets are all Lie algebras; cf. Exercise VI.1.4.) [§1.2, 2.4, 5.9, VI.1.2, VI.1.4]

1.5. Let R be a ring. If a, b are zero-divisors in R , is $a + b$ necessarily a zero-divisor?

1.6. ¬ An element a of a ring R is *nilpotent* if $a^n = 0$ for some n .

- Prove that if a and b are nilpotent in R and $ab = ba$, then $a + b$ is also nilpotent.
- Is the hypothesis $ab = ba$ in the previous statement necessary for its conclusion to hold?

[3.12]

1.7. Prove that $[m]$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$ if and only if m is divisible by all prime factors of n .

1.8. Prove that $x = \pm 1$ are the only solutions to the equation $x^2 = 1$ in an integral domain. Find a ring in which the equation $x^2 = 1$ has more than 2 solutions.

1.9. ▷ Prove Proposition 1.12. [§1.2]

1.10. Let R be a ring. Prove that if $a \in R$ is a right-unit and has two or more left-inverses, then a is *not* a left-zero-divisor and *is* a right-zero-divisor.

1.11. ▷ Construct a field with 4 elements: as mentioned in the text, the underlying abelian group will have to be $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$; $(0, 0)$ will be the zero element, and $(1, 1)$ will be the multiplicative identity. The question is what $(0, 1) \cdot (0, 1)$, $(0, 1) \cdot (1, 0)$, $(1, 0) \cdot (1, 0)$ must be, in order to get a *field*. [§1.2, §V.5.1]

1.12. ▷ Just as complex numbers may be viewed as combinations $a + bi$, where $a, b \in \mathbb{R}$ and i satisfies the relation $i^2 = -1$ (and commutes with $\mathbb{R}$), we may construct a ring³ $\mathbb{H}$ by considering linear combinations $a + bi + cj + dk$ where $a, b, c, d \in \mathbb{R}$ and i, j, k commute with $\mathbb{R}$ and satisfy the following relations:

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

Addition in $\mathbb{H}$ is defined componentwise, while multiplication is defined by imposing distributivity and applying the relations. For example,

$$(1 + i + j) \cdot (2 + k) = 1 \cdot 2 + i \cdot 2 + j \cdot 2 + 1 \cdot k + i \cdot k + j \cdot k = 2 + 2i + 2j + k - j + i = 2 + 3i + j + k.$$

- Verify that this prescription does indeed define a ring.
- Compute $(a + bi + cj + dk)(a - bi - cj - dk)$, where $a, b, c, d \in \mathbb{R}$.
- Prove that $\mathbb{H}$ is a division ring.

Elements of $\mathbb{H}$ are called *quaternions*. Note that $Q_8 := \{\pm 1, \pm i, \pm j, \pm k\}$ forms a subgroup of the group of units of $\mathbb{H}$; it is a noncommutative group of order 8, called the *quaternionic group*.

- List all subgroups of Q_8 , and prove that they are all normal.
- Prove that Q_8, D_8 are not isomorphic.
- Prove that Q_8 admits the presentation $(x, y \mid x^2y^{-2}, y^4, xyx^{-1}y)$.

[§II.7.1, 2.4, IV.1.12, IV.5.16, IV.5.17, V.6.19]

1.13. ▷ Verify that the multiplication defined in $R[x]$ is associative. [§1.3]

³The letter $\mathbb{H}$ is chosen in honor of William Rowan Hamilton.

1.14. ▷ Let R be a ring, and let $f(x), g(x) \in R[x]$ be nonzero polynomials. Prove that

$$\deg(f(x) + g(x)) \leq \max(\deg(f(x)), \deg(g(x))).$$

Assuming that R is an integral domain, prove that

$$\deg(f(x) \cdot g(x)) = \deg(f(x)) + \deg(g(x)).$$

[§1.3]

1.15. ▷ Prove that $R[x]$ is an integral domain if and only if R is an integral domain.

[§1.3]

1.16. Let R be a ring, and consider the ring of power series $R[[x]]$ (cf. §1.3).

(i) Prove that a power series $a_0 + a_1x + a_2x^2 + \cdots$ is a unit in $R[[x]]$ if and only if a_0 is a unit in R . What is the inverse of $1 - x$ in $R[[x]]$?

(ii) Prove that $R[[x]]$ is an integral domain if and only if R is.

1.17. ▷ Explain in what sense $R[x]$ agrees with the monoid ring $R[\mathbb{N}]$. [§1.4]

2. The category Ring

2.1. Ring homomorphisms. *Ring homomorphisms* are defined in the natural way: if R, S are rings, a function $\varphi : R \rightarrow S$ is a ring homomorphism if it preserves *both* operations and the identity element. That is, φ must be a homomorphism of the underlying abelian groups,

$$(\forall a, b \in R) : \quad \varphi(a + b) = \varphi(a) + \varphi(b),$$

it must preserve the operation of multiplication,

$$(\forall a, b \in R) : \quad \varphi(ab) = \varphi(a)\varphi(b),$$

and finally

$$\varphi(1_R) = 1_S.$$

It is evident that rings form a category, with ring homomorphisms as morphisms. I will denote this category by **Ring**.

The zero-ring is clearly final in **Ring**. However, note that it is *not* initial: because of the requirement that ring homomorphisms send 1 to 1, the only rings to which zero-rings map homomorphically are the zero-rings (Exercise 2.1).

The category **Ring** *does* have initial objects: the ring of integers $\mathbb{Z}$ (with the usual operations $+, \cdot$) is initial in **Ring**. Indeed, for every ring R we can define a *group* homomorphism $\varphi : \mathbb{Z} \rightarrow R$ by

$$(\forall n \in \mathbb{Z}) : \quad \varphi(n) = n \cdot 1_R,$$

that is, as the ‘exponential map’ ϵ_{1_R} corresponding to $1_R \in R$; cf. §II.4.1. But φ is in fact a *ring* homomorphism, since $\varphi(1) = 1_R$, and

$$\varphi(mn) = (mn)1_R = m(n1_R) \stackrel{!}{=} (m1_R) \cdot (n1_R) = \varphi(m) \cdot \varphi(n),$$

where the equality $\stackrel{!}{=}$ holds by the distributivity axiom⁴. This ring homomorphism is unique, since it is determined by the requirement that $\varphi(1) = 1_R$ and by the fact that φ preserves addition. Thus for every ring R there exists one and only one ring homomorphism $\mathbb{Z} \rightarrow R$, showing that $\mathbb{Z}$ is initial in $\mathbf{Ring}$.

This should already convince the reader that $\mathbb{Z}$ is a *very special* ring. There is in fact nothing arbitrary about $\mathbb{Z}$: knowledge of the *group* $\mathbb{Z}$ determines the *ring* structure of $\mathbb{Z}$, as will be underscored below. This is one of the main reasons why I included the ‘identity’ axiom in the list in Definition 1.1: there are many nonisomorphic structures of ‘ring without identity’ on the group $(\mathbb{Z}, +)$ (cf. Exercise 2.15), but only one structure of ring *with* identity (Exercise 2.16).

Ring homomorphisms preserve units: that is, if u is a (left-, resp., right-) unit in R and $\varphi : R \rightarrow S$ is a ring homomorphism, then $\varphi(u)$ is a (left-, resp., right-) unit. Indeed, if v is a (right-, say) inverse of u , then

$$\varphi(u)\varphi(v) = \varphi(uv) = \varphi(1_R) = 1_S,$$

so that $\varphi(v)$ is a (right-) inverse of $\varphi(u)$.

On the other hand, the image of a non-zero-divisor by a ring homomorphism may well be a zero-divisor: for example, the canonical projection $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$ is a ring homomorphism, and 2 is a *non-zero-divisor* in $\mathbb{Z}$, yet $\pi(2) = [2]_6$ is a zero-divisor.

2.2. Universal property of polynomial rings. Polynomial rings satisfy a universal property not unlike the one for free groups explored in §II.5.2. The simplest (and possibly most useful) case is for the polynomial rings $\mathbb{Z}[x_1, \dots, x_n]$, and with respect to *commutative* rings; I will leave the reader the pleasure of stating and proving fancier notions.

Let $A = \{a_1, \dots, a_n\}$ be a set of order n . Consider the category $\mathcal{R}_A$ whose objects are pairs (j, R) , where R is a commutative ring⁵ and

$$j : A \rightarrow R$$

is a set-function (cf. §II.5.2!); morphisms

$$(j_1, R_1) \rightarrow (j_2, R_2)$$

are commutative diagrams

$$\begin{array}{ccc} R_1 & \xrightarrow{\varphi} & R_2 \\ j_1 \uparrow & \nearrow j_2 & \\ A & & \end{array}$$

in which φ is a *ring* homomorphism.

For example, $(i, \mathbb{Z}[x_1, \dots, x_n])$ is an object of $\mathcal{R}_A$, where $i : A \rightarrow \mathbb{Z}[x_1, \dots, x_n]$ sends a_k to x_k .

⁴The reader should parse this display carefully, as there is a potentially confusing mix of two operations: multiples (such as $m1_R$) and multiplication in R (explicitly denoted here by $\cdot$).

⁵For the reader interested in generalizations: only the requirement that $j(a_1), \dots, j(a_n)$ commute with one another is needed here.

Proposition 2.1. $(i, \mathbb{Z}[x_1, \dots, x_n])$ is initial in $\mathcal{R}_A$.

Proof. Let (j, R) be an arbitrary object of $\mathcal{R}_A$; we have to show that there is a unique morphism $(i, \mathbb{Z}[x_1, \dots, x_n]) \rightarrow (j, R)$, that is, there exists exactly one ring homomorphism $\varphi : \mathbb{Z}[x_1, \dots, x_n] \rightarrow R$ such that

$$\begin{array}{ccc} \mathbb{Z}[x_1, \dots, x_n] & \xrightarrow{\varphi} & R \\ \uparrow i & \nearrow j & \\ A & & \end{array}$$

commutes.

As usual in these verifications, the key point is that the requirements posed on φ force its definition. The postulated commutativity of the diagram means that $\varphi(x_k) = j(a_k)$ for $k = 1, \dots, n$. Then, since φ must be a ring homomorphism, necessarily

$$\begin{aligned} \varphi\left(\sum m_{i_1 \dots i_n} x_1^{i_1} \dots x_n^{i_n}\right) &= \sum \varphi(m_{i_1 \dots i_n}) \varphi(x_1)^{i_1} \dots \varphi(x_n)^{i_n} \\ &= \sum \iota(m_{i_1 \dots i_n}) j(a_1)^{i_1} \dots j(a_n)^{i_n}, \end{aligned}$$

where $\iota : \mathbb{Z} \rightarrow R$ is the unique ring homomorphism (as $\mathbb{Z}$ is initial in Ring).

Thus, if φ exists, then it is unique. On the other hand, the formula we just obtained clearly⁶ preserves the operations and sends 1 to 1, so it does define a ring homomorphism, concluding the proof. $\square$

Example 2.2. For $n = 1$, Proposition 2.1 says that if s is any element of a ring S , then there is a unique ring homomorphism $\mathbb{Z}[x] \rightarrow S$ sending x to s and ‘extending’ the unique ring homomorphism $\iota : \mathbb{Z} \rightarrow S$. In this case the commutativity of S is immaterial (why?). $\lrcorner$

Example 2.3. More generally, let $\alpha : R \rightarrow S$ be a fixed ring homomorphism, and let $s \in S$ be an element commuting with $\alpha(r)$ for all $r \in R$. Then there is a unique ring homomorphism $\bar{\alpha} : R[x] \rightarrow S$ extending α and sending x to s (Exercise 2.6).

In particular, we get an ‘evaluation map’ for polynomials over commutative rings as follows. Given a polynomial $f(x) = \sum_{i \geq 0} a_i x^i \in R[x]$, every $r \in R$ determines an element

$$f(r) = \sum_{i \geq 0} a_i r^i :$$

this may be viewed as $\bar{\alpha}(f(x))$, where $\bar{\alpha}$ is obtained as above with $\alpha = \text{id}_R : R \rightarrow R$ and $s = r$.

Thus, every polynomial $f(x)$ determines a *polynomial function* $f : R \rightarrow R$, defined by $r \mapsto f(r)$. It is a good idea to keep the two concepts of ‘polynomial’ and ‘polynomial function’ well distinct (cf. Exercise 2.7). $\lrcorner$

⁶Well, it is really clear that it preserves addition; multiplication requires a bit of work, which the reader would be well advised to perform. This is where the commutativity of R is used.

2.3. Monomorphisms and epimorphisms. The *kernel* of a homomorphism $\varphi : R \rightarrow S$ of rings is

$$\ker \varphi := \{r \in R \mid \varphi(r) = 0\} :$$

that is, it is nothing but the kernel of φ when the latter is viewed as a homomorphism of *groups*. As such, $\ker \varphi$ is a subgroup of $(R, +)$; it satisfies an even stronger requirement⁷, which we will explore at great length in §3.

The reader may hope that an analog to Proposition II.6.12 holds in Ring, and this is indeed the case:

Proposition 2.4. *For a ring homomorphism $\varphi : R \rightarrow S$, the following are equivalent:*

- (a) φ is a monomorphism;
- (b) $\ker \varphi = \{0\}$;
- (c) φ is injective (as a set-function).

Proof. We prove (a) $\implies$ (b), leaving the rest to the reader. Assume $\varphi : R \rightarrow S$ is a monomorphism and $r \in \ker \varphi$. Applying the extension property of Example 2.2, we obtain unique ring homomorphisms $\text{ev}_r : \mathbb{Z}[x] \rightarrow R$ such that $\text{ev}_r(x) = r$ and $\text{ev}_0 : \mathbb{Z}[x] \rightarrow R$ such that $\text{ev}_0(x) = 0$. Consider the parallel ring homomorphisms:

$$\mathbb{Z}[x] \begin{array}{c} \xrightarrow{\text{ev}_r} \\ \xrightarrow{\text{ev}_0} \end{array} R \xrightarrow{\varphi} S :$$

since $\varphi(r) = 0 = \varphi(0)$, the two compositions $\varphi \circ \text{ev}_r$, $\varphi \circ \text{ev}_0$ agree (because they agree on $\mathbb{Z}$ and they agree on x); hence $\text{ev}_r = \text{ev}_0$ since φ is a monomorphism. Therefore

$$r = \text{ev}_r(x) = \text{ev}_0(x) = 0.$$

This proves $r \in \ker \varphi \implies r = 0$, that is, (b). □

By Proposition 2.4, if $S \rightarrow R$ is a monomorphism, then S may be identified with a subset of R ; the following definition formalizes this situation.

Definition 2.5. A *subring* S of a ring R is a ring whose underlying set is a subset of R and such that the inclusion function $S \hookrightarrow R$ is a ring homomorphism. ┘

Equivalently, a subring of R is a subset $S \subseteq R$ which contains 1_R and satisfies the ring axioms with respect to the operations $+$, $\cdot$ induced from R . It is immediately checked that $S \subseteq R$ is a subring if it is a subgroup of $(R, +)$, it is closed with respect to $\cdot$, and it contains 1_R . As a nonexample, the zero-ring is *not* a subring of a nonzero ring R (because it does not contain 1_R).

Proposition 2.4 may induce the reader to believe that Ring and Grp are rather similar from this general categorical standpoint. But a new phenomenon occurs concerning *epimorphisms*. A *surjective* map of rings is certainly an epimorphism in Ring, since it is already an epimorphism in Set—we have run into this argument earlier (e.g., ‘(c) $\implies$ (a)’ in Proposition II.8.18); but unlike as in Set, Grp, and Ab,

⁷Note that $\ker \varphi$ cannot be a subring in any reasonable sense, according to our definition of ring, since it does not contain a multiplicative identity in all but very pathological situations. It is a subring if the identity requirement is omitted.

epimorphisms *need not* be surjective⁸ in Ring. Indeed, consider the inclusion homomorphism of rings

$$\iota : \mathbb{Z} \hookrightarrow \mathbb{Q} :$$

ι is not surjective, and hence it is not an epimorphism in Set or Ab (can the reader ‘describe’ $\text{coker } \iota$? Exercise 2.12); but it *is* an epimorphism of rings. Indeed, if α_1, α_2 are parallel ring homomorphisms

$$\mathbb{Z} \xrightarrow{\iota} \mathbb{Q} \begin{matrix} \xrightarrow{\alpha_1} \\ \xrightarrow{\alpha_2} \end{matrix} R$$

and α_1, α_2 agree⁹ on $\mathbb{Z}$, say $\alpha = \alpha_1|_{\mathbb{Z}} = \alpha_2|_{\mathbb{Z}}$, then they must agree on $\mathbb{Q}$: because for $p, q \in \mathbb{Z}, q \neq 0$,

$$\alpha_i \left(\frac{p}{q} \right) = \alpha_i(p)\alpha_i(q^{-1}) = \alpha(p)\alpha(q)^{-1} \quad (i = 1, 2)$$

is the same for both¹⁰. Thus, ι satisfies the categorical requirement for epimorphisms (cf. §I.2.6).

Warning: Thus, in Ring, a homomorphism may well be both a monomorphism and an epimorphism *without* being an isomorphism!

2.4. Products. *Products* exist in Ring: if R_1, R_2 are rings, then $R_1 \times R_2$ may be defined by endowing the direct product of *groups* $R_1 \times R_2$ (cf. §II.3.4) with componentwise multiplication. Thus, both operations on $R_1 \times R_2$ are defined componentwise: $\forall (a_1, a_2), (b_1, b_2) \in R_1 \times R_2$,

$$\begin{aligned} (a_1, a_2) + (b_1, b_2) &:= (a_1 + b_1, a_2 + b_2), \\ (a_1, a_2) \cdot (b_1, b_2) &:= (a_1 \cdot b_1, a_2 \cdot b_2). \end{aligned}$$

The identity in $R_1 \times R_2$ is $(1_{R_1}, 1_{R_2})$. The reader will check (Exercise 2.13) that $R_1 \times R_2$ is indeed a *categorical* product in Ring.

The reader should keep in mind that in general this is *not* the only ring structure one can define on the direct product of underlying groups. For example, as mentioned in §1.2, one can define a *field* whose underlying group is $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, while the product ring $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ is very far from being a field (why?).

The situation with *coproducts* is more unfortunate—dealing with this in any generality (even for the case of *commutative* rings) requires *tensor products*, and by the time we develop tensor products (§VIII.2), we will have almost forgotten this question. But the universal property reviewed in §2.2 suffices to deal with simple examples, and the reader should work out one template case now, for fun (Exercise 2.14).

Incidentally—with the case of abelian groups in mind (§II.3.5), the reader may be tempted to consider a ‘direct sum of rings’, agreeing with the product in the finite

⁸Unfortunately, some references *define* ring epimorphisms as ‘surjective ring homomorphisms’; this should be discouraged.

⁹As they must, since $\mathbb{Z}$ is initial.

¹⁰Note that $\alpha(q)$ must have a double-sided inverse in R since q has one in $\mathbb{Q}$, namely $1/q$. In particular, $\alpha_1(q^{-1}) = \alpha_2(q^{-1})$ must agree, since they must equal this unique inverse of $\alpha(q)$; cf. Proposition 1.12.

case and maybe giving something interesting in general. This is less promising than it looks: it does *not* satisfy the universal property of coproducts in **Ring** (why?), and, further, the ‘infinite version’ is not a ring because it is missing the identity.

2.5. $\text{End}_{\text{Ab}}(G)$. This is as good a place as any to expand a little on one of the main examples of rings, and one that is not quite as special as the examples reviewed in §1.2.

For every abelian group G , the group $\text{End}_{\text{Ab}}(G) := \text{Hom}_{\text{Ab}}(G, G)$ of endomorphisms of G is a *ring*, under the operations of addition and composition (as discussed in the paragraph preceding Definition 1.1). Indeed, associativity follows from the category axioms, and distributivity is immediately checked.

It is instructive to examine carefully the endomorphism ring of the *abelian group* $(\mathbb{Z}, +)$:

Proposition 2.6. $\text{End}_{\text{Ab}}(\mathbb{Z}) \cong \mathbb{Z}$ as rings.

Proof. Consider the function

$$\varphi : \text{End}_{\text{Ab}}(\mathbb{Z}) \rightarrow \mathbb{Z}$$

defined by

$$\varphi(\alpha) = \alpha(1)$$

for all group homomorphisms $\alpha : \mathbb{Z} \rightarrow \mathbb{Z}$. Then φ is a group homomorphism: the addition in $\text{End}_{\text{Ab}}(\mathbb{Z})$ is defined so that $\forall n \in \mathbb{Z}$

$$(\alpha + \beta)(n) = \alpha(n) + \beta(n)$$

(cf. §II.4.4); in particular

$$\varphi(\alpha + \beta) = (\alpha + \beta)(1) = \alpha(1) + \beta(1) = \varphi(\alpha) + \varphi(\beta).$$

Further, φ is a *ring* homomorphism. Indeed, for α, β in $\text{End}_{\text{Ab}}(\mathbb{Z})$ denote $\alpha(1)$ by a ; then

$$\alpha(n) = n\alpha(1) = na = an$$

for all $n \in \mathbb{Z}$; in particular,

$$\alpha(\beta(1)) = a\beta(1) = \alpha(1)\beta(1).$$

Therefore,

$$\varphi(\alpha \circ \beta) = (\alpha \circ \beta)(1) = \alpha(\beta(1)) = \alpha(1)\beta(1) = \varphi(\alpha)\varphi(\beta)$$

as needed. Also, $\varphi(\text{id}_{\mathbb{Z}}) = \text{id}_{\mathbb{Z}}(1) = 1$.

Finally, φ has an inverse: for $a \in \mathbb{Z}$, let $\psi(a)$ be the homomorphism $\alpha : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by

$$(\forall n \in \mathbb{Z}) : \quad \alpha(n) = an;$$

the reader will easily check that ψ is a ring homomorphism and inverse to φ .

Therefore φ is a ring isomorphism, verifying the statement. $\square$

Proposition 2.6 gives a sense in which the ring structure on $\mathbb{Z}$ is truly ‘natural’: this structure arises naturally from categorical considerations, there is nothing ‘arbitrary’ about it.

More generally, the rings $\text{End}_{\text{Ab}}(G)$ and other rings arising as endomorphism rings of other structures (e.g., vector spaces) are arguably the most important class of examples of rings. The entire theory of *modules* is based on the observation that $\text{End}_{\text{Ab}}(G)$ is a ring for every abelian group G .

Some of the notions reviewed in §1.2 are expressed very concretely in these endomorphism rings; for example, the *group of units* in $\text{End}_{\text{Ab}}(G)$ is nothing but $\text{Aut}_{\text{Ab}}(G)$.

Every ring R interacts with the ring of endomorphisms of the underlying abelian group $(R, +)$. For $r \in R$, define left- and right-multiplication by r by λ_r, μ_r , respectively. That is, $\forall a \in R$

$$\lambda_r(a) = ra, \quad \mu_r(a) = ar.$$

The following observation is nothing more than easy notation juggling, but it is useful; it is a ‘ring analog’ of Cayley’s theorem (Theorem II.9.5):

Proposition 2.7. *Let R be a ring. Then the function $r \mapsto \lambda_r$ is an injective ring homomorphism*

$$\lambda : R \rightarrow \text{End}_{\text{Ab}}(R).$$

Proof. For any $r \in R$ and for all $a, b \in R$, distributivity gives

$$\lambda_r(a + b) = r(a + b) = ra + rb = \lambda_r(a) + \lambda_r(b) :$$

this shows that λ_r is indeed an endomorphism of the *group* $(R, +)$, that is, $\lambda_r \in \text{End}_{\text{Ab}}(R)$.

The function $\lambda : R \rightarrow \text{End}_{\text{Ab}}(R)$ defined by the assignment $r \mapsto \lambda_r$ is clearly injective, since if $r \neq s$, then

$$\lambda_r(1) = r \neq s = \lambda_s(1),$$

so that $\lambda_r \neq \lambda_s$.

We have to verify that λ is in fact a homomorphism of rings. Recall that the addition in $\text{End}_{\text{Ab}}(R)$ is inherited from R (cf. §II.4.4): for all $r, s \in R$, $\lambda_r + \lambda_s$ is defined by

$$(\forall a \in R) : (\lambda_r + \lambda_s)(a) = \lambda_r(a) + \lambda_s(a).$$

Thus, the fact that λ preserves addition is an immediate consequence of distributivity: for all $r, s, a \in R$,

$$\lambda_{r+s}(a) = (r + s)a = ra + sa = \lambda_r(a) + \lambda_s(a).$$

As for multiplication, it is associativity’s turn to do its job:

$$\lambda_{rs}(a) = (rs)a = r(sa) = r\lambda_s(a) = \lambda_r(\lambda_s(a)) = (\lambda_r \circ \lambda_s)(a).$$

Of course λ_1 is the identity, completing the verification. □

The function $\mu : R \rightarrow \text{End}_{\text{Ab}}(R)$ defined by $r \mapsto \mu_r$ is ‘almost’ a ring homomorphism: the reader will check (Exercise 2.18) that

$$\mu_{r+s} = \mu_r + \mu_s,$$

$$\mu_{rs} = \mu_s \circ \mu_r,$$

and $\mu_1 = \text{id}_R$. That is, μ would in fact be a ring homomorphism if we ‘reversed’ multiplication¹¹ in R . Of course this issue disappears if R happens to be commutative (since then $\lambda = \mu$).

Exercises

2.1. ▷ Prove that if there is a homomorphism from a zero-ring to a ring R , then R is a zero-ring [§2.1]

2.2. Let R and S be rings, and let $\varphi : R \rightarrow S$ be a function preserving both operations $+$, $\cdot$.

- Prove that if φ is *surjective*, then necessarily $\varphi(1_R) = 1_S$.
- Prove that if $\varphi \neq 0$ and S is an integral domain, then $\varphi(1_R) = 1_S$.

(Therefore, in both cases φ is in fact a ring homomorphism).

2.3. Let S be a set, and consider the power set ring $\mathcal{P}(S)$ (Exercise 1.2) and the ring $(\mathbb{Z}/2\mathbb{Z})^S$ you constructed in Exercise 1.3. Prove that these two rings are isomorphic. (Cf. Exercise I.2.11.)

2.4. Define functions $\mathbb{H} \rightarrow \mathfrak{gl}_4(\mathbb{R})$ and $\mathbb{H} \rightarrow \mathfrak{gl}_2(\mathbb{C})$ (cf. Exercises 1.4 and 1.12) by

$$a + bi + cj + dk \mapsto \begin{pmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{pmatrix},$$

$$a + bi + cj + dk \mapsto \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix}$$

for all $a, b, c, d \in \mathbb{R}$. Prove that both functions are injective ring homomorphisms. Thus, quaternions may be viewed as real or complex matrices.

2.5. ¬ The *norm* of a quaternion $w = a + bi + cj + dk$, with $a, b, c, d \in \mathbb{R}$, is the real number $N(w) = a^2 + b^2 + c^2 + d^2$.

Prove that the function from the multiplicative group $\mathbb{H}^*$ of nonzero quaternions to the multiplicative group $\mathbb{R}^+$ of positive real numbers, defined by assigning to each nonzero quaternion its norm, is a homomorphism. Prove that the kernel of this homomorphism is isomorphic to $\text{SU}(2)$ (cf. Exercise II.6.3). [4.10, IV.5.17, V.6.19]

¹¹This issue is entirely analogous to the business of right-actions vs. left-actions; cf. §II.9.3, especially Exercise II.9.3.

2.6. $\triangleright$ Verify the ‘extension property’ of polynomial rings, stated in Example 2.3. [§2.2]

2.7. $\triangleright$ Let $R = \mathbb{Z}/2\mathbb{Z}$, and let $f(x) = x^2 - x$; note $f(x) \neq 0$. What is the polynomial function $R \rightarrow R$ determined by $f(x)$? [§2.2, §V.4.2, §V.5.1]

2.8. Prove that every subring of a field is an integral domain.

2.9. $\neg$ The *center* of a ring R consists of the elements a such that $ar = ra$ for all $r \in R$. Prove that the center is a subring of R .

Prove that the center of a division ring is a field. [2.11, IV.2.17, VII.5.14, VII.5.16]

2.10. $\neg$ The *centralizer* of an element a of a ring R consists of the elements $r \in R$ such that $ar = ra$. Prove that the centralizer of a is a subring of R , for every $a \in R$.

Prove that the center of R is the intersection of all its centralizers.

Prove that every centralizer in a division ring is a division ring. [2.11, IV.2.17, VII.5.16]

2.11. $\neg$ Let R be a division ring consisting of p^2 elements, where p is a prime. Prove that R is commutative, as follows:

- If R is not commutative, then its center C (Exercise 2.9) is a proper subring of R . Prove that C would then consist of p elements.
- Let $r \in R$, $r \notin C$. Prove that the centralizer of r (Exercise 2.10) contains both r and C .
- Deduce that the centralizer of r is the whole of R .
- Derive a contradiction, and conclude that R had to be commutative (hence, a field).

This is a particular case of Wedderburn’s theorem: every finite division ring is a field. [IV.2.17, VII.5.16]

2.12. $\triangleright$ Consider the inclusion map $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$. Describe the cokernel of ι in **Ab** and its cokernel in **Ring** (as defined by the appropriate universal property in the style of the one given in §II.8.6). [§2.3, §5]

2.13. $\triangleright$ Verify that the ‘componentwise’ product $R_1 \times R_2$ of two rings satisfies the universal property for products in a category, given in §I.5.4. [§2.4]

2.14. $\triangleright$ Verify that $\mathbb{Z}[x_1, x_2]$ (along with the evident morphisms) satisfies the universal property for the coproduct of two copies of $\mathbb{Z}[x]$ in the category of *commutative* rings. Explain why it does not satisfy it in **Ring**. [§2.4]

2.15. $\triangleright$ For $m > 1$, the abelian groups $(\mathbb{Z}, +)$ and $(m\mathbb{Z}, +)$ are manifestly isomorphic: the function $\varphi : \mathbb{Z} \rightarrow m\mathbb{Z}$, $n \mapsto mn$ is a group isomorphism. Use this isomorphism to transfer the structure of ‘ring without identity’ $(m\mathbb{Z}, +, \cdot)$ back onto $\mathbb{Z}$: give an explicit formula for the ‘multiplication’ $\bullet$ this defines on $\mathbb{Z}$ (that is, such that $\varphi(a \bullet b) = \varphi(a) \cdot \varphi(b)$). Explain why structures induced by different positive integers m are nonisomorphic as ‘rings without 1’.

(This shows that there are many different ways to give a structure of ring without identity to the *group* $(\mathbb{Z}, +)$. Compare this observation with Exercise 2.16.) [§2.1]

2.16. ▷ Prove that there is (up to isomorphism) only one structure of ring *with identity* on the abelian group $(\mathbb{Z}, +)$. (Hint: Let R be a ring whose underlying group is $\mathbb{Z}$. By Proposition 2.7, there is an injective ring homomorphism $\lambda : R \rightarrow \text{End}_{\text{Ab}}(R)$, and the latter is isomorphic to $\mathbb{Z}$ by Proposition 2.6. Prove that λ is surjective.) [§2.1, 2.15]

2.17. ¬ Let R be a ring, and let $E = \text{End}_{\text{Ab}}(R)$ be the ring of endomorphisms of the underlying abelian group $(R, +)$. Prove that the center of E is isomorphic to a subring of the center of R . (Prove that if $\alpha \in E$ commutes with all right-multiplications by elements of R , then α is left-multiplication by an element of R ; then use Proposition 2.7.)

2.18. ▷ Verify the statements made about right-multiplication μ , following Proposition 2.7. [§2.5]

2.19. Prove that for $n \in \mathbb{Z}$ a positive integer, $\text{End}_{\text{Ab}}(\mathbb{Z}/n\mathbb{Z})$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ as a ring.

3. Ideals and quotient rings

3.1. Ideals. In both **Set** and **Grp** we have been able to ‘classify’ surjective morphisms: in both cases, a surjective morphism is, up to natural identifications, a *quotient* by a (suitable) equivalence relation. In **Grp**, we have seen that such equivalence relations arise in fact from certain substructures: normal subgroups.

The situation in **Ring** is analogous. We will establish a canonical decomposition for rings, modeled after Theorems I.2.7 and II.8.1; the corresponding version of the ‘first isomorphism theorem’ (Corollary II.8.2) will identify every surjective ring homomorphism with a quotient by a suitable substructure. The role of normal subgroups will be played by *ideals*.

Definition 3.1. Let R be a ring. A subgroup I of $(R, +)$ is a *left-ideal* of R if $rI \subseteq I$ for all $r \in R$; that is,

$$(\forall r \in R)(\forall a \in I) : \quad ra \in I;$$

it is a *right-ideal* if $Ir \subseteq I$ for all $r \in R$; that is,

$$(\forall r \in R)(\forall a \in I) : \quad ar \in I.$$

A *two-sided ideal* is a subgroup I which is both a left- and a right-ideal. ┐

Of course in a commutative ring there is no distinction between left- and right-ideals. Even in the general setting, we will almost exclusively be concerned with two-sided ideals; thus I will omit qualifiers, and an *ideal* of a ring will implicitly be two-sided.

Remark 3.2. As seen in §2.3, a *subring* of R is a subset $S \subseteq R$ which contains 1_R and satisfies the ring axioms with respect to the operations $+$, $\cdot$ induced from R .

Ideals are close to being subrings: they are subgroups, and they are closed with respect to multiplication. *But* the only ideal of a ring R containing 1_R is R itself: this is an immediate consequence of the ‘absorption properties’ stated in Definition 3.1. Thus ideals are in general *not* subrings; they are ‘rngs’. $\perp$

Ideals are considerably more important than subrings in the development of the theory of rings¹². Of course the *image* of a ring homomorphism is necessarily a subring of the target; but a lesson learned from §II.8.1 and following is that *kernels* really capture the structure of a homomorphism, and kernels are ideals:

Example 3.3. Let $\varphi : R \rightarrow S$ be any ring homomorphism. Then $\ker \varphi$ is an ideal of R .

Indeed, we know already that $\ker \varphi$ is a subgroup; we have to verify the absorption properties. These are an immediate consequence of Lemma 1.2: for all $r \in R$, all $a \in \ker \varphi$, we have

$$\varphi(ra) = \varphi(r)\varphi(a) = \varphi(r) \cdot 0 = 0,$$

$$\varphi(ar) = \varphi(a)\varphi(r) = 0 \cdot \varphi(r) = 0.$$

More generally, it is easy to verify that the inverse image of an ideal is an ideal (Exercise 3.2) and $\{0_S\}$ is clearly an ideal of S .

Similarly to the situation with normal subgroups in the context of groups, we will soon see that ‘kernels of ring homomorphisms’ and ‘ideals’ are in fact equivalent concepts. $\perp$

3.2. Quotients. Let I be a subgroup of the abelian group $(R, +)$ of a ring R . Subgroups of abelian groups are automatically normal, so we have a quotient *group* R/I , whose elements are the cosets of I :

$$r + I$$

(written, of course, in additive notation). Further, we have a surjective *group* homomorphism

$$\pi : R \rightarrow \frac{R}{I}, \quad r \mapsto r + I.$$

As we have explored in great detail for groups, this construction satisfies a suitable universal property with respect to group homomorphisms (Theorem II.7.12). *Of course* we are now going to ask under what circumstances this construction can be performed in **Ring**, satisfying the analogous universal property with respect to ring homomorphisms.

That is, what should we ask of I , in order to have a ring structure on R/I , so that π becomes a ring homomorphism? Go figure this out for yourself, before reading ahead!

¹²Arguably, the reason is that ideals are precisely the *submodules* of a ring R .

As so often is the case, the requirement is precisely what tells us the answer. Since π will have to be a ring homomorphism, the multiplication in R/I *must* be as follows: for all $(a + I), (b + I)$ in R/I ,

$$(a + I) \cdot (b + I) = \pi(a) \cdot \pi(b) \stackrel{!}{=} \pi(ab) = ab + I,$$

where the equality $\stackrel{!}{=}$ is forced if π is to be a ring homomorphism.

This says that there is only one sensible ring structure on R/I , given by

$$(\forall a, b \in R) : \quad (a + I)(b + I) := ab + I.$$

The reader should realize right away that *if* this operation is well-defined, then it does make R/I into a ring: associativity will be inherited from the associativity in R , and the identity will simply be the coset $1 + I$ of 1.

So all is well, if the proposed operation is well-defined. *But* is it going to be well-defined?

Example 3.4. It need not be, if I is an arbitrary subgroup of R . For example, take $\mathbb{Z}$ as a subgroup of $\mathbb{Q}$; then

$$0 + \mathbb{Z} = 1 + \mathbb{Z}$$

($= \mathbb{Z}$) as elements of the group $\mathbb{Q}/\mathbb{Z}$, and $\frac{1}{2} + \mathbb{Z}$ is another coset, yet

$$0 \cdot \frac{1}{2} + \mathbb{Z} = \mathbb{Z} \neq \frac{1}{2} + \mathbb{Z} = 1 \cdot \frac{1}{2} + \mathbb{Z}. \quad \lrcorner$$

Assume then that the operation *is* well-defined, so that R/I is a ring and $\pi : R \rightarrow R/I$ is a ring homomorphism. What does this say about I ?

Answer: I is the kernel of $R \rightarrow R/I$, so necessarily I must be an ideal, as seen in Example 3.3!

Conversely, let us assume I is an ideal of R , and verify that the proposed prescription for the operation in R/I is well-defined. For this, suppose

$$a' + I = a'' + I \quad \text{and} \quad b' + I = b'' + I;$$

recall that this means that $a'' - a' \in I$, $b'' - b' \in I$; then

$$a''b'' - a'b' = a''b'' - a''b' + a''b' - a'b' = a''(b'' - b') + (a'' - a')b' \in I,$$

using both the left-absorption and right-absorption properties of Definition 3.1. This says precisely that

$$a'b' + I = a''b'' + I,$$

proving that the operation is well-defined.

Summarizing, we have verified that R/I is a ring, in such a way that the canonical projection $\pi : R \rightarrow R/I$ is a *ring* homomorphism, if and only if I is an *ideal* of R .

Definition 3.5. This ring R/I is called the *quotient ring* of R modulo I . $\lrcorner$

Example 3.6. We know that all subgroups of $(\mathbb{Z}, +)$ are of the form $n\mathbb{Z}$ for a nonnegative integer n (Proposition II.6.9). It is immediately verified that all subgroups of $\mathbb{Z}$ are in fact *ideals* of the ring $(\mathbb{Z}, +, \cdot)$. The quotients $\mathbb{Z}/n\mathbb{Z}$ are of course nothing but the rings so-denoted in §1.2 (and earlier).

The fact that $\mathbb{Z}$ is initial in **Ring** now prompts a natural definition. For a ring R , let $f : \mathbb{Z} \rightarrow R$ be the unique ring homomorphism, defined by $a \mapsto a \cdot 1_R$. Then $\ker f = n\mathbb{Z}$ for a well-defined nonnegative integer n determined by R .

Definition 3.7. The *characteristic* of R is this nonnegative integer n . $\lrcorner$

Thus, the characteristic of R is $n > 0$ if the order of 1_R as an element of $(R, +)$ is a positive integer n , while the characteristic is 0 if the order of 1_R is ∞ . $\lrcorner$

I have now fulfilled my promise to identify the notions of ideal and kernel of ring homomorphisms: every kernel is an ideal (cf. Example 3.3); and on the other hand every ideal I is the kernel of the ring homomorphisms $R \rightarrow R/I$. We have the slogan:

$$\text{kernel} \iff \text{ideal}$$

in the context of ring theory.

The key universal property holds in this context as it does for groups; cf. Theorem II.7.12. I reproduce the statement here for reference, but the reader should realize that very little needs to be proven at this point: the needed (group) homomorphism exists and is unique by Theorem II.7.12, and verifying it is a *ring* homomorphism is immediate.

Theorem 3.8. Let I be a two-sided ideal of a ring R . Then for every ring homomorphism $\varphi : R \rightarrow S$ such that $I \subseteq \ker \varphi$ there exists a unique ring homomorphism $\tilde{\varphi} : R/I \rightarrow S$ so that the diagram

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ & \searrow \pi & \nearrow \exists! \tilde{\varphi} \\ & R/I & \end{array}$$

commutes.

As a reminder to the lazy reader, $\tilde{\varphi}$ is defined by

$$\tilde{\varphi}(r + I) := \varphi(r);$$

(part of) the content of the theorem is that this function is well-defined (if $I \subseteq \ker \varphi$), and it is a ring homomorphism.

3.3. Canonical decomposition and consequences. As the reader should now expect, Theorem 3.8 is the key element in a standard decomposition of every ring homomorphism. This is entirely analogous to the decomposition of set-functions studied in §I.2.8 and the decomposition of group homomorphisms obtained in §II.8.1. Here is the statement:

Theorem 3.9. Every ring homomorphism $\varphi : R \rightarrow S$ may be decomposed as follows:

$$\begin{array}{ccccc} & & \varphi & & \\ & \curvearrowright & & \curvearrowleft & \\ R & \twoheadrightarrow & R/\ker \varphi & \xrightarrow[\tilde{\varphi}]{\sim} & \text{im } \varphi \hookrightarrow S \end{array}$$

where the isomorphism $\tilde{\varphi}$ in the middle is the homomorphism induced by φ (as in Theorem 3.8).

The reader will realize that this statement requires no proof at this point: the decomposition holds at the level of groups (by Theorem II.8.1) and the maps are all ring homomorphisms as observed earlier in this section.

The ‘first isomorphism theorem’ for rings is an immediate corollary:

Corollary 3.10. *Suppose $\varphi : R \rightarrow S$ is a surjective ring homomorphism. Then*

$$S \cong \frac{R}{\ker \varphi}.$$

As in the group case, the reader should develop the healthy instinctive reaction of viewing every surjective homomorphism of rings as a quotient (by an ideal), up to a natural identification.

Equally instinctive should be the realization that *ideals of a quotient R/I* are in one-to-one correspondence with *ideals of R containing I* . Once more, not a whole lot is new here: we already know (cf. §II.8.3) that the function

$$u : \{\text{subgroups } J \text{ of } R \text{ containing } I\} \rightarrow \{\text{subgroups of } R/I\}$$

defined by $u(J) = J/I$ is a bijection preserving inclusions; it takes a moment to check that J/I is an *ideal* of R/I if and only if J is an ideal of R . The main content of this observation is best packaged in the corresponding ‘third isomorphism theorem’:

Proposition 3.11. *Let I be an ideal of a ring R , and let J be an ideal of R containing I . Then J/I is an ideal of R/I , and*

$$\frac{R/I}{J/I} \cong \frac{R}{J}.$$

Proof. Since $I \subseteq J = \ker(R \rightarrow R/J)$, we have an induced ring homomorphism

$$\varphi : R/I \rightarrow R/J$$

by Theorem 3.8. Explicitly, $\varphi(r + I) = r + J$; φ is manifestly surjective. Since

$$\ker \varphi = \{r + I \mid \varphi(r + I) = J\} = \{r + I \mid r + J = J\} = \{r + I \mid r \in J\} = J/I,$$

we see that J/I is an ideal (since it is a kernel), and the stated isomorphism follows from Corollary 3.10. $\square$

What about the ‘second’ isomorphism theorem? This would be a relation between the ideals

$$\frac{I + J}{I}, \quad \frac{J}{I \cap J}$$

of the rings R/I , $R/(I \cap J)$, respectively, assuming I and J are ideals of R (and where it is immediately checked that $I + J$ and $I \cap J$ are indeed ideals¹³).

The reader may want to go back to §II.8.4 for the version of this story for groups. My feeling is that Ring is not the best place to play this game, since

¹³The notation $I + J$ should not surprise the reader: I , J are subgroups of the abelian group R , so we know what it means to ‘add’ them; cf. §II.7.1.

$(I + J)/I$, $J/(I \cap J)$ are not even rings according to my conventions. *Modules* will be a more natural context for this result.

In any case, the reader will benefit the most from exploring the matter on his/her own; cf. Exercise 3.17.

Exercises

3.1. Prove that the image of a ring homomorphism $\varphi : R \rightarrow S$ is a subring of S . What can you say about φ if its image is an ideal of S ? What can you say about φ if its kernel is a subring of R ?

3.2. $\triangleright$ Let $\varphi : R \rightarrow S$ be a ring homomorphism, and let J be an ideal of S . Prove that $I = \varphi^{-1}(J)$ is an ideal of R . [§3.1]

3.3. $\neg$ Let $\varphi : R \rightarrow S$ be a ring homomorphism, and let J be an ideal of R .

- Show that $\varphi(J)$ need not be an ideal of S .
- Assume that φ is surjective; then prove that $\varphi(J)$ is an ideal of S .
- Assume that φ is surjective, and let $I = \ker \varphi$; thus we may identify S with R/I . Let $\overline{J} = \varphi(J)$, an ideal of R/I by the previous point. Prove that

$$\frac{R/I}{\overline{J}} \cong \frac{R}{I+J}.$$

(Of course this is just a rehash of Proposition 3.11.) [4.11]

3.4. Let R be a ring such that every subgroup of $(R, +)$ is in fact an ideal of R . Prove that $R \cong \mathbb{Z}/n\mathbb{Z}$, where n is the characteristic of R .

3.5. $\neg$ Let J be a *two-sided* ideal of the ring $\mathcal{M}_n(R)$ of $n \times n$ matrices over a ring R . Prove that a matrix $A \in \mathcal{M}_n(R)$ belongs to J if and only if the matrices obtained by placing any entry of A in any position, and 0 elsewhere, belong to J . (Hint:

Carefully contemplate the operation $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ b & 0 & 0 \end{pmatrix}.$) [3.6]

3.6. $\neg$ Let J be a two-sided ideal of the ring $\mathcal{M}_n(R)$ of $n \times n$ matrices over a ring R , and let $I \subseteq R$ be the set of $(1, 1)$ entries of matrices in J . Prove that I is a two-sided ideal of R and J consists precisely of those matrices whose entries all belong to I . (Hint: Exercise 3.5.) [3.9]

3.7. Let R be a ring, and let $a \in R$. Prove that Ra is a left-ideal of R and aR is a right-ideal of R . Prove that a is a left-, resp. right-, unit if and only if $R = aR$, resp. $R = Ra$.

3.8. $\triangleright$ Prove that a ring R is a division ring if and only if its only left-ideals and right-ideals are $\{0\}$ and R .

In particular, a commutative ring R is a field if and only if the only ideals of R are $\{0\}$ and R . [3.9, §4.3]

3.9. $\neg$ Counterpoint to Exercise 3.8: It is *not* true that a ring R is a division ring if and only if its only two-sided ideals are $\{0\}$ and R . A nonzero ring with this property is said to be *simple*; by Exercise 3.8, fields are the only simple *commutative* rings.

Prove that $\mathcal{M}_n(\mathbb{R})$ is simple. (Use Exercise 3.6.) [4.20]

3.10. $\triangleright$ Let $\varphi : k \rightarrow R$ be a ring homomorphism, where k is a field and R is a nonzero ring. Prove that φ is *injective*. [§V.4.2, §V.5.2]

3.11. Let R be a ring containing $\mathbb{C}$ as a subring. Prove that there are no ring homomorphisms $R \rightarrow \mathbb{R}$.

3.12. $\triangleright$ Let R be a *commutative* ring. Prove that the set of nilpotent elements of R is an ideal of R . (Cf. Exercise 1.6. This ideal is called the *nilradical* of R .)

Find a noncommutative ring in which the set of nilpotent elements is *not* an ideal. [3.13, 4.18, V.3.13, §VII.2.3]

3.13. $\neg$ Let R be a commutative ring, and let N be its nilradical (cf. Exercise 3.12). Prove that R/N contains no nonzero nilpotent elements. (Such a ring is said to be *reduced*.) [4.6, VII.2.8]

3.14. $\neg$ Prove that the characteristic of an integral domain is either 0 or a prime integer. Do you know any ring of characteristic 1? [V.4.17]

3.15. $\neg$ A ring R is¹⁴ *Boolean* if $a^2 = a$ for all $a \in R$. Prove that $\mathcal{P}(S)$ is Boolean, for every set S (cf. Exercise 1.2). Prove that every nonzero Boolean ring is commutative and has characteristic 2. Prove that if an integral domain R is Boolean, then $R \cong \mathbb{Z}/2\mathbb{Z}$. [4.23, V.6.3]

3.16. $\neg$ Let S be a set and $T \subseteq S$ a subset. Prove that the subsets of S contained in T form an ideal of the power set ring $\mathcal{P}(S)$. Prove that if S is finite, then *every* ideal of $\mathcal{P}(S)$ is of this form. For S infinite, find an ideal of $\mathcal{P}(S)$ that is *not* of this form. [V.1.5]

3.17. $\triangleright$ Let I, J be ideals of a ring R . State and prove a precise result relating the ideals $(I + J)/I$ of R/I and $J/(I \cap J)$ of $R/(I \cap J)$. [§3.3]

4. Ideals and quotients: Remarks and examples. Prime and maximal ideals

4.1. Basic operations. It is often convenient to define ideals in terms of a set of *generators*.

Let $a \in R$ be any element of a ring. Then the subset $I = Ra$ of R is a left-ideal of R . Indeed, for all $r \in R$ we have

$$rI = rRa \subseteq Ra$$

as needed. Similarly, aR is right-ideal.

¹⁴After George Boole.

In the commutative case, these two subsets coincide and are denoted (a) . This is the *principal ideal generated by a* . For example, the zero-ideal $\{0\} = (0)$ and the whole ring $R = (1)$ are both principal ideals.

In general (Exercise 4.1), if $\{I_\alpha\}_{\alpha \in A}$ is a family of ideals of a ring R , then the sum $\sum_\alpha I_\alpha$ is an ideal of R . If a_α is any collection of elements of a commutative ring R , then

$$(a_\alpha)_{\alpha \in A} := \sum_{\alpha \in A} (a_\alpha)$$

is the ideal *generated by the elements a_α* . In particular,

$$(a_1, \dots, a_n) = (a_1) + \dots + (a_n)$$

is the smallest ideal of R containing $a_1, \dots, a_n$; this ideal consists of the elements of R that may be written as

$$r_1 a_1 + \dots + r_n a_n$$

for $r_1, \dots, r_n \in R$. An ideal I of a commutative ring R is *finitely generated* if $I = (a_1, \dots, a_n)$ for some $a_1, \dots, a_n \in R$.

Example 4.1. It is a good idea to get used to a bit of ‘calculus’ of ideals and quotients in terms of generators; judicious use of the isomorphism theorems yields convenient statements. For example, let R be a commutative ring, and let $a, b \in R$; denote by $\bar{b}$ the class of b in $R/(a)$. Then

$$(R/(a))/(\bar{b}) \cong R/(a, b).$$

Indeed, this is a particular case of Proposition 3.11, since

$$(\bar{b}) = \frac{(a, b)}{(a)}$$

as ideals of $R/(a)$. ┘

Note that principal ideals are (very special) finitely generated ideals. These notions are so important that we give special names to rings in which they are satisfied by *every* ideal.

Definition 4.2. A commutative ring R is *Noetherian* if every ideal of R is finitely generated. ┘

Definition 4.3. An integral domain R is a *PID* (‘Principal Ideal Domain’) if every ideal of R is principal. ┘

Thus, PIDs are (very special) Noetherian rings. In due time we will deal at length with these classes of rings (cf. Chapter V); Noetherian rings are very important in number theory and algebraic geometry.

The reader is already familiar with an important PID:

Proposition 4.4. $\mathbb{Z}$ is a PID.

Proof. Let $I \subseteq \mathbb{Z}$ be an ideal. Since I is a subgroup, $I = n\mathbb{Z}$ for some $n \in \mathbb{Z}$, by Proposition II.6.9. Since $n\mathbb{Z} = (n)$, this shows that I is principal. □

The fact that $\mathbb{Z}$ is a PID captures precisely ‘why’ greatest common divisors behave as they do in $\mathbb{Z}$: if m, n are integers, then the ideal (m, n) must be principal, and hence

$$(m, n) = (d)$$

for some (positive) integer d . This integer is manifestly the gcd of m and n : since $m \in (d)$ and $n \in (d)$, then $d \mid m$ and $d \mid n$, etc.

If k is a field, the ring of polynomials $k[x]$ is also a PID; proving this is easy, using the ‘division with remainder’ that we will run into very soon (§4.2); the reader should work this out on his/her own (Exercise 4.4) now. This fact will be absorbed in the general theory when we review the general notion of ‘Euclidean domain’, in §V.2.4.

By contrast, the ring $\mathbb{Z}[x]$ is *not* a PID: indeed, the reader should be able to verify that the ideal $(2, x)$ cannot be generated by a single element. As we will see in due time, greatest common divisors make good sense in a ring such as $\mathbb{Z}[x]$, but the matter is a little more delicate, since this ring is not a PID¹⁵.

There are several more basic operations involving ideals; for now, the following two will suffice.

- Again assume that $\{I_\alpha\}_{\alpha \in A}$ is a collection of ideals of a ring R . Then the intersection $\bigcap_{\alpha \in A} I_\alpha$ is (clearly) an ideal of R ; it is the largest ideal contained in all of the ideals I_α .
- If I, J are ideals of R , then IJ denotes the ideal *generated* by all products ij with $i \in I, j \in J$. More generally, if $I_1, \dots, I_n$ are ideals in R , then the ‘product’ $I_1 \cdots I_n$ denotes the ideal generated by all products $i_1 \cdots i_n$ with $i_k \in I_k$.

The reader should note the clash of notation: in the context of groups (especially §II.8.4) IJ would mean *something else*. Watch out!

It is clear that $IJ \subseteq I \cap J$: every element ij with $i \in I$ and $j \in J$ is in I (because I is a right-ideal) and in J (because J is a left-ideal); therefore $I \cap J$ contains all products ij , and hence it must contain the ideal IJ they generate. Sometime the product agrees with the intersection:

$$(4) \cap (3) = (12) = (4) \cdot (3) \quad \text{in } \mathbb{Z};$$

and sometime it does not:

$$(4) \cap (6) = (12) \neq (24) = (4) \cdot (6).$$

The matter of whether $IJ = I \cap J$ is often subtle; a prototype situation in which this equality holds is given in Exercise 4.5.

4.2. Quotients of polynomial rings. I have already observed that the quotient $\mathbb{Z}/n\mathbb{Z}$ is our familiar ring of congruence classes modulo n . Quotients of *polynomial* rings by principal ideals are a good source of ‘concrete’, but maybe less familiar, examples.

¹⁵It is, however, a ‘UFD’, that is, a ‘unique factorization domain’. This suffices for a good notion of gcd; cf. §V.2.1.

Let R be a (nonzero) ring, and let

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1x + a_0 \in R[x]$$

be a polynomial; for convenience, I am assuming that $f(x)$ is *monic*, that is, its leading coefficient (the coefficient of the highest power of x appearing in $f(x)$) is 1. In terms of ideals, this is not a serious requirement if the coefficient ring R is a field (Exercise 4.7), but it may be substantial otherwise: for example, $(2x) \subseteq \mathbb{Z}[x]$ cannot be generated by a monic polynomial. Also note that a monic polynomial is necessarily a non-zero-divisor (Exercise 4.8) and that if $f(x)$ is monic, then $\deg(f(x)q(x)) = \deg f(x) + \deg q(x)$ for all polynomials $q(x)$.

It is convenient to assume that $f(x)$ is monic because we can then *divide* by $f(x)$, with remainder. That is, if $g(x) \in R[x]$ is another polynomial, then there exist polynomials $q(x), r(x) \in R[x]$ such that

$$g(x) = f(x)q(x) + r(x)$$

and¹⁶ $\deg r(x) < \deg f(x)$. This is simply the process of ‘long division’ of polynomials, which is surely familiar to the reader, and can be performed over any ring when dividing by *monic* polynomials¹⁷.

The situation appears then to be similar to the situation in $\mathbb{Z}$, where we also have division with remainder. Quotients and remainders are uniquely¹⁸ determined by $g(x)$ and $f(x)$:

Lemma 4.5. *Let $f(x)$ be a monic polynomial, and assume*

$$f(x)q_1(x) + r_1(x) = f(x)q_2(x) + r_2(x)$$

with both $r_1(x)$ and $r_2(x)$ polynomials of degree $< \deg f(x)$. Then $q_1(x) = q_2(x)$ and $r_1(x) = r_2(x)$.

Proof. Indeed, we have

$$f(x)(q_1(x) - q_2(x)) = r_2(x) - r_1(x);$$

if $r_2(x) \neq r_1(x)$, then $r_2(x) - r_1(x)$ has degree $< \deg f(x)$, while $f(x)(q_1(x) - q_2(x))$ has degree $\geq \deg f(x)$, giving a contradiction. Therefore $r_1(x) = r_2(x)$, and $q_1(x) = q_2(x)$ follows right away since monic polynomials are non-zero-divisors. $\square$

The preceding considerations may be summarized in a rather efficient way in the language of ideals and cosets. We will now restrict ourselves to the *commutative* case, mostly for notational convenience, but also because this will guarantee that ideals are two-sided ideals, so that quotients are defined *as rings* (cf. §3.2).

¹⁶Note: With the convention that the degree of the polynomial 0 is $-\infty$, the condition $\deg r(x) < \deg f(x)$ is satisfied by $r(x) = 0$.

¹⁷The key point is that if $n > d$, then for all $a \in R$ we have $ax^n = ax^{n-d}f(x) + h(x)$ for some polynomial $h(x)$ of degree $< n$. Arguing inductively, this shows that we may perform division by $f(x)$ with remainder for all ‘monomials’ ax^n , and hence (by linearity) for all polynomials $g(x) \in R[x]$.

¹⁸This assertion has to be taken with a grain of salt in the noncommutative case, as different quotients and remainders may arise if we divide ‘on the left’ rather than ‘on the right’.

Assume then that R is a commutative ring. What we have shown is that, if $f(x)$ is monic, then for every $g(x) \in R[x]$ there exists a *unique* polynomial $r(x)$ of degree $< \deg f(x)$ and such that

$$g(x) + (f(x)) = r(x) + (f(x))$$

as cosets of the principal ideal $(f(x))$ in $R[x]$.

Refining this observation leads to a useful group-theoretic statement. Note that polynomials of degree $< d$ may be seen as elements of a direct sum

$$R^{\oplus d} = \underbrace{R \oplus \cdots \oplus R}_{d \text{ times}}$$

indeed, the function $\psi : R^{\oplus d} \rightarrow R[x]$ defined by

$$\psi((r_0, r_1, \dots, r_{d-1})) = r_0 + r_1x + \cdots + r_{d-1}x^{d-1}$$

is clearly an injective homomorphism of abelian groups, hence an isomorphism onto its image, and this consists precisely of the polynomials of degree $< d$. I will glibly identify $R^{\oplus d}$ with this set of polynomials for the purpose of the discussion that follows.

The next result may be seen as a way to concoct many different and interesting *ring* structures on the direct sum $R^{\oplus d}$:

Proposition 4.6. *Let R be a commutative ring, and let $f(x) \in R[x]$ be a monic polynomial of degree d . Then the function*

$$\varphi : R[x] \rightarrow R^{\oplus d}$$

defined by sending $g(x) \in R[x]$ to the remainder of the division of $g(x)$ by $f(x)$ induces an isomorphism of abelian groups

$$\frac{R[x]}{(f(x))} \cong R^{\oplus d}.$$

Proof. The given function φ is well-defined by Lemma 4.5, and it is surjective since it has a right inverse (that is, the function $\psi : R^{\oplus d} \rightarrow R[x]$ defined above).

I claim that φ is a homomorphism of abelian groups. Indeed, if

$$g_1(x) = f(x)q_1(x) + r_1(x) \quad \text{and} \quad g_2(x) = f(x)q_2(x) + r_2(x)$$

with $\deg r_1(x) < d$, $\deg r_2(x) < d$, then

$$g_1(x) + g_2(x) = f(x)(q_1(x) + q_2(x)) + (r_1(x) + r_2(x))$$

and $\deg(r_1(x) + r_2(x)) < d$: this implies (again by Lemma 4.5)

$$\varphi(g_1(x) + g_2(x)) = r_1(x) + r_2(x) = \varphi(g_1(x)) + \varphi(g_2(x)).$$

By the first isomorphism theorem for abelian groups, then, φ induces an isomorphism

$$\frac{R[x]}{\ker \varphi} \cong R^{\oplus d}.$$

On the other hand, $\varphi(g(x)) = 0$ if and only if $g(x) = f(x)q(x)$ for some $q(x) \in R[x]$, that is, if and only if $g(x)$ is in the principal ideal generated by $f(x)$. This shows $\ker \varphi = (f(x))$, concluding the proof. $\square$

Example 4.7. Assume $f(x)$ is monic of degree 1: $f(x) = x - a$ for some $a \in R$. Then the remainder of $g(x)$ after division by $f(x)$ is simply the ‘evaluation’ $g(a)$ (cf. Example 2.3): indeed,

$$g(x) = (x - a)q(x) + r$$

for some $r \in R$ (the remainder must have degree < 1 ; hence it is a constant); evaluating at a gives

$$g(a) = (a - a)q(a) + r = 0 \cdot q(a) + r = r$$

as claimed. In particular, $g(a) = 0$ if and only if $g(x) \in (x - a)$.

The content of Proposition 4.6 in this case is that the evaluation map

$$R[x] \rightarrow R, \quad g(x) \mapsto g(a)$$

induces an isomorphism

$$\frac{R[x]}{(x - a)} \cong R$$

of abelian groups; the reader will verify (either by hand or by invoking Corollary 3.10) that this is in fact an isomorphism of rings. $\square$

Example 4.8. It is fun to analyze higher-degree examples. For every monic $f(x) \in R[x]$ of degree d , Proposition 4.6 gives a potentially *different* ring (that is, $R[x]/(f(x))$) isomorphic to $R^{\oplus d}$ as a group; one can then use this isomorphism to define a new ring structure onto the group $R^{\oplus d}$.

For $d = 1$ all these structures are isomorphic (as seen in Example 4.7); but interesting structures already arise in degree 2. For a concrete example, apply this procedure with $f(x) = x^2 + 1$: Proposition 4.6 gives an isomorphism of groups

$$R \oplus R \cong \frac{R[x]}{(x^2 + 1)};$$

what multiplication does this isomorphism induce on $R \oplus R$? Take two elements $(a_0, a_1), (b_0, b_1)$ of $R \oplus R$. With the notation used in Proposition 4.6, we have

$$(a_0, a_1) = \varphi(a_0 + a_1x), \quad (b_0, b_1) = \varphi(b_0 + b_1x).$$

Now a bit of high-school algebra gives

$$\begin{aligned} (a_0 + a_1x)(b_0 + b_1x) &= a_0b_0 + (a_0b_1 + a_1b_0)x + a_1b_1x^2 \\ &= (x^2 + 1)a_1b_1 + ((a_0b_0 - a_1b_1) + (a_0b_1 + a_1b_0)x) \end{aligned}$$

which shows

$$\varphi((a_0 + a_1x)(b_0 + b_1x)) = (a_0b_0 - a_1b_1, a_0b_1 + a_1b_0).$$

Therefore, the multiplication induced on $R \oplus R$ by this procedure is defined by

$$(a_0, a_1) \cdot (b_0, b_1) = (a_0b_0 - a_1b_1, a_0b_1 + a_1b_0).$$

This recipe may seem somewhat arbitrary, but note that upon taking $R = \mathbb{R}$, the ring of real numbers, and identifying pairs $(x, y) \in \mathbb{R} \oplus \mathbb{R}$ with *complex numbers*

$x + iy$, the multiplication we obtained on $\mathbb{R} \oplus \mathbb{R}$ matches precisely the ordinary multiplication in $\mathbb{C}$. Therefore,

$$\frac{\mathbb{R}[x]}{(x^2 + 1)} \cong \mathbb{C}$$

as rings. In other words, this procedure constructs the ring $\mathbb{C}$ ‘from scratch’, starting from $\mathbb{R}[x]$. $\lrcorner$

The point is that the polynomial equation $x^2 + 1 = 0$ has no solutions in $\mathbb{R}$; the quotient $\mathbb{R}[x]/(x^2 + 1)$ produces a ring containing a copy of $\mathbb{R}$ and in which the polynomial *does* have roots (that is, $\pm$ the class of x in the quotient). The fact that this turns out to be isomorphic to $\mathbb{C}$ may not be too surprising, considering that $\mathbb{C}$ is precisely a ring containing a copy of $\mathbb{R}$ and in which $x^2 + 1$ does have roots (that is, $\pm i$).

Such constructions are the “algebraist’s way” to solve equations. We will come back to them in §V.5.2, and in a sense the whole of Chapter VII will be devoted to this topic.

4.3. Prime and maximal ideals. Other ‘qualities’ of ideals are best expressed in terms of quotients. I am still assuming that our rings are commutative—both due to unforgivable laziness and because that is the only context in which we will use these notions.

Definition 4.9. Let $I \neq (1)$ be an ideal of a commutative ring R .

- I is a *prime ideal* if R/I is an integral domain.
- I is a *maximal ideal* if R/I is a field. $\lrcorner$

Example 4.10. For all $a \in R$, the ideal $(x - a)$ is prime in $R[x]$ if and only if R is an integral domain; it is maximal if and only if R is a field. Indeed, $R[x]/(x - a) \cong R$, as we have seen in Example 4.7.

The ideal $(2, x)$ is maximal in $\mathbb{Z}[x]$, since

$$\frac{\mathbb{Z}[x]}{(2, x)} \stackrel{!}{\cong} \frac{\mathbb{Z}[x]/(x)}{(2)} \cong \frac{\mathbb{Z}}{(2)} = \mathbb{Z}/2\mathbb{Z}$$

is a field (for the isomorphism $\stackrel{!}{\cong}$, cf. Example 4.1). $\lrcorner$

Of course these notions may be translated into terms not involving quotients at all, and it is largely a matter of æsthetic preference whether prime and maximal ideals should be defined as in Definition 4.9 or by the following equivalent conditions:

Proposition 4.11. Let $I \neq (1)$ be an ideal of a commutative ring R . Then

- I is prime if and only if for all $a, b \in R$

$$ab \in I \implies (a \in I \text{ or } b \in I);$$
- I is maximal if and only if for all ideals J of R

$$I \subseteq J \implies (I = J \text{ or } J = R).$$

Proof. The ring R/I is an integral domain if and only if $\forall \bar{a}, \bar{b} \in R/I$

$$\bar{a} \cdot \bar{b} = 0 \implies (\bar{a} = 0 \text{ or } \bar{b} = 0).$$

This condition translates immediately to the given condition in R , with $\bar{a} = a + I$, $\bar{b} = b + I$, since the 0 in R/I is I .

As for maximality, the given condition follows from the correspondence between ideals of R/I and ideals of R containing I (§3.3) and the observation that a commutative ring is a field if and only if its only ideals are (0) and (1) (Exercise 3.8). $\square$

From the formulation in terms of quotients, it is completely clear that

$$\text{maximal} \implies \text{prime};$$

indeed, fields are integral domains. This fact is of course easy to check in terms of the other description, but the argument is a little more cumbersome (Exercise 4.14). Prime ideals are not necessarily maximal, but note the following:

Proposition 4.12. *Let I be an ideal of a commutative ring R . If R/I is finite, then I is prime if and only if it is maximal.*

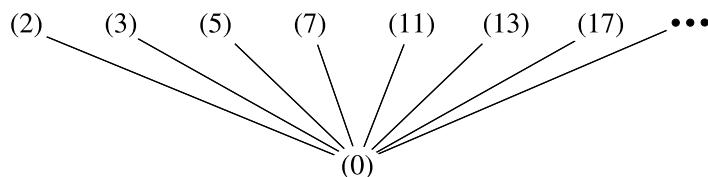
Proof. This follows immediately from Proposition 1.15. $\square$

For example, let (n) be an ideal of $\mathbb{Z}$, with $n > 0$; then

$$(n) \text{ prime} \iff (n) \text{ maximal} \iff n \text{ is prime as an integer.}$$

Indeed, for nonzero n the ring $\mathbb{Z}/n\mathbb{Z}$ is finite, so Proposition 4.12 applies; cf. Example 1.17.

In general, the set of prime ideals of a commutative ring R is called¹⁹ the *spectrum* of R , denoted $\text{Spec } R$. We can ‘draw’ $\text{Spec } \mathbb{Z}$ as follows:



Actually, attributing the fact that nonzero prime ideals of $\mathbb{Z}$ are maximal to the finiteness of the quotients (as I have just done) is slightly misleading; a better ‘explanation’ is that $\mathbb{Z}$ is a PID, and this phenomenon is common to all PIDs:

Proposition 4.13. *Let R be a PID, and let I be a nonzero ideal in R . Then I is prime if and only if it is maximal.*

Proof. Maximal ideals are prime in every ring, so we only need to verify that nonzero prime ideals are maximal in a PID; we will use the characterization of prime and maximal ideals obtained in Proposition 4.11. Let $I = (a)$ be a prime ideal in R , with $a \neq 0$, and assume $I \subseteq J$ for an ideal of R . As R is a PID, $J = (b)$

¹⁹Believe it or not, the term is borrowed from functional analysis.

for some $b \in R$. Since $I = (a) \subseteq (b) = J$, we have that $a = bc$ for some $c \in R$. But then $b \in (a)$ or $c \in (a)$, since $I = (a)$ is prime.

If $b \in (a)$, then $(b) \subseteq (a)$; and $I = J$ follows. If $c \in (a)$, then $c = da$ for some $d \in R$. But then

$$a = bc = bda,$$

from which $bd = 1$ since cancellation by the nonzero a holds in R (since R is an integral domain). This implies that b is a unit, and hence $J = (b) = R$.

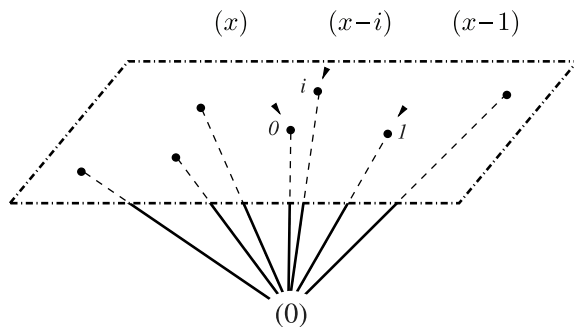
That is, we have shown that if $I \subseteq J$, then either $I = J$ or $J = R$: thus I is maximal, by Proposition 4.11. $\square$

Example 4.14. Let k be a field. Then nonzero prime ideals in $k[x]$ are maximal, since $k[x]$ is a PID (as the reader has hopefully checked by now; cf. Exercise 4.4). Therefore, a picture of $\text{Spec } k[x]$ would look pretty much like the picture of $\text{Spec } \mathbb{Z}$ shown above: with the maximal ideals at one level, all containing the prime (and nonmaximal) ideal (0) .

This picture is particularly appealing for fields such as $k = \mathbb{C}$, which are *algebraically closed*, that is, in which for every nonconstant $f(x) \in k[x]$ there exists $r \in k$ such that $f(r) = 0$. It would take us too far afield to discuss this notion at any length now; but the reader should be aware that $\mathbb{C}$ is algebraically closed. We will come back to this²⁰. Assuming this fact, it is easy to verify (Exercise 4.21) that the maximal ideals in $\mathbb{C}[x]$ are all and only the ideals

$$(x - z)$$

where z ranges over all complex numbers. That is, stretching our imagination a little, we could come up with the following picture for $\text{Spec } \mathbb{C}[x]$:



There is a ‘complex line’²¹ worth of maximal ideals: for each $z \in \mathbb{C}$ we have the maximal $(x - z)$; the prime (0) is contained in all the maximal ideals; and there are no other prime ideals.

The picture for $\text{Spec } \mathbb{C}[x]$ may serve as justification for the fact that, in algebraic geometry, $\mathbb{C}[x]$ is the ring corresponding to the ‘affine line’ $\mathbb{C}$; it is the ring of an algebraic *curve*. It turns out that the fact that there is exactly ‘one level’ of maximal

²⁰A particularly pleasant proof may be given using elementary complex analysis, as a consequence of Liouville’s theorem, or the maximum modulus principle; cf. §V.5.3.

²¹I know: it looks like a *plane*. But it is a *line*, as a complex entity.

ideals over (0) in $\mathbb{C}[x]$ reflects precisely the fact that the corresponding geometric object has dimension 1.

In general, the (*Krull*) *dimension* of a commutative ring R is the length of the longest chain of prime ideals in R . Thus, Proposition 4.13 tells us that PIDs other than fields, such as $\mathbb{Z}$, have ‘dimension 1’. In the lingo of algebraic geometry, they all correspond to curves. $\lrcorner$

Example 4.15. For examples of rings of higher dimension, consider

$$k[x_1, \dots, x_n],$$

where k is a field. Note that there are chains of prime ideals of length n

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \dots, x_n)$$

in this ring. (Why are these ideals prime? Cf. Exercise 4.13.) This says that $k[x_1, \dots, x_n]$ has dimension $\geq n$. One can show that n is in fact the longest length of a chain of prime ideals in $k[x_1, \dots, x_n]$; that is, the Krull dimension of $k[x_1, \dots, x_n]$ is precisely n . In algebraic geometry, the ring $\mathbb{C}[x_1, \dots, x_n]$ corresponds to the n -dimensional complex space $\mathbb{C}^n$.

Dealing precisely with these notions is not so easy, however. Even the seemingly simple statement that the maximal ideals of $\mathbb{C}[x_1, \dots, x_n]$ are all and only the ideals $(x_1 - z_1, \dots, x_n - z_n)$, for $(z_1, \dots, z_n) \in \mathbb{C}^n$, requires a rather deep result, known as *Hilbert’s Nullstellensatz*. $\lrcorner$

We will come back to all of this and get a very small taste of algebraic geometry in §VII.2, after we develop (much) more machinery.

Exercises

4.1. $\triangleright$ Let R be a ring, and let $\{I_\alpha\}_{\alpha \in A}$ be a family of ideals of R . We let

$$\sum_{\alpha \in A} I_\alpha := \left\{ \sum_{\alpha \in A} r_\alpha \text{ such that } r_\alpha \in I_\alpha \text{ and } r_\alpha = 0 \text{ for all but finitely many } \alpha \right\}.$$

Prove that $\sum_{\alpha} I_\alpha$ is an ideal of R and that it is the smallest ideal containing all of the ideals I_α . [§4.1]

4.2. $\triangleright$ Prove that the homomorphic image of a Noetherian ring is Noetherian. That is, prove that if $\varphi : R \rightarrow S$ is a surjective ring homomorphism and R is Noetherian, then S is Noetherian. [§6.4]

4.3. Prove that the ideal $(2, x)$ of $\mathbb{Z}[x]$ is not principal.

4.4. $\triangleright$ Prove that if k is a field, then $k[x]$ is a PID. (Hint: Let $I \subseteq k[x]$ be any ideal. If $I = (0)$, then I is principal. If $I \neq (0)$, let $f(x)$ be a monic polynomial in I of minimal degree. Use division with remainder to construct a proof that $I = (f(x))$, arguing as in the proof of Proposition II.6.9.) [§4.1, §4.3, §V.2.4, §V.4.1, §VI.7.2, §VII.1.2]

4.5. $\triangleright$ Let I, J be ideals in a commutative ring R , such that $I + J = (1)$. Prove that $IJ = I \cap J$. [§4.1, §V.6.1]

4.6. Let I, J be ideals in a commutative ring R . Assume that $R/(IJ)$ is reduced (that is, it has no nonzero nilpotent elements; cf. Exercise 3.13). Prove that $IJ = I \cap J$.

4.7. $\triangleright$ Let $R = k$ be a field. Prove that every nonzero (principal) ideal in $k[x]$ is generated by a unique *monic* polynomial. [§4.2, §VI.7.2]

4.8. $\triangleright$ Let R be a ring and $f(x) \in R[x]$ a *monic* polynomial. Prove that $f(x)$ is not a (left- or right-) zero-divisor. [§4.2, 4.9]

4.9. Generalize the result of Exercise 4.8, as follows. Let R be a commutative ring, and let $f(x)$ be a zero-divisor in $R[x]$. Prove that $\exists b \in R, b \neq 0$, such that $f(x)b = 0$. (Hint: Let $f(x) = a_d x^d + \cdots + a_0$, and let $g(x) = b_e x^e + \cdots + b_0$ be a nonzero polynomial of minimal degree e such that $f(x)g(x) = 0$. Deduce that $a_d g(x) = 0$, and then prove $a_{d-i}g(x) = 0$ for all i . What does this say about b_e ?)

4.10. $\neg$ Let d be an integer that is not the square of an integer, and consider the subset of $\mathbb{C}$ defined by²²

$$\mathbb{Q}(\sqrt{d}) := \{a + b\sqrt{d} \mid a, b \in \mathbb{Q}\}.$$

- Prove that $\mathbb{Q}(\sqrt{d})$ is a subring of $\mathbb{C}$.
- Define a function $N : \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Q}$ by $N(a + b\sqrt{d}) := a^2 - b^2d$. Prove that $N(zw) = N(z)N(w)$ and that $N(z) \neq 0$ if $z \in \mathbb{Q}(\sqrt{d}), z \neq 0$.

The function N is a ‘norm’; it is very useful in the study of $\mathbb{Q}(\sqrt{d})$ and of its subrings. (Cf. also Exercise 2.5.)

- Prove that $\mathbb{Q}(\sqrt{d})$ is a field and in fact the smallest subfield of $\mathbb{C}$ containing both $\mathbb{Q}$ and $\sqrt{d}$. (Use N .)
- Prove that $\mathbb{Q}(\sqrt{d}) \cong \mathbb{Q}[t]/(t^2 - d)$. (Cf. Example 4.8.)

[V.1.17, V.2.18, V.6.13, VII.1.12]

4.11. Let R be a commutative ring, $a \in R$, and $f_1(x), \dots, f_r(x) \in R[x]$.

- Prove the equality of ideals

$$(f_1(x), \dots, f_r(x), x - a) = (f_1(a), \dots, f_r(a), x - a).$$

- Prove the useful substitution trick

$$\frac{R[x]}{(f_1(x), \dots, f_r(x), x - a)} \cong \frac{R}{(f_1(a), \dots, f_r(a))}.$$

(Hint: Exercise 3.3.)

4.12. $\triangleright$ Let R be a commutative ring and $a_1, \dots, a_n$ elements of R . Prove that

$$\frac{R[x_1, \dots, x_n]}{(x_1 - a_1, \dots, x_n - a_n)} \cong R.$$

[§VII.2.2]

²²Of course there are *two* ‘square roots of d ’; but the definition of $\mathbb{Q}(\sqrt{d})$ does not depend on which one is used.

4.13. $\triangleright$ Let R be an integral domain. For all $k = 1, \dots, n$ prove that $(x_1, \dots, x_k)$ is prime in $R[x_1, \dots, x_n]$. [§4.3]

4.14. $\triangleright$ Prove ‘by hand’ that maximal ideals are prime, *without* using quotient rings. [§4.3]

4.15. Let $\varphi : R \rightarrow S$ be a homomorphism of commutative rings, and let $I \subseteq S$ be an ideal. Prove that if I is a prime ideal in S , then $\varphi^{-1}(I)$ is a prime ideal in R . Show that $\varphi^{-1}(I)$ is not necessarily maximal if I is maximal.

4.16. Let R be a commutative ring, and let P be a prime ideal of R . Suppose 0 is the only zero-divisor of R contained in P . Prove that R is an integral domain.

4.17. $\neg$ (If you know a little topology...) Let K be a compact topological space, and let R be the ring of continuous real-valued functions on K , with addition and multiplication defined pointwise.

- (i) For $p \in K$, let $M_p = \{f \in R \mid f(p) = 0\}$. Prove that M_p is a maximal ideal in R .
- (ii) Prove that if $f_1, \dots, f_r \in R$ have no common zeros, then $(f_1, \dots, f_r) = (1)$. (Hint: Consider $f_1^2 + \dots + f_r^2$.)
- (iii) Prove that every maximal ideal M in R is of the form M_p for some $p \in K$. (Hint: You will use the compactness of K and (ii).)

Conclude that $p \mapsto M_p$ defines a bijection from K to the set of maximal ideals of R . (The set of maximal ideals of a commutative ring R is called the *maximal spectrum* of R ; it is contained in the (prime) spectrum $\text{Spec } R$ defined in §4.3. Relating commutative rings and ‘geometric’ entities such as topological spaces is the business of *algebraic geometry*.)

The compactness hypothesis is necessary: cf. Exercise V.3.10. [V.3.10]

4.18. Let R be a commutative ring, and let N be its nilradical (Exercise 3.12). Prove that N is contained in every prime ideal of R . (Later on the reader will check that the nilradical is in fact the intersection of all prime ideals of R : Exercise V.3.13.)

4.19. Let R be a commutative ring, let P be a prime ideal in R , and let I_j be ideals of R .

- (i) Assume that $I_1 \cdots I_r \subseteq P$; prove that $I_j \subseteq P$ for some j .
- (ii) By (i), if $P \supseteq \bigcap_{j=1}^r I_j$, then P contains one of the ideals I_j . Prove or disprove: if $P \supseteq \bigcap_{j=1}^\infty I_j$, then P contains one of the ideals I_j .

4.20. Let M be a two-sided ideal in a (not necessarily commutative) ring R . Prove that M is maximal if and only if R/M is a simple ring (cf. Exercise 3.9).

4.21. $\triangleright$ Let k be an algebraically closed field, and let $I \subseteq k[x]$ be an ideal. Prove that I is maximal if and only if $I = (x - c)$ for some $c \in k$. [§4.3, §V.5.2, §VII.2.1, §VII.2.2]

4.22. Prove that $(x^2 + 1)$ is maximal in $\mathbb{R}[x]$.

4.23. A ring R has Krull dimension 0 if every prime ideal in R is maximal. Prove that fields and Boolean rings (Exercise 3.15) have Krull dimension 0.

4.24. Prove that the ring $\mathbb{Z}[x]$ has Krull dimension ≥ 2 . (It is in fact exactly 2; thus it corresponds to a *surface* from the point of view of algebraic geometry.)

5. Modules over a ring

I have emphasized the parallels between the basic theory of groups and the basic theory of rings; there are also important differences. In **Grp**, one takes the quotient of a *group*, by a (normal sub)*group*, and the results is a *group*: throughout the process one never leaves **Grp**. The situation is in a sense even better in **Ab**, where the normality condition is automatic; one simply takes the quotient of an abelian group by an abelian (sub)group, obtaining an abelian group.

The situation in **Ring** is not nearly as neat. One of the three characters in the story is an *ideal*: which is not a ring according to the axioms listed in Definition 1.1, in all but the most pathological cases. In other words, the kernel of a homomorphism of rings is (usually) not a ring. Also, cokernels do not behave as one would hope (cf. Exercise 2.12). Even if one relaxes the ring definition, giving up the identity and making ideals subrings, there is no reasonably large class of examples in which the ‘ideal’ condition is automatically satisfied by substructures. In short, **Ring** is not a particularly pleasant category.

Modules will fix all these problems. If R is a ring and $I \subseteq R$ is a two-sided ideal, then all three structures R , I , and R/I are *modules over R* . The category of R -modules is the prime example of a well-behaved category: in this category kernels and cokernels exist and do precisely what they ought to. The category **Ab** is a particular case of this construction, since it is the category of modules over $\mathbb{Z}$ in disguise (Example 5.4). The category of modules over a ring R will share many of the excellent properties of **Ab**; and we will get a brand new, well-behaved category for each ring R . These are all examples²³ of the important notion of *abelian category*.

5.1. Definition of (left-) R -module. In short, R -modules are abelian groups endowed with an action of R . To flesh out this idea, recall that ‘actions’ in general denote homomorphisms into some kind of endomorphism structure: for example, we defined *group actions* in §II.9.1 as group homomorphisms from a fixed group to the groups of automorphisms of objects of a category.

We can give an analogous definition of the action of a *ring* on an *abelian group*. Indeed, recall that if M is an abelian group, then $\text{End}_{\mathbf{Ab}}(M) := \text{Hom}_{\mathbf{Ab}}(M, M)$ is a ring in a natural way (cf. §2.5). A *left-action* of a ring R on M is then simply a homomorphism of rings

$$\sigma : R \rightarrow \text{End}_{\mathbf{Ab}}(M);$$

²³In fact, it can be shown (‘Freyd-Mitchell’s embedding theorem’) that every small abelian category is equivalent to a subcategory of the category of left-modules over a ring. So to some extent we can understand abelian categories by understanding categories of modules well enough. We will come back to this in Chapter IX.

we will say that σ makes M into a *left- R -module*.

Similarly to the situation with group actions, it is convenient to spell out this definition.

Claim 5.1. *The datum of a homomorphism σ as above is precisely the same as the datum of a function*

$$\rho : R \times M \rightarrow M$$

satisfying the following requirements: $(\forall r, s \in R) (\forall m, n \in M)$

- $\rho(r, m + n) = \rho(r, m) + \rho(r, n)$;
- $\rho(r + s, m) = \rho(r, m) + \rho(s, m)$;
- $\rho(rs, m) = \rho(r, \rho(s, m))$;
- $\rho(1, m) = m$.

The proof of this claim follows precisely the pattern of the discussion in the beginning of §II.9.2, so it is left to the reader (Exercise 5.2). Of course the relation between ρ and σ is given by

$$\rho(r, m) = \sigma(r)(m).$$

As always, carrying ρ around is inconvenient, so it is common to omit mention of it: $\rho(r, m)$ is often just denoted rm ; this makes the requirements listed above a little more readable. Summarizing and adopting this shorthand,

Definition 5.2. A left- R -module structure on an abelian group M consists of a map $R \times M \rightarrow M$, $(r, m) \mapsto rm$, such that

- $r(m + n) = rm + rn$;
- $(r + s)m = rm + sm$;
- $(rs)m = r(sm)$;
- $1m = m$.

┘

Right- R -modules are defined analogously. The reader can glance back at §II.9, especially Exercise II.9.3, for a reminder on right- vs. left-actions; the issues here are analogous. Thus, for example, a right- R -module structure may be identified with a left- R° -module structure, where R° is the ‘opposite’ ring obtained by reversing the order of multiplication (Exercise 5.1). However, note that R and R° have no good reasons to be isomorphic in general (while every group is isomorphic to its opposite).

These issues become immaterial if R is *commutative*: then the identity $R \rightarrow R^\circ$ is an isomorphism, and left-modules/right-modules are identical concepts. The reader will not miss much by adopting the blanket assumption that all rings mentioned in this section are commutative. It is occasionally important to make this hypothesis explicit (for example in dealing with *algebras*, cf. Example 5.6), but most of the material we are going to review works *verbatim* for, say, left-modules over an arbitrary ring as for modules over a commutative ring. I will write ‘module’ for ‘left-module’, for convenience; it will be the reader’s responsibility to take care of appropriate changes, if necessary, to adapt the various concepts to *right-modules*.

5.2. The category $R\text{-Mod}$. The reader should spend some time getting familiar with the notion of module, by proving simple properties such as

$$\begin{aligned} (\forall m \in M) : \quad 0 \cdot m &= 0, \\ (\forall m \in M) : \quad (-1) \cdot m &= -m, \end{aligned}$$

where M is a module over some ring (Exercise 5.3). The following fancier-sounding (but equally trivial) property is also useful in developing a feel for modules:

Proposition 5.3. *Every abelian group is a $\mathbb{Z}$ -module, in exactly one way.*

Proof. Let G be an abelian group. A $\mathbb{Z}$ -module structure on G is a ring homomorphism

$$\mathbb{Z} \rightarrow \text{End}_{\text{Ab}}(G).$$

Since $\mathbb{Z}$ is initial in Ring (§2.1), there exists exactly one such homomorphism, proving the statement. $\square$

Thus, ‘abelian group’ and ‘ $\mathbb{Z}$ -module’ are one and the same notion. Quite concretely, the action of $n \in \mathbb{Z}$ on an element a of an abelian group simply yields the ordinary ‘multiple’ na ; this operation is trivially compatible with the operations in $\mathbb{Z}$.

A *homomorphism of R -modules* is a homomorphism of (abelian) groups which is compatible with the module structure. That is, if M, N are R -modules and $\varphi : M \rightarrow N$ is a function, then φ is a homomorphism of R -modules if and only if

- $(\forall m_1 \in M)(\forall m_2 \in M) : \quad \varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2);$
- $(\forall r \in R)(\forall m \in M) : \quad \varphi(rm) = r\varphi(m).$

It is hopefully clear that the composition of two R -module homomorphisms is an R -module homomorphism and that the identity is an R -module homomorphism:

R -modules form a category

which I will denote²⁴ ‘ $R\text{-Mod}$ ’.

Example 5.4. The category $\mathbb{Z}\text{-Mod}$ of $\mathbb{Z}$ -modules is ‘the same as’ the category Ab : indeed, every abelian group is a $\mathbb{Z}$ -module in exactly one way (Proposition 5.3), and $\mathbb{Z}$ -module homomorphisms are simply homomorphisms of abelian groups. $\lrcorner$

Example 5.5. If $R = k$ is a field, R -modules are called *k -vector spaces*. I will call the category of vector spaces over a field k ‘ $k\text{-Vect}$ ’; this is just another name for $k\text{-Mod}$. Morphisms in $k\text{-Vect}$ are often called²⁵ *linear maps*. ‘Linear algebra’ is the study of $k\text{-Vect}$ (extended to $R\text{-Mod}$ when possible); Chapters VI and VIII will be devoted to this subject. $\lrcorner$

²⁴If R is not commutative, we should agree on whether $R\text{-Mod}$ denotes the category of *left*-modules or of *right*-modules. I will mean ‘left-modules’.

²⁵This term is also used for homomorphisms of R modules for more general rings R , but not as frequently.

Example 5.6. Any homomorphism of rings $\alpha : R \rightarrow S$ may be used to define an interesting R -module: define $\rho : R \times S \rightarrow S$ by

$$\rho(r, s) := \alpha(r)s$$

for all $r \in R$ and $s \in S$. The operation on the right is simply multiplication in S , and the axioms of Definition 5.2 are immediate consequence of the ring axioms and of the fact that α is a homomorphism. For instance, taking $S = R$ and $\alpha = \text{id}_R$ makes R a (left-) module over itself.

It is common to write rs rather than $\alpha(r)s$.

The ring operation in S and the R -module structure induced by α are linked even more tightly if we require R to be *commutative* and α to map R to the *center* of S : that is, if we require $\alpha(r)$, s to commute for every $r \in R$, $s \in S$. Indeed, in this case the left-module structure defined above and the right-module structure defined analogously would coincide; further, with these requirements the ring operation in S

$$(s_1, s_2) \mapsto s_1 s_2$$

is compatible with the R -module structure in the sense that²⁶

$$(r_1 s_1)(r_2 s_2) = \alpha(r_1)s_1\alpha(r_2)s_2 = \alpha(r_1)\alpha(r_2)s_1 s_2 = (r_1 r_2)(s_1 s_2)$$

$\forall r_1, r_2 \in R, \forall s_1, s_2 \in S$: that is, we can ‘move’ the action of R at will through products in S .

Due to their importance, these examples deserve their own official name:

Definition 5.7. Let R be a commutative ring. An R -algebra is a ring homomorphism $\alpha : R \rightarrow S$ such that $\alpha(R)$ is contained in the center of S . $\lrcorner$

The usual abuse of language leads us to refer to an R -algebra by the name of the target S of the homomorphism. Thus, an R -algebra ‘is’ an R -module S with a compatible ring structure, or, if you prefer, a ring S with a compatible R -module structure. An R -algebra S is a *division algebra* if S is a division ring.

There is an evident notion of ‘ R -algebra homomorphism’ (preserving both the ring and module structure), and we thus get a category $R\text{-Alg}$. The situation simplifies substantially if S is itself commutative, in which case the condition on the center is unnecessary. ‘Commutative R -algebras’ form a category, which the attentive reader will recognize as a coslice category (Example I.3.7) in the category of commutative rings.

Also, note that $\mathbb{Z}\text{-Alg}$ is just another name for Ring (why?).

The polynomial rings $R[x_1, \dots, x_n]$, as well as all their quotients, are commutative R -algebras. This is a particularly important class of examples; for $R = k$ an algebraically closed field, these are the rings used in ‘classical’ affine algebraic geometry (cf. §VII.2.3). $\lrcorner$

The trivial group 0 has a unique module structure over any ring R and is a zero-object in $R\text{-Mod}$, that is, it is both initial and final. As in the other main categories we have encountered, a bijective homomorphism of R -modules is automatically an

²⁶More generally, this choice makes the multiplication in S ‘ R -bilinear’.

isomorphism in $R\text{-Mod}$ (Exercise 5.12). In these and many other respects, the category $R\text{-Mod}$ (for any commutative ring R) and $\mathbf{Ab}$ are similar.

If R is commutative, the similarity goes further: just as in the category $\mathbf{Ab}$, each set $\text{Hom}_{R\text{-Mod}}(M, N)$ may itself be seen as an object of the category (cf. §II.4.4)²⁷. Indeed, let M and N be R -modules. Since homomorphisms of R -modules are in particular homomorphisms of abelian groups,

$$\text{Hom}_{R\text{-Mod}}(M, N) \subseteq \text{Hom}_{\mathbf{Ab}}(M, N)$$

as sets (up to natural identifications). The operation making $\text{Hom}_{\mathbf{Ab}}(M, N)$ into an abelian group, as in §II.4.4, clearly preserves $\text{Hom}_{R\text{-Mod}}(M, N)$; it follows that the latter is an abelian group. For $r \in R$ and $\varphi \in \text{Hom}_{R\text{-Mod}}(M, N)$, the prescription

$$(\forall m \in M) : (r\varphi)(m) := r\varphi(m)$$

defines a function²⁸ $r\varphi : M \rightarrow N$. This function is an R -module homomorphism if R is commutative, because $(\forall a \in R), (\forall m \in M)$

$$(r\varphi)(am) = r\varphi(am) = (ra)\varphi(m) \stackrel{!}{=} (ar)\varphi(m) = a(r\varphi(m)).$$

Thus, we have a natural action of R on the abelian group $\text{Hom}_{R\text{-Mod}}(M, N)$, and it is immediate to verify that this makes $\text{Hom}_{R\text{-Mod}}(M, N)$ into an R -module.

Watch out: if R is not commutative, then in general $\text{Hom}_{R\text{-Mod}}(M, N)$ is ‘just’ an abelian group. More structure is available if M, N are *bimodules*; cf. §VIII.3.2.

5.3. Submodules and quotients. Since R -modules are an ‘enriched’ version of abelian groups, we can progress through the usual constructions very quickly, by just pointing out that the analogous constructions in $\mathbf{Ab}$ are preserved by the R -module structure.

A *submodule* N of an R -module M is a subgroup preserved by the action of R . That is, for all $r \in R$ and $n \in N$, the element rn (defined by the R -module structure of M) is in fact in N . Put otherwise, and perhaps more transparently, N is itself an R -module, and the inclusion $N \subseteq M$ is an R -module homomorphism.

Example 5.8. We can view R itself as a (left-) R -module (cf. Example 5.6); the submodules of R are then precisely the (left-) *ideals* of R . ┘

Example 5.9. Both the kernel and the image of a homomorphism $\varphi : M \rightarrow M'$ of R -modules are submodules (of M, M' , respectively). ┘

Example 5.10. If r is in the center of R and M is an R -module, then $rM = \{rm \mid m \in M\}$ is a submodule of M . If I is any (left-)ideal of R , then $IM = \{\sum_i r_i m_i \mid r_i \in I, m_i \in M\}$ is a submodule of M .

If N is a submodule of M , then it is in particular a (normal) subgroup of the abelian group $(M, +)$; thus we may define the quotient M/N as an abelian group.

²⁷Notational convention: One often writes $\text{Hom}_R(M, N)$ for $\text{Hom}_{R\text{-Mod}}(M, N)$. I will not adopt this convention here but I will use it freely in later chapters.

²⁸Parse the notation carefully: $r\varphi$ is the name of a function on the left, while $r\varphi(m)$ on the right is the action of the element $r \in R$ on the element $\varphi(m) \in N$, defined by the R -module structure on N .

Of course it would be desirable to see this as a module, and as usual there is only one reasonable way to do so: we will want the canonical projection

$$\pi : M \rightarrow M/N$$

to be an R -module homomorphism, and this forces

$$r(m + N) = r\pi(m) = \pi(rm) = rm + N$$

for all $m \in M$. That is, we are led to define the action of R on M/N by

$$r(m + N) := rm + N.$$

Claim 5.11. *For all submodules N , this prescription does define a structure of R -module on M/N .*

The proof of this claim is immediate and is left to the reader. The R -module M/N is (of course) called the *quotient of M by N* .

Example 5.12. If R is a ring and I is a two-sided ideal of R , then all three of I , R , and the quotient ring R/I are R -modules: I is a submodule of R , and the rings R and R/I are in fact R -algebras if R is commutative (cf. Example 5.6). $\square$

Example 5.13. If R is not commutative and I is just a (say) left-ideal, then the quotient R/I is not defined as a *ring*, but it is defined as a *left-module* (the quotient of the module R by the submodule I). The action of R on R/I is given by left-multiplication: $r(a + I) = ra + I$.

The reader should now expect a universal property for quotients, and here it is:

Theorem 5.14. *Let N be a submodule of an R -module M . Then for every homomorphism of R -modules $\varphi : M \rightarrow P$ such that $N \subseteq \ker \varphi$ there exists a unique homomorphism of R -modules $\tilde{\varphi} : M/N \rightarrow P$ so that the diagram*

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & P \\ & \searrow \pi & \nearrow \exists! \tilde{\varphi} \\ & M/N & \end{array}$$

commutes.

As in previous appearances of such statements, this is an immediate consequence of the set-theoretic version (§I.5.3) and of easy notation matching and compatibility checks. For an even faster proof, one can just apply Theorem II.7.12 and verify that $\tilde{\varphi}$ is an R -module homomorphism.

Since every submodule N is then the kernel of the canonical projection $M \rightarrow M/N$, our recurring slogan becomes, in the context of $R\text{-Mod}$

$$\text{kernel} \iff \text{submodule} :$$

unlike as in **Grp** or **Ring**, being a kernel poses no restriction on the relevant substructures. Put otherwise, ‘every monomorphism in $R\text{-Mod}$ is a kernel’; this is one of the distinguishing features of an abelian category.

5.4. Canonical decomposition and isomorphism theorems. The discussion now proceeds along the same lines as for (abelian) groups; the statements of the key facts and a few comments should suffice, as the proofs are nothing but a rehashing of the proofs of analogous statements we have encountered previously. *Of course* the reader should take the following statements as assignments and provide all the needed details.

In the context of R -modules, the canonical decomposition takes the following form:

Theorem 5.15. *Every R -module homomorphism $\varphi : M \rightarrow M'$ may be decomposed as follows:*

$$\begin{array}{ccccc} & & \varphi & & \\ & \searrow & \curvearrowright & \searrow & \\ M & \twoheadrightarrow & M/\ker \varphi & \xrightarrow[\tilde{\varphi}]{\sim} & \operatorname{im} \varphi \hookrightarrow M' \end{array}$$

where the isomorphism $\tilde{\varphi}$ in the middle is the homomorphism induced by φ (as in Theorem 5.14).

The ‘first isomorphism theorem’ is the following consequence:

Corollary 5.16. *Suppose $\varphi : M \rightarrow M'$ is a surjective R -module homomorphism. Then*

$$M' \cong \frac{M}{\ker \varphi}.$$

If M is an R -module and N is a submodule of M , then there is a bijection (cf. §II.8.3)

$$u : \{\text{submodules } P \text{ of } M \text{ containing } N\} \rightarrow \{\text{submodules of } M/N\}$$

preserving inclusions, and the ‘third isomorphism theorem’ holds:

Proposition 5.17. *Let N be a submodule of an R -module M , and let P be a submodule of M containing N . Then P/N is a submodule of M/N , and*

$$\frac{M/N}{P/N} \cong \frac{M}{P}.$$

We also have a version of the ‘second isomorphism theorem’ (cf. Proposition II.8.11), with simplifications due to the fact that normality is not an issue in the theory of modules:

Proposition 5.18. *Let N, P be submodules of an R -module M . Then*

- $N + P$ is a submodule of M ;
- $N \cap P$ is a submodule of P , and

$$\frac{N + P}{N} \cong \frac{P}{N \cap P}.$$

More generally, it is hopefully clear that the sum $\sum_{\alpha} N_{\alpha}$ and the intersection $\bigcap_{\alpha} N_{\alpha}$ of any family $\{N_{\alpha}\}_{\alpha}$ of submodules of an R -module M (which are defined as *subgroups* of the abelian group M ; cf. for example Lemma II.6.3) are submodules of M .

Exercises

5.1. $\triangleright$ Let R be a ring. The *opposite* ring R° is obtained from R by reversing the multiplication: that is, the product $a \bullet b$ in R° is defined to be $ba \in R$. Prove that the identity map $R \rightarrow R^\circ$ is an isomorphism if and only if R is commutative. Prove that $\mathcal{M}_n(\mathbb{R})$ is isomorphic to its opposite (*not* via the identity map!). Explain how to turn right- R -modules into left- R -modules and conversely, if $R \cong R^\circ$. [§5.1, VIII.5.19]

5.2. $\triangleright$ Prove Claim 5.1. [§5.1]

5.3. $\triangleright$ Let M be a module over a ring R . Prove that $0 \cdot m = 0$ and that $(-1) \cdot m = -m$, for all $m \in M$. [§5.2]

5.4. $\neg$ Let R be a ring. A nonzero R -module M is *simple* (or *irreducible*) if its only submodules are $\{0\}$ and M . Let M, N be simple modules, and let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that either $\varphi = 0$ or φ is an isomorphism. (This rather innocent statement is known as *Schur's lemma*.) [5.10, 6.16, VI.1.16]

5.5. Let R be a commutative ring, viewed as an R -module over itself, and let M be an R -module. Prove that $\text{Hom}_{R\text{-Mod}}(R, M) \cong M$ as R -modules.

5.6. Let G be an abelian group. Prove that if G has a structure of $\mathbb{Q}$ -vector space, then it has only one such structure. (Hint: First prove that every element of G has necessarily infinite order. Alternative hint: The unique ring homomorphism $\mathbb{Z} \rightarrow \mathbb{Q}$ is an epimorphism.)

5.7. Let K be a field, and let $k \subseteq K$ be a subfield of K . Show that K is a vector space over k (and in fact a k -algebra) in a natural way. In this situation, we say that K is an *extension* of k .

5.8. What is the initial object of the category $R\text{-Alg}$?

5.9. $\neg$ Let R be a commutative ring, and let M be an R -module. Prove that the operation of composition on the R -module $\text{End}_{R\text{-Mod}}(M)$ makes the latter an R -algebra in a natural way.

Prove that $\mathcal{M}_n(R)$ (cf. Exercise 1.4) is an R -algebra, in a natural way. [VI.1.12, VI.2.3]

5.10. Let R be a commutative ring, and let M be a simple R -module (cf. Exercise 5.4). Prove that $\text{End}_{R\text{-Mod}}(M)$ is a division R -algebra.

5.11. $\triangleright$ Let R be a commutative ring, and let M be an R -module. Prove that there is a natural bijection between the set of $R[x]$ -module structures on M and $\text{End}_{R\text{-Mod}}(M)$. [§VI.7.1]

5.12. $\triangleright$ Let R be a ring. Let M, N be R -modules, and let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Assume φ is a bijection, so that it has an inverse φ^{-1} as a set-function. Prove that φ^{-1} is a homomorphism of R -modules. Conclude that a bijective R -module homomorphism is an isomorphism of R -modules. [§5.2, §VI.2.1, §IX.1.3]

5.13. Let R be an integral domain, and let I be a nonzero *principal* ideal of R . Prove that I is isomorphic to R as an R -module.

5.14. ▷ Prove Proposition 5.18. [§5.4]

5.15. Let R be a commutative ring, and let I, J be ideals of R . Prove that $I \cdot (R/J) \cong (I + J)/J$ as R -modules.

5.16. ¬ Let R be a commutative ring, M an R -module, and let $a \in R$ be a nilpotent element, determining a submodule aM of M . Prove that $M = 0 \iff aM = M$. (This is a particular case of *Nakayama's lemma*, Exercise VI.3.8.) [VI.3.8]

5.17. ▷ Let R be a commutative ring, and let I be an ideal of R . Noting that $I^j \cdot I^k \subseteq I^{j+k}$, define a ring structure on the direct sum

$$\text{Rees}_R(I) := \bigoplus_{j \geq 0} I^j = R \oplus I \oplus I^2 \oplus I^3 \oplus \cdots.$$

The homomorphism sending R identically to the first term in this direct sum makes $\text{Rees}_R(I)$ into an R -algebra, called the *Rees algebra* of I . Prove that if $a \in R$ is a non-zero-divisor, then the Rees algebra of (a) is isomorphic to the polynomial ring $R[x]$ (as an R -algebra). [5.18]

5.18. With notation as in Exercise 5.17 let $a \in R$ be a non-zero-divisor, let I be any ideal of R , and let J be the ideal aI . Prove that $\text{Rees}_R(J) \cong \text{Rees}_R(I)$.

6. Products, coproducts, etc., in $R\text{-Mod}$

I have stated several times that categories such as $R\text{-Mod}$ are ‘well-behaved’. We will explore in this section the sense in which this can be formalized at this stage. The bottom line is that these categories enjoy the same nice properties that we have noted along the way for the category Ab .

I will also include here some general considerations on finitely generated modules and algebras.

As in the previous section, I will write ‘module’ for ‘left-module’; the reader should make appropriate adaptations to the case of right-modules. Little will be lost by assuming that all rings appearing here are commutative (thereby removing the distinction between left- and right-modules).

6.1. Products and coproducts. As in Ab , products and coproducts exist, and *finite* products and coproducts coincide, in $R\text{-Mod}$. Indeed, recall the construction of the *direct sum* of two abelian groups (§II.3.5): if M and N are abelian groups, then $M \oplus N$ denotes their product, with componentwise operation. If M and N are R -modules, we can give an R -module structure to $M \oplus N$ by prescribing $\forall r \in R$

$$r(m, n) := (rm, rn).$$

This defines the *direct sum* of M, N , as an R -module. Note that $M \oplus N$ comes together with several homomorphisms of R -modules:

$$\pi_M : M \oplus N \rightarrow M, \quad \pi_N : M \oplus N \rightarrow N$$

sending (m, n) to m , n , respectively, and

$$i_M : M \rightarrow M \oplus N, \quad i_N : N \rightarrow M \oplus N$$

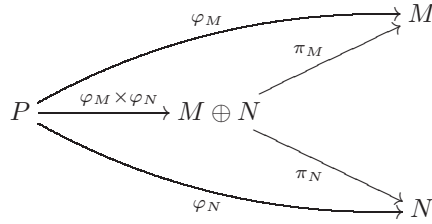
sending m to $(m, 0)$ and n to $(0, n)$.

Proposition 6.1. *The direct sum $M \oplus N$ satisfies the universal properties of both the product and the coproduct of M and N .*

Proof. *Product:* Let P be an R -module, and let $\varphi_M : P \rightarrow M$, $\varphi_N : P \rightarrow N$ be two R -module homomorphisms. The definition of an R -module homomorphism

$$\varphi_M \times \varphi_N : P \rightarrow M \oplus N$$

is forced by the needed commutativity of the diagram



That is,

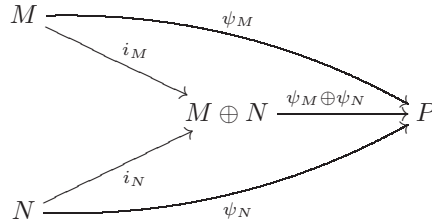
$$(\forall p \in P) : (\varphi_M \times \varphi_N)(p) := (\varphi_M(p), \varphi_N(p)).$$

This *is* an R -module homomorphism, and it is the unique one making the diagram commute; therefore $M \oplus N$ works as a product of M and N .

Coproduct: View the preceding argument through a mirror! Let P be an R -module, and let $\psi_M : M \rightarrow P$, $\psi_N : N \rightarrow P$ be two R -module homomorphisms. The definition of an R -module homomorphism

$$\psi_M \oplus \psi_N : M \oplus N \rightarrow P$$

is forced by the needed commutativity of the diagram



That is²⁹, $(\forall m \in M)(\forall n \in N)$

$$\begin{aligned} (\psi_M \oplus \psi_N)(m, n) &= (\psi_M \oplus \psi_N)(m, 0) + (\psi_M \oplus \psi_N)(0, n) \\ &= (\psi_M \oplus \psi_N) \circ i_M(m) + (\psi_M \oplus \psi_N) \circ i_N(n) \\ &= \psi_M(m) + \psi_N(n). \end{aligned}$$

This *is* an R -module homomorphism: it is a homomorphism of abelian groups by virtue of the commutativity of addition in P , and it clearly preserves the action of R .

²⁹This should look familiar; cf. Exercise II.3.3.

Since it is the unique R -module homomorphism making the diagram commute, this verifies that $M \oplus N$ works as a coproduct of M and N . $\square$

It may seem like a good idea to write $M \times N$ rather than $M \oplus N$ when viewing the latter as the *product* of M and N ; but in due time (§VIII.2) we will encounter $M \times N$ again, in the context of ‘bilinear maps’, and in that context $M \times N$ is *not* viewed as an R -module.

The reader should also work out the *fibred* versions of these constructions; cf. Exercises 6.10 and 6.11.

The fact that finite products and coproducts agree in $R\text{-Mod}$ does not extend to the infinite case (Exercise 6.7).

6.2. Kernels and cokernels. The facts that $R\text{-Mod}$ has a zero-object (the 0-module), its Hom sets are abelian groups (§5.2), and it has (finite) products and coproducts make $R\text{-Mod}$ an *additive* category. The fact that $R\text{-Mod}$ has well-behaved kernels and cokernels, which we review next, upgrades it to the status of *abelian category*. (We will come back to these general definitions in §IX.1.)

In general, monomorphisms and epimorphisms do not automatically satisfy good properties, even when objects of a given category are realized by adding structure to sets. For example, we have seen that the precise relationship between ‘surjective morphism’ and ‘epimorphism’ may be rather subtle: epimorphisms are surjective in $\mathbf{Grp}$, but for complicated reasons (§II.8.6); and there are epimorphisms that are not surjective in $\mathbf{Ring}$ (§2.3).

The situation in $R\text{-Mod}$ is as simple as it can be. Recall that we have identified universal properties for kernels and cokernels (cf. §II.8.6); in the category $R\text{-Mod}$ these would go as follows: if

$$\begin{array}{ccccc} & & 0 & & \\ & \searrow & \text{curved arrow} & \nearrow & \\ P & \xrightarrow{\alpha} & M & \xrightarrow{\varphi} & N \\ & \searrow \exists! \bar{\alpha} & \downarrow & & \\ & & \ker \varphi & & \end{array}$$

while $\text{coker } \varphi$ is initial with respect to the property of factoring R -module homomorphisms $\beta : N \rightarrow P$ such that $\beta \circ \varphi = 0$:

$$\begin{array}{ccccc} & & 0 & & \\ & \searrow & \text{curved arrow} & \nearrow & \\ M & \xrightarrow{\varphi} & N & \xrightarrow{\beta} & P \\ & & \downarrow \pi & \nearrow \exists! \bar{\beta} & \\ & & \text{coker } \varphi & & \end{array}$$

Proposition 6.2. *The following hold in $R\text{-Mod}$:*

- *kernels and cokernels exist;*

- φ is a monomorphism $\iff \ker \varphi$ is trivial $\iff \varphi$ is injective as a set-function;
- φ is an epimorphism $\iff \operatorname{coker} \varphi$ is trivial $\iff \varphi$ is surjective as a set-function.

Further, every monomorphism identifies its source with the kernel of some morphism, and every epimorphism identifies its target with the cokernel of some morphism.

This proposition of course simply generalizes to $R\text{-Mod}$ facts we know already from our study of $\mathbf{Ab}$, and a quick review should suffice for the careful reader. Kernels exist: indeed, the ‘standard’ definition of kernel satisfies the universal properties spelled out above (same argument as in Proposition II.6.6). Cokernels exist: indeed, let

$$\operatorname{coker} \varphi = \frac{N}{\operatorname{im} \varphi};$$

if $\beta : N \rightarrow P$ is such that $\beta \circ \varphi = 0$, then $\operatorname{im} \varphi \subseteq \ker \beta$; so β must factor uniquely through $N/\operatorname{im} \varphi$ by the universal property of quotients, Theorem 5.14. That is, $N/\operatorname{im} \varphi$ does satisfy the universal property for cokernels.

The proofs of all the implications in the second and third points in Proposition 6.2 follow familiar patterns from (for example) Proposition II.6.12 and Proposition II.8.18. The last sentence of Proposition 6.2 simply reiterates the slogan *submodule $\iff$ kernel* and its mirror statement (which is just as true). Further details are left to the reader.

By the way, whatever happened to the conditions characterizing monomorphisms and epimorphisms in $\mathbf{Set}$ (Proposition I.2.1)? In $\mathbf{Set}$, a function with non-empty source is a monomorphism if and only if it has a left-inverse, and it is an epimorphism if and only if it has a right-inverse. We have learned not to expect any of this to happen in more general categories. Modules are not an exception: the function $\mathbb{Z} \rightarrow \mathbb{Z}$ defined by ‘multiplication by 2’ is a monomorphism without a left-inverse, and the projection $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is an epimorphism without a right-inverse. We will come back to this point in §7.2.

6.3. Free modules and free algebras. The universal property of *free R -modules* is modeled after the properties defining the other free objects we have encountered: the goal is to define the R -module containing a given set A ‘in the most efficient way’. Again, the situation will match the case of abelian groups closely, so the reader may want to refer back to §II.5.4.

The universal property formalizing the heuristic requirement goes as follows: given a set A , we are seeking an R -module $F^R(A)$, called a *free R -module* on the set A , together with a set-function $j : A \rightarrow F^R(A)$, such that for all R -modules M and set-functions $f : A \rightarrow M$ there exists a unique R -module homomorphism

$\varphi : F^R(A) \rightarrow M$ such that the diagram

$$\begin{array}{ccc} F^R(A) & \xrightarrow{\varphi} & M \\ \uparrow j & \nearrow f & \\ A & & \end{array}$$

commutes. Abstract nonsense guarantees that such an R -module is unique up to isomorphism if it exists at all (Proposition I.5.4) and that the function $j : A \rightarrow F^R(A)$ is necessarily injective (cf. Exercise II.5.3). The question is, does it exist?

The answer will not appear to be very exciting, since it generalizes directly the case of abelian groups (that is, $\mathbb{Z}$ -modules). Given any set A , define the (possibly infinite) direct sum $N^{\oplus A}$ of an R -module N as follows:

$$N^{\oplus A} := \{\alpha : A \rightarrow N \mid \alpha(a) \neq 0 \text{ for only finitely many elements } a \in A\}.$$

Of course this agrees with the definition given for abelian groups in §II.5.4; $N^{\oplus A}$ has an evident R -module structure, obtained by defining, for all $r \in R$ and $a \in A$,

$$(r\alpha)(a) := r(\alpha(a)).$$

For $N = R$ we may define a function $j : A \rightarrow R^{\oplus A}$ by sending $a \in A$ to the function $j_a : A \rightarrow R$:

$$(\forall x \in A) : \quad j_a(x) := \begin{cases} 1 & \text{if } x = a, \\ 0 & \text{if } x \neq a. \end{cases}$$

Claim 6.3. $F^R(A) \cong R^{\oplus A}$.

The proof of this claim matches precisely the proof of Proposition II.5.6 and is left to the reader (Exercise 6.1). The key is that every element of $R^{\oplus A}$ may be written uniquely as a *finite* sum

$$\sum_{a \in A} r_a a$$

(shorthand for $\sum_{a \in A} r_a j(a)$); incidentally, this is how elements of ‘the free R -module on A ’ are often written—this is legal, by virtue of Claim 6.3.

In particular, for $A = \{1, \dots, n\}$ a *finite* set, Claim 6.3 states that the R -module $R^{\oplus n}$, with $j : A \rightarrow R^{\oplus n}$ defined by

$$j(i) := (0, \dots, 0, \underset{i\text{-th place}}{1}, 0, \dots, 0) \in R^{\oplus n},$$

satisfies the universal property for $F^R(\{1, \dots, n\})$.

This is all entirely analogous to the story for $\mathbb{Z}$ -modules. The situation becomes a little more interesting if we switch from R -modules to *commutative R -algebras*; the category is different, so we should expect a different answer. The finite case is essentially the only one we will need in these notes, so assume $A = \{\mathbf{1}, \dots, \mathbf{n}\}$ is a finite set. In this case, we write $R[A]$ for the polynomial ring $R[x_1, \dots, x_n]$; we have a set-function $j : A \rightarrow R[A]$, defined by $j(\mathbf{i}) = x_{\mathbf{i}}$.

Proposition 6.4. $R[A]$ is a free commutative R -algebra on the set A .

Proof. The statement translates into the following: for every commutative R -algebra S and every set-function $f : A \rightarrow S$, there exists a unique R -algebra homomorphism $\varphi : R[A] \rightarrow S$ such that the diagram

$$\begin{array}{ccc} R[A] & \xrightarrow{\varphi} & S \\ \uparrow j & \nearrow f & \\ A & & \end{array}$$

commutes. Since S is an R -algebra, we have a fixed homomorphism of rings $\alpha : R \rightarrow S$ (cf. Example 5.6). Then we may construct $\varphi : R[A] = R[x_1, \dots, x_n] \rightarrow S$ by applying n times the ‘extension property’ of Example 2.3: extend α to $R[x_1]$ so as to map x_1 to $f(\mathbf{1})$, then to $R[x_1, x_2] = R[x_1][x_2]$ so as to map x_2 to $f(\mathbf{2})$, etc. Note that each extension is *uniquely* determined by its requirements.

This gives φ as a ring homomorphism and shows that it is unique. The reader will verify that φ is also (automatically) an R -module homomorphism, and hence a homomorphism of R -algebras, concluding the proof. $\square$

After the fact, the reader may want to revisit §2.2 and recognize that the ‘universal property of polynomial rings $\mathbb{Z}[x_1, \dots, x_n]$ ’ given there was really a version of their role as free objects in the category of commutative rings, a.k.a. commutative $\mathbb{Z}$ -algebras.

It is not difficult to identify free objects in the larger category $R\text{-Alg}$: they consist of ‘noncommutative polynomial rings’ $R\langle A \rangle$, with variables from the set A but without any condition relating ab and ba for $a \neq b$ in A . More precisely, $R\langle A \rangle$ is isomorphic to the *monoid ring* (cf. §1.4) over the free *monoid* on A , consisting of all finite strings of elements in A , with operation defined by concatenation³⁰. We will encounter this ring again, but only in the distant future (Example VIII.4.17).

6.4. Submodule generated by a subset; Noetherian modules. Let M be an R -module, and let $A \subseteq M$ be a subset of M . By the universal property of free modules, there is a unique homomorphism of R -modules

$$\varphi_A : R^{\oplus A} \rightarrow M.$$

The image of this homomorphism is a submodule of M , the *submodule generated by* A in M , usually denoted $\langle A \rangle$ (or $\langle a_1, \dots, a_n \rangle$ if $A = \{a_1, \dots, a_n\}$ is finite). Thus,

$$\langle A \rangle = \left\{ \sum_{a \in A} r_a a \mid r_a \neq 0 \text{ for only finitely many elements } a \in A \right\}.$$

It is hopefully clear that $\langle A \rangle$ is the smallest submodule of M containing A .

The module M is *finitely generated* if $M = \langle A \rangle$ for a *finite* set A , that is, if and only if there is a surjective homomorphism of R -modules

$$R^{\oplus n} \twoheadrightarrow M$$

for some n . One of the highlights of Chapter VI will be the classification of *finitely generated modules over PIDs* (Theorem VI.5.6). I have already briefly mentioned

³⁰This construction is similar to the free *group* on A , but without the complication of the presence of inverses and of possible cancellations.

the case for $\mathbb{Z}$ (recall that $\mathbb{Z}$ is a PID and $\mathbb{Z}$ -modules are nothing but abelian groups!) back in §II.6.3.

Finitely generated R -modules are tremendously important, but they are not as well-behaved as one might hope at first. For example, it may be that a module M is finitely generated, but some *submodule* of M is *not* finitely generated!

Example 6.5. Let $R = \mathbb{Z}[x_1, x_2, \dots]$, a polynomial ring on infinitely many indeterminates. Then R is finitely generated as an R -module: indeed, 1 generates it. However, the ideal

$$(x_1, x_2, \dots)$$

of R generated by all indeterminates is *not* finitely generated as an R -module (Exercise 6.14). $\lrcorner$

Definition 6.6. An R -module M is *Noetherian* if every submodule of M is finitely generated as an R -module. $\lrcorner$

Thus, a ring R is Noetherian in the sense of Definition 4.2 if and only if it is Noetherian ‘as a module over itself’. The ring in Example 6.5 is not Noetherian.

We will study the Noetherian condition more carefully later on (§V.1.1); but we can already see one reason why this is a good, ‘solid’ notion.

Proposition 6.7. *Let M be an R -module, and let N be a submodule of M . Then M is Noetherian if and only if both N and M/N are Noetherian.*

Proof. If M is Noetherian, then so is M/N (same proof as for Exercise 4.2), and so is N (because every submodule of N is a submodule of M , so it is finitely generated because M is Noetherian). This proves the ‘only if’ part of the statement.

For the converse, assume N and M/N are Noetherian, and let P be a submodule of M ; we have to prove that P is finitely generated. Since $P \cap N$ is a submodule of N and N is Noetherian, $P \cap N$ is finitely generated. By the ‘second isomorphism theorem’, Proposition 5.18,

$$\frac{P}{P \cap N} \cong \frac{P + N}{N},$$

and hence $P/(P \cap N)$ is isomorphic to a submodule of M/N . Since M/N is Noetherian, this shows that $P/(P \cap N)$ is finitely generated.

It follows that P itself is finitely generated, by Exercise 6.18. $\square$

Corollary 6.8. *Let R be a Noetherian ring, and let M be a finitely generated R -module. Then M is Noetherian (as an R -module).*

Proof. Indeed, by hypothesis there is an onto homomorphism $R^{\oplus n} \twoheadrightarrow M$ of R -modules; hence (by the first isomorphism theorem, Corollary 5.16) M is isomorphic to a quotient of $R^{\oplus n}$. By Proposition 6.7, it suffices to prove that $R^{\oplus n}$ is Noetherian.

This may be done by induction. The statement is true for $n = 1$ by hypothesis. For $n > 1$, assume we know that $R^{\oplus(n-1)}$ is Noetherian; since $R^{\oplus(n-1)}$ may be viewed as a submodule of $R^{\oplus n}$, in such a way that

$$\frac{R^{\oplus n}}{R^{\oplus(n-1)}} \cong R$$

(Exercise 6.4), and R is Noetherian, it follows that $R^{\oplus n}$ is Noetherian, again by applying Proposition 6.7. $\square$

6.5. Finitely generated vs. finite type. If S is an R -algebra, it may be ‘finitely generated’ in two very different ways: as an R -module and as an R -algebra. It is important to keep these two concepts well distinct, although unfortunately the language used to express them is very similar.

The following definitions differ in three small details...

“ S is finitely generated as a module over R if there is an onto homomorphism of R -modules from the free R -module on a finite set to S .”

“ S is finitely generated as an algebra over R if there is an onto homomorphism of R -algebras from the free R -algebra on a finite set to S .”

The mathematical difference is more substantial than it may appear. As we have seen in §6.3, the free R -module over a finite set $A = \{1, \dots, n\}$ is isomorphic to $R^{\oplus n}$; the free commutative R -algebra over A is isomorphic to $R[x_1, \dots, x_n]$. Thus, a commutative³¹ ring S is finitely generated *as an R -module* if there is an onto homomorphism of R -modules

$$R^{\oplus n} \twoheadrightarrow S$$

for some n ; it is finitely generated *as an R -algebra* if there is an onto homomorphism of R -algebras

$$R[x_1, \dots, x_n] \twoheadrightarrow S$$

for some n . In other words, S is finitely generated as an R module if and only if $S \cong R^{\oplus n}/M$ for some n and a submodule M of $R^{\oplus n}$; it is a finite-type R -algebra if and only if $S \cong R[x_1, \dots, x_n]/I$ for some n and an ideal I of $R[x_1, \dots, x_n]$.

We say that S is *finite* in the first case³² and *of finite type* in the second. It is clear that ‘finite’ $\implies$ ‘finite type’; it should be just as clear that the converse does not hold.

Example 6.9. The polynomial ring $R[x]$ is a finite-type R -algebra, but it is not finite as an R -module. $\lrcorner$

The distinction, while macroscopic in general, may evaporate in special, important cases. For example, one can prove that if k and K are fields and $k \subseteq K$, then K is of finite type over k if and only if it is in fact finite as a k -module (that is, it is a *finite-dimensional k -vector space*). This is one version of *Hilbert’s Nullstellensatz*, a deep result we already mentioned in Example 4.15 and that we will prove (in an important class of examples) in §VII.2.2.

David Hilbert’s name is associated to another important result concerning finite-type R -algebras: if R is Noetherian (as a ring, that is, as an R -module)

³¹We are mostly interested in the commutative case, so I will make this hypothesis here; the only change in the general case is typographical: $\langle \dots \rangle$ rather than $[\dots]$. Also, note that a commutative ring is finitely generated as an algebra if and only if it is finitely generated as a *commutative algebra*; cf. Exercise 6.15.

³²This is particularly unfortunate, since S may very well be an *infinite* set.

and S is a finite-type R -algebra, then S is also Noetherian (as a ring, that is, as an S -module). This is an immediate consequence of the so-called *Hilbert's basis theorem*.

The proof of Hilbert's basis theorem is completely elementary: it could be given here as an exercise, with a few key hints; we will see it in §V.1.1.

Exercises

6.1. ▷ Prove Claim 6.3. [§6.3]

6.2. Prove or disprove that if R is ring and M is a nonzero R -module, then M is not isomorphic to $M \oplus M$.

6.3. Let R be a ring, M an R -module, and $p : M \rightarrow M$ an R -module homomorphism such that $p^2 = p$. (Such a map is called a *projection*.) Prove that $M \cong \ker p \oplus \operatorname{im} p$.

6.4. ▷ Let R be a ring, and let $n > 1$. View $R^{\oplus(n-1)}$ as a submodule of $R^{\oplus n}$, via the injective homomorphism $R^{\oplus(n-1)} \hookrightarrow R^{\oplus n}$ defined by

$$(r_1, \dots, r_{n-1}) \mapsto (r_1, \dots, r_{n-1}, 0).$$

Give a one-line proof that

$$\frac{R^{\oplus n}}{R^{\oplus(n-1)}} \cong R.$$

[§6.4]

6.5. ▷ (Notation as in §6.3.) For any ring R and any two sets A_1, A_2 , prove that $(R^{\oplus A_1})^{\oplus A_2} \cong R^{\oplus(A_1 \times A_2)}$. [§VIII.2.2]

6.6. ¬ Let R be a ring, and let $F = R^{\oplus n}$ be a finitely generated free R -module. Prove that $\operatorname{Hom}_{R\text{-Mod}}(F, R) \cong F$. On the other hand, find an example of a ring R and a nonzero R -module M such that $\operatorname{Hom}_{R\text{-Mod}}(M, R) = 0$. [6.8]

6.7. ▷ Let A be any set.

- For any family $\{M_a\}_{a \in A}$ of modules over a ring R , define the *product* $\prod_{a \in A} M_a$ and coproduct $\bigoplus_{a \in A} M_a$. If $M_a \cong R$ for all $a \in A$, these are denoted $R^A, R^{\oplus A}$, respectively.
- Prove that $\mathbb{Z}^{\mathbb{N}} \not\cong \mathbb{Z}^{\oplus \mathbb{N}}$. (Hint: Cardinality.)

[§6.1, 6.8]

6.8. Let R be a ring. If A is any set, prove that $\operatorname{Hom}_{R\text{-Mod}}(R^{\oplus A}, R)$ satisfies the universal property for the *product* of the family $\{R_a\}_{a \in A}$, where $R_a \cong R$ for all a ; thus, $\operatorname{Hom}_{R\text{-Mod}}(R^{\oplus A}, R) \cong R^A$. Conclude that $\operatorname{Hom}_{R\text{-Mod}}(R^{\oplus A}, R)$ is *not* isomorphic to $R^{\oplus A}$ in general (cf. Exercises 6.6 and 6.7.)

6.9. ¬ Let R be a ring, F a nonzero free R -module, and let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that φ is onto if and only if for all R -module homomorphisms $\alpha : F \rightarrow N$ there exists an R -module homomorphism $\beta : F \rightarrow M$

such that $\alpha = \varphi \circ \beta$. (Free modules are *projective*, as we will see in Chapter VIII.) [7.8, VI.5.5]

6.10. $\triangleright$ (Cf. Exercise I.5.12.) Let M , N , and Z be R -modules, and let $\mu : M \rightarrow Z$, $\nu : N \rightarrow Z$ be homomorphisms of R -modules.

Prove that $R\text{-Mod}$ has ‘fibered products’: there exists an R -module $M \times_Z N$ with R -module homomorphisms $\pi_M : M \times_Z N \rightarrow M$, $\pi_N : M \times_Z N \rightarrow N$, such that $\mu \circ \pi_M = \nu \circ \pi_N$, and which is universal with respect to this requirement. That is, for every R -module P and R -module homomorphisms $\varphi_M : P \rightarrow M$, $\varphi_N : P \rightarrow N$ such that $\mu \circ \varphi_M = \nu \circ \varphi_N$, there exists a unique R -module homomorphism $P \rightarrow M \times_Z N$ making the diagram

$$\begin{array}{ccccc}
 P & & & & \\
 \varphi_M \searrow & & \varphi_N \searrow & & \\
 & M \times_Z N & \xrightarrow{\pi_N} & N & \\
 & \downarrow \pi_M & & \downarrow \nu & \\
 & M & \xrightarrow{\mu} & Z &
 \end{array}$$

(A dotted arrow from P to $M \times_Z N$ is labeled $\exists!$)

commute. The module $M \times_Z N$ may be called the *pull-back* of M along ν (or of N along μ , since the construction is symmetric). ‘Fiber diagrams’

$$\begin{array}{ccc}
 M \times_Z N & \longrightarrow & N \\
 \downarrow & \square & \downarrow \nu \\
 M & \xrightarrow{\mu} & Z
 \end{array}$$

are commutative, but ‘even better’ than commutative; they are often decorated by a square, as shown here. [§6.1, 6.11, §IX.1.4]

6.11. $\triangleright$ Define a notion of *fibered coproduct* of two R -modules M , N , along an R -module A , in the style of Exercise 6.10 (and cf. Exercise I.5.12)

$$\begin{array}{ccc}
 A & \xrightarrow{\nu} & N \\
 \mu \downarrow & & \downarrow \\
 M & \longrightarrow & M \oplus_A N
 \end{array}$$

Prove that fibered coproducts exist in $R\text{-Mod}$. The fibered coproduct $M \oplus_A N$ is called the *push-out* of M along ν (or of N along μ). [§6.1]

6.12. Prove Proposition 6.2.

6.13. Prove that every homomorphic image of a finitely generated module is finitely generated.

6.14. $\triangleright$ Prove that the ideal $(x_1, x_2, \dots)$ of the ring $R = \mathbb{Z}[x_1, x_2, \dots]$ is not finitely generated (as an ideal, i.e., as an R -module). [§6.4]

6.15. $\triangleright$ Let R be a commutative ring. Prove that a *commutative* R -algebra S is finitely generated *as an algebra* over R if and only if it is finitely generated *as a commutative algebra* over R . (Cf. §6.5.) [§6.5]

6.16. $\triangleright$ Let R be a ring. A (left-) R -module M is *cyclic* if $M = \langle m \rangle$ for some $m \in M$. Prove that simple modules (cf. Exercise 5.4) are cyclic. Prove that an R -module M is cyclic if and only if $M \cong R/I$ for some (left-)ideal I . Prove that every quotient of a cyclic module is cyclic. [6.17, §VI.4.1]

6.17. $\neg$ Let M be a cyclic R -module, so that $M \cong R/I$ for a (left-)ideal I (Exercise 6.16), and let N be another R -module.

- Prove that $\text{Hom}_{R\text{-Mod}}(M, N) \cong \{n \in N \mid (\forall a \in I), an = 0\}$.
- For $a, b \in \mathbb{Z}$, prove that $\text{Hom}_{\text{Ab}}(\mathbb{Z}/a\mathbb{Z}, \mathbb{Z}/b\mathbb{Z}) \cong \mathbb{Z}/\gcd(a, b)\mathbb{Z}$.

[7.7]

6.18. $\triangleright$ Let M be an R -module, and let N be a submodule of M . Prove that if N and M/N are both finitely generated, then M is finitely generated. [§6.4]

7. Complexes and homology

In many contexts, modules arise not ‘one at a time’ but in whole series: for example, a real manifold of dimension d has one ‘homology’ group for each dimension from 0 to d . It is necessary to develop a language capable of dealing with whole sequences of modules at once. This is the language of *homological algebra*, of which we will get a tiny taste in this section, and a slightly heartier course in Chapter IX.

7.1. Complexes and exact sequences. A *chain complex* of R -modules (or, for simplicity, a *complex*) is a sequence of R -modules and R -module homomorphisms

$$\cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots$$

such that $(\forall i) : d_i \circ d_{i+1} = 0$.

The notation $(M_\bullet, d_\bullet)$ may be used to denote a complex, or simply $M_\bullet$ for simplicity (but do not forget that the homomorphisms d_i are part of the information carried by a complex).

A complex may be infinite in both directions; ‘tails’ of 0’s are (usually) omitted. Several possible alternative conventions may be used: for example, indices may be increasing rather than decreasing, giving a *cochain complex* (whose *homology* is called *cohomology*; this will be our choice in Chapter IX). Such choices are clearly mathematically immaterial, at least for the simple considerations which follow.

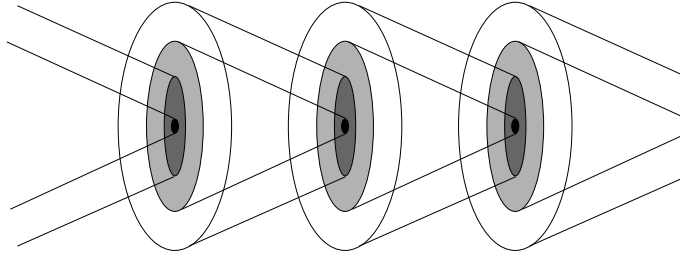
The homomorphisms d_i are called *boundary*, or *differentials*, due to important examples from geometry. Note that the defining condition

$$d_i \circ d_{i+1} = 0$$

is equivalent to the requirement

$$\text{im } d_{i+1} \subseteq \ker d_i.$$

I carry in my mind an image such as

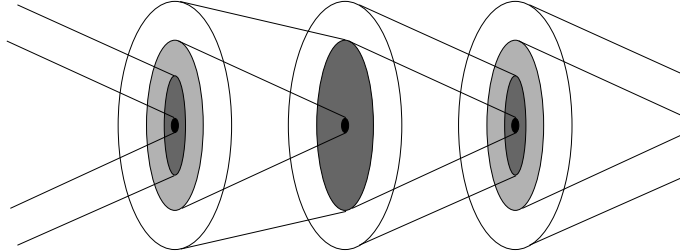


when I think of a complex. The ovals are the modules M_i ; the fat black dots are the 0 elements; the gray ovals, getting squashed to zero at each step, are the kernels; and I thus visualize the fact that the image of ‘the preceding homomorphism’ falls inside the kernel of ‘the next homomorphism’.

The picture is inaccurate in that it hints that the ‘difference’ between the image of d_{i+1} and the kernel of d_i (that is, the areas colored in a lighter shade of gray) should be the same for all i ; this is of course not the case in general. In fact, almost the whole point about complexes is to ‘measure’ this difference, which is called the *homology* of the complex (cf. §7.3). We say that a complex is *exact* ‘at M_i ’ if it has no homology there; that is,

$$\operatorname{im} d_{i+1} = \ker d_i.$$

Visually,



This complex appears to be exact at the oval in the middle.

For example, if M_i is a trivial module (usually denoted simply by 0), then the complex is necessarily exact at M_i , since then $\operatorname{im} d_{i+1} = \ker d_i = 0$.

A complex is *exact* and is often called an *exact sequence* if it is exact at all its modules.

Example 7.1. A complex

$$\cdots \longrightarrow 0 \longrightarrow L \xrightarrow{\alpha} M \longrightarrow \cdots$$

is exact at L if and only if α is a monomorphism.

Indeed, exactness at L is equivalent to $\ker \alpha = \operatorname{image}$ of the trivial homomorphism $0 \rightarrow L$, that is, to

$$\ker \alpha = 0.$$

This is equivalent to the injectivity of α (Proposition 6.2). ┘

Example 7.2. A complex

$$\cdots \longrightarrow M \xrightarrow{\beta} N \longrightarrow 0 \longrightarrow \cdots$$

is exact at N if and only if β is an epimorphism.

Indeed, the complex is exact at N if and only if $\text{im } \beta = \text{kernel of the trivial homomorphism } N \rightarrow 0$, that is, $\text{im } \beta = N$. $\lrcorner$

Definition 7.3. A *short exact sequence* is an exact complex of the form

$$0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0.$$

As seen in the previous two examples, exactness at L and N is equivalent to α being injective and β being surjective. The extra piece of data carried by a short exact sequence is the exactness at M , that is,

$$\text{im } \alpha = \ker \beta;$$

by the first isomorphism theorem (Corollary 5.16), we then have

$$N \cong \frac{M}{\ker \beta} = \frac{M}{\text{im } \alpha}.$$

All in all, we have good material to work on some more Pavlovian conditioning: at the sight of a short exact sequence as above, the reader should instinctively identify L with a submodule of M (via the injective map α) and N with the quotient M/L (via the isomorphism induced by the surjective map β , under the auspices of the first isomorphism theorem).

Short exact sequences abound in nature. For example, a single homomorphism $\varphi : M \rightarrow M'$ gives rise immediately to a short exact sequence

$$0 \longrightarrow \ker \varphi \longrightarrow M \longrightarrow \text{im } \varphi \longrightarrow 0.$$

In fact, one important reason to focus on short exact sequences is that this observation allows us to break up every exact complex into a large number of short exact sequences: contemplate the impressive diagram

$$\begin{array}{ccccccc}
 & & 0 & & & & 0 \\
 & & \searrow & & \nearrow & & \\
 & & \text{im } d_{i+1} = \ker d_i & & & & \\
 & & \nearrow & & \searrow & & \\
 M_{i+2} & \xrightarrow{d_{i+2}} & M_{i+1} & \xrightarrow{d_{i+1}} & M_i & \xrightarrow{d_i} & M_{i-1} \\
 & \searrow & \nearrow & & \searrow & \nearrow & \\
 & \text{im } d_{i+2} = \ker d_{i+1} & & & \text{im } d_i = \ker d_{i-1} & & \\
 & \nearrow & \searrow & & \nearrow & \searrow & \\
 0 & & 0 & & 0 & & 0
 \end{array}$$

The diagonal sequences are short exact sequences, and they interlock nicely by the exactness of the horizontal complex.

This observation simplifies many arguments; cf. for example Exercise 7.5.

7.2. Split exact sequences. A particular case of short exact sequence arises by considering the second projection from a direct sum: $M_1 \oplus M_2 \rightarrow M_2$; there is then an exact sequence

$$0 \longrightarrow M_1 \longrightarrow M_1 \oplus M_2 \longrightarrow M_2 \longrightarrow 0 ,$$

obtained by identifying M_1 with the kernel of the projection. These short exact sequences are said to ‘split’; more generally, a short exact sequence

$$0 \longrightarrow M_1 \longrightarrow N \longrightarrow M_2 \longrightarrow 0$$

‘splits’ if it is isomorphic to one of these sequences in the sense that there is a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_1 & \longrightarrow & N & \longrightarrow & M_2 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & M'_1 & \longrightarrow & M'_1 \oplus M'_2 & \longrightarrow & M'_2 \longrightarrow 0 \end{array}$$

in which the vertical maps are all isomorphisms³³.

Example 7.4. The exact sequence of $\mathbb{Z}$ -modules

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \frac{\mathbb{Z}}{2\mathbb{Z}} \longrightarrow 0$$

is *not* split. ┘

Splitting sequences give us the opportunity to go back to a question we left dangling at the end of §6.2: what should we make of the condition of ‘having a left- (resp., right-) inverse’ for a homomorphism? We realized that this condition is stronger than the requirement of being a monomorphisms (resp., an epimorphism); can we give a more explicit description of such morphisms?

Proposition 7.5. *Let $\varphi : M \rightarrow N$ be an R -module homomorphism. Then*

- *φ has a left-inverse if and only if the sequence*

$$0 \longrightarrow M \xrightarrow{\varphi} N \longrightarrow \operatorname{coker} \varphi \longrightarrow 0$$

splits.

- *φ has a right-inverse if and only if the sequence*

$$0 \longrightarrow \ker \varphi \longrightarrow M \xrightarrow{\varphi} N \longrightarrow 0$$

splits.

Proof. I will prove the first part and leave the other as an exercise to the reader (Exercise 7.6).

³³In fact, this last requirement is somewhat redundant; cf. Exercise 7.11.

If the sequence splits, then φ may be identified with the embedding of M into a direct sum $M \oplus M'$, and the projection $M \oplus M' \rightarrow M$ gives a left-inverse of φ . Conversely, assume that φ has a left-inverse ψ :

$$\begin{array}{ccccc} 0 & \longrightarrow & M & \xrightarrow{\varphi} & N \\ & & \searrow \text{id} & & \downarrow \psi \\ & & & & M \end{array}$$

Then I claim that N is isomorphic to $M \oplus \ker \psi$ and that φ corresponds to the identification of M with the first factor: $M \rightarrow M \oplus \ker \psi \cong N$. The isomorphism $M \oplus \ker \psi \rightarrow N$ is given by

$$(m, k) \mapsto \varphi(m) + k;$$

its inverse $N \rightarrow M \oplus \ker \psi$ is

$$n \mapsto (\psi(n), n - \varphi\psi(n)).$$

The element $n - \varphi\psi(n)$ is in $\ker \psi$ as it should be, since

$$\psi(n - \varphi\psi(n)) = \psi(n) - \psi\varphi\psi(n) = \psi(n) - \psi(n) = 0.$$

All necessary verifications are immediate and are left to the reader. $\square$

Because of Proposition 7.5, R -module homomorphisms with a left-inverse are called *split monomorphisms*, and homomorphisms with a right-inverse are called *split epimorphisms*.

We will come back to split exact sequences (in the more demanding context of Grp) in §IV.5.2 and then later again when we return to modules and, more generally, to abelian categories (Chapters VIII and IX).

7.3. Homology and the snake lemma.

Definition 7.6. The i -th homology of a complex

$$M_{\bullet} : \cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots$$

of R -modules is the R -module

$$H_i(M_{\bullet}) := \frac{\ker d_i}{\operatorname{im} d_{i+1}}. \quad \lrcorner$$

That is, $H_i(M_{\bullet})$ is a module capturing the ‘light gray annulus’ in my heuristic picture of a complex. Of course

$$H_i(M_{\bullet}) = 0 \iff \operatorname{im} d_{i+1} = \ker d_i \iff \text{the complex } M_{\bullet} \text{ is exact at } M_i :$$

that is, the homology modules are a measure of the ‘failure of a complex from being exact’.

Example 7.7. In fact, homology should be thought of as a (vast) generalization of the notions of kernel and cokernel. Indeed, consider the (very) particular case in which $M_{\bullet}$ is the complex

$$0 \longrightarrow M_1 \xrightarrow{\varphi} M_0 \longrightarrow 0.$$

Then

$$H_1(M_\bullet) \cong \ker \varphi, \quad H_0(M_\bullet) \cong \operatorname{coker} \varphi. \quad \lrcorner$$

I will end this very brief excursion into more abstract territories by indicating how a commutative diagram involving *two* short exact sequences generates a ‘long exact sequence’ in homology. This is actually a particular case of a more general construction—according to which a suitable commutative diagram involving *three* complexes yields a *really long* ‘long exact homology sequence’. We will come back to this general construction when we deal more extensively with homological algebra in Chapter IX. The reader is also likely to learn about it in a course on algebraic topology, where this fact is put to impressive use in studying invariants of manifolds.

In the simple form we will analyze, this is affectionately known as the *snake lemma*. Consider two short exact sequences linked by homomorphisms, so as to form a commutative diagram³⁴:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L_1 & \xrightarrow{\alpha_1} & M_1 & \xrightarrow{\beta_1} & N_1 & \longrightarrow & 0 \\ & & \downarrow \lambda & & \downarrow \mu & & \downarrow \nu & & \\ 0 & \longrightarrow & L_0 & \xrightarrow{\alpha_0} & M_0 & \xrightarrow{\beta_0} & N_0 & \longrightarrow & 0 \end{array}$$

Lemma 7.8 (The snake lemma). *With notation as above, there is an exact sequence*

$$0 \longrightarrow \ker \lambda \longrightarrow \ker \mu \longrightarrow \ker \nu \xrightarrow{\delta} \operatorname{coker} \lambda \longrightarrow \operatorname{coker} \mu \longrightarrow \operatorname{coker} \nu \longrightarrow 0.$$

Remark 7.9. Most of the homomorphisms in this sequence are induced in a completely straightforward way from the corresponding homomorphisms λ, μ, ν . The one ‘surprising’ homomorphism is the one denoted δ ; I will discuss its definition below. $\lrcorner$

Remark 7.10. In view of Example 7.7, we could have written the sequence in this statement as

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_1(L_\bullet) & \longrightarrow & H_1(M_\bullet) & \longrightarrow & H_1(N_\bullet) \\ & & & & \searrow \delta & & \uparrow \\ & & H_0(L_\bullet) & \longrightarrow & H_0(M_\bullet) & \longrightarrow & H_0(N_\bullet) \longrightarrow 0 \end{array}$$

where $L_\bullet$ is the complex $0 \longrightarrow L_1 \xrightarrow{\lambda} L_0 \longrightarrow 0$, etc. The snake lemma generalizes to arbitrary complexes $L_\bullet, M_\bullet, N_\bullet$, producing a ‘long exact homology sequence’ of which this is just the tail end. As mentioned above, we will discuss this rather straightforward generalization later (§IX.3.3). $\lrcorner$

Remark 7.11. A popular version of the snake lemma does not assume that α_1 is injective and β_0 is surjective: that is, we could consider a commutative diagram of

³⁴In fact, it is better to view this diagram as *three* (very short) complexes linked by R -module homomorphisms α_i, β_i so that ‘the rows are exact’. In fact, one can define a category of *complexes*, and this diagram is nothing but a ‘short exact sequence of complexes’; this is the approach we will take in Chapter IX.

exact sequences

$$\begin{array}{ccccccc}
 & L_1 & \longrightarrow & M_1 & \longrightarrow & N_1 & \longrightarrow 0 \\
 & \downarrow \lambda & & \downarrow \mu & & \downarrow \nu & \\
 0 & \longrightarrow & L_0 & \longrightarrow & M_0 & \longrightarrow & N_0
 \end{array}$$

The lemma will then state that there is ‘only’ an exact sequence

$$\ker \lambda \longrightarrow \ker \mu \longrightarrow \ker \nu \xrightarrow{\delta} \operatorname{coker} \lambda \longrightarrow \operatorname{coker} \mu \longrightarrow \operatorname{coker} \nu . \quad \sqcup$$

PROVING the snake lemma is something that should not be done in public, and it is notoriously useless to write down the details of the verification for others to read: the details are all essentially obvious, but they lead quickly to a notational quagmire. Such proofs are collectively known as the sport of *diagram chase*, best executed by pointing several fingers at different parts of a diagram on a blackboard, while enunciating the elements one is manipulating and stating their fate³⁵.

Nevertheless, I should explain where the ‘connecting’ homomorphism δ comes from, since this is the heart of the statement of the snake lemma and of its proof. Here is the whole diagram, including kernels and cokernels; thus, columns are exact (as well as the two original sequences, placed horizontally):

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & \ker \lambda & \longrightarrow & \ker \mu & \longrightarrow & \ker \nu & \xrightarrow{\delta} & 0 \\
 & \downarrow & & \downarrow & & \downarrow & & \uparrow & \\
 0 & \longrightarrow & L_1 & \xrightarrow{\alpha_1} & M_1 & \xrightarrow{\beta_1} & N_1 & \longrightarrow & 0 \\
 & \downarrow \lambda & & \downarrow \mu & & \downarrow \nu & & \nearrow & \\
 0 & \longrightarrow & L_0 & \xrightarrow{\alpha_0} & M_0 & \xrightarrow{\beta_0} & N_0 & \longrightarrow & 0 \\
 & \downarrow & & \downarrow & & \downarrow & & \nwarrow & \\
 & \operatorname{coker} \lambda & \longrightarrow & \operatorname{coker} \mu & \longrightarrow & \operatorname{coker} \nu & \longrightarrow & 0 \\
 & \downarrow & & \downarrow & & \downarrow & & \\
 & 0 & & 0 & & 0 & &
 \end{array}$$

By the way, I trust that the reader now sees why this lemma is called the *snake* lemma.

Definition of the snaking homomorphism δ . Let $a \in \ker \nu$. I claim that a can be mapped through the diagram all the way to $\operatorname{coker} \lambda$, along the solid arrows marked

³⁵Real purists chase diagrams in arbitrary categories, thus without the benefit of talking about ‘elements’, and we will practice this skill later on (Chapter IX). For example, the snake lemma can be proven by appealing to universal property after universal property of kernels and cokernels, without ever choosing elements anywhere. But the performing technique of pointing fingers at a board while monologuing through the argument remains essentially the same.

here:

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \vdots & & \vdots & & \vdots \\
 0 & \cdots & & & & & a \\
 & & \vdots & & \vdots & & \downarrow \\
 0 & \cdots & & c & \xrightarrow{\beta_1} & b & \cdots 0 \\
 & & \vdots & & \vdots & & \downarrow \nu \\
 0 & \cdots & e & \xrightarrow{\alpha_0} & d & \cdots * & \cdots 0 \\
 & & \vdots & & \vdots & & \downarrow \beta_0 \\
 & & f & & & & \cdots 0 \\
 & & \vdots & & \vdots & & \vdots \\
 & & 0 & & 0 & & 0
 \end{array}$$

Indeed,

- $\ker \nu \subseteq N_1$; so view a as an element b of N_1 .
- β_1 is surjective, so $\exists c \in M_1$, mapping to b .
- Let $d = \mu(c)$ be the image of c in M_0 .
- What is the image of d in the spot marked $*$? By the commutativity of the diagram, it must be the same as $\nu(b)$. However, b was the image in N_1 of $a \in \ker \nu$, so $\nu(b) = 0$. Thus, $d \in \ker \beta_0$. Since rows are exact, $\ker \beta_0 = \text{im } \alpha_0$; therefore, $\exists e \in L_0$, mapping to d .
- Finally, let $f \in \text{coker } \lambda$ be the image of e .

I want to set $\delta(a) := f$.

Is this legal? At two steps in the chase we have taken preimages:

- $\exists c \in M_1$ such that $\beta_1(c) = b$,
- $\exists e \in L_0$ such that $\alpha_0(e) = d$.

The second step does not involve a choice: because α_0 is injective by assumption, so the element e mapping to d is uniquely determined by d . But there *was* a choice involved in the first step: in order to verify that δ is well-defined, we have to show that choosing *some other* c would not affect the proposed value f for $\delta(a)$.

This is proved by another chase. Here is the relevant part of the diagram:

$$\begin{array}{ccccccc}
 & & & \alpha_1 & & \beta_1 & \\
 0 & \cdots & & c & \xrightarrow{\beta_1} & b & \cdots 0 \\
 & & \lambda \downarrow & \downarrow \mu & & & \\
 0 & \cdots & e & \xrightarrow{\alpha_0} & d & & \\
 & & \downarrow & & & & \\
 & & f & & & &
 \end{array}$$

Suppose we choose a different c' mapping to the same b :

$$0 \cdots \cdots \alpha_1 \cdots c' \xrightarrow{\beta_1} b \cdots \cdots 0.$$

Then $\beta_1(c' - c) = 0$; by exactness, $\exists g \in L_1$ such that $(c' - c) = \alpha_1(g)$:

$$0 \cdots \cdots g \xrightarrow{\alpha_1} (c' - c) \xrightarrow{\beta_1} 0 \cdots \cdots 0.$$

Now the point is that, since columns form complexes, g dies in $\text{coker } \lambda$:

$$\begin{array}{ccccccc} 0 & \cdots & g & \xrightarrow{\alpha_1} & (c' - c) & \xrightarrow{\beta_1} & 0 \cdots \cdots 0 \\ & & \downarrow \lambda & & \downarrow \mu & & \downarrow \\ 0 & \cdots & \lambda(g) & \xrightarrow{\alpha_0} & & & \\ & & \downarrow & & & & \\ & & 0 & & & & \end{array}$$

and it follows (by the commutativity of the diagram and the injectivity of α_0) that changing c to c' modifies e to $e + \lambda(g)$ and f to $f + 0 = f$. That is, f is indeed independent of the choice.

Thus δ is well-defined!

This is a tiny part of the proof of the snake lemma, but it probably suffices to demonstrate why reading a written-out version of a diagram chase may be supremely uninformative.

The rest of the proof (left to the reader(!) but I am not listing this as an official exercise for fear that someone might actually turn a solution in for grading) amounts to many, many similar arguments. The definition of the maps induced on kernels and cokernels is substantially less challenging than the definition of the connecting morphism δ described above. Exactness at most spots in the sequence

$$0 \longrightarrow \ker \lambda \longrightarrow \ker \mu \longrightarrow \ker \nu \xrightarrow{\delta} \text{coker } \lambda \longrightarrow \text{coker } \mu \longrightarrow \text{coker } \nu \longrightarrow 0$$

is also reasonably straightforward; most of the work will go into proving exactness at $\ker \nu$ and $\text{coker } \lambda$.

Dear reader: don't shy away from trying this, for it is excellent, indispensable practice. Miss this opportunity and you will forever feel unsure about such manipulations.

The snake lemma streamlines several facts, which would not be hard to prove individually, but become really straightforward once the lemma is settled. For example,

Corollary 7.12. *In the same situation presented in the snake lemma (notation as in §7.3), assume that μ is surjective and ν is injective. Then λ is surjective and ν is an isomorphism.*

Proof. Indeed, μ surjective $\implies \text{coker } \mu = 0$; ν injective $\implies \ker \nu = 0$ (Proposition 6.2). Feeding this information into the sequence of the snake lemma gives an exact sequence

$$0 \longrightarrow \ker \lambda \longrightarrow \ker \mu \longrightarrow 0 \longrightarrow \text{coker } \lambda \longrightarrow 0 \longrightarrow \text{coker } \nu \longrightarrow 0$$

Exactness implies $\text{coker } \lambda = \text{coker } \nu = 0$ (Exercise 7.1); hence λ and ν are surjective, with the stated consequences. $\square$

Several more such statements may be proved just as easily; the reader should experiment to his or her heart's content.

Exercises

7.1. $\triangleright$ Assume that the complex

$$\cdots \longrightarrow 0 \longrightarrow M \longrightarrow 0 \longrightarrow \cdots$$

is exact. Prove that $M \cong 0$. [§7.3]

7.2. Assume that the complex

$$\cdots \longrightarrow 0 \longrightarrow M \longrightarrow M' \longrightarrow 0 \longrightarrow \cdots$$

is exact. Prove that $M \cong M'$.

7.3. Assume that the complex

$$\cdots \longrightarrow 0 \longrightarrow L \longrightarrow M \xrightarrow{\varphi} M' \longrightarrow N \longrightarrow 0 \longrightarrow \cdots$$

is exact. Show that, up to natural identifications, $L = \ker \varphi$ and $N = \text{coker } \varphi$.

7.4. Construct short exact sequences of $\mathbb{Z}$ -modules

$$0 \longrightarrow \mathbb{Z}^{\oplus \mathbb{N}} \longrightarrow \mathbb{Z}^{\oplus \mathbb{N}} \longrightarrow \mathbb{Z} \longrightarrow 0$$

and

$$0 \longrightarrow \mathbb{Z}^{\oplus \mathbb{N}} \longrightarrow \mathbb{Z}^{\oplus \mathbb{N}} \longrightarrow \mathbb{Z}^{\oplus \mathbb{N}} \longrightarrow 0.$$

(Hint: David Hilbert's Grand Hotel.)

7.5. $\triangleright$ Assume that the complex

$$\cdots \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow \cdots$$

is exact and that L and N are Noetherian. Prove that M is Noetherian. [§7.1]

7.6. $\triangleright$ Prove the 'split epimorphism' part of Proposition 7.5. [§7.2]

7.7. $\triangleright$ Let

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

be a short exact sequence of R -modules, and let L be an R -module.

(i) Prove that there is an exact sequence³⁶

$$0 \longrightarrow \text{Hom}_{R\text{-Mod}}(P, L) \longrightarrow \text{Hom}_{R\text{-Mod}}(N, L) \longrightarrow \text{Hom}_{R\text{-Mod}}(M, L).$$

(ii) Redo Exercise 6.17. (Use the exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$.)

³⁶In general, this will be a sequence of abelian groups; if R is commutative, so that each $\text{Hom}_{R\text{-Mod}}$ is an R -module (§5.2), then it will be an exact sequence of R -modules.

- (iii) Construct an example showing that the rightmost homomorphism in (i) need not be onto.
- (iv) Show that if the original sequence splits, then the rightmost homomorphism in (i) *is* onto.

[7.9, VIII.3.14, §VIII.5.1]

7.8. ▷ Prove that every exact sequence

$$0 \longrightarrow M \longrightarrow N \longrightarrow F \longrightarrow 0$$

of R modules, with F free, splits. (Hint: Exercise 6.9.) [§VIII.5.4]

7.9. Let

$$0 \longrightarrow M \longrightarrow N \longrightarrow F \longrightarrow 0$$

be a short exact sequence of R -modules, with F free, and let L be an R -module. Prove that there is an exact sequence

$$0 \longrightarrow \text{Hom}_{R\text{-Mod}}(F, L) \longrightarrow \text{Hom}_{R\text{-Mod}}(N, L) \longrightarrow \text{Hom}_{R\text{-Mod}}(M, L) \longrightarrow 0.$$

(Cf. Exercise 7.7.)

7.10. ▷ In the situation of the snake lemma, assume that λ and ν are isomorphisms. Use the snake lemma and prove that μ is an isomorphism. This is called the ‘short five-lemma,’ as it follows immediately from the five-lemma (cf. Exercise 7.14), as well as from the snake lemma. [VIII.6.21, IX.2.4]

7.11. ▷ Let

$$(*) \quad 0 \longrightarrow M_1 \longrightarrow N \longrightarrow M_2 \longrightarrow 0$$

be an exact sequence of R -modules. (This may be called an ‘extension’ of M_2 by M_1 .) Suppose there is *any* R -module homomorphism $N \rightarrow M_1 \oplus M_2$ making the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_1 & \longrightarrow & N & \longrightarrow & M_2 \longrightarrow 0 \\ & & \parallel & & \downarrow & & \parallel \\ 0 & \longrightarrow & M_1 & \longrightarrow & M_1 \oplus M_2 & \longrightarrow & M_2 \longrightarrow 0 \end{array}$$

commute, where the bottom sequence is the standard sequence of a direct sum. Prove that $(*)$ splits. [§7.2]

7.12. ¬ Practice your diagram chasing skills by proving the ‘four-lemma’: if

$$\begin{array}{ccccccc} A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 & \longrightarrow & D_1 \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A_0 & \longrightarrow & B_0 & \longrightarrow & C_0 & \longrightarrow & D_0 \end{array}$$

is a commutative diagram of R -modules with exact rows, α is an epimorphism, and β, δ are monomorphisms, then γ is a monomorphism. [7.13, IX.2.3]

7.13. Prove another³⁷ version of the ‘four-lemma’ of Exercise 7.12: if

$$\begin{array}{ccccccc} B_1 & \longrightarrow & C_1 & \longrightarrow & D_1 & \longrightarrow & E_1 \\ \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \epsilon \\ B_0 & \longrightarrow & C_0 & \longrightarrow & D_0 & \longrightarrow & E_0 \end{array}$$

is a commutative diagram of R -modules with exact rows, β and δ are epimorphisms, and ϵ is a monomorphism, then γ is an epimorphism.

7.14. $\neg$ Prove the ‘five-lemma’: if

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 & \longrightarrow & D_1 & \longrightarrow & E_1 \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \epsilon \\ A_0 & \longrightarrow & B_0 & \longrightarrow & C_0 & \longrightarrow & D_0 & \longrightarrow & E_0 \end{array}$$

is a commutative diagram of R -modules with exact rows, β and δ are isomorphisms, α is an epimorphism, and ϵ is a monomorphism, then γ is an isomorphism. (You can avoid the needed diagram chase by pasting together results from previous exercises.) [7.10]

7.15. $\neg$ Consider the following commutative diagram of R -modules:

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & L_2 & \longrightarrow & M_2 & \longrightarrow & N_2 \longrightarrow 0 \\ & & \downarrow & & \downarrow \alpha & & \downarrow \\ 0 & \longrightarrow & L_1 & \longrightarrow & M_1 & \longrightarrow & N_1 \longrightarrow 0 \\ & & \downarrow & & \downarrow \beta & & \downarrow \\ 0 & \longrightarrow & L_0 & \longrightarrow & M_0 & \longrightarrow & N_0 \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

Assume that the three rows are exact and the two rightmost columns are exact. Prove that the left column is exact. Second version: assume that the three rows are exact and the two leftmost columns are exact; prove that the right column is exact. This is the ‘nine-lemma’. (You can avoid a diagram chase by applying the snake lemma; for this, you will have to turn the diagram by 90° .) [7.16]

7.16. In the same situation as in Exercise 7.15, assume that the three rows are exact and that the leftmost and rightmost columns are exact.

- Prove that α is a monomorphism and β is an epimorphism.
- Is the central column necessarily exact?

³⁷It is in fact unnecessary to prove *both* versions, but to realize this one has to view the matter from the more general context of abelian categories; cf. Exercise IX.2.3.

(Hint: No. Place $\mathbb{Z} \oplus \mathbb{Z}$ in the middle, and surround it artfully with six copies of $\mathbb{Z}$ and two 0's.)

- Assume further that the central column is a complex (that is, $\beta \circ \alpha = 0$); prove that it is then necessarily exact.

7.17. $\neg$ Generalize the previous two exercises as follows. Consider a (possibly infinite) commutative diagram of R -modules:

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & L_{i+1} & \longrightarrow & M_{i+1} & \longrightarrow & N_{i+1} \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & L_i & \longrightarrow & M_i & \longrightarrow & N_i \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & L_{i-1} & \longrightarrow & M_{i-1} & \longrightarrow & N_{i-1} \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

in which the central column is a complex and every row is exact. Prove that the left and right columns are also complexes. Prove that if any two of the columns are exact, so is the third. (The first part is straightforward. The second part will take you a couple of minutes now due to the needed diagram chases, and a couple of seconds later, once you learn about the long exact (co)homology sequence in §IX.3.3.) [IX.3.12]

Groups, second encounter

In this chapter we return to $\mathbf{Grp}$ and study several topics of a less ‘general’ nature than those considered in Chapter II. Most of what we do here will apply exclusively to *finite* groups; this is an important example in its own right, as it has spectacular applications (for example, in Galois theory; cf. §VII.7), and it is a good subject from the expository point of view, since it gives us the opportunity to see several general concepts at work in a context that is complex enough to carry substance, but simple enough (in this tiny selection of elementary topics) to be appreciated easily.

1. The conjugation action

1.1. Actions of groups on sets, reminder. Groups really shine when you let them act on something. This section will make this point very effectively, since we will get surprisingly precise results on finite groups by extremely simple-minded applications of the elementary facts concerning group actions that we established back in §II.9.

Recall that we proved (Proposition II.9.9) that every *transitive* (left-) action of a group G on a set S is, up to a natural notion of isomorphism, ‘left-multiplication on the set of left-cosets G/H ’. Here, H may be taken to be the stabilizer $\text{Stab}_G(a)$ of any element $a \in S$, that is (Definition II.9.8) the subgroup of G fixing a . This fact applies to the orbits of *every* left-action of G on a set; in particular, the number of elements in a finite orbit O equals the index of the stabilizer of any $a \in O$; in particular (Corollary II.9.10) the number of elements $|O|$ of an orbit must divide the order $|G|$ of G , if G is finite.

These considerations may be packaged into a useful ‘counting’ formula, which we could call the *class formula* for that action; this name is usually reserved to the particular case of the action of G onto itself by conjugation, which we will explore more carefully below.

In order to state the formula, assume G acts on a set S ; for $a \in S$, let G_a denote the stabilizer $\text{Stab}_G(a)$. Also, let Z be the set of *fixed points* of the action:

$$Z = \{a \in S \mid (\forall g \in G) : ga = a\}.$$

Note that $a \in Z \iff G_a = G$; we could say that $a \in Z$ if and only if the orbit of a is ‘trivial’, in the sense that it consists of a alone.

Proposition 1.1. *Let S be a finite set, and let G be a group acting on S . With notation as above,*

$$|S| = |Z| + \sum_{a \in A} [G : G_a],$$

where $A \subseteq S$ has exactly one element for each nontrivial orbit of the action.

Proof. The orbits form a partition of S , and Z collects the trivial orbits; hence

$$|S| = |Z| + \sum_{a \in A} |O_a|,$$

where O_a denotes the orbit of a . By Proposition II.9.9, the order $|O_a|$ equals the index of the stabilizer of a , yielding the statement. $\square$

The main strength of Proposition 1.1 rests in the fact that, if G is finite, each summand $[G : G_a]$ divides the order of G (and is > 1). This can be a strong constraint, when some information is known about $|G|$. For example, let’s see what this says when G is a p -group:

Definition 1.2. A p -group is a finite group whose order is a power of a prime integer p . $\lrcorner$

Corollary 1.3. *Let G be a p -group acting on a finite set S , and let Z be the fixed point set of the action. Then*

$$|Z| \equiv |S| \pmod{p}.$$

Proof. Indeed, each summand $[G : G_a]$ in Proposition 1.1 is a number larger than 1, and a power of p ; hence it is $0 \pmod{p}$. $\square$

For instance, in certain situations this can be used to establish¹ that $Z \neq \emptyset$: see Exercise 1.1. Such immediate consequences of Proposition 1.1 will assist us below, in the proof of Sylow’s theorems.

¹In this sense, Proposition 1.1 is an instance of a class of results known as ‘fixed point theorems’. The reader will likely encounter a few such theorems in topology courses, where the role of the ‘size’ of a set may be played by (for example) the Euler characteristic of a topological space.

1.2. Center, centralizer, conjugacy classes. Recall (Example II.9.3) that every group G acts on itself in at least two interesting ways: by (left-) multiplication and by *conjugation*. The latter action is defined by the following $\rho : G \times G \rightarrow G$:

$$\rho(g, a) = gag^{-1}.$$

As we know (§II.9.2), this datum is equivalent to the datum of a certain group homomorphism:

$$\sigma : G \rightarrow S_G$$

from G to the permutation group on G .

This action highlights several interesting objects:

Definition 1.4. The *center* of G , denoted $Z(G)$, is the subgroup $\ker \sigma$ of G . $\square$

Concretely, the center² of G is

$$Z(G) = \{g \in G \mid (\forall a \in G) : ga = ag\}.$$

Indeed, $\sigma(g)$ is the identity in S_G if and only if $\sigma(g)$ acts as the identity on G ; that is, if and only if $gag^{-1} = a$ for all $a \in G$; that is, if and only if g commutes with all elements of G . In other words, the center is the set of fixed points in G under the conjugation action.

Note that the center of a group G is automatically normal in G : this is nearly immediate to check ‘by hand’, but there is no need to do so since it is a kernel by definition and kernels are normal.

A group G is commutative if and only if $Z(G) = G$, that is, if and only if the conjugation action is trivial on G .

In general, feel happy when you discover that the center of a group is not trivial: this will often allow you to set up proofs by induction on the number of elements of the group, by mod-ing out by the center (this is, roughly, how we will prove the first Sylow theorem). Or note the following useful fact, which comes in handy when trying to prove that a group is commutative:

Lemma 1.5. *Let G be a finite group, and assume $G/Z(G)$ is cyclic. Then G is commutative (and hence $G/Z(G)$ is in fact trivial).*

Proof. (Cf. Exercise 1.5.) As $G/Z(G)$ is cyclic, there exists an element $g \in G$ such that the class $gZ(G)$ generates $G/Z(G)$. Then $\forall a \in G$

$$aZ(G) = (gZ(G))^r$$

for some $r \in \mathbb{Z}$; that is, there is an element $z \in Z(G)$ of the center such that $a = g^r z$.

If now a, b are in G , use this fact to write

$$a = g^r z, \quad b = g^s w$$

for some $s \in \mathbb{Z}$ and $w \in Z(G)$; but then

$$ab = (g^r z)(g^s w) = g^{r+s} zw = (g^s w)(g^r z) = ba,$$

²Why ‘ Z ’? ‘Center’ iff Zentrum auf Deutsch.

where I have used the fact that z and w commute with every element of G . As a and b were arbitrary, this proves that G is commutative. $\square$

Next, the stabilizer of $a \in G$ under conjugation has a special name:

Definition 1.6. The *centralizer* (or *normalizer*) $Z_G(a)$ of $a \in G$ is its stabilizer under conjugation. $\lrcorner$

Thus,

$$Z_G(a) = \{g \in G \mid gag^{-1} = a\} = \{g \in G \mid ga = ag\}$$

consists of those elements in G which commute with a . In particular, $Z(G) \subseteq Z_G(a)$ for all $a \in G$; in fact, $Z(G) = \bigcap_{a \in G} Z_G(a)$. Clearly $a \in Z(G) \iff Z_G(a) = G$.

If there is no ambiguity concerning the group G containing a , the index G may be dropped.

Definition 1.7. The *conjugacy class* of $a \in G$ is the orbit $[a]$ of a under the conjugation action. Two elements a, b of G are *conjugate* if they belong to the same conjugacy class. $\lrcorner$

The notation $[a]$ is not standard; $C(a)$ is used more frequently, but I am not fond of it. Using $[a]$ reminds us that these are nothing but the equivalence classes of elements of G under a certain interesting equivalence relation.

Note that $[a] = \{a\}$ if and only if $gag^{-1} = a$ for all $g \in G$; that is, if and only if $ga = ag$ for all $g \in G$; that is, if and only if $a \in Z(G)$.

1.3. The Class Formula. The ‘official’ Class Formula for a finite group G is the particular case of Proposition 1.1 for the conjugation action.

Proposition 1.8 (Class formula). *Let G be a finite group. Then*

$$|G| = |Z(G)| + \sum_{a \in A} [G : Z(a)],$$

where $A \subseteq G$ is a set containing one representative for each nontrivial conjugacy class in G .

Proof. The set of fixed points is $Z(G)$, and the stabilizer of a is the centralizer $Z(a)$; apply Proposition 1.1. $\square$

The class formula is surprisingly useful. In applying it, keep in mind that every summand on the right (that is, both $|Z(G)|$ and each $[G : Z(a)]$) is a divisor of $|G|$; this fact alone often suffices to draw striking conclusions about G .

Possibly the most famous such application is to p -groups, via Corollary 1.3:

Corollary 1.9. *Let G be a nontrivial p -group. Then G has a nontrivial center.*

Proof. Since $|Z(G)| \equiv |G| \pmod{p}$ and $|G| > 1$ is a power of p , necessarily $|Z(G)|$ is a multiple of p . As $Z(G) \neq \emptyset$ (since $e_G \in Z(G)$), this implies $|Z(G)| \geq p$. $\square$

For example, it follows immediately (from Corollary 1.9 and Lemma 1.5; cf. Exercise 1.6) that if p is prime, then every group of order p^2 is commutative.

In general, the class formula poses a strong constraint on what can go on in a group.

Example 1.10. Consider a group G of order 6; what are the possibilities for its class formula?

If G is commutative, then the class formula will tell us very little:

$$6 = 6.$$

If G is not commutative, then its center must be trivial (as a consequence of Lagrange's theorem and Lemma 1.5); so the class formula is $6 = 1 + \cdots$, where $\cdots$ collects the sizes of the nontrivial conjugacy classes. But each of these summands must be larger than 1, smaller than 6, and must divide 6; that is, there are no choices:

$$6 = 1 + 2 + 3$$

is the only possibility. The reader should check that this is indeed the class formula for S_3 ; in fact, S_3 is the only noncommutative group of order 6 up to isomorphism (Exercise 1.13). $\lrcorner$

Another useful observation is that *normal* subgroups must be unions of conjugacy classes: because if H is a normal subgroup, $a \in H$, and $b = gag^{-1}$ is conjugate to a , then

$$b \in gHg^{-1} = H.$$

To stick with the $|G| = 6$ example, note that every subgroup of a group must contain the identity and its size must divide the order of the group; it follows that a normal subgroup of a noncommutative group of order 6 cannot have order 2, since 2 cannot be written as sums of orders of conjugacy classes (including the class of the identity).

1.4. Conjugation of subsets and subgroups. We may also act by conjugation on the power set of G : if $A \subseteq G$ is a subset and $g \in G$, the *conjugate* of A is the subset gAg^{-1} . By cancellation, the conjugation map $a \mapsto gag^{-1}$ is a *bijection* between A and gAg^{-1} .

This leads to terminology analogous to the one introduced in §1.2.

Definition 1.11. The *normalizer* $N_G(A)$ of A is its stabilizer under conjugation. The *centralizer* of A is the subgroup $Z_G(A) \subseteq N_G(A)$ fixing each element of A . $\lrcorner$

Thus, $g \in N_G(A)$ if and only if³ $gAg^{-1} = A$, and $g \in Z_G(A)$ if and only if $\forall a \in A, gag^{-1} = a$.

For $A = \{a\}$ a singleton, we have $N_G(\{a\}) = Z_G(\{a\}) = Z_G(a)$. In general, $Z_G(A) \subsetneq N_G(A)$.

If H is a subgroup of G , every conjugate gHg^{-1} of H is also a subgroup of G ; conjugate subgroups have the same order.

³If A is finite (but not in general), this condition is equivalent to $gAg^{-1} \subseteq A$.

Remark 1.12. The definition implies immediately that $H \subseteq N_G(H)$ and that H is normal in G if and only if $N_G(H) = G$. More generally, the normalizer $N_G(H)$ of H in G is (clearly) the *largest subgroup of G in which H is normal*. $\lrcorner$

One could apply Proposition 1.1 to the conjugation action on subsets or subgroup; however, there are too many subsets, and one has little control over the number of subgroups. Other numerical considerations involving the number of conjugates of a given subset or subgroups may be very useful.

Lemma 1.13. *Let $H \subseteq G$ be a subgroup. Then (if finite) the number of subgroups conjugate to H equals the index $[G : N_G(H)]$ of the normalizer of H in G .*

Proof. This is again an immediate consequence of Proposition II.9.9. $\square$

Corollary 1.14. *If $[G : H]$ is finite, then the number of subgroups conjugate to H is finite and divides $[G : H]$.*

Proof.

$$[G : H] = [G : N_G(H)] \cdot [N_G(H) : H]$$

(cf. §II.8.5). $\square$

One of the celebrated Sylow theorems will strengthen this statement substantially in the case in which H is a *maximal* p -group contained in a finite group G . For a statement concerning the size of the normalizer of an arbitrary p -subgroup of a group, see Lemma 2.9.

Another useful numerical tool is the observation that if H and K are subgroups of a group G and $H \subseteq N_G(K)$ —so that $gKg^{-1} = K$ for all $g \in H$ —then conjugation by $g \in H$ gives an *automorphism* of K . Indeed, I have already observed that conjugation is a bijection, and it is immediate to see that it is a homomorphism: $\forall k_1, k_2 \in K$

$$(gk_1g^{-1})(gk_2g^{-1}) = gk_1(g^{-1}g)k_2g^{-1} = g(k_1k_2)g^{-1}.$$

Thus, conjugation gives a set-function

$$\gamma : H \rightarrow \text{Aut}_{\text{Grp}}(K).$$

The reader will check that this is a group homomorphism and will determine $\ker \gamma$ (Exercise 1.21).

This is especially useful if H is finite and some information is available concerning $\text{Aut}_{\text{Grp}}(K)$ (for an example, see Exercise 4.14). A classic application is presented in Exercise 1.22.

Exercises

- 1.1.** $\triangleright$ Let p be a prime integer, let G be a p -group, and let S be a set such that $|S| \not\equiv 0 \pmod{p}$. If G acts on S , prove that the action must have fixed points. [§1.1, §2.3]
- 1.2.** Find the center of D_{2n} . (The answer depends on the parity of n . You have actually done this already: Exercise II.2.7. This time, use a presentation.)
- 1.3.** Prove that the center of S_n is trivial for $n \geq 3$. (Suppose that $\sigma \in S_n$ sends a to $b \neq a$, and let $c \neq a, b$. Let τ be the permutation that acts solely by swapping b and c . Then compare the action of $\sigma\tau$ and $\tau\sigma$ on a .)
- 1.4.** $\triangleright$ Let G be a group, and let N be a subgroup of $Z(G)$. Prove that N is normal in G . [§2.2]
- 1.5.** $\triangleright$ Let G be a group. Prove that $G/Z(G)$ is isomorphic to the group $\text{Inn}(G)$ of inner automorphisms of G . (Cf. Exercise II.4.8.) Then prove Lemma 1.5 again by using the result of Exercise II.6.7. [§1.2]
- 1.6.** $\triangleright$ Let p, q be prime integers, and let G be a group of order pq . Prove that either G is commutative or the center of G is trivial. Conclude (using Corollary 1.9) that every group of order p^2 , for a prime p , is commutative. [§1.3]
- 1.7.** Prove or disprove that if p is prime, then every group of order p^3 is commutative.
- 1.8.** $\triangleright$ Let p be a prime number, and let G be a p -group: $|G| = p^r$. Prove that G contains a normal subgroup of order p^k for every nonnegative $k \leq r$. [§2.2]
- 1.9.** $\neg$ Let p be a prime number, G a p -group, and H a nontrivial normal subgroup of G . Prove that $H \cap Z(G) \neq \{e\}$. (Hint: Use the class formula.) [3.11]
- 1.10.** Prove that if G is a group of odd order and $g \in G$ is conjugate to g^{-1} , then $g = e_G$.
- 1.11.** Let G be a finite group, and suppose there exist representatives $g_1, \dots, g_r$ of the r distinct conjugacy classes in G , such that $\forall i, j, g_i g_j = g_j g_i$. Prove that G is commutative. (Hint: What can you say about the sizes of the conjugacy classes?)
- 1.12.** Verify that the class formula for both D_8 and Q_8 (cf. Exercise III.1.12) is $8 = 2 + 2 + 2 + 2$. (Also note that $D_8 \not\cong Q_8$.)
- 1.13.** $\triangleright$ Let G be a noncommutative group of order 6. As observed in Example 1.10, G must have trivial center and exactly two conjugacy classes, of order 2 and 3.
- Prove that if every element of a group has order ≤ 2 , then the group is commutative. Conclude that G has an element y of order 3.
 - Prove that $\langle y \rangle$ is normal in G .
 - Prove that $[y]$ is the conjugacy class of order 2 and $[y] = \{y, y^2\}$.
 - Prove that there is an $x \in G$ such that $yx = xy^2$.

- Prove that x has order 2.
- Prove that x and y generate G .
- Prove that $G \cong S_3$.

[§1.3, §2.5]

1.14. Let G be a group, and assume $[G : Z(G)] = n$ is finite. Let $A \subseteq G$ be any subset. Prove that the number of conjugates of A is at most n .

1.15. Suppose that the class formula for a group G is $60 = 1 + 15 + 20 + 12 + 12$. Prove that the only *normal* subgroups of G are $\{e\}$ and G .

1.16. $\triangleright$ Let G be a finite group, and let $H \subseteq G$ be a subgroup of index 2. For $a \in H$, denote by $[a]_H$, resp., $[a]_G$, the conjugacy class of a in H , resp., G . Prove that either $[a]_H = [a]_G$ or $[a]_H$ is half the size of $[a]_G$, according to whether the centralizer $Z_G(a)$ is not or is contained in H . (Hint: Note that H is normal in G , by Exercise II.8.2; apply Proposition II.8.11.) [§4.4]

1.17. $\neg$ Let H be a proper subgroup of a finite group G . Prove that G is *not* the union of the conjugates of H . (Hint: You know the number of conjugates of H ; keep in mind that any two subgroups overlap, at least at the identity.) [1.18, 1.20]

1.18. Let S be a set endowed with a transitive action of a finite group G , and assume $|S| \geq 2$. Prove that there exists a $g \in G$ without fixed points in S , that is, such that $gs \neq s$ for all $s \in S$. (Hint: By Proposition II.9.9, you may assume $S = G/H$, with H proper in G . Use Exercise 1.17.)

1.19. Let H be a proper subgroup of a finite group G . Prove that there exists a $g \in G$ whose conjugacy class is disjoint from H .

1.20. Let $G = \text{GL}_2(\mathbb{C})$, and let H be the subgroup consisting of upper triangular matrices (Exercise II.6.2). Prove that G is the union of the conjugates of H . Thus, the finiteness hypothesis in Exercise 1.17 is necessary. (Hint: Equivalently, prove that every 2×2 matrix is conjugate to a matrix in H . You will use the fact that $\mathbb{C}$ is algebraically closed; see Example III.4.14.)

1.21. $\triangleright$ Let H, K be subgroups of a group G , with $H \subseteq N_G(K)$. Verify that the function $\gamma : H \rightarrow \text{Aut}_{\text{Grp}}(K)$ defined by conjugation is a homomorphism of groups and that $\ker \gamma = H \cap Z_G(K)$, where $Z_G(K)$ is the centralizer of K . [§1.4, 1.22]

1.22. $\triangleright$ Let G be a finite group, and let H be a cyclic subgroup of G of order p . Assume that p is the smallest prime dividing the order of G and that H is normal in G . Prove that H is contained in the center of G .

(Hint: By Exercise 1.21 there is a homomorphism $\gamma : G \rightarrow \text{Aut}_{\text{Grp}}(H)$; by Exercise II.4.14, $\text{Aut}_{\text{Grp}}(H)$ has order $p - 1$. What can you say about γ ?) [§1.4]

2. The Sylow theorems

2.1. Cauchy's theorem. The 'Sylow theorems' consist of three statements concerning p -subgroups (cf. Definition 1.2) of a given finite group G . The form I will

give for the first of these statements will tell us that G contains p -groups of all sizes allowed by Lagrange's theorem: if p is a prime and p^k divides $|G|$, then G contains a subgroup of order p^k . The proof of this statement is an easy induction, provided the statement for $k = 1$ is known: that is, provided that one has established

Theorem 2.1 (Cauchy's theorem). *Let G be a finite group, and let p be a prime divisor of $|G|$. Then G contains an element of order p .*

As it happens, only the *abelian* version of this statement is needed for the proof of the first Sylow theorem; then the full statement of Cauchy's theorem follows from the first Sylow theorem itself. Since the (diligent) reader has already proved Cauchy's theorem for abelian groups (in Exercise II.8.17), we could directly move on to Sylow theorems.

However, there is a quick proof⁴ of the full statement of Cauchy's theorem which does not rely on Sylow and is a good illustration of the power of the general 'class formula for arbitrary actions' (Proposition 1.1). I will present this proof, while also encouraging the reader to go back and (re)do Exercise II.8.17 now.

Proof of Theorem 2.1. Consider the set S of ordered p -tuples of elements of G :

$$(a_1, \dots, a_p)$$

such that $a_1 \cdots a_p = e$. I claim that $|S| = |G|^{p-1}$: indeed, once $a_1, \dots, a_{p-1}$ are chosen (arbitrarily), then a_p is determined as it is the inverse of $a_1 \cdots a_{p-1}$.

Therefore, p divides the order of S as it divides the order of G .

Also note that if $a_1 \cdots a_p = e$, then

$$a_2 \cdots a_p a_1 = e$$

(even if G is not commutative): because if a_1 is a left-inverse to $a_2 \cdots a_p$, then it is also a right-inverse to it.

Therefore, we may act with the group $\mathbb{Z}/p\mathbb{Z}$ on S : given $[m]$ in $\mathbb{Z}/p\mathbb{Z}$, with $0 \leq m < p$, act by $[m]$ on

$$(a_1, \dots, a_p)$$

by sending it to

$$(a_{m+1}, \dots, a_p, a_1, \dots, a_m) :$$

as we just observed, this is still an element of S .

Now Corollary 1.3 implies

$$|Z| \equiv |S| \equiv 0 \pmod{p},$$

where Z is the set of fixed points of this action. Fixed points are p -tuples of the form

$$(*) \quad (a, \dots, a);$$

and note that $Z \neq \emptyset$, since $\{e, \dots, e\} \in Z$. Since $p \geq 2$ and p divides $|Z|$, we conclude that $|Z| > 1$; therefore there exists some element in Z of the form $(*)$, with $a \neq e$.

⁴This argument is apparently due to James McKay.

This says that there exists an $a \in G$, $a \neq e$, such that $a^p = e$, proving the statement. $\square$

We should remark that the proof given here proves a more precise result than the raw statement of Theorem 2.1: every element of order p in G generates a cyclic subgroup of G of order p , and we are able to say something about the number of such subgroups.

Claim 2.2. *Let G be a finite group, let p be a prime divisor of $|G|$, and let N be the number of cyclic subgroups of G of order p . Then $N \equiv 1 \pmod{p}$.*

The proof of this fact is left to the reader (as an incentive to really understand the proof of Theorem 2.1).

Claim 2.2, coupled with the simple observation that if there is *only* 1 cyclic subgroup H of order p , then that subgroup must be normal (Exercise 2.2), suffices for interesting applications.

Definition 2.3. A group G is *simple* if it is nontrivial and its only normal subgroups are $\{e\}$ and G itself. $\lrcorner$

Simple groups occupy a special place in the theory of groups: one can ‘break up’ any finite group into basic constituents which are simple groups; we will see how this is done in §3.1. Thus, it is important to be able to tell whether a group is simple or not⁵.

Example 2.4. Let p be a positive prime integer. If $|G| = mp$, with $1 < m < p$, then G is not simple.

Indeed, consider the subgroups of G with p elements. By Claim 2.2, the number of such subgroups is $\equiv 1 \pmod{p}$. Thus, if there is more than one such subgroup, then there must be at least $p + 1$. Any two distinct subgroups of prime order can only meet at the identity (why?); therefore this would account for at least

$$1 + (p + 1)(p - 1) = p^2$$

elements in G . Since $|G| = mp < p^2$, this is impossible. Therefore there is only one cyclic subgroup of order p in G , which must be normal as mentioned above, proving that G is not simple. $\lrcorner$

2.2. Sylow I. Let p be a prime integer. A *p -Sylow subgroup* of a finite group G is a subgroup of order p^r , where $|G| = p^r m$ and $\gcd(p, m) = 1$. That is, $P \subseteq G$ is a p -Sylow subgroup if it is a p -group and p does not divide $[G : P]$.

If p does *not* divide the order of G , then G contains a p -Sylow subgroup: namely, $\{e\}$. This is not very interesting; what is interesting is that G contains a p -Sylow subgroup even when p *does* divide the order of G :

Theorem 2.5 (First Sylow theorem). *Every finite group contains a p -Sylow subgroup, for all primes p .*

⁵In fact, a complete list of all finite simple groups is known: this is the classification result mentioned at the end of §II.6.3, arguably one of the deepest and hardest results in mathematics.

The first Sylow theorem follows from the seemingly stronger statement:

Proposition 2.6. *If p^k divides the order of G , then G has a subgroup of order p^k .*

The statements are actually easily seen to be equivalent, by Exercise 1.8; in any case, the standard argument proving Theorem 2.5 proves Proposition 2.6, and I see no reason to hide this fact. Here is the argument:

Proof of Proposition 2.6. If $k = 0$, there is nothing to prove, so we may assume $k \geq 1$ and in particular that $|G|$ is a multiple of p .

Argue by induction on $|G|$: if $|G| = p$, again there is nothing to prove; if $|G| > p$ and G contains a proper subgroup H such that $[G : H]$ is relatively prime to p , then p^k divides the order of H , and hence H contains a subgroup of order p^k by the induction hypothesis, and thus so does G .

Therefore, we may assume that all proper subgroups of G have index divisible by p . By the class formula (Proposition 1.8), p divides the order of the center $Z(G)$. By Cauchy's theorem⁶, $\exists a \in Z(G)$ such that a has order p . The cyclic subgroup $N = \langle a \rangle$ is contained in $Z(G)$, and hence it is normal in G (Exercise 1.4). Therefore we can consider the quotient G/N .

Since $|G/N| = |G|/p$ and p^k divides $|G|$ by hypothesis, we have that p^{k-1} divides the order of G/N . By the induction hypothesis, we may conclude that G/N contains a subgroup of order p^{k-1} . By the structure of the subgroups of a quotient (§II.8.3, especially Proposition II.8.9), this subgroup must be of the form P/N , for P a subgroup of G .

But then $|P| = |P/N| \cdot |N| = p^{k-1} \cdot p = p^k$, as needed. $\square$

There are slicker ways to prove Theorem 2.5. We will see a pretty (and insightful) alternative in §2.3; but the proof given above is easy to remember and is a good template for similar arguments.

Remark 2.7. The diligent reader worked out in Exercise II.8.20 a *stronger* statement than Proposition 2.6, for *abelian* groups. The arguments are similar; the advantage in the abelian case is that any cyclic subgroup produced by Cauchy's theorem is automatically normal, while ensuring normality requires a few twists and turns in the general case (and, as a result, yields a weaker statement). $\lrcorner$

2.3. Sylow II. Theorem 2.5 tells us that *some* maximal p -group in G attains the largest size allowed by Lagrange's theorem, that is, the maximal power of the prime p dividing $|G|$.

One can be more precise: the second Sylow theorem tells us that *every* maximal p -group in $|G|$ is in fact a p -Sylow subgroup. It is as large as is allowed by Lagrange's theorem.

The situation is in fact even better: all p -Sylow subgroups are conjugates of each other⁷. Moreover, even better than this, *every* p -group inside G must be contained in a conjugate of any fixed p -Sylow subgroup.

⁶Note that, as mentioned in §2.1, we only need the *abelian* case of this theorem.

⁷Of course if P is a p -Sylow subgroup of G , then so are all conjugates gPg^{-1} of P .

The proof of this very precise result is very easy!

Theorem 2.8 (Second Sylow theorem). *Let G be a finite group, let P be a p -Sylow subgroup, and let $H \subseteq G$ be a p -group. Then H is contained in a conjugate of P : there exists $g \in G$ such that $H \subseteq gPg^{-1}$.*

Proof. Act with H on the set of left-cosets of P , by left-multiplication. Since there are $[G : P]$ cosets and p does not divide $[G : P]$, we know this action must have fixed points (Exercise 1.1): let gP be one of them. This means that $\forall h \in H$:

$$hgP = gP;$$

that is, $g^{-1}hgP = P$ for all h in H ; that is, $g^{-1}Hg \subseteq P$; that is, $H \subseteq gPg^{-1}$, as needed. $\square$

We can obtain an even more complete picture of the situation. Suppose we have constructed a chain

$$H_0 = \{e\} \subseteq H_1 \subseteq \cdots \subseteq H_k$$

of p -subgroups of a group G , where $|H_i| = p^i$. By Theorem 2.8 we know that H_k is contained in some p -Sylow subgroup, of order p^r = the maximum power of p dividing the order of G . But I claim that the chain can in fact be continued one step at a time all the way up to the Sylow subgroup:

$$H_0 = \{e\} \subseteq H_1 \subseteq \cdots \subseteq H_k \subseteq H_{k+1} \subseteq \cdots \subseteq H_r;$$

and, further, H_k may be assumed to be normal in H_{k+1} . The following lemma will simplify the proof of this fact considerably and will also help us prove the *third* Sylow theorem.

Lemma 2.9. *Let H be a p -group contained in a finite group G . Then*

$$[N_G(H) : H] \equiv [G : H] \pmod{p}.$$

Proof. If H is trivial, then $N_G(H) = G$ and the two numbers are equal.

Assume then that H is nontrivial, and act with H on the set of left-cosets of H in G , by left-multiplication. The fixed points of this action are the cosets gH such that $\forall h \in H$

$$hgH = gH,$$

that is, such that $g^{-1}hg \in H$ for all $h \in H$; in other words, $H \subseteq gHg^{-1}$, and hence (by order considerations) $gHg^{-1} = H$. This means precisely that $g \in N_G(H)$. Therefore, the set of fixed points of the action consists of the set of cosets of H in $N_G(H)$.

The statement then follows immediately from Corollary 1.3. $\square$

As a consequence, if H_k is not a p -Sylow subgroup ‘already’, in the sense that p ‘still’ divides $[G : H_k]$, then p must also divide $[N_G(H_k) : H_k]$. Another application of Cauchy’s theorem tells us how to obtain the next subgroup H_{k+1} in the chain. More precisely, we have the following result.

Proposition 2.10. *Let H be a p -subgroup of a finite group G , and assume that H is not a p -Sylow subgroup. Then there exists a p -subgroup H' of G containing H , such that $[H' : H] = p$ and H is normal in H' .*

Proof. Since H is not a p -Sylow subgroup of G , p divides $[N_G(H) : H]$, by Lemma 2.9. Since H is normal in $N_G(H)$, we may consider the quotient group $N_G(H)/H$, and p divides the order of this group. By Theorem 2.1, $N_G(H)/H$ has an element of order p ; this generates a subgroup of order p of $N_G(H)/H$, which must be (cf. §II.8.3) of the form H'/H for a subgroup H' of $N_G(H)$.

It is straightforward to verify that H' satisfies the stated requirements. $\square$

The statement about ‘chains of p -subgroups’ follows immediately from this result.

Note that Cauchy’s theorem and Proposition 2.10 provide a new proof of Proposition 2.6 and hence of the first Sylow theorem.

2.4. Sylow III. The third (and last) Sylow theorem gives a good handle on the number of p -Sylow subgroups of a given finite group G . This is especially useful in establishing the existence of normal subgroups of G : since all p -Sylow subgroups of a group are conjugates of each other (by the second Sylow theorem), if there is *only one* p -Sylow subgroup, then that subgroup must be normal⁸.

Theorem 2.11 (Third Sylow theorem). *Let p be a prime integer, and let G be a finite group of order $|G| = p^r m$. Assume that p does not divide m . Then the number of p -Sylow subgroups of G divides m and is congruent to 1 modulo p .*

Proof. Let N_p denote the number of p -Sylow subgroups of G .

By Theorem 2.8, the p -Sylow subgroups of G are the conjugates of any given p -Sylow subgroup P . By Lemma 1.13, N_p is the index of the normalizer $N_G(P)$ of P ; thus (Corollary 1.14) it divides the index m of P . In fact,

$$m = [G : P] = [G : N_G(P)] \cdot [N_G(P) : P] = N_p \cdot [N_G(P) : P].$$

Now, by Lemma 2.9 we have

$$m = [G : P] \equiv [N_G(P) : P] \pmod{p};$$

multiplying by N_p , we get

$$mN_p \equiv m \pmod{p}.$$

Since $m \not\equiv 0 \pmod{p}$ and p is prime, this implies

$$N_p \equiv 1 \pmod{p},$$

as needed. $\square$

Of course there are other ways to prove Theorem 2.11: see for example Exercise 2.11.

⁸For an alternative viewpoint, see Exercise 2.2.

2.5. Applications. Consequences stemming from the group actions we have encountered, and especially the Sylow theorems, may be applied to establish exquisitely precise facts about individual groups as well as whole classes of groups; this is often based on some simple but clever numerology.

The following examples are exceedingly simple-minded but will hopefully convey the flavor of what can be done with the tools we have built in the previous two sections. More examples may be found among the exercises at the end of this section.

2.5.1. More nonsimple groups.

Claim 2.12. *Let G be a group of order mp^r , where p is a prime integer and $1 < m < p$. Then G is not simple.*

(Cf. Example 2.4.)

Proof. By the third Sylow theorem, the number N_p of p -Sylow subgroups divides m and is of the form $1 + kp$. Since $m < p$, this forces $k = 0$, $N_p = 1$. Therefore G has a normal subgroup of order p^r ; hence it is not simple. $\square$

Of course the same argument gives the same conclusion for every group of order mp^r , where $(m, p) = 1$ and the only divisor d of m such that $d \equiv 1 \pmod p$ is $d = 1$.

Example 2.13. There are no simple groups of order 2002.

Indeed⁹,

$$2002 = 2 \cdot 7 \cdot 11 \cdot 13;$$

the divisors of $2 \cdot 7 \cdot 13$ are

$$1, 2, 7, 13, 14, 26, 91, 182;$$

of these, only 1 is congruent to 1 mod 11. Thus there is a normal subgroup of order 11 in every group of order 2002. $\lrcorner$

The reader should not expect the third Sylow theorem to always yield its fruits so readily, however.

Example 2.14. There are no simple groups of order 12.

Note that $3 \equiv 1 \pmod 2$ and $4 \equiv 1 \pmod 3$: thus the argument used above does not guarantee the existence of either a normal 2-Sylow subgroup or a normal 3-Sylow subgroup.

However, suppose that there is more than one 3-Sylow subgroup. Then there must be 4, by the third Sylow theorem. Since any two such subgroups must intersect in the identity, this accounts for exactly 8 elements of order 3. Excluding these leaves us with the identity and 3 elements of order 2 or 4; that is just enough room to fit *one* 2-Sylow subgroup. This subgroup will then have to be normal.

Thus, either there is a 3-Sylow normal subgroup or there is a 2-Sylow normal subgroup—either way, the group is not simple. $\lrcorner$

⁹It is safe to guess that this statement has been assigned on hundreds of algebra tests across the world in the year 2002.

Even this more refined counting will often fail, and one has to dig deeper.

Example 2.15. There are no simple groups of order 24.

Indeed, let G be a group of order 24, and consider its 2-Sylow subgroups; by the third Sylow theorem, there are either 1 or 3 such subgroups. If there is 1, the 2-Sylow subgroup is normal and G is not simple. Otherwise, G acts (nontrivially) by conjugation on this set of three 2-Sylow subgroups; this action gives a nontrivial homomorphism $G \rightarrow S_3$, whose kernel is a proper, nontrivial normal subgroup of G —thus again G is not simple. $\lrcorner$

The reader should practice by selecting a random number n and trying to say as much as he/she can, in general, about groups of order n . Beware: such problems are a common feature of qualifying exams.

2.5.2. Groups of order pq , $p < q$ prime.

Claim 2.16. Assume $p < q$ are prime integers and $q \not\equiv 1 \pmod{p}$. Let G be a group of order pq . Then G is cyclic.

Proof. By the third Sylow theorem, G has a unique (hence normal) subgroup H of order p . Indeed, the number N_p of p -Sylow subgroups must divide q , and q is prime, so $N_p = 1$ or q . Necessarily $N_p \equiv 1 \pmod{p}$, and $q \not\equiv 1 \pmod{p}$ by hypothesis; therefore $N_p = 1$.

Since H is normal, conjugation gives an action of G on H , hence (by Exercise 1.21) a homomorphism $\gamma : G \rightarrow \text{Aut}(H)$. Now H is cyclic of order p , so $|\text{Aut}(H)| = p - 1$ (Exercise II.4.14); the order of $\gamma(G)$ must divide both pq and $p - 1$, and it follows that γ is the trivial map.

Therefore, conjugation is *trivial* on H : that is, $H \subseteq Z(G)$. Lemma 1.5 implies that G is abelian.

Finally, an abelian group of order pq , with $p < q$ primes, is necessarily cyclic: indeed it must contain elements g, h of order p, q , respectively (for example by Cauchy's theorem), and then $|gh| = pq$ by Exercise II.1.14. $\square$

For example, this statement ‘classifies’ all groups of order 15, 33, 35, 51, $\dots$: such groups are necessarily cyclic.

The argument given in the proof is rather ‘high-brow’, as it involves the automorphism group of H ; that is precisely why I gave it. For low-brow alternatives, see Exercise 2.18 or Remark 5.4.

The condition $q \not\equiv 1 \pmod{p}$ in Claim 2.16 is clearly necessary: indeed, $|S_3| = 2 \cdot 3$ is the product of two distinct primes, and yet S_3 is *not* cyclic. The argument given in the proof shows that if $|G| = pq$, with $p < q$ prime, and G has a normal subgroup of order p , then G is cyclic. If $q \equiv 1 \pmod{p}$, it can be shown that there is in fact a unique noncommutative group of order pq up to isomorphism: the reader will work this out after learning about semidirect products (Exercise 5.12). But we are in fact already in the position of obtaining rather sophisticated information about this group, even without knowing its construction in general (Exercise 2.19).

For fun, let's tackle the case in which $p = 2$.

Claim 2.17. *Let q be an odd prime, and let G be a noncommutative group of order $2q$. Then $G \cong D_{2q}$, the dihedral group.*

Proof. By Cauchy's theorem, $\exists y \in G$ such that y has order q . By the third Sylow theorem, $\langle y \rangle$ is the unique subgroup of order q in G (and is therefore normal). Since G is not commutative and in particular it is not cyclic, it has no elements of order $2q$; therefore, every element in the complement of $\langle y \rangle$ has order 2; let x be any such element.

The conjugate xyx^{-1} of y by x is an element of order q , so $xyx^{-1} \in \langle y \rangle$. Thus, $xyx^{-1} = y^r$ for some r between 0 and $q-1$.

Now observe that

$$(y^r)^r = (xyx^{-1})^r = xy^r x^{-1} = x^2 y (x^{-1})^2 = y$$

since $|x| = 2$. Therefore, $y^{r^2-1} = e$, which implies

$$q \mid (r^2 - 1) = (r - 1)(r + 1)$$

by Corollary II.1.11. Since q is prime, this says that $q \mid (r - 1)$ or $q \mid (r + 1)$; since $0 \leq r \leq q - 1$, it follows that $r = 1$ or $r = q - 1$.

If $r = 1$, then $xyx^{-1} = y$; that is, $xy = yx$. But then the order of xy is $2q$ (by Exercise II.1.14), and G is cyclic, a contradiction.

Therefore $r = q - 1$, and we have established the relations

$$\begin{cases} x^2 = e, \\ y^q = e, \\ yx = xy^{q-1}. \end{cases}$$

These are the relations satisfied by generators x, y of D_{2q} , as the reader hopefully verified in Exercise II.2.5; the statement follows. $\square$

Claim 2.17 yields a classification of groups of order $2q$, for q an odd prime: such a group must be either abelian (and hence cyclic, by the usual considerations) or isomorphic to a dihedral group. For $q = 3$, we recover the result of Exercise 1.13: every noncommutative group of order 6 is isomorphic to $D_6 \cong S_3$.

Exercises

2.1. $\triangleright$ Prove Claim 2.2. [§2.1]

2.2. $\triangleright$ Let G be a group. A subgroup H of G is *characteristic* if $\varphi(H) \subseteq H$ for every automorphism φ of G .

- Prove that characteristic subgroups are normal.
- Let $H \subseteq K \subseteq G$, with H characteristic in K and K normal in G . Prove that H is normal in G .
- Let G, K be groups, and assume that G contains a single subgroup H isomorphic to K . Prove that H is normal in G .

- Let K be a normal subgroup of a finite group G , and assume that $|K|$ and $|G/K|$ are relatively prime. Prove that K is characteristic in G .

[§2.1, §2.4, 2.13, §3.3]

2.3. Prove that a nonzero abelian group G is simple if and only if $G \cong \mathbb{Z}/p\mathbb{Z}$ for some positive prime integer p .

2.4. $\triangleright$ Prove that a nontrivial group G is simple if and only if its only homomorphic images (i.e., groups G' such that there is an onto homomorphism $G \rightarrow G'$) are the trivial group and G itself (up to isomorphism). [§3.2]

2.5. Let G be a *simple* group, and assume $\varphi : G \rightarrow G'$ is a nontrivial group homomorphism. Prove that φ is injective.

2.6. Prove that there are no simple groups of order 4, 8, 9, 16, 25, 27, 32, or 49. In fact, prove that no p -group of order $\geq p^2$ is simple.

2.7. Prove that there are no simple groups of order 6, 10, 14, 15, 20, 21, 22, 26, 28, 33, 34, 35, 38, 39, 42, 44, 46, 51, 52, 55, 57, or 58. (Hint: Example 2.4.)

2.8. Let G be a finite group, p a prime integer, and let N be the intersection of the p -Sylow subgroups of G . Prove that N is a *normal* p -subgroup of G and that every normal p -subgroup of G is contained in N . (In other words, G/N is final with respect to the property of being a homomorphic image of G of order $|G|/p^\alpha$ for some α .)

2.9. $\neg$ Let P be a p -Sylow subgroup of a finite group G , and let $H \subseteq G$ be a p -subgroup. Assume $H \subseteq N_G(P)$. Prove that $H \subseteq P$. (Hint: P is normal in $N_G(P)$, so PH is a subgroup of $N_G(P)$ by Proposition II.8.11, and $|PH/P| = |H/(P \cap H)|$. Show that this implies that PH is a p -group, and hence $PH = P$ since P is a maximal p -subgroup of G . Deduce that $H \subseteq P$.) [2.10]

2.10. $\neg$ Let P be a p -Sylow subgroup of a finite group G , and act with P by conjugation on the set of p -Sylow subgroups of G . Show that P is the unique fixed point of this action. (Hint: Use Exercise 2.9.) [2.11]

2.11. $\triangleright$ Use the second Sylow theorem, Corollary 1.14, and Exercise 2.10 to paste together an alternative proof of the third Sylow theorem. [§2.4]

2.12. Let P be a p -Sylow subgroup of a finite group G , and let $H \subseteq G$ be a subgroup containing the normalizer $N_G(P)$. Prove that $[G : H] \equiv 1 \pmod{p}$.

2.13. $\neg$ Let P be a p -Sylow subgroup of a finite group G .

- Prove that if P is normal in G , then it is in fact characteristic in G (cf. Exercise 2.2).
- Let $H \subseteq G$ be a subgroup containing the Sylow subgroup P . Assume P is normal in H and H is normal in G . Prove that P is normal in G .
- Prove that $N_G(N_G(P)) = N_G(P)$.

[3.12]

2.14. Prove that there are no simple groups of order 18, 40, 45, 50, or 54.

2.15. Classify all groups of order $n \leq 15$, $n \neq 8, 12$: that is, produce a list of nonisomorphic groups such that every group of order $n \neq 8, 12$, $n \leq 15$ is isomorphic to one group in the list.

2.16. $\neg$ Let G be a noncommutative group of order 8.

- Prove that G contains elements of order 4 and no elements of order 8.
- Let y be an element of order 4. Prove that G is generated by y and by an element $x \notin \langle y \rangle$, such that $x^2 = e$ or $x^2 = y^2$.
- In either case, $G = \{e, y, y^2, y^3, x, yx, y^2x, y^3x\}$. Prove that the multiplication table of G is determined by whether $x^2 = e$ or $x^2 = y^2$, and by the value of xy .
- Prove that necessarily $xy = y^3x$. (Hint: To eliminate $xy = y^2x$, multiply on the right by y .)
- Prove that $G \cong D_8$ or $G \cong Q_8$.

[6.2, VII.6.6]

2.17. $\neg$ Let R be a *division ring* (Definition III.1.13), and assume $|R| = 64$. Prove that R is necessarily commutative (hence, a field), as follows:

- The group of units of R has order 63. Prove it has a commutative subgroup G of order 9. (Sylow.)
- Prove that R is the only sub-division ring of R containing G .
- Prove that the set of elements of R commuting with every element of G is a sub-division ring of R containing G . (Cf. Exercise III.2.10.)
- Conclude that G is contained in the center of R . Recall that the center of R is a sub-division ring of R (cf. Exercise III.2.9), and conclude that R is commutative.

Like Exercise III.2.11, this is a particular case of a theorem of Wedderburn, according to which *every* finite division ring is a field. [VII.5.16]

2.18. $\triangleright$ Give an alternative proof of Claim 2.16 as follows: use the third Sylow theorem to count the number of elements of order p and q in G ; use this to show that there are elements in G of order neither 1 nor p nor q ; deduce that G is cyclic. [§2.5]

2.19. $\triangleright$ Let G be a noncommutative group of order pq , where $p < q$ are primes.

- Show that $q \equiv 1 \pmod{p}$.
- Show that the center of G is trivial.
- Draw the lattice of subgroups of G .
- Find the number of elements of each possible order in G .
- Find the number and size of the conjugacy classes in G .

[§2.5]

2.20. How many elements of order 7 are there in a simple group of order 168?

2.21. Let $p < q < r$ be prime integers, and let G be a group of order pqr . Prove that G is not simple.

2.22. Let G be a finite group, $n = |G|$, and p be a prime divisor of n . Assume that the only divisor of n that is congruent to 1 modulo p is 1. Prove that G is not simple.

2.23. $\neg$ Let N_p denote the number of p -Sylow subgroups of a group G . Prove that if G is simple, then $|G|$ divides $N_p!$ for all primes p in the factorization of $|G|$. More generally, prove that if G is simple and H is a subgroup of G of index $N > 1$, then $|G|$ divides $N!$. (Hint: Exercise II.9.12.) This problem capitalizes on the idea behind Example 2.15. [2.25]

2.24. $\triangleright$ Prove that there are no noncommutative simple groups of order less than 60. If you have sufficient stamina, prove that the next possible order for a noncommutative simple group is 168. (Don't feel too bad if you have to cheat and look up a few particularly troublesome orders > 60 .) [§4.4]

2.25. $\neg$ Assume that G is a *simple* group of order 60.

- Use Sylow's theorems and simple numerology to prove that G has either five or fifteen 2-Sylow subgroups, accounting for fifteen elements of order 2 or 4. (Exercise 2.23 will likely be helpful.)
- If there are fifteen 2-Sylow subgroups, prove that there exists an element $g \in G$ of order 2 contained in at least two of them. Prove that the centralizer of g has index 5.

Conclude that every simple group¹⁰ of order 60 contains a subgroup of index 5. [4.22]

3. Composition series and solvability

I have claimed that *simple* groups (in the sense of Definition 2.3) are the 'basic constituents' of all finite groups. Among other things, the material in this section will (partially) justify this claim.

3.1. The Jordan-Hölder theorem. A *series* of subgroups G_i of a group G is a decreasing sequence of subgroups starting from G :

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots.$$

The length of a series is the number of strict inclusions.

A series is *normal* if G_{i+1} is normal in G_i for all i . We will be interested in the *maximal length of a normal series* in G ; if finite, I will denote this number¹¹ by $\ell(G)$. The number $\ell(G)$ is a measure of how far G is from being simple. Indeed, $\ell(G) = 0$ if and only if G is trivial, and $\ell(G) = 1$ if and only if G is *simple*: for a simple group, the only maximal normal series is

$$G \supsetneq \{e\}.$$

¹⁰The reader will prove later (Exercise 4.22) that there is in fact *only one* simple group of order 60 up to isomorphism and that this group contains exactly five 2-Sylow subgroups. The result obtained here will be needed to establish this fact.

¹¹There does not appear to be a standard notation for this concept.

Definition 3.1. A *composition series* for G is a normal series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

such that the successive quotients G_i/G_{i+1} are simple. $\lrcorner$

It is clear (by induction on the order) that finite groups have composition series, while infinite groups do not necessarily have one (Exercise 3.3). It is also clear that if a normal series has maximal length $\ell(G)$, then it is a composition series. What is *not* clear is that the converse holds: conceivably, there could exist composition series of different lengths (the longest ones having length $\ell(G)$). For example, why can't there be a finite group G with $\ell(G) = 3$ and two different composition series

$$G \supsetneq G_1 \supsetneq G_2 \supsetneq \{e\}$$

and

$$G \supsetneq G'_1 \supsetneq \{e\}$$

(that is: a finite group G with $\ell(G) = 3$ and a simple normal subgroup G'_1 such that G/G'_1 is simple)?

Part of the content of the *Jordan-Hölder* theorem is that (luckily) this cannot happen. In fact, the theorem is much more precise: not only do all composition series have the same length, but they also have the same quotients (appearing, however, in possibly different orders).

Theorem 3.2 (Jordan-Hölder). *Let G be a group, and let*

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\},$$

$$G = G'_0 \supsetneq G'_1 \supsetneq G'_2 \supsetneq \cdots \supsetneq G'_m = \{e\}$$

be two composition series for G . Then $m = n$, and the lists of quotient groups $H_i = G_i/G_{i+1}$, $H'_i = G'_i/G'_{i+1}$ agree (up to isomorphism) after a permutation of the indices.

Proof. Let

$$(*) \quad G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

be a composition series. Argue by induction on n : if $n = 0$, then G is trivial, and there is nothing to prove. Assume $n > 0$, and let

$$(**) \quad G = G'_0 \supsetneq G'_1 \supsetneq G'_2 \supsetneq \cdots \supsetneq G'_m = \{e\}$$

be another composition series for G . If $G_1 = G'_1$, then the result follows from the induction hypothesis, since G_1 has a composition series of length $n - 1 < n$.

We may then assume $G_1 \neq G'_1$. Note that $G_1 G'_1 = G$: indeed, $G_1 G'_1$ is normal in G (Exercise 3.5), and $G_1 \subsetneq G_1 G'_1$; but there are no proper normal subgroups between G_1 and G since G/G_1 is simple.

Let $K = G_1 \cap G'_1$. The *distinct* subgroups $G_i \cap K$ determine a composition series

$$K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

of K : this is not difficult to see, and will be verified more formally in the proof of Proposition 3.4. By Proposition II.8.11 (the “second isomorphism theorem”),

$$\frac{G_1}{K} = \frac{G_1}{G_1 \cap G'_1} \cong \frac{G_1 G'_1}{G'_1} = \frac{G}{G'_1} \quad \text{and} \quad \frac{G'_1}{K} \cong \frac{G}{G_1}$$

are simple. Therefore, we have two new composition series for G :

$$\begin{array}{ccccccccccc} G & \supsetneq & G_1 & \supsetneq & K & \supsetneq & K_1 & \supsetneq & \cdots & \supsetneq & \{e\} \\ \parallel & & & & \parallel & & \parallel & & & & \parallel \\ G & \supsetneq & G'_1 & \supsetneq & K & \supsetneq & K_1 & \supsetneq & \cdots & \supsetneq & \{e\} \end{array}$$

which only differ at the first step. These two series trivially have the same length and the same quotients (the first two quotients get switched from one series to the other).

Now I claim that the first of these two series has the same length and quotients as the series (*). Indeed,

$$G_1 \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

is a composition series for G_1 : by the induction hypothesis, it must have the same length and quotients as the composition series

$$G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\};$$

verifying my claim (and note that, in particular, $r = n - 2$).

By the same token, applying the induction hypothesis to the series (of length $n - 1$)

$$G'_1 \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_{n-2} = \{e\},$$

shows that the second series has the same length and quotients as (**), and the statement follows. $\square$

3.2. Composition factors; Schreier’s theorem. Two normal series are *equivalent* if they have the same length and the same quotients (up to order). The Jordan-Hölder theorem shows that any two *maximal* finite series of a group are equivalent. That is, the (isomorphism classes of the) quotients of a composition series depend only on the group, not on the chosen series. These are the *composition factors* of the group. They form a multiset¹² of *simple* groups: the ‘basic constituents’ of our loose comment back in §2.

It is clear that two isomorphic groups must have the same composition factors. Unfortunately, it is not possible to reconstruct a group from its composition factors alone (Exercise 3.4). One has to take into account the way the simple groups are ‘glued’ together; we will come back to this point in §5.2.

The intuition that the composition factors of a group are its basic constituents is reinforced by the following fact: if G is a group with a composition series, then the composition factors of every normal subgroup N of G are composition factors of G and the remaining ones are the composition factors of the quotient G/N .

¹²See §I.2.2 for a reminder on *multisets*: they are sets of elements counted with multiplicity. For example, the composition factors of $\mathbb{Z}/4\mathbb{Z}$ form the multiset consisting of *two* copies of $\mathbb{Z}/2\mathbb{Z}$.

Example 3.3. Let $G = \mathbb{Z}/6\mathbb{Z} = \{[0], [1], [2], [3], [4], [5]\}$. Then

$$\{[0], [1], [2], [3], [4], [5]\} \supsetneq \{[0], [3]\} \supsetneq \{[0]\}$$

is a composition series for G ; the quotients are $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z}$, respectively. The (normal) subgroup $N = \{[0], [2], [4]\}$ ‘turns off’ the second factor: indeed, intersecting the series with N gives

$$\{[0], [2], [4]\} \supsetneq \{[0]\} = \{[0]\},$$

a series with composition factor $\mathbb{Z}/3\mathbb{Z}$. On the other hand, ‘mod-ing out by N ’ turns off the first factor: keeping in mind $[3] + N = [1] + N$, etc., we find

$$\{[0] + N, [1] + N\} = \{[0] + N, [1] + N\} \supsetneq \{[0] + N\},$$

a series with lone composition factor $\mathbb{Z}/2\mathbb{Z}$. ┘

This phenomenon holds in complete generality:

Proposition 3.4. *Let G be a group, and let N be a normal subgroup of G . Then G has a composition series if and only if both N and G/N have composition series. Further, if this is the case, then*

$$\ell(G) = \ell(N) + \ell(G/N),$$

and the composition factors of G consist of the collection of composition factors of N and of G/N .

Proof. If G/N has a composition series, the subgroups appearing in it correspond to subgroups of G containing N , with isomorphic quotients, by Proposition II.8.10 (the “third isomorphism theorem”). Thus, if both G/N and N have composition series, juxtaposing them produces a composition series for G , with the stated consequence on composition factors.

The converse is a little trickier. Assume that G has a composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

and that N is a normal subgroup of G . Intersecting the series with N gives a sequence of subgroups of the latter:

$$N = G \cap N \supseteq G_1 \cap N \supseteq \cdots \supseteq \{e\} \cap N = \{e\}$$

such that $G_{i+1} \cap N$ is normal in $G_i \cap N$, for all i . I claim that this becomes a composition series for N once repetitions are eliminated. Indeed, this follows once we establish that

$$\frac{G_i \cap N}{G_{i+1} \cap N}$$

is either trivial (so that $G_{i+1} \cap N = G_i \cap N$, and the corresponding inclusion may be omitted) or isomorphic to G_i/G_{i+1} (hence simple, and one of the composition factors of G). To see this, consider the homomorphism

$$G_i \cap N \hookrightarrow G_i \twoheadrightarrow \frac{G_i}{G_{i+1}} :$$

the kernel is clearly $G_{i+1} \cap N$; therefore (by the first isomorphism theorem) we have an injective homomorphism

$$\frac{G_i \cap N}{G_{i+1} \cap N} \hookrightarrow \frac{G_i}{G_{i+1}}$$

identifying $(G_i \cap N)/(G_{i+1} \cap N)$ with a subgroup of G_i/G_{i+1} . Now, this subgroup is *normal* (because N is normal in G) and G_i/G_{i+1} is simple; our claim follows.

As for G/N , obtain a sequence of subgroups from a composition series for G :

$$\frac{G}{N} \supseteq \frac{G_1 N}{N} \supseteq \frac{G_2 N}{N} \supseteq \cdots \supseteq \frac{\{e_G\}N}{N} = \{e_{G/N}\},$$

such that $(G_{i+1}N)/N$ is normal in $(G_i N)/N$. As above, we have to check that

$$\frac{(G_i N)/N}{(G_{i+1} N)/N}$$

is either trivial or isomorphic to G_i/G_{i+1} . By the third isomorphism theorem, this quotient is isomorphic to $(G_i N)/(G_{i+1} N)$. This time, consider the homomorphism

$$G_i \hookrightarrow G_i N \twoheadrightarrow \frac{G_i N}{G_{i+1} N} :$$

this is *surjective* (check!), and the subgroup G_{i+1} of the source is sent to the identity element in the target; hence (by Theorem II.7.12) there is an onto homomorphism

$$\frac{G_i}{G_{i+1}} \twoheadrightarrow \frac{G_i N}{G_{i+1} N}.$$

Since G_i/G_{i+1} is simple, it follows that $(G_i N)/(G_{i+1} N)$ is either trivial or isomorphic to it (Exercise 2.4), as needed.

Summarizing, we have shown that if G has a composition series and N is normal in G , then both N and G/N have composition series. The first part of the argument yields the statement on lengths and composition factors, concluding the proof. $\square$

One nice consequence of the Jordan-Hölder theorem is the following observation. A series is a *refinement* of another series if all terms of the first appear in the second.

Proposition 3.5. *Any two normal series of a finite group ending with $\{e\}$ admit equivalent refinements.*

Proof. Refine the series to a composition series; then apply the Jordan-Hölder theorem. $\square$

In fact, *Schreier's theorem* asserts that this holds for *all* groups (while the argument given here only works for groups admitting a composition series, e.g., finite groups). Proving this in general is reasonably straightforward, from judicious applications of the second isomorphism theorem (cf. Exercise 3.7).

3.3. The commutator subgroup, derived series, and solvability. It has been a while since we have encountered a universal object; here is one. For any group G , consider the category whose objects are group homomorphisms $\alpha : G \rightarrow A$ from G to a *commutative* group and whose morphisms $\alpha \rightarrow \beta$ are (as the reader should expect) commutative diagrams

$$\begin{array}{ccc} & G & \\ \alpha \swarrow & & \searrow \beta \\ A & \xrightarrow{\varphi} & B \end{array}$$

where φ is a homomorphism.

Does this category have an initial object? That is, given a group G , does there exist a *commutative* group which is universal with respect to the property of being a homomorphic image of G ?

Yes.

Such a group may well be thought of as the closest ‘commutative approximation’ of the given group G . To verify that this universal object exists, we introduce the following important notion. (The diligent reader has begun exploring this territory already, in Exercise II.7.11.)

Definition 3.6. Let G be a group. The *commutator* subgroup of G is the subgroup generated by all elements

$$ghg^{-1}h^{-1}$$

with $g, h \in G$. ┘

The element $ghg^{-1}h^{-1}$ is often denoted $[g, h]$ and is called the *commutator* of g and h . Thus, g, h commute with each other if and only if $[g, h] = e$.

In the same notational style, the commutator subgroup of G should be denoted $[G, G]$; this is a bit heavy, and the common shorthand for it is G' , which offers the possibility of iterating the notation. Thus, G'' may be used to denote the commutator subgroup of the commutator subgroup of G , and $G^{(i)}$ denotes the i -th iterate. I will adopt this notation in this subsection for convenience, but not elsewhere in this book (as I want to be able to ‘prime’ any letter I wish, for any reason).

First we record the following trivial, but useful, remark:

Lemma 3.7. Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. Then $\forall g, h \in G_1$ we have

$$\varphi([g, h]) = [\varphi(g), \varphi(h)]$$

and $\varphi(G'_1) \subseteq G'_2$.

This simple observation makes the key properties of the commutator subgroup essentially immediate (cf. Exercise II.7.11):

Proposition 3.8. Let G' be the commutator subgroup of G . Then

- G' is normal in G ;
- the quotient G/G' is commutative;

- if $\alpha : G \rightarrow A$ is a homomorphism of G to a commutative group, then $G' \subseteq \ker \alpha$;
- the natural projection $G \rightarrow G/G'$ is universal in the sense explained above.

Proof. These are all easy consequences of Lemma 3.7:

—By Lemma 3.7, the commutator subgroup is *characteristic*, hence normal (cf. Exercise 2.2).

—By Lemma 3.7, the commutator of any two cosets gG' , hG' is the coset of the commutator $[g, h]$; hence it is the identity in G/G' . As noted above, this implies that G/G' is commutative.

—Let $\alpha : G \rightarrow A$ be a homomorphism to a commutative group. By Lemma 3.7, $\alpha(G') \subseteq A' = \{e\}$: that is, $G' \subseteq \ker \alpha$.

—The universality follows from the previous point and from the universal property of quotients (Theorem II.7.12). $\square$

Taking successive commutators of a group produces a descending sequence of subgroups,

$$G \supseteq G' \supseteq G'' \supseteq G''' \supseteq \cdots,$$

which is ‘normal’ in the sense indicated in §3.1.

Definition 3.9. Let G be a group. The *derived series* of G is the sequence of subgroups

$$G \supseteq G' \supseteq G'' \supseteq G''' \supseteq \cdots. \quad \lrcorner$$

The derived series may or may not end with the identity of G . For example, if G is commutative, then the sequence gets there right away:

$$G \supseteq G' = \{e\};$$

however, if G is simple and noncommutative, then it gets stuck at the first step:

$$G = G' = G'' = \cdots$$

(indeed, G' is normal and $\neq \{e\}$ as G is noncommutative; but then $G' = G$ since G is simple).

Definition 3.10. A group is *solvable* if its derived series terminates with the identity. $\lrcorner$

For example, abelian groups are solvable.

The importance of this notion will be most apparent in the relatively distant future because of a brilliant application of Galois theory (§VII.7.4). But we can already appreciate it in the way it relates to the material we just covered. A normal series is *abelian*, resp., *cyclic*, if all quotients are abelian, resp., cyclic¹³.

Proposition 3.11. For a finite group G , the following are equivalent:

- All composition factors of G are cyclic.
- G admits a cyclic series ending in $\{e\}$.

¹³Thus, a composition series should be called ‘simple’; to our knowledge, it is not.

(iii) G admits an abelian series ending in $\{e\}$.

(iv) G is solvable.

Proof. (i) $\implies$ (ii) $\implies$ (iii) are trivial. (iii) $\implies$ (i) is obtained by refining an abelian series to a composition series (keeping in mind that the simple abelian groups are cyclic p -groups).

(iv) $\implies$ (iii) is also trivial, since the derived series is abelian (by the second point in Proposition 3.8).

Thus, we only have to prove (iii) $\implies$ (iv). For this, let

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

be an abelian series. Then I claim that $G^{(i)} \subseteq G_i$ for all i , where $G^{(i)}$ denotes the i -th ‘iterated’ commutator subgroup.

This can be verified by induction. For $i = 1$, G/G_1 is commutative; thus $G' \subseteq G_1$, by the third point in Proposition 3.8. Assuming we know $G^{(i)} \subseteq G_i$, the fact that G_i/G_{i+1} is abelian implies $G'_i \subseteq G_{i+1}$, and hence

$$G^{(i+1)} = (G^{(i)})' \subseteq G'_i \subseteq G_{i+1},$$

as claimed.

In particular we obtain that $G^{(n)} \subseteq G_n = \{e\}$: that is, the derived series terminates at $\{e\}$, as needed. $\square$

Example 3.12. All p -groups are solvable. Indeed, the composition factors of a p -group are simple p -groups (what else could they be?), hence cyclic. $\lrcorner$

Corollary 3.13. *Let N be a normal subgroup of a group G . Then G is solvable if and only if both N and G/N are solvable.*

Proof. This follows immediately from Proposition 3.4 and the formulation of solvability in terms of composition factors given in Proposition 3.11. $\square$

It is worth mentioning that *any* subgroup H of a solvable group is solvable: indeed, the commutator H' of H is a subgroup of the commutator G' of G , hence $H'' \subseteq G''$, $H''' \subseteq G'''$, and so on.

The *Feit-Thompson* theorem asserts that every finite group of *odd* order is solvable. This result is many orders of magnitude beyond the scope of this book: the original 1963 proof runs about 250 pages.

Exercises

3.1. Prove that $\mathbb{Z}$ has normal series of arbitrary lengths. (Thus, $\ell(\mathbb{Z})$ is not finite.)

3.2. Let G be a finite *cyclic* group. Compute $\ell(G)$ in terms of $|G|$. Generalize to finite solvable groups.

3.3. $\triangleright$ Prove that every finite group has a composition series. Prove that $\mathbb{Z}$ does not have a composition series. [§3.1]

3.4. $\triangleright$ Find an example of two nonisomorphic groups with the same composition factors. [§3.2]

3.5. $\triangleright$ Show that if H, K are *normal* subgroups of a group G , then HK is a normal subgroup of G . [§3.1]

3.6. Prove that $G_1 \times G_2$ has a composition series if and only if both G_1 and G_2 do, and explain how the corresponding composition factors are related.

3.7. $\triangleright$ Locate and understand a proof of (the general form of) Schreier's theorem that does not use the Jordan-Hölder theorem. Then obtain an alternative proof of the Jordan-Hölder theorem, using Schreier's. [§3.2]

3.8. $\triangleright$ Prove Lemma 3.7. [§3.3]

3.9. Let G be a nontrivial p -group. Construct explicitly an abelian series for G , using the fact that the center of a nontrivial p -group is nontrivial (Corollary 1.9). This gives an alternative proof of the fact that p -groups are solvable (Example 3.12).

3.10. $\neg$ Let G be a group. Define inductively an increasing sequence $Z_0 = \{e\} \subseteq Z_1 \subseteq Z_2 \subseteq \cdots$ of subgroups of G as follows: for $i \geq 1$, Z_i is the subgroup of G corresponding (as in Proposition II.8.9) to the center of G/Z_{i-1} .

- Prove that each Z_i is normal in G , so that this definition makes sense.

A group is¹⁴ *nilpotent* if $Z_m = G$ for some m .

- Prove that G is nilpotent if and only if $G/Z(G)$ is nilpotent.
- Prove that p -groups are nilpotent.
- Prove that nilpotent groups are solvable.
- Find a solvable group that is not nilpotent.

[3.11, 3.12, 5.1]

3.11. $\neg$ Let H be a nontrivial normal subgroup of a nilpotent group G (cf. Exercise 3.10). Prove that H intersects $Z(G)$ nontrivially. (Hint: Let $r \geq 1$ be the smallest index such that $\exists h \neq e, h \in H \cap Z_r$. Contemplate a well-chosen commutator $[g, h]$.) Since p -groups are nilpotent, this strengthens the result of Exercise 1.9. [3.14]

¹⁴There are many alternative characterizations for this notion that are equivalent to the one given here but not too trivially so.

3.12. Let H be a proper subgroup of a finite nilpotent group G (cf. Exercise 3.10). Prove that $H \subsetneq N_G(H)$. (Hint: $Z(G)$ is nontrivial. First dispose of the case in which H does not contain $Z(G)$, and then use induction to deal with the case in which H does contain $Z(G)$.) Deduce that every Sylow subgroup of a finite nilpotent group is normal¹⁵. (Use Exercise 2.13.)

3.13. $\neg$ For a group G , let $G^{(i)}$ denote the iterated commutator, as in §3.3. Prove that each $G^{(i)}$ is characteristic (hence normal) in G . [3.14]

3.14. Let H be a nontrivial normal subgroup of a solvable group G .

- Prove that H contains a nontrivial *commutative* subgroup that is normal in G . (Hint: Let r be the largest index such that $K = H \cap G^{(r)}$ is nontrivial. Prove that K is commutative, and use Exercise 3.13 to show it is normal in G .)
- Find an example showing that H need not intersect the center of G nontrivially (cf. Exercise 3.11).

3.15. Let p, q be prime integers, and let G be a group of order p^2q . Prove that G is solvable. (This is a particular case of *Burnside's theorem*: for p, q primes, every group of order $p^a q^b$ is solvable.)

3.16. $\triangleright$ Prove that every group of order < 120 and $\neq 60$ is solvable. [§4.4, §VII.7.4]

3.17. Prove that the Feit-Thompson theorem is equivalent to the assertion that every noncommutative finite simple group has even order.

4. The symmetric group

4.1. Cycle notation. It is time to give a second look at *symmetric groups*. Recall that S_n denotes the group of permutations (i.e., automorphisms in **Set**) of the set $\{1, \dots, n\}$. In §II.2 we denoted elements of S_n in a straightforward but inconvenient way:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 1 & 2 & 7 & 5 & 3 & 4 & 6 \end{pmatrix}$$

would stand for the element in S_8 sending **1** to **8**, **2** to **1**, etc.

There is clearly “too much” information here (the first row should be implicit), and at the same time it seems hard to find out anything interesting about a permutation from this notation. For example, can the reader say anything about the conjugates of σ in S_8 ? For maximal enlightenment, try to do Exercise 4.1 now, and then try again after absorbing the material in §4.2. In short, we should be able to do better.

As often is the case, thinking in terms of actions helps. By its very definition, the group S_n acts on the set $\{1, \dots, n\}$; so does every subgroup of S_n . Given a permutation $\sigma \in S_n$, consider the cyclic group $\langle \sigma \rangle$ generated by σ and its action on $\{1, \dots, n\}$. The orbits of this action form a partition of $\{1, \dots, n\}$; therefore, every

¹⁵This property characterizes finite nilpotent groups; cf. Exercise 5.1.

$\sigma \in S_n$ determines a partition of $\{1, \dots, n\}$. For example, the element $\sigma \in S_8$ given above splits $\{1, \dots, 8\}$ into three orbits:

$$\{1, 2, 3, 6, 8\}, \quad \{4, 7\}, \quad \{5\}.$$

The action of $\langle \sigma \rangle$ is transitive on each orbit. This means that one can get from any element of the orbit to any other element and then back to the original one by applying σ enough times. In the example,

$$1 \mapsto 8 \mapsto 6 \mapsto 3 \mapsto 2 \mapsto 1, \quad 4 \mapsto 7 \mapsto 4, \quad 5 \mapsto 5.$$

Definition 4.1. A (nontrivial) *cycle* is an element of S_n with exactly one nontrivial orbit. For distinct $a_1, \dots, a_r$ in $\{1, \dots, n\}$, the notation

$$(a_1 a_2 \dots a_r)$$

denotes the cycle in S_n with nontrivial orbit $\{a_1, \dots, a_r\}$, acting as

$$a_1 \mapsto a_2 \mapsto \dots \mapsto a_r \mapsto a_1.$$

In this case, r is the *length* of the cycle. A cycle of length r is called an *r-cycle*. $\square$

The identity is considered a cycle of length 1 in a trivial way and is denoted by (1) (and could just as well be denoted by (i) for any i).

Note that $(a_1 a_2 \dots a_r) = (a_2 \dots a_r a_1)$ according to the notation introduced in Definition 4.1: the notation determines the cycle, but a nontrivial cycle only determines the notation ‘up to a cyclic permutation’.

Two cycles are *disjoint* if their nontrivial orbits are. The following observation deserves to be highlighted, but it does not seem to deserve a proof:

Lemma 4.2. *Disjoint cycles commute.*

The next one gives us the alternative notation we were looking for.

Lemma 4.3. *Every $\sigma \in S_n$, $\sigma \neq e$, can be written as a product of disjoint nontrivial cycles, in a unique way up to permutations of the factors.*

Proof. As we have seen, every $\sigma \in S_n$ determines a partition of $\{1, \dots, n\}$ into orbits under the action of $\langle \sigma \rangle$. If $\sigma \neq e$, then $\langle \sigma \rangle$ has nontrivial orbits. As σ acts as a cycle on each orbit, it follows that σ may be written as a product of cycles.

The proof of the uniqueness is left to the reader (Exercise 4.2). $\square$

The *cycle notation* for $\sigma \in S_n$ is the (essentially) unique expression of σ as a product of disjoint cycles found in Lemma 4.3 (or (1) for $\sigma = e$). In our running example,

$$\sigma = (18632)(47),$$

and keep in mind that this expression is unique *cum grano salis*: we could write

$$\sigma = (63218)(47) = (74)(21863) = (32186)(74) = \dots,$$

and all of these are ‘the same’ cycle notation for σ .

4.2. Type and conjugacy classes in S_n . The cycle notation has obviously annoying features—such as the not-too-unique uniqueness pointed out a moment ago, or the fact that

$$(123)$$

can be an element of S_3 just as well as of S_{100000} , and only the context can tell. However, it is invaluable as it gives easy access to quite a bit of important information on a given permutation. In fact, much of this information is carried already by something even simpler than the cycle decomposition.

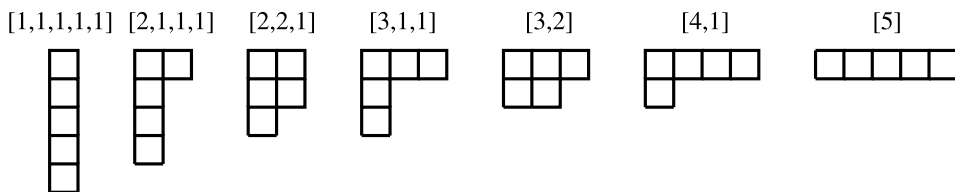
A *partition* of an integer $n > 0$ is a nonincreasing¹⁶ sequence of positive integers whose sum is n . It is easy to enumerate partitions for small values of n . For example, 5 has 7 distinct partitions:

$$\begin{aligned} 5 &= 1 + 1 + 1 + 1 + 1 \\ &= 2 + 1 + 1 + 1 \\ &= 2 + 2 + 1 \\ &= 3 + 1 + 1 \\ &= 3 + 2 \\ &= 4 + 1 \\ &= 5. \end{aligned}$$

The partition $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$ may be denoted

$$[\lambda_1, \dots, \lambda_r];$$

for example, the fourth partition listed above would be denoted $[3, 1, 1]$. A nicer ‘visual’ representation is by means of the corresponding *Young* (or *Ferrers*) *diagram*, obtained by stacking λ_1 boxes on top of λ_2 boxes on top of λ_3 boxes on top of $\dots$. For example, the diagrams corresponding to the seven partitions listed above are



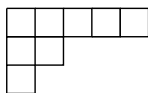
Definition 4.4. The *type* of $\sigma \in S_n$ is the partition of n given by the sizes of the orbits of the action of $\langle \sigma \rangle$ on $\{\mathbf{1}, \dots, \mathbf{n}\}$. ┘

It is hopefully clear (from the argument proving Lemma 4.3) that the type of $\sigma \in S_n$ is simply given by the lengths of the cycles in the decomposition of σ as the product of disjoint cycles, together with as many 1’s as needed. In our running example,

$$\sigma = (18632)(47) \in S_8$$

has type $[5, 2, 1]$:

¹⁶Of course this choice is arbitrary, and nondecreasing sequences would do just as well.



The main reason why ‘types’ are introduced is a consequence of the following simple observation.

Lemma 4.5. *Let $\tau \in S_n$, and let $(a_1 \dots a_r)$ be a cycle. Then*

$$\tau(a_1 \dots a_r)\tau^{-1} = (a_1\tau^{-1} \dots a_r\tau^{-1}).$$

The funny notation $a_1\tau^{-1}$ stands for the action of the permutation τ^{-1} on $a_1 \in \{\mathbf{1}, \dots, \mathbf{n}\}$; recall that we agreed in §II.2.1 that we would let our permutations act *on the right*, for consistency with the usual notation for products in groups.

Proof. This is verified by checking that both sides act in the same way on $\{\mathbf{1}, \dots, \mathbf{n}\}$. For example, for $1 \leq i < r$

$$(a_i\tau^{-1})(\tau(a_1 \dots a_r)\tau^{-1}) = a_i(a_1 \dots a_r)\tau^{-1} = a_{i+1}\tau^{-1}$$

as it should; the other cases are left to the reader. $\square$

By the usual trick of judiciously inserting identity factors $\tau^{-1}\tau$, this formula for computing conjugates extends immediately to any product of cycles:

$$\tau(a_1 \dots a_r) \cdots (b_1 \dots b_s)\tau^{-1} = (a_1\tau^{-1} \dots a_r\tau^{-1}) \cdots (b_1\tau^{-1} \dots b_s\tau^{-1}).$$

This holds whether the cycles are disjoint or not. However, since τ is a bijection, disjoint cycles remain disjoint after conjugation. This is essentially all there is to the following important observation:

Proposition 4.6. *Two elements of S_n are conjugate in S_n if and only if they have the same type.*

Proof. The ‘only if’ part of this statement follows immediately from the preceding considerations: conjugating a permutation yields a permutation of the same type. As for the ‘if’ part, suppose

$$\sigma_1 = (a_1 \dots a_r)(b_1 \dots b_s) \cdots (c_1 \dots c_t)$$

and

$$\sigma_2 = (a'_1 \dots a'_r)(b'_1 \dots b'_s) \cdots (c'_1 \dots c'_t)$$

are two permutations with the same type, written in cycle notation, with $r \geq s \geq \dots \geq t$ (so the type is $[r, s, \dots, t]$). Let τ be any permutation such that $a_i = a'_i\tau$, $b_j = b'_j\tau$, $\dots$, $c_k = c'_k\tau$ for all $i, j, \dots, k$. Then Lemma 4.5 implies $\sigma_2 = \tau\sigma_1\tau^{-1}$, so σ_1 and σ_2 are conjugate, as needed. $\square$

Example 4.7. In S_8 ,

$$(18632)(47) \quad \text{and} \quad (12345)(67)$$

must be conjugate, since they have the same type. The proof of Proposition 4.6 tells us that

$$\tau(18632)(47)\tau^{-1} = (12345)(67)$$

for

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 8 & 6 & 3 & 2 & 4 & 7 & 5 \end{pmatrix},$$

and of course this may be checked by hand in a second. Running this check, and especially staring at the second row in τ *vis-à-vis* the cycle notation of the first permutation, should clarify everything. $\lrcorner$

Summarizing, the *type* (or the corresponding Young diagram) tells us everything about conjugation in S_n .

Corollary 4.8. *The number of conjugacy classes in S_n equals the number of partitions of n .*

For example, there are 7 conjugacy classes in S_5 , indexed by the Tetris look-alikes drawn above.

It is also reasonably straightforward to compute the number of elements in each conjugacy class, in terms of the type. For example, in order to count the number of permutations of type $[2, 2, 1]$ in S_5 , note that there are $5! = 120$ ways to fill the corresponding Young diagram with the numbers $1, \dots, 5$:

a_1	a_2
a_3	a_4
a_5	

that is, 120 ways to write a permutation as a product of two 2-cycles:

$$(a_1 a_2)(a_3 a_4);$$

but switching $a_1 \leftrightarrow a_2$ and $a_3 \leftrightarrow a_4$, as well as switching the two cycles, gives the same permutation. Therefore there are

$$\frac{120}{2 \cdot 2 \cdot 2} = 15$$

permutations of type $[2, 2, 1]$. Performing this computation for all Young diagrams for S_5 gives us the size of each conjugacy class, that is, the class formula (cf. §1.2) for S_5 :

$$120 = 1 + 10 + 15 + 20 + 20 + 30 + 24.$$

Example 4.9. There are no normal subgroups of size 30 in S_5 .

Indeed, normal subgroups are unions of conjugacy classes (§1.3); since the identity is in every subgroup and $30 - 1 = 29$ cannot be written as a sum of the numbers appearing in the class formula for S_5 , there is no such subgroup. $\lrcorner$

This observation will be dwarfed by much stronger results that we will prove soon (such as Theorem 4.20, Corollary 4.21); but it is remarkable that such precise statements can be established with so little work.

4.3. Transpositions, parity, and the alternating group. For $n \geq 1$, consider the polynomials

$$\Delta_n = \prod_{1 \leq i < j \leq n} (x_i - x_j) \in \mathbb{Z}[x_1, \dots, x_n],$$

that is,

$$\begin{aligned} \Delta_1 &= 1, \\ \Delta_2 &= x_1 - x_2, \\ \Delta_3 &= (x_1 - x_2)(x_1 - x_3)(x_2 - x_3), \\ \Delta_4 &= (x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_2 - x_3)(x_2 - x_4)(x_3 - x_4), \\ &\dots \end{aligned}$$

We can act with any $\sigma \in S_n$ on Δ_n , by permuting the indices according to σ :

$$\Delta_n \sigma := \prod_{1 \leq i < j \leq n} (x_{i\sigma} - x_{j\sigma}).$$

For example,

$$\Delta_4(1234) = (x_2 - x_3)(x_2 - x_4)(x_2 - x_1)(x_3 - x_4)(x_3 - x_1)(x_4 - x_1) = -\Delta_4.$$

In general, it is clear that $\Delta_n \sigma$ is still the product of all binomials $(x_i - x_j)$, where the factors are permuted and some factors may change sign in the process. Hence, $\Delta_n \sigma = \pm \Delta_n$, where the factor ± 1 depends on σ .

Definition 4.10. The *sign* of a permutation $\sigma \in S_n$, denoted $(-1)^\sigma$, is determined by the action of σ on Δ_n :

$$\Delta_n \sigma = (-1)^\sigma \Delta_n.$$

We say that a permutation is *even* if its sign is $+1$ and *odd* if its sign is -1 . □

Note that $\forall \sigma, \tau \in S_n$ we have

$$\Delta_n(\sigma\tau) = (\Delta_n \sigma)\tau :$$

it follows that $(-1)^{\sigma\tau} = (-1)^\sigma (-1)^\tau$. Viewing $\{-1, +1\}$ as a group under multiplication¹⁷, we see that the ‘sign’ function

$$\epsilon : S_n \rightarrow \{-1, +1\}, \quad \epsilon(\sigma) := (-1)^\sigma$$

is a *homomorphism*.

Here is a different viewpoint on this sign function. A *transposition* is a cycle of length 2. Every permutation is a product of transpositions:

Lemma 4.11. *Transpositions generate S_n .*

Proof. Indeed, by Lemma 4.3 it suffices to show that every *cycle* is a product of transpositions, and indeed

$$(a_1 \dots a_r) = (a_1 a_2)(a_1 a_3) \cdots (a_1 a_r),$$

as may be checked by applying¹⁸ both sides to every element of $\{1, \dots, n\}$. □

¹⁷This is the group of units in $\mathbb{Z}$; of course it is isomorphic to C_2 .

¹⁸Don’t forget that permutations act *on the right*.

Of course a given permutation may be written as a product of transpositions in many different ways. However, whether an *even* number of transpositions or an *odd* number is needed only depends on σ . Indeed,

Lemma 4.12. *Let $\sigma = \tau_1 \cdots \tau_r$ be a product of transpositions. Then σ is even, resp., odd, according to whether r is even, resp., odd.*

Proof. This follows immediately from the facts that ϵ is a homomorphism and the sign of a transposition is -1 : indeed, (ij) acts on Δ_n by permuting its factors and changing the sign of an odd number of factors (for $i < j$, the factor $(x_i - x_j)$ and the pairs of factors $(x_i - x_k)$, $(x_k - x_j)$ for all $i < k < j$). $\square$

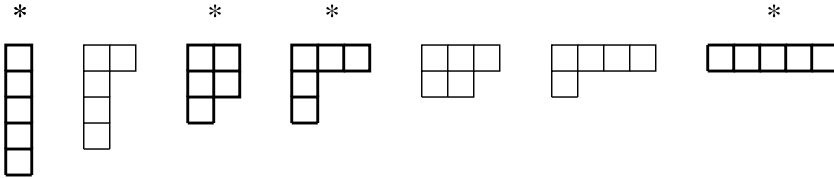
Definition 4.13. The *alternating group* on $\{1, \dots, n\}$, denoted A_n , consists of all *even* permutations $\sigma \in S_n$. $\lrcorner$

The alternating group is a *normal* subgroup of S_n , and

$$[S_n : A_n] = 2$$

for $n \geq 2$: indeed, $A_n = \ker \epsilon$, and ϵ is surjective for $n \geq 2$.

It is very easy to tell whether a permutation σ belongs to A_n in terms of the type of σ . Indeed (by the argument proving Lemma 4.11) a cycle is *even*, resp., *odd*, if it has *odd*, resp., *even*, length¹⁹. Pictorially, a permutation $\sigma \in S_n$ is even if and only if n and the number of rows in the Young diagram of σ have the same parity. Take $n = 5$ for example:



The starred types correspond to even permutations. Adding up the sizes of the corresponding conjugacy classes,

$$1 + 15 + 20 + 24 = 60,$$

confirms that A_5 has index 2 in S_5 .

However, do not read too much into this computation: I am not claiming that these are the sizes of the conjugacy classes *in* A_5 , only that these are the conjugacy classes *in* S_5 making up the normal subgroup A_5 . Conjugacy in A_n is rather interesting and ties in nicely with the issue of the *simplicity* of A_n .

4.4. Conjugacy in A_n ; simplicity of A_n and solvability of S_n . Denote by $[\sigma]_{S_n}$, resp., $[\sigma]_{A_n}$, the conjugacy class of an even permutation σ in S_n , resp., A_n . Clearly $[\sigma]_{A_n} \subseteq [\sigma]_{S_n}$; we proceed to compare these two sets.

Lemma 4.14. *Let $n \geq 2$, and let $\sigma \in A_n$. Then $[\sigma]_{A_n} = [\sigma]_{S_n}$ or the size of $[\sigma]_{A_n}$ is half the size of $[\sigma]_{S_n}$, according to whether the centralizer $Z_{S_n}(\sigma)$ is not or is contained in A_n .*

¹⁹Note the unfortunate terminology clash. For example, 3-cycles are *even* permutations.

Proof. (Cf. Exercise 1.16.) Note that

$$Z_{A_n}(\sigma) = A_n \cap Z_{S_n}(\sigma) :$$

this follows immediately from the definition of centralizer (Definition 1.6). Now recall that the centralizer of σ is its stabilizer under conjugation, and therefore the size of the conjugacy class of σ equals the index of its centralizer.

If $Z_{S_n}(\sigma) \subseteq A_n$, then $Z_{A_n}(\sigma) = Z_{S_n}(\sigma)$, so that

$$[S_n : Z_{S_n}(\sigma)] = [S_n : Z_{A_n}(\sigma)] = [S_n : A_n][A_n : Z_{A_n}(\sigma)] = 2 \cdot [A_n : Z_{A_n}(\sigma)];$$

therefore, $[\sigma]_{A_n}$ is half the size of $[\sigma]_{S_n}$ in this case.

If $Z_{S_n}(\sigma) \not\subseteq A_n$, then note that $A_n Z_{S_n}(\sigma) = S_n$: indeed, $A_n Z_{S_n}(\sigma)$ is a subgroup of S_n (because A_n is normal; cf. Proposition II.8.11), and it properly contains A_n , so it must equal S_n as A_n has index 2 in S_n . By index considerations (cf. Exercise II.8.21)

$$[A_n : Z_{A_n}(\sigma)] = [A_n : A_n \cap Z_{S_n}(\sigma)] = [A_n Z_{S_n}(\sigma) : Z_{S_n}(\sigma)] = [S_n : Z_{S_n}(\sigma)],$$

so the classes have the same size. Since $[\sigma]_{A_n} \subseteq [\sigma]_{S_n}$ in any case, it follows that $[\sigma]_{A_n} = [\sigma]_{S_n}$, completing the proof. $\square$

Therefore, conjugacy classes of even permutations either are preserved from S_n to A_n or they split into two distinct, equal-sized classes. We are now in a position to give precise conditions determining which happens when.

Proposition 4.15. *Let $\sigma \in A_n$, $n \geq 2$. Then the conjugacy class of σ in S_n splits into two conjugacy classes in A_n precisely if the type of σ consists of distinct odd numbers.*

Proof. By Lemma 4.14, we have to verify that $Z_{S_n}(\sigma)$ is contained in A_n precisely when the stated condition is satisfied; that is, we have to show that

$$\sigma = \tau \sigma \tau^{-1} \implies \tau \text{ is even}$$

precisely when the type of σ consists of distinct odd numbers.

Write σ in cycle notation (including cycles of length 1):

$$\sigma = (a_1 \dots a_\lambda)(b_1 \dots b_\mu) \cdots (c_1 \dots c_\nu),$$

and recall (Lemma 4.5) that

$$\tau \sigma \tau^{-1} = (a_1 \tau^{-1} \dots a_\lambda \tau^{-1})(b_1 \tau^{-1} \dots b_\mu \tau^{-1}) \cdots (c_1 \tau^{-1} \dots c_\nu \tau^{-1}).$$

Assume that $\lambda, \mu, \dots, \nu$ are odd and distinct. If $\tau \sigma \tau^{-1} = \sigma$, then conjugation by τ must preserve each cycle in σ , as all cycle lengths are distinct:

$$\tau(a_1 \dots a_\lambda) \tau^{-1} = (a_1 \dots a_\lambda), \quad \text{etc.},$$

that is,

$$(a_1 \tau^{-1} \dots a_\lambda \tau^{-1}) = (a_1 \dots a_\lambda), \quad \text{etc.}$$

This means that τ acts as a cyclic permutation on (e.g.) $a_1, \dots, a_\lambda$ and therefore in the same way as a power of $(a_1 \dots a_\lambda)$. It follows that

$$\tau = (a_1 \dots a_\lambda)^r (b_1 \dots b_\mu)^s \cdots (c_1 \dots c_\nu)^t$$

for suitable $r, s, \dots, t$. Since all cycles have odd lengths, each cycle is an even permutation; and τ must then be even as it is a product of even permutations. This proves that $Z_{S_n}(\sigma) \subseteq A_n$ if the stated condition holds.

Conversely, assume that the stated condition does *not* hold: that is, either some of the cycles in the cycle decomposition have even length or all have odd length but two of the cycles have the same length.

In the first case, let τ be an even-length cycle in the cycle decomposition of σ . Note that $\tau\sigma\tau^{-1} = \sigma$: indeed, τ commutes with itself and with all cycles in σ other than τ . Since τ has even length, then it is odd as a permutation: this shows that $Z_{S_n}(\sigma) \not\subseteq A_n$, as needed.

In the second case, without loss of generality assume $\lambda = \mu$, and consider the odd permutation

$$\tau = (a_1b_1)(a_2b_2)\cdots(a_\lambda b_\lambda) :$$

conjugating by τ simply interchanges the first two cycles in σ ; hence $\tau\sigma\tau^{-1} = \sigma$. As τ is odd, this again shows that $Z_{S_n}(\sigma) \not\subseteq A_n$, and we are done. $\square$

Example 4.16. Looking again at A_5 , we have noted in §4.3 that the types of the even permutations in S_5 are $[1, 1, 1, 1, 1]$, $[2, 2, 1]$, $[3, 1, 1]$, and $[5]$. By Proposition 4.15 the conjugacy classes corresponding to the first three types are preserved in A_5 , while the last one splits.

Therefore there are exactly 5 conjugacy classes in A_5 , and the class formula for A_5 is

$$60 = 1 + 15 + 20 + 12 + 12. \quad \lrcorner$$

Finally! We can now complete a circle of thought begun in the first section of this chapter. Along the way, the reader has hopefully checked that every simple group of order < 60 is commutative (Exercise 2.24); and we now see why 60 is special:

Corollary 4.17. *The alternating group A_5 is a simple noncommutative group of order 60.*

Proof. A normal subgroup of A_5 is necessarily the union of conjugacy classes, contains the identity, and has order equal to a divisor of 60 (by Lagrange's theorem). The divisors of 60 other than 1 and 60 are

$$2, 3, 4, 5, 6, 10, 12, 15, 20, 30;$$

counting the elements other than the identity would give one of

$$1, 2, 3, 4, 5, 9, 11, 14, 19, 29$$

as a sum of numbers $\neq 1$ from the class formula for A_5 . But this simply does not happen. $\square$

The reader will check that A_6 is simple, by the same method (Exercise 4.21). It is in fact the case that *all* groups A_n , $n \geq 5$, are simple (and noncommutative), implying that S_n is not solvable for $n \geq 5$, and these facts are rather important in view of applications to Galois theory (cf., e.g., Corollary VII.7.16). Note that A_2 is trivial, $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ is simple and abelian, and A_4 is *not* simple (Exercise 2.24).

The alternating group A_5 is also called the *icosahedral (rotation) group*: indeed, it is the group of symmetries of an icosahedron obtained through rigid motions. (Can the reader verify²⁰ this fact?)

The simplicity of A_n for $n \geq 5$ may be established by studying 3-cycles. First of all, it is natural to wonder whether every even permutation may be written as a product of 3-cycles, and this is indeed so:

Lemma 4.18. *The alternating group A_n is generated by 3-cycles.*

Proof. Since every even permutation is a product of an even number of 2-cycles, it suffices to show that every product of two 2-cycles may be written as product of 3-cycles. Therefore, consider a product

$$(ab)(cd)$$

with $a \neq b, c \neq d$. If $(ab) = (cd)$, then this product is the identity, and there is nothing to prove. If $\{a, b\}, \{c, d\}$ have exactly one element in common, then we may assume $c = a$ and observe

$$(ab)(ad) = (abd).$$

If $\{a, b\}, \{c, d\}$ are disjoint, then

$$(ab)(cd) = (abc)(adc),$$

and we are done. □

Now we can capitalize on our study of conjugacy in A_n :

Claim 4.19. *Let $n \geq 5$. If a normal subgroup of A_n contains a 3-cycle, then it contains all 3-cycles.*

Proof. Normal subgroups are unions of conjugacy classes, so we just need to verify that 3-cycles form a conjugacy class in A_n , for $n \geq 5$. But they do in S_n , and the type of a 3-cycle is $[3, 1, 1, \dots]$ for $n \geq 5$; hence the conjugacy class does not split in A_n , by Proposition 4.15. □

The general statement now follows easily by tying up loose ends:

Theorem 4.20. *The alternating group A_n is simple for $n \geq 5$.*

Proof. We have already checked this for $n = 5$, and the reader has checked it for $n = 6$. For $n > 6$, let N be a nontrivial normal subgroup of A_n ; we will show that necessarily $N = A_n$, by proving that N contains 3-cycles.

Let $\tau \in N$, $\tau \neq (1)$, and let $\sigma \in A_n$ be a 3-cycle. Since the center of A_n is trivial (Exercise 4.14) and 3-cycles generate A_n , we may assume that τ and σ do not commute, that is, the commutator

$$[\tau, \sigma] = \tau(\sigma\tau^{-1}\sigma^{-1}) = (\tau\sigma\tau^{-1})\sigma^{-1}$$

²⁰It is good practice to start with smaller examples: for instance, the tetrahedral rotation group is isomorphic to A_4 .

is not the identity. This element is in N (as is evident from the first expression, as N is normal) and is a product of two 3-cycles (as is evident from the second expression, since the conjugate of a 3-cycle is a 3-cycle).

Therefore, replacing τ by $[\tau, \sigma]$ if necessary, we may assume that $\tau \in N$ is a nonidentity permutation acting on ≤ 6 elements: that is, on a subset of a set $T \subseteq \{1, \dots, \mathbf{n}\}$ with $|T| = 6$. Now we may view A_6 as a subgroup of A_n , by letting it act on T . The subgroup $N \cap A_6$ of A_6 is then normal (because N is normal) and nontrivial (because $\tau \in N \cap A_6$ and $\tau \neq (1)$). Since A_6 is simple (Exercise 4.21), this implies $N \cap A_6 = A_6$. In particular, N contains 3-cycles.

By Claim 4.19, this implies that N contains *all* 3-cycles. By Lemma 4.18, it follows that $N = A_n$, as needed. $\square$

Corollary 4.21. *For $n \geq 5$, the group S_n is not solvable.*

Proof. Since A_n is simple, the sequence

$$S_n \supsetneq A_n \supsetneq \{(1)\}$$

is a composition series for S_n . It follows that the composition factors of S_n are $\mathbb{Z}/2\mathbb{Z}$ and A_n . By Proposition 3.11, S_n is not solvable. $\square$

In particular, S_5 is a nonsolvable group of order 120. This is in fact the smallest order of a nonsimple, nonsolvable group; cf. Exercise 3.16.

Exercises

4.1. $\triangleright$ Compute the number of elements in the conjugacy class of

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 1 & 2 & 7 & 5 & 3 & 4 & 6 \end{pmatrix}$$

in S_8 . [§4.1]

4.2. $\triangleright$ Suppose

$$(a_1 \dots a_r)(b_1 \dots b_s) \cdots (c_1 \dots c_t) = (d_1 \dots d_u)(e_1 \dots e_v) \cdots (f_1 \dots f_w)$$

are two products of disjoint cycles. Prove that the factors agree up to order.

(Hint: The two corresponding partitions of $\{1, \dots, \mathbf{n}\}$ must agree.) [§4.1]

4.3. Assume σ has type $[\lambda_1, \dots, \lambda_r]$ and that the λ_i 's are pairwise relatively prime. What is $|\sigma|$? What can you say about $|\sigma|$, without the additional hypothesis on the numbers λ_i ?

4.4. Make sense of the 'Taylor series' of the infinite product

$$\frac{1}{(1-x)} \cdot \frac{1}{(1-x^2)} \cdot \frac{1}{(1-x^3)} \cdot \frac{1}{(1-x^4)} \cdot \frac{1}{(1-x^5)} \cdots$$

Prove that the coefficient of x^n in this series is the number of partitions of n .

4.5. Find the class formula for S_n , $n \leq 6$.

4.6. Let N be a *normal* subgroup of S_4 . Prove that $|N| = 1, 4, 12$, or 24 .

4.7. $\triangleright$ Prove that S_n is generated by (12) and $(12 \dots n)$.

(Hint: It is enough to get all transpositions. What is the conjugate of (12) by $(12 \dots n)$?) [4.9, §VII.7.5]

4.8. $\neg$ For $n > 1$, prove that the subgroup H of S_n consisting of permutations fixing $\mathbf{1}$ is isomorphic to S_{n-1} . Prove that there are no proper subgroups of S_n properly containing H . [VII.7.17]

4.9. By Exercise 4.7, S_4 is generated by (12) and (1234) . Prove that (13) and (1234) generate a copy of D_8 in S_4 . Prove that every subgroup of S_4 of order 8 is conjugate to $\langle (13), (1234) \rangle$. Prove there are exactly 3 such subgroups. For all $n \geq 3$ prove that S_n contains a copy of the dihedral group D_{2n} , and find generators for it.

4.10. $\neg \bullet$ Prove that there are exactly $(n-1)!$ n -cycles in S_n .

$\bullet$ More generally, find a formula for the size of the conjugacy class of a permutation of given type in S_n . [4.11]

4.11. Let p be a prime integer. Compute the number of p -Sylow subgroups of S_p . (Use Exercise 4.10.) Use this result and Sylow's third theorem to prove again the 'only if' implication in Wilson's theorem (cf. Exercise II.4.16.)

4.12. $\triangleright$ A subgroup G of S_n is *transitive* if the induced action of G on $\{\mathbf{1}, \dots, \mathbf{n}\}$ is transitive.

- $\bullet$ Prove that if $G \subseteq S_n$ is transitive, then $|G|$ is a multiple of n .
- $\bullet$ List the transitive subgroups of S_3 .
- $\bullet$ Prove that the following subgroups of S_4 are all transitive:
 - $\langle (1234) \rangle \cong C_4$ and its conjugates,
 - $\langle (12)(34), (13)(24) \rangle \cong C_2 \times C_2$,
 - $\langle (12)(34), (1234) \rangle \cong D_8$ and its conjugates,
 - A_4 , and S_4 .

With a bit of stamina, you can prove that these are the *only* transitive subgroups of S_4 .

[§VII.7.5]

4.13. (If you know about determinants.) Prove that the sign of a permutation σ , as defined in Definition 4.10, equals the determinant of the matrix M_σ defined in Exercise II.2.1.

4.14. $\triangleright$ Prove that the center of A_n is trivial for $n \geq 4$. [§4.4]

4.15. Justify the 'pictorial' recipe given in §4.3 to decide whether a permutation is even.

4.16. The number of conjugacy classes in A_n , $n \geq 2$, is (allegedly)

$$1, 3, 4, 5, 7, 9, 14, 18, 24, 31, 43, \dots$$

Check the first several numbers in this list by finding the class formulas for the corresponding alternating groups.

4.17. ▷ • Find the class formula for A_4 .

- Use it to prove that A_4 has no subgroup of order 6. [§II.8.5]

4.18. For $n \geq 5$, let H be a proper subgroup of A_n . Prove that $[A_n : H] \geq n$. Prove that A_n does have a subgroup of index n for all $n \geq 3$.

4.19. Prove that for $n \geq 5$ there are no nontrivial actions of A_n on any set S with $|S| < n$. Construct²¹ a nontrivial action of A_4 on a set S , $|S| = 3$. Is there a nontrivial action of A_4 on a set S with $|S| = 2$?

4.20. − Find all fifteen elements of order 2 in A_5 , and prove that A_5 has exactly five 2-Sylow subgroups. [4.22]

4.21. ▷ Prove that A_6 is simple, by using its class formula (as is done for A_5 in the proof of Corollary 4.17). [§4.4]

4.22. − Verify that A_5 is the *only* simple group of order 60, up to isomorphism. (Hint: By Exercise 2.25, a simple group G of order 60 contains a subgroup of index 5. Use this fact to construct a homomorphism $G \rightarrow S_5$, and prove that the image of this homomorphism must be A_5 .) Note that A_5 has exactly five 2-Sylow subgroups; cf. Exercise 4.20. Thus, the other possibility contemplated in Exercise 2.25 does not occur. [2.25]

5. Products of groups

We already know that products exist in **Grp** (see §II.3.4); here we analyze this notion further and explore variations on the same theme, with an eye towards the question of determining the information needed to reconstruct a group from its composition factors.

5.1. The direct product. Recall from §II.3.4 that the (*direct*) *product* of two groups H, K is the group supported on the set $H \times K$, with operation defined componentwise. We have checked (Proposition II.3.4) that the direct product satisfies the universal property defining products in the category **Grp**.

There are situations in which the direct product of two subgroups N, H of a group G may be realized as a subgroup of G . Recall (Proposition II.8.11) that if one of the subgroups is *normal*, then the subset NH of G is in fact a subgroup of G . The relation between NH and $N \times H$ depends on how N and H intersect in G , so we take a look at this intersection.

The ‘commutator’ $[A, B]$ of two subsets A, B of G (see §3.3) is the subgroup generated by all commutators $[a, b]$ with $a \in A, b \in B$.

Lemma 5.1. *Let N, H be normal subgroups of a group G . Then*

$$[N, H] \subseteq N \cap H.$$

²¹You can think algebraically if you want; if you prefer geometry, visualize pairs of opposite sides on a tetrahedron.

Proof. It suffices to verify this on generators; that is, it suffices to check that

$$[n, h] = n(hn^{-1}h^{-1}) = (nhn^{-1})h^{-1} \in N \cap H$$

for all $n \in N$, $h \in H$. But the first expression and the normality of N show that $[n, h] \in N$; the second expression and the normality of H show that $[n, h] \in H$. $\square$

Corollary 5.2. *Let N, H be normal subgroups of a group G . Assume $N \cap H = \{e\}$. Then N, H commute with each other:*

$$(\forall n \in N) (\forall h \in H) \quad nh = hn.$$

Proof. By Lemma 5.1, $[N, H] = \{e\}$ if $N \cap H = \{e\}$; the result follows immediately. $\square$

In fact, under the same hypothesis more is true:

Proposition 5.3. *Let N, H be normal subgroups of a group G , such that $N \cap H = \{e\}$. Then $NH \cong N \times H$.*

Proof. Consider the function

$$\varphi : N \times H \rightarrow NH$$

defined by $\varphi(n, h) = nh$. Under the stated hypothesis, φ is a group homomorphism: indeed

$$\begin{aligned} \varphi((n_1, h_1) \cdot (n_2, h_2)) &= \varphi((n_1n_2, h_1h_2)) \\ &= n_1n_2h_1h_2 \\ &= n_1h_1n_2h_2 \end{aligned}$$

since N, H commute by Corollary 5.2

$$= \varphi((n_1, h_1)) \cdot \varphi((n_2, h_2)).$$

The homomorphism φ is surjective by definition of NH . To verify it is injective, consider its kernel:

$$\ker \varphi = \{(n, h) \in N \times H \mid nh = e\}.$$

If $nh = e$, then $n \in N$ and $n = h^{-1} \in H$; thus $n = e$ since $N \cap H = \{e\}$. Using the same token for h , we conclude $h = e$; hence (n, h) is the identity in $N \times H$, proving that φ is injective.

Thus φ is an isomorphism, as needed. $\square$

Remark 5.4. This result gives an alternative argument for the proof of Claim 2.16: if $|G| = pq$, with $p < q$ prime integers, and G contains normal subgroups H, K of order p, q , respectively (as is the case if $q \not\equiv 1 \pmod{p}$, by Sylow), then $H \cap K = \{e\}$ necessarily, and then Proposition 5.3 shows $HK \cong H \times K$. As $|HK| = |G| = pq$, this proves $G \cong H \times K \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$. Finally, $(1, 1)$ has order pq in this group, so G is cyclic, with the same conclusion we obtained in Claim 2.16. $\lrcorner$

5.2. Exact sequences of groups; extension problem. Of course, the hypothesis that both subgroups N , H are normal is necessary for the result of Proposition 5.3: for example, the permutations (123) and (12) generate subgroups N , H of S_3 meeting only at $\{e\}$, and N is normal in S_3 , but $S_3 = NH$ is *not* isomorphic to the direct product of N and H . It is natural to examine this more general situation.

Let N , H be subgroups of a group G , with N normal (but with no *a priori* assumptions on H) and such that $N \cap H = \{e\}$; assume $G = NH$. We are after a description of the structure of G in terms of the structure of N and H .

It is notationally convenient to use the language of *exact sequences*, introduced for modules in §III.7.1. A (short) exact sequence of groups is a sequence of groups and group homomorphisms

$$1 \longrightarrow N \xrightarrow{\varphi} G \xrightarrow{\psi} H \longrightarrow 1$$

where ψ is surjective and φ identifies N with $\ker \psi$. In other words (by the first isomorphism theorem), use φ to identify N with a subgroup of G ; then the sequence is exact if N is normal in G and ψ induces an isomorphism $G/N \rightarrow H$.

The reader should pause a moment and check that if G , N , H are *abelian*, then this notion matches precisely the notion of short exact sequence of abelian groups (i.e., $\mathbb{Z}$ -modules) from §III.7.1; a notational difference is that here the trivial group is denoted²² ‘1’ rather than ‘0’.

Of course there always is an exact sequence

$$1 \longrightarrow N \longrightarrow N \times H \longrightarrow H \longrightarrow 1 :$$

map $n \in N$ to (n, e_H) and $(n, h) \in N \times H$ to h . However, keep in mind that this is a very special case: to reiterate the example mentioned above, there also is an exact sequence

$$1 \longrightarrow C_3 \longrightarrow S_3 \longrightarrow C_2 \longrightarrow 1 ,$$

yet $S_3 \not\cong C_3 \times C_2$.

Definition 5.5. Let N , H be groups. A group G is an *extension of H by N* if there is an exact sequence of groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1 .$$

┘

The *extension problem* aims to describe all extensions of two given groups, up to isomorphism. For example, there are two extensions of C_2 by C_3 : namely $C_6 \cong C_3 \times C_2$ and S_3 ; we will soon be able to verify that, up to isomorphism, there are no other extensions.

The extension problem is the ‘second half’ of the classification problem: the first half consists of determining all simple groups, and the second half consists of figuring out how these can be put together to construct any group²³. For example,

²²This is not unreasonable, since groups are more often written ‘multiplicatively’ rather than additively, so the identity element is more likely to be denoted 1 rather than 0.

²³As mentioned earlier, the first half has been settled, although the complexity of the work leading to its solution justifies some skepticism concerning the absolute correctness of the proof. The status of the second half is, as far as I know, (even) murkier.

if

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq G_3 \supsetneq G_4 = \{e\}$$

is a composition series, with (simple) quotients $H_i = G_i/G_{i+1}$, then G is an *extension of H_0 by an extension of H_1 by an extension of H_2 by H_3* : knowing the composition factors of G and the extension process, it should in principle be possible to reconstruct G .

We are going to ‘solve’ the extension problem *in the particular case in which H is also a subgroup of G , intersecting N at $\{e\}$* .

Definition 5.6. An exact sequence of groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1$$

(or the corresponding extension) is said to *split* if H may be identified with a subgroup of G , so that $N \cap H = \{e\}$.

We encountered this terminology in §III.7.2 for modules, thus for *abelian* groups. Note that the notion examined there appears to be more restrictive than Definition 5.6, since it requires G to be isomorphic to a direct product $N \times H$. This apparent mismatch evaporates because of Proposition 5.3: in the abelian case, every split extension (according to Definition 5.6) is in fact a direct product.

Of course split extensions are anyway very special, since quotients of a group G are usually not isomorphic to subgroups of G , even in the abelian case (cf. Exercise 5.4).

Lemma 5.7. *Let N be a normal subgroup of a group G , and let H be a subgroup of G such that $G = NH$ and $N \cap H = \{e\}$. Then G is a split extension of H by N .*

Proof. We have to construct an exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1 ;$$

we let $N \rightarrow G$ be the inclusion map, and we prove that $G/N \cong H$. For this, consider the composition

$$\alpha : H \hookrightarrow G \twoheadrightarrow G/N.$$

Then α is surjective: indeed, since $G = NH$, $\forall g \in G$ we have $g = nh$ for some $n \in N$ and $h \in H$, and then

$$gN = nhN = h(h^{-1}nh)N = hN = \alpha(h).$$

Further, $\ker \alpha = \{h \in H \mid hN = N\} = N \cap H = \{e\}$; therefore α is also injective, as needed. $\square$

To recap, if in the situation of Lemma 5.7 we also require that H be *normal* in G , then G is necessarily isomorphic to the ‘trivial’ extension $N \times H$: this is what we have proved in Proposition 5.3. We are seeking to describe the extension ‘even if’ H is not normal in G .

5.3. Internal/semidirect products. The attentive reader should have noticed that the key to Proposition 5.3 is really Corollary 5.2: if both N and H are normal and $N \cap H = \{e\}$, then N and H commute with each other. This is what ultimately causes the extension NH to be trivial. Now, recall that as soon as N is normal, then every subgroup H of G acts on N by conjugation: in fact (cf. Exercise 1.21) conjugation determines a homomorphism

$$\gamma : H \rightarrow \text{Aut}_{\text{Grp}}(N), \quad h \mapsto \gamma_h.$$

(Explicitly, for $h \in H$ the automorphism $\gamma_h : N \rightarrow N$ acts by $\gamma_h(n) := hnh^{-1}$.) The subgroups H and N commute precisely when γ is *trivial*. Corollary 5.2 shows that if N and H are both normal and $N \cap H = \{e\}$, then γ is indeed trivial.

This is the crucial remark. The next several considerations may be summarized as follows: if N is normal in G , H is a subgroup of G , $N \cap H = \{e\}$ and $G = NH$, then the extension G of H by N may be reconstructed from the conjugation action $\gamma : H \rightarrow \text{Aut}_{\text{Grp}}(N)$. The reader is advised to stare at the following triviality, which is the motivating observation for the general discussion:

$$(*) \quad (\forall n_1, n_2 \in N), (\forall h_1, h_2 \in H) \quad n_1 h_1 n_2 h_2 = (n_1 (h_1 n_2 h_1^{-1})) (h_1 h_2).$$

This says that if we know the conjugation action of H on N , then we can recover the operation in G from this information and from the operations in N and H .

Here is the general discussion. It is natural to abstract the situation and begin with *any two groups N , H and an arbitrary homomorphism*²⁴

$$\theta : H \rightarrow \text{Aut}_{\text{Grp}}(N), \quad h \mapsto \theta_h.$$

Define an operation $\bullet_\theta$ on the set $N \times H$ as follows: for $n_1, n_2 \in N$ and $h_1, h_2 \in H$, let

$$(n_1, h_1) \bullet_\theta (n_2, h_2) := (n_1 \theta_{h_1}(n_2), h_1 h_2).$$

This will look more reasonable once it is compared with $(*)$!

Lemma 5.8. *The resulting structure $(N \times H, \bullet_\theta)$ is a group, with identity element (e_N, e_H) .*

Proof. The reader should carefully verify this. For example, inverses exist because $(n_1, h_1) \bullet_\theta (\theta_{h_1}^{-1}(n_1^{-1}), h_1^{-1}) = (n_1 \theta_{h_1}(\theta_{h_1}^{-1}(n_1^{-1})), h_1 h_1^{-1}) = (n_1 n_1^{-1}, e_H) = (e_N, e_H)$ and similarly in the reverse order. $\square$

Definition 5.9. The group $(N \times H, \bullet_\theta)$ is a *semidirect product* of N and H and is denoted by $N \rtimes_\theta H$. $\lrcorner$

For example, the ordinary direct product is a semidirect product and corresponds to $\theta =$ the trivial map. If the reader feels a little uneasy about giving one name (*semidirect product*) for a whole host of different groups supported on the Cartesian product, welcome to the club. In fact, it gets worse still: it is not

²⁴The reason why I am not denoting the image of h by θ as $\theta(h)$ is that this is an automorphism of N and I dislike the notation $\theta(h)(n)$ for the image of $n \in N$ obtained by applying the automorphism corresponding to h . The alternative $\theta_h(n)$ looks a little easier to parse.

uncommon to omit ‘ θ ’ from the notation and simply write $N \rtimes H$ for a semidirect product²⁵.

In any case, the notation $\bullet_\theta$ is too heavy to carry around, so we generally revert back to the usual simple juxtaposition of elements in order to denote multiplication in $N \rtimes_\theta H$. The following proposition checks that semidirect products are split extensions:

Proposition 5.10. *Let N, H be groups, and let $\theta : H \rightarrow \text{Aut}_{\text{Grp}}(N)$ be a homomorphism; let $G = N \rtimes_\theta H$ be the corresponding semidirect product. Then*

- G contains isomorphic copies of N and H ;
- the natural projection $G \rightarrow H$ is a surjective homomorphism, with kernel N ; thus N is normal in G , and the sequence

$$1 \longrightarrow N \longrightarrow N \rtimes_\theta H \longrightarrow H \longrightarrow 1$$

is (split) exact;

- $N \cap H = \{e_G\}$;
- $G = NH$;
- the homomorphism θ is realized by conjugation in G : that is, for $h \in H$ and $n \in N$ we have

$$\theta_h(n) = hnh^{-1}$$

in G .

Proof. The functions $N \rightarrow G, H \rightarrow G$ defined for $n \in N, h \in H$ by

$$n \mapsto (n, e_H), \quad h \mapsto (e_N, h)$$

are manifestly injective homomorphisms, allowing us to identify N, H with the corresponding subgroups of G . It is clear that $N \cap H = \{(e_N, e_H)\} = \{e_G\}$, and

$$(n, e_H) \bullet_\theta (e_N, h) = (n, h)$$

shows that $G = NH$.

The projection $G \rightarrow H$ defined by

$$(n, h) \mapsto h$$

is a surjective homomorphism, with kernel N ; therefore N is normal in G . Finally,

$$(e_N, h) \bullet_\theta (n, e_H) \bullet_\theta (e_N, h)^{-1} = (\theta_h(n), h) \bullet_\theta (e_N, h^{-1}) = (\theta_h(n), e_H),$$

as claimed in the last point. □

Our original goal of ‘reconstructing’ a given split extension of a group H by a group N is a sort of converse to this proposition. More precisely,

²⁵This is actually OK, if N and H are given as subgroups of a common group G , in which case the implicit action is just conjugation.

Proposition 5.11. *Let N, H be subgroups of a group G , with N normal in G . Assume that $N \cap H = \{e\}$, and $G = NH$. Let $\gamma : H \rightarrow \text{Aut}_{\text{Grp}}(N)$ be defined by conjugation: for $h \in H, n \in N$,*

$$\gamma_h(n) = hnh^{-1}.$$

Then $G \cong N \rtimes_{\gamma} H$.

Proof. Define a function

$$\varphi : N \rtimes_{\gamma} H \rightarrow G$$

by $\varphi(n, h) = nh$; this is clearly a bijection. We need to verify that φ is a homomorphism, and indeed $(\forall n_1, n_2 \in N), (\forall h_1, h_2 \in H)$:

$$\begin{aligned} \varphi((n_1, h_1) \bullet_{\gamma} (n_2, h_2)) &= \varphi((n_1 \gamma_{h_1}(n_2), h_1 h_2)) \\ &= \varphi((n_1 (h_1 n_2 h_1^{-1}), h_1 h_2)) \\ &= n_1 h_1 n_2 (h_1^{-1} h_1) h_2 = (n_1 h_1) (n_2 h_2) \\ &= \varphi((n_1, h_1)) \varphi((n_2, h_2)) \end{aligned}$$

as needed. $\square$

When realized ‘within’ a group, as in the previous proposition, the semidirect product is sometime called an *internal* product.

Remark 5.12. If N and H commute, then the conjugation action of H on N is trivial; therefore γ is the trivial map, and the semidirect product $N \rtimes_{\gamma} H$ is the direct product $N \times H$. Thus, Proposition 5.11 recovers the result of Proposition 5.3 in this case. $\lrcorner$

Example 5.13. The automorphism group of C_3 is isomorphic to the cyclic group C_2 : if $C_3 = \{e, y, y^2\}$, then the two automorphisms of C_3 are

$$\text{id} : \begin{cases} e \mapsto e, \\ y \mapsto y, \\ y^2 \mapsto y^2, \end{cases} \quad \sigma : \begin{cases} e \mapsto e, \\ y \mapsto y^2, \\ y^2 \mapsto y. \end{cases}$$

Therefore, there are two homomorphisms $C_2 \rightarrow \text{Aut}_{\text{Grp}}(C_3)$: the trivial map, and the isomorphism sending the identity to id and the nonidentity element to σ . The semidirect product corresponding to the trivial map is the direct product $C_3 \times C_2 \cong C_6$; the other semidirect product $C_3 \rtimes C_2$ is isomorphic to S_3 . This can of course be checked by hand (and you should do it, for fun); but it also follows immediately from Proposition 5.11, since $N = \langle (123) \rangle, H = \langle (12) \rangle \subseteq S_3$ satisfy the hypotheses of this result. $\lrcorner$

The reader should contemplate carefully the slightly more general case of dihedral groups (Exercise 5.11); this enriches (and in a sense explains) the discussion presented in Claim 2.17.

In fact, semidirect products shed light on all groups of order pq , for $p < q$ primes; the reader should be able to complete the classification of these groups begun in §2.5.2 and show that if $q \equiv 1 \pmod{p}$, then there is exactly one such non-commutative group up to isomorphism (Exercise 5.12).

The reader would in fact be well-advised to try to use semidirect products to classify groups of small order: if a nontrivial normal subgroup N is found (typically by applying Sylow's theorems), with some luck the classification is reduced to the study of possible homomorphisms from known groups to $\text{Aut}_{\text{Grp}}(N)$ and can be carried out. See Exercise 5.15 for a further illustration of this technique.

Exercises

5.1. $\triangleright$ Let G be a finite group, and let $P_1, \dots, P_r$ be its nontrivial Sylow subgroups. Assume all P_i are normal in G .

- Prove that $G \cong P_1 \times \cdots \times P_r$. (Induction on r ; use Proposition 5.3.)
- Prove that G is nilpotent. (Hint: Mod out by the center, and work by induction on $|G|$. What is the center of a direct product of groups?)

Together with Exercise 3.10, this shows that a finite group is nilpotent if and only if each of its Sylow subgroups is normal. [3.12, §6.1]

5.2. Let G be an extension of H by N . Prove that the composition factors of G are the collection of the composition factors of H and those of N .

5.3. Let

$$G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_r = \{e\}$$

be a normal series. Show how to 'connect' $\{e\}$ to G by means of r exact sequences of groups using the groups G_i and the quotients $H_i = G_i/G_{i+1}$.

5.4. $\triangleright$ Prove that the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

is exact but does not split. [§5.2]

5.5. In Proposition III.7.5 we have seen that if an exact sequence

$$0 \longrightarrow M \xrightarrow{\varphi} N \longrightarrow N/(\varphi(M)) \longrightarrow 0$$

of *abelian* groups splits, then φ has a left-inverse. Is this necessarily the case for split sequences of *groups*?

5.6. Prove Lemma 5.8.

5.7. Let N be a group, and let $\alpha : N \rightarrow N$ be an automorphism of N . Prove that α may be realized as conjugation, in the sense that there exists a group G containing N as a normal subgroup and such that $\alpha(n) = gng^{-1}$ for some $g \in G$.

5.8. Prove that any semidirect product of two solvable groups is solvable. Show that semidirect products of nilpotent groups need not be nilpotent.

5.9. $\triangleright$ Prove that if $G = N \rtimes H$ is commutative, then $G \cong N \times H$. [§6.1]

5.10. Let N be a normal subgroup of a finite group G , and assume that $|N|$ and $|G/N|$ are relatively prime. Assume there is a subgroup H in G such that $|H| = |G/N|$. Prove that G is a semidirect product of N and H .

5.11. $\triangleright$ For all $n > 0$ express D_{2n} as a semidirect product $C_n \rtimes_{\theta} C_2$, finding θ explicitly. [§5.3]

5.12. $\triangleright$ Classify groups G of order pq , with $p < q$ prime: show that if $|G| = pq$, then either G is cyclic or $q \equiv 1 \pmod{p}$ and there is exactly one isomorphism class of noncommutative groups of order pq in this case. (You will likely have to use the fact that $\text{Aut}_{\text{Grp}}(C_q) \cong C_{q-1}$ if q is prime; cf. Exercise II.4.15.) [§2.5, §5.3]

5.13. $\neg$ Let $G = N \rtimes_{\theta} H$ be a semidirect product, and let K be the subgroup of G corresponding to $\ker \theta \subseteq H$. Prove that K is the kernel of the action of G on the set G/H of left-cosets of H . [5.14]

5.14. Recall that $S_3 \cong \text{Aut}_{\text{Grp}}(C_2 \times C_2)$ (Exercise II.4.13). Let ι be this isomorphism. Prove that $(C_2 \times C_2) \rtimes_{\iota} S_3 \cong S_4$. (Hint: Exercise 5.13.)

5.15. $\triangleright$ Let G be a group of order 28.

- Prove that G contains a normal subgroup N of order 7.
- Recall (or prove again) that, up to isomorphism, the only groups of order 4 are C_4 and $C_2 \times C_2$. Prove that there are two homomorphisms $C_4 \rightarrow \text{Aut}_{\text{Grp}}(N)$ and two homomorphisms $C_2 \times C_2 \rightarrow \text{Aut}_{\text{Grp}}(N)$ up to the choice of generators for the sources.
- Conclude that there are four groups of order 28 up to isomorphism: the two direct products $C_4 \times C_7$, $C_2 \times C_2 \times C_7$, and two noncommutative groups.
- Prove that $D_{28} \cong C_2 \times D_{14}$. The other noncommutative group of order 28 is a *generalized quaternionic group*.

[§5.3]

5.16. Prove that the quaternionic group Q_8 (cf. Exercise III.1.12) cannot be written as a semidirect product of two nontrivial subgroups.

5.17. Prove that the multiplicative group $\mathbb{H}^*$ of nonzero quaternions (cf. Exercise III.1.12) is isomorphic to a semidirect product $\text{SU}(2) \rtimes \mathbb{R}^+$. (Hint: Exercise III.2.5.) Is this semidirect product in fact direct?

6. Finite abelian groups

I will end this chapter by treating in some detail the classification theorem for finite abelian groups mentioned in §II.6.3.

6.1. Classification of finite abelian groups. Now that we have acquired more familiarity with products, we are in a position to classify all finite *abelian* groups²⁶.

²⁶Of course fancier *semidirect* products will not be needed here; cf. Exercise 5.9.

In due time (Proposition VI.2.11, Exercise VI.2.19) we will in fact be able to classify all *finitely generated* abelian groups: as mentioned in Example II.6.3, all such groups are products of cyclic groups²⁷. In particular, this is the case for finite abelian groups: this is what we prove in this section.

Since in this section we exclusively deal with abelian groups, we revert to the abelian style of notations: thus the operation will be denoted $+$; the identity will be 0 ; direct *products* will be called direct *sums* (and denoted $\oplus$); and so on.

First of all, I will congeal into an explicit statement a simple observation that has been with us in one form or another since at least as far back as²⁸ Exercise II.4.9.

Lemma 6.1. *Let G be an abelian group, and let H, K be subgroups such that $|H|, |K|$ are relatively prime. Then $H + K \cong H \oplus K$.*

Proof. By Lagrange's theorem (Corollary II.8.14), $H \cap K = \{0\}$. Since subgroups of abelian groups are automatically normal, the statement follows from Proposition 5.3. $\square$

Now let G be a *finite* abelian group. For each prime p , the p -Sylow subgroup of G is unique, since it is automatically normal in G . Since the distinct nontrivial Sylow subgroups of G are p -groups for different primes p , Lemma 6.1 immediately implies the following result.

Corollary 6.2. *Every finite abelian group is the direct sum of its nontrivial Sylow subgroups.*

(The diligent reader knew already that this had to be the case, since abelian groups are nilpotent; cf. Exercise 5.1.) Thus, we already know that every finite abelian group is a direct sum of p -groups, and our main task amounts to classifying *abelian* p -groups for a fixed prime p . This is somewhat technical; we will get there by a seemingly roundabout path.

Lemma 6.3. *Let G be an abelian p -group, and let $g \in G$ be an element of maximal order. Then the exact sequence*

$$0 \longrightarrow \langle g \rangle \longrightarrow G \longrightarrow G/\langle g \rangle \longrightarrow 0$$

splits.

Put otherwise, there is a subgroup L of G such that L maps isomorphically to $G/\langle g \rangle$ via the canonical projection, that is, such that $\langle g \rangle \cap L = \{0\}$ and $\langle g \rangle + L = G$. Note that it will follow that $G \cong \langle g \rangle \oplus L$, by Proposition 5.3.

The main technicality needed in order to prove this lemma is the following particular case:

Lemma 6.4. *Let p be a prime integer and $r \geq 1$. Let G be a noncyclic abelian group of order p^{r+1} , and let $g \in G$ be an element of order p^r . Then there exists an element $h \in G$, $h \notin \langle g \rangle$, such that $|h| = p$.*

²⁷The natural context to prove this more general result is that of modules over Euclidean rings or even principal ideal domains.

²⁸In fact, this observation will really find its most appropriate resting place when we prove the *Chinese Remainder Theorem*, Theorem V.6.1.

Lemma 6.4 is a special case of Lemma 6.3 in the sense that, with notation as in the statement, necessarily $\langle h \rangle \cong G/\langle g \rangle$, and in fact $G \cong \langle g \rangle \oplus \langle h \rangle$ (and the reader is warmly encouraged to understand this before proceeding!). That is, we can split off the ‘large’ cyclic subgroup $\langle g \rangle$ as a direct summand of G , provided that G is not cyclic and not much larger than $\langle g \rangle$. Lemma 6.3 claims that this can be done whenever $\langle g \rangle$ is a maximal cyclic subgroup of G . We will be able to prove this more general statement easily once the particular case is settled.

Proof of Lemma 6.4. Denote $\langle g \rangle$ by K , and let h' be *any* element of G , $h' \notin K$. The subgroup K is normal in G since G is abelian; the quotient group G/K has order p . Since $h' \notin K$, the coset $h' + K$ has order p in G/K ; that is, $ph' \in K$. Let $k = ph'$.

Note that $|k|$ divides p^r ; hence it is a power of p . Also $|k| \neq p^r$, otherwise $|h'| = p^{r+1}$ and G would be cyclic, contrary to the hypothesis.

Therefore $|k| = p^s$ for some $s < r$; k generates a subgroup $\langle k \rangle$ of the cyclic group K , of order p^s . By Proposition II.6.11, $\langle k \rangle = \langle p^{r-s}g \rangle$. Since $s < r$, $\langle k \rangle \subseteq \langle pg \rangle$; thus, $k = mpg$ for some $m \in \mathbb{Z}$.

Then let $h = h' - mg$: $h \neq 0$ (since $h' \notin K$), and

$$ph = ph' - p(mg) = k - k = 0,$$

showing that $|h| = p$, as stated. $\square$

Proof of Lemma 6.3. Argue by induction on the order of G ; the case $|G| = p^0 = 1$ requires no proof. Thus we will assume that G is nontrivial and that the statement is true for every p -group smaller than G .

Let $g \in G$ be an element of maximal order, say p^r , and denote by K the subgroup $\langle g \rangle$ generated by g ; this subgroup is normal, as G is abelian. If $G = K$, then the statement holds trivially. If not, G/K is a nontrivial p -group, and hence it contains an element of order p by Cauchy’s theorem (Theorem 2.1). This element generates a subgroup of order p in G/K , corresponding to a subgroup G' of G of order p^{r+1} , containing K . This subgroup is not cyclic (otherwise the order of g is not maximal).

That is, we are in the situation of Lemma 6.4: hence we can conclude that there is an element $h \in G'$ (and hence $h \in G$) with $h \notin K$ and $|h| = p$. Let $H = \langle h \rangle \subseteq G$ be the subgroup generated by h , and note that $K \cap H = \{0\}$.

Now work modulo H . The quotient group G/H has smaller size than G , and $g + H$ generates a cyclic subgroup $K' = (K + H)/H \cong K/(K \cap H) \cong K$ of maximal order in G/H . By the induction hypothesis, there is a subgroup L' of G/H such that $K' + L' = G/H$ and $K' \cap L' = \{0_{G/H}\}$. This subgroup L' corresponds to a subgroup L of G containing H .

Now I claim that (i) $K + L = G$ and (ii) $K \cap L = \{0\}$. Indeed, we have the following:

- (i) For any $a \in G$, there exist $mg + H \in K'$, $\ell + H \in L'$ such that $a + H = mg + \ell + H$ (since $K' + L' = G/H$). This implies $a - mg \in L$, and hence $a \in K + L$ as needed.

- (ii) If $a \in K \cap L$, then $a + H \in K' \cap L' = \{0_{G/H}\}$, and hence $a \in H$. In particular, $a \in K \cap H = \{0\}$, forcing $a = 0$, as needed.
- (i) and (ii) imply the lemma, as observed in the comments following the statement. $\square$

Now we are ready to state the classification theorem; the proof is quite straightforward after all this preparation work. We first give the statement in a somewhat coarse form, as a corollary of the previous considerations:

Corollary 6.5. *Let G be a finite abelian group. Then G is a direct sum of cyclic groups, which may be assumed to be cyclic p -groups.*

Proof. As noted in Corollary 6.2, G is a direct sum of p -groups (as a consequence of the Sylow theorems). I claim that every abelian p -group P is a direct sum of cyclic p -groups.

To establish this, argue by induction on $|P|$. There is nothing to prove if P is trivial. If P is not trivial, let g be an element of P of maximal order. By Lemma 6.3

$$P = \langle g \rangle \oplus P'$$

for some subgroup P' of P ; by the induction hypothesis P' is a direct sum of cyclic p -groups, concluding the proof. $\square$

6.2. Invariant factors and elementary divisors. Here is a more precise version of the classification theorem. It is common to state the result in two equivalent forms.

Theorem 6.6. *Let G be a finite nontrivial abelian group. Then*

- *there exist prime integers $p_1, \dots, p_r$ and positive integers n_{ij} such that $|G| = \prod_{i,j} p_i^{n_{i,j}}$ and*

$$G \cong \bigoplus_{i,j} \frac{\mathbb{Z}}{p_i^{n_{i,j}} \mathbb{Z}};$$

- *there exist positive integers $1 < d_1 \mid \dots \mid d_s$ such that $|G| = d_1 \cdots d_s$ and*

$$G \cong \frac{\mathbb{Z}}{d_1 \mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{d_s \mathbb{Z}}.$$

Further, these decompositions are uniquely determined by G .

The first form is nothing but a more explicit version of the statement of Corollary 6.5, so it has already been proven. I will explain how to obtain the second form from the first. The uniqueness statement²⁹ is left to the reader (Exercise 6.1).

The prime powers appearing in the first form of Theorem 6.6 are called the *elementary divisors* of G ; the integers d_i appearing in the second form are called *invariant factors*. To go from elementary divisors to invariant factors, collect the

²⁹Of course the ‘uniqueness’ statement only holds up to trivial manipulation such as a permutation of the factors. The claim is that the factors themselves are determined by G , in the sense that two direct sums of either form given in the statement are isomorphic only if their factors match.

elementary divisors in a table, listing (for example) prime powers according to increasing primes in the horizontal direction and decreasing exponents in the vertical direction; then the invariant factors are obtained as products of the factors in each row:

$d_r =$	$p_1^{n_{11}}$	$p_2^{n_{21}}$	$p_3^{n_{31}}$	$\dots$
$d_{r-1} =$	$p_1^{n_{12}}$	$p_2^{n_{22}}$	$p_3^{n_{32}}$	$\dots$
$d_{r-2} =$	$p_1^{n_{13}}$	$p_2^{n_{23}}$	$p_3^{n_{33}}$	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

Conversely, given the invariant factors d_i , obtain the rows of this table by factoring d_i into prime powers: the condition $d_1 \mid \dots \mid d_r$ guarantees that these will be decreasing.

Repeated applications of Lemma 6.1 show that if $d = p_1^{n_1} \dots p_r^{n_r}$ for *distinct* primes p_i and positive n_i (as is the case in each row of the table), then

$$\frac{\mathbb{Z}}{d\mathbb{Z}} \cong \frac{\mathbb{Z}}{p_1^{n_1}\mathbb{Z}} \oplus \dots \oplus \frac{\mathbb{Z}}{p_r^{n_r}\mathbb{Z}},$$

proving that the two decompositions given in Theorem 6.6 are indeed equivalent.

This will likely be much clearer once the reader works through a few examples.

Example 6.7. Here are the two decompositions for a (random) group of order $29160 = 2^3 \cdot 3^6 \cdot 5$:

$$\left(\frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3^2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3^2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{5\mathbb{Z}} \right) \cong \left(\frac{\mathbb{Z}}{3\mathbb{Z}} \oplus \frac{\mathbb{Z}}{6\mathbb{Z}} \oplus \frac{\mathbb{Z}}{18\mathbb{Z}} \oplus \frac{\mathbb{Z}}{90\mathbb{Z}} \right)$$

and here is the corresponding table of invariant factors/elementary divisors:

$90 =$	2	3^2	5
$18 =$	2	3^2	
$6 =$	2	3	
$3 =$		3	

Example 6.8. There are exactly 6 isomorphism classes of abelian groups of order 360. Indeed, $360 = 2^3 \cdot 3^2 \cdot 5$; the six possible tables of elementary divisors are shown below. In terms of invariant factors, the six distinct abelian groups of order 360 (up to isomorphism, by the uniqueness part of Theorem 6.6) are therefore

$$\begin{array}{ccc} \frac{\mathbb{Z}}{360\mathbb{Z}}, & \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{180\mathbb{Z}}, & \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{90\mathbb{Z}}, \\ \frac{\mathbb{Z}}{3\mathbb{Z}} \oplus \frac{\mathbb{Z}}{120\mathbb{Z}}, & \frac{\mathbb{Z}}{6\mathbb{Z}} \oplus \frac{\mathbb{Z}}{60\mathbb{Z}}, & \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{6\mathbb{Z}} \oplus \frac{\mathbb{Z}}{30\mathbb{Z}}. \end{array}$$

360=	2 ³	3 ²	5

180=	2 ²	3 ²	5
2=	2		

90=	2	3 ²	5
2=	2		
2=	2		

120=	2 ³	3	5
3=		3	

60=	2 ²	3	5
6=	2	3	

30=	2	3	5
6=	2	3	
2=	2		

6.3. Application: Finite subgroups of multiplicative groups of fields. Any classification theorem is useful in that it potentially reduces the proof of general facts to explicit verifications. Here is one example illustrating this strategy:

Lemma 6.9. *Let G be a finite abelian group, and assume that for every integer $n > 0$ the number of elements $g \in G$ such that $ng = 0$ is at most n . Then G is cyclic.*

The reader should try to prove this ‘by hand’, to appreciate the fact that it is not entirely trivial. It does become essentially immediate once we take the classification of finite abelian groups into account. Indeed, by Theorem 6.6

$$G \cong \frac{\mathbb{Z}}{d_1\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{d_s\mathbb{Z}}$$

for some positive integers $1 < d_1 \mid \cdots \mid d_s$. But if $s > 1$, then $|G| > d_s$ and $d_s g = 0$ for all $g \in G$ (so that the order of g divides d_s), contradicting the hypothesis. Therefore $s = 1$; that is, G is cyclic.

Lemma 6.9 is the key to a particularly nice proof of the following important fact, a weak form of which³⁰ we ran across back in Example II.4.6. Recall that the set F^* of nonzero elements of a field F is a commutative group under multiplication. Also recall (Example III.4.7) that a polynomial $f(x) \in F[x]$ is divisible by $(x - a)$ if and only if $f(a) = 0$; since a nonzero polynomial of degree n over a field can have at most n linear factors, this shows³¹ that if $f(x) \in F[x]$ has degree n , then $f(a) = 0$ for at most n distinct elements $a \in F$.

Theorem 6.10. *Let F be a field, and let G be a finite subgroup of the multiplicative group $(F^*, \cdot)$. Then G is cyclic.*

Proof. By the considerations preceding the statement, for every n there are at most n elements $a \in F$ such that $a^n - 1 = 0$, that is, at most n elements $a \in G$ such that $a^n = 1$. Lemma 6.9 implies then that G is cyclic. $\square$

³⁰The diligent reader has proved that particular case in Exercise II.4.11. The proof hinted at in that exercise upgrades easily to the general case presented here. The point is not that the classification theorem is *necessary* in order to prove statements such as Theorem 6.10; the point is that it makes such statements nearly evident.

³¹Unique factorization in $F[x]$ is secretly needed here. We will deal with this issue more formally later; cf. Lemma V.5.1.

As a (very) particular case, the multiplicative group $((\mathbb{Z}/p\mathbb{Z})^*, \cdot)$ is cyclic: this is the fact pointed to in Example II.4.6.

Preview of coming attractions: Finitely generated (as opposed to just finite) abelian groups are also direct sums of cyclic groups. The only difference between the classification of finitely generated abelian groups and the classification of finite abelian groups explored here is the possible presence of a ‘free’ factor $\mathbb{Z}^{\oplus r}$ in the decomposition. The reader will first prove this fact in Exercise VI.2.19, as a consequence of ‘Gaussian elimination over integral domains’, and then recover it again as a particular case of the classification theorem for finitely generated modules over PIDs, Theorem VI.5.6. Neither Gaussian elimination nor the very general Theorem VI.5.6 are any harder to prove than the particular case of *finite abelian groups* laboriously worked out by hand in this section—a common benefit of finding the right general point of view is that, as a rule, proofs simplify. Technical work such as that performed in order to prove Lemma 6.3 is absorbed into the work necessary to build up the more general apparatus; the need for such technicalities evaporates in the process.

Exercises

6.1. ▷ Prove that the decomposition of a finite abelian group G as a direct sum of cyclic p -groups is unique. (Hint: The prime factorization of $|G|$ determines the primes, so it suffices to show that if

$$\frac{\mathbb{Z}}{p^{r_1}\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{p^{r_m}\mathbb{Z}} \cong \frac{\mathbb{Z}}{p^{s_1}\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{p^{s_n}\mathbb{Z}},$$

with $r_1 \geq \cdots \geq r_m$ and $s_1 \geq \cdots \geq s_n$, then $m = n$ and $r_i = s_i$ for all i . Do this by induction, by considering the group pG obtained as the image of the homomorphism $G \rightarrow G$ defined by $g \mapsto pg$.) [§6.2, §VI.5.3, VI.5.12]

6.2. Complete the classification of groups of order 8 (cf. Exercise 2.16).

6.3. Let G be a noncommutative group of order p^3 , where p is a prime integer. Prove that $Z(G) \cong \mathbb{Z}/p\mathbb{Z}$ and $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

6.4. Classify abelian groups of order 400.

6.5. Let p be a prime integer. Prove that the number of distinct isomorphism classes of abelian groups of order p^r equals the number of partitions of the integer r .

6.6. ▷ How many *abelian* groups of order 1024 are there, up to isomorphism? [§II.6.3]

6.7. ¬ Let $p > 0$ be a prime integer, G a finite abelian group, and denote by $\rho: G \rightarrow G$ the homomorphism defined by $\rho(g) = pg$.

- Let A be a finite abelian group such that $pA = 0$. Prove that $A \cong \mathbb{Z}/p\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p\mathbb{Z}$.
- Prove that $p \ker \rho$ and $p(\operatorname{coker} \rho)$ are both 0.

- Prove that $\ker \rho \cong \operatorname{coker} \rho$.
- Prove that every subgroup of G of order p is contained in $\ker \rho$ and that every subgroup of G of index p contains $\operatorname{im} \rho$.
- Prove that the number of subgroups of G of order p equals the number of subgroups of G of index p .

[6.8]

6.8. $\neg$ Let G be a finite abelian p -group, with elementary divisors $p^{n_1}, \dots, p^{n_r}$ ($n_1 \geq n_2 \geq \dots$). Prove that G has a subgroup H with invariant divisors $p^{m_1}, \dots, p^{m_s}$ ($m_1 \geq m_2 \geq \dots$) if and only if $s \leq r$ and $m_i \leq n_i$ for $i = 1, \dots, s$. (Hint: One direction is immediate. For the other, with notation as in Exercise 6.7, compare $\ker \rho$ for H and G to establish $s \leq r$; this also proves the statement if all $n_i = 1$. For the general case use induction, noting that if $G \cong \bigoplus_i \mathbb{Z}/p^{n_i} \mathbb{Z}$, then $\rho(G) \cong \bigoplus_i \mathbb{Z}/p^{n_i-1} \mathbb{Z}$.)

Prove that the same description holds for the homomorphic images of G . [6.9]

6.9. Let H be a subgroup of a finite abelian group G . Prove that G contains a subgroup isomorphic to G/H . (Reduce to the case of p -groups; then use Exercise 6.8.) Show that both hypotheses ‘finite’ and ‘abelian’ are needed for this result. (Hint: Q_8 has a unique subgroup of order 2.)

6.10. The *dual* of a finite group G is the abelian group $G^\vee := \operatorname{Hom}_{\operatorname{Grp}}(G, \mathbb{C}^*)$, where $\mathbb{C}^*$ is the multiplicative group of $\mathbb{C}$.

- Prove that the image of every $\sigma \in G^\vee$ consists of *roots of 1* in $\mathbb{C}$, that is, roots of polynomials $x^n - 1$ for some n .
- Prove that if G is a finite abelian group, then $G \cong G^\vee$. (Hint: First prove this for cyclic groups; then use the classification theorem to generalize to the arbitrary case.)

In §VIII.6.5 we will encounter another notion of ‘dual’ of a group.

6.11. • Use the classification theorem for finite abelian groups (Theorem 6.6) to classify all finite modules over the ring $\mathbb{Z}/n\mathbb{Z}$.

- Prove that if p is prime, all finite modules over $\mathbb{Z}/p\mathbb{Z}$ are free³².

6.12. Let G, H, K be finite abelian groups such that $G \oplus H \cong G \oplus K$. Prove that $H \cong K$.

6.13. $\neg$ Let G, H be finite abelian groups such that, for all positive integers n , G and H have the same number of elements of order n . Prove that $G \cong H$. (Note: The ‘abelian’ hypothesis is necessary! $C_4 \times C_4$ and $Q_8 \times C_2$ are nonisomorphic groups both with 1 element of order 1, 3 elements of order 2, and 12 elements of order 4.) [§II.4.3]

6.14. Let G be a finite abelian p -group, and assume G has only one subgroup of order p . Prove that G is cyclic. (This is in some sense a converse to Proposition II.6.11. You are welcome to try to prove it ‘by hand’, but use of the classification theorem will simplify the argument considerably.)

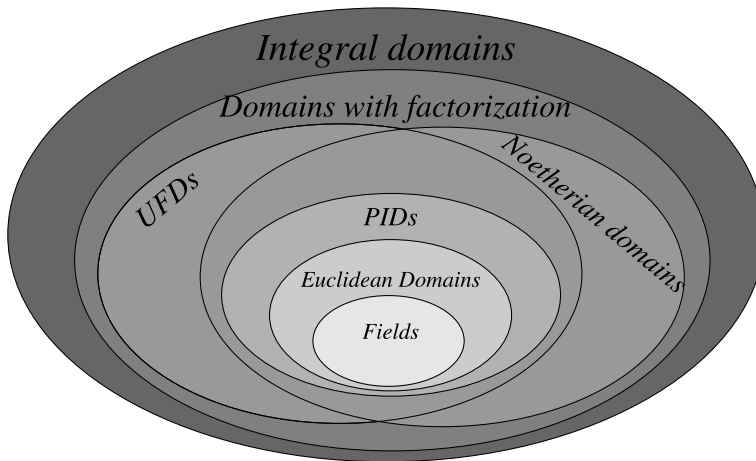
³²As we will see in Proposition VI.4.10, this property characterizes fields.

6.15. Let G be a finite abelian group, and let $a \in G$ be an element of *maximal* order in G . Prove that the order of every $b \in G$ divides $|a|$. (This essentially reproduces the result of Exercise II.1.15.)

6.16. Let G be an abelian group of order n , and assume that G has at most one subgroup of order d for all $d \mid n$. Prove that G is cyclic.

Irreducibility and factorization in integral domains

We move our attention back to *rings* and analyze several useful classes of integral domains. One guiding theme in this chapter is the issue of *factorization*: we will address the problem of existence and uniqueness of factorizations of elements in a ring, abstracting good factorization properties of rings such as $\mathbb{Z}$ or $k[x]$ (for k a field) to whole classes of integral domains. The reader may want to associate the following picture with the first part of this chapter:



Blanket assumption: all rings considered in this chapter will be *commutative*¹. In fact, most of the special classes of rings we will consider will be *integral domains*,

¹Also, recall that all our rings have 1; cf. Definition III.1.1.

that is, commutative rings with 1 and with no nonzero zero-divisors (cf. Definition III.1.10).

1. Chain conditions and existence of factorizations

1.1. Noetherian rings revisited. Let R be a commutative ring. Recall that R is said to be *Noetherian* if every ideal of R is finitely generated (Definition III.4.2). In fact, this is a special case of the corresponding definition for *modules*: a module M over a ring R is Noetherian if every submodule of M is finitely generated (Definition III.6.6). In §III.6.4 we have verified that this condition is preserved through exact sequences: if M, N, P are R -modules and

$$0 \longrightarrow N \longrightarrow M \longrightarrow P \longrightarrow 0$$

is an exact sequence of R -modules, then M is Noetherian if and only if both N and P are Noetherian (Proposition III.6.7). An easy and useful consequence of this fact is that every finitely generated module over a Noetherian ring is Noetherian (Corollary III.6.8).

The Noetherian condition may be expressed in alternative ways, and it is useful to acquire some familiarity with them.

Proposition 1.1. *Let R be a commutative ring, and let M be an R -module. Then the following are equivalent:*

- (1) *M is Noetherian; that is, every submodule of M is finitely generated.*
- (2) *Every ascending chain of submodules of M stabilizes; that is, if*

$$N_1 \subseteq N_2 \subseteq N_3 \subseteq \cdots$$

is a chain of submodules of M , then $\exists i$ such that $N_i = N_{i+1} = N_{i+2} = \cdots$

- (3) *Every nonempty family of submodules of M has a maximal element w.r.t. inclusion.*

The second condition listed here is called the *ascending chain condition* (a.c.c.) for submodules. For $M = R$, Proposition 1.1 tells us (among other things) that a ring is Noetherian if and only if the ascending chain condition holds for its *ideals*.

Proof. (1) $\implies$ (2): Assume that M is Noetherian, and let

$$N_1 \subseteq N_2 \subseteq N_3 \subseteq \cdots$$

be a chain of submodules of M . Consider the union

$$N = \bigcup_i N_i :$$

the reader will verify that N is a submodule of M . Since M is Noetherian, N is finitely generated, say $N = \langle n_1, \dots, n_r \rangle$. Now $n_k \in N \implies n_k \in N_i$ for some i ; by picking the largest such i , we see that $\exists i$ such that all $n_1, \dots, n_r$ are contained in N_i . But then $N \subseteq N_i$, and since $N_i \subseteq N_{i+1} \subseteq \cdots$ are all contained in $N = N_i$, it follows that $N_i = N_{i+1} = N_{i+2} = \cdots$ as needed.

(2) $\implies$ (3): Arguing contrapositively, assume that M admits a family $\mathcal{F}$ of submodules that does *not* have a maximal element. Construct an infinite ascending

chain as follows: let N_1 be any element of $\mathcal{F}$; since N_1 is not maximal in $\mathcal{F}$, there exists an element N_2 of $\mathcal{F}$ such that $N_1 \subsetneq N_2$; since N_2 is not maximal in $\mathcal{F}$, there exists an element N_3 of $\mathcal{F}$ such that $N_2 \subsetneq N_3$; etc. The chain

$$N_1 \subsetneq N_2 \subsetneq N_3 \subsetneq \cdots$$

does not stabilize, showing that (2) does not hold.

(3) $\implies$ (1): Assume (3) holds, and let N be a submodule of M . Then the family $\mathcal{F}$ of *finitely generated* submodules of N is nonempty (as $(0) \in \mathcal{F}$); hence it has a maximal element N' . Say that $N' = \langle n_1, \dots, n_r \rangle$. Now I claim that $N' = N$: indeed, let $n \in N$; the submodule $\langle n_1, \dots, n_r, n \rangle$ is finitely generated, and therefore it is in $\mathcal{F}$; as it contains N' and N' is maximal, necessarily $\langle n_1, \dots, n_r, n \rangle = N'$; in particular $n \in N'$, as needed.

This shows that $N = N'$ is finitely generated, and since $N \subseteq M$ was arbitrary, this implies that M is Noetherian. $\square$

Noetherian rings are a very useful and flexible class of rings. In §III.6.5 I mentioned the important fact that *every finite-type algebra over a Noetherian ring is Noetherian*. ‘Finite-type (commutative) algebra’ is just a fancy name for a quotient of a polynomial ring (§III.6.5), so this is what the fact states:

Theorem 1.2. *Let R be a Noetherian ring, and let J be an ideal of the polynomial ring $R[x_1, \dots, x_n]$. Then the ring $R[x_1, \dots, x_n]/J$ is Noetherian.*

Note that finite-type R algebras are (in general) very far from being finitely generated *as modules over R* (cf. again §III.6.5), so it would be foolish to expect them to be Noetherian *as R -modules*. The fact that they turn out to be Noetherian *as rings* (that is, as modules over themselves) provides us with a huge class of examples of Noetherian rings, among which are the rings of (classical) algebraic geometry and number theory. Thus, entire fields of mathematics are a little more manageable thanks to Theorem 1.2.

The proof of this deep fact is surprisingly easy. By Exercise 1.1, it suffices to prove that

$$R \text{ Noetherian} \implies R[x_1, \dots, x_n] \text{ Noetherian};$$

and an immediate induction reduces the statement to the following particular case, which carries a distinguished name:

Lemma 1.3 (Hilbert’s basis theorem). *R Noetherian $\implies R[x]$ Noetherian.*

Proof. Assume R is Noetherian, and let I be an ideal of $R[x]$. We have to prove that I is finitely generated.

Recall that if $f(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_0 \in R[x]$ and $a_d \neq 0$, then a_d is called the *leading coefficient* of $f(x)$. Consider the following subset of R :

$$A = \{0\} \cup \{a \in R \mid a \text{ is a leading coefficient of an element of } I\}.$$

It is clear that A is an *ideal* of R (Exercise 1.6); since R is Noetherian, A is finitely generated. Thus there exist elements $f_1(x), \dots, f_r(x) \in I$ whose leading coefficients $a_1, \dots, a_r$ generate A as an ideal of R .

Now let d_i be the degree of $f_i(x)$, and let d be the maximum among these degrees. Consider the sub- R -module

$$M = \langle 1, x, x^2, \dots, x^{d-1} \rangle \subseteq R[x],$$

that is, the R -module consisting of polynomials of degree $< d$. Since M is finitely generated as a module over R , it is Noetherian as an R -module (by Corollary III.6.8). Therefore, the submodule

$$M \cap I$$

of M is finitely generated over R , say by $g_1(x), \dots, g_s(x) \in I$.

Claim 1.4.

$$I = (f_1(x), \dots, f_r(x), g_1(x), \dots, g_s(x)).$$

This claim implies the statement of the theorem. To prove the claim, we only need to prove the $\subseteq$ inclusion; to this end, let $\alpha(x) \in I$ be an arbitrary polynomial in I . If $\deg \alpha(x) \geq d$, let a be the leading coefficient of $\alpha(x)$. Then $a \in A$, so $\exists b_1, \dots, b_r \in R$ such that

$$a = b_1 a_1 + \dots + b_r a_r.$$

Letting $e = \deg \alpha(x)$, so that $e \geq d_i$ for all i , this says that

$$\alpha(x) - b_1 x^{e-d_1} f_1(x) - \dots - b_r x^{e-d_r} f_r(x)$$

has degree $< e$. Iterating this procedure, we obtain a finite list of polynomials $\beta_1(x), \dots, \beta_r(x) \in R[x]$ such that

$$\alpha(x) - \beta_1(x) f_1(x) - \dots - \beta_r(x) f_r(x)$$

has degree $< d$. But this places this element in $M \cap I$; therefore $\exists c_1, \dots, c_s \in R$ such that

$$\alpha(x) - \beta_1(x) f_1(x) - \dots - \beta_r(x) f_r(x) = c_1 g_1(x) + \dots + c_s g_s(x),$$

and we are done, since this verifies that

$$\begin{aligned} \alpha(x) &= \beta_1(x) f_1(x) + \dots + \beta_r(x) f_r(x) + c_1 g_1(x) + \dots + c_s g_s(x) \\ &\in (f_1(x), \dots, f_r(x), g_1(x), \dots, g_s(x)), \end{aligned}$$

completing the proof of Claim 1.4, hence of Lemma 1.3, hence of Theorem 1.2. $\square$

1.2. Prime and irreducible elements. Let R be a (commutative) ring, and let $a, b \in R$. We say that a divides b , or that a is a *divisor* of b , or that b is a *multiple* of a , if $b \in (a)$, that is,

$$(\exists c \in R), \quad b = ac.$$

We use the notation $a \mid b$.

Two elements a, b are *associates* if $(a) = (b)$, that is, if $a \mid b$ and $b \mid a$.

Lemma 1.5. *Let a, b be nonzero elements of an integral domain R . Then a and b are associates if and only if $a = ub$, for u a unit in R .*

Proof. Assume a and b are associates. Then $\exists c, d \in R$ such that

$$b = ac, \quad a = bd;$$

therefore $a = bd = acd$, i.e.,

$$a(1 - cd) = 0.$$

Since cancellation by nonzero elements hold in integral domains, this implies $cd = 1$. Thus c is a unit, as needed.

The converse is left to the reader. □

Incidentally, here the reader sees why it is convenient to restrict our attention to integral domains. This argument really shows that if $(a) = (b) \neq (0)$ in an integral domain, and $b = ca$, then c is necessarily a unit. Away from the comfortable environment of integral domains, even such harmless-looking statements may fail: in $\mathbb{Z}/6\mathbb{Z}$ the classes $[2]_6, [4]_6$ of 2 and 4 are associates according to our definition, and $[4]_6 = [2]_6 \cdot [2]_6$, yet $[2]_6$ is not a unit. However, $[4]_6 = [5]_6 \cdot [2]_6$ and $[5]_6$ is a unit, so this is not a counterexample to Lemma 1.5. In fact, Lemma 1.5 may fail over rings with ‘non-harmless’ zero-divisors (yes, there is such a notion).

The notions reviewed above generalize directly the corresponding notions in $\mathbb{Z}$. We are going to explore analogues of other common notions in $\mathbb{Z}$, such as ‘primality’ and ‘irreducibility’, in more general integral domains.

Definition 1.6. Let R be an integral domain.

- An element $a \in R$ is *prime* if the ideal (a) is prime; that is, a is not a unit and (cf. Proposition III.4.11)

$$a \mid bc \implies (a \mid b \text{ or } a \mid c).$$

- An element $a \in R$ is *irreducible* if a is not a unit and

$$a = bc \implies (b \text{ is a unit or } c \text{ is a unit}).$$

Note that 0 is always *reducible* (integral domains are nonzero rings!). For nonzero elements, there are useful alternative ways to think about the notion of ‘irreducible’: a nonunit $a \neq 0$ is irreducible if and only if

- $a = bc$ implies that a is an associate of b or of c ;
- $a = bc$ implies that $(a) = (b)$ or $(a) = (c)$ (Lemma 1.5);
- $(a) \subseteq (b) \implies (b) = (a)$ or $(b) = (1)$ (Exercise 1.12);
- (a) is maximal among proper principal ideals (rephrasing the previous point!).

It is important to realize that primality and irreducibility are *not equivalent*, even for nonzero elements; this is somewhat counterintuitive since they *are* equivalent in $\mathbb{Z}$, as the reader should verify² (Exercise 1.13). What *is* true in general is that *prime* is stronger than *irreducible*:

Lemma 1.7. Let R be an integral domain, and let $a \in R$ be a nonzero prime element. Then a is irreducible.

²This fact will be fully explained by the general theory, so the reader should work this out right away.

Proof. Since (a) is prime, $(a) \neq (1)$; hence a is not a unit. If $a = bc$, then $bc = a \in (a)$; therefore $b \in (a)$ or $c \in (a)$ since (a) is prime. Assuming without loss of generality $b \in (a)$, we have $(b) \subseteq (a)$. On the other hand $a = bc$ implies $(a) \subseteq (b)$: hence $(a) = (b)$, that is, a and b are associates, as needed. $\square$

We will soon see under what circumstances the converse statement holds.

1.3. Factorization into irreducibles; domains with factorizations.

Definition 1.8. Let R be an integral domain. An element $r \in R$ has a *factorization* (or *decomposition*) into irreducibles if there exist irreducible elements $q_1, \dots, q_n$ such that $r = q_1 \cdots q_n$.

This factorization is *unique* if the elements q_i are determined by r up to order and associates, that is, if whenever

$$r = q'_1 \cdots q'_m$$

is another factorization of r into irreducibles, then $m = n$ and q'_i is an associate of q_i after (possibly) shuffling the factors. $\lrcorner$

Definition 1.9. An integral domain R is a *domain with factorizations* (or ‘*factorizations exist in R* ’) if every nonzero, nonunit element $r \in R$ has a factorization into irreducibles. $\lrcorner$

Definition 1.10. An integral domain R is *factorial*, or a *unique factorization domain* (abbreviated *UFD*), if every nonzero, nonunit element $r \in R$ has a unique factorization into irreducibles. $\lrcorner$

The terminology introduced in Definition 1.9 does not appear to be too standard; by contrast, UFDs are famous.

We will study the unique factorization condition in §2. For now, it seems worth spending a little time contemplating the mere existence of factorizations. Interestingly, this condition is implied by an *ascending chain condition*, for a special class of ideals.

Proposition 1.11. Let R be an integral domain, and let r be a nonzero, nonunit element of R . Assume that every ascending chain of principal ideals

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq (r_3) \subseteq \cdots$$

stabilizes. Then r has a factorization into irreducibles.

Proof. Assume that r does *not* have a factorization into irreducible elements. In particular, r is itself not irreducible; thus $\exists r_1, s_1 \in R$ such that $r = r_1 s_1$ and $(r) \subsetneq (r_1)$, $(r) \subsetneq (s_1)$. If both r_1, s_1 have factorizations into irreducibles, the product of these factorizations gives a factorization of r ; thus we may assume that (e.g.) r_1 does not have a factorization into irreducibles. Therefore we have

$$(r) \subsetneq (r_1)$$

and r_1 does not have a factorization; iterating this argument constructs an infinitely increasing chain

$$(r) \subsetneq (r_1) \subsetneq (r_2) \subsetneq (r_3) \subsetneq \cdots,$$

contradicting our hypothesis. $\square$

Thus, factorizations exist in integral domains in which the ascending chain condition holds *for principal ideals*. This has the following immediate and convenient consequence:

Corollary 1.12. *Let R be a Noetherian domain. Then factorizations exist in R .*

Proof. By Proposition 1.1, Noetherian domains satisfy the ascending chain condition for *all* ideals. $\square$

Corollary 1.12 verifies part of the picture presented at the beginning of the chapter: the class of Noetherian domains is contained in the class of domains with factorization. This inclusion is proper: for instance, the standard example of a non-Noetherian ring,

$$\mathbb{Z}[x_1, x_2, x_3, \dots],$$

(Example III.6.5) *does* have factorizations. Indeed, every given polynomial $f \in \mathbb{Z}[x_1, x_2, \dots]$ involves only finitely many variables³, so it belongs to a subring $\mathbb{Z}[x_1, \dots, x_n]$ isomorphic to an ordinary polynomial ring over $\mathbb{Z}$; further, this subring contains every divisor of f . It follows easily (cf. Exercise 1.15) that the ascending chain condition for principal ideals holds in $\mathbb{Z}[x_1, x_2, x_3, \dots]$, because it holds in $\mathbb{Z}[x_1, \dots, x_n]$ (since this ring is Noetherian, by Hilbert's basis theorem).

Exercises

Remember that in this section all rings are taken to be *commutative*.

1.1. $\triangleright$ Let R be a Noetherian ring, and let I be an ideal of R . Prove that R/I is a Noetherian ring. [§1.1]

1.2. Prove that if $R[x]$ is Noetherian, so is R . (This is a 'converse' to Hilbert's basis theorem.)

1.3. Let k be a field, and let $f \in k[x]$, $f \notin k$. For every subring R of $k[x]$ containing k and f , define a homomorphism $\varphi : k[t] \rightarrow R$ by extending the identity on k and mapping t to f . This makes every such R a $k[t]$ -algebra (Example III.5.6).

- Prove that $k[x]$ is finitely generated as a $k[t]$ -module.
- Prove that every subring R as above is finitely generated as a $k[t]$ -module.
- Prove that every subring of $k[x]$ containing k is a Noetherian ring.

1.4. Let R be the ring of real-valued continuous functions on the interval $[0, 1]$. Prove that R is not Noetherian.

1.5. Determine for which sets S the power set ring $\mathcal{P}(S)$ is Noetherian. (Cf. Exercise III.3.16.)

³Remember that polynomials are *finite* linear combinations of monomials; cf. §III.1.3.

1.6. $\triangleright$ Let I be an ideal of $R[x]$, and let $A \subseteq R$ be the set defined in the proof of Theorem 1.2. Prove that A is an ideal of R . [§1.1]

1.7. Prove that if R is a Noetherian ring, then the ring of power series $R[[x]]$ (cf. §III.1.3) is also Noetherian. (Hint: The order of a power series $\sum_{i=0}^{\infty} a_i x^i$ is the smallest i for which $a_i \neq 0$; the *dominant coefficient* is then a_i . Let $A_i \subseteq R$ be the set of dominant coefficients of series of order i in I , together with 0. Prove that A_i is an ideal of R and $A_0 \subseteq A_1 \subseteq A_2 \subseteq \cdots$. This sequence stabilizes since R is Noetherian, and each A_i is finitely generated for the same reason. Now adapt the proof of Lemma 1.3.)

1.8. Prove that every ideal in a Noetherian ring R contains a finite product of prime ideals. (Hint: Let $\mathcal{F}$ be the family of ideals that do not contain finite products of prime ideals. If $\mathcal{F}$ is nonempty, it has a maximal element M since R is Noetherian. Since $M \in \mathcal{F}$, M is not itself prime, so $\exists a, b \in R$ s.t. $a \notin M$, $b \notin M$, yet $ab \in M$. What's wrong with this?)

1.9. $\neg$ Let R be a commutative ring, and let $I \subseteq R$ be a proper ideal. The reader will prove in Exercise 3.12 that the set of prime ideals containing I has minimal elements (the *minimal primes* of I). Prove that if R is Noetherian, then the set of minimal primes of I is finite. (Hint: Let $\mathcal{F}$ be the family of ideals that do *not* have finitely many minimal primes. If $\mathcal{F} \neq \emptyset$, note that $\mathcal{F}$ must have a maximal element I , and I is not prime itself. Find ideals J_1, J_2 strictly larger than I , such that $J_1 J_2 \subseteq I$, and deduce a contradiction.) [VI.4.10]

1.10. $\neg$ By Proposition 1.1, a ring R is Noetherian if and only if it satisfies the a.c.c. for ideals. A ring is *Artinian* if it satisfies the d.c.c. (descending chain condition) for ideals. Prove that if R is Artinian and $I \subseteq R$ is an ideal, then R/I is Artinian. Prove that if R is an Artinian integral domain, then it is a field. (Hint: Let $r \in R$, $r \neq 0$. The ideals (r^n) form a descending sequence; hence $(r^n) = (r^{n+1})$ for some n . Therefore...) Prove that Artinian rings have Krull dimension 0 (that is, prime ideals are maximal in Artinian rings⁴). [2.11]

1.11. Prove that the ‘associate’ relation is an equivalence relation.

1.12. $\triangleright$ Let R be an integral domain. Prove that a nonzero $a \in R$ is irreducible if and only if (a) is maximal among proper principal ideals of R . [§1.2, §2.3]

1.13. $\triangleright$ Prove that, for nonzero elements, prime $\iff$ irreducible in $\mathbb{Z}$. [§1.2, §2.3]

1.14. For a, b in a commutative ring R , prove that the class of a in $R/(b)$ is prime if and only if the class of b in $R/(a)$ is prime.

1.15. $\triangleright$ Identify $S = \mathbb{Z}[x_1, \dots, x_n]$ in the natural way with a subring of the polynomial ring in countably infinitely many variables $R = \mathbb{Z}[x_1, x_2, x_3, \dots]$. Prove that if $f \in S$ and $(f) \subseteq (g)$ in R , then $g \in S$ as well. Conclude that the ascending chain condition for principal ideals holds in R , and hence R is a domain with factorizations. [§1.3, §4.3]

⁴One can prove that Artinian rings are necessarily Noetherian; in fact, a ring is Artinian if and only if it is Noetherian and has Krull dimension 0. Thus, the d.c.c. implies the a.c.c., while the a.c.c. implies the d.c.c. if and only if all prime ideals are maximal.

1.16. Let

$$R = \frac{\mathbb{Z}[x_1, x_2, x_3, \dots]}{(x_1 - x_2^2, x_2 - x_3^2, \dots)}.$$

Does the ascending chain condition for principal ideals hold in R ?

1.17. $\triangleright$ Consider the subring of $\mathbb{C}$:

$$\mathbb{Z}[\sqrt{-5}] := \{a + bi\sqrt{5} \mid a, b \in \mathbb{Z}\}.$$

- Prove that this ring is isomorphic to $\mathbb{Z}[t]/(t^2 + 5)$.
- Prove that it is a Noetherian integral domain.
- Define a ‘norm’ N on $\mathbb{Z}[\sqrt{-5}]$ by setting $N(a + bi\sqrt{5}) = a^2 + 5b^2$. Prove that $N(zw) = N(z)N(w)$. (Cf. Exercise III.4.10.)
- Prove that the units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1 . (Use the preceding point.)
- Prove that 2, 3, $1 + i\sqrt{5}$, $1 - i\sqrt{5}$ are all irreducible nonassociate elements of $\mathbb{Z}[\sqrt{-5}]$.
- Prove that no element listed in the preceding point is prime. (Prove that the rings obtained by mod-ing out the ideals generated by these elements are not integral domains.)
- Prove that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

[§2.2, 2.18, 6.14]

2. UFDs, PIDs, Euclidean domains

2.1. Irreducible factors and greatest common divisor. An integral domain R is a UFD if factorizations exist in R and are *unique* in the sense of Definition 1.8.

Thus, in a UFD all elements (other than 0 and the units) determine a multiset (a set of elements ‘with multiplicity’; cf. §I.1.1) of irreducible *factors*, determined up to the associate relation. We can also agree that units have *no* factors; that is, the corresponding multiset is $\emptyset$.

The following trivial remark is at the root of most elementary facts about UFDs, such as the characterization of Theorem 2.5:

Lemma 2.1. *Let R be a UFD, and let a, b, c be nonzero elements of R . Then*

- $(a) \subseteq (b) \iff$ *the multiset of irreducible factors of b is contained in the multiset of irreducible factors of a ;*
- a and b are associates (that is, $(a) = (b)$) $\iff$ *the two multisets coincide;*
- *the irreducible factors of a product bc are the collection of all irreducible factors of b and of c .*

The proof is left to the reader (Exercise 2.1). The advantage of working in a UFD resides in the fact that ring-theoretic statements about elements of the ring often reduce to straightforward set-theoretic statements about multisets of irreducible elements, by means of Lemma 2.1.

One important instance of this mechanism is the existence of *greatest common divisors*. I have liberally used this notion (at least for integers) in previous chapters; now we can appreciate it from a more technical perspective.

Definition 2.2. Let R be an integral domain, and let $a, b \in R$. An element $d \in R$ is a *greatest common divisor* (often abbreviated ‘gcd’) of a and b if $(a, b) \subseteq (d)$ and (d) is the smallest principal ideal in R with this property. $\square$

In other words, d is a gcd of a and b if $d \mid a$, $d \mid b$, and

$$c \mid a, c \mid b \implies c \mid d.$$

This definition is immediately extended to any finite number of elements.

Note that greatest common divisors are not defined uniquely by this prescription: if d is a greatest common divisor of a and b , so is every associate of d . Thus, the notation ‘gcd(a, b)’ should only be used for the associate class formed by all greatest common divisors of a, b . Of course, language is often (harmlessly) abused on this point. For example, the fact that we can talk about *the* greatest common divisor of two integers is due to the fact that in $\mathbb{Z}$ there is a convenient way to choose a distinguished element in each class of associate integers (that is, the nonnegative one).

Also note that greatest common divisors need not *exist* (cf. Exercise 2.5); but they do exist in UFDs:

Lemma 2.3. *Let R be a UFD, and let a, b be nonzero elements of R . Then a, b have a greatest common divisor.*

Proof. We can write

$$a = uq_1^{\alpha_1} \cdots q_r^{\alpha_r}, \quad b = vq_1^{\beta_1} \cdots q_r^{\beta_r}$$

where u and v are units, the elements q_i are irreducible, q_i is not an associate of q_j for $i \neq j$, and $\alpha_i \geq 0$, $\beta_i \geq 0$ (so that the multisets of irreducible factors of a , resp., b , consist of those q_i for which $\alpha_i > 0$, resp., $\beta_i > 0$; the units u, v are included since the irreducible factors are only defined up to the associate relation).

I claim that

$$d = q_1^{\min(\alpha_1, \beta_1)} \cdots q_r^{\min(\alpha_r, \beta_r)}$$

is a gcd of a and b . Indeed, d is clearly a divisor of a and b ; and if c also divides a and b , then the multiset of factors of c must be contained in both multisets of factors for a and b (by Lemma 2.1); that is,

$$c = wq_1^{\gamma_1} \cdots q_r^{\gamma_r}$$

with w a unit and $\gamma_i \leq \alpha_i$, $\gamma_i \leq \beta_i$. This implies $\gamma_i \leq \min(\alpha_i, \beta_i)$, and hence $c \mid d$ (again by Lemma 2.1), as needed. $\square$

Of course the argument given in the proof generalizes one of the standard ways to compute greatest common divisors in $\mathbb{Z}$: find smallest exponents in prime factorizations. But note that this is not the only way to compute the gcd in $\mathbb{Z}$; we will come back to this point in a moment. In fact, greatest common divisors in $\mathbb{Z}$ have properties that should not be expected in a more general domain: for

example, the result of Exercise II.2.13 does *not* generalize to arbitrary domains (and not even to arbitrary UFDs), as the reader will check in Exercise 2.4.

2.2. Characterization of UFDs. It is easy to construct integral domains where unique factorization fails. The diligent reader has already analyzed one example (Exercise 1.17); for another, in the domain⁵

$$R = \frac{\mathbb{C}[x, y, z, w]}{(xw - yz)}$$

the (classes of the) elements x, y, z, w are irreducible and not associates of one another; since $xw - yz = 0$ in R , the element $r = xw$ has two distinct factorizations into irreducibles: $r = xw = yz$.

Note that this ring is Noetherian, by Theorem 1.2 (and in particular factorizations do exist in R). Thus, there are Noetherian integral domains that are *not* UFDs.

Also note that this ring provides an example in which the converse to Lemma 1.7 does *not* hold: indeed, (the class of) x is irreducible, but the quotient

$$\left(\frac{\mathbb{C}[x, y, z, w]}{(xw - yz)} \bigg/ (x) \right) \cong \frac{\mathbb{C}[x, y, z, w]}{(x, xw - yz)} = \frac{\mathbb{C}[x, y, z, w]}{(x, yz)}$$

is not an integral domain (because $y \neq 0, z \neq 0$, and yet $yz = 0$ in this ring); that is, x is not prime.

In fact, and maybe a little surprisingly, the issue of unique factorization is inextricably linked with the relation between primality and irreducibility. Indeed, the ‘converse’ to Lemma 1.7 does hold in UFDs:

Lemma 2.4. *Let R be a UFD, and let a be an irreducible element of R . Then a is prime.*

Proof. The element a is not a unit, by definition of irreducible. Assume $bc \in (a)$: thus $(bc) \subseteq (a)$, and by Lemma 2.1 the irreducible factors of a , that is, a itself, must be among the factors of b or of c . We have $b \in (a)$ in the first case and $c \in (a)$ in the second. This shows that (a) is a prime ideal, as needed. $\square$

In fact, more is true. Provided that the ascending chain condition for principal ideals holds, then UFDs are *characterized* by the equivalence between irreducibility and primality.

Theorem 2.5. *An integral domain R is a UFD if and only if*

- *the a.c.c. for principal ideals holds in R and*
- *every irreducible element of R is prime.*

Proof. ($\implies$) Assume that R is a UFD. Lemma 2.4 shows that irreducible elements of R are prime. To prove that the a.c.c. for principal ideals holds, consider an ascending chain

$$(r_1) \subsetneq (r_2) \subsetneq (r_3) \subsetneq \cdots$$

⁵In algebraic geometry, this is the ring of a ‘quadric cone in $\mathbb{A}^4$ ’. The vertex of this cone (at the origin) is a *singular* point, and this has to do with the fact that R is not a UFD.

By Lemma 2.1, this chain determines a corresponding *descending* chain of multisets of irreducible factors. A descending chain of finite multisets clearly stabilizes, and it follows (by Lemma 2.1 again) that $(r_i) = (r_{i+1}) = (r_{i+2}) = \dots$ for large enough i .

($\Leftarrow$) Now assume that R satisfies the a.c.c. for principal ideals and irreducibles are prime. Proposition 1.11 implies that factorizations exist in R ; we have to verify uniqueness. Let $q_1, \dots, q_m$ and $q'_1, \dots, q'_n$ be irreducible elements of R , and assume

$$q_1 \cdots q_m = q'_1 \cdots q'_n.$$

Then $q'_1 \cdots q'_n \in (q_1)$, and (q_1) is a prime ideal (by hypothesis); thus $q'_i \in (q_1)$ for some i , which we may assume to be 1 after changing the order of the factors. Therefore $q'_1 = uq_1$ for some $u \in R$. Since q'_1 is irreducible and q_1 is not a unit, necessarily u is a unit. Thus q_1 and q'_1 are associates. Canceling q_1 and replacing q'_2 by uq'_2 , we find

$$q_2 \cdots q_m = q'_2 \cdots q'_n.$$

Repeating this process matches all factors one by one. It is clear that $m = n$, because otherwise we will obtain $1 =$ a product of irreducibles, contradicting the fact that irreducibles are not units. $\square$

Theorem 2.5 is a pleasant statement, but it does not make it particularly easy to check whether a given ring is in fact a UFD. The situation is not unlike that with Noetherian rings: the a.c.c. is a sharp equivalent formulation of the Noetherian condition, but in practice one relies more often on other tools, such as Hilbert's basis theorem, in order to establish that a given ring is Noetherian. Does an analog of Hilbert's basis theorem hold for unique factorization domains?

Answering this question requires some preparatory work, and we will come back to it in §4.

2.3. PID $\implies$ UFD. There are simple ways to produce examples of UFDs. Recall (from §III.4) that a *principal ideal domain* (abbreviated PID) is an integral domain in which every ideal is principal. In §III.4 we have observed that

- PIDs are Noetherian.
- $\mathbb{Z}$ and $k[x]$ (where k is a field) are PIDs.
- If R is a PID and $a, b \in R$, then d is a greatest common divisor for a and b if and only if $(a, b) = (d)$. In particular, if d is a greatest common divisor of a and b in a PID R , then d is a linear combination of a and b : $\exists r, s \in R$ such that $d = ra + sb$.
- If I is a nonzero ideal in a PID, then I is prime if and only if it is maximal.

We now add one important point to this list:

Proposition 2.6. *If R is a PID, then it is a UFD.*

Proof. Let R be a PID. The a.c.c. (for principal ideals, as all ideals in R are principal!) holds in R since PIDs are Noetherian. We verify that irreducible elements are prime in R , which implies that R is a UFD by Theorem 2.5.

Let $a \in R$ be an irreducible element. Ideals generated by irreducible elements are maximal among principal ideals (Exercise 1.12), hence (a) is a maximal ideal in R as all ideals in R are principal. Since maximal ideals are prime it follows that (a) is prime, as needed. $\square$

In particular, $k[x]$ is a UFD if k is a field, and $\mathbb{Z}$ is a UFD—which hopefully will not come as a big surprise to our readers⁶. Note that at this point we have recovered the fact stated in Exercise 1.13 in full glory.

Proposition 2.6 justifies another feature of the picture given at the beginning of this chapter: the class of PIDs is contained in the class of UFDs. We will soon see that the inclusion is proper, that is, that there are UFDs which are not PIDs; for example, $\mathbb{Z}[x]$ is not a PID (Exercise 2.12), yet

$\mathbb{Z}[x]$ is a unique factorization domain

as we will soon be able to prove (Theorem 4.14). In fact, there are UFDs which are not Noetherian, as represented in the picture; however, these examples are best discussed after more material has been developed (Example 4.18).

The reader can already get a feel for the gap between UFD and PID, by contemplating the fact (recalled above and essentially tautological) that, *in a PID*, greatest common divisors of a, b are linear combinations of a and b . This is a very strong requirement: for example, it characterizes PIDs among Noetherian domains, as the reader will check (Exercise 2.7); it does not hold in general UFDs (Exercise 2.4).

2.4. Euclidean domain $\implies$ PID. The excellent properties of $\mathbb{Z}$ and $k[x]$ (where k is a field) make these rings even more special than PIDs: they are *Euclidean domains*.

Informally, Euclidean domains are rings in which one may perform a ‘division with remainder’: this is the case for $\mathbb{Z}$ and for $k[x]$, as observed in §III.4. The point is that in both $\mathbb{Z}$ and $k[x]$ one can define a notion of ‘size’ of an element: $|n|$ for an integer n and $\deg f(x)$ for a polynomial $f(x)$. In both cases, one has control over the size of the ‘remainder’ in a division. The definition of Euclidean domain simply abstracts this mechanism.

For the purpose of this discussion⁷, a *valuation* on an integral domain R is any function $v : R \setminus \{0\} \rightarrow \mathbb{Z}^{\geq 0}$.

Definition 2.7. A *Euclidean valuation* on an integral domain R is a valuation satisfying the following property⁸: for all $a \in R$ and all nonzero $b \in R$ there exist $q, r \in R$ such that

$$a = qb + r,$$

with either $r = 0$ or $v(r) < v(b)$. An integral domain R is a *Euclidean domain* if it admits a Euclidean valuation. $\lrcorner$

⁶This fact is known as the *fundamental theorem of arithmetic*.

⁷Entire libraries have been written on the subject of *valuations*, studying a more precise notion than what is needed here.

⁸It is not uncommon to also require that $v(ab) \geq v(b)$ for all nonzero $a, b \in R$; but this is not needed in the considerations that follow, and cf. Exercise 2.15.

We say that q is the *quotient* of the division and r is the *remainder*. Division with remainder in $\mathbb{Z}$ and in $k[x]$ (where k is a field) provide examples, so that $\mathbb{Z}$ and $k[x]$ are Euclidean domains.

Proposition 2.8. *Let R be a Euclidean domain. Then R is a PID.*

The proof is modeled after the instances encountered for $\mathbb{Z}$ (Proposition III.4.4) and $k[x]$ (which the reader has hopefully worked out in Exercise III.4.4).

Proof. Let I be an ideal of R ; we have to prove that I is principal. If $I = \{0\}$, there is nothing to show; therefore, assume $I \neq \{0\}$. The valuation maps the nonzero elements of I to a subset of $\mathbb{Z}^{\geq 0}$; let $b \in I$ be an element with the smallest valuation. Then I claim that $I = (b)$; therefore I is principal, as needed. Since clearly $(b) \subseteq I$, we only need to verify that $I \subseteq (b)$.

For this, let $a \in I$ and apply division with remainder: we have

$$a = qb + r$$

for some q, r in R , with $r = 0$ or $v(r) < v(b)$. But

$$r = a - qb \in I :$$

by the minimality of $v(b)$ among nonzero elements of I , we cannot have $v(r) < v(b)$. Therefore $r = 0$, showing that $a = qb \in (b)$, as needed. $\square$

Proposition 2.8 justifies one more feature of the picture at the beginning of the chapter: the class of Euclidean domains is contained in the class of principal ideal domains.

This inclusion is proper, as suggested in the picture. Producing an explicit example of a PID which is not a Euclidean domain is not so easy, but the gap between PIDs and Euclidean domains can in fact be described very sharply: PID may be characterized as domains satisfying a weaker requirement than ‘division with remainder’.

More precisely, a ‘Dedekind-Hasse valuation’ is a valuation v such that $\forall a, b$, either $(a, b) = (b)$ (that is, b divides a) or there exists $r \in (a, b)$ such that $v(r) < v(b)$. This latter condition amounts to requiring that there exist $q, s \in R$ such that $as = bq + r$ with $v(r) < v(b)$; hence a Euclidean valuation (for which we may in fact choose $s = 1$) is a Dedekind-Hasse valuation. It is not hard to show that an integral domain is a PID if and only if it admits a Dedekind-Hasse valuation (Exercise 2.21). For example, this can be used to show that the ring $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID: the norm considered in Exercise 2.18 in order to prove that this ring is *not* a Euclidean domain turns out to be a Dedekind-Hasse valuation⁹. Thus, this ring gives an example of a PID that is not a Euclidean domain.

One excellent feature of Euclidean domains, and the one giving them their names, is the presence of an effective algorithm computing greatest common divisors: the *Euclidean algorithm*. As Euclidean domains are PIDs, and hence UFDs, we know that they do have greatest common divisors. However, the ‘algorithm’

⁹This boils down to a case-by-case analysis, which I am happily leaving to my most patient readers.

obtained by distilling the proof of Lemma 2.3 is highly impractical: if we had to factor two integers a, b in order to compute their gcd, this would make it essentially impossible (with current technologies and factorization algorithms) for integers of a few hundred digits. The Euclidean algorithm *bypasses* the necessity of factorization: greatest common divisors of thousand-digit integers may be computed in a fraction of a second.

The key lemma on which the algorithm is based is the following trivial general fact:

Lemma 2.9. *Let $a = bq + r$ in a ring R . Then $(a, b) = (b, r)$.*

Proof. Indeed, $r = a - bq \in (a, b)$, proving $(b, r) \subseteq (a, b)$; and $a = bq + r \in (b, r)$, proving $(a, b) \subseteq (b, r)$. $\square$

In particular,

$$(\forall c \in R), \quad (a, b) \subseteq (c) \iff (b, r) \subseteq (c);$$

that is, the set of common divisors of a, b and the set of common divisors of b, r coincide. Therefore,

Corollary 2.10. *Assume $a = bq + r$. Then a, b have a gcd if and only if b, r have a gcd, and in this case $\gcd(a, b) = \gcd(b, r)$.*

Of course ‘ $\gcd(a, b) = \gcd(b, r)$ ’ means that the two classes of associate elements coincide.

These considerations hold over any integral domain; assume now that R is a Euclidean domain. Then we can use division with remainder to gain some control over the remainders r . Given two elements a, b in R , with $b \neq 0$, we can apply division with remainder repeatedly:

$$\begin{aligned} a &= bq_1 + r_1, \\ b &= r_1q_2 + r_2, \\ r_1 &= r_2q_3 + r_3, \\ &\dots \end{aligned}$$

as long as the remainder r_i is nonzero.

Claim 2.11. *This process terminates: that is, $r_N = 0$ for some N .*

Proof. Each line in the table is a division with remainder. If no r_i were zero, we would have an infinite decreasing sequence

$$v(b) > v(r_1) > v(r_2) > v(r_3) > \dots$$

of *nonnegative* integers, which is nonsense. $\square$

Thus the table of divisions with remainders must be as follows: letting $r_0 = b$,

$$\begin{aligned} a &= r_0 q_1 + r_1, \\ b &= r_1 q_2 + r_2, \\ r_1 &= r_2 q_3 + r_3, \\ &\dots \\ r_{N-3} &= r_{N-2} q_{N-1} + r_{N-1}, \\ r_{N-2} &= r_{N-1} q_N \end{aligned}$$

with $r_{N-1} \neq 0$.

Proposition 2.12. *With notation as above, r_{N-1} is a gcd of a, b .*

Proof. By Corollary 2.10,

$$\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{N-2}, r_{N-1}).$$

But $r_{N-2} = r_{N-1} q_{N-1}$ gives $r_{N-2} \in (r_{N-1})$; hence $(r_{N-2}, r_{N-1}) = (r_{N-1})$. Therefore r_{N-1} is a gcd for r_{N-2} and r_{N-1} , hence for a and b , as needed. $\square$

The ring of integers and the polynomial ring over a field are both Euclidean domains. Fields are Euclidean domains (as represented in the picture at the beginning of the chapter), but not for a very interesting reason: the remainder of the division by a nonzero element in a field is *always* zero, so every function qualifies as a ‘Euclidean valuation’ for trivial reasons.

We will study another interesting Euclidean domain later in this chapter (§6.2).

Exercises

2.1. $\triangleright$ Prove Lemma 2.1. [§2.1]

2.2. Let R be a UFD, and let a, b, c be elements of R such that $a \mid bc$ and $\gcd(a, b) = 1$. Prove that a divides c .

2.3. Let n be a positive integer. Prove that there is a one-to-one correspondence preserving multiplicities between the irreducible factors of n (as an integer) and the composition factors of $\mathbb{Z}/n\mathbb{Z}$ (as a group). (In fact, the Jordan-Hölder theorem may be used to prove that $\mathbb{Z}$ is a UFD.)

2.4. $\triangleright$ Consider the elements x, y in $\mathbb{Z}[x, y]$. Prove that 1 is a gcd of x and y , and yet 1 is *not* a linear combination of x and y . (Cf. Exercise II.2.13.) [§2.1, §2.3]

2.5. $\triangleright$ Let R be the subring of $\mathbb{Z}[t]$ consisting of polynomials with no term of degree 1: $a_0 + a_2 t^2 + \cdots + a_d t^d$.

- Prove that R is indeed a subring of $\mathbb{Z}[t]$, and conclude that R is an integral domain.
- List all common divisors of t^5 and t^6 in R .

- Prove that t^5 and t^6 have no gcd in R .

[§2.1]

2.6. Let R be a domain with the property that the intersection of any family of principal ideals in R is necessarily a principal ideal.

- Show that greatest common divisors exist in R .
- Show that UFDs satisfy this property.

2.7. $\triangleright$ Let R be a Noetherian domain, and assume that for all nonzero a, b in R , the greatest common divisors of a and b are linear combinations of a and b . Prove that R is a PID. [§2.3]

2.8. Let R be a UFD, and let $I \neq (0)$ be an ideal of R . Prove that every descending chain of principal ideals containing I must stabilize.

2.9. $\neg$ The *height* of a prime ideal P in a ring R is (if finite) the maximum length h of a chain of prime ideals $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_h = P$ in R . (Thus, the Krull dimension of R , if finite, is the maximum height of a prime ideal in R .) Prove that if R is a UFD, then every prime ideal of height 1 in R is principal. [2.10]

2.10. $\neg$ It is a consequence of a theorem known as *Krull's Hauptidealsatz* that every nonzero, nonunit element in a Noetherian domain is contained in a prime ideal of height 1. Assuming this, prove a converse to Exercise 2.9, and conclude that a Noetherian domain R is a UFD if and only if every prime ideal of height 1 in R is principal. [4.16]

2.11. Let R be a PID, and let I be a nonzero ideal of R . Show that R/I is an artinian ring (cf. Exercise 1.10), by proving explicitly that the d.c.c. holds in R/I .

2.12. $\triangleright$ Prove that if $R[x]$ is a PID, then R is a field. [§2.3, §VI.7.1]

2.13. $\triangleright$ For a, b, c positive integers with $c > 1$, prove that $c^a - 1$ divides $c^b - 1$ if and only if $a \mid b$. Prove that $x^a - 1$ divides $x^b - 1$ in $\mathbb{Z}[x]$ if and only if $a \mid b$. (Hint: For the interesting implications, write $b = ad + r$ with $0 \leq r < a$, and take 'size' into account.) [§VII.5.1, VII.5.13]

2.14. $\triangleright$ Prove that if k is a field, then $k[[x]]$ is a Euclidean domain. [§4.3]

2.15. $\triangleright$ Prove that if R is a Euclidean domain, then R admits a Euclidean valuation $\bar{v}$ such that $\bar{v}(ab) \geq \bar{v}(b)$ for all nonzero $a, b \in R$. (Hint: Since R is a Euclidean domain, it admits a valuation v as in Definition 2.7. For $a \neq 0$, let $\bar{v}(a)$ be the minimum of all $v(ab)$ as $b \in R, b \neq 0$. To see that R is a Euclidean domain with respect to $\bar{v}$ as well, let a, b be nonzero in R , with $b \nmid a$; choose q, r so that $a = bq + r$, with $v(r)$ minimal; assume that $\bar{v}(r) \geq \bar{v}(b)$, and get a contradiction.) [§2.4, 2.16]

2.16. Let R be a Euclidean domain with Euclidean valuation v ; assume that $v(ab) \geq v(b)$ for all nonzero $a, b \in R$ (cf. Exercise 2.15). Prove that associate elements have the same valuation and that units have minimum valuation.

2.17. $\neg$ Let R be a Euclidean domain that is not a field. Prove that there exists a nonzero, nonunit element c in R such that $\forall a \in R, \exists q, r \in R$ with $a = qc + r$ and either $r = 0$ or r a unit. [2.18]

2.18. $\triangleright$ For an integer d , denote by $\mathbb{Q}(\sqrt{d})$ the smallest subfield of $\mathbb{C}$ containing $\mathbb{Q}$ and $\sqrt{d}$, with norm N defined as in Exercise III.4.10. See Exercise 1.17 for the case $d = -5$; in this problem, you will take $d = -19$.

Let $\delta = (1 + i\sqrt{19})/2$, and consider the following subring of $\mathbb{Q}(\sqrt{-19})$:

$$\mathbb{Z}[\delta] := \left\{ a + b \frac{1 + i\sqrt{19}}{2} \mid a, b \in \mathbb{Z} \right\}.$$

- Prove that the smallest values of $N(z)$ for $z = a + b\delta \in \mathbb{Z}[\delta]$ are 0, 1, 4, 5. Prove that $N(a + b\delta) \geq 5$ if $b \neq 0$.
- Prove that the units in $\mathbb{Z}[\delta]$ are ± 1 .
- If $c \in \mathbb{Z}[\delta]$ satisfies the condition specified in Exercise 2.17, prove that c must divide 2 or 3 in $\mathbb{Z}[\delta]$, and conclude that $c = \pm 2$ or $c = \pm 3$.
- Now show that $\nexists q \in \mathbb{Z}[\delta]$ such that $\delta = qc + r$ with $c = \pm 2, \pm 3$ and $r = 0, \pm 1$.

Conclude that $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is not a Euclidean domain. [§2.4, 6.14]

2.19. $\neg$ A *discrete valuation* on a field k is a surjective homomorphism of abelian groups $v : (k^*, \cdot) \rightarrow (\mathbb{Z}, +)$ such that $v(a + b) \geq \min(v(a), v(b))$ for all $a, b \in k^*$ such that $a + b \in k^*$.

- Prove that the set $R := \{a \in k^* \mid v(a) \geq 0\} \cup \{0\}$ is a subring of k .
- Prove that R is a Euclidean domain.

Rings arising in this fashion are called *discrete valuation rings*, abbreviated DVR. They arise naturally in number theory and algebraic geometry. Note that the Krull dimension of a DVR is 1 (Example III.4.14); in algebraic geometry, DVRs correspond to particularly nice points on a ‘curve’.

- Prove that the ring of rational numbers a/b with b not divisible by a fixed prime integer p is a DVR.

[2.20, VIII.1.19]

2.20. $\neg$ As seen in Exercise 2.19, DVRs are Euclidean domains. In particular, they must be PIDs. Check this directly, as follows. Let R be a DVR, and let $t \in R$ be an element such that $v(t) = 1$. Prove that if $I \subseteq R$ is any nonzero ideal, then $I = (t^k)$ for some $k \geq 1$. (The element t is called a ‘local parameter’ of R .) [4.13, VII.2.18]

2.21. $\triangleright$ Prove that an integral domain is a PID if and only if it admits a Dedekind-Hasse valuation. (Hint: For the $\Leftarrow$ implication, adapt the argument in Proposition 2.8; for $\Rightarrow$, let $v(a)$ be the size of the multiset of irreducible factors of a .) [§2.4]

2.22. $\neg$ Suppose $R \subseteq S$ is an inclusion of integral domains, and assume that R is a PID. Let $a, b \in R$, and let $d \in R$ be a gcd for a and b in R . Prove that d is also a gcd for a and b in S . [5.2]

2.23. Compute $d = \gcd(5504227617645696, 2922476045110123)$. Further, find a, b such that $d = 5504227617645696a + 2922476045110123b$.

2.24. $\triangleright$ Prove that there are infinitely many prime integers. (Hint: Assume by contradiction that $p_1, \dots, p_N$ is a complete list of all positive prime integers. What can you say about $p_1 \cdots p_N + 1$? This argument was already known to Euclid, more than 2,000 years ago.) [2.25, §5.2, 5.11]

2.25. $\neg$ Variation on the theme of Euclid from Exercise 2.24: Let $f(x) \in \mathbb{Z}[x]$ be a nonconstant polynomial such that $f(0) = 1$. Prove that infinitely many primes divide the numbers $f(n)$, as n ranges in $\mathbb{Z}$. (If $p_1, \dots, p_N$ were a complete list of primes dividing the numbers $f(n)$, what could you say about $f(p_1 \cdots p_N x)$?)

Once you are happy with this, show that the hypothesis $f(0) = 1$ is unnecessary. (If $f(0) = a \neq 0$, consider $f(p_1 \cdots p_N ax)$. Finally, note that there is nothing special about 0.) [VII.5.18]

3. Intermezzo: Zorn's lemma

3.1. Set theory, reprise. We leave ring theory for a moment and take a little detour to contemplate an issue from *set theory*. As remarked at the very outset, only naive set theory is used in this book; all set-theoretic operations we have used so far are nothing more than a formalization of intuitive ideas regarding collections of objects. However, I will occasionally need to refer to a less ‘intuitively obvious’ set-theoretic statement: for example, this statement is needed in order to show that every proper ideal in a ring is contained in a maximal ideal (Proposition 3.5).

This set-theoretic fact is *Zorn's lemma*. An *order relation* on a set Z is a relation $\preceq$ which is reflexive, transitive, and *antisymmetric*: the first two terms are familiar to the reader, and the third means that

$$(\forall a, b \in Z), \quad a \preceq b \text{ and } b \preceq a \implies a = b.$$

Typical prototypes are the $\leq$ relation on $\mathbb{Z}$ or the inclusion relation $\subseteq$ among subsets of a given set. We use $a \prec b$ to denote $a \preceq b$ and $b \neq a$.

A pair $(Z, \preceq)$, consisting of a set Z and an order relation $\preceq$ on Z , is called a *poset*, for *partially ordered set*. The qualifier ‘partially’ is not necessary, but convenient, as it reminds us that if $a, b \in Z$, then it is not necessarily the case that $a \preceq b$ or $b \preceq a$: for example, $\subseteq$ does not satisfy this additional requirement in general (while $(\mathbb{Z}, \leq)$ does). An order is *total* if it does satisfy this additional requirement. A *totally ordered set* is not called a ‘toset’, as would seem reasonable, but rather a *chain*.

An element m of a poset Z is *maximal* if nothing comes properly ‘after’ it in the order:

$$(\forall a \in Z), \quad m \preceq a \implies m = a.$$

For example, maximal ideals in a ring are maximal elements in the set of *proper* ideals, that is, ideals $\neq (1)$ (by Proposition III.4.11). An *upper bound* for a subset S of a poset Z is an element $u \in Z$ coming after every element of S :

$$(\forall a \in S), \quad a \preceq u.$$

Notions of ‘smallest’ (or, rather, ‘least’) or ‘largest’ are defined in the evident way.

Posets may or may not have maximal elements, upper bounds, etc.: for example, the family of *finite* subsets of an infinite set does *not* have maximal elements.

All this terminology comes together in the following statement:

Lemma 3.1 (Zorn’s lemma). *Let Z be a nonempty poset. Assume that every chain in Z has an upper bound in Z ; then there exists a maximal element in Z .*

The status of Zorn’s lemma is peculiar: on the one hand, it is complex enough that no one I know of finds it ‘intuitively clear’; on the other hand, it turns out to be logically *equivalent*¹⁰ to the axiom of choice, which *does* look reasonable to most people, and to the ‘well-ordering theorem’, which I find intuitively *unreasonable*. I will not prove all these equivalences (the diligent reader will prove them in the exercises and should in any case have no trouble locating more detailed proofs); but I will attempt to describe the situation in general terms and explain why the unreasonable statement implies the other two.

The *axiom of choice* states that if $\mathcal{F}$ is a family of disjoint nonempty subsets of a set Z , then we can form a new set by selecting one element x from each $X \in \mathcal{F}$. This may sound reasonable, but it raises rather subtle points: for example, it can be shown that this axiom is independent of the other axioms of (Zermelo-Fränkel) set theory. Also, it has disturbingly counterintuitive consequences, such as the Banach-Tarski paradox¹¹.

The subtlety of the axiom of choice boils down to the following question: *how does one choose the element x ?* This is not controversial if Z (and hence $\mathcal{F}$) is finite, but, not surprisingly, it becomes an issue if Z is infinite.

A suitable order relation on Z would come in handy here: if $\preceq$ is an order relation on Z such that every nonempty subset of Z has a least element, then we could simply let x be the least element of X for each $X \in \mathcal{F}$. We say that Z is *well-ordered* by $\preceq$, or that $\preceq$ is a *well-ordering* on Z , if this is the case. (The abbreviation *woset* is also used in the literature, but not very often.) For example, the set $\mathbb{Z}^{>0}$ of positive numbers *is* well-ordered by $\leq$; this fact is called the *well-ordering principle*.

Thus, the statement of the axiom of choice is *really* ‘clear’ if the set Z is the set $\mathbb{Z}^{>0}$ of positive numbers. It may seem somewhat less so for the set $\mathbb{Z}$, since $\mathbb{Z}$ is not well-ordered by $\leq$; however, it takes a moment (Exercise 3.4) to construct a *different* relation on $\mathbb{Z}$, making $\mathbb{Z}$ a well-ordered set. Thus, the axiom of choice is also completely transparent for $Z = \mathbb{Z}$ or any countable set (like $\mathbb{Q}$) for that matter.

The well-ordering principle is at the basis of proofs by induction: in fact (cf. Exercise 3.5) it is equivalent to the so-called *principle of induction*. More is true, however. If $(Z, \preceq)$ is *any* woset, then we can consider the following ‘principle of induction’ on Z :

¹⁰ *Equivalent* in the sense that each can be derived from the other together with the other axioms of ‘Zermelo-Fränkel’ set theory.

¹¹ One can subdivide a solid ball of radius 1 into finitely many pieces, then reassemble them after rotations and translations in 3-space and obtain *two* balls of radius 1.

Let $S \subseteq Z$ be a subset such that $\forall a \in Z$

$$((\forall b \in Z), \quad b \prec a \implies b \in S) \implies a \in S;$$

then $S = Z$.

That is, Z is the only set S with the property that $a \in S$ if all $b \prec a$ are in S . (Note that the least element a of Z is automatically in S , since the condition on all $b \prec a$ is vacuously true in this case.)

The reader will recognize that for $Z = \mathbb{Z}^{>0}$ with the relation $\leq$, this is the ordinary principle of induction.

Claim 3.2. *Let Z be any woset. Then the principle of induction holds for Z .*

Proof. Let $S \subseteq Z$ be a subset with the property given above, and assume $S \subsetneq Z$. Then the complement T of S in Z is nonempty; hence it has a least element a . Now if $b \prec a$, then necessarily $b \in S$ (otherwise $b \in T$, contradicting the fact that a is the least element in T). But the property of S then forces $a \in S$, contradiction. Therefore T is empty; i.e., $S = Z$. $\square$

This is remarkable, since it extends induction to uncountable sets¹², provided a well-ordering is available.

For example, if we had a well-ordering on $\mathbb{R}$, then we could prove a statement about all reals ‘by induction’. Many people find this somewhat counterintuitive, and hence so seems the following amazing claim:

Theorem 3.3 (Well-ordering theorem). *Every set admits a well-ordering.*

As argued above, this implies that the axiom of choice holds on every set: if you accept the well-ordering theorem, the statement of the axiom of choice becomes as evident for every set as it is for the set of positive integers. The catch is of course that the axiom of choice is *used* in the proof of the well-ordering theorem; in fact, the statements are equivalent.

In any case, the statement of the well-ordering theorem is easy to absorb (even if perhaps less ‘intuitively clear’ than the axiom of choice). The well-ordering theorem is equivalent to Zorn’s lemma, since they are both equivalent to the axiom of choice. The good news is that the derivation of Zorn’s lemma from the well-ordering theorem is reasonably straightforward:

Well-ordering theorem $\implies$ Zorn’s lemma. Let $(Z, \leq)$ be a nonempty poset such that every chain in Z has an upper bound in Z . By the well-ordering theorem, there is a well-ordering¹³ $\preceq$ on Z . Define a function f from Z to the power set of Z as

¹²Even beyond; in the context of *ordinals* this induction principle is called *transfinite induction*.

¹³Watch out: we are considering *two* orderings on Z : $\leq$ (about which we want to say something) and $\preceq$ (about which we know something already).

follows:

$$f(a) = \begin{cases} \{a\} & \text{if } (\{a\} \cup \bigcup_{b \prec a} f(b)) \text{ is totally ordered by } \leq; \\ \emptyset & \text{if } (\{a\} \cup \bigcup_{b \prec a} f(b)) \text{ is not totally ordered by } \leq. \end{cases}$$

The fact that Z is well-ordered by $\preceq$ implies that f is defined for every $a \in Z$. Indeed, let T be the set of elements of Z for which f is not defined; if $T \neq \emptyset$, T has a least element a w.r.t. $\preceq$; but then $f(b)$ is defined for all $b \prec a$, and the prescription given for $f(a)$ defines f at a , a contradiction.

Let $S = \bigcup_{a \in Z} f(a)$. It is clear that S is totally ordered by $\leq$: if $a, b \in S$, and (say) $b \prec a$, then both a and b belong to

$$\{a\} \cup \bigcup_{b \prec a} f(b),$$

which is totally ordered by $\leq$ by construction.

I claim that S is a *maximal* totally ordered subset¹⁴ of Z :

Claim 3.4. *If $S \subseteq S' \subseteq Z$ and S' is totally ordered by $\leq$, then $S = S'$.*

Indeed, let T be the complement of S in S' . If T is nonempty, let $a \in T$ and observe that

$$\{a\} \cup \bigcup_{b \prec a} f(b)$$

is totally ordered by $\leq$, since it is a subset of $\{a\} \cup S \subseteq S'$. But then $f(a) = \{a\}$, that is, $a \in S$, a contradiction.

Since S is a chain, by the hypothesis of Zorn's lemma S has an upper bound, m . It is now clear that m must be maximal in Z w.r.t. $\leq$, verifying the statement of Zorn's lemma. Indeed, if $m' \geq m$, then $S \cup \{m'\}$ is totally ordered; hence $S = S \cup \{m'\}$ by the claim. This means $m' \in S$, and hence $m' = m$ since m is an upper bound for S . $\square$

Zorn's lemma is the key to several basic results in algebra, the first of which we are about to encounter; the reader will sample a few more results in the exercises. The reader will also encounter equally basic applications of Zorn's lemma (or of other manifestations of the axiom of choice) in other fields: for example Tychonoff's theorem in topology and the Hahn-Banach theorem in functional analysis.

3.2. Application: Existence of maximal ideals. Recall (§III.4.3) that an ideal $\mathfrak{m}$ of a ring R is *maximal* if and only if $R/\mathfrak{m}$ is a field, if and only if no other ideal stands between $\mathfrak{m}$ and $R = (1)$, that is, if and only if $\mathfrak{m}$ is maximal with respect to inclusion, in the family of proper ideals of R . It is not obvious from this definition that maximal ideals *exist*, but they do:

Proposition 3.5. *Let $I \neq (1)$ be a proper ideal of a commutative ring R . Then there exists a maximal ideal $\mathfrak{m}$ of R containing I .*

¹⁴The existence of maximal totally ordered subsets is known as the *Hausdorff maximal principle*; it is also equivalent to the axiom of choice.

This is immediate if R is Noetherian, by applying condition (3) from Proposition 1.1 to the family of proper ideals of R containing I . The argument for arbitrary rings is a classic application of Zorn's lemma. In fact, this is the first application given by M. Zorn in his 1935 article introducing a 'maximum principle' (now known as Zorn's lemma). The result had earlier been proven by Krull, using the well-ordering theorem.

Proof. The set $\mathcal{J}$ of proper ideals of R containing I is ordered by inclusion. Then let $\mathcal{C}$ be a chain of proper ideals, and consider

$$U := \bigcup_{J \in \mathcal{C}} J.$$

I claim that U is a proper ideal containing I ; hence it is an upper bound for $\mathcal{C}$ in $\mathcal{J}$. This proves that every chain in $\mathcal{J}$ has an upper bound, and it follows that $\mathcal{J}$ has maximal elements, by Zorn's lemma.

To verify my claim, it is clear that U contains I and that it is an ideal (for example, if $a, b \in U$, then $\exists J \in \mathcal{C}$ such that $a, b \in J$; hence $a \pm b \in J$, and therefore $a \pm b \in U$). We have to check that U is *proper*. But if $U = (1)$, then $1 \in J$ for some $J \in \mathcal{J}$, contradicting the fact that $\mathcal{J}$ consists of proper ideals. $\square$

Note that this argument relies crucially on the fact that R has a multiplicative unit 1, and indeed the stated fact is *not* true in general for rings without 1 (Exercise 3.9). Also, the use of Zorn's lemma is not just a convenient trick; the statement is known to be equivalent to the axiom of choice, by work of W. Hodges.

Exercises

3.1. Prove that every well-ordering is total.

3.2. Prove that a totally ordered set $(Z, \preceq)$ is a woset if and only if every descending chain

$$z_1 \succeq z_2 \succeq z_3 \succeq \cdots$$

in Z stabilizes.

3.3. Prove that the axiom of choice is equivalent to the statement that a set-function is surjective if and only if it has a right-inverse (cf. Exercise I.2.2).

3.4. $\triangleright$ Construct explicitly a well-ordering on $\mathbb{Z}$. Explain why you know that $\mathbb{Q}$ can be well-ordered, even without performing an explicit construction. [§3.1]

3.5. $\triangleright$ Prove that the (ordinary) principle of induction is equivalent to the statement that $\leq$ is a well-ordering on $\mathbb{Z}^{>0}$. (To prove by induction that $(\mathbb{Z}^{>0}, \leq)$ is well-ordered, assume it is known that 1 is the least element of $\mathbb{Z}^{>0}$ and that $\forall n \in \mathbb{Z}^{>0}$ there are no integers between n and $n + 1$.) [§3.1]

3.6. In this exercise assume the truth of Zorn's lemma and the conventional set-theoretic constructions; you will be proving the well-ordering theorem.

Let Z be a nonempty set, and let $\mathcal{Z}$ be the set of pairs $(S, \leq)$ consisting of a subset S of Z and of a *well-ordering* $\leq$ on S . Note that $\mathcal{Z}$ is not empty (singletons can be well-ordered). Define a relation $\preceq$ on $\mathcal{Z}$ by prescribing

$$(S, \leq) \preceq (T, \leq')$$

if and only if $S \subseteq T$, $\leq$ is the restriction of $\leq'$ to S , and every element of S precedes every element of $T \setminus S$ w.r.t. $\leq'$.

- Prove that $\preceq$ is an order relation in $\mathcal{Z}$.
- Prove that every chain in $\mathcal{Z}$ has an upper bound in $\mathcal{Z}$.
- Use Zorn's lemma to obtain a maximal element $(M, \leq)$ in $\mathcal{Z}$. Prove that $M = Z$.

Thus every set admits a well-ordering, as stated in Theorem 3.3.

3.7. In this exercise assume the truth of the axiom of choice and the conventional set-theoretic constructions; you will be proving the well-ordering theorem¹⁵.

Let Z be a nonempty set. Use the axiom of choice to choose an element $\gamma(S) \notin S$ for each proper subset $S \subsetneq Z$. Call a pair $(S, \leq)$ a γ -*woset* if $S \subseteq Z$, $\leq$ is a well-ordering on S , and for every $a \in S$, $a = \gamma(\{b \in S, b < a\})$.

- Show how to begin constructing a γ -woset, and show that all γ -wosets must begin in the same way.

Define an ordering on γ -wosets by prescribing that $(U, \leq'') \preceq (T, \leq')$ if and only if $U \subseteq T$ and $\leq''$ is the restriction of $\leq'$.

- Prove that if $(U, \leq'') \prec (T, \leq')$, then $\gamma(U) \in T$.
- For two γ -wosets $(S, \leq)$ and $(T, \leq')$, prove that there is a maximal γ -woset $(U, \leq'')$ preceding both w.r.t. $\preceq$. (Note: There is no need to use Zorn's lemma!)
- Prove that the maximal γ -woset found in the previous point in fact equals $(S, \leq)$ or $(T, \leq')$. Thus, $\preceq$ is a total ordering.
- Prove that there is a maximal γ -woset $(M, \leq)$ w.r.t. $\preceq$. (Again, Zorn's lemma need not and should not be invoked.)
- Prove that $M = Z$.

Thus every set admits a well-ordering, as stated in Theorem 3.3.

3.8. Prove that every nontrivial finitely generated group has a maximal proper subgroup. Prove that $(\mathbb{Q}, +)$ has no maximal proper subgroup.

3.9. $\triangleright$ Consider the rng (= ring without 1; cf. §III.1.1) consisting of the abelian group $(\mathbb{Q}, +)$ endowed with the trivial multiplication $qr = 0$ for all $q, r \in \mathbb{Q}$. Prove that this rng has no maximal ideals. [§3.2]

3.10. $\neg$ As shown in Exercise III.4.17, every maximal ideal in the ring of continuous real-valued functions on a *compact* topological space K consists of the functions vanishing at a point of K .

¹⁵I learned this proof from notes of Dan Grayson.

Prove that there are maximal ideals in the ring of continuous real-valued functions on the *real line* that do not correspond to points of the real line in the same fashion. (Hint: Produce a proper ideal that is not contained in any maximal ideal corresponding to a point, and apply Proposition 3.5.) [III.4.17]

3.11. Prove that a UFD R is a PID if and only if every nonzero prime ideal in R is maximal. (Hint: One direction is Proposition III.4.13. For the other, assume that every nonzero prime ideal in a UFD R is maximal, and prove that every maximal ideal in R is principal; then use Proposition 3.5 to relate arbitrary ideals to maximal ideals, and prove that every ideal of R is principal.)

3.12. $\neg$ Let R be a commutative ring, and let $I \subseteq R$ be a proper ideal. Prove that the set of prime ideals containing I has minimal elements. (These are the *minimal primes* of I .) [1.9]

3.13. $\neg$ Let R be a commutative ring, and let N be its nilradical (Exercise III.3.12). Let $r \notin N$.

- Consider the family $\mathcal{F}$ of ideals of R that do not contain any power r^k of r for $k > 0$. Prove that $\mathcal{F}$ has maximal elements.
- Let I be a maximal element of $\mathcal{F}$. Prove that I is prime.
- Conclude $r \notin N \implies r$ is not in the intersection of all prime ideals of R .

Together with Exercise III.4.18, this shows that the nilradical of a commutative ring R equals the intersection of all prime ideals of R . [III.4.18, VII.2.8]

3.14. $\neg$ The *Jacobson radical* of a commutative ring R is the intersection of the maximal ideals in R . (Thus, the Jacobson radical contains the nilradical.) Prove that r is in the Jacobson radical if and only if $1 + rs$ is invertible for every $s \in R$. [VI.3.8]

3.15. Recall that a (commutative) ring R is Noetherian if every ideal of R is finitely generated. Assume the seemingly weaker condition that every *prime* ideal of R is finitely generated. Let $\mathcal{F}$ be the family of ideals that are not finitely generated in R . You will prove $\mathcal{F} = \emptyset$.

- If $\mathcal{F} \neq \emptyset$, prove that it has a maximal element I .
- Prove that R/I is Noetherian.
- Prove that there are ideals J_1, J_2 containing I , such that $J_1 J_2 \subseteq I$.
- Give a structure of R/I module to $I/J_1 J_2$ and $J_1/J_1 J_2$.
- Prove that $I/J_1 J_2$ is a finitely generated R/I -module.
- Prove that I is finitely generated, thereby reaching a contradiction.

Thus, a ring is Noetherian if and only if its *prime* ideals are finitely generated.

4. Unique factorization in polynomial rings

We now return to regular programming and study unique factorization in polynomial rings; we will finally establish the fact (already hinted at) that $R[x]$ is a UFD if R is a UFD.

Among the necessary preparatory work we also discuss the *field of fractions* of an integral domain; this is a particular instance of the process of *localization* of a ring (or a module) at a multiplicative subset, which the diligent reader will explore a little more in the exercises.

4.1. Primitivity and content; Gauss's lemma. One issue that we have encountered already, and will encounter again, is the description of the *ideals* of the polynomial ring $R[x]$, given enough information about R . For example, we have proved that the ideals of $k[x]$ are principal if k is a field, and Hilbert's basis theorem shows that all ideals of $R[x]$ are finitely generated if all ideals of R are (Exercise III.4.4 and Lemma 1.3).

An even more naive observation is simply that every ideal I of R generates an ideal of $R[x]$:

$$IR[x] := \{a_0 + a_1x + \cdots + a_dx^d \in R[x] \mid \forall i, a_i \in I\}.$$

Lemma 4.1. *Let R be a ring, and let I be an ideal of R . Then*

$$\frac{R[x]}{IR[x]} \cong \frac{R}{I}[x].$$

The proof of this lemma is a standard application of the first isomorphism theorem and is left to the reader (Exercise 4.1).

Corollary 4.2. *If I is a prime ideal of R , then $IR[x]$ is prime in $R[x]$.*

Proof. If I is prime in R , then R/I is an integral domain; hence so is $R[x]/IR[x] \cong (R/I)[x]$, and therefore $IR[x]$ is prime in $R[x]$. $\square$

We will use this fact in a moment.

The following definitions can be studied for every commutative ring; our main application will be to UFDs.

Definition 4.3. Let R be a commutative ring, and let

$$f = a_0 + a_1x + \cdots + a_dx^d \in R[x]$$

be a polynomial.

- f is *very primitive* if for all prime ideals $\mathfrak{p}$ of R , $f \notin \mathfrak{p}R[x]$.
- f is *primitive* if for all *principal* prime ideals $\mathfrak{p}$ of R , $f \notin \mathfrak{p}R[x]$. $\lrcorner$

The notion of ‘primitive’ polynomial is standard, but it is not usually presented in this way; the standard definition is the equivalent formulation given in Lemma 4.5. As for ‘very primitive’, this term is essentially a joke—my excuse for bringing up this notion is that it is very natural and that unfortunately some references blur the distinction between this notion and the notion of ‘primitive’. I hope to ward off any possible confusion by being rather explicit on this point. *Very primitive* polynomials are primitive, but the converse does not hold in general, even for UFDs (cf. Exercise 4.3).

Perhaps the most important fact about ‘primitivity’ is the following easy remark.

Lemma 4.4. *Let R be a commutative ring. Then for $f, g \in R[x]$*

$$fg \text{ is primitive} \iff \text{both } f \text{ and } g \text{ are primitive.}$$

Proof. This is an easy consequence of Corollary 4.2:

$$\begin{aligned} fg \text{ primitive} &\iff \forall \mathfrak{p} \text{ prime and principal in } R, fg \notin \mathfrak{p}R[x] \\ &\iff \forall \mathfrak{p} \text{ prime and principal in } R, f \notin \mathfrak{p}R[x] \text{ and } g \notin \mathfrak{p}R[x] \\ &\iff f \text{ is primitive and } g \text{ is primitive} \end{aligned}$$

since $\mathfrak{p}R[x]$ is prime if $\mathfrak{p}$ is a prime ideal. □

The analogous equivalence holds for *very* primitive polynomials (Exercise 4.4). Lemma 4.4 will be ultimately responsible for the fact that $R[x]$ is a UFD if R is a UFD, which is my main objective in this section. If R is a UFD, the notion of ‘primitive’ has the following interpretation; I accompany it with the ‘very primitive’ case, for comparison.

Lemma 4.5. *Let R be a commutative ring and $f = a_0 + a_1x + \cdots + a_dx^d \in R[x]$ as above.*

- *f is very primitive if and only if $(a_0, \dots, a_d) = (1)$.*
- *If R is a UFD, then f is primitive if and only if $\gcd(a_0, \dots, a_d) = 1$.*

Proof. If $(a_0, \dots, a_d) = (1)$, then no prime ideal can contain all coefficients a_i , and it follows that f is very primitive. Conversely, if f is very primitive, then the coefficients of f are not all contained in any one prime ideal, and in particular they are not all contained in any maximal ideal (since maximal ideals are prime). Thus, $(a_0, \dots, a_d) = (1)$ in this case, since every proper ideal is contained in a maximal ideal by Proposition 3.5. This proves the first point.

For the second point note that, in a UFD, $\gcd(a_0, \dots, a_d) \neq 1$ if and only if there exists an irreducible element $q \in R$ such that $(a_0, \dots, a_d) \subseteq (q)$. As (q) is then prime (by Lemma 2.4), the second point follows as well. □

With this in mind, in order to capitalize on Lemma 4.4, it is convenient to give a name to the gcd of the coefficients of a polynomial. I now assume that R is a UFD, since this is the case of greatest interest in the applications.

Definition 4.6. Let R be a UFD. The *content* of a nonzero polynomial $f \in R[x]$, denoted cont_f , is the gcd of its coefficients. ┘

Of course the content of a polynomial is only defined up to units. I find this ambiguity distasteful. What is uniquely determined is the *principal ideal* generated by the content of f , which I will denote

$$(\text{cont}_f) :$$

thus f is primitive precisely when $(\text{cont}_f) = (1)$. I will take a somewhat stubborn approach and deal with principal ideals throughout, rather with individual (but only defined up to unit) elements. The reader should keep in mind that ideals may be multiplied (cf. §III.4.1), and if (a) , (b) are principal ideals, then their product $(a)(b)$ is the principal ideal (ab) .

The proof of the following remarks is immediate from the definition; hence it is left to the reader (Exercise 4.5):

Lemma 4.7. *Let R be a UFD, and let $f \in R[x]$. Then*

- $(f) = (\text{cont}_f)(\underline{f})$, where $\underline{f}$ is primitive;
- if $(f) = (c)(g)$, with $c \in R$ and g primitive, then $(c) = (\text{cont}_f)$.

In view of its consequences, the following fact is rather profound, and it therefore deserves a name. Some references call the result of Lemma 4.4, or its main consequence (Theorem 4.14), *Gauss's lemma*. We prefer to use this name for the following statement.

Proposition 4.8 (Gauss's lemma). *Let R be a UFD, and let $f, g \in R[x]$. Then*

$$(\text{cont}_{fg}) = (\text{cont}_f)(\text{cont}_g).$$

Proof. This follows easily from our preparatory work. Write

$$(fg) = ((\text{cont}_f)(\underline{f}))((\text{cont}_g)(\underline{g})) = (\text{cont}_f)(\text{cont}_g)(\underline{fg}),$$

with $\underline{f}, \underline{g}$ primitive. By Lemma 4.4, $\underline{fg}$ is primitive; by Lemma 4.7 it follows that $(\text{cont}_f)(\text{cont}_g)$ is the content of fg , as needed. $\square$

Note the following immediate consequence:

Corollary 4.9. *Let R be a UFD, and let $f, g \in R[x]$. Assume $(f) \subseteq (g)$. Then $(\text{cont}_f) \subseteq (\text{cont}_g)$.*

4.2. The field of fractions of an integral domain. Gauss's lemma is the key to the important observation that if R is a UFD, then so is $R[x]$. However, we need one more tool before we can prove this fact; this is one instance of the important process of *localization*.

In the case we need for our immediate goal, the process starts from any integral domain and produces a *field*, in precisely the same way the field $\mathbb{Q}$ of rational numbers may be obtained from the integral domain $\mathbb{Z}$. This construction is known as the *field of fractions* (or *field of quotients*) of the integral domain R . It satisfies the following beautiful universal property.

Given an integral domain R , consider the category $\mathcal{R}$ whose objects are pairs

$$(i, K),$$

where K is a field and $i : R \hookrightarrow K$ is an injective ring homomorphism. Morphisms are defined in the style of every analogous construction we have encountered: that is, a morphism $(i, K) \rightarrow (j, L)$ is determined by a homomorphism of fields $\alpha : K \rightarrow L$ making the following diagram

$$\begin{array}{ccc} K & \xrightarrow{\alpha} & L \\ & \swarrow i \quad \searrow j & \\ & R & \end{array}$$

commute.

Note that α , as every homomorphism of *fields*, is necessarily injective (Exercise III.3.10); this is compatible with the requirement that $i : R \hookrightarrow K$ be injective to begin with. The injectivity of $R \hookrightarrow K$ also forces R to be an integral domain, since subrings of fields are necessarily integral domains.

Definition 4.10. The *field of fractions* $K(R)$ of R is an initial object of the category $\mathcal{R}$. ┘

Thus, $K(R)$ is the ‘smallest field containing R ’.

The usual caveats apply to any such definition: the initial object of $\mathcal{R}$ carries not only the information of a field K (the field of fractions proper), but also of a specific realization of R as a subring of K ; this distinction is blurred in common use of the language. Also, of course this prescription only defines the field of fractions up to isomorphism (as is true of any universal object; cf. Proposition I.5.4).

Finally, just stating the universal property does not guarantee that the sought-for initial object exists. Thus, our next task is the construction of an initial object for $\mathcal{R}$. Fortunately, the construction is essentially straightforward: we just need to formalize a notion of ‘fraction’ of elements of R .

Consider the set $R \times (R^*)$ of pairs (a, r) of elements of R , where $r \neq 0$. The pair (a, r) will be associated with a ‘fraction’ $\frac{a}{r}$; the reader would be well-advised to put this book away now and carry out the construction of $K(R)$ on his/her own, profiting from this major hint.

Here is the construction. Denote by $\frac{a}{r}$ the equivalence class of $(a, r) \in R \times (R^*)$ with respect to the following equivalence relation:

$$(a, r) \sim (b, s) \iff as - br = 0.$$

The reader will verify that this is indeed an equivalence relation. As a set, $K(R)$ is defined by

$$K(R) := \left\{ \frac{a}{r} \mid a \in R, r \in R, r \neq 0 \right\}.$$

It is clear that these ‘fractions’ behave much as ordinary fractions. For example,

$$\frac{as}{rs} = \frac{a}{r}$$

if $s \neq 0$: indeed,

$$(as)r = a(rs)$$

by associativity and commutativity in R , and this shows

$$(as, rs) \sim (a, r)$$

as needed.

We define operations on $K(R)$ as follows:

$$\begin{aligned} \frac{a}{r} + \frac{b}{s} &= \frac{as + br}{rs}, \\ \frac{a}{r} \cdot \frac{b}{s} &= \frac{ab}{rs}. \end{aligned}$$

Of course one must verify that these operations are well-defined; this is also left to the reader. The fact that R is an integral domain is used in these definitions: it guarantees that $rs \neq 0$ if $r \neq 0$ and $s \neq 0$.

Now I claim that $K(R)$ is made into a field by these operations. This is a straightforward verification; for example, distributivity amounts to the following computation:

$$\begin{aligned} \frac{a}{r} \left(\frac{b}{s} + \frac{c}{t} \right) &= \frac{a}{r} \frac{(bt + cs)}{st} = \frac{a(bt + cs)}{r(st)} = \frac{a(bt)}{r(st)} + \frac{a(cs)}{r(st)} = \frac{ab}{rs} + \frac{ac}{rt} \\ &= \frac{a}{r} \frac{b}{s} + \frac{a}{r} \frac{c}{t}. \end{aligned}$$

The zero element is the fraction $\frac{0}{1}$ (or in fact $\frac{0}{r}$ for any nonzero $r \in R$); the multiplicative identity is the fraction $\frac{1}{1}$ (or in fact $\frac{r}{r}$ for any nonzero $r \in R$). Since

$$\frac{r}{s} \frac{s}{r} = \frac{rs}{rs} = \frac{1}{1}$$

for all $\frac{r}{s} \neq \frac{0}{1}$ (that is, for all $r \neq 0$), every nonzero element in $K(R)$ has an inverse, as promised.

An ‘inclusion map’ $i : R \hookrightarrow K(R)$ is defined by

$$a \mapsto \frac{a}{1};$$

it is immediately checked that this is an injective ring homomorphism. It is common to use this map to identify R with its isomorphic copy inside $K(R)$ and simply view R as a subring of $K(R)$.

Claim 4.11. $(i, K(R))$ is initial in $\mathcal{R}$.

Proof. Let $j : R \hookrightarrow L$ be any injective ring homomorphism from R to a field L . We need to define an induced homomorphism $\hat{j} : K(R) \rightarrow L$ so that the diagram

$$\begin{array}{ccc} K & \xrightarrow{\hat{j}} & L \\ & \searrow i \quad \nearrow j & \\ & R & \end{array}$$

commutes, and we must show that $\hat{j}$ is unique. Now, the definition of $\hat{j}$ is in fact forced upon us: if $\hat{j}$ exists as a homomorphism, then necessarily

$$\begin{aligned} \hat{j} \left(\frac{a}{r} \right) &= \hat{j} \left(\frac{a}{1} \right) \hat{j} \left(\left(\frac{r}{1} \right)^{-1} \right) = \hat{j} \left(\frac{a}{1} \right) \hat{j} \left(\frac{r}{1} \right)^{-1} = (\hat{j} \circ i(a))(\hat{j} \circ i(r)^{-1}) \\ &= j(a)j(r)^{-1}. \end{aligned}$$

Thus $\hat{j}$ is indeed unique, if it exists. On the other hand, the prescription

$$\hat{j} \left(\frac{a}{r} \right) := j(a)j(r)^{-1}$$

does define a function $K(R) \rightarrow L$: indeed, if $(a, r) \sim (b, s)$, then

$$as = br$$

in R , hence

$$j(a)j(s) = j(b)j(r)$$

in L , and (note that $j(r), j(s)$ are nonzero in L since r, s are nonzero in R and j is injective)

$$j(a)j(r)^{-1} = j(b)j(s)^{-1},$$

showing that the proposed $\hat{j}$ is well-defined. The reader will verify that it is a ring homomorphism, concluding the proof of the claim. $\square$

Example 4.12. With the notation introduced above, $K(\mathbb{Z}) = \mathbb{Q}$.

The universal property implies immediately that $F \hookrightarrow K(F)$ is an isomorphism if F is itself a field. Thus, the construction adds nothing to $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/p\mathbb{Z}$, etc.

If R is any integral domain, so that $R[x]$ is also an integral domain, $K(R[x])$ is a famous field:

Definition 4.13. The field of *rational functions* with coefficients in R is the field of fractions of the ring $R[x]$. This field is denoted $R(x)$. $\lrcorner$

Elements of $R(x)$ are fractions of polynomials

$$\frac{p(x)}{q(x)}$$

with $p(x), q(x) \in R[x]$ and $q(x) \neq 0$. The term *function* is inaccurate, since the ‘function’ $R \rightarrow R$ given by $a \mapsto \frac{p(a)}{q(a)}$ is not defined for all $a \in R$ (not for those for which $q(a) = 0$, that is); and the function itself does not suffice to determine the element in $R(x)$ (cf. Exercise III.2.7). $\lrcorner$

4.3. R UFD $\implies R[x]$ UFD. We are now in a position to prove the analogue of Hilbert’s basis theorem for unique factorization domains, that is,

Theorem 4.14. *Let R be a UFD; then $R[x]$ is a UFD.*

For example, this result (and an immediate induction) shows that the rings $\mathbb{Z}[x_1, \dots, x_n]$ and $k[x_1, \dots, x_n]$ (for k a field) are UFDs. Theorem 4.14 is also often called *Gauss’s lemma*.

As a measure of how delicate the statement of Theorem 4.14 is, note that the power series ring $R[[x]]$ is *not* necessarily a UFD if R is a UFD; examples of this phenomenon are however not easy to construct. Of course $k[[x]]$ is a UFD if k is a field, since $k[[x]]$ is a Euclidean domain in this case (Exercise 2.14).

By Theorem 2.5, in order to prove Theorem 4.14, we have to verify that $R[x]$ satisfies the a.c.c. for principal ideals and that every irreducible element in $R[x]$ is prime, provided that R is itself a UFD. The general idea is to reduce these questions to matters in $K[x]$, where $K = K(R)$ is the field of fractions of R : as we know, $K[x]$ is a UFD (in fact it is a Euclidean domain, and Euclidean domain $\implies$ PID $\implies$ UFD, as shown in §2).

The following lemma captures the most crucial ingredient of the interaction between $R[x]$ and $K[x]$:

Lemma 4.15. *Let R be a UFD, and let $K = K(R)$ be its field of fractions. For nonzero $f, g \in R[x]$, denote by (f) , (g) the principal ideals $fR[x]$, $gR[x]$ in $R[x]$, and denote by $(f)_K$, $(g)_K$ the principal ideals $fK[x]$, $gK[x]$ in $K[x]$. Assume*

- $(\text{cont}_g) \subseteq (\text{cont}_f)$ and
- $(g)_K \subseteq (f)_K$.

Then $(g) \subseteq (f)$.

Proof. Since $(g)_K \subseteq (f)_K$, we have $g = fh$, where $h \in K[x]$. Write $h = \frac{a}{b}\underline{h}$, where $a, b \in R$ and $\underline{h} \in R[x]$ is a *primitive* polynomial: this can be done by collecting common denominators in h , then applying the first point of Lemma 4.7. We then have

$$bg = af\underline{h}$$

in $R[x]$. By Gauss's lemma and since $\underline{h}$ is primitive,

$$(a \text{cont}_f) = (b \text{cont}_g);$$

and since $(\text{cont}_g) \subseteq (\text{cont}_f)$ by hypothesis, we obtain

$$(a \text{cont}_f) \subseteq (b \text{cont}_f).$$

Since R is an integral domain and $(\text{cont}_f) \neq (0)$, this implies

$$a = bc$$

for some $c \in R$. But then $h = \frac{a}{b}\underline{h} = c\underline{h} \in R[x]$, and $g = fh \in (f)$; that is, $(g) \subseteq (f)$ in $R[x]$, as needed. $\square$

The first application is the following description of the irreducible elements of $R[x]$; this will also be used in the proof of Theorem 4.14 and is independently interesting.

Proposition 4.16. *Let R be a UFD, and let K be its field of fractions. Let $f \in R[x]$ be a nonconstant, irreducible polynomial. Then f is irreducible as an element of $K[x]$.*

Proof. First note that f is primitive: otherwise we could factor out its content, and f would not be irreducible.

Next, assume $f = gh$, with $g, h \in K[x]$; we have to prove that either g or h is a unit in $K[x]$. Let $c, d \in K$ such that

$$g = c\underline{g}, \quad h = d\underline{h},$$

and $\underline{g}$, $\underline{h}$ are *primitive* polynomials in $R[x]$. By Lemma 4.4, $\underline{gh}$ is also primitive; thus $(\text{cont}_{\underline{gh}}) = (1) = (\text{cont}_f)$; further,

$$(f)_K = (\underline{gh})_K$$

as $cd \neq 0$ is a unit in K . By Lemma 4.15 we obtain

$$(f) = (\underline{gh})$$

as ideals of $R[x]$; that is, $f = u\underline{gh}$ with $u \in R[x]$ a unit. As f is irreducible in $R[x]$, this implies that either $\underline{g}$ or $\underline{h}$ is a unit in $R[x]$. But then g or h were units in $K[x]$, verifying that f is irreducible in $K[x]$. $\square$

I have always found this fact almost counterintuitive: K is ‘larger’ than R , so one may expect that it should be ‘easier’ to factor polynomials over K than over R . Proposition 4.16 tells us that this is not the case: with due attention to special cases, irreducibility in $R[x]$ is ‘the same as’ irreducibility in $K[x]$. To be precise,

Corollary 4.17. *Let R be a UFD and K the field of fractions of R . Let $f \in R[x]$ be a nonconstant polynomial. Then f is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$ and primitive.*

The proof amounts to tying up loose ends, and I leave it to the reader (Exercise 4.21).

We can now prove the main result of this section. We will make systematic use of the characterization of UFDs found in Theorem 2.5.

Proof of Theorem 4.14. We begin by verifying the a.c.c. for principal ideals in $R[x]$. Let

$$(f_1) \subseteq (f_2) \subseteq (f_3) \subseteq \cdots$$

be an ascending chain of principal ideals of $R[x]$. By Corollary 4.9, this induces an ascending chain of principal ideals

$$(\text{cont}_{f_1}) \subseteq (\text{cont}_{f_2}) \subseteq (\text{cont}_{f_3}) \subseteq \cdots$$

in R ; since R is a UFD, this chain stabilizes: that is, $(\text{cont}_{f_i}) = (\text{cont}_{f_{i+1}})$ for¹⁶ $i \gg 0$. On the other hand, with notation as in Lemma 4.15 we have

$$(f_1)_K \subseteq (f_2)_K \subseteq (f_3)_K \subseteq \cdots$$

as a sequence of ideals in $K[x]$; since $K[x]$ is a UFD (because it is a PID; cf. §2.3) this sequence stabilizes. Therefore, $(f_i)_K = (f_{i+1})_K$ for $i \gg 0$.

By Lemma 4.15, $(f_i) = (f_{i+1})$ for $i \gg 0$; that is, the given chain of principal ideals stabilizes, as needed.

Next, we consider an irreducible element f of $R[x]$. We verify that (f) is a prime ideal; by Theorem 2.5 it then follows that $R[x]$ is a UFD, as stated.

If f is irreducible and constant, then f is prime in R as R is a UFD, and it follows that f is prime in $R[x]$ (by Corollary 4.2). Thus we may assume that f is nonconstant and irreducible (and in particular primitive) in $R[x]$.

By Proposition 4.16, f is irreducible as an element of $K[x]$; since $K[x]$ is a PID, $(f)_K$ is prime in $K[x]$. Consider the composition

$$\rho : R[x] \hookrightarrow K[x] \twoheadrightarrow \frac{K[x]}{(f)_K} :$$

I claim that $\ker \rho = (f)$. Indeed, the inclusion $\supseteq$ is trivial; for the other inclusion, note that $\rho(g) = 0$ implies that g is divisible by f in $K[x]$: that is, $(g)_K \subseteq (f)_K$; and we have $(\text{cont}_g) \subseteq (\text{cont}_f)$ since $(\text{cont}_f) = (1)$ as f is primitive. By Lemma 4.15, we obtain that $(g) \subseteq (f)$, i.e., g is divisible by f in $R[x]$, as needed. Since $\ker \rho = (f)$, we find that ρ induces an *injective* homomorphism

$$\frac{R[x]}{(f)} \hookrightarrow \frac{K[x]}{(f)_K}.$$

¹⁶‘For $i \gg 0$ ’ is shorthand for $(\exists N \geq 0)(\forall i \geq N) \dots$, that is, ‘for all sufficiently large $i \dots$ ’.

Since the ring on the right is an integral domain (as $(f)_K$ is prime in $K[x]$), so is the ring on the left. This proves that (f) is prime in $R[x]$, and we are done. $\square$

Summarizing, if R is a UFD, then factorization in $R[x]$ is ‘the same as’ factorization in the polynomial ring $K[x]$ over the field of quotients of R . If $f(x) \in R[x]$, then $f(x)$ has a prime factorization in $K[x]$ for the simpler reason that $K[x]$ is a PID, hence a UFD; but if R itself is a UFD, then we know that each of the factors may be assumed to be in $R[x]$ to begin with (cf. Exercise 4.23).

Example 4.18. As mentioned already, Theorem 4.14 implies that several important rings, such as $\mathbb{Z}[x_1, \dots, x_n]$ or $\mathbb{C}[x_1, \dots, x_n]$, are UFDs; for example $\mathbb{Z}[x]$ is a UFD, as announced in §2.3. Further, arguing as in Exercise 1.15 to reduce to the case of finitely many indeterminates, it follows that $\mathbb{Z}[x_1, x_2, \dots]$ is a UFD: this is an example of a non-Noetherian UFD, promised a while back (and illustrating the last missing feature of the picture presented at the beginning of the chapter). $\dashv$

Exercises

4.1. $\triangleright$ Prove Lemma 4.1. [§4.1]

4.2. Let R be a ring, and let I be an ideal of R . Prove or disprove that if I is maximal in R , then $IR[x]$ is maximal in $R[x]$.

4.3. $\triangleright$ Let R be a PID, and let $f \in R[x]$. Prove that f is primitive if and only if it is very primitive. Prove that this is not necessarily the case in an arbitrary UFD. [§4.1]

4.4. $\triangleright$ Let R be a commutative ring, and let $f, g \in R[x]$. Prove that

$$fg \text{ is very primitive} \iff \text{both } f \text{ and } g \text{ are very primitive.}$$

[§4.1]

4.5. $\triangleright$ Prove Lemma 4.7. [§4.1]

4.6. Let R be a PID, and let K be its field of fractions.

- Prove that every element $c \in K$ can be written as a finite sum

$$c = \sum_i \frac{a_i}{p_i^{r_i}}$$

where the p_i are nonassociate irreducible elements in R , $r_i \geq 0$, and a_i, p_i are relatively prime.

- If $\sum_i \frac{a_i}{p_i^{r_i}} = \sum_j \frac{b_j}{q_j^{s_j}}$ are two such expressions, prove that (up to reshuffling) $p_i = q_i$, $r_i = s_i$, and $a_i \equiv b_i \pmod{p_i^{r_i}}$.
- Relate this to the process of integration by ‘partial fractions’ you learned about when you took calculus.

4.7. $\triangleright$ A subset S of a commutative ring R is a *multiplicative subset* (or *multiplicatively closed*) if (i) $1 \in S$ and (ii) $s, t \in S \implies st \in S$. Define a relation on the set of pairs (a, s) with $a \in R, s \in S$ as follows:

$$(a, s) \sim (a', s') \iff (\exists t \in S), t(s'a - sa') = 0.$$

Note that if R is an integral domain and $S = R \setminus \{0\}$, then S is a multiplicative subset, and the relation agrees with the relation introduced in §4.2.

- Prove that the relation $\sim$ is an *equivalence* relation.
- Denote by $\frac{a}{s}$ the equivalence class of (a, s) , and define the same operations $+$, $\cdot$ on such ‘fractions’ as the ones introduced in the special case of §4.2. Prove that these operations are well-defined.
- The set $S^{-1}R$ of fractions, endowed with the operations $+$, $\cdot$, is the *localization*¹⁷ of R at the multiplicative subset S . Prove that $S^{-1}R$ is a commutative ring and that the function $a \mapsto \frac{a}{1}$ defines a ring homomorphism $\ell : R \rightarrow S^{-1}R$.
- Prove that $\ell(s)$ is invertible for every $s \in S$.
- Prove that $R \rightarrow S^{-1}R$ is initial among ring homomorphisms $f : R \rightarrow R'$ such that $f(s)$ is invertible in R' for every $s \in S$.
- Prove that $S^{-1}R$ is an integral domain if R is an integral domain.
- Prove that $S^{-1}R$ is the zero-ring if and only if $0 \in S$.

[4.8, 4.9, 4.11, 4.15, VII.2.16, VIII.1.4, VIII.2.5, VIII.2.6, VIII.2.12, §IX.9.1]

4.8. $\neg$ Let S be a multiplicative subset of a commutative ring R , as in Exercise 4.7. For every R -module M , define a relation $\sim$ on the set of pairs (m, s) , where $m \in M$ and $s \in S$:

$$(m, s) \sim (m', s') \iff (\exists t \in S), t(s'm - sm') = 0.$$

Prove that this is an equivalence relation, and define an $S^{-1}R$ -module structure on the set $S^{-1}M$ of equivalence classes, compatible with the R -module structure on M . The module $S^{-1}M$ is the *localization* of M at S . [4.9, 4.11, 4.14, VIII.1.4, VIII.2.5, VIII.2.6]

4.9. $\neg$ Let S be a multiplicative subset of a commutative ring R , and consider the localization operation introduced in Exercises 4.7 and 4.8.

- Prove that if I is an ideal of R such that $I \cap S = \emptyset$, then¹⁸ $I^e := S^{-1}I$ is a proper ideal of $S^{-1}R$.
- If $\ell : R \rightarrow S^{-1}R$ is the natural homomorphism, prove that if J is a proper ideal of $S^{-1}R$, then $J^c := \ell^{-1}(J)$ is an ideal of R such that $J^c \cap S = \emptyset$.
- Prove that $(J^c)^e = J$, while $(I^e)^c = \{a \in R \mid (\exists s \in S) sa \in I\}$.
- Find an example showing that $(I^e)^c$ need not equal I , even if $I \cap S = \emptyset$. (Hint: Let $S = \{1, x, x^2, \dots\}$ in $R = \mathbb{C}[x, y]$. What is $(I^e)^c$ for $I = (xy)$?)

[4.10, 4.14]

¹⁷The terminology is motivated by applications to algebraic geometry; see Exercise VII.2.17.

¹⁸The superscript e stands for ‘extension’ (of the ideal from a smaller ring to a larger ring); the superscript c in the next point stands for ‘contraction’.

4.10. $\neg$ With notation as in Exercise 4.9, prove that the assignment $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ gives an inclusion-preserving bijection between the set of *prime* ideals of R disjoint from S and the set of prime ideals of $S^{-1}R$. (Prove that $(\mathfrak{p}^e)^c = \mathfrak{p}$ if $\mathfrak{p}$ is a prime ideal disjoint from S .) [4.16]

4.11. $\neg$ (Notation as in Exercise 4.7 and 4.8.) A ring is said to be *local* if it has a single maximal ideal.

Let R be a commutative ring, and let $\mathfrak{p}$ be a prime ideal of R . Prove that the set $S = R \setminus \mathfrak{p}$ is multiplicatively closed. The localizations $S^{-1}R$, $S^{-1}M$ are then denoted $R_{\mathfrak{p}}$, $M_{\mathfrak{p}}$.

Prove that there is an inclusion-preserving bijection between the prime ideals of $R_{\mathfrak{p}}$ and the prime ideals of R contained in $\mathfrak{p}$. Deduce that $R_{\mathfrak{p}}$ is a local ring¹⁹. [4.12, 4.13, VI.5.5, VII.2.17, VIII.2.21]

4.12. $\neg$ (Notation as in Exercise 4.11.) Let R be a commutative ring, and let M be an R -module. Prove that the following are equivalent²⁰:

- $M = 0$.
- $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p}$.
- $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

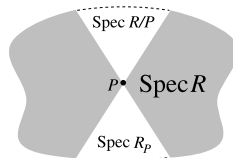
(Hint: For the interesting implication, suppose that $m \neq 0$ in M ; then the ideal $\{r \in R \mid rm = 0\}$ is proper. By Proposition 3.5, it is contained in a maximal ideal $\mathfrak{m}$. What can you say about $M_{\mathfrak{m}}$?) [VIII.1.26, VIII.2.21]

4.13. $\neg$ Let k be a field, and let v be a discrete valuation on k . Let R be the corresponding DVR, with local parameter t (see Exercise 2.20).

- Prove that R is local (Exercise 4.11), with maximal ideal $\mathfrak{m} = (t)$. (Hint: Note that every element of $R \setminus \mathfrak{m}$ is invertible.)
- Prove that k is the field of fractions of R .
- Now let A be a PID, and let $\mathfrak{p}$ be a prime ideal in A . Prove that the localization $A_{\mathfrak{p}}$ (cf. Exercise 4.11) is a DVR. (Hint: If $\mathfrak{p} = (p)$, define a valuation on the field of fractions of A in terms of ‘divisibility by p ’.)

[VII.2.18]

¹⁹We have the following picture in mind associated with the two operations R/P , $R_{\mathfrak{p}}$ for a prime ideal P :



Can you make any sense of this?

²⁰The way to think of this type of results is that a module M over R is zero if and only if it is zero ‘at every point of $\text{Spec } R$ ’. Working in the localization $M_{\mathfrak{p}}$ amounts to looking ‘near’ the point $\mathfrak{p}$ of $\text{Spec } R$; this is what is local about localization. As in this exercise, one can often detect ‘global’ features by looking locally at every point.

4.14. With notation as in Exercise 4.8, define operations $N \mapsto N^e$ and $\hat{N} \mapsto \hat{N}^c$ for submodules $N \subseteq M$, $\hat{N} \subseteq S^{-1}M$, respectively, analogously to the operations defined in Exercise 4.9. Prove that $(\hat{N}^c)^e = \hat{N}$. Prove that every localization of a Noetherian module is Noetherian.

In particular, all localizations $S^{-1}R$ of a Noetherian ring are Noetherian.

4.15. $\neg$ Let R be a UFD, and let S be a multiplicatively closed subset of R (cf. Exercise 4.7).

- Prove that if q is irreducible in R , then $q/1$ is either irreducible or a unit in $S^{-1}R$.
- Prove that if a/s is irreducible in $S^{-1}R$, then a/s is an associate of $q/1$ for some irreducible element q of R .
- Prove that $S^{-1}R$ is also a UFD.

[4.16]

4.16. Let R be a Noetherian integral domain, and let $s \in R$, $s \neq 0$, be a prime element. Consider the multiplicatively closed subset $S = \{1, s, s^2, \dots\}$. Prove that R is a UFD if and only if $S^{-1}R$ is a UFD. (Hint: By Exercise 2.10, it suffices to show that every prime of height 1 is principal. Use Exercise 4.10 to relate prime ideals in R to prime ideals in the localization.)

On the basis of results such as this and of Exercise 4.15, one might suspect that being factorial is a local property, that is, that R is a UFD if and only if $R_{\mathfrak{p}}$ is a UFD for all primes $\mathfrak{p}$, if and only if $R_{\mathfrak{m}}$ is a UFD for all maximal $\mathfrak{m}$. This is regrettably not the case. A ring R is *locally factorial* if $R_{\mathfrak{m}}$ is a UFD for all maximal ideals $\mathfrak{m}$; factorial implies locally factorial by Exercise 4.15, but locally factorial rings that are not factorial do exist.

4.17. $\triangleright$ Let F be a field, and recall the notion of *characteristic* of a ring (Definition III.3.7); the characteristic of a field is either 0 or a prime integer (Exercise III.3.14.)

- Show that F has characteristic 0 if and only if it contains a copy of $\mathbb{Q}$ and that F has characteristic p if and only if it contains a copy of the field $\mathbb{Z}/p\mathbb{Z}$.
- Show that (in both cases) this determines the smallest subfield of F ; it is called the *prime subfield* of F .

[§5.2, §VII.1.1]

4.18. $\neg$ Let R be an integral domain. Prove that the invertible elements in $R[x]$ are the units of R , viewed as constant polynomials. [4.20]

4.19. $\triangleright$ An element $a \in R$ in a ring is said to be *nilpotent* if $a^n = 0$ for some $n \geq 0$. Prove that if a is nilpotent, then $1 + a$ is a unit in R . [VI.7.11, §VII.2.3]

4.20. Generalize the result of Exercise 4.18 as follows: let R be a commutative ring, and let $f = a_0 + a_1x + \dots + a_dx^d \in R[x]$; prove that f is a unit in $R[x]$ if and only if a_0 is a unit in R and $a_1, \dots, a_d$ are nilpotent. (Hint: If $b_0 + b_1x + \dots + b_ex^e$ is the inverse of f , show by induction that $a_d^{i+1}b_{e-i} = 0$ for all $i \geq 0$, and deduce that a_d is nilpotent.)

4.21. ▷ Establish the characterization of irreducible polynomials over a UFD given in Corollary 4.17. [§4.3]

4.22. Let k be a field, and let f, g be two polynomials in $k[x, y] = k[x][y]$. Prove that if f and g have a nontrivial common factor in $k(x)[y]$, then they have a nontrivial common factor in $k[x, y]$.

4.23. ▷ Let R be a UFD, K its field of fractions, $f(x) \in R[x]$, and assume $f(x) = \alpha(x)\beta(x)$ with $\alpha(x), \beta(x)$ in $K[x]$. Prove that there exists a $c \in K$ such that $c\alpha(x) \in R[x]$, $c^{-1}\beta(x) \in R[x]$, so that

$$f(x) = (c\alpha(x))(c^{-1}\beta(x))$$

splits as a product of factors in $R[x]$.

Deduce that if $\alpha(x)\beta(x) = f(x) \in R[x]$ is monic and $\alpha(x) \in K[x]$ is monic, then $\alpha(x), \beta(x)$ are both in $R[x]$ and $\beta(x)$ is also monic. [§4.3, 4.24, §VII.5.2]

4.24. In the same situation as in Exercise 4.23, prove that the product of any coefficient of α with any coefficient of β lies in R .

4.25. Prove *Fermat's last theorem for polynomials*: the equation

$$f^n + g^n = h^n$$

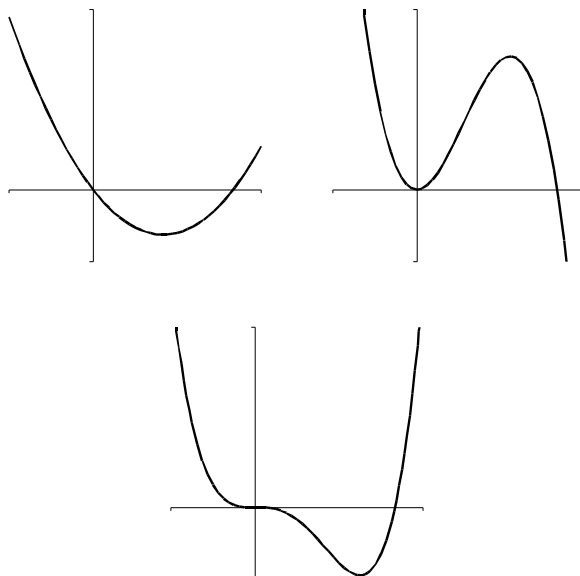
has no solutions in $\mathbb{C}[t]$ for $n > 2$ and f, g, h not all constant. (Hint: First, prove that f, g, h may be assumed to be relatively prime. Next, the polynomial $1 - t^n$ factorizes in $\mathbb{C}[t]$ as $\prod_{i=1}^n (1 - \zeta^i t)$ for $\zeta = e^{2\pi i/n}$; deduce that $f^n = \prod_{i=1}^n (h - \zeta^i g)$. Use unique factorization in $\mathbb{C}[t]$ to conclude that each of the factors $h - \zeta^i g$ is an n -th power. Now let $h - g = a^n$, $h - \zeta g = b^n$, $h - \zeta^2 g = c^n$ (this is where the $n > 2$ hypothesis enters). Use this to obtain a relation $(\lambda a)^n + (\mu b)^n = (\nu c)^n$, where λ, μ, ν are suitable complex numbers. What's wrong with this?)

The same pattern of proof would work in any environment where unique factorization is available; if adjoining to $\mathbb{Z}$ an n -th root of 1 lead to a unique factorization domain, the full-fledged Fermat's last theorem would be as easy to prove as indicated in this exercise. This is not the case, a fact famously missed by G. Lamé as he announced a 'proof' of Fermat's last theorem to the Paris Academy on March 1, 1847.

5. Irreducibility of polynomials

Our work in §4.3, especially Corollary 4.17, links the irreducibility of polynomials over a UFD with their irreducibility over the corresponding field of fractions. For example, irreducibility of polynomials in $\mathbb{Z}[x]$ is 'essentially the same' as irreducibility in $\mathbb{Q}[x]$. This can be useful in both directions, provided we have ways to effectively verify irreducibility of polynomials over one kind of ring or the other. I collect here several remarks aimed at establishing whether a given polynomial is irreducible and briefly discuss the notion of algebraically closed field.

5.1. Roots and reducibility. Let R be a ring and $f \in R[x]$. An element $a \in R$ is a *root* of f if $f(a) = 0$. Recall (Example III.4.7) that a polynomial $f(x) \in R[x]$ is divisible by $(x - a)$ if and only if a is a root of f . More generally, we say that a is a root of f with *multiplicity* r if $(x - a)^r$ divides f and $(x - a)^{r+1}$ does *not* divide f . The reader is probably familiar with this notion from experience with calculus. For example, the graph of polynomials with roots of multiplicities 1, 2, and 3 at 0 look (near the origin), respectively, like



Lemma 5.1. *Let R be an integral domain, and let $f \in R[x]$ be a polynomial of degree n . Then the number of roots of f , counted with multiplicity, is at most n .*

Proof. The number of roots of f in R is less than or equal to the number of roots of f viewed as a polynomial over the field of fractions K of R ; so we may replace R by K .

Now, $K[x]$ is a UFD, and the roots of f correspond to the irreducible factors of f of degree 1. Since the product of *all* irreducible factors of f has degree n , the number of factors of degree 1 can be at most n , as claimed. $\square$

It is worth remembering the trick of replacing an integral domain R by its field of fractions K , used in this proof: one is often able to use convenient properties due to the fact that K is a field (the fact that $K[x]$ is a UFD, in this case), even if R is very far from satisfying such properties ($R[x]$ may not be a UFD, since R itself need not be a UFD).

Also note that the statement of Lemma 5.1 may fail if R is not an integral domain. For example, the degree 2 polynomial $x^2 + x$ has *four* roots over $\mathbb{Z}/6\mathbb{Z}$.

The innocent Lemma 5.1 has important applications: for example, we used it (without much fanfare) in the proof of Theorem IV.6.10. Also, recall that a polynomial $f \in R[x]$ determines an ‘evaluation function’ (cf. Example III.2.3) $R \rightarrow R$, namely $r \mapsto f(r)$. The reader checked in Exercise III.2.7 that in general the function does *not* determine the polynomial; but polynomials *are* determined by the corresponding functions over *infinite* integral domains:

Corollary 5.2. *Let R be an infinite integral domain, and let $f, g \in R[x]$ be polynomials. Then $f = g$ if and only if the evaluation functions $r \mapsto f(r)$, $r \mapsto g(r)$ agree.*

Proof. Indeed, the two functions agree if and only if every $a \in R$ is a root of $f - g$; but a nonzero polynomial over R cannot have infinitely many roots, by Lemma 5.1. $\square$

Clearly, an irreducible polynomial of degree ≥ 2 can have no roots. The converse holds for degree 2 and 3, over a field:

Proposition 5.3. *Let k be a field. A polynomial $f \in k[x]$ of degree 2 or 3 is irreducible if and only if it has no roots.*

Proof. Exercise 5.5. $\square$

Example 5.4. Let $\mathbb{F}_2$ be the field $\mathbb{Z}/2\mathbb{Z}$.

The polynomial $f(t) = t^2 + t + 1 \in \mathbb{F}_2[t]$ is irreducible, since it has no roots: $f(0) = f(1) = 1$. Therefore the ideal $(t^2 + t + 1)$ is prime in $\mathbb{F}_2[t]$, hence maximal (because $\mathbb{F}_2[t]$ is a PID: don’t forget Proposition III.4.13). This gives a one-second construction of a field with four elements:

$$\frac{\mathbb{F}_2[t]}{(t^2 + t + 1)}.$$

The reader (hopefully) constructed this field in Exercise III.1.11, by the tiresome process of searching by hand for a suitable multiplication table. The reader will now have no difficulty constructing much larger examples (cf. Exercise 5.6). $\lrcorner$

Proposition 5.3 is pleasant and can help to decide irreducibility over more general rings: for example, a primitive polynomial $f \in \mathbb{Z}[x]$ of degree 2 or 3 is irreducible if and only if it has no roots in $\mathbb{Q}$. Indeed, f is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$ (by Corollary 4.17). However, note that e.g. $4x^2 - 1 = (2x - 1)(2x + 1)$ is primitive and reducible in $\mathbb{Z}[x]$, although it has no *integer* roots. Also, keep in mind that the statement of Proposition 5.3 may fail for polynomials of degree ≥ 4 : for example, $x^4 + 2x^2 + 1 = (x^2 + 1)^2$ is *reducible* in $\mathbb{Q}[x]$, but it has *no* rational roots.

Looking for rational roots of a polynomial in $\mathbb{Z}[x]$ is in principle a finite endeavor, due to the following observation (which holds over every UFD). This is often called the ‘rational root test’.

Proposition 5.5. *Let R be a UFD, and let K be its field of fractions. Let*

$$f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x],$$

and let $c = \frac{p}{q} \in K$ be a root of f , with $p, q \in R$, $\gcd(p, q) = 1$. Then $p \mid a_0$ and $q \mid a_n$ in R .

Proof. By hypothesis,

$$a_0 + a_1 \frac{p}{q} + \cdots + a_n \frac{p^n}{q^n} = 0;$$

that is,

$$a_0 q^n + a_1 p q^{n-1} + \cdots + a_n p^n = 0.$$

Therefore

$$a_0 q^n = -p(a_1 q^{n-1} + \cdots + a_n p^{n-1}),$$

proving that $p \mid (a_0 q^n)$. Since the $\gcd$ of p and q is one, this implies that the multiset of factors of p is contained in the multiset of irreducible factors of a_0 , that is, $p \mid a_0$.

An entirely similar argument proves that $q \mid a_n$. $\square$

Example 5.6. Looking for *rational* roots of the polynomial

$$3 - 2x + 3x^2 - 2x^3 + 3x^4 - 2x^5$$

is therefore reduced to trying fractions $\frac{p}{q}$ with $q = \pm 1, \pm 2$, $p = \pm 1, \pm 3$. As it happens, $\frac{3}{2}$ is the only root found among these possibilities, and it follows that it is the only rational root of the polynomial. $\lrcorner$

5.2. Adding roots; algebraically closed fields. A homomorphism of *fields*

$$i : k \rightarrow F$$

is necessarily injective (cf. Exercise III.3.10): indeed, its kernel is a proper ideal of k , and the only proper ideal of a field is (0) . In this situation we say that F (or more properly the homomorphism i) is an *extension* of k . Abusing language a little, we then think of k as contained in F : $k \subseteq F$. Keep in mind that this is the case as soon as there is *any* ring homomorphism $k \rightarrow F$.

There is a standard procedure for constructing extensions in which a given polynomial acquires a root; we have encountered an instance of this process in Example III.4.8. In fact, the resulting field is almost universal with respect to this requirement.

Proposition 5.7. *Let k be a field, and let $f(t) \in k[t]$ be a nonzero irreducible polynomial. Then*

$$F := \frac{k[t]}{(f(t))}$$

is a field, endowed with a natural homomorphism $i : k \hookrightarrow F$ (obtained as the composition $k \rightarrow k[x] \rightarrow F$) realizing it as an extension of k . Further,

- $f(x) \in k[x] \subseteq F[x]$ has a root in F , namely the coset of t ;

- if $k \subseteq K$ is any extension in which f has a root, then there exists a homomorphism $j : F \rightarrow K$ such that the diagram

$$\begin{array}{ccc} k & \xrightarrow{\iota} & K \\ & \searrow i \quad \nearrow j & \\ & F & \end{array}$$

commutes.

Proof. Since k is a field, $k[t]$ is a PID; hence $(f(t))$ is a maximal ideal of $k[t]$, by Proposition III.4.13. Therefore F is indeed a field. Denoting cosets in $k[t]/(f(t)) = F$ by underlining, we have

$$f(\underline{t}) = \underline{f(t)} = 0,$$

as claimed.

To verify the second part of the statement, suppose $k \subseteq K$ is an extension and $f(u) = 0$, with $u \in K$. This means that the evaluation homomorphism

$$\epsilon : k[t] \rightarrow K$$

defined by $\epsilon(g(t)) := g(u)$ vanishes at $f(t)$; hence $(f(t)) \subseteq \ker(\epsilon)$, and the universal property of quotients gives a unique homomorphism

$$j : F = \frac{k[t]}{(f(t))} \rightarrow K$$

satisfying the stated requirement. $\square$

I have stopped short of claiming that the extension constructed in Proposition 5.7 is universal with respect to the requirement of containing a root of f because the homomorphism j appearing in the statement is not *unique*: in fact, the proof shows that there are as many such homomorphisms as there are roots of f in the larger field K . We could say that F is *versal*, meaning ‘universal without uniqueness’. We would have full universality if we included the information of the root of f ; we will come back to this construction in Chapter VII, when we analyze field extensions in greater depth.

Example 5.8. For $k = \mathbb{R}$ and $f(x) = x^2 + 1$, the field constructed in Proposition 5.7 is (isomorphic to) $\mathbb{C}$: this was checked carefully in Example III.4.8.

Similarly, $\mathbb{Q}[t]/(t^2 - 2)$ produces a field containing $\mathbb{Q}$ and in which there is a ‘square root of 2’. There are two embeddings of this field in $\mathbb{R}$, because $\mathbb{R}$ contains two distinct square roots of 2: $\pm\sqrt{2}$. $\lrcorner$

The fact that the irreducible polynomial $f \in k[x]$ acquires a root in the extension F constructed in Proposition 5.7 implies that f has a linear (i.e., degree 1) factor over F ; in particular, if $\deg(f) > 1$, then f is no longer irreducible over F .

Given any polynomial $f \in k[x]$, it is easy to construct an extension of k in which f factors completely as a product of linear factors (Exercise 5.13). The case in which this already happens in k itself for every nonzero f is very important, and hence it is given a name.

Definition 5.9. A field k is *algebraically closed* if all irreducible polynomials in $k[x]$ have degree 1. $\lrcorner$

We have encountered this notion in passing, back in Example III.4.14 (and Exercise III.4.21). The following lemma is left to the reader:

Lemma 5.10. *A field k is algebraically closed if and only if every nonconstant polynomial $f \in k[x]$ factors completely as a product of linear factors, if and only if every nonconstant polynomial $f \in k[x]$ has a root in k .*

In other words, a field k is algebraically closed if the only polynomial equations with coefficients in k and *no* solutions are equations ‘of degree 0’, such as $1 = 0$. The result known as *Nullstellensatz* (also due to David Hilbert; I have mentioned this result in Chapter III) is a vast generalization of this observation and one of the pillars of (old-fashioned) algebraic geometry. We will come back to all of this in §VII.2.

Is any of the finite fields we have encountered (such as $\mathbb{Z}/p\mathbb{Z}$, for p prime) algebraically closed? No. Adapting Euclid’s argument proving the infinitude of prime integers (Exercise 2.24) reveals that algebraically closed fields are infinite.

Proposition 5.11. *Let k be an algebraically closed field. Then k is infinite.*

Proof. By contradiction, assume that k is algebraically closed and finite; let the elements of k be $c_1, \dots, c_N$.

Then there are exactly N irreducible monic polynomials in $k[x]$, namely $(x - c_1), \dots, (x - c_N)$. Consider the polynomial

$$f(x) = (x - c_1) \cdots (x - c_N) + 1 :$$

for all $c \in k$ we have

$$f(c) = (c - c_1) \cdots (c - c_N) + 1 = 1 \neq 0,$$

since c equals one of the c_i ’s. Thus $f(x)$ is a nonconstant polynomial with no roots, contradicting Lemma 5.10. $\square$

Since finite fields are not algebraically closed, the reader may suspect that algebraically closed fields necessarily have characteristic 0 (cf. Exercise 4.17). This is not so: there are algebraically closed fields of any characteristic. In fact, as we will see in due time, *every* field F may be embedded into an algebraically closed field; the smallest such extension is called the ‘algebraic closure’ of F and is denoted $\overline{F}$. Thus, for example, $\overline{\mathbb{Z}/2\mathbb{Z}}$ is an algebraically closed field of characteristic 2.

The algebraic closure $\overline{\mathbb{Q}}$ is a *countable* subfield of $\mathbb{C}$. The extension $\mathbb{Q} \subseteq \overline{\mathbb{Q}}$, and especially its ‘Galois group’, cf. §VII.6, is considered one of the most important objects of study in mathematics (cf. the discussion following Corollary VII.7.6).

We will construct the algebraic closure of a field rather explicitly, in §VII.2.1.

5.3. Irreducibility in $\mathbb{C}[x]$, $\mathbb{R}[x]$, $\mathbb{Q}[x]$. Every polynomial $f \in \mathbb{C}[x]$ factors completely over $\mathbb{C}$. Indeed,

Theorem 5.12. *$\mathbb{C}$ is algebraically closed.*

Gauss is credited with providing the first proof²¹ of this fundamental theorem (which is indeed known as the *fundamental theorem of algebra*.)

‘Algebraic’ proofs of the fundamental theorem of algebra require more than we know at this point (we will encounter one in §VII.7.1, after we have seen a little Galois theory); curiously, a little *complex analysis* makes the statement nearly trivial. Here is a sketch of such an argument. Let $f \in \mathbb{C}[x]$ be a nonconstant polynomial; the task (cf. Lemma 5.10) consists of proving that f has a root in $\mathbb{C}$. Whatever $f(0)$ is, we can find an $r \in \mathbb{R}$ large enough that $|f(z)| > |f(0)|$ for all z on the circle $|z| = r$ (since f is nonconstant, $\lim_{z \rightarrow \infty} |f(z)| = +\infty$). The disk $|z| \leq r$ is compact, so the continuous function $|f(z)|$ has a minimum on it; by the choice of r , it must be somewhere in the interior of the disk, say at $z = a$. The *minimum modulus principle* (cf. Exercise 5.16) then implies that $f(a) = 0$, q.e.d.

By Theorem 5.12, irreducibility of polynomials in $\mathbb{C}[x]$ is as simple as it can be: a nonconstant polynomial $f \in \mathbb{C}[x]$ is irreducible if and only if it has degree 1. Every $f \in \mathbb{C}[x]$ of degree ≥ 2 is reducible.

Over $\mathbb{R}$, the situation is almost as simple:

Proposition 5.13. *Every polynomial $f \in \mathbb{R}[x]$ of degree ≥ 3 is reducible.*

The nonconstant irreducible polynomials in $\mathbb{R}[x]$ are precisely the polynomials of degree 1 and the quadratic polynomials

$$f = ax^2 + bx + c$$

with $b^2 - 4ac < 0$.

Proof. Let $f \in \mathbb{R}[x]$ be a nonconstant polynomial:

$$f = a_0 + a_1x + \cdots + a_nx^n,$$

with all $a_i \in \mathbb{R}$. By Theorem 5.12, f has a *complex* root z :

$$a_0 + a_1z + \cdots + a_nz^n = 0.$$

Applying complex conjugation $z \mapsto \bar{z}$ and noting that $\bar{a}_i = a_i$ since $a_i \in \mathbb{R}$,

$$a_0 + a_1\bar{z} + \cdots + a_n\bar{z}^n = \overline{a_0 + a_1z + \cdots + a_nz^n} = \overline{0} = 0 :$$

this says that $\bar{z}$ is also a root of f . There are two possibilities:

- either $z = \bar{z}$, that is, $z = r$ was real to begin with, and hence $(x - r)$ is a factor of f in $\mathbb{R}[x]$; or
- $z \neq \bar{z}$, and then $(x - z)$ and $(x - \bar{z})$ are nonassociate irreducible factors of f in $\mathbb{C}[x]$. In this case (since $\mathbb{C}[x]$ is a UFD!)

$$x^2 - (z + \bar{z})x + z\bar{z} = (x - z)(x - \bar{z})$$

divides f .

²¹ Actually, Gauss’s first proof apparently had a gap; the first rigorous proof is due to Argand. Later, Gauss fixed the gap in his proof and provided several other proofs.

Since $z + \bar{z}$ and $z\bar{z}$ are both real numbers, this analysis shows that every nonconstant $f \in \mathbb{R}[x]$ has an irreducible factor of degree ≤ 2 , proving the first statement.

The second statement then follows immediately from Proposition 5.3 and the fact that the quadratic polynomials in $\mathbb{R}[x]$ with no real roots are precisely the polynomials $ax^2 + bx + c$ with $b^2 - 4ac < 0$. $\square$

The following remark is an immediate consequence of Proposition 5.13 and is otherwise evident from elementary calculus (Exercise 5.17):

Corollary 5.14. *Every polynomial $f \in \mathbb{R}[x]$ of odd degree has a real root.*

Summarizing, the issue of irreducibility of polynomials in $\mathbb{C}[x]$ or $\mathbb{R}[x]$ is very straightforward. By contrast, irreducibility in $\mathbb{Q}[x]$ is a complicated business: as we have seen (Proposition 4.16), it is just as subtle as irreducibility in $\mathbb{Z}[x]$. There is no sharp characterization of irreducibility in $\mathbb{Z}[x]$; but the following simple-minded remarks (and Eisenstein's criterion, discussed in §5.4) suffice in many interesting cases.

One simple-minded remark is that if $\varphi : R \rightarrow S$ is a homomorphism, and a is *reducible* in R , then $\varphi(a)$ is likely to be *reducible* in S . This is just because if $a = bc$, then $\varphi(a) = \varphi(b)\varphi(c)$.

Of course this is not quite right: for example because $\varphi(a)$ and/or $\varphi(b)$ and/or $\varphi(c)$ may be units in S . But with due care for special cases this observation may be used to great effect, especially in the contrapositive form: if $\varphi(a)$ is *irreducible* in S , then a is (likely...) irreducible in R .

Here is a simple statement formalizing these remarks, for $R = \mathbb{Z}[x]$. The natural projection $\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ induces a surjective homomorphism $\pi : \mathbb{Z}[x] \rightarrow \mathbb{Z}/p\mathbb{Z}[x]$: quite simply, $\pi(f)$ is obtained from f by reading all coefficients modulo p ; I will write ' $f \bmod p$ ' for the result of this operation.

Proposition 5.15. *Let $f \in \mathbb{Z}[x]$ be a primitive polynomial, and let p be a prime integer. Assume $f \bmod p$ has the same degree as f and is irreducible in $\mathbb{Z}/p\mathbb{Z}[x]$. Then f is irreducible in $\mathbb{Z}[x]$.*

Proof. Argue contrapositively: if f is primitive and reducible in $\mathbb{Z}[x]$ and $\deg f = n$, then $f = gh$ with $\deg g = d$, $\deg h = e$, $d + e = n$, and both d, e , positive. But then the same can be said of $f \bmod p$, so $f \bmod p$ is also reducible. $\square$

The hypothesis on degrees is necessary, precisely to take care of the 'special cases' mentioned in the discussion preceding the statement. For example, $(2x^3 + 3x^2 + 3x + 1)$ equals $(x^2 + x + 1) \bmod 2$, and the latter is irreducible in $\mathbb{Z}/2\mathbb{Z}[x]$; yet $(2x + 1)$ is a factor of the former. The point is of course that $2x + 1$ is a unit $\bmod 2$.

Warning: As we will see in a fairly distant future (Example VII.5.3), there are irreducible polynomials in $\mathbb{Z}[x]$ which are reducible modulo all primes! So Proposition 5.15 cannot be turned into a characterization of irreducibility in $\mathbb{Z}[x]$. It is however rather useful in producing examples. For instance,

Corollary 5.16. *There are irreducible polynomials in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ of arbitrarily large degree.*

Proof. By Proposition 4.16, the statement for $\mathbb{Z}[x]$ implies the one for $\mathbb{Q}[x]$. By Proposition 5.15, it suffices to verify that there are irreducible polynomials in $\mathbb{Z}/p\mathbb{Z}[x]$ of arbitrarily large degree, for any prime integer p . This is a particular case of Exercise 5.11. $\square$

5.4. Eisenstein's criterion. I am giving this result in a separate subsection only on account of the fact that it is very famous; it is also very simple-minded, like the considerations leading up to Proposition 5.15, and further it is not really a 'criterion', in the sense that it also does not give a characterization of irreducibility.

The result is usually stated for $\mathbb{Z}$, but it holds for every ring R .

Proposition 5.17. *Let R be a (commutative) ring, and let $\mathfrak{p}$ be a prime ideal of R . Let*

$$f = a_0 + a_1x + \cdots + a_nx^n \in R[x]$$

be a polynomial, and assume that

- $a_n \notin \mathfrak{p}$;
- $a_i \in \mathfrak{p}$ for $i = 0, \dots, n-1$;
- $a_0 \notin \mathfrak{p}^2$.

Then f is not the product of polynomials of degree $< n$ in $R[x]$.

Proof. Argue by contradiction. Assume $f = gh$ in $R[x]$, with both $d = \deg g$ and $e = \deg h$ less than $n = \deg f$; write

$$g = b_0 + b_1x + \cdots + b_dx^d, \quad h = c_0 + c_1x + \cdots + c_ex^e,$$

and note that necessarily $d > 0$ and $e > 0$. Consider f modulo $\mathfrak{p}$: thus

$$\underline{f} = \underline{g}\underline{h} \quad \text{in } (R/\mathfrak{p})[x],$$

where $\underline{f}$ denotes f modulo $\mathfrak{p}$, etc.

By hypothesis, $\underline{f} = \underline{a}_n x^n$ modulo $\mathfrak{p}$, where $\underline{a}_n \neq 0$ in $R/\mathfrak{p}$. Since $R/\mathfrak{p}$ is an integral domain, factors of $\underline{f}$ must also be monomials: that is, necessarily

$$\underline{g} = \underline{b}_d x^d, \quad \underline{h} = \underline{c}_e x^e.$$

Since $d > 0$, $e > 0$, this implies $b_0 \in \mathfrak{p}$, $c_0 \in \mathfrak{p}$.

But then $a_0 = b_0 c_0 \in \mathfrak{p}^2$, contradicting the hypothesis. $\square$

For example, $x^4 + 2x^2 + 2$ must be irreducible in $\mathbb{Z}[x]$ (and hence in $\mathbb{Q}[x]$): apply Eisenstein's criterion with $R = \mathbb{Z}$, $\mathfrak{p} = (2)$. More exciting applications involve whole classes of polynomials:

Example 5.18. For all n and all primes p , the polynomial $x^n - p$ is irreducible in $\mathbb{Z}[x]$. This follows immediately from Eisenstein's criterion and gives an alternative proof of Corollary 5.16. $\lrcorner$

Example 5.19. This is probably the most famous application of Eisenstein's criterion. Let p be a prime integer, and let

$$f(x) = 1 + x + x^2 + \cdots + x^{p-1} \in \mathbb{Z}[x].$$

These polynomials are called *cyclotomic*; we will encounter them again in §VII.5.2.

It may not look like it, but Eisenstein's criterion may be used to prove that $f(x)$ is irreducible. The trick (which the reader should endeavor to remember) is to apply the shift $x \rightarrow x + 1$. It is hopefully clear that $f(x)$ is irreducible if and only if $f(x + 1)$ is; thus we are reduced to showing that

$$f(x + 1) = 1 + (x + 1) + (x + 1)^2 + \cdots + (x + 1)^{p-1}$$

is irreducible. This is better than it looks: since $f(x) = (x^p - 1)/(x - 1)$,

$$f(x + 1) = \frac{(x + 1)^p - 1}{(x + 1) - 1} = x^{p-1} + \binom{p}{p-1}x^{p-2} + \cdots + \binom{p}{3}x^2 + \binom{p}{2}x + \binom{p}{1};$$

Eisenstein's criterion proves that this is irreducible, since $\binom{p}{1} = p$ and

Claim 5.20. For p prime and $k = 1, \dots, p - 1$, p divides $\binom{p}{k}$.

There is nothing to this, because

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}$$

and p divides the numerator and does not divide the denominator; the claim follows as p is prime. It may be worthwhile noting that this fact does *not* hold if p is not prime: for example, 4 does not divide $\binom{4}{2}$. $\square$

Exercises

5.1. $\neg$ Let $f(x) \in \mathbb{C}[x]$. Prove that $a \in \mathbb{C}$ is a root of f with multiplicity r if and only if $f(a) = f'(a) = \cdots = f^{(r-1)}(a) = 0$ and $f^{(r)}(a) \neq 0$, where $f^{(k)}(a)$ denotes the value of the k -th derivative of f at a . Deduce that $f(x) \in \mathbb{C}[x]$ has multiple roots if and only if $\gcd(f(x), f'(x)) \neq 1$. [5.2]

5.2. Let F be a subfield of $\mathbb{C}$, and let $f(x)$ be an irreducible polynomial in $F[x]$. Prove that $f(x)$ has no multiple roots in $\mathbb{C}$. (Use Exercises 2.22 and 5.1.)

5.3. Let R be a ring, and let $f(x) = a_{2n}x^{2n} + a_{2n-2}x^{2n-2} + \cdots + a_2x^2 + a_0 \in R[x]$ be a polynomial only involving *even* powers of x . Prove that if $g(x)$ is a factor of $f(x)$, so is $g(-x)$.

5.4. Show that $x^4 + x^2 + 1$ is reducible in $\mathbb{Z}[x]$. Prove that it has *no* rational roots, without finding its (complex) roots.

5.5. $\triangleright$ Prove Proposition 5.3. [§5.1]

5.6. $\triangleright$ Construct fields with 27 elements and with 121 elements. [§5.1]

5.7. Let R be an integral domain, and let $f(x) \in R[x]$ be a polynomial of degree d . Prove that $f(x)$ is determined by its value at any $d + 1$ distinct elements of R .

5.8. $\neg$ Let K be a field, and let $a_0, \dots, a_d$ be distinct elements of K . Given any elements $b_0, \dots, b_d$ in K , construct explicitly a polynomial $f(x) \in K[x]$ of degree at most d such that $f(a_0) = b_0, \dots, f(a_d) = b_d$, and show that this polynomial is unique. (Hint: First solve the problem assuming that only one b_i is not equal to zero.) This process is called *Lagrange interpolation*. [5.9]

5.9. $\neg$ Pretend you can factor integers, and then use Lagrange interpolation (cf. Exercise 5.8) to give a finite algorithm to factor *polynomials*²² with integer coefficients over $\mathbb{Q}[x]$. Use your algorithm to factor $(x - 1)(x - 2)(x - 3)(x - 4) + 1$. [5.10]

5.10. Prove that the polynomial $(x - 1)(x - 2) \cdots (x - n) - 1$ is irreducible in $\mathbb{Q}[x]$ for all $n \geq 1$. (Hint: Think along the lines of Exercise 5.9.)

5.11. $\triangleright$ Let F be a finite field. Prove that there are irreducible polynomials in $F[x]$ of arbitrarily high degree. (Hint: Exercise 2.24.) [§5.3]

5.12. Prove that applying the construction in Proposition 5.7 to an irreducible *linear* polynomial in $k[x]$ produces a field isomorphic to k .

5.13. $\triangleright$ Let k be a field, and let $f \in k[x]$ be any polynomial. Prove that there is an extension $k \subseteq F$ in which f factors completely as a product of linear terms. [§5.2, §VI.7.3]

5.14. How many different embeddings of the field $\mathbb{Q}[t]/(t^3 - 2)$ are there in $\mathbb{R}$? How many in $\mathbb{C}$?

5.15. Prove Lemma 5.10.

5.16. $\triangleright$ If you know about the ‘maximum modulus principle’ in complex analysis: formulate and prove the ‘minimum modulus principle’ used in the sketch of the proof of the fundamental theorem of algebra. [§5.3]

5.17. $\triangleright$ Let $f \in \mathbb{R}[x]$ be a polynomial of *odd* degree. Use the intermediate value theorem to give an ‘algebra-free’ proof of the fact that f has real roots. [§5.3, §VII.7.1]

5.18. Let $f \in \mathbb{Z}[x]$ be a cubic polynomial such that $f(0)$ and $f(1)$ are odd and with odd leading coefficient. Prove that f is irreducible in $\mathbb{Q}[x]$.

5.19. Give a proof of the fact that $\sqrt{2}$ is not rational by using Eisenstein’s criterion.

5.20. Prove that $x^6 + 4x^3 + 1$ is irreducible by using Eisenstein’s criterion.

5.21. Prove that $1 + x + x^2 + \cdots + x^{n-1}$ is reducible over $\mathbb{Z}$ if n is *not* prime.

5.22. Let R be a UFD, and let $a \in R$ be an element that is not divisible by the square of some irreducible element in its factorization. Prove that $x^n - a$ is irreducible for every integer $n \geq 1$.

5.23. Decide whether $y^5 + x^2y^3 + x^3y^2 + x$ is reducible or irreducible in $\mathbb{C}[x, y]$.

²²It is in fact much harder to factor integers than integer polynomials.

5.24. Prove that $\mathbb{C}[x, y, z, w]/(xw - yz)$ is an integral domain, by using Eisenstein's criterion. (I used this ring as an example in §2.2, but I did not have the patience to prove that it was a domain back then!)

6. Further remarks and examples

6.1. Chinese remainder theorem. Suppose all you know about an integer is its class modulo several numbers; can you reconstruct the integer? Also, if you are given arbitrary classes in $\mathbb{Z}/n\mathbb{Z}$ for several integers n , can you find an $N \in \mathbb{Z}$ satisfying all these congruences simultaneously?

The answer is no in both cases, for trivial reasons. If an integer N satisfies given congruences modulo $n_1, \dots, n_k$, then adding any multiple of $n_1 \cdots n_k$ to N produces an integer satisfying the same congruences (so N cannot be entirely reconstructed from the given data); and there is no integer N such that $N \equiv 1 \pmod{2}$ and $N \equiv 2 \pmod{4}$, so there are plenty of congruences which cannot be simultaneously satisfied.

The *Chinese remainder theorem* (CRT) sharpens these questions so that they can in fact be answered affirmatively, and (in its modern form) it generalizes them to a much broader setting. Its statement is most pleasant for PIDs; it is very impressive²³ even in the limited context of $\mathbb{Z}$. Protoversions of the theorem have already appeared here and there in this book, e.g., Lemma IV.6.1.

I will first give the more general version, which is in some way simpler. Let R be any commutative ring.

Theorem 6.1. *Let $I_1, \dots, I_k$ be ideals of R such that $I_i + I_j = (1)$ for all $i \neq j$. Then the natural homomorphism*

$$\varphi : R \longrightarrow \frac{R}{I_1} \times \cdots \times \frac{R}{I_k}$$

is surjective and induces an isomorphism

$$\tilde{\varphi} : \frac{R}{I_1 \cdots I_k} \xrightarrow{\sim} \frac{R}{I_1} \times \cdots \times \frac{R}{I_k}.$$

The ‘natural’ homomorphism φ is determined by the canonical projections $R \rightarrow R/I_j$ and the universal property of products; the homomorphism $\tilde{\varphi}$ is induced by virtue of the universal property of quotients, since $I_1 \cdots I_k \subseteq I_j$ for all j , hence $I_1 \cdots I_k \subseteq \ker \varphi$. Theorem 6.1 is proven by an induction relying on the following lemma.

Lemma 6.2. *Let $I_1, \dots, I_k$ be ideals of R such that $I_i + I_k = (1)$ for all $i = 1, \dots, k-1$. Then $(I_1 \cdots I_{k-1}) + I_k = (1)$.*

Proof. By hypothesis, for $i = 1, \dots, k-1$ there exists $a_i \in I_k$ such that $1 - a_i \in I_i$. Then

$$(1 - a_1) \cdots (1 - a_{k-1}) \in I_1 \cdots I_{k-1},$$

²³Allegedly, a large number of ‘Putnam’ problems can be solved by clever applications of the CRT.

and

$$1 - (1 - a_1) \cdots (1 - a_{k-1}) \in I_k,$$

because it is a combination of $a_1, \dots, a_{k-1} \in I_k$. $\square$

Since clearly $\ker \varphi = I_1 \cap \cdots \cap I_k$, the second part of the statement of Theorem 6.1 follows immediately from the first part, the ‘first isomorphism theorem’, and the following (independently interesting) observation:

Lemma 6.3. *Let $I_1, \dots, I_k$ be ideals of R such that $I_i + I_j = (1)$ for all $i \neq j$. Then $I_1 \cdots I_k = I_1 \cap \cdots \cap I_k$.*

Proof. By Lemma 6.2, under the stated hypotheses we have that $I_1 \cdots I_{k-1} + I_k = (1)$ for $k \geq 3$. Thus, the general statement is reduced by induction to the case $k = 2$. (By the way, this case is Exercise III.4.5!) Assume I and J are ideals of R such that $I + J = (1)$. The inclusion $IJ \subseteq I \cap J$ holds for all ideals I, J , so the task amounts to proving $I \cap J \subseteq IJ$ when $I + J = (1)$. If $I + J = (1)$, then there exist elements $a \in I, b \in J$ such that $a + b = 1$. But if $r \in I \cap J$, then

$$r = r \cdot 1 = r(a + b) = ra + rb \in IJ :$$

because $ra \in IJ$ as $r \in J$ and $a \in I$, while $rb \in IJ$ as $r \in I$ and $b \in J$. The statement follows. $\square$

Thus, we only need to prove the first part of Theorem 6.1.

Proof of Theorem 6.1. Argue by induction on k . For $k = 1$, there is nothing to show. For $k > 1$, assume the statement is known for fewer ideals. Thus, we may assume that the natural projection induces an isomorphism

$$\frac{R}{I_1 \cdots I_{k-1}} \cong \frac{R}{I_1} \times \cdots \times \frac{R}{I_{k-1}};$$

and all that we have left to prove is that the natural homomorphism

$$R \rightarrow \frac{R}{I_1 \cdots I_{k-1}} \times \frac{R}{I_k}$$

is surjective. By Lemma 6.2, $(I_1 \cdots I_{k-1}) + I_k = (1)$; thus we are reduced to the case of *two* ideals.

Let then I, J be ideals of a commutative ring R , such that $I + J = (1)$, and let $r_I, r_J \in R$; we have to verify that $\exists r \in R$ such that $r \equiv r_I \pmod{I}$ and $r \equiv r_J \pmod{J}$. Since $I + J = (1)$, there are $a \in I, b \in J$ such that $a + b = 1$. Let $r = ar_J + br_I$: then

$$r = ar_J + (1 - a)r_I = r_I + a(r_J - r_I) \equiv r_I \pmod{I}$$

as $a \in I$, and

$$r = (1 - b)r_J + br_I = r_J + b(r_I - r_J) \equiv r_J \pmod{J}$$

as $b \in J$, as needed, and completing the proof. $\square$

In a PID, the CRT takes the following form:

Corollary 6.4. *Let R be a PID, and let $a_1, \dots, a_k \in R$ be elements such that $\gcd(a_i, a_j) = 1$ for all $i \neq j$. Let $a = a_1 \cdots a_k$. Then the function*

$$\varphi: \frac{R}{(a)} \rightarrow \frac{R}{(a_1)} \times \cdots \times \frac{R}{(a_k)}$$

defined by $r + (a) \mapsto (r + (a_1), \dots, r + (a_k))$ is an isomorphism.

This is an immediate consequence of Theorem 6.1, since (in a PID!) $\gcd(a, b) = 1$ if and only if $(a, b) = (1)$ as ideals. This is not the case for arbitrary UFDs, and indeed the natural map

$$\mathbb{Z}[x] \rightarrow \frac{\mathbb{Z}[x]}{(2)} \times \frac{\mathbb{Z}[x]}{(x)}$$

is not surjective (check this!), even though $\gcd(2, x) = 1$. But the kernel of this map is $(2x)$; and this is what can be expected in general from the CRT over UFDs (Exercise 6.6).

As $\mathbb{Z}$ is a PID, Corollary 6.4 holds for $\mathbb{Z}$ and gives the promised answer to a revised version of the questions posed at the beginning of this subsection. In fact, tracing the proof in this case gives an effective procedure to solve simultaneous congruences over $\mathbb{Z}$ (or in fact over any Euclidean domain, as will be apparent from the argument).

To see this more explicitly, let $n_1, \dots, n_k$ be pairwise relatively prime integers, and let $n = n_1 \cdots n_k$; for each i , let $m_i = n/n_i$. Then n_i and m_i are relatively prime²⁴, and we can use the Euclidean algorithm to explicitly find integers a_i, b_i such that

$$a_i n_i + b_i m_i = 1.$$

The numbers $q_i = b_i m_i$ have the property that

$$q_i \equiv 1 \pmod{n_i}, \quad q_i \equiv 0 \pmod{n_j} \quad \forall j \neq i$$

(why?). These integers may be used to solve any given system of congruences modulo $n_1, \dots, n_k$: indeed, if $r_1, \dots, r_k \in \mathbb{Z}$ are given, then

$$N := r_1 q_1 + \cdots + r_k q_k$$

satisfies

$$N \equiv r_1 \cdot 0 + \cdots + r_i \cdot 1 + \cdots + r_k \cdot 0 \equiv r_i \pmod{n_i}$$

for all i .

The same procedure can be applied over any Euclidean domain R ; see Exercise 6.7 for an example.

6.2. Gaussian integers. The rings $\mathbb{Z}$ and $k[x]$ (where k is a field) may be the only examples of Euclidean domains known to our reader. The next most famous example is the ring of *Gaussian integers*; this is a very pretty ring, and it has elementary but striking applications in number theory (one instance of which we will see in §6.3).

²⁴This is a particular case of Lemma 6.2 and is clear anyway from gcd considerations.

Abstractly, we may define this ring as

$$\mathbb{Z}[i] := \frac{\mathbb{Z}[x]}{(x^2 + 1)};$$

and we are going to verify that $\mathbb{Z}[i]$ is a Euclidean domain.

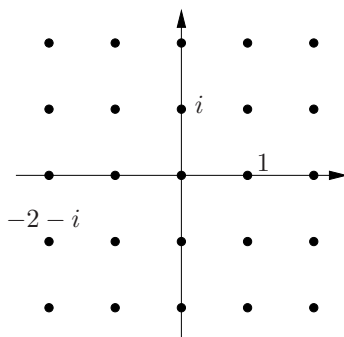
The notation $\mathbb{Z}[i]$ is justified by the discussion in §5.2 (cf. especially Proposition 5.7): $\mathbb{Z}[i]$ may be viewed as the ‘smallest’ ring containing $\mathbb{Z}$ and a root of $x^2 + 1$, that is, a square root i of -1 . The natural embedding

$$\frac{\mathbb{Z}[x]}{(x^2 + 1)} \subseteq \frac{\mathbb{R}[x]}{(x^2 + 1)}$$

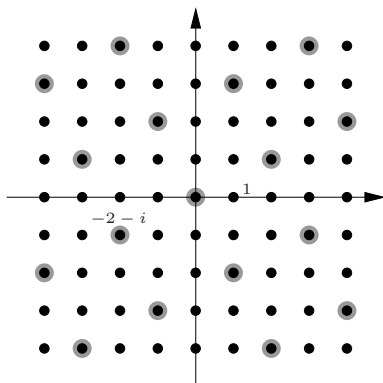
and the identification of the rightmost ring with $\mathbb{C}$ (Example III.4.8) realize $\mathbb{Z}[i]$ as a subring of $\mathbb{C}$; tracing this embedding shows that we could equivalently define

$$\mathbb{Z}[i] = \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}.$$

Thus, the reader should think of $\mathbb{Z}[i]$ as consisting of the complex numbers whose real and imaginary parts are *integers*; this may be referred to as the ‘integer lattice’ in $\mathbb{C}$:



This picture is particularly compelling, because it allows us to ‘visualize’ principal ideals in $\mathbb{Z}[i]$: simple properties of complex multiplication show that the multiples of a fixed $w \in \mathbb{Z}[i]$ form a regular lattice superimposed on $\mathbb{Z}[i]$. For example, the fattened dots in the picture



represent the ideal $(-2 - i)$ in $\mathbb{Z}[i]$: an enlarged, tilted lattice superimposed on the integer lattice in $\mathbb{C}$. The reader should stop reading now and make sure to understand why this works out so neatly.

A Gaussian integer is a complex number, and as such it has a *norm*

$$N(a + bi) := (a + bi)(a - bi) = a^2 + b^2.$$

Geometrically, this is the square of the distance from the origin to $a + bi$.

The norm of a Gaussian integer is a nonnegative integer; thus N is a function

$$\mathbb{Z}[i] \rightarrow \mathbb{Z}^{\geq 0}.$$

Lemma 6.5. *The function N is a Euclidean valuation on $\mathbb{Z}[i]$; further, N is multiplicative in the sense that $\forall z, w \in \mathbb{Z}[i]$*

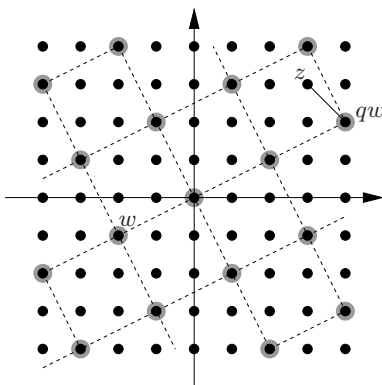
$$N(zw) = N(z)N(w).$$

Proof. The multiplicativity is an immediate consequence of the elementary properties of complex conjugation:

$$N(zw) = (zw)(\overline{zw}) = (z\overline{z})(w\overline{w}) = N(z)N(w).$$

To see that N is a Euclidean valuation, we have to show how to perform a ‘division with remainder’ in $\mathbb{Z}[i]$. It is not hard, but it is a bit messy, to do this algebraically; I will attempt to convince the reader that this is in fact visually evident (and then the reader can have fun producing the needed algebraic computations).

Let $z, w \in \mathbb{Z}[i]$, and assume $w \neq 0$. The ideal (w) is a lattice superimposed on $\mathbb{Z}[i]$. The given z is either one of the vertices of this lattice (in which case z is a multiple of w , so that the division z/w can be performed in $\mathbb{Z}[i]$, with remainder 0) or it sits inside one of the ‘boxes’ of the lattice. In the latter case, pick any of the vertices of that box, that is, a multiple qw of w , and let $r = z - qw$. The situation may look as follows:



Then we have obtained

$$z = qw + r,$$

and the norm of r is the square of the length of the segment between qw and z . Since this segment is contained in a box, and the square of the size of the box is $N(w)$, we have achieved

$$N(r) < N(w),$$

completing the proof. $\square$

As we know, all sorts of amazing properties hold for the ring of Gaussian integers as a consequence of Lemma 6.5: $\mathbb{Z}[i]$ is a PID and a UFD; irreducible elements are prime in $\mathbb{Z}[i]$; greatest common divisors exist and may be found by applying the Euclidean algorithm, etc. Any one of these facts would seem fairly challenging in itself, but they are all immediate consequences of the existence of a Euclidean valuation and of the general considerations in §2.

The fact that the norm is multiplicative simplifies further the analysis of the ring $\mathbb{Z}[i]$.

Lemma 6.6. *The units of $\mathbb{Z}[i]$ are $\pm 1, \pm i$.*

Proof. If u is a unit in $\mathbb{Z}[i]$, then there exists $v \in \mathbb{Z}[i]$ such that $uv = 1$. But then $N(u)N(v) = N(uv) = N(1) = 1$ by multiplicativity, so $N(u)$ is a unit in $\mathbb{Z}$. This implies $N(u) = 1$, and the only elements in $\mathbb{Z}[i]$ with norm 1 are $\pm 1, \pm i$. The statement follows. $\square$

Lemma 6.7. *Let $q \in \mathbb{Z}[i]$ be a prime element. Then there is a prime integer $p \in \mathbb{Z}$ such that $N(q) = p$ or $N(q) = p^2$.*

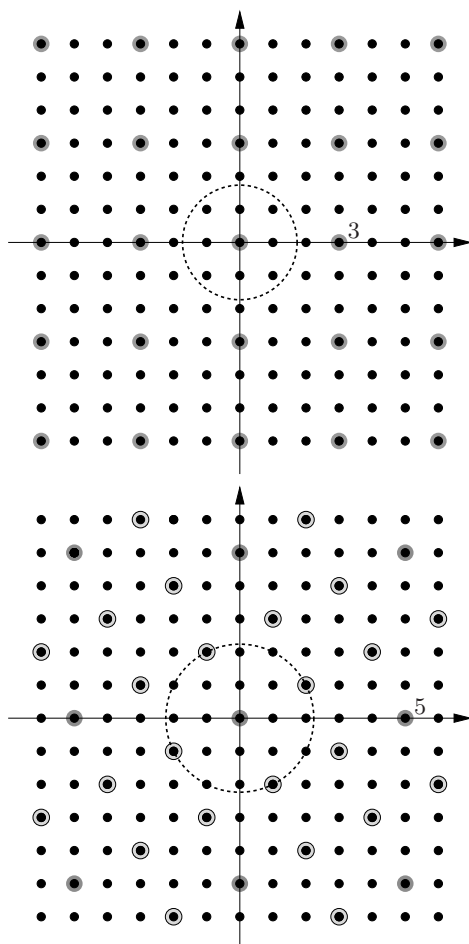
Proof. Since q is not a unit, $N(q) \neq 1$ (by Lemma 6.6). Thus $N(q)$ is a nontrivial product of (integer) primes, and since q is prime in $\mathbb{Z}[i] \supseteq \mathbb{Z}$, q must divide one of the prime integer factors of $N(q)$; let p be this integer prime. But then $q \mid p$ in $\mathbb{Z}[i]$, and it follows (by multiplicativity of the norm) that $N(q) \mid N(p) = p^2$. Since $N(q) \neq 1$, the only possibilities are $N(q) = p$ and $N(q) = p^2$, as claimed. $\square$

Both possibilities presented in Lemma 6.7 occur, and studying this dichotomy further will be key to the result in §6.3. But we can already work out several cases ‘by hand’:

Example 6.8. The prime integer 3 is a prime element of $\mathbb{Z}[i]$; this can be verified by proving that 3 is irreducible in $\mathbb{Z}[i]$ (since $\mathbb{Z}[i]$ is a UFD). For this purpose, note that since $N(3) = 9$, the norm of a factor of 3 would have to be a divisor of 9, that is, 1, 3, or 9. Gaussian integers with norm 1 are units, and those with norm 9 are associates of 3 (Exercise 6.10); thus a nontrivial factor of 3 would necessarily have norm equal to 3. But there are no such elements in $\mathbb{Z}[i]$; hence 3 is indeed irreducible.

The prime integer 5 is *not* a prime element of $\mathbb{Z}[i]$: running through the same argument as we just did for 3, we find that a factor of 5 should have norm 5, and $2 + i$ does. In fact, $5 = (2 + i)(2 - i)$ is a prime factorization of 5 in $\mathbb{Z}[i]$.

Visually, what happens is that the lattice generated by 3 in $\mathbb{Z}[i]$ cannot be refined further into a tighter lattice, while the one generated by 5 does admit a refinement:



(the circles represent complex numbers with norm 3, 5, respectively). $\square$

The reader is encouraged to work out several other examples and to try to figure out what property of an integer prime makes it split in $\mathbb{Z}[i]$ (like 5 and unlike 3). But this should be done before reading on!

6.3. Fermat's theorem on sums of squares. Once armed with our knowledge about rings, playing with $\mathbb{Z}[i]$ a little further teaches us something new about $\mathbb{Z}$ —this is a beautiful and famous example of the success achieved by judicious use of generalization over brute force.

First, let us complete the circle of thought begun with Lemma 6.7. Say that a prime integer p *splits in* $\mathbb{Z}[i]$ if it is *not* a prime element of $\mathbb{Z}[i]$; we have seen in Example 6.8 that 5 splits in $\mathbb{Z}[i]$, while 3 does not.

Lemma 6.9. *A positive integer prime $p \in \mathbb{Z}$ splits in $\mathbb{Z}[i]$ if and only if it is the sum of two squares in $\mathbb{Z}$.*

Proof. First assume that $p = a^2 + b^2$, with $a, b \in \mathbb{Z}$. Then

$$p = (a + bi)(a - bi)$$

in $\mathbb{Z}[i]$, and $N(a \pm bi) = a^2 + b^2 = p \neq 1$, so neither of the two factors is a unit in $\mathbb{Z}[i]$ (Lemma 6.6). Thus p is not irreducible, hence not prime, in $\mathbb{Z}[i]$.

Conversely, assume that p is not irreducible in $\mathbb{Z}[i]$: then it has an irreducible factor $q \in \mathbb{Z}[i]$, which is not an associate of p . Since $q \mid p$, by multiplicativity of the norm we have $N(q) \mid N(p) = p^2$, and hence $N(q) = p$ since q and p are not associates (thus $N(q) \neq p^2$) and q is not a unit (thus $N(q) \neq 1$). If $q = a + bi$, we find

$$p = N(q) = a^2 + b^2,$$

verifying that p is the sum of two squares and completing the proof. $\square$

Thus, 2 splits (as $2 = 1^2 + 1^2$), and so do

$$5 = 1^2 + 2^2, \quad 13 = 2^2 + 3^2, \quad 17 = 1^2 + 4^2, \quad \dots$$

while

$$3, \quad 7, \quad 11, \quad 19, \quad 23, \quad \dots$$

remain prime if viewed as elements of $\mathbb{Z}[i]$.

The next puzzle is as follows: what *else* distinguishes the primes in the first list from the primes in the second list? Again, the reader who does not know the answer already should pause and try to come up with a conjecture and then prove that conjecture! Do not read the next statement before trying this on your own²⁵!

Lemma 6.10. *A positive odd prime integer p splits in $\mathbb{Z}[i]$ if and only if it is congruent to 1 modulo 4.*

Proof. The question is whether p is prime as an element of $\mathbb{Z}[i]$, that is, whether $\mathbb{Z}[i]/(p)$ is an integral domain. But we have isomorphisms

$$\frac{\mathbb{Z}[i]}{(p)} \cong \frac{\mathbb{Z}[x]/(x^2 + 1)}{(p)} \cong \frac{\mathbb{Z}[x]}{(p, x^2 + 1)} \cong \frac{\mathbb{Z}[x]/(p)}{(x^2 + 1)} \cong \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2 + 1)}$$

by virtue of the usual isomorphism theorems (including an appearance of Lemma 4.1) and taking some liberties with the language (for example, (p) means three different

²⁵The point I am trying to make is that these are not difficult facts, and a conscientious reader really *is* in the position of discovering and proving them on his or her own. This is extremely remarkable: the theorem we are heading towards was stated by Fermat, without proof, in 1640, and had to wait about one hundred years before being proven rigorously (by Euler, using ‘infinite descent’). Proofs using Gaussian integers are due to Dedekind and had to wait another hundred years. The moral is, of course, that the modern algebraic apparatus acts as a tremendous amplifier of our skills.

things, hopefully self-clarifying through the context). Therefore,

$$\begin{aligned}
 p \text{ splits in } \mathbb{Z}[i] &\iff \mathbb{Z}[i]/(p) \text{ is not an integral domain} \\
 &\iff \mathbb{Z}/p\mathbb{Z}[x]/(x^2 + 1) \text{ is not an integral domain} \\
 &\iff x^2 + 1 \text{ is not irreducible in } \mathbb{Z}/p\mathbb{Z}[x] \\
 &\iff x^2 + 1 \text{ has a root in } \mathbb{Z}/p\mathbb{Z} \\
 &\iff \text{there is an integer } n \text{ such that } n^2 \equiv -1 \pmod{p},
 \end{aligned}$$

and we are reduced to verifying that this last condition is equivalent to $p \equiv 1 \pmod{4}$, provided that p is an odd prime.

For this purpose, recall (Theorem IV.6.10) that the multiplicative group G of $\mathbb{Z}/p\mathbb{Z}$ is *cyclic*; let $g \in G$ be a generator of G . Since p is assumed to be odd, $p - 1$ is an even number, say 2ℓ : thus g has order $|G| = 2\ell$. Also, denote the class of an integer $n \pmod{p}$ by $\underline{n}$. Since g is a generator of G , for every integer $n \not\equiv 0 \pmod{p}$ there is an integer m such that $\underline{n} = g^m$.

The class $\underline{-1}$ generates the unique subgroup of order 2 of G (unique by the classification of subgroups of a cyclic group, Proposition II.6.11); since g^ℓ does the same, we have

$$g^\ell = \underline{-1}.$$

Therefore, with $\underline{n} = g^m$, we see that $n^2 \equiv -1 \pmod{p}$ if and only if $g^{2m} = g^\ell$, that is, if and only if

$$2m \equiv \ell \pmod{2\ell}.$$

Summarizing, $\exists n \in \mathbb{Z}$ such that $n^2 \equiv -1 \pmod{p}$ if and only if $\exists m \in \mathbb{Z}$ such that $2m \equiv \ell \pmod{2\ell}$. Now this is clearly the case if and only if ℓ is even, that is, if and only if $(p - 1) = 2\ell$ is a multiple of 4, that is, if and only if $p \equiv 1 \pmod{4}$; and we are done. $\square$

Putting together Lemma 6.9 and Lemma 6.10 gives the following beautiful number-theoretic statement:

Theorem 6.11 (Fermat). *A positive odd prime $p \in \mathbb{Z}$ is a sum of two squares if and only if $p \equiv 1 \pmod{4}$.*

Remark 6.12. Lagrange proved (in 1770) that every positive integer is the sum of *four* squares. One way to prove this is not unlike the proof given for Theorem 6.11: it boils down to analyzing the splitting of (integer) primes in a ring; the role of $\mathbb{Z}[i]$ is taken here by a ring of ‘integral quaternions’. $\lrcorner$

The reader should pause and note that there is no mention of UFDs, Euclidean domains, complex numbers, cyclic groups, etc., in the *statement* of Theorem 6.11, although a large selection of these tools was used in its *proof*. This is of course the dream of the generalizer: to set up an abstract machinery making interesting facts (nearly) evident—facts that would be extremely mysterious or that would require Fermat-grade cleverness to be understood without that machinery.

Exercises

6.1. Generalize the CRT for two ideals, as follows. Let I, J be ideals in a commutative ring R ; prove that there is an exact sequence of R -modules

$$0 \longrightarrow I \cap J \longrightarrow R \xrightarrow{\varphi} \frac{R}{I} \times \frac{R}{J} \longrightarrow \frac{R}{I+J} \longrightarrow 0$$

where φ is the natural map. (Also, explain why this implies the first part of Theorem 6.1, for $k = 2$.)

6.2. Let R be a commutative ring, and let $a \in R$ be an element such that $a^2 = a$. Prove that $R \cong R/(a) \times R/(1-a)$.

Show that the multiplication in R endows the ideal (a) with a *ring* structure, with a as the identity²⁶. Prove that $(a) \cong R/(1-a)$ as rings. Prove that $R \cong (a) \times (1-a)$ as rings.

6.3. Recall (Exercise III.3.15) that a ring R is called *Boolean* if $a^2 = a$ for all $a \in R$. Let R be a finite Boolean ring; prove that $R \cong \mathbb{Z}/2\mathbb{Z} \times \cdots \times \mathbb{Z}/2\mathbb{Z}$.

6.4. Let R be a finite commutative ring, and let p be the smallest prime dividing $|R|$. Let $I_1, \dots, I_k$ be proper ideals such that $I_i + I_j = (1)$ for $i \neq j$. Prove that $k \leq \log_p |R|$. (Hint: Prove $|R|^{k-1} \leq |I_1| \cdots |I_k| \leq (|R|/p)^k$.)

6.5. Show that the map $\mathbb{Z}[x] \rightarrow \mathbb{Z}[x]/(2) \times \mathbb{Z}[x]/(x)$ is not surjective.

6.6. $\triangleright$ Let R be a UFD.

- Let $a, b \in R$ such that $\gcd(a, b) = 1$. Prove that $(a) \cap (b) = (ab)$.
- Under the hypotheses of Corollary 6.4 (but only assuming that R is a UFD) prove that the function φ is injective.

[§6.1]

6.7. $\triangleright$ Find a polynomial $f \in \mathbb{Q}[x]$ such that $f \equiv 1 \pmod{x^2+1}$ and $f \equiv x \pmod{x^{100}}$. [§6.1]

6.8. $\neg$ Let $n \in \mathbb{Z}$ be a positive integer and $n = p_1^{a_1} \cdots p_r^{a_r}$ its prime factorization. By the classification theorem for finite abelian groups (or, in fact, simpler considerations; cf. Exercise II.4.9)

$$\frac{\mathbb{Z}}{(n)} \cong \frac{\mathbb{Z}}{(p_1^{a_1})} \times \cdots \times \frac{\mathbb{Z}}{(p_r^{a_r})}$$

as abelian groups.

- Use the CRT to prove that this is in fact a *ring* isomorphism.

²⁶This is an extremely unusual situation. Note that this ring (a) is *not* a subring of R if $a \neq 1$ according to Definition III.2.5, since the identities in (a) and R differ.

- Prove that

$$\left(\frac{\mathbb{Z}}{(n)}\right)^* \cong \left(\frac{\mathbb{Z}}{(p_1^{a_1})}\right)^* \times \cdots \times \left(\frac{\mathbb{Z}}{(p_r^{a_r})}\right)^*$$

(recall that $(\mathbb{Z}/n\mathbb{Z})^*$ denotes the group of units of $\mathbb{Z}/n\mathbb{Z}$).

- Recall (Exercise II.6.14) that Euler's ϕ -function $\phi(n)$ denotes the number of positive integers $\leq n$ that are relatively prime to n . Prove that

$$\phi(n) = p_1^{a_1-1}(p_1 - 1) \cdots p_r^{a_r-1}(p_r - 1).$$

[II.2.15, VII.5.19]

6.9. Let I be a nonzero ideal of $\mathbb{Z}[i]$. Prove that $\mathbb{Z}[i]/I$ is finite.

6.10. $\triangleright$ Let $z, w \in \mathbb{Z}[i]$. Show that if z and w are associates, then $N(z) = N(w)$. Show that if $w \in (z)$ and $N(z) = N(w)$, then z and w are associates. [§6.2]

6.11. Prove that the irreducible elements in $\mathbb{Z}[i]$ are, up to associates: $1 + i$; the integer primes congruent to 3 mod 4; and the elements $a \pm bi$ with $a^2 + b^2$ an integer prime congruent to 1 mod 4.

6.12. $\neg$ Prove Lemma 6.5 without any 'visual' aid. (Hint: Let $z = a + bi$, $w = c + di$ be Gaussian integers, with $w \neq 0$. Then $z/w = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$. Find integers e, f such that $|e - \frac{ac+bd}{c^2+d^2}| \leq \frac{1}{2}$ and $|f - \frac{bc-ad}{c^2+d^2}| \leq \frac{1}{2}$, and set $q = e + if$. Prove that $|\frac{z}{w} - q| < 1$. Why does this do the job?) [6.13]

6.13. $\neg$ Consider the set $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$.

- Prove that $\mathbb{Z}[\sqrt{2}]$ is a ring, isomorphic to $\mathbb{Z}[t]/(t^2 - 2)$.
- Prove that the function $N : \mathbb{Z}[\sqrt{2}] \rightarrow \mathbb{Z}$ defined by $N(a + b\sqrt{2}) = a^2 - 2b^2$ is multiplicative: $N(zw) = N(z)N(w)$. (Cf. Exercise III.4.10.)
- Prove that $\mathbb{Z}[\sqrt{2}]$ has infinitely many units.
- Prove that $\mathbb{Z}[\sqrt{2}]$ is a Euclidean domain, by using the absolute value of N as valuation. (Hint: Follow the same steps as in Exercise 6.12.)

[6.14]

6.14. Working as in Exercise 6.13, prove that $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain. (Use the norm $N(a + b\sqrt{-2}) = a^2 + 2b^2$.)

If you are particularly adventurous, prove that $\mathbb{Z}[(1 + \sqrt{d})/2]$ is also a Euclidean domain²⁷ for $d = -3, -7, -11$. (You can still use the norm defined by $N(a + b\sqrt{d}) = a^2 - db^2$; note that this is still an integer on $\mathbb{Z}[(1 + \sqrt{d})/2]$, if $d \equiv 1 \pmod{4}$.)

The five values $d = -1, -2$, resp., $-3, -7, -11$, are the only ones for which $\mathbb{Z}[\sqrt{d}]$, resp., $\mathbb{Z}[(1 + \sqrt{d})/2]$, is Euclidean. For the values $d = -19, -43, -67, -163$, the ring $\mathbb{Z}[(1 + \sqrt{d})/2]$ is still a PID (cf. §2.4 and Exercise 2.18 for $d = -19$); the fact that there are no other negative values for which the ring of integers in $\mathbb{Q}(\sqrt{d})$ is a PID was conjectured by Gauss and only proven by Alan Baker and Harold

²⁷You are probably wondering why we switched from $\mathbb{Z}[\sqrt{d}]$ to $\mathbb{Z}[(1 + \sqrt{d})/2]$. These rings are the 'rings of integers' in $\mathbb{Q}(\sqrt{d})$; the form they take depends on the class of d modulo 4. Their study is a cornerstone of algebraic number theory.

Stark around 1966. Also, keep in mind that $\mathbb{Z}[\sqrt{-5}]$ is not even a UFD, as you have proved all by yourself in Exercise 1.17.

6.15. Give an elementary proof (using modular arithmetic) of the fact that if an integer n is congruent to 3 modulo 4, then it is not the sum of two squares.

6.16. Prove that if m and n are two integers both of which can be written as sums of two squares, then mn can also be written as the sum of two squares.

6.17. Let n be a positive integer.

- Prove that n is a sum of two squares if and only if it is the norm of a Gaussian integer $a + bi$.
- By factoring $a^2 + b^2$ in $\mathbb{Z}$ and $a + bi$ in $\mathbb{Z}[i]$, prove that n is a sum of two squares if and only if each integer prime factor p of n such that $p \equiv 3 \pmod{4}$ appears with an even power in n .

6.18. $\neg$ One ingredient in the proof of Lagrange's theorem on four squares is the following result, which can be proven by completely elementary means. Let $p > 0$ be an odd prime integer. Then there exists an integer n , $0 < n < p$, such that np may be written as $1 + a^2 + b^2$ for two integers a, b . Prove this result, as follows:

- Prove that the numbers a^2 , $0 \leq a \leq (p-1)/2$, represent $(p+1)/2$ distinct congruence classes mod p .
- Prove the same for numbers of the form $-1 - b^2$, $0 \leq b \leq (p-1)/2$.
- Now conclude, using the pigeon-hole principle.

[6.21]

6.19. $\neg$ Let $\mathbb{I} \subseteq \mathbb{H}$ be the set of quaternions (cf. Exercise III.1.12) of the form $\frac{a}{2}(1 + i + j + k) + bi + cj + dk$ with $a, b, c, d \in \mathbb{Z}$.

- Prove that $\mathbb{I}$ is a (noncommutative) subring of the ring of quaternions.
- Prove that the norm $N(w)$ (Exercise III.2.5) of an integral quaternion $w \in \mathbb{I}$ is an integer and $N(w_1 w_2) = N(w_1)N(w_2)$.
- Prove $\mathbb{I}$ has exactly 24 units in $\mathbb{I}$: $\pm 1, \pm i, \pm j, \pm k$, and $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$.
- Prove that every $w \in \mathbb{I}$ is an associate of an element $a + bi + cj + dk \in \mathbb{I}$ with $a, b, c, d \in \mathbb{Z}$.

The ring $\mathbb{I}$ is called the ring of *integral quaternions*. [6.20, 6.21]

6.20. $\neg$ Let $\mathbb{I}$ be as in Exercise 6.19. Prove that $\mathbb{I}$ shares most good properties of a Euclidean domain, notwithstanding the fact that it is noncommutative.

- Let $z, w \in \mathbb{I}$, with $w \neq 0$. Prove that $\exists q, r \in \mathbb{I}$ such that $z = qw + r$, with $N(r) < N(w)$. (This is a little tricky; don't feel too bad if you have to cheat and look it up somewhere.)
- Prove that every left-ideal in $\mathbb{I}$ is of the form $\mathbb{I}w$ for some $w \in \mathbb{I}$.
- Prove that every $z, w \in \mathbb{I}$, not both zero, have a 'greatest common right-divisor' d in $\mathbb{I}$, of the form $\alpha z + \beta w$ for $\alpha, \beta \in \mathbb{I}$.

[6.21]

6.21. Prove Lagrange's theorem on four squares. Use notation as in Exercises 6.19 and 6.20.

- Let $z \in \mathbb{I}$ and $n \in \mathbb{Z}$. Prove that the greatest common right-divisor of z and n in $\mathbb{I}$ is 1 if and only if $(N(z), n) = 1$ in $\mathbb{Z}$. (If $\alpha z + \beta n = 1$, then $N(\alpha)N(z) = N(1 - \beta n) = (1 - \beta n)(1 - \overline{\beta}n)$, where $\overline{\beta}$ is obtained by changing the signs of the coefficients of i, j, k . Expand, and deduce that $(N(z), n) \mid 1$.)
 - For an odd prime integer p , use Exercise 6.18 to obtain an integral quaternion $z = 1 + ai + bj$ such that $p \mid N(z)$. Prove that z and p have a common right-divisor that is not a unit and not an associate of p .
 - Say that $w \in \mathbb{I}$ is *irreducible* if $w = \alpha\beta$ implies that either α or β is a unit. Prove that integer primes are *not* irreducible in $\mathbb{I}$. Deduce that every positive prime integer is the norm of some integral quaternion.
 - Prove that every positive integer is the norm of some integral quaternion.
 - Finally, use the last point of Exercise 6.19 to deduce that every positive integer may be written as the sum of four perfect squares.
-

